JUNIPER NETWORKS

Junos® OS

SNMP MIBs and Traps Reference

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YEAR 2000 NOTICE

Juniper Networks hardware and software products are Year 2000 compliant. Junos OS has no known time-related limitations through the year 2038. However, the NTP application is known to have some difficulty in the year 2036.

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About This Guide

This preface provides the following guidelines for using the *Junos® OS SNMP MIBs and Traps Reference*:

- Junos OS Documentation and Release Notes on page xxix
- Objectives on page xxix
- Audience on page xxx
- Supported Platforms on page xxx
- Documentation Conventions on page xxxi
- Documentation Feedback on page xxxii
- Requesting Technical Support on page xxxiii

Junos OS Documentation and Release Notes

For a list of related Junos OS documentation, see http://www.juniper.net/techpubs/software/junos/ .

If the information in the latest release notes differs from the information in the documentation, follow the *Junos OS Release Notes.*

To obtain the most current version of all Juniper Networks® technical documentation, see the product documentation page on the Juniper Networks website at http://www.juniper.net/techpubs/ .

Juniper Networks supports a technical book program to publish books by Juniper Networks engineers and subject matter experts with book publishers around the world. These books go beyond the technical documentation to explore the nuances of network architecture, deployment, and administration using the Junos operating system (Junos OS) and Juniper Networks devices. In addition, the Juniper Networks Technical Library, published in conjunction with O'Reilly Media, explores improving network security, reliability, and availability using Junos OS configuration techniques. All the books are for sale at technical bookstores and book outlets around the world. The current list can be viewed at http://www.juniper.net/books .

Objectives

This guide is designed to provide information about the SNMP MIBs and traps supported by the Junos OS.

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NOTE: For additional information about the Junos OS—either corrections to or information that might have been omitted from this guide—see the software release notes at http://www.juniper.net/ .

Audience

This guide is designed for network administrators who are configuring and monitoring a Juniper Networks M Series, MX Series, T Series, EX Series, or J Series router or switch.

To use this guide, you need a broad understanding of networks in general, the Internet in particular, networking principles, and network configuration. You must also be familiar with one or more of the following Internet routing protocols:

- Border Gateway Protocol (BGP)
- Distance Vector Multicast Routing Protocol (DVMRP)
- Intermediate System-to-Intermediate System (IS-IS)
- Internet Control Message Protocol (ICMP) router discovery
- Internet Group Management Protocol (IGMP)
- Multiprotocol Label Switching (MPLS)
- Open Shortest Path First (OSPF)
- Protocol-Independent Multicast (PIM)
- Resource Reservation Protocol (RSVP)
- Routing Information Protocol (RIP)
- Simple Network Management Protocol (SNMP)

Personnel operating the equipment must be trained and competent; must not conduct themselves in a careless, willfully negligent, or hostile manner; and must abide by the instructions provided by the documentation.

Supported Platforms

For the features described in this manual, Junos OS currently supports the following platforms:

- J Series
- M Series
- MX Series
- T Series
- EX Series

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Documentation Conventions

Table 1 on page xxxi defines notice icons used in this guide.

Table 1: Notice Icons

Icon	Meaning	Description
	Informational note	Indicates important features or instructions.
	Caution	Indicates a situation that might result in loss of data or hardware damage.
	Warning	Alerts you to the risk of personal injury or death.
	Laser warning	Alerts you to the risk of personal injury from a laser.

Table 2 on page xxxi defines the text and syntax conventions used in this guide.

Table 2: Text and Syntax Conventions

Convention	Description	Examples
Bold text like this	Represents text that you type.	To enter configuration mode, type the **configure** command: user@host> **configure**
`Fixed-width text like this`	Represents output that appears on the terminal screen.	`user@host>` **show chassis alarms** `No alarms currently active`
Italic text like this	• Introduces important new terms. • Identifies book names. • Identifies RFC and Internet draft titles.	• A policy *term* is a named structure that defines match conditions and actions. • *Junos OS System Basics Configuration Guide* • RFC 1997, *BGP Communities Attribute*
Italic text like this	Represents variables (options for which you substitute a value) in commands or configuration statements.	Configure the machine's domain name: [edit] root@# **set system domain-name** *domain-name*

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Table 2: Text and Syntax Conventions *(continued)*

Convention	Description	Examples
Text like this	Represents names of configuration statements, commands, files, and directories; interface names; configuration hierarchy levels; or labels on routing platform components.	• To configure a stub area, include the **stub** statement at the **[edit protocols ospf area area-id]** hierarchy level. • The console port is labeled **CONSOLE**.
< > (angle brackets)	Enclose optional keywords or variables.	**stub** <**default-metric** *metric*>**;**
| (pipe symbol)	Indicates a choice between the mutually exclusive keywords or variables on either side of the symbol. The set of choices is often enclosed in parentheses for clarity.	**broadcast | multicast** (***string1*** **|** ***string2*** **|** ***string3***)
# (pound sign)	Indicates a comment specified on the same line as the configuration statement to which it applies.	**rsvp { # Required for dynamic MPLS only**
[ ] (square brackets)	Enclose a variable for which you can substitute one or more values.	**community name members [** ***community-ids*** **]**

Indention and braces ({ })	Identify a level in the configuration hierarchy.	[edit]
		routing-options {
		  static {
		  route default {
		  nexthop *address*;
		  retain;
		  }
		  }
		}

; (semicolon)	Identifies a leaf statement at a configuration hierarchy level.	
J-Web GUI Conventions		
Bold text like this	Represents J-Web graphical user interface (GUI) items you click or select.	• In the Logical Interfaces box, select **All Interfaces**. • To cancel the configuration, click **Cancel**.
> (bold right angle bracket)	Separates levels in a hierarchy of J-Web selections.	In the configuration editor hierarchy, select **Protocols>Ospf**.

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- Document or topic name
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- Software release version (if applicable)

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Technical product support is available through the Juniper Networks Technical Assistance Center (JTAC). If you are a customer with an active J-Care or JNASC support contract, or are covered under warranty, and need postsales technical support, you can access our tools and resources online or open a case with JTAC.

- JTAC policies—For a complete understanding of our JTAC procedures and policies, review the JTAC User Guide located at http://www.juniper.net/us/en/local/pdf/resource-guides/7100059-en.pdf .
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For quick and easy problem resolution, Juniper Networks has designed an online self-service portal called the Customer Support Center (CSC) that provides you with the following features:

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- Find solutions and answer questions using our Knowledge Base: http://kb.juniper.net/
- Download the latest versions of software and review release notes: http://www.juniper.net/customers/csc/software/
- Search technical bulletins for relevant hardware and software notifications: https://www.juniper.net/alerts/
- Join and participate in the Juniper Networks Community Forum: http://www.juniper.net/company/communities/
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Opening a Case with JTAC

You can open a case with JTAC on the Web or by telephone.

- Use the Case Management tool in the CSC at http://www.juniper.net/cm/ .
- Call 1-888-314-JTAC (1-888-314-5822 toll-free in the USA, Canada, and Mexico).

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PART 1

Overview

- Understanding Junos OS MIB Support on page 3

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CHAPTER 1

Understanding Junos OS MIB Support

- Understanding the SNMP Implementation in Junos OS on page 3
- Standard SNMP MIBs Supported by Junos OS on page 7
- Juniper Networks Enterprise-Specific MIBs on page 30
- Enterprise-Specific MIBs and Supported Devices on page 42
- Loading MIB Files to a Network Management System on page 51

Understanding the SNMP Implementation in Junos OS

SNMP enables the monitoring of network devices from a central location. This topic provides an overview of SNMP and describes how SNMP is implemented in the Junos OS.

This topic includes the following sections:

- SNMP Architecture on page 3
- Junos OS SNMP Agent Features on page 6

SNMP Architecture

The SNMP agent exchanges network management information with SNMP manager software running on a network management system (NMS), or host. The agent responds to requests for information and actions from the manager. The agent also controls access to the agent's MIB, the collection of objects that can be viewed or changed by the SNMP manager.

The SNMP manager collects information about network connectivity, activity, and events by polling managed devices.

Communication between the agent and the manager occurs in one of the following forms:

- **Get**, **GetBulk**, and **GetNext** requests—The manager requests information from the agent; the agent returns the information in a **Get** response message.
- **Set** requests—The manager changes the value of a MIB object controlled by the agent; the agent indicates status in a **Set** response message.

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- **Traps** notification—The agent sends traps to notify the manager of significant events that occur on the network device.

This topic contains the following sections:

- SNMP MIBs on page 4
- SNMP Traps and Informs on page 4

SNMP MIBs

A MIB is a hierarchy of information used to define managed objects in a network device. The MIB structure is based on a tree structure, which defines a grouping of objects into related sets. Each object in the MIB is associated with an object identifier (OID), which names the object. The "leaf" in the tree structure is the actual managed object instance, which represents a resource, event, or activity that occurs in your network device.

MIBs are either standard or enterprise-specific. Standard MIBs are created by the Internet Engineering Task Force (IETF) and documented in various RFCs. Depending on the vendor, many standard MIBs are delivered with the NMS software. You can also download the standard MIBs from the IETF website, www.ietf.org, and compile them into your NMS, if necessary.

For a list of standard supported MIBs, see "Standard SNMP MIBs Supported by Junos OS" on page 7 in the *Junos OS SNMP MIBs and Traps Reference*.

Enterprise-specific MIBs are developed and supported by a specific equipment manufacturer. If your network contains devices that have enterprise-specific MIBs, you must obtain them from the manufacturer and compile them into your network management software.

For a list of Juniper Networks enterprise-specific supported MIBs, see "Juniper Networks Enterprise-Specific MIBs" on page 30 in the *Junos OS SNMP MIBs and Traps Reference*.

SNMP Traps and Informs

Routers can send notifications to SNMP managers when significant events occur on a network device, most often errors or failures. SNMP notifications can be sent as traps or inform requests. SNMP traps are unconfirmed notifications. SNMP informs are confirmed notifications.

SNMP traps are defined in either standard or enterprise-specific MIBs. Standard traps are created by the IETF and documented in various RFCs. The standard traps are compiled into the network management software. You can also download the standard traps from the IETF website, www.ietf.org .

For more information about standard traps supported by the Junos OS, see "Standard SNMP Traps Supported on Devices Running Junos OS" on page 653 in the *Junos OS SNMP MIBs and Traps Reference*.

Enterprise-specific traps are developed and supported by a specific equipment manufacturer. If your network contains devices that have enterprise-specific traps, you

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must obtain them from the manufacturer and compile them into your network management software.

For more information about enterprise-specific traps supported by the Junos OS, see "Juniper Networks Enterprise-Specific SNMP Traps" on page 633 in the *Junos OS SNMP MIBs and Traps Reference*. For information about system logging severity levels for SNMP traps, see "System Logging Severity Levels for SNMP Traps" on page 6.

With traps, the receiver does not send any acknowledgment when it receives a trap, and the sender cannot determine if the trap was received. To increase reliability, SNMP informs are supported in SNMPv3. An SNMP manager that receives an inform acknowledges the message with a response. For information about SNMP informs, see Configuring SNMP Informs.

SNMP Trap Queuing

The Junos OS supports trap queuing to ensure that traps are not lost because of temporary unavailability of routes. Two types of queues, destination queues and a throttle queue, are formed to ensure delivery of traps and to control the trap traffic.

The Junos OS forms a destination queue when a trap to a particular destination is returned because the host is not reachable, and adds the subsequent traps to the same destination to the queue. The Junos OS checks for availability of routes every 30 seconds and sends the traps from the destination queue in a round-robin fashion. If the trap delivery fails, the trap is added back to the queue, and the delivery attempt counter and the next delivery attempt timer for the queue are reset. Subsequent attempts occur at progressive intervals of 1 minute, 2 minutes, 4 minutes, and 8 minutes. The maximum delay between the attempts is 8 minutes, and the maximum number of attempts is 10. After 10 unsuccessful attempts, the destination queue and all the traps in the queue are deleted.

The Junos OS also has a throttle mechanism to control the number of traps (**throttle threshold**; default value of 500 traps) sent during a particular time period (**throttle interval**; default of 5 seconds) and to ensure consistency in trap traffic, especially when a large number of traps are generated because of interface status changes. The throttle interval period begins when the first trap arrives at the throttle. All traps within the trap threshold are processed, and the traps beyond the threshold limit are queued. The maximum size of trap queues (that is, the throttle queue and the destination queue combined) is 40,000 traps. However, on EX Series switches, the maximum size of the trap queue is 1000 traps. The maximum size of any one queue is 20,000 traps for devices other than EX Series switches. On EX Series switches, the maximum size of one queue is 500 traps. If a trap is sent from a destination queue when the throttle queue has exceeded the maximum size, the trap is added back to the top of the destination queue, and all subsequent attempts from the destination queue are stopped for a 30-second period, after which the destination queue restarts sending the traps.



NOTE: Users cannot configure the Junos OS for trap queuing. Users cannot view any information about trap queues except what is available in the syslog.

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System Logging Severity Levels for SNMP Traps

For some traps, when a trap condition occurs, regardless of whether the SNMP agent sends a trap to an NMS, the trap is logged if the system logging is configured to log an event with that system logging severity level. For more information about system logging severity levels, see the *Junos OS System Basics Configuration Guide*.

For more information about system logging severity levels for standard traps, see "Standard SNMP Version 1 Traps" on page 654 and "Standard SNMP Version 2 Traps" on page 657 in the *Junos OS SNMP MIBs and Traps Reference*. For more information about system logging severity levels for enterprise-specific traps, see "Juniper Networks Enterprise-Specific SNMP Version 1 Traps" on page 634 and "Juniper Networks Enterprise-Specific SNMP Version 2 Traps" on page 641 in the *Junos OS SNMP MIBs and Traps Reference*.

Junos OS SNMP Agent Features

The Junos OS SNMP agent software consists of an SNMP master agent that delegates all SNMP requests to subagents. Each subagent is responsible for the support of a specific set of MIBs.

The Junos OS supports the following versions of SNMP:

- SNMPv1—The initial implementation of SNMP that defines the architecture and framework for SNMP.
- SNMPv2c—The revised protocol, with improvements to performance and manager-to-manager communications. Specifically, SNMPv2c implements community strings, which act as passwords when determining who, what, and how the SNMP clients can access the data in the SNMP agent. The community string is contained in SNMP **Get**, **GetBulk**, **GetNext**, and **Set** requests. The agent may require a different community string for **Get**, **GetBulk**, and **GetNext** requests (**read-only** access) than it does for **Set** requests (**read-write** access).
- SNMPv3—The most up-to-date protocol focuses on security. SNMPv3 defines a security model, user-based security model (USM), and a view-based access control model (VACM). SNMPv3 USM provides data integrity, data origin authentication, message replay protection, and protection against disclosure of the message payload. SNMPv3 VACM provides access control to determine whether a specific type of access (read or write) to the management information is allowed.

In addition, the Junos OS SNMP agent software accepts IPv4 and IPv6 addresses for transport over IPv4 and IPv6. For IPv6, the Junos OS supports the following features:

- SNMP data over IPv6 networks
- IPv6-specific MIB data
- SNMP agents for IPv6

Related Documentation

- SNMPv3 Overview
- Configuring SNMP on a Device Running Junos OS

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- Configuration Statements at the [edit snmp] Hierarchy Level

Standard SNMP MIBs Supported by Junos OS

Table 3 on page 7 contains the list of standard SNMP MIBs and RFCs that are supported on various devices running Junos OS. RFCs can be found at http://www.ietf.org .

NOTE: In this table, a value of 1 in any of the platform columns (M, T, J, MX, EX, and SRX) denotes that the corresponding MIB is supported on that particular platform, and a value of 0 denotes that the MIB is not supported on the platform.

Table 3: Standard MIBs Supported on Devices Running Junos OS

MIB/RFC	Platforms							
	M	T	J	MX	EX	SRX		
						Low-End	Mid-Range	High-End
IEEE 802.1ab section 12.1, *Link Layer Discovery Protocol (LLDP) MIB* Supported tables and objects: • **lldpRemManAddrOID, lldpLocManAddrOID, lldpReinitDelay, lldpNotificationInterval, lldpStatsRxPortFramesDiscardedTotal, lldpStatsRxPortFramesErrors, lldpStatsRxPortTLVsDiscardedTotal, lldpStatsRxPortTLVsUnrecognizedTotal, lldpStatsRxPortAgeoutsTotal**	0	0	0	0	1	0		

Table 3: Standard MIBs Supported on Devices Running Junos OS *(continued)*

MIB/RFC	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
IEEE, 802.3ad, *Aggregation of Multiple Link Segments* Supported tables and objects: • **dot3adAggPortTable**, **dot3adAggPortListTable**, **dot3adAggTable**, and **dot3adAggPortStatsTable** NOTE: EX Series switches do not support the **dot3adAggPortTable** and **dot3adAggPortStatsTable**. • **dot3adAggPortDebugTable** (only **dot3adAggPortDebugRxState**, **dot3adAggPortDebugMuxState**, **dot3adAggPortDebugActorSyncTransitionCount**, **dot3adAggPortDebugPartnerSyncTransitionCount**, **dot3adAggPortDebugActorChangeCount**, and **dot3adAggPortDebugPartnerChangeCount**) NOTE: EX Series switches do not support the **dot3adAggPortDebugTable**. • **dot3adTablesLastChanged** NOTE: Gigabit Ethernet interfaces on J Series Services Routers do not support the 802.3ad MIB.	1	1	1	1	1	1	1	1
RFC 1155, *Structure and Identification of Management Information for TCP/IP-based Internets*	1	1	1	1	1	1	1	1
RFC 1157, *A Simple Network Management Protocol (SNMP)*	1	1	1	1	1	1	1	1

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Table 3: Standard MIBs Supported on Devices Running Junos OS *(continued)*

MIB/RFC	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
RFC 1195, *Use of OSI IS-IS for Routing in TCP/IP and Dual Environments* (only the objects **isisSystem**, **isisMANAreaAddr**, **isisAreaAddr**, **isisSysProtSupp**, **isisSummAddr**, **isisCirc**, **isisCircLevel**, **isisPacketCount**, **isisISAdj**, **isisISAdjAreaAddr**, **isisAdjIPAddr**, **isisISAdjProtSupp**, **isisRa**, and **isisIPRA** are supported)	1	1	1	1	1	1	1	1
RFC 1212, *Concise MIB Definitions*	1	1	1	1	1	0	0	1

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Table 3: Standard MIBs Supported on Devices Running Junos OS *(continued)*

MIB/RFC	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
RFC 1213, *Management Information Base for Network Management of TCP/IP-Based Internets: MIB-II*. Junos OS supports the following areas: • MIB II and its SNMP version 2 derivatives, including: ◦ Statistics counters ◦ IP, except for **ipRouteTable**, which has been replaced by **ipCidrRouteTable** (RFC 2096, *IP Forwarding Table MIB*) ◦ SNMP management ◦ Interface management • SNMPv1 **Get**, **GetNext** requests, and version 2 **GetBulk** request • Junos OS-specific secured access list • Master configuration keywords • Reconfigurations upon SIGHUP	1	1	1	1	1	0	0	1
RFC 1215, *A Convention for Defining Traps for use with the SNMP* (only MIB II SNMP version 1 traps and version 2 notifications)	1	1	1	1	1	0	0	1
RFC 1406, *Definitions of Managed Objects for the DS1 and E1 Interface Types* (T1 MIB is supported)	1	1	1	0	0	1	0	0

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Table 3: Standard MIBs Supported on Devices Running Junos OS *(continued)*

MIB/RFC	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
RFC 1407, *Definitions of Managed Objects for the DS3/E3 Interface Type* (T3 MIB is supported)	1	1	1	0	0	0	0	0
RFC 1471, *Definitions of Managed Objects for the Link Control Protocol of the Point-to-Point Protocol* (only pppLink group is supported. The pppLink group consists of the **pppLcp 1** object and the tables **pppLinkStatustable** and **pppLinkConfigTable**).	1	0	0	1	0	0	0	0
RFC 1657, *Definitions of Managed Objects for the Fourth Version of the Border Gateway Protocol (BGP-4) using SMIv2*	1	1	1	1	1	0	0	0
RFC 1695, *Definitions of Managed Objects for ATM Management Version 8.0 Using SMIv2*	1	1	1	0	0	0	0	0
RFC 1850, *OSPF Version 2 Management Information Base* (except for the **ospfOriginateNewLsas** and **ospfRxNewLsas** objects, the Host Table, and the traps **ospfOriginateLSA**, **ospfLsdbOverflow**, and **ospfLsdbApproachingOverflow**)	1	1	1	1	1	1	0	0
RFC 1901, *Introduction to Community-based SNMPv2*	1	1	1	1	1	1	1	1

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Table 3: Standard MIBs Supported on Devices Running Junos OS *(continued)*

MIB/RFC	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
RFC 2011, *SNMPv2 Management Information Base for the Internet Protocol Using SMIv2*	1	1	1	1	1	0	0	0
RFC 2012, *SNMPv2 Management Information Base for the Transmission Control Protocol Using SMIv2*	1	1	1	1	1	1	0	1
RFC 2013, *SNMPv2 Management Information Base for the User Datagram Protocol Using SMIv2*	1	1	1	1	1	1	0	1
RFC 2024, *Definitions of Managed Objects for Data Link Switching Using SMIv2* (except for the **dlswInterface** and **dlswSdlc** object groups; the **dlswDirLocateMacTable**, **dlswDirNBTable**, and **dlswDirLocateNBTable** tables; the **dlswCircuitDiscReasonLocal** and **dlswCircuitDiscReasonRemote** tabular objects; and the **dlswDirMacCacheNextIndex** and **dlswDirNBCacheNextIndex** scalar objects; read-only access)	1	1	1	1	0	0	0	0
RFC 2096, *IP Forwarding Table MIB* (The **ipCidrRouteTable** has been extended to include the tunnel name when the next hop is through an RSVP-signaled LSP.)	1	1	1	1	1	0	0	1

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Table 3: Standard MIBs Supported on Devices Running Junos OS *(continued)*

MIB/RFC	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
RFC 2115, *Management Information Base for Frame Relay DTEs Using SMIv2* (**frDlcmiTable** only; **frCircuitTable** and **frErrTable** are not supported)	1	1	1	1	0	1	0	0
RFC 2233, *The Interfaces Group MIB Using SMIv2* NOTE: RFC 2233 has been replaced by RFC 2863, IF MIB. However, Junos OS supports both RFC 2233 and RFC 2863.	1	1	1	1	1	1	0	1
RFC 2287, *Definitions of System-Level Managed Objects for Applications* (only the objects **sysApplInstallPkgTable**, **sysApplInstallElmtTable**, **sysApplElmtRunTable**, and **sysApplMapTable**)	1	1	1	1	1	1	0	1
RFC 2465, *Management Information Base for IP Version 6: Textual Conventions and General Group* (except for IPv6 interface statistics)	1	1	1	1	0	1	0	0
RFC 2495, *Definitions of Managed Objects for the DS1, E1, DS2, and E2 Interface Types* (except for **dsx1FarEndConfigTable**, **dsx1FarEndCurrentTable**, **dsx1FarEndIntervalTable**, **dsx1FarEndTotalTable**, and **dsx1FracTable**)	1	1	1	0	0	1	0	0

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Table 3: Standard MIBs Supported on Devices Running Junos OS *(continued)*

MIB/RFC	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
RFC 2515, *Definitions of Managed Objects for ATM Management* (except **atmVpCrossConnectTable**, **atmVcCrossConnectTable**, and **aal5VccTable**)	1	1	1	0	0	0	0	0
RFC 2570, *Introduction to Version 3 of the Internet-standard Network Management Framework*	1	1	1	1	1	0	0	1
RFC 2571, *An Architecture for Describing SNMP Management Frameworks* (read-only access) NOTE: RFC 2571 has been replaced by RFC 3411. However, Junos OS supports both RFC 2571 and RFC 3411.	1	1	1	1	1	1	0	1
RFC 2572, *Message Processing and Dispatching for the Simple Network Management Protocol (SNMP)* (read-only access) NOTE: RFC 2572 has been replaced by RFC 3412. However, Junos OS supports both RFC 2572 and RFC 3412.	1	1	1	1	1	1	0	1

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Table 3: Standard MIBs Supported on Devices Running Junos OS *(continued)*

MIB/RFC	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
RFC 2576, *Coexistence between Version 1, Version 2, and Version 3 of the Internet-standard Network Management Framework* NOTE: RFC 2576 has been replaced by RFC 3584. However, Junos OS supports both RFC 2576 and RFC 3584.	1	1	1	1	1	1	0	1
RFC 2578, *Structure of Management Information Version 2 (SMIv2)*	1	1	1	1	1	0	0	1
RFC 2579, *Textual Conventions for SMIv2*	1	1	1	1	1	0	0	1
RFC 2580, *Conformance Statements for SMIv2*	1	1	1	1	1	0	0	1
RFC 2662, *Definitions of Managed Objects for ADSL Lines* (J Series Services Routers. All MIB tables, objects, and traps are applicable for the ADSL ATU-R agent.)	1	1	1	1	0	1	0	0
RFC 2665, *Definitions of Managed Objects for the Ethernet-like Interface Types*	1	1	1	1	1	1	0	1
RFC 2787, *Definitions of Managed Objects for the Virtual Router Redundancy Protocol* (except row creation, the **Set** operation, and the object **vrrpStatsPacketLengthErrors**)	1	1	1	1	1	1	0	1

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Table 3: Standard MIBs Supported on Devices Running Junos OS *(continued)*

MIB/RFC	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
RFC 2790, *Host Resources MIB* • Only the **hrStorageTable**. The file systems **/**, **/config**, **/var**, and **/tmp** always return the same index number. When SNMP restarts, the index numbers for the remaining file systems might change. • Only the objects of the **hrSystem** and **hrSWInstalled** groups.	1	1	1	1	1	1	0	1
RFC 2819, *Remote Network Monitoring Management Information Base* • **etherStatsTable** (for Ethernet interfaces only),**alarmTable**, **eventTable**, and **logTable** are supported on all devices running Junos OS. • **historyControlTable** and **etherHistoryTable** (except **etherHistoryUtilization** object) are supported only on EX Series switches.	1	1	1	1	1	1	0	1
RFC 2863, *The Interfaces Group MIB* NOTE: RFC 2863 replaces RFC 2233. However, Junos OS supports both RFC 2233 and RFC 2863.	1	1	1	1	1	0	0	1

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Table 3: Standard MIBs Supported on Devices Running Junos OS *(continued)*

MIB/RFC	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
RFC 2864, *The Inverted Stack Table Extension to the Interfaces Group MIB*	1	1	1	1	0	0	0	1
RFC 2922, *The Physical Topology (PTOPO) MIB* Supported objects: **ptopoConnDiscAlgorithm, ptopoConnAgentNetAddrType, ptopoConnAgentNetAddr, ptopoConnMultiMacSASeen, ptopoConnMultiNetSASeen, ptopoConnIsStatic, ptopoConnLastVerifyTime, ptopoConnRowStatus**	0	0	0	0	1	1	0	1
RFC 2925, *Definitions of Managed Objects for Remote Ping, Traceroute, and Lookup Operations* (only the objects **pingCtlTable**, **pingResultsTable**, **pingProbeHistoryTable**, **pingMaxConcurrentRequests**, **traceRouteCtlTable**, **traceRouteResultsTable**, **traceRouteProbeHistoryTable**, and **traceRouteHopsTable**)	1	1	1	1	1	1	0	1
RFC 2932, *IPv4 Multicast Routing MIB*	1	1	1	1	1	1	0	1
RFC 2933, *Internet Group Management Protocol (IGMP) MIB*	1	1	1	1	1	1	0	0

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Table 3: Standard MIBs Supported on Devices Running Junos OS *(continued)*

MIB/RFC	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
RFC 2934, *Protocol Independent Multicast MIB for IPv4* NOTE: In Junos OS, RFC 2934 is implemented based on a draft version, *pimmib.mib*, of the now standard RFC. Support for the **pimNeighborLoss** trap was added in Release 11.4.	1	1	1	1	1	1	0	0
RFC 2981, *Event MIB*	1	1	1	1	0	0	0	0
RFC 3014, *Notification Log MIB*	1	1	1	1	0	0	0	0
RFC 3019, *IP Version 6 Management Information Base for The Multicast Listener Discovery Protocol*	1	1	1	1	0	0	0	1
RFC 3410 *Introduction and Applicability Statements for Internet-Standard Management Framework*	1	1	1	1	1	0	0	1
RFC 3411, *An Architecture for Describing Simple Network Management Protocol (SNMP) Management Frameworks* NOTE: RFC 3411 replaces RFC 2571. However, Junos OS supports both RFC 3411 and RFC 2571.	1	1	1	1	1	0	0	1

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Table 3: Standard MIBs Supported on Devices Running Junos OS *(continued)*

MIB/RFC	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
RFC 3412, *Message Processing and Dispatching for the Simple Network Management Protocol (SNMP)* NOTE: RFC 3412 replaces RFC 2572. However, Junos OS supports both RFC 3412 and RFC 2572.	1	1	1	1	1	0	0	1
RFC 3413, *Simple Network Management Protocol (SNMP) Applications* (except for the Proxy MIB)	1	1	1	1	1	1	0	1
RFC 3414, *User-based Security Model (USM) for version 3 of the Simple Network Management Protocol (SNMPv3)*	1	1	1	1	1	0	0	1
RFC 3415, *View-based Access Control Model (VACM) for the Simple Network Management Protocol (SNMP)*	1	1	1	1	1	0	0	1
RFC 3416, *Version 2 of the Protocol Operations for the Simple Network Management Protocol (SNMP)* NOTE: RFC 3416 replaces RFC 1905, which was supported in earlier versions of Junos OS.	1	1	1	1	1	0	0	1
RFC 3417, *Transport Mappings for the Simple Network Management Protocol (SNMP)*	1	1	1	1	1	1	0	1

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Table 3: Standard MIBs Supported on Devices Running Junos OS *(continued)*

MIB/RFC	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
RFC 3418, *Management Information Base (MIB) for the Simple Network Management Protocol (SNMP)* NOTE: RFC 3418 replaces RFC 1907, which was supported in earlier versions of Junos OS.	1	1	1	1	1	0	0	1
RFC 3498, *Definitions of Managed Objects for Synchronous Optical Network (SONET) Linear Automatic Protection Switching (APS) Architectures* (implemented under the Juniper Networks enterprise branch [**jnxExperiment**])	1	1	1	0	0	0	0	0
RFC 3584 *Coexistence between Version 1, Version 2, and Version 3 of the Internet-standard Network Management Framework*	1	1	1	1	1	0	0	1

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Table 3: Standard MIBs Supported on Devices Running Junos OS *(continued)*

MIB/RFC	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
RFC 3591 *Managed Objects for the Optical Interface Type* **optIfOTMnTable** (except **optIfOTMnOpticalReach**, **optIfOTMnInterfaceType**, and **optIfOTMnOrder**), **optIfOChConfigTable** (except **optIfOChDirectionality** and **optIfOChCurrentStatus**), **optIfOTUkConfigTable** (except **optIfOTUkTraceIdentifierAccepted**, **optIfOTUkTIMDetMode**, **optIfOTUkTIMActEnabled**, **optIfOTUkTraceIdentifierTransmitted**, **optIfOTUkDEGThr**, **optIfOTUkDEGM**, **optIfOTUkSinkAdaptActive**, and **optIfOTUkSourceAdaptActive**), and **optIfODUkConfigTable** (except **optIfODUkPositionSeqCurrentSize** and **optIfODUkTtpPresent**)	1	1	1	0	0	0	0	0
RFC 3592, *Definitions of Managed Objects for the Synchronous Optical Network/Synchronous Digital Hierarchy (SONET/SDH) Interface Type*	1	1	1	1	0	0	0	0
RFC 3621, *Power Ethernet MIB*	0	0	0	0	1	0	0	0

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Table 3: Standard MIBs Supported on Devices Running Junos OS *(continued)*

MIB/RFC	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
RFC 3637, *Definitions of Managed Objects for the Ethernet WAN Interface Sublayer* (except **etherWisDeviceTable**, **etherWisSectionCurrentTable**, and **etherWisFarEndPathCurrentTable**)	1	1	1	1	0	0	0	0
RFC 3811, *Definitions of Textual Conventions (TCs) for Multiprotocol Label Switching (MPLS) Management*	1	1	1	1	0	1	0	0

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Table 3: Standard MIBs Supported on Devices Running Junos OS *(continued)*

MIB/RFC	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
RFC 3812, *Multiprotocol Label Switching (MPLS) Traffic Engineering (TE) Management Information Base (MIB)* (read only access) • MPLS tunnels as interfaces are not supported. • The following objects in the **TunnelResource** table are not supported: **mplsTunnelResourceMeanRate**, **mplsTunnelResourceMaxBurstSize**, **mplsTunnelResourceMeanBurstSize**, **mplsTunnelResourceExBurstSize**, **mplsTunnelResourceWeight**. • **mplsTunnelPerfTable** and **mplsTunnelCRLDPResTable** are not supported. • **mplsTunnelCHopTable** is supported on ingress routers only. NOTE: The branch used by the proprietary LDP MIB (**ldpmib.mib**) conflicts with RFC 3812. **ldpmib.mib** has been deprecated and replaced by **jnx-mpls-ldp.mib**.	1	1	1	1	0	0	0	0
RFC 3813, *Multiprotocol Label Switching (MPLS) Label Switching Router (LSR) Management Information Base (MIB)* (read-only access). **mplsInterfacePerfTable**, **mplsInSegmentPerfTable**, **mplsOutSegmentPerfTable**, **mplsInSegmentMapTable**, **mplsXCUp**, and **mplsXCDown** are not supported.	1	1	1	1	0	1	0	0

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Table 3: Standard MIBs Supported on Devices Running Junos OS *(continued)*

MIB/RFC	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
RFC 3826, *The Advanced Encryption Standard (AES) Cipher Algorithm in the SNMP User-based Security Model*	1	1	1	1	1	0	0	1
RFC 3896, *Definitions of Managed Objects for the DS3/E3 Interface Type* (except **dsx3FarEndConfigTable**, **dsx3FarEndCurrentTable**, **dsx3FarEndIntervalTable**, **dsx3FarEndTotalTable**, and **dsx3FracTable**)	1	1	1	0	0	0	0	0
RFC 4188, *Definitions of Managed Objects for Bridges*—Supports 802.1D STP(1998). Supports only the following subtrees and objects: • **dot1dStp** subtree is supported on MX Series 3D Universal Edge Routers . • **dot1dTpFdbAddress**, **dot1dTpFdbPort**, and **dot1dTpFdbStatus** objects from the **dot1dTpFdbTable** of the **dot1dTp** subtree are supported on EX Series Ethernet Switches. NOTE: **dot1dTpLearnedEntryDiscards** and **dot1dTpAgingTime** objects are supported on M and T Series routers.	0	0	0	1	1	0	0	0

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Table 3: Standard MIBs Supported on Devices Running Junos OS *(continued)*

MIB/RFC	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
RFC 4318, *Definitions of Managed Objects for Bridges with Rapid Spanning Tree Protocol*—Supports 802.1w and 802.1t extensions for RSTP.	1	1	1	1	1	0	0	0
RFC 4363b, *Q-Bridge VLAN MIB*	0	0	0	1	1	0	0	0
RFC 4382 *MPLS/BGP Layer 3 Virtual Private Network (VPN) MIB* The Junos OS support for RFC 4382 includes the following scalar objects and tables: • **mplsL3VpnConfiguredVrfs** • **mplsL3VpnActiveVrfs** • **mplsL3VpnConnectedInterfaces** • **mplsL3VpnNotificationEnable** • **mplsL3VpnMaxPossRts** • **mplsL3VpnMaxThshTime** • **mplsL3VpnIllLblRcvThrsh** • **mplsL3VpnVrfTable** • **mplsL3VpnIfConfTable** • **mplsL3VpnVrfPerfTable** • **mplsL3VpnVrfRteTable** • **mplsVpnVrfRTTable**	1	1	1	1	1	0	0	0
RFC 4444, *IS-IS MIB*	1	1	1	1	1	1	0	0
RFC 4668, *RADIUS Accounting Client Management Information Base (MIB) for IPv6* (read-only access)	0	0	0	1	0	0	0	0
RFC 4670, *RADIUS Accounting Client Management Information Base (MIB)* (read-only access)	0	0	0	1	0	0	0	0

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Table 3: Standard MIBs Supported on Devices Running Junos OS *(continued)*

MIB/RFC	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
RFC 4801, *Definitions of Textual Conventions for Generalized Multiprotocol Label Switching (GMPLS) Management Information Base (MIB)* (read-only access)	1	1	1	1	0	0	0	0
RFC 4802, *Generalized Multiprotocol Label Switching (GMPLS) Traffic Engineering (TE) Management Information Base (MIB)* (read-only access). **gmplsTunnelReversePerfTable**, **gmplsTeScalars**, **gmplsTunnelTable**, **gmplsTunnelARHopTable**, **gmplsTunnelCHopTable**, and **gmplsTunnelErrorTable** are not supported.)	1	1	1	1	0	0	0	0
RFC 4803, *Generalized Multiprotocol Label Switching (GMPLS) Label Switching Router (LSR) Management Information Base (MIB)* (read-only access). **gmplsLabelTable** and **gmplsOutsegmentTable** are not supported. NOTE: The tables in GMPLS TE (RFC 4802) and LSR (RFC 4803) MIBs are extensions of the corresponding tables from the MPLS TE (RFC 3812) and LSR (RFC 3813) MIBs and use the same index as the MPLS MIB tables.	1	1	1	1	0	0	0	0

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Table 3: Standard MIBs Supported on Devices Running Junos OS *(continued)*

MIB/RFC	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
Internet Assigned Numbers Authority, *IANAiftype Textual Convention MIB* (referenced by RFC 2233, available at http://www.iana.org/assignments/ianaiftype-mib)	1	1	1	1	1	1	0	0
Internet draft draft-ietf-atommib-sonetaps-mib-01.txt, *Definitions of Managed Objects for SONET Linear APS Architectures* (as defined under the Juniper Networks enterprise branch [**jnxExperiment**] only)	1	1	1	1	0	0	0	0
Internet draft draft-ieft-bfd-mib-02.txt, *Bidirectional Forwarding Detection Management Information Base* (Represented by **mib-jnx-bfd-exp.txt** and implemented under the Juniper Networks enterprise branch [**jnxExperiment**]. Read only. Includes **bfdSessUp** and **bfdSessDown** traps. Does not support **bfdSessPerfTable** and **bfdSessMapTable**.)	1	1	1	1	1	0	0	1
Internet draft draft-ietf-idmr-igmp-mib-13.txt, *Internet Group Management Protocol (IGMP) MIB*	1	1	1	1	1	0	0	1

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Table 3: Standard MIBs Supported on Devices Running Junos OS *(continued)*

MIB/RFC	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
Internet draft draft-ietf-idr-bgp4-mibv2-04.txt, *Definitions of Managed Objects for the Fourth Version of Border Gateway Protocol (BGP-4), Second Version* (only **jnxBgpM2PrefixInPrefixes**, **jnxBgpM2PrefixInPrefixesAccepted**, and **jnxBgpM2PrefixInPrefixesRejected** objects)	1	1	1	1	1	0	0	1
Internet draft draft-reeder-snmpv3-usm-3desede-00.txt, *Extension to the User-Based Security Model (USM) to Support Triple-DES EDE in 'Outside' CBC Mode*	1	1	1	1	1	0	0	1
Internet draft draft-ietf-isis-wg-mib-07.txt, *Management Information Base for IS-IS* (only **isisISAdjTable**, **isisISAdjAreaAddrTable**, **isisISAdjIPAddrTable**, and **isisISAdjProtSuppTable)** NOTE: Replaced with RFC 4444, *IS-IS MIB* in Junos OS Release 11.3 and later.	1	1	1	1	1	1	0	0
Internet draft draft-ietf-ppvpn-mpls-vpn-mib-04.txt, *MPLS/BGP Virtual Private Network Management Information Base Using SMIv2* (only **mplsVpnScalars**, **mplsVpnVrfTable,** **mplsVpnPerTable**, and **mplsVpnVrfRouteTargetTable)**	1	1	1	1	0	0	0	0

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Table 3: Standard MIBs Supported on Devices Running Junos OS *(continued)*

MIB/RFC	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
Internet draft draft-ietf-ospf-ospfv3-mib-11.txt, *Management Information Base for OSPFv3* (Represented by **mib-jnx-ospfv3mib.txt** and implemented under the Juniper Networks enterprise branch **{jnxExperiment}**. Support for **ospfv3NbrTable** only. Read only. Object names are prefixed by **jnx**. For example, **jnxOspfv3NbrTable**, **jnxOspfv3NbrAddressType**, and **jnxOspfv3NbrPriority**.)	1	1	1	1	0	0	0	1
Internet draft draft-ietf-idmr-pim-mib-09.txt, *Protocol Independent Multicast (PIM) MIB*	1	1	1	1	1	0	0	1
ESO Consortium MIB, which can be found at http://www.snmp.com/eso/ NOTE: The ESO Consortium MIB has been replaced by RFC 3826.	1	1	1	1	1	1	0	0
Internet Draft P2MP MPLS-TE MIB (draft-ietf-mpls-p2mp-te-mib-09.txt) (read-only access) (except **mplsTeP2mpTunnelBandwidthTable**).	1	1	1	1	0	0	0	0

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Loading MIB Files to a Network Management System on page 51

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Juniper Networks Enterprise-Specific MIBs

The Junos OS supports the following enterprise-specific MIBs:

- AAA Objects MIB—Provides support for monitoring user authentication, authorization, and accounting through the RADIUS, LDAP, SecurID, and local authentication servers. This MIB is currently supported by Junos OS for J Series and SRX Series devices only. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/reference/mibs/mib-jnx-user-aaa.txt

  For more information, see "AAA Objects MIB" on page 441.

- Access Authentication Objects MIB—Provides support for monitoring firewall authentication, including data about the users trying to access firewall-protected resources and the firewall authentication service itself. This MIB is currently supported by Junos OS for J Series and SRX Series devices only. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/reference/mibs/mib-jnx-js-auth.txt.

  For more information, see "Access Authentication Objects MIB" on page 449.

- Alarm MIB—Provides support for alarms from the router. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-chassis-alarm.txt .

  For more information, see "Alarm MIB" on page 395.

- Analyzer MIB—Contains analyzer and remote analyzer data related to port mirroring on the EX Series Ethernet Switches. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-analyzer.txt .

  For more information, see "Analyzer MIB" on page 529.

- Antivirus Objects MIB—Provides information about the antivirus engine, antivirus scans, and antivirus scan-related traps. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-utm-av.txt .

  For more information, see "Antivirus Objects MIB" on page 65.

- ATM Class-of-Service MIB—Provides support for monitoring Asynchronous Transfer Mode, version 2 (ATM2) virtual circuit (VC) class-of-service (CoS) configurations. It also provides CoS queue statistics for all VCs that have CoS configured. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-atm-cos.txt

  For more information, see "ATM Class-of-Service MIB" on page 365.

- ATM MIB—Provides support for ATM interfaces and virtual connections. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-atm.txt.

  For more information, see "ATM MIB" on page 377.

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- BGP4 V2 MIB—Contains objects used to monitor BGP peer-received prefix counters. It is based upon similar objects in the MIB documented in Internet draft draft-ietf-idr-bgp4-mibv2-03.txt, *Definitions of Managed Objects for the Fourth Version of BGP (BGP-4), Second Version*. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-bgpmib2.txt .

  For more information, see "BGP4 V2 MIB" on page 189.

- Bidirectional Forwarding Detection MIB—Provides support for monitoring Bidirectional Forwarding Detection (BFD) sessions. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-bfd.txt .

  For more information, see "Bidirectional Forwarding Detection MIB" on page 325.

- Chassis Definitions for Router Model MIB—Contains the object identifiers (OIDs) that are used by the Chassis MIB to identify platform and chassis components. The Chassis MIB provides information that changes often, whereas the Chassis Definitions for Router Model MIB provides information that changes less often. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-chas-defines.txt .

  For more information, see "Chassis MIBs" on page 70.

- Chassis Forwarding MIB—Enables J Series Services Routers to fully support the Junos OS health monitor. This MIB extends the scope of health monitoring to include Junos forwarding process (**fwdd**) components. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-chassis-fwdd.txt .

  For more information, see "Chassis Forwarding MIB" on page 307.

- Chassis MIB—Provides support for environmental monitoring (power supply state, board voltages, fans, temperatures, and air flow) and inventory support for the chassis, System Control Board (SCB), System and Switch Board (SSB), Switching and Forwarding Module (SFM), Flexible PIC Concentrators (FPCs), and PICs. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-chassis.txt .

  For more information, see "Chassis MIBs" on page 70.

- Chassis Cluster MIB—Provides information about objects that are used whenever the state of the control link interfaces or fabric link interfaces changes (up to down or down to up) in a chassis cluster deployment. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-jsrpd.txt .

  For more information, see Chassis Cluster MIBs .

- Class-of-Service MIB—Provides support for monitoring interface output queue statistics per interface and per forwarding class. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-cos.txt .

  For more information, see "Class-of-Service MIB" on page 351.

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- Configuration Management MIB—Provides notification for configuration changes as SNMP traps. Each trap contains the time at which the configuration change was committed, the name of the user who made the change, and the method by which the change was made. A history of the last 32 configuration changes is kept in **jnxCmChgEventTable**. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-cfgmgmt.txt .

  For more information, see "Configuration Management MIB" on page 387.

- Destination Class Usage MIB—Provides support for monitoring packet counts based on the ingress and egress points for traffic transiting your networks. Ingress points are identified by the input interface. Egress points are identified by destination prefixes grouped into one or more sets, known as destination classes. One counter is managed per interface per destination class, up to a maximum of 16 counters per interface. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-dcu.txt .

  For more information, see "Destination Class Usage MIB" on page 185.

- DHCP Objects MIB— Provides SNMP support (get and trap) for DHCP local server and relay configurations. It also provides support for bindings and leases tables, and for statistics. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-jdhcp.txt .

  For more information, see "DHCP MIB" on page 191.

- DHCPv6 MIB—Provides SNMP support (get and trap) for DHCPv6 local server and relay configurations. It also provides support for bindings and leases tables, and for statistics. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-jdhcpv6.txt .

  For more information, see "DHCPv6 MIB" on page 203.

- Digital Optical Monitoring MIB—Provides support for the **SNMP Get** request for statistics and **SNMP Trap** notifications for alarms. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-dom.txt .

  For more information, see "Digital Optical Monitoring MIB" on page 561.

- DNS Objects MIB—Provides support for monitoring DNS proxy queries, requests, responses, and failures. This MIB is currently supported by Junos OS for J Series and SRX Series devices only. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-dns.txt .

  For more information, see "DNS Objects MIB" on page 453.

- Dynamic Flow Capture MIB—Provides support for monitoring the operational status of dynamic flow capture (DFC) PICs. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-dfc.txt .

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For more information, see "Dynamic Flow Capture MIB" on page 299.

- Ethernet MAC MIB—Monitors media access control (MAC) statistics on Gigabit Ethernet intelligent queuing (IQ) interfaces. It collects MAC statistics; for example, **inoctets**, **inframes**, **outoctets**, and **outframes** on each source MAC address and virtual LAN (VLAN) ID for each Ethernet port. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/jnx-mac.txt .

  For more information, see "Ethernet MAC MIB" on page 265.

- Event MIB—Defines a generic trap that can be generated using an op script or event policy. This MIB provides the ability to specify a system log string and raise a trap if that system log string is found. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-event.txt .

  For more information, see "Event MIB" on page 323.

- Experimental MIB—Contains object identifiers for experimental MIBs. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-exp.txt .

  For more information, see "jnxExperiment" on page 64.

- Firewall MIB—Provides support for monitoring firewall filter counters. Routers must have the Internet Processor II ASIC to perform firewall monitoring. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-firewall.txt .

  For more information, see "Firewall MIB" on page 373.

- Flow Collection Services MIB—Provides statistics on files, records, memory, FTP, and error states of a monitoring services interface. It also provides SNMP traps for unavailable destinations, unsuccessful file transfers, flow overloading, and memory overloading. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-coll.txt .

  For more information, see "Flow Collection Services MIB" on page 287.

- Host Resources MIB—Extends the **hrStorageTable** object, providing a measure of the usage of each file system on the router in percentage format. Previously, the objects in the **hrStorageTable** measured the usage in allocation units—**hrStorageUsed** and **hrStorageAllocationUnits**—only. Using the percentage measurement, you can more easily monitor and apply thresholds on usage. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-hostresources.txt .

  For more information, see "Host Resources MIB" on page 411.

- IDP Objects MIB—Provides support for monitoring SNMP IDP queries, requests, responses, and failures. This MIB defines the key monitoring and threshold crossing trap support, IDP database update status and trap support, attack-related monitoring and trap support for SRX100, SRX210, SRX220, SRX240, and SRX650 Services

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Gateways. This MIB models IDP attributes specific to the appropriate Juniper Networks implementation. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-idp.txt .

For more information, see "IDP MIB" on page 211.

- Interface MIB—Extends the standard **ifTable** (RFC 2863) with additional statistics and Juniper Networks enterprise-specific chassis information. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-if-extensions.txt .

  For more information, see "Interface MIB" on page 267.

- IP Forward MIB—Extends the standard IP Forwarding Table MIB (RFC 2096) to include CIDR forwarding information. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ipforward.txt .

  For more information, see "IP Forward MIB" on page 355.

- IPsec Generic Flow Monitoring Object MIB—Based on **jnx-ipsec-monitor-mib**, this MIB provides support for monitoring IPsec and IPsec VPN management objects. This MIB is currently supported by Junos OS for J Series and SRX Series devices only. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ipsec-flow-mon.txt .

  For more information, see "IPsec Generic Flow Monitoring Object MIB" on page 455.

- IPsec Monitoring MIB—Provides operational and statistical information related to the IPsec and IKE tunnels on Juniper Networks routers. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ipsec-monitor-asp.txt .

  For more information, see "IPSec Monitoring MIB" on page 257.

- IPsec VPN Objects MIB—Provides support for monitoring IPsec and IPsec VPN management objects for Juniper security product lines. This MIB is an extension of **jnx-ipsec-flow-mon.mib**. This MIB is currently supported by Junos OS for J Series and SRX Series devices only. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-ipsec-vpn.txt .

  For more information, see "IPsec VPN Objects MIB" on page 471.

- IPv4 MIB—Provides additional Internet Protocol version 4 (IPv4) address information, supporting the assignment of identical IPv4 addresses to separate interfaces. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ipv4.txt .

  For more information, see "IPv4 MIB" on page 393.

- IPv6 and ICMPv6 MIB—Provides IPv6 and Internet Control Message Protocol version 6 (ICMPv6) statistics. For a downloadable version of this MIB, see

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http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ipv6.txt .

For more information, see "IPv6 MIB" on page 357.

- L2ALD MIB—Contains information about the Layer 2 Address Learning Daemon (L2ALD) and related traps, such as the routing instance MAC limit trap and the interface MAC limit trap. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-l2ald.txt .

  For more information, see "L2ALD MIB" on page 433.

- L2CP MIB—Provides information about Layer 2 Control Protocols (L2CP) based features on MX Series 3D Universal Edge Routers . Currently, Junos OS supports only the **jnxDot1dStpPortRootProtectEnabled**, **jnxDot1dStpPortRootProtectState**, and **jnxPortRootProtectStateChangeTrap** objects. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-l2cp-features.txt .

  For more information, see "L2CP MIB" on page 413.

- L2TP MIB—Provides information about Layer 2 Transport Protocol (L2TP) tunnels and sessions. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-l2tp.txt .

  For more information, see "L2TP MIB" on page 327.

- LDP MIB—Provides LDP statistics and defines LDP label-switched path (LSP) notifications. LDP traps support only IPv4 standards. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ldp.txt .

  For more information, see "LDP MIB" on page 517.

- License MIB—Extends SNMP support to licensing information, and introduces SNMP traps that alert users when the licenses are about to expire, expire, or when the total number of users exceeds the number specified in the license. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-license.txt .

  For more information, see "License MIB" on page 567.

- MIMSTP MIB—Provides information about MSTP instances (that is, routing instances of type Virtual Switch/Layer 2 control, also known as virtual contexts), MSTIs within the MSTP instance, and VLANs associated with the MSTI. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-mimstp.txt .

  For more information, see "MIMSTP MIB" on page 415.

- MPLS MIB—Provides MPLS information and defines MPLS notifications. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-mpls.txt .

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NOTE: To collect information about MPLS statistics on transit routers, use the enterprise-specific RSVP MIB (mib-jnx-rsvp.txt) instead of the enterprise-specific MPLS MIB (mib-jnx-mpls.txt).

For more information, see "MPLS MIB" on page 403.

- MPLS LDP MIB—Contains object definitions as described in RFC 3815, *Definitions of Managed Objects for the Multiprotocol Label Switching (MPLS), Label Distribution Protocol (LDP)*. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-mpls-ldp.txt .

  NOTE: Objects in the MPLS LDP MIB were supported in earlier releases of the Junos OS as a proprietary LDP MIB (mib-ldpmib.txt). Because the branch used by the proprietary LDP (mib-ldpmib.txt) conflicts with RFC 3812, the proprietary LDP MIB (mib-ldpmib.txt) has been deprecated and replaced by the enterprise-specific MPLS LDP MIB (mib-jnx-mpls-ldp.txt).

  For more information, see "MPLS LDP MIB" on page 313.

- NAT Objects MIB—Provides support for monitoring network address translation (NAT). This MIB is currently supported by Junos OS for J Series and SRX Series devices only. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-nat.txt .

  For more information, see "NAT Objects MIB" on page 475.

- NAT Resources-Monitoring MIB—Provides support for monitoring NAT pools usage and NAT rules. Notifications of usage of NAT resources are also provided by this MIB. This MIB is currently supported on the Multiservices PIC and Multiservices DPC on M Series and MX Series routers only. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-sp-nat.txt .

  For more information, see "Network Address Translation Resources–Monitoring MIB" on page 483.

- OTN Interface Management MIB—Defines objects for managing Optical Transport Network (OTN) interfaces on devices running the Junos OS. For a downloadable version of the MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-otn.txt .

  For more information, see "OTN Interface Management MIB" on page 557.

- Packet Forwarding Engine MIB—Provides notification statistics for Packet Forwarding Engines. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-pfe.txt .

  For more information, see "Packet Forwarding Engine MIB" on page 315.

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- Packet Mirror MIB—Enables you to capture and view packet mirroring-related information. This MIB is currently supported by Junos OS for MX Series routers only. Packet mirroring traps are an extension of the standard SNMP implementation and are only available to SNMPv3 users. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-packet-mirror.txt.

  For more information, see "Packet Mirror MIB Overview" on page 319.

- PAE Extension MIB—Extends the standard IEEE802.1x PAE Extension MIB, and contains information for Static MAC Authentication. The enterprise-specific PAE Extension MIB is supported only on EX Series Ethernet Switches. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-pae-extension.txt .

  For more information, see "PAE Extension MIB" on page 543.

- Passive Monitoring MIB—Performs traffic flow monitoring and lawful interception of packets transiting between two routers. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-pmon.txt .

  For more information, see "Passive Monitoring MIB" on page 241.

- Ping MIB—Extends the standard Ping MIB control table (RFC 2925). Items in this MIB are created when entries are created in **pingCtlTable** of the Ping MIB. Each item is indexed exactly as it is in the Ping MIB. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ping.txt .

  For more information, see "PING MIB" on page 215.

- Policy Objects MIB—Provides support for monitoring the security policies that control the flow of traffic from one zone to another. This MIB is currently supported by Junos OS for J Series and SRX Series devices only. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-policy.txt .

  For more information, see "Policy Objects MIB" on page 489.

- Power Supply Unit MIB—Enables monitoring and managing of the power supply on a device running the Junos OS. This MIB is currently supported only on EX Series Ethernet Switches. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-power-supply-unit.txt .

  For more information, see "Power Supply Unit MIB" on page 547.

- PPP MIB—Provides SNMP support for PPP-related information such as the type of authentication used, interface characteristics, status, and statistics. This MIB is currently supported only on M Series and MX Series routers. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ppp.txt .

  For more information, see "PPP MIB Overview" on page 589.

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- PPPoE MIB—Provides SNMP support for PPPoE-related information such as the type of authentication used, interface characteristics, status, and statistics. This MIB is currently supported only on M Series and MX Series routers. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-pppoe.txt .

  For more information, see "PPPoE MIB Overview" on page 619.

- Pseudowire TDM MIB—Extends the standard Pseudowire MIB, and contains information about configuration and statistics for specific pseudowire types. The enterprise-specific Pseudowire TDM MIB is the Juniper Networks implementation of the standard Managed Objects for TDM over Packet Switched Network MIB (draft-ietf-pwe3-tdm-mib-08.txt). For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-pwtdm.txt .

- Real-Time Performance Monitoring MIB—Provides real-time performance-related data and enables you to access jitter measurements and calculations using SNMP. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-rpm.txt .

  For more information, see "Real-Time Performance Monitoring MIB" on page 341.

- Reverse-Path-Forwarding MIB—Monitors statistics for traffic that is rejected because of reverse-path-forwarding (RPF) processing. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-rpf.txt .

  

  NOTE: **The enterprise-specific RPF MIB is not supported on EX Series Ethernet Switches.**

  For more information, see "Reverse-Path-Forwarding MIB" on page 237.

- RMON Events and Alarms MIB—Supports the Junos OS extensions to the standard Remote Monitoring (RMON) Events and Alarms MIB (RFC 2819). The extension augments **alarmTable** with additional information about each alarm. Two new traps are also defined to indicate when problems are encountered with an alarm. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-rmon.txt .

  For more information, see "RMON Events and Alarms MIB" on page 233.

- RSVP MIB—Provides information about RSVP-traffic engineering sessions that correspond to MPLS LSPs on transit routers in the service provider core network. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-rsvp.txt .

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NOTE: To collect information about MPLS statistics on transit routers, use the enterprise-specific RSVP MIB (mib-jnx-rsvp.txt) instead of the enterprise-specific MPLS MIB (mib-jnx-mpls.txt).

For more information, see "RSVP MIB" on page 399.

- Security Interface Extension Objects MIB—Provides support for the security management of interfaces. This MIB is currently supported by Junos OS for J Series and SRX Series devices only. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-if-ext.txt .

  For more information, see "Security Interface Extension Objects MIB" on page 495.

- Security Screening Objects MIB—Defines the MIB for the Juniper Networks Enterprise Firewall screen functionality. This MIB is currently supported by Junos OS for J Series and SRX Series devices only. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-screening.txt .

  For more information, see "Security Screening Objects MIB" on page 503.

- Services PIC MIB—Provides statistics for Adaptive Services (AS) PICs and defines notifications for AS PICs. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-sp.txt .

  For more information, see "Services PIC MIB" on page 293.

- SONET APS MIB—Monitors any SONET interface that participates in Automatic Protection Switching (APS). For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-sonetaps.txt .

  For more information, see "SONET APS MIB" on page 247.

- SONET/SDH Interface Management MIB—Monitors the current alarm for each SONET/SDH interface. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-sonet.txt .

  For more information, see "SONET/SDH Interface Management MIB" on page 243.

- Source Class Usage MIB—Counts packets sent to customers by performing a lookup on the IP source address and the IP destination address. The Source Class Usage (SCU) MIB makes it possible to track traffic originating from specific prefixes on the provider core and destined for specific prefixes on the customer edge. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-scu.txt .

  For more information, see "Source Class Usage MIB" on page 239.

- SPU Monitoring MIB—Provides support for monitoring SPUs on SRX5600 and SRX5800 devices. For a downloadable version of this MIB, see

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http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-spu-monitoring.txt .

For more information, see "SPU Monitoring Objects MIB" on page 553.

- Structure of Management Information MIB—Explains how the Juniper Networks enterprise-specific MIBs are structured. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-smi.txt .

  For more information, see "Structure of Management Information MIB" on page 59.

- Structure of Management Information MIB for EX Series Ethernet Switches—Defines a MIB branch for switching-related MIB definitions for the EX Series Ethernet Switches. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ex-smi.txt .

  For more information, see "EX Series SMI MIB" on page 523.

- Structure of Management Information MIB—Contains object identifiers (OIDs) for the security branch of the MIBs used in Junos OS for J Series and SRX Series devices, services, and traps. This MIB is currently supported by Junos OS for J Series and SRX Series devices only. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-smi.txt .

- Subscriber MIB—Provides SNMP support for subscriber-related information. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-subscriber.txt .

  For more information, see "Subscriber MIB" on page 571.

- System Log MIB—Enables notification of an SNMP trap-based application when an important system log message occurs. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-syslog.txt .

  For more information, see "System Log MIB" on page 309.

- Traceroute MIB—Supports the Junos OS extensions of traceroute and remote operations. Items in this MIB are created when entries are created in the **traceRouteCtlTable** of the Traceroute MIB. Each item is indexed exactly the same way as it is in the Traceroute MIB. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-traceroute.txt .

  For more information, see "Traceroute MIB" on page 231.

- Utility MIB—Provides SNMP support for exposing Junos OS data and has tables that contain information about each type of data, such as integer and string. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-util.txt .

  For more information, see "Utility MIB" on page 437.

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- Virtual Chassis MIB—Contains information about the virtual chassis on the EX Series Ethernet Switches. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-virtualchassis.txt .

  For more information, see "Virtual Chassis MIB" on page 541.

- VLAN MIB—Contains information about prestandard IEEE 802.10 VLANs and their association with LAN emulation clients. The enterprise-specific VLAN MIB is supported only on EX Series Ethernet Switches. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-vlan.txt .

  For more information, see "VLAN MIB" on page 533.

- VPLS MIBs—Provides information about generic, BGP-based, and LDP-based VPLS and pseudowires associated with the VPLS networks. The enterprise-specific VPLS MIBs are Juniper Networks extensions of the following IETF standard MIBs defined in Internet draft draft-ietf-l2vpn-vpls-mib-05.txt, and are implemented as part of the **jnxExperiment** branch:

  - **VPLS-Generic-Draft-01-MIB** implemented as **mib-jnx-vpls-generic.txt**
  - **VPLS-BGP-Draft-01-MIB** implemented as **mib-jnx-vpls-bgp.txt**
  - **VPLS-LDP-Draft-01-MIB** implemented as **mib-jnx-vpls-ldp.txt**

  For downloadable versions of these MIBs, see:

  - http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-vpls-generic.txt
  - http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-vpls-bgp.txt
  - http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-vpls-ldp.txt

- VPN Certificate Objects MIB—Provides support for monitoring the local and CA certificates loaded on the router. This MIB is currently supported by Junos OS for J Series and SRX Series devices only. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-cert.txt .

  For more information, see "VPN Certificate Objects MIB" on page 499.

- VPN MIB—Provides monitoring for Layer 3 VPNs, Layer 2 VPNs, and virtual private LAN service (VPLS) (read access only). For a downloadable version of the MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-vpn.txt .

  For more information, see "VPN MIB" on page 273.

Related Documentation

- Standard SNMP MIBs Supported by Junos OS on page 7

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- Enterprise-Specific MIBs and Supported Devices on page 42
- Loading MIB Files to a Network Management System on page 51

Enterprise-Specific MIBs and Supported Devices

Table 4 on page 42 lists the enterprise-specific MIBs that are supported on various devices running the Junos OS.

NOTE: In this table, a value of 1 in any of the platform columns (M, MX, T, EX, J, and SRX) denotes that the corresponding MIB is supported on that particular platform. A value of 0 denotes that the MIB is not supported on the platform.

NOTE: This topic uses the following classification for SRX devices: Low-End (SRX100, SRX210, SRX220, and SRX240), Mid-Range (SRX650 and SRX1400), and High-End (SRX3400, SRX3600, SRX5600, and SRX5800).

Table 4: Enterprise-Specific MIBs and Supported Devices

Enterprise-Specific MIB	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
AAA Objects MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-user-aaa.txt	1	1	0	0	0	0	1	1
Access Authentication Objects MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-auth.txt	0	0	0	0	1	1	1	1
Alarm MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-chassis-alarm.txt	1	1	1	1	1	1	1	1
Analyzer MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-analyzer.txt	0	0	0	1	0	0	0	0

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Table 4: Enterprise-Specific MIBs and Supported Devices *(continued)*

Enterprise-Specific MIB	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
Antivirus Objects MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-utm-av.txt	0	0	0	0	0	1	0	0
ATM Class-of-Service MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-atm-cos.txt	1	1	1	0	0	1	0	1
ATM MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-atm.txt	1	1	1	0	0	0	0	0
BGP4 V2 MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-bgpmib2.txt	1	1	1	1	1	1	1	1
Bidirectional Forwarding Detection MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-bfd.txt	1	1	1	1	1	1	1	1
Chassis Forwarding MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-chassis-fwdd.txt	0	0	0	0	1	1	0	0
Chassis MIBs http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-chassis.txt http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-chas-defines.txt	1	1	1	1	1	1	1	1
Chassis Cluster MIBs http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-jspd.txt	0	0	0	0	0	0	1	1
Class-of-Service MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-cos.txt	1	1	1	1	1	0	0	1

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Table 4: Enterprise-Specific MIBs and Supported Devices *(continued)*

Enterprise-Specific MIB	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
Configuration Management MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-cfgmgmt.txt	1	1	1	1	1	1	1	1
Destination Class Usage MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-dcu.txt	1	1	1	0	1	0	1	1
DHCP MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-jdhcp.txt	1	1	1	0	0	0	0	0
DHCPv6 MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-jdhcpv6.txt	1	1	1	0	0	0	0	0
Digital Optical Monitoring MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-dom.txt	1	0	1	0	0	0	0	0
DNS Objects MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-dns.txt	0	0	0	0	0	0	1	1
Dynamic Flow Capture MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-dfc.txt	1	1	1	0	0	0	0	0
Ethernet MAC MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/jnx-mac.txt	1	1	1	1	1	0	0	1
Event MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-event.txt	1	1	1	1	1	1	1	1

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Table 4: Enterprise-Specific MIBs and Supported Devices *(continued)*

Enterprise-Specific MIB	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
EX Series MAC Notification MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ex-mac-notification.txt	0	0	0	1	0	0	0	0
EX Series SMI MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ex-smi.txt	0	0	0	1	0	0	0	0
Experimental MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-exp.txt	1	1	1	1	1	0	0	0
Firewall MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-firewall.txt	1	1	1	1	1	1	1	1
Flow Collection Services MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-coll.txt	1	1	1	0	0	0	0	0
Host Resources MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-hostresources.txt	1	1	1	1	1	1	1	1
Interface MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-if-extensions.txt	1	1	1	1	1	1	1	1
IP Forward MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ipforward.txt	1	1	1	1	1	1	1	1
IPsec Generic Flow Monitoring Object MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ipsec-flow-mon.txt	0	0	0	0	1	0	0	1

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Table 4: Enterprise-Specific MIBs and Supported Devices *(continued)*

Enterprise-Specific MIB	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
IPsec Monitoring MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ipsec-monitor-asp.txt	1	1	1	0	1	0	0	1
IPsec VPN Objects MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-ipsec-vpn.txt	0	0	0	0	1	1	0	0
IPv4 MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ipv4.txt	1	1	1	1	1	1	1	1
IPv6 and ICMPv6 MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ipv6.txt	1	1	1	1	0	1	1	1
L2ALD MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-l2ald.txt	0	1	0	1	0	0	0	
L2CP MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-l2cp-features.txt	0	0	0	1	0	0	0	
L2TP MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-l2tp.txt	1	1	0	0	0	0	0	0
LDP MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ldp.txt	1	1	1	0	0	0	0	1
License MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-license.txt	1	1	1	0	0	1	1	1

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Table 4: Enterprise-Specific MIBs and Supported Devices *(continued)*

Enterprise-Specific MIB	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
MIMSTP MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-mimstp.txt	0	1	0	1	0	0	0	0
MPLS LDP MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-mpls-ldp.txt	1	1	1	0	1	0	0	0
MPLS MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-mpls.txt	1	1	1	1	1	0	0	1
NAT Objects MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-nat.txt	0	0	0	0	1	1	1	1
NAT Resources-Monitoring MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/ mibs/mib-jnx-sp-nat.txt	1	1	1	0	0	0	0	0
OTN Interface Management MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-otn.txt	1	1	1	0	0	0	0	0
Packet Forwarding Engine MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-pfe.txt	1	1	1	0	1	1	1	1
Packet Mirror MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-packet-mirror.txt	0	1	0	0	0	0	0	0
PAE Extension MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-pae-extension.txt	0	0	0	1	0	0	0	0

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Table 4: Enterprise-Specific MIBs and Supported Devices *(continued)*

Enterprise-Specific MIB	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
Passive Monitoring MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-pmon.txt	1	1	1	0	0	0	0	0
Ping MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ping.txt	1	1	1	1	1	1	1	1
Policy Objects MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-policy.txt	0	0	0	0	1	1	1	1
Power Supply Unit MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-power-supply-unit.txt	0	0	0	1	0	0	0	0
PPP MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ppp.txt.	1	1	0	0	0	0	0	0
PPPoE MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-pppoe.txt.	1	1	0	0	0	0	0	0
Psuedowire TDM MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-pwtdm.txt .	1	1	1	0	0	0	0	0
Real-Time Performance Monitoring MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-rpm.txt	1	1	1	1	1	1	0	0
Reverse-Path-Forwarding MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-rpf.txt	1	1	1	0	1	1	1	1

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Table 4: Enterprise-Specific MIBs and Supported Devices *(continued)*

Enterprise-Specific MIB	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
RMON Events and Alarms MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-rmon.txt	1	1	1	0	1	1	1	1
RSVP MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-rsvp.txt	1	1	1	0	0	0	0	0
Security Interface Extension Objects MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-if-ext.txt	0	0	0	0	1	1	1	1
Security Screening Objects MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-screening.txt	0	0	0	0	0	0	0	1
Services PIC MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-sp.txt	1	1	1	0	0	0	0	0
SNMP IDP MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-idp.txt.	0	0	0	0	0	1	1	1
SONET APS MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-sonetaps.txt	1	1	1	0	0	0	0	0
SONET/SDH Interface Management MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-sonet.txt	1	1	1	0	0	0	0	0
Source Class Usage MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-scu.txt	1	1	1	0	0	0	0	1

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Table 4: Enterprise-Specific MIBs and Supported Devices *(continued)*

Enterprise-Specific MIB	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
SPU Monitoring MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-spu-monitoring.txt	0	0	0	0	0	1	1	1
Structure of Management Information MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-smi.txt	1	1	1	1	1	1	1	1
Subscriber MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-subscriber.txt	0	1	0	0	0	0	0	0
System Log MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-syslog.txt	1	1	1	1	1	1	1	1
Traceroute MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-traceroute.txt	1	1	1	1	1	1	1	1
Utility MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-util.txt	1	1	1	1	1	1	1	1
Virtual Chassis MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-virtualchassis.txt	0	0	0	1	0	0	0	0
VLAN MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-vlan.txt	0	0	0	1	0	0	0	0

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Table 4: Enterprise-Specific MIBs and Supported Devices *(continued)*

Enterprise-Specific MIB	Platforms							
						SRX		
	M	T	J	MX	EX	Low-End	Mid-Range	High-End
VPLS MIBs • http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-vpls-generic.txt • http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-vpls-ldp.txt • http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-vpls-bgp.txt	1	1	1	1	0	0	0	0
VPN Certificate Objects MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-cert.txt	0	0	0	0	1	1	1	1
VPN MIB http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-vpn.txt	1	1	1	0	1	0	0	0

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Juniper Networks Enterprise-Specific SNMP Traps on page 633
- Standard SNMP MIBs Supported by Junos OS on page 7
- Loading MIB Files to a Network Management System on page 51

Loading MIB Files to a Network Management System

For your network management system (NMS) to identify and understand the MIB objects used by the Junos OS, you must first load the MIB files to your NMS using a MIB compiler. A MIB compiler is a utility that parses the MIB information such as the MIB object name, IDs, and data type for the NMS.

You can download the Junos MIB package from the **Enterprise-Specific MIBs and Traps** section of the Junos OS Technical Publications index page at http://www.juniper.net/techpubs/software/junos/index.html . The Junos MIB package is available in **.zip** and **.tar** packages. You can download the appropriate format based on your requirements.

The Junos MIB package contains two folders: **StandardMibs** and **JuniperMibs**. The **StandardMibs** folder contains the standard MIBs and RFCs that are supported on devices running the Junos OS, whereas the **JuniperMibs** folder contains the Juniper Networks enterprise-specific MIBs.

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To load MIB files that are required for managing and monitoring devices running the Junos OS:

1. Go to the Junos OS Technical Publications index page (http://www.juniper.net/techpubs/software/junos/index.html).
2. Click the tab that corresponds to the Junos OS Release for which you want to download the MIB files.
3. On the selected tab, click the **+** (plus) sign that corresponds to the **Enterprise-Specific MIBs and Traps** section to expand the section.
4. Click the **TAR** or **ZIP** link that corresponds to the **Enterprise MIBs** link under the **Enterprise-Specific MIBs and Traps** section to download the Junos MIB package.
5. Decompress the file (**.tar** or **.zip**) using an appropriate utility.
6. Load the standard MIB files (from the **StandardMibs** folder) in the following order:

   NOTE: Some of the MIB compilers that are commonly used have the standard MIBs preloaded on them. If the standard MIBs are already loaded on the MIB compiler that you are using, skip this step and proceed to Step 7.

   a. **mib-SNMPv2-SMI.txt**
   b. **mib-SNMPv2-TC.txt**
   c. **mib-IANAifType-MIB.txt**
   d. **mib-IANA-RTPROTO-MIB.txt**
   e. **mib-rfc1907.txt**
   f. **mib-rfc2011a.txt**
   g. **mib-rfc2012a.txt**
   h. **mib-rfc2013a.txt**
   i. **mib-rfc2863a.txt**
7. Load the remaining standard MIB files.

   NOTE: You must follow the order specified in this procedure, and ensure that all standard MIBs are loaded before you load the enterprise-specific MIBs. There might be dependencies that require a particular MIB to be present on the compiler before loading some other MIB. You can find such dependencies listed in the IMPORT section of the MIB file.
8. Load the Juniper Networks enterprise-specific SMI MIB, **mib-jnx-smi.txt**, and the following optional SMI MIBs based on your requirements:

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- **mib-jnx-js-smi.txt**—(Optional) For Juniper Security MIB tree objects
- **mib-jnx-ex-smi.txt**—(Optional) For EX Series Ethernet Switches
- **mib-jnx-exp.txt**—(Recommended) For Juniper Networks experimental MIB objects

9. Load the remaining enterprise-specific MIBs from the **JuniperMibs** folder.



TIP: While loading a MIB file, if the compiler returns an error message saying that any of the objects is undefined, open the MIB file using a text editor and ensure that all the MIB files listed in the IMPORT section are loaded on the compiler. If any of the MIB files listed in the IMPORT section is not loaded on the compiler, load that MIB file, and then try to load the MIB file that failed to load.

For example, the enterprise-specific PING MIB, **mib-jnx-ping.txt**, has dependencies on RFC 2925, DiSMAN-PING-MIB, **mib-rfc2925a.txt**. If you try to load **mib-jnx-ping.txt** before loading **mib-rfc2925a.txt**, the compiler returns an error message saying that certain objects in **mib-jnx-ping.txt** are undefined. Load **mib-rfc2925a.txt**, and then try to load **mib-jnx-ping.txt**. The enterprise-specific PING MIB, **mib-jnx-ping.txt**, then loads without any issue.

Related Documentation

- Standard SNMP MIBs Supported by Junos OS on page 7
- Juniper Networks Enterprise-Specific MIBs on page 30

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PART 2

Juniper Networks Enterprise-Specific MIBs

- Interpreting the Enterprise-Specific Structure of Management Information MIB on page 59
- Interpreting the Enterprise-Specific Antivirus Objects MIB on page 65
- Interpreting the Enterprise-Specific Chassis MIBs on page 69
- Interpreting the Enterprise-Specific Destination Class Usage MIB on page 185
- Interpreting the Enterprise-Specific BGP4 V2 MIB on page 189
- Interpreting the Enterprise-Specific DHCP MIB on page 191
- Interpreting the Enterprise-Specific DHCPv6 MIB on page 203
- Interpreting the Enterprise-Specific SNMP IDP MIB on page 211
- Interpreting the Enterprise-Specific Ping MIB on page 215
- Interpreting the Enterprise-Specific Traceroute MIB on page 231
- Interpreting the Enterprise-Specific RMON Events and Alarms MIB on page 233
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- Interpreting the Enterprise-Specific IPsec Monitoring MIB on page 257
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- Interpreting the Enterprise-Specific Packet Forwarding Engine MIB on page 315
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- Interpreting the Enterprise-Specific Event MIB on page 323
- Interpreting the Enterprise-Specific Bidirectional Forwarding Detection MIB on page 325
- Interpreting the Enterprise-Specific L2TP MIB on page 327
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- Interpreting the Enterprise-Specific Class-of-Service MIB on page 351
- Interpreting the Enterprise-Specific IP Forward MIB on page 355
- Interpreting the Enterprise-Specific IPv6 MIB on page 357
- Interpreting the Enterprise-Specific ATM Class-of-Service MIB on page 365
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- Interpreting the Enterprise-Specific Alarm MIB on page 395
- Interpreting the Enterprise-Specific RSVP MIB on page 399
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- Interpreting the Enterprise-Specific Layer 2 Control Protocol (L2CP) MIB on page 413
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- Interpreting the Enterprise-Specific Utility MIB on page 437
- Interpreting the Enterprise-Specific AAA Objects MIB on page 441
- Interpreting the Enterprise-Specific Access Authentication Objects MIB on page 449
- Interpreting the Enterprise-Specific DNS Objects MIB on page 453
- Interpreting the Enterprise-Specific IPsec Generic Flow Monitoring Object MIB on page 455
- Interpreting the Enterprise-Specific IPsec VPN Objects MIB on page 471
- Interpreting the Enterprise-Specific Network Address Translation Objects MIB on page 475
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- Interpreting the Enterprise-Specific Policy Objects MIB on page 489
- Interpreting the Enterprise-Specific Security Interface Extension Objects MIB on page 495
- Interpreting the Enterprise-Specific VPN Certificate Objects MIB on page 499
- Interpreting the Enterprise-Specific Security Screening Objects MIB on page 503
- Interpreting the Enterprise-Specific LDP MIB on page 517
- Interpreting the Enterprise-Specific EX Series SMI MIB on page 523
- Interpreting the Enterprise-Specific EX MAC Notification MIB on page 525
- Interpreting the Enterprise-Specific Analyzer MIB on page 529

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- Interpreting the Enterprise-Specific VLAN MIB on page 533
- Interpreting the Enterprise-Specific Virtual Chassis MIB on page 541
- Interpreting the Enterprise-Specific PAE Extension MIB on page 543
- Interpreting the Enterprise-Specific Power Supply Unit MIB on page 547
- Interpreting the Enterprise-Specific SPU Monitoring MIB on page 553
- Interpreting the Enterprise-Specific OTN Interface Management MIB on page 557
- Interpreting the Enterprise-Specific Digital Optical Monitoring MIB on page 561
- Interpreting the Enterprise-Specific License MIB on page 567
- Interpreting the Enterprise-Specific Subscriber MIB on page 571
- Interpreting the Enterprise-Specific VPLS MIBs on page 579
- Interpreting the Enterprise-Specific PPP MIB on page 589
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CHAPTER 2

Interpreting the Enterprise-Specific Structure of Management Information MIB

- Structure of Management Information MIB on page 59
- jnxProducts on page 60
- jnxServices on page 60
- jnxMibs on page 61
- jnxTraps on page 63
- jnxExperiment on page 64

Structure of Management Information MIB

The Juniper Networks enterprise-specific Structure of Management Information MIB defines the top-level structure of the Juniper Networks enterprise-specific MIB space. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-smi.txt .

The Structure of Management Information MIB space has five root branches:

- jnxProducts on page 60
- jnxServices on page 60
- jnxMibs on page 61
- jnxTraps on page 63
- jnxExperiment on page 64

Related Documentation

- Juniper Networks Enterprise-Specific SNMP Traps on page 633
- Standard SNMP Traps Supported on Devices Running Junos OS on page 653
- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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jnxProducts

The object identifier for the **jnxProducts** root branch of the Structure of Management Information MIB is **{juniperMIB 1}**. This branch of the MIB describes the Juniper Networks routers and their components, such as product line, product name, model, number of slots, and media space for holding PICs. It also provides information about the system's power supply state, board voltages, fans, temperatures, and air flow. In general, this branch of the Structure of Management Information MIB is rarely polled for information because it is descriptive. However, you can poll this branch of the Structure of Management Information MIB to determine the **sysObjectId** of a router as defined by MIB-II.

jnxProducts 1 is reserved for Junos OS based products.

Related Documentation

- Structure of Management Information MIB on page 59
- jnxServices on page 60
- jnxMibs on page 61
- jnxTraps on page 63
- jnxExperiment on page 64

jnxServices

The object identifier for the **jnxServices** root branch is **{juniperMIB 2}**. This MIB file added the nodes to create the Juniper Networks security tree structure under the object node **jnxJsObjects**. In general, the prefix **jnxJs** is used to name the object identifiers and to designate them. This branch of the network describes the Juniper Networks services objects that provide enhanced network security. This MIB is currently supported by Junos OS for J Series and SRX Series devices only.

The **jnxJsSecurity** node is designed to provide a branch for the security-related MIB definitions specific to the Juniper Networks security products. The next level object identifiers under **jnxJsSecurity** are:

- **jnxJsIf**—Whose object identifier is **{jnxJsSecurity 1}**.
- **jnxJsAuth**—Whose object identifier is **{jnxJsSecurity 2}**.
- **jnxJsCertificates**—Whose object identifier is **{jnxJsSecurity 3}**.
- **jnxJsPolicies**—Whose object identifier is **{jnxJsSecurity 4}**.
- **jnxJsIPSecVpn**—Whose object identifier is **{jnxJsSecurity 5}**.
- **jnxJsNAT**—Whose object identifier is **{jnxJsSecurity 7}**.
- **jnxJsScreening**—Whose object identifier is **{jnxJsSecurity 8}**.
- **jnxJsDhcp**—Whose object identifier is **{jnxJsSecurity 9}**.
- **jnxJsDnsRoot**—Whose object identifier is **{jnxJsSecurity 10}**.

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The Juniper Networks enterprise-specific security MIBs include:

- AAA Objects MIB—Whose object identifier is **{jnxUserAAAMibRoot 1}**.
- Access Authentication Objects MIB—Whose object identifier is **{jnxJsAuth 1}**.
- DNS Objects MIB—Whose object identifier is **{jnxJsDns 1}**.
- IPsec Generic Flow Monitoring Objects MIB—Whose object identifier is **{jnxIpSecMibRoot 1}**.
- IPsec VPN Objects MIB—Whose object identifier is **{jnxJsIPSecVpn 1}**.
- NAT Objects MIB—Whose object identifier is **{jnxJsNAT 1}**.
- Policy Objects MIB—Whose object identifier is **{jnxJsPolicies 1}**.
- Security Interface Extension Objects MIB—Whose object identifier is **{jnxJsIf 1}**.
- Security Screening Objects MIB—Whose object identifier is **{jnxJsScreening 1}**.
- VPN Certificate Objects MIB—Whose object identifier is **{jnxJsCertificates 1}**.

For more information about these MIBs, see "Juniper Networks Enterprise-Specific MIBs" on page 30.

Related Documentation

- Structure of Management Information MIB on page 59
- jnxProducts on page 60
- jnxMibs on page 61
- jnxTraps on page 63
- jnxExperiment on page 64

jnxMibs

The object identifier for the **jnxMibs** root branch is **{juniperMIB 3}** and includes one main subbranch, **jnxBoxAnatomy**, whose object identifier is **{jnxMibs 1}**. The other Juniper Networks enterprise-specific MIBs are also branches of **jnxMibs**. These Juniper Networks enterprise-specific MIBs include:

- MPLS MIB—Whose object identifier is **{jnxMibs 2}**.
- Juniper Networks enterprise-specific extensions to the Interface MIB—Whose object identifier is **{jnxMibs 3}**.
- Alarm MIB—Whose object identifier is **{jnxMibs 4}**.
- Firewall MIB—Whose object identifier is **{jnxMibs 5}**.
- Destination Class Usage MIB—Whose object identifier is **{jnxMibs 6}**.
- Juniper Networks enterprise-specific extensions to the Ping MIB—Whose object identifier is **{jnxMibs 7}**.
- Juniper Networks enterprise-specific extensions to the Traceroute MIB—Whose object identifier is **{jnxMibs 8}**.

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- ATM MIB—Whose object identifier is **{jnxMibs 10}**.
- IPv6 and ICMPv6 MIB—Whose object identifier is **{jnxMibs 11}**.
- IPv4 MIB—Whose object identifier is **{jnxMibs 12}**.
- Juniper Networks enterprise-specific extensions to the RMON Events and Alarms MIB—Whose object identifier is **{jnxMIBs 13}**.
- Juniper Networks enterprise-specific extensions to the LDP Traps MIB—Whose object identifier is **{jnxMibs 14}**.
- Class-of-Service MIB—Whose object identifier is **{jnxMibs 15}**.
- Source Class Usage MIB—Whose object identifier is **{jnxMibs 16}**.
- Reverse-Path-Forwarding MIB—Whose object identifier is **{jnxMibs 17}**.
- Configuration Management MIB—Whose object identifier is **{jnxMibs 18}**.
- Passive Monitoring MIB—Whose object identifier is **{jnxMibs 19}**.
- SONET/SDH Interface Management MIB—Whose object identifier is **{jnxMibs 20}**.
- ATM Class-of-Service MIB—Whose object identifier is **{jnxMibs 21}**.
- IPsec Monitoring MIB—Whose object identifier is **{jnxMibs 22}**.
- Ethernet MAC MIB—Whose object identifier is **{jnxMibs 23}**.
- SONET APS MIB—Whose object identifier is **{jnxMibs 24}**.
- Chassis Definitions for Router Model MIB—Whose object identifier is **{jnxMibs 25}**.
- VPN MIB—Whose object identifier is **{jnxMibs 26}**.
- Flow Collection Services MIB—Whose object identifier is **{jnxMibs 28}** .
- RSVP Traffic Engineering (TE) MIB—Whose object identifier is **{jnxMibs 30}**.
- Host Resources MIB—Whose object identifier is **{jnxMibs 31}**.
- Services PIC MIB—Whose object identifier is **{jnxMibs 32}**.
- Dynamic Flow Capture (DFC) MIB—Whose object identifier is **{jnxMibs 33}**.
- Chassis Forwarding MIB—Whose object identifier is **{jnxMibs 34}**.
- System Log MIB—Whose object identifier is **{jnxMibs 35}**.
- MPLS LDP MIB—Whose object identifier is **{jnxMibs 36}**.
- Event MIB—Whose object identifier is **{jnxMibs 37}**.
- IP Forward MIB—Whose object identifier is **{jnxMibs 38}**.
- **jnxJSMibRoot**—Whose object identifier is **{jnxMibs 39}**.
- **jnxEXMibRoot**—Whose object identifier is **{jnxMibs 40}**.
- **jnxWXMibRoot**—Whose object identifier is **{jnxMibs 41}**.
- Packet Forwarding Engine MIB—Whose object identifier is **{jnxMibs 44}**.
- BFD MIB—Whose object identifier is **{jnxMibs 45}**.

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- Utility MIB—Whose object identifier is **{jnxMibs 47}**.
- L2ALD MIB—Whose object identifier is **{jnxMibs 48}**.
- L2TP MIB—Whose object identifier is **{jnxMibs 49}**.
- RPM MIB—Whose object identifier is **{jnxMibs 50}**.
- User AAA MIB—Whose object identifier is **{jnxMibs 51}**.
- Generic IPsec MIB—Whose object identifier is **{jnxMibs 52}**.
- L2CP MIB—Whose object identifier is **{jnxMibs 53}**.
- Pseudowire TDM MIB—Whose object identifier is **{jnxMibs 54}**.
- Pseudowire TC MIB—Whose object identifier is **{jnxMibs 55}**.
- OTN MIB—Whose object identifier is **{jnxMibs 56}**.

For more information about these MIBs, see "Juniper Networks Enterprise-Specific MIBs" on page 30.

Related Documentation

- Structure of Management Information MIB on page 59
- jnxProducts on page 60
- jnxServices on page 60
- jnxTraps on page 63
- jnxExperiment on page 64

jnxTraps

The object identifier for the **jnxTraps** root branch of the Structure of Management Information MIB is **{juniperMIB 4}**. The **jnxTraps** root branch contains the enterprise-specific SNMP traps supported by the Junos OS. These Juniper Networks enterprise-specific SNMP traps include:

- **jnxChassisTraps**—Whose object identifier is **{jnxTraps 1}**.
- **jnxChassisOKTraps**—Whose object identifier is **{jnxTraps 2}**.
- **jnxRmonTraps**—Whose object identifier is **{jnxTraps 3}**.
- **jnxLdpTraps**—Whose object identifier is **{jnxTraps 4}**.
- **jnxCmNotifications**—Whose object identifier is **{jnxTraps 5}**.
- **jnxSonetNotifications**—Whose object identifier is **{jnxTraps 6}**.
- **jnxPMonNotifications**— Whose object identifier is **{jnxTraps 7}**
- **jnxCollectorNotifications**—Whose object identifier is **{jnxTraps 8}**.
- **jnxPingNotification**—Whose object identifier is **{jnxTraps 9}**.
- **jnxSpNotification**—Whose object identifier is **{jnxTraps10}**.

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- **jnxDfcNotifications**—Whose object identifier is **{jnxTraps11}**.
- **jnxSyslogNotifications**—Whose object identifier is **{jnxTraps12}**.
- **jnxEventNotifications**—Whose object identifier is **{jnxTraps13}**.
- **jnxVccpNotifications**—Whose object identifier is **{jnxTraps14}**.
- **jnxOtnNotifications**—Whose object identifier is **{jnxTraps15}**.

Related Documentation

- Structure of Management Information MIB on page 59
- jnxProducts on page 60
- jnxServices on page 60
- jnxMibs on page 61
- jnxExperiment on page 64
- Configuring SNMP Trap Options and Groups on a Device Running Junos OS
- Managing Traps and Informs

jnxExperiment

The object identifier for the **jnxExperiment** root branch of the Structure of Management Information MIB is **{juniperMIB 5}**. The **jnxExperiment root** branch contains experimental Juniper Networks enterprise-specific MIBs. This is the top-level object identifier registry used by Juniper Networks products for SNMP modules containing experimental MIB definitions.

jnxExperiment MIBs are defined as the following MIBs:

- IETF work-in-process MIBs that have not been assigned a permanent object identifier by the IANA.
- Juniper Networks work-in-process MIBs that have not achieved final production quality or field experience.

The following draft supports the **jnxExperiment** MIB space: Internet draft draft-ietf-idr-bgp4-mibv2-03.txt, *Definitions of Managed Objects for the Fourth Version of Border Gateway Protocol (BGP-4), Second Version* (**jnxBgpM2PrefixInPrefixes**, **jnxBgpM2PrefixInPrefixesAccepted**, and **jnxBgpM2PrefixInPrefixesRejected** objects only).

Related Documentation

- Structure of Management Information MIB on page 59
- jnxProducts on page 60
- jnxServices on page 60
- jnxMibs on page 61
- jnxTraps on page 63

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CHAPTER 3

Interpreting the Enterprise-Specific Antivirus Objects MIB

- Antivirus Objects MIB on page 65
- Scan Engine Objects Table on page 66
- Scan Statistics Objects Table on page 66
- Antivirus Traps Objects Table on page 67

Antivirus Objects MIB

The Juniper Networks enterprise-specific Antivirus Objects MIB, **jnxJsUtmAvMIB**, whose object ID is **{jnxJsUTMRoot 1}**, defines the MIB for antivirus monitoring.

The Antivirus Objects MIB is part of unified threat management (UTM).

Related MIB objects include the following:

- **jnxJsAntiVirusObjects—{jnxJsAntiVirus 1}**
- **jnxJsAntiVirusTraps—{jnxJsAntiVirus 2}**
- **jnxJsAntiVirusTrapVars—{jnxJsAntiVirus 3}**

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-utm-av.txt .

For information about the enterprise-specific Antivirus Objects MIB, see the following topics:

- Scan Engine Objects Table on page 66
- Scan Statistics Objects Table on page 66
- Antivirus Traps Objects Table on page 67

Related Documentation

- Juniper Networks Enterprise-Specific SNMP Traps on page 633
- Standard SNMP Traps Supported on Devices Running Junos OS on page 653
- Juniper Networks Enterprise-Specific MIBs on page 30

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- Standard SNMP MIBs Supported by Junos OS on page 7

Scan Engine Objects Table

The **jnxJsAntiVirusEngine** object, whose object ID is **{jnxJsAntiVirusObjects 1}**, describes the objects in the Scan Engine Objects Table. See Table 5 on page 66.

Table 5: Scan Engine Objects Table

Object	Object ID	Description
jnxJsAVEngineType	**jnxJsAntiVirusEngine 1**	The antivirus scan engine type in use. If antivirus is not configured, the engine type is not known. • unknown-engine (1) • kaspersky-lab-engine (2) • juniper-express-engine (3)
jnxJsAVCurrentPatternVersionString	**jnxJsAntiVirusEngine 2**	The antivirus pattern database version currently in use.
jnxJsAVDatabaseType	**jnxJsAntiVirusEngine 3**	The database type in use: • full (1) • express (2) • unknown (3)

Related Documentation

- Antivirus Objects MIB on page 65
- Scan Statistics Objects Table on page 66
- Antivirus Traps Objects Table on page 67

Scan Statistics Objects Table

The **jnxJsAntiVirusStats** object, whose object ID is **{jnxJsAntiVirusObjects 2}**, describes the objects in the Scan Statistics Objects Table. See Table 6 on page 66.

Table 6: Scan Statistics Objects Table

Object	Object ID	Description
jnxJsAVScanCodeClean	**jnxJsAntiVirusStats 1**	Number of clean requests passed by the antivirus scan engine.
jnxJsAVScanCodeInfected	**jnxJsAntiVirusStats 2**	Number of infected requests found by the antivirus scan engine.
jnxJsAVScanCodeProtected	**jnxJsAntiVirusStats 3**	Number of requests that the antivirus scan engine cannot scan because of password protection.

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Table 6: Scan Statistics Objects Table *(continued)*

Object	Object ID	Description
jnxJsAVScanCodeDecompress	jnxJsAntiVirusStats 4	Number of requests that the antivirus scan engine cannot scan because of those requests that exceeded the maximum decompression layer limit.
jnxJsAVScanCodeCorrupted	jnxJsAntiVirusStats 5	Number of requests that the antivirus scan engine cannot scan because of file corruption.
jnxJsAVScanCodeNoResource	jnxJsAntiVirusStats 6	Number of requests that the antivirus scan engine cannot scan because of a lack of resources.
jnxJsAVScanCodeInternalError	jnxJsAntiVirusStats 7	Number of requests that the antivirus scan engine cannot scan because of an internal error.
jnxJsAVScanCodeMaxContentSize	jnxJsAntiVirusStats 8	Number of requests that the antivirus scan engine cannot scan because of those requests that exceeded the maximum content size limit.
jnxJsAVScanCodeTooManyReq	jnxJsAntiVirusStats 9	Number of requests that the antivirus scan engine cannot scan because of those requests that exceeded the maximum request limit.
jnxJsAVScanCodeTimeout	jnxJsAntiVirusStats 10	Number of requests that the antivirus scan engine cannot scan because of a scanning timeout.
jnxJsAVScanCodeEngineNotReady	jnxJsAntiVirusStats 11	Number of requests that the antivirus scan engine cannot scan because of the scan engine is not ready.

Related Documentation

- Antivirus Objects MIB on page 65
- Scan Engine Objects Table on page 66
- Antivirus Traps Objects Table on page 67

Antivirus Traps Objects Table

The **jnxJsAvPatternUpdateTrap** object, whose object ID is **{jnxJsAntiVirusTrapsPrefix 1}** , generates a trap when the antivirus pattern database is updated.

Table 7 on page 68 describes the objects in the Antivirus Traps Objects Table.

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Table 7: Antivirus Traps Objects Table

Object	Object ID	Description
jnxAVPatternVersionString	jnxJsAntiVirusTrapVars 1	The last successful update of the antivirus pattern database version.
jnxAVPatternTimestamp	jnxJsAntiVirusTrapVars 2	The last successful update of the antivirus pattern database timestamp.

Related Documentation

- Antivirus Objects MIB on page 65
- Scan Engine Objects Table on page 66
- Scan Statistics Objects Table on page 66
- Configuring SNMP Trap Options and Groups on a Device Running Junos OS
- Managing Traps and Informs

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CHAPTER 4

Interpreting the Enterprise-Specific Chassis MIBs

- Chassis MIBs on page 70
- Chassis MIB Textual Conventions on page 71
- jnxBoxAnatomy on page 72
- Chassis Traps on page 158
- Chassis Definitions for Router Model MIB on page 162
- MIB Objects for the M120 Router on page 165
- MIB Objects for the MX960 3D Universal Edge Router on page 166
- MIB Objects for the MX480 3D Universal Edge Router on page 167
- MIB Objects for the MX240 3D Universal Edge Router on page 167
- MIB Objects for the MX80 3D Universal Edge Router on page 168
- MIB Objects for the MX40 3D Universal Edge Router on page 169
- MIB Objects for the MX10 3D Universal Edge Router on page 169
- MIB Objects for the MX5 3D Universal Edge Router on page 170
- MIB Objects for the EX Series Ethernet Switches on page 171
- MIB Objects for the QFX Series on page 175
- MIB Objects for the SRX1400 Services Gateway on page 176
- MIB Objects for the SRX3400 Services Gateway on page 177
- MIB Objects for the SRX3600 Services Gateway on page 178
- MIB Objects for the SRX5600 Services Gateway on page 178
- MIB Objects for the SRX5800 Services Gateway on page 179
- MIB Objects for the SRX100 Services Gateway on page 180
- MIB Objects for the SRX210 Services Gateway on page 181
- MIB Objects for the SRX220 Services Gateway on page 182
- MIB Objects for the SRX240 Services Gateway on page 182
- MIB Objects for the SRX650 Services Gateway on page 183

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Chassis MIBs

The Juniper Networks enterprise-specific Chassis MIB provides information about the router and its components. MIB objects represent each component and the status of the components. The enterprise-specific Chassis Definitions for Router Model MIB contains the object identifiers (OIDs) that are used by the Chassis MIB to identify platform and chassis components. The Chassis MIB provides information that changes often. The Chassis Definitions for Router Model MIB provides information that changes less often.

You can retrieve information from the MIB using any network management system. For a downloadable version of the Chassis Definitions for Router Model MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-chas-defines.txt .

For more information about the enterprise-specific Chassis MIBs, see the following topics:

- Chassis MIB Textual Conventions on page 71
- jnxBoxAnatomy on page 72
- Chassis Traps on page 158
- Chassis Definitions for Router Model MIB on page 162
- MIB Objects for the M120 Router on page 165
- MIB Objects for the MX960 3D Universal Edge Router on page 166
- MIB Objects for the MX480 3D Universal Edge Router on page 167
- MIB Objects for the MX240 3D Universal Edge Router on page 167
- MIB Objects for the MX80 3D Universal Edge Router on page 168
- MIB Objects for the MX40 3D Universal Edge Router on page 169
- MIB Objects for the MX10 3D Universal Edge Router on page 169
- MIB Objects for the MX5 3D Universal Edge Router on page 170
- MIB Objects for the EX Series Ethernet Switches on page 171
- MIB Objects for the QFX Series on page 175
- MIB Objects for the SRX1400 Services Gateway on page 176
- MIB Objects for the SRX3400 Services Gateway on page 177
- MIB Objects for the SRX3600 Services Gateway on page 178
- MIB Objects for the SRX5600 Services Gateway on page 178
- MIB Objects for the SRX5800 Services Gateway on page 179
- MIB Objects for the SRX100 Services Gateway on page 180
- MIB Objects for the SRX210 Services Gateway on page 181
- MIB Objects for the SRX220 Services Gateway on page 182

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- MIB Objects for the SRX240 Services Gateway on page 182
- MIB Objects for the SRX650 Services Gateway on page 183

Related Documentation

- Juniper Networks Enterprise-Specific SNMP Traps on page 633
- Standard SNMP Traps Supported on Devices Running Junos OS on page 653
- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

Chassis MIB Textual Conventions

The enterprise-specific Chassis MIB uses the **jnxChassisId** object to denote the router chassis type. **jnxChassisId** can be one of the following integer values:

- **1**—Unknown
- **2**—Single chassis
- **3**—scc (TX Matrix router)
- **4**—lcc0 (T640 router)
- **5**—lcc1 (T640 router)
- **6**—lcc2 (T640 router)
- **7**—lcc3 (T640 router)
- **8**—jcs1
- **9**—jcs2
- **10**—jcs3
- **11**—jcs4
- **12**—sfc0
- **13**—sfc1
- **14**—sfc2
- **15**—sfc3
- **16**—sfc4
- **17**—lcc4
- **18**—lcc5
- **19**—lcc6
- **20**—lcc7
- **21**—lcc8
- **22**—lcc9

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- **23**—lcc10
- **24**—lcc11
- **25**—lcc12
- **26**—lcc13
- **27**—lcc14
- **28**—lcc15

Related Documentation

- Chassis MIBs on page 70

jnxBoxAnatomy

The object identifier for the **jnxMIBs** root branch is **{juniperMIB 3}** and includes one main subbranch, **jnxBoxAnatomy**, whose object identifier is **{jnxMibs 1}**.

The **jnxBoxAnatomy** MIB is explained in the following sections:

- Top-Level Objects on page 72
- jnxContainersTable on page 73
- jnxContentsLastChange on page 80
- jnxContentsTable on page 80
- jnxLEDLastChange on page 92
- jnxLEDTable on page 92
- jnxFilledLastChange on page 95
- jnxFilledTable on page 95
- jnxOperatingTable on page 106
- jnxRedundancyTable on page 115
- jnxFruTable on page 121
- jnxBoxKernelMemoryUsedPercent on page 157
- jnxBoxSystemDomainType on page 157

Top-Level Objects

The following branches of the **jnxBoxAnatomy** MIB are top-level objects:

- **jnxBoxClass**—The object identifier for the **jnxBoxClass** object is **{jnxBoxAnatomy 1}**. This object classifies the chassis product line.
- **jnxBoxDescr**—The object identifier for the **jnxBoxDescr** object is **{jnxBoxAnatomy 2}**. This object describes the chassis name and model.
- **jnxBoxSerialNo**—The object identifier for the **jnxBoxSerialNo** object is **{jnxBoxAnatomy 3}**. This object indicates the serial number of the chassis. **jnxBoxSerialNo** remains blank if the serial number is unknown or unavailable.

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- **jnxBoxRevision**—The object identifier for the **jnxBoxRevision** object is **{jnxBoxAnatomy 4}**. This object indicates the last revision of the chassis.
- **jnxBoxInstalled**—The object identifier for the **jnxBoxInstalled** object is **{jnxBoxAnatomy 5}**. This object indicates the last time the box was installed and operational, represented by the **sysUpTime** value.

jnxContainersTable

The object identifier for the **jnxContainersTable** object is **{jnxBoxAnatomy 6}**. This object shows the structure of the chassis.

You can use the **jnxContainersTable** object to retrieve specific information about the router, such as how many of each component the router can contain. For example, the **jnxContainersTable** object of an M20 router indicates that the router can accommodate four Flexible PIC Concentrators (FPCs). However, it does not describe how many FPCs the router actually has.

For more information about how many FPCs are actually on a router, see "jnxContentsTable" on page 80.

Entries within the **jnxContainersTable** object are represented by the **jnxContainersEntry** object, whose object identifier is **{jnxContainersTable 1}**. The **jnxContainersEntry** object contains the following objects that describe the contents of a particular router:

- **jnxContainersIndex**—The index value of an entry in the **jnxContainersEntry** object that corresponds to **jnxContainersType** and **jnxContainersDescr**, and whose object identifier is **{jnxContainersEntry 1}**.
- **jnxContainersView**—The orientation of a container from the front of the router and whose object identifier is **{jnxContainersEntry 2}**. This object also indicates that the container is embedded in the router and how it is accessible from corresponding views. The value of this object is a bitmap represented as a sum. If multiple bits are set, you can access the container from that set of views. The values represent the bit positions and their corresponding views as follows:
  - 1—Front
  - 2—Rear
  - 4—Top
  - 8—Bottom
  - 16—Left side
  - 32—Right side

  For each view plane, if specified counters are scattered in various views, the numbering sequence starts from left to right and then from top to bottom, as follows:
  - Left side
  - Right side

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- Top
- Bottom
- Front
- Rear



NOTE: References to left and right sides are based on the view from the front of the chassis.

In accordance with network management conventions, all indexes in the MIB begin with 1, not 0, although the slot number might be labeled 0.

- **jnxContainersLevel**—The abstraction level of the box or components for the **jnxContainersEntry** object and whose object identifier is **{jnxContainersEntry 3}**. The level is enumerated from the outside to the inside, and from the outer layer to the inner layer.

  For example, if the top level (level 0) of the box refers to the chassis frame, then the next level (level 1) refers to the FPC slot within the chassis frame. Finally, the PIC space within the FPC slot of the chassis corresponds to level 2.

- **jnxContainersWithin**—The container housing the entry at the next-higher level of the **jnxContainersEntry** object and whose object identifier is **{jnxContainersEntry 4}**.

  For example, the within value for **jnxMediaCardSpacePIC.0** is 7. Because the **jnxM20SlotFPC.0** retains an index value of 7, the FPC houses the PIC.

- **jnxContainersType**—The component of the Chassis MIB at a specific index, view, level, and within value for the **jnxContainersEntry** object and whose object identifier is **{jnxContainersEntry 5}**.

- **jnxContainersDescr**—The description of the component in the **jnxContainersEntry** object and whose object identifier is **{jnxContainersEntry 6}**.

- **jnxContainersCount**—The maximum number of a given component that the router can accommodate within the **jnxContainersEntry** object and whose object identifier is **{jnxContainersEntry 7}**.

  For example, the M20 router can house a specific maximum number of FPCs within the chassis frame. The maximum number is not necessarily the actual number of FPCs; this can change dynamically.

Table 8 on page 75 through Table 15 on page 79 provide examples of **jnxContainersEntry** objects in the **jnxContainersTable**. The following column headings for each table are abbreviated to correspond to the parts of the **jnxContainersEntry** objects:

- Index—**jnxContainersIndex**
- View—**jnxContainersView**
- Level—**jnxContainersLevel**

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- Within—**jnxContainersWithin**
- Type—**jnxContainersType**
- Description—**jnxContainersDescr**
- Count—**jnxContainersCount**

Table 8 on page 75 describes objects contained in a **jnxContainersEntry** in the **jnxContainersTable** of an M40 router.

Table 8: jnxContainersEntry Objects in the jnxContainersTable of an M40 Router

Index	View	Level	Within	Type	Description	Count
1	1	0	0	**jnxChassisM40.0**	Chassis frame compartment	1
2	2	1	1	**jnxSlotPowerSupply.0**	Power supply compartment	2
3	3	1	1	**jnxSlotCoolingImpeller.0**	Impeller compartment	2
4	2	1	1	**jnxSlotCoolingFan.0**	Fan compartment	3
5	2	1	1	**jnxSlotHostCtrl.0**	Host controller compartment	1
6	1	1	1	**jnxSlotSCB.0**	SCB slot	1
7	1	1	1	**jnxSlotFPC.0**	FPC slot	8
8	1	2	7	**jnxMediaSlotCardPIC.0**	PIC space	4
9	2	1	1	**jnxSlotRoutingEngine.0**	Routing Engine compartment	1

Table 9 on page 75 describes objects in the **jnxContainersTable** of an M20 router.

Table 9: jnxContainersEntry Objects in the jnxContainersTable of an M20 Router

Index	View	Level	Within	Type	Description	Count
1	1	0	0	**jnxChassisM20.0**	Chassis frame compartment	1
2	2	1	1	**jnxM20SlotPower.0**	Power supply compartment	2
4	3	1	1	**jnxSlotFan.0**	Fan compartment	4
6	2	1	1	**jnxM20SlotSSB.0**	SSB slot	2
7	1	1	1	**jnxM20SlotFPC.0**	FPC slot	4

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Table 9: jnxContainersEntry Objects in the jnxContainersTable of an M20 Router *(continued)*

Index	View	Level	Within	Type	Description	Count
8	1	2	7	**jnxM20MediaCardSpacePIC.0**	PIC space	4
9	2	1	1	**jnxM20RE.0**	Routing Engine compartment	2
10	1	1	1	**JNXM20FrontPanel.0**	Front display slot	1

Table 10 on page 76 describes objects contained in a **jnxContainersEntry** in the **jnxContainersTable** of an M160 router.

Table 10: jnxContainersEntry Objects in the jnxContainersTable of an M160 Router

Index	View	Level	Within	Type	Description	Count
1	1	0	0	**jnxChassisM160.0**	Chassis frame compartment	1
2	2	1	1	**Jnx160SlotPower.0**	Power supply compartment	2
4	3	1	1	**jnxM160SlotFan.0**	Fan compartment	4
6	2	1	1	**jnxM160SlotSFM.0**	SFM slot	4
7	1	1	1	**jnxM160SlotFPC.0**	FPC slot	8
8	1	2	7	**jnxM160MediaCardSlotPIC.0**	PIC space	4
9	2	1	1	**jnxM160SlotHM.0**	Host slot	2
10	1	1	1	**jnxM160SlotFPM.0**	FPM slot	1
11	2	1	1	**jnxM160SlotPCG.0**	PCG slot	2
12	2	1	1	**jnxM160SlotMCS.0**	MCS slot	2
13	1	1	1	**jnxM160SlotCIP.0**	CIP slot	1

Table 11 on page 77 describes objects contained in a **jnxContainersEntry** in the **jnxContainersTable** of an M10 router.

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Table 11: jnxContainersEntry Objects in the jnxContainersTable of an M10 Router

Index	View	Level	Within	Type	Description	Count
1	1	0	0	**jnxChassisM10.0**	Chassis frame compartment	1
2	2	1	1	**jnxM10SlotPower.0**	Power supply compartment	2
4	2	1	1	**jnxM10SlotFan.0**	Fan compartment	1
6	2	1	1	**jnxM10SlotFEB.0**	FEB slot	1
7	1	1	1	**jnxM10SlotFPC.0**	FPC slot	2
8	1	2	7	**jnxM10MediaCardSpacePIC.0**	PIC space	4
9	2	1	1	**jnxM10SlotRE.0**	Routing Engine compartment	1

Table 12 on page 77 describes objects contained in a **jnxContainersEntry** in the **jnxContainersTable** of an M5 router.

Table 12: jnxContainersEntry Objects in the jnxContainersTable of an M5 Router

Index	View	Level	Within	Type	Description	Count
1	1	0	0	**jnxChassisM5.0**	Chassis frame compartment	1
2	2	1	1	**jnxM5SlotPower.0**	Power supply compartment	2
4	3	1	1	**jnxM5SlotFan.0**	Fan compartment	4
6	2	1	1	**jnxM5SlotFEB.0**	FEB slot	1
7	1	1	1	**jnxM5SlotFPC.0**	FPC slot	1
8	1	2	7	**jnxM5MediaCardSlotPIC.0**	PIC space	4
9	2	1	1	**jnxM5SlotRE.0**	Routing Engine compartment	1

Table 13 on page 78 describes objects contained in a **jnxContainersEntry** in the **jnxContainersTable** of a T640 router.

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Table 13: jnxContainersEntry Objects in the jnxContainersTable of a T640 Router

Index	View	Level	Within	Type	Description	Count
1	1	0	0	**jnxChassisT640.0**	Chassis frame	1
2	2	1	1	**jnxT640SlotPower.0**	PEM slot	2
4	3	1	1	**jnxT640SlotFan.0**	Fan slot	3
7	1	1	1	**jnxT640SlotFPC.0**	FPC slot	8
8	1	2	7	**jnxT640MediaCardSpacePIC.0**	PIC slot	4
9	2	1	1	**jnxT640SlotHM.0**	Host slot	2
10	1	1	1	**jnxT640SlotFPB.0**	FPM slot	1
11	2	1	1	**jnxT640SlotSCG.0**	SCG slot	2
12	2	1	1	**jnxT640SlotCB.0**	CG slot	2
13	1	1	1	**jnxT640SlotCIP.0**	CIP slot	1
14	2	1	1	**jnxT640SlotSPMB.0**	SPMB slot	2
15	2	1	1	**jnxT640SlotSIB.0**	SIB slot	5

Table 14 on page 78 describes objects contained in a **jnxContainersEntry** in the **jnxContainersTable** of a T320 router.

Table 14: jnxContainersEntry Objects in the jnxContainersTable of a T320 Router

Index	View	Level	Within	Type	Description	Count
1	1	0	0	**jnxChassisT320.0**	Chassis frame	1
2	2	1	1	**jnxT320SlotPower.0**	PEM slot	2
4	3	1	1	**jnx320SlotFan.0**	Fan slot	3
7	1	1	1	**jnxT320SlotFPC.0**	FPC slot	8
8	1	2	7	**jnxT320MediaCardSpacePIC.0**	PIC slot	2
9	2	1	1	**jnxT320SlotHM.0**	Host slot	2
10	1	1	1	**jnxT320SlotFPB.0**	FPM slot	1

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Table 14: jnxContainersEntry Objects in the jnxContainersTable of a T320 Router *(continued)*

Index	View	Level	Within	Type	Description	Count
11	2	1	1	**jnxT320SlotSCG.0**	SCG slot	2
12	2	1	1	**jnxT320SlotCB.0**	CB slot	2
13	1	1	1	**jnxT320SlotCIP.0**	CIP slot	1
14	2	1	1	**jnxT320SlotSPMB.0**	SPMB slot	2
15	2	1	1	**jnxT320SlotSIB.0**	SIB slot	3

Table 15 on page 79 describes objects contained in a **jnxContainersEntry** in the **jnxContainersTable** of an M40e router.

Table 15: jnxContainersEntry Objects in the jnxContainersTable of an M40e Router

Index	View	Level	Within	Type	Description	Count
1	1	0	0	**jnxChassisM40e.0**	Chassis frame compartment	1
2	2	1	1	**jnxM40eSlotPower.0**	Power supply compartment	2
4	3	1	1	**jnxM40eSlotFan.0**	Fan compartment	4
6	2	1	1	**jnxM40eSlotSFM.0**	SFM slot	2
7	1	1	1	**jnxM40eSlotFPC.0**	FPC slot	8
8	1	2	7	**jnxM40eMediaCardSpacePIC.0**	PIC space	4
9	2	1	1	**jnxM40eSlotHM.0**	Host slot	2
10	1	1	1	**jnxM40eSlotFPM.0**	FPM slot	1
11	2	1	1	**jnxM40eSlotPCG.0**	PCG slot	2
12	2	1	1	**jnxM40eSlotMCS.0**	MCS slot	2
13	1	1	1	**jnxM40eSlotCIP.0**	CIP slot	1

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jnxContentsLastChange

The object identifier for **jnxContentsLastChange** object is **{jnxBoxAnatomy 7}**. This object indicates the time at which the box contents last changed, represented by the **sysUpTime** value.

jnxContentsTable

The object identifier for **jnxContentsTable** object is **{jnxBoxAnatomy 8}**. This object specifies the contents of the chassis.

The **jnxContentsTable** lists the contents of an entry, which are defined as follows:

- **jnxContentsContainerIndex**—Associates the **jnxContainersIndex** with the **jnxContainersTable**. The object identifier is **{jnxContentsEntry 1}**.
- **jnxContentsL1Index**—Denotes the level-one index of the container housing the component. The object identifier is **{jnxContentsEntry 2}**. It indicates the position of the component within different levels of the containers. This value is 0 if the position is unavailable or not applicable.

  NOTE: MIBs start with a value of 1, whereas the physical count on the router starts with a value of 0. To find the actual location of a component within a router, you must subtract 1 from the L1, L2, or L3 index.

- **jnxContentsL2Index**—Denotes the level-two index of the container housing the component. The object identifier is **{jnxContentsEntry 3}**. It indicates the position of the component within different levels of the containers. This value is 0 if the position is unavailable or not applicable.
- **jnxContentsL3Index**—Denotes the level-three index of the container housing the component. The object identifier is **{jnxContentsEntry 4}**. It indicates the position of the component within different levels of the containers. This value is 0 if the position is unavailable or not applicable.
- **jnxContentsType**—Indicates the type of the component at a specific container index or L1, L2, or L3 index. The object identifier is **{jnxContentsEntry 5}**.
- **jnxContentsDescr**—Specifies the type of the component in plain English. The object identifier is **{jnxContentsEntry 6}**.
- **jnxContentsSerialNo**—Specfies the serial number of the component. The object identifier is **{jnxContentsEntry 7}**.
- **jnxContentsRevision**—Contains the revision level of the component. The object identifier is **{jnxContentsEntry 8}**.
- **jnxContentsInstalled**—Indicates the time at which the component was last installed and operational, represented by the **sysUpTime** value. The object identifier is **{jnxContentsEntry 9}**.

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- **jnxContentsPartNo**—Contains the part number of the component (blank if unknown or unavailable). The object identifier is **{jnxContentsEntry 10}**.
- **jnxContentsChassisId** —Contains the ID of the chassis that contains the components. The object identifier is **{jnxContentsEntry 11}**.
- **jnxContentsChassisDescr**—Specifies the description of the chassis in plain English. The object identifier is **{jnxContentsEntry 12}**.
- **jnxContentsChassisCleiCode**—Contains the CLEI code of the chassis that contains the components listed in the row. A CLEI code is an intelligent code that consists of 10 alphanumeric characters with 4 data elements. The first data element is considered the basic code, with the first two characters indicating the technology or equipment type, and the third and fourth characters denoting the functional subcategory. The second data element represents the features, and its three characters denote functional capabilities or changes. The third data element has one character and denotes a reference to a manufacturer, system ID, specification, or drawing. The fourth data element consists of two characters and contains complementary data. These two characters provide a means of differentiating or providing uniqueness between the eight character CLEI codes by identifying the manufacturing vintage of the product. For more information about CLEI code, see http://www.commonlanguage.com/resources/commonlang/productshowroom/showroom/equip_id/carriers/overview.html . The object identifier is **{jnxContentsEntry 13}**.

Table 16 on page 82 through Table 18 on page 89 provide examples of **jnxContentEntry** objects. The following column headings for each table are abbreviated to correspond to the parts of the **jnxContentsEntry** objects:

- Container index— **jnxContentsContainerIndex**
- L1 Index—**jnxContentsL1Index**
- L2 Index—**jnxContentsL2Index**
- L3 Index—**jnxContentsL3Index**
- Type—**jnxContentsType**
- Description—**jnxContentsDescr**
- Serial Number—**jnxContentsSerialNo**
- Revision—**jnxContentsRevision**
- Installed—j**nxContentsInstalled**
- Part Number—**jnxContentsPartNo**

Table 16 on page 82 provides an example of **jnxContentEntry** objects in the **jnxContentTable** of an M20 router.

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Table 16: jnxContentsEntry Objects in the jnxContentsTable of an M20 Router

Container Index	L1 Index	L2 Index	L3 Index	Type	Description	Serial Number	Revision	Installed	Part Number
1	1	1	0	**jnxBackplaneM20.0**	Midplane	AL3280	REV07	0:0:00:00.00	710-00157
2	1	0	0	**jnxM20PowerDC.0**	DC power supply A	001652	REV 05	0:0:00:00.00	740-00146
2	2	0	0	**jnxM20PowerDC.0**	DC power supply B	001652	REV 05	0:0:00:00.00	740-00146
4	1	0	0	**jnxM20Fan.0**	Front top fan	–	–	0:0:00:00.00	–
4	2	0	0	**jnxM20Fan**	Middle fan	–	–	0:0:00:00.00	–
4	3	0	0	**jnxM20Fan**	Bottom fan	–	–	0:0:00:00.00	–
4	4	4	0	**jnxM20Fan**	Rear fan	–	–	0:0:00:00.00	–
6	1	0	0	**jnxM20SSB.0**	SSB 0 Internet Processor II	AG0809	REV 01	0:0:00:35.17	710-001951
7	1	0	0	**jnxM20FPC.0**	FPC @ 0/*/*	AN1335	REV 01	0:0:01:01.80	710-001292
7	2	0	0	**jnxM20FPC.0**	FPC @ 1/*/*	AN1124	REV 01	0:0:01:07:96	710-001292
7	3	0	0	**jnxM20FPC.0**	FPC @ 2/*/*	AN1726	REV 01	0:0:01:14:12	710-001292
7	4	0	0	**jnxM20FPC.0**	FPC @ 3/*/*	AN1691	REV 01	0:0:01:20.28	710-001292
8	1	1	0	**jnxM20QuadEther.0**	PIC: 4x, F/E, 100BASE-TX @ 0/0/*	HD4313	REV 04	0:0:00:00.00	750-002992
8	1	2	0	**jnxM20QuadEther.0**	PIC: 4x, F/E, 100BASE-TX @ 0/1/*	AJ5844	REV 04	0:0:00:00.00	750-002992
8	1	3	0	**jnxM20QuadEther.0**	PIC: 4x, F/E, 100BASE-TX @ 0/2/*	HD4518	REV 04	0:0:00:00.00	750-002992
8	1	4	0	**jnxM20QuadEther.0**	PIC: 4x, F/E, 100BASE-TX @ 0/3/*	HD4515	REV 04	0:0:00:00.00	750-002992

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Table 16: jnxContentsEntry Objects in the jnxContentsTable of an M20 Router *(continued)*

Container Index	L1 Index	L2 Index	L3 Index	Type	Description	Serial Number	Revision	Installed	Part Number
8	2	1	0	**jnxM20QuadEther.0**	PIC: 4x, F/E, 100BASE-TX @ 1/0/*	HD4296	REV 04	0:0:00:00.00	750-002992
8	2	2	0	**jnxM20QuadEther.0**	PIC: 4x, F/E, 100BASE-TX @ 1/1/*	HD4323	REV 04	0:0:00:00.00	750-002992
8	2	3	0	**jnxM20QuadEther.0**	PIC: 4x, F/E, 100BASE-TX @ 1/2/*	HD4129	REV 04	0:0:00:00.00	750-002992
8	2	4	0	**jnxM20QuadEther.0**	PIC: 4x, F/E, 100BASE-TX @ 1/3/*	HD4341	REV 04	0:0:00:00.00	750-002992
8	3	1	0	**jnxM20QuadEther.0**	PIC: 4x, F/E, 100BASE-TX@ 2/0/*	AH4147	REV 07	0:0:00:00.00	750-002303
8	3	2	0	**jnxM20QuadEther.0**	PIC: 4x, F/E, 100BASE-TX @ 2/1/*	AH4238	REV 07	0:0:00:00.00	750-002303
8	3	3	0	**jnxM20QuadEther.0**	PIC: 4x, F/E, 100BASE-TX @ 2/2/*	AH4116	REV 07	0:0:00:00.00	750-002303
8	3	4	0	**jnxM20QuadEther.0**	PIC: 4x, F/E, 100BASE-TX @ 2/3/*	AH4208	REV 07	0:0:00:00.00	750-002303
8	4	1	0	**jnxM20GigEther.0**	PIC: 1x G/E, 1000BASE-SX @ 3/0/*	AS3697	REV 07	0:0:00:00.00	750-001072
8	4	2	0	**jnxM20ChOc12toDS3.0**	PIC: 1x COC12SMIR @ 3/1/*	AE1110	REV 08	0:0:00:00.00	750-001190
8	4	4	0	**jnxM20ChStml1.0**	PIC: 1x CSTM1SMIR @ 3/3/*	AD9599	REV 04	0:0:00:00.00	750-003250
9	1	0	0	**jnxM20RE.0**	Routing Engine	–	–	3:16:16:53.21	–
10	1	0	0	**jnxM20FrontPanel.0**	Front panel display	–	–	0:0:00:00.00	–

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To verify the L1, L2, and L3 indexes, use the **show chassis hardware** command. Sample command output from an M20 router is as follows:

```
user@host> show chassis hardware
Item    Version  Part number   Serial Number    Description
Chassis53711     M20
Backplane        REV 07   710-001517       AL3280
Power Supply A   REV 05   740-001466          001652      DC
Power Supply B   REV 05   740-001466          001632      DC
Display         REV 04   710-001519      AP9225
Host 0 c900000619e6ba01 teknor
SSB slot 0        REV 01  710-001951        AG0809      Internet Processor
II
FPC 0       REV 01  710-001292  AN1335
 PIC 0      REV 04  750-002992  HD4313      4x F/E, 100 BASE-TX
 PIC 1      REV 04  750-002992  AJ5844      4x F/E, 100 BASE-TX
 PIC 2      REV 04  750-002992  HD4518      4x F/E, 100 BASE-TX
 PIC 3      REV 04  750-002992  HD4515      4x F/E, 100 BASE-TX
FPC 1       REV 01  710-001292  AN1124
 PIC 0      REV 04  750-002992  HD4296      4x F/E, 100 BASE-TX
 PIC 1      REV 04  750-002992  HD4323      4x F/E, 100 BASE-TX
 PIC 2      REV 04  750-002992  HD4129      4x F/E, 100 BASE-TX
 PIC 3      REV 04  750-002992  HD4341      4x F/E, 100 BASE-TX
FPC 2       REV 01  710-001292  AN1726
 PIC 0      REV 07  750-002303  AH4147      4x F/E, 100 BASE-TX
 PIC 1      REV 07  750-002303  AH4238      4x F/E, 100 BASE-TX
 PIC 2      REV 07  750-002303  AH4116      4x F/E, 100 BASE-TX
 PIC 3      REV 07  750-002303  AH4208      4x F/E, 100 BASE-TX
FPC 3       REV 01  710-001292  AN1691
 PIC 0      REV 08  750-001072  AS3697      1x G/E, 1000
BASE-SX
 PIC 1      REV 03  750-001190  AE1110      1x COC12, SMIR
 PIC 3      REV 04  750-003250  AD9599      1x CSTM1, SMIR
```

Table 17 on page 84 provides an example of **jnxContentEntry** objects in the **jnxContentTable** of a T640 router.

Table 17: jnxContentsEntry Objects in the jnxContentsTable of a T640 Router

Container Index	L1 Index	L2 Index	L3 Index	Type	Description	Serial Number	Revision	Installed	Part Number
1	1	0	0	**jnxMidplaneT640.0**	Midplane	AX5633	REV 04	0:0:00:00.00	710-002726
2	2	0	0	**jnxT640Power.0**	PEM 1	MD21815	RevX02	0:0:00:00.00	740-002595
4	1	1	0	**jnxT640Fan.0**	Top left front fan	–	–	0:0:00:00.00	–
4	1	2	0	**jnxT640Fan.0**	Top left middle fan	–	–	0:0:00:00.00	–
4	1	3	0	**jnxT640Fan.0**	Top left rear fan	–	–	0:0:00:00.00	–
4	1	4	0	**jnxT640Fan.0**	Top right front fan	–	–	0:0:00:00.00	–

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Table 17: jnxContentsEntry Objects in the jnxContentsTable of a T640 Router *(continued)*

Container Index	L1 Index	L2 Index	L3 Index	Type	Description	Serial Number	Revision	Installed	Part Number
4	1	5	0	**jnxT640Fan.0**	Top right middle fan	–	–	0:0:00:00.00	–
4	1	6	0	**jnxT640Fan.0**	Top right rear fan	–	–	0:0:00:00.00	–
4	2	1	0	**jnxT640Fan.0**	Bottom left front fan	–	–	0:0:00:00.00	–
4	2	2	0	**jnxT640Fan.0**	Bottom left middle fan	–	–	0:0:00:00.00	–
4	2	3	0	**jnxT640Fan.0**	Bottom left rear fan	–	–	0:0:00:00.00	–
4	2	4	0	**jnxT640Fan.0**	Bottom right front fan	–	–	0:0:00:00.00	–
4	2	5	0	**jnxT640Fan.0**	Bottom right middle fan	–	–	0:0:00:00.00	–
4	2	6	0	**jnxT640Fan.0**	Bottom right rear fan	–	–	0:0:00:00.00	–
4	3	1	0	**jnxT640Fan.0**	Fourth blower from top	–	–	0:0:00:00.00	–
4	3	2	0	**jnxT640Fan.0**	Bottom blower	–	–	0:0:00:00.00	–
4	3	3	0	**jnxT640Fan.0**	Middle blower	–	–	0:0:00:00.00	–
4	3	4	0	**jnxT640Fan.0**	Top blower	–	–	0:0:00:00.00	–
4	3	5	0	**jnxT640Fan.0**	Second blower from top	–	–	0:0:00:00.00	–
7	2	0	0	**jnxT640FPC.0**	FPC @ 1/*/*	HE3009	REV 01	0:18:56:48.81	710-002385
7	2	1	0	**jnxT640FPC.0**	FPC @ 1/0/* top temperature sensor	HE3009	REV 01	0:18:56:48.81	710-002385
7	2	2	0	**jnxT640FPC.0**	FPC @ 1/1/* bottom temperature sensor	HE3009	REV 01	0:18:56:48.81	710-002385

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Table 17: jnxContentsEntry Objects in the jnxContentsTable of a T640 Router *(continued)*

Container Index	L1 Index	L2 Index	L3 Index	Type	Description	Serial Number	Revision	Installed	Part Number
7	6	0	0	**jnxT640FPC.0**	FPC @ 5/*/*	HD5001	REV 03	0:18:57:02.71	710-001721
7	6	1	0	**jnxT640FPC.0**	FPC @ 5/0/* top temperature sensor	HD5001	REV 03	0:18:57:02.71	710-001721
7	6	2	0	**jnxT640FPC.0**	FPC @ 5/1/* bottom temperature sensor	HD5001	REV 03	0:18:57:02.71	710-001721
7	8	0	0	**jnxT640FPC.0**	FPC @ 7/*/*	HE3179	REV 01	0:18:56:52.85	710-002385
7	8	1	0	**jnxT640FPC.0**	FPC @ 7/0/* top temperature sensor	HE3179	REV 01	0:18:56:52.85	710-002385
7	8	2	0	**jnxT640FPC.0**	FPC @ 7/1/* bottom temperature sensor	HE3179	REV 01	0:18:56:52.85	710-002385
8	2	1	0	**jnxT640PIC3.0**	PIC: 1x G/E, 1000BASE-SX @ 1/0/*	AP5542	REV 08	0:18:56:50.91	750-001072
8	2	2	0	**jnxT640PIC3.0**	PIC: 1x OC-12 ATM, SMIR @ 1/1/*	AK6894	REV 02	0:18:56:55.24	750-002983
8	2	3	0	**jnxT640PIC3.0**	PIC: 1x G/E, 1000 BASE-SX @ 1/2/*	HD4968	REV 04	0:18:56:55.64	750-001894
8	6	1	0	**jnxT640PIC3.0**	PIC: 1x OC-192 SM SR1 @ 5/0/*	HC0273	REV 01	0:18:57:04.47	750-004535
8	6	2	0	**jnxT640PIC3.0**	PIC: 1x OC-192 SM SR1 @ 5/1/*	HC0271	REV 01	0:18:57:04.55	750-004535
8	6	3	0	**jnxT640PIC3.0**	PIC: 1x OC-192 SM SR1 @ 5/2/*	HC0254	REV 01	0:18:57:04.64	750-004535

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Table 17: jnxContentsEntry Objects in the jnxContentsTable of a T640 Router *(continued)*

Container Index	L1 Index	L2 Index	L3 Index	Type	Description	Serial Number	Revision	Installed	Part Number
8	8	1	0	**jnxT640PIC3.0**	PIC: 2x G/E, 1000Base-SX @ 7/0/*	AD3632	REV 01	0:18:56:55.16	710-002381
8	8	2	0	**jnxT640PIC3.0**	PIC: 4x OC-12 SONET, SMIR @ 7/1/*	AD3831	REV 05	0:18:56:55.18	750-001901
8	8	3	0	**jnxT640PIC3.0**	PIC: 1x OC-48 SONET, SMIR @ 7/2/*	AA9603	REV 01	0:18:56:55.21	750-001900
8	8	4	0	**jnxT640PIC3.0**	PIC: 1x OC-48 SONET, SMSR @ 7/3/*	AD5724	REV 05	0:18:56:55.24	750-001900
9	1	0	0	**jnxT640HM.0**	Host 0	–	–	0:19:19:30.95	–
9	2	0	0	**jnxT640HM.0**	Host 1	2108 6570 0292	REV 01	2:19:45:51.00	740-005022
10	1	0	0	**jnxT640FPB.0**	FPM	HE3245	REV 02	0:0:00:00.00	710-002901
11	1	0	0	**jnxT640SCG.0**	SCG 0	HF6023	REV 04	0:0:00:00.00	710-003423
11	2	0	0	**jnxT640SCG.0**	SCG 1	HF6061	REV 04	0:0:00:00.00	710-003423
12	2	0	0	**jnxT640CB.0**	CB 0	HE3614	REV 06	0:0:00:00.00	710-002728
12	2	0	0	**jnxT640CB.0**	CB 1	HE3627	REV 06	0:0:00:00.00	710-002728
13	1	0	0	**jnxT640CIP.0**	CIP	HA4729	REV 05	0:0:00:00.00	710-002895
14	1	0	0	**jnxT640SPMB.0**	SPMB 0	HF6876	REV 02	0:18:56:06.72	710-003229
14	2	0	0	**jnxT640SPMB.0**	SPMB 1	HG6237	REV 02	0:18:56:08.01	710-003229
15	1	0	0	**jnxT640SIB.0**	SIB 0	HJ9669	REV 02	0:0:00:00.00	710-005157
15	2	0	0	**jnxT640SIB.0**	SIB 1	HJ9668	REV 02	0:0:00:00.00	710-005157
15	3	0	0	**jnxT640SIB.0**	SIB 2	HH3039	REV 02	0:0:00:00.00	710-005157
15	4	0	0	**jnxT640SIB.0**	SIB 3	HH3041	REV 02	0:0:00:00.00	710-005157
15	5	0	0	**jnxT640SIB.0**	SIB 4	HJ9657	REV 02	0:0:00:00.00	710-005157

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To verify the L1, L2, and L3 indexes, use the **show chassis hardware** command. Sample command output from a T640 router is as follows:

```
user@host> show chassis hardware
Hardware inventory:
Item       Version Part number Serial number   Description
Chassis    T640
Midplane    REV 04  710-002726  AX5633
FPM GBUS    REV 02  710-002901  HE3245
FPM Display    REV 02  710-002897  HA4873
CIP          REV 05  710-002895  HA4729
PEM 1         RevX02  740-002595  MD21815         Power Entry Module
SCG 0         REV 04  710-003423  HF6023
SCG 1         REV 04  710-003423  HF6061
Host 0    unknown
Host 1    REV 01  740-005022  210865700292   RE-3.0
CB 0          REV 06  710-002728  HE3614
CB 1          REV 06  710-002728  HE3627
FPC 1         REV 01  710-002385  HE3009      FPC Type 1
 CPU          REV 06  710-001726  HC0010
 PIC 0        REV 08  750-001072  AP5542      1x G/E, 1000Base-SX
 PIC 1        REV 02  750-002983  AK6894      1x OC-12 ATM, SMIR
 PIC 2        REV 04  750-001894  HD4968      1x G/E, 1000Base-SX
 MMB 1        REV 03  710-001723  HE7264      MMB-144mbit
 ICBM         REV 01  710-003384  HE3042
 PPB 0        REV 01  710-003758  HE7173      PPB Type 2
 PPB 1        REV 01  710-003758  HE7170      PPB Type 2
FPC 5         REV 03  710-001721  HD5001      FPC Type 3
 CPU         REV 06  710-001726          HA5080
 PIC 0        REV 01  750-004535  HC0273      1x OC-192 SM SR1
 PIC 1        REV 01  750-004535  HC0271      1x OC-192 SM SR1
 PIC 2        REV 01  750-004535  HC0254      1x OC-192 SM SR1
 MMB 0        REV 03  710-001723  HE7263      MMB-144mbit
 MMB 1        REV 03  710-001723  HE7266      MMB-144mbit
 ICBM         REV 01  710-003384  HE3044
 PPB 0        REV 02  710-002845  HD6027      PPB Type 3
 PPB 1        REV 02  710-002845  HD6039      PPB Type 3
FPC 7         REV 01  710-002385  HE3179      FPC Type 2
 CPU         REV 06  710-001726          HE7915
 PIC 0        REV 01  710-002381  AD3632      2x G/E, 1000 BASE-SX
 PIC 1        REV 05  750-001901  AD3831      4x OC-12 SONET, SMIR
 PIC 2        REV 01  750-001900  AA9603      1x OC-48 SONET, SMIR
 PIC 3        REV 05  750-001900  AD5724      1x OC-48 SONET, SMSR
 MMB 1        REV 02  710-004047  HE3424      MMB-288mbit
 ICBM         REV 04  710-003384  HA4480
 PPB 0        REV 02  710-003758  HE3169      PPB Type 2
 PPB 1        REV 02  710-003758  HA4535      PPB Type 2
SPMB 0       REV 02  710-003229 HF6876
SPMB 1       REV 02  710-003229 HG6237
SIB 0         REV 02  710-005157  HJ9669      SIB-I8-F16
SIB 1         REV 02  710-005157  HJ9668      SIB-I8-F16
SIB 2         REV 02  710-005157  HH3039      SIB-I8-F16
SIB 3         REV 02  710-005157  HH3041      SIB-I8-F16
SIB 4         REV 02  710-005157  HJ9657      SIB-I8-F16
```

Table 18 on page 89 provides an example of **jnxContentEntry** objects in the **jnxContentTable** of a T320 router.

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Table 18: jnxContentsEntry Objects in the jnxContentsTable of a T320 Router

Container Index	L1 Index	L2 Index	L3 Index	Type	Description	Serial Number	Revision	Installed	Part Number
1	1	0	0	**jnxMidplaneT320.0**	Midplane	AY4527	Rev 01	(0) 0:00:00.00	710-004339
2	1	0	0	**jnxT320Power.0**	PEM 0	ML14099	Rev 01	(0) 0:00:00.00	–
4	1	1	0	**jnxT320Fan.0**	Top left front fan	–	–	(0) 0:00:00.00	–
4	1	2	0	**jnxT320Fan.0**	Top left middle fan	–	–	(0) 0:00:00.00	–
4	1	3	0	**jnxT320Fan.0**	Top left rear fan	–	–	(0) 0:00:00.00	–
4	1	4	0	**jnxT320Fan.0**	Top right front fan	–	–	(0) 0:00:00.00	–
4	1	5	0	**jnxT320Fan.0**	Top right middle fan	–	–	(0) 0:00:00.00	–
4	1	6	0	**jnxT320Fan.0**	Top right rear fan	–	–	(0) 0:00:00.00	–
4	2	1	0	**jnxT320Fan.0**	Bottom left front fan	–	–	(0) 0:00:00.00	–
4	2	2	0	**jnxT320Fan.0**	Bottom left middle fan	–	–	(0) 0:00:00.00	–
4	2	3	0	**jnxT320Fan.0**	Bottom left rear fan	–	–	(0) 0:00:00.00	–
4	2	4	0	**jnxT320Fan.0**	Bottom right front fan	–	–	(0) 0:00:00.00	–
4	2	5	0	**jnxT320Fan.0**	Bottom right middle fan	–	–	(0) 0:00:00.00	–
4	2	6	0	**jnxT320Fan.0**	Bottom right rear fan	–	–	(0) 0:00:00.00	–
4	3	1	0	**jnxT320Fan.0**	Rear tray top fan	–	–	(0) 0:00:00.00	–
4	3	2	0	**jnxT320Fan.0**	Rear tray second fan	–	–	(0) 0:00:00.00	–

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Table 18: jnxContentsEntry Objects in the jnxContentsTable of a T320 Router *(continued)*

Container Index	L1 Index	L2 Index	L3 Index	Type	Description	Serial Number	Revision	Installed	Part Number
4	3	3	0	**jnxT320Fan.0**	Rear tray middle fan	–	–	(0) 0:00:00.00	–
4	3	4	0	**jnxT320Fan.0**	Rear tray fourth fan	–	–	(0) 0:00:00.00	–
4	3	5	0	**jnxT320Fan.0**	Rear tray bottom fan	–	–	(0) 0:00:00.00	–
7	4	0	0	**jnxT320FPC.0**	FPC @ 3/*/*	AY4706	REV 01	(26190949) 3 days, 0:45:09.49	710-004333
7	4	1	0	**jnxT320FPC.0**	FPC @ 3/0/* top temperature sensor	AY4706	REV 01	(26190949) 3 days, 0:45:09.49	710-004333
7	4	2	0	**jnxT320FPC.0**	FPC @ 3/1/* bottom temperature sensor	AY4706	REV 01	(26190949) 3 days, 0:45:09.49	710-004333
8	1	1	0	**jnxT320PIC3**	PIC: 1x OC-192 SM SR2 @ 0/0/*	HJ9283	REV 06	(6378) 0:01:03.78	750-004535
8	1	2	0	**jnxT320PIC3**	PIC: 1x OC-192 SM SR2 @ 0/1/*	HJ9298	REV 06	(6434) 0:01:04.34	750-004535
9	1	0	0	**jnxT320HM.0**	Host 0	2108 6570 0286	REV 01	(32762924) 3 days, 19:00:29.24	740-005022
9	2	0	0	**jnxT320HM.0**	Host 1	2109 2900 0186	REV 01	(110269900) 12 days, 18:18:19.00	740-005022
10	1	0	0	**jnxT320FPB.0**	FPM	AY4514	REV 02	(0) 0:00:00.00	710-004461
11	1	0	0	**jnxT320SCG.0**	SCG 0	AY4520	REV 06	(0) 0:00:00.00	710-004455
11	2	0	0	**jnxT320SCG.0**	SCG 1	AY4526	REV 06	(0) 0:00:00.00	710-004455
12	1	0	0	**jnxT320CB.0**	CB 0	AY4765	REV 11	(0) 0:00:00.00	710-002728

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Table 18: jnxContentsEntry Objects in the jnxContentsTable of a T320 Router *(continued)*

Container Index	L1 Index	L2 Index	L3 Index	Type	Description	Serial Number	Revision	Installed	Part Number
12	2	0	0	**jnxT320CB.0**	CB 1	HG6051	REV 06	(0) 0:00:00.00	710-002728
13	1	0	0	**jnxT320CIP.0**	CIP	HC0476	REV 05	(0) 0:00:00.00	710-002895
14	1	0	0	**jnxT320SPMB.0**	SPMB 0	HB1893	REV 02	(26186997) 3 days, 0:44:29.97	710-003229
14	2	0	0	**jnxT320SPMB.0**	SPMB 1	HD5520	REV 02	(26186913) 3 days, 0:44:29.13	710-003229
15	1	0	0	**jnxT320SIB.0**	SIB 0	BC1509	REV 02	(0) 0:00:00.00	710-005157
15	2	0	0	**jnxT320SIB.0**	SIB 1	BC1512	REV 02	(0) 0:00:00.00	710-005157
15	3	0	0	**jnxT320SIB.0**	SIB 2	BC1494	REV 02	(0) 0:00:00.00	710-005157

To verify the L1, L2, and L3 indexes, use the **show chassis hardware** command. Sample command output from a T320 router is as follows:

```
user@host> show chassis hardware
Hardware inventory:
Item             Version  Part number  Serial number     Description
Chassis  T320
Midplane         REV 01   710-004339   AY4527
FPM GBUS         REV 02   710-004461   AY4514
FPM Display      REV 02   710-002897   HF6097
CIP              REV 05   710-002895   HC0476
PEM 0            Rev 01   740-004359   ML14099           Power Entry
Module
SCG 0            REV 06   710-004455   AY4520
SCG 1            REV 06   710-004455   AY4526
RE 0               REV 01   740-005022   210865700286      RE-3.0
RE 1               REV 01   740-005022   210929000186      RE-3.0
CB 0             REV 11   710-002728   AY4765
CB 1             REV 06   710-002728   HG6051
FPC 1            REV 01   710-004333   AY4507            FPC Type 3
  CPU            REV 06   710-001726   HA4719
  MMB 1          REV 03   710-004047   HD5738            MMB-288mbit
  PPB 0          REV 02   710-002845   HC0988            PPB Type 3
FPC 3            REV 01   710-004333   AY4706            FPC Type 3
  CPU            REV 06   710-001726   HE7916
  MMB 1          REV 03   710-004047   HG6326            MMB-288mbit
  PPB 0          REV 02   710-002845   HC0958            PPB Type 3
SPMB 0           REV 02   710-003229   HB1893
```

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```
SPMB 1          REV 02   710-003229   HD5520
SIB 0           REV 02   710-005157   BC1509          SIB-I8-F16
SIB 1           REV 02   710-005157   BC1512          SIB-I8-F16
SIB 2           REV 02   710-005157   BC1494          SIB-I8-F16
```

jnxLEDLastChange

The object identifier for the **jnxLEDLastChange** object is **{jnxBoxAnatomy 9}**. This object indicates when the LED last changed state. Its value is 0 if the **sysUpTime** value is unknown or if it already existed when the agent was active.

jnxLEDTable

The object identifier for the **jnxLEDTable** object is **{jnxBoxAnatomy 10}**. This object indicates the LED status of the router and lists the contents of an entry. Entries in the **jnxLEDTable** are represented by the **jnxLEDEntry** object, whose object identifier is **{jnxLEDTable 1}**.

The **jnxLEDTable** describes the components of the LED Box Indicators, whose elements are described as follows:

- **jnxLEDAssociateTable**—The associate table to which the entry is related, whose object identifier is **{jnxLEDEntry 1}**.
- **jnxLEDAssociateIndex**—The index of the subject in the associated table to which the entry is related, whose object identifier is **{jnxLEDEntry 2}**. The associate index is the index of the subject in the associated table, which returns you to the **jnxContainersTable**.
- **jnxLEDL1Index**—The level-one index of the associate table to which an entry is related, whose object identifier is **{jnxLEDEntry 3}**. It indicates the position of the component within the different levels of the containers. This value is 0 if the position is unavailable or not applicable.

  NOTE: MIBs start with a value of 1, while the physical count on the router starts with a value of 0. To find the actual location of a component within a router, you must subtract 1 from the L1, L2, or L3 index.

- **jnxLEDL2Index**—The level-two index of the associate table to which an entry is related, whose object identifier is **{jnxLEDEntry 4}**. It indicates the position of the component within the different levels of the containers. This value is 0 if the position is unavailable or not applicable.
- **jnxLEDL3Index**—The level-three index of the associate table to which an entry is related, whose object identifier is **{jnxLEDEntry 5}**. It indicates the position of the component within the different levels of the containers. This value is 0 if the position is unavailable or not applicable.
- **jnxLEDOriginator**—The chassis component that originated the update, whose object identifier is **{jnxLEDEntry 6}**.
- **jnxLEDDescr**—The name or detailed description of the entry, whose object identifier is **{jnxLEDEntry 7}**.

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- **jnxLEDState**—The state of the LED indicator, whose object identifier is **{jnxLEDEntry 8}**. The state is represented by one of the following values:
  - Amber—Alarm, offline, not working
  - Blue—Online as the active primary
  - Green—Working normally online as a standby backup if there is an active primary
  - Other—Unknown or unavailable
  - Red—Alert, component failed
  - Yellow—Alarm, warning
- **jnxLEDStateOrdered**—The state of the LED indicator, whose object identifier is **{jnxLEDEntry 9}**. **jnxLEDStateOrdered** provides the same information as **jnxLEDState** but lists the states in a different order. The states are defined as follows:
  - Blue—Online as the active primary
  - Green—Working normally online as a standby backup if there is an active primary
  - Amber—Alarm, offline, not working
  - Yellow—Alarm, warning
  - Red—Alert, component failed
  - Other—Unknown or unavailable

Table 19 on page 94 through Table 21 on page 95 provide examples of **jnxLEDEntry** objects. The following column headings for each table are abbreviated to correspond to the parts of the **jnxLEDEntry** objects:

- Associate table—**jnxLEDAssociateTable**
- Associate index—**jnxLEDAssociateIndex**
- L1 Index—**jnxLEDL1Index**
- L2 Index—**jnxLEDL2Index**
- L3 Index—**jnxLEDL3Index**
- Originator—**jnxLEDOriginator**
- Description—**jnxLEDDescr**
- State—**jnxLEDState**

Table 19 on page 94 provides an example of **jnxLEDEntry** objects in the **jnxLEDTable** of an M20 router.

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Table 19: jnxLEDEntry Objects in the jnxLEDTable of an M20 Router

Associate Table	Associate Index	L1 Index	L2 Index	L3 Index	Originator	Description	State
jnxContentsTable	1	1	0	0	**jnxChassisM20.0**	Chassis alarm LED	Other
jnxContentsTable	6	1	0	0	**jnxM20SSB.0**	SSB 1 LED	Blue
jnxContentsTable	6	2	0	0	**jnxM20SSB.0**	SSB 2 LED	Green
jnxContentsTable	7	1	0	0	**jnxM20FPC.0**	FPC 1 LED	Amber
jnxContentsTable	7	2	0	0	**jnxM20FPC.0**	FPC 2 LED	Blue
jnxContentsTable	7	3	0	0	**jnxM20FPC.0**	FPC 3 LED	Blue
jnxContentsTable	7	4	0	0	**jnxM20FPC.0**	FPC 4 LED	Amber
jnxContentsTable	9	1	0	0	**jnxM20RE.0**	Routing Engine 1 LED	Blue
jnxContentsTable	9	2	0	0	**jnxM20RE.0**	Routing Engine 2 LED	Other

Table 20 on page 94 provides an example of **jnxLEDEntry** objects in the **jnxLEDTable** of a T640 router.

Table 20: jnxLEDEntry Objects in the jnxLEDTable of a T640 Router

Associate Table	Associate Index	L1 Index	L2 Index	L3 Index	Originator	Description	State
jnxContentsTable	1	1	0	0	**jnxChassisT640.0**	Chassis alarm LED	Other
jnxContentsTable	7	1	0	0	**jnxT640FPC.0**	FPC slot 0 LED	Other
jnxContentsTable	7	2	0	0	**jnxT640FPC.0**	FPC slot 1 LED	Green
jnxContentsTable	7	3	0	0	**jnxT640FPC.0**	FPC slot 2 LED	Other
jnxContentsTable	7	4	0	0	**jnxT640FPC.0**	FPC slot 3 LED	Other
jnxContentsTable	7	5	0	0	**jnxT640FPC.0**	FPC slot 4 LED	Other
jnxContentsTable	7	6	0	0	**jnxT640FPC.0**	FPC slot 5 LED	Green
jnxContentsTable	7	7	0	0	**jnxT640FPC.0**	FPC slot 6 LED	Other
jnxContentsTable	7	8	0	0	**jnxT640FPC.0**	FPC slot 7 LED	Green
jnxContentsTable	9	1	0	0	**jnxT640HM.0**	Host 0 LED	Blue

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Table 20: jnxLEDEntry Objects in the jnxLEDTable of a T640 Router *(continued)*

Associate Table	Associate Index	L1 Index	L2 Index	L3 Index	Originator	Description	State
jnxContentsTable	9	2	0	0	**jnxT640HM.0**	Host 1 LED	Green

Table 21 on page 95 provides an example of **jnxLEDEntry** objects in the **jnxLEDTable** of a T320 router.

Table 21: jnxLEDEntry Objects in the jnxLEDTable of a T320 Router

Associate Table	Associate Index	L1 Index	L2 Index	L3 Index	Originator	Description	State
jnxContentsTable(3)	1	1	0	0	**jnxChassisT320.0**	Chassis alarm LED	Other
jnxContentsTable(3)	7	1	0	0	**jnxT320FPC.0**	FPC slot 0 LED	Other
jnxContentsTable(3)	7	2	0	0	**jnxT320FPC.0**	FPC slot 1 LED	Other
jnxContentsTable(3)	7	3	0	0	**jnxT320FPC.0**	FPC slot 2 LED	Other
jnxContentsTable(3)	7	4	0	0	**jnxT320FPC.0**	FPC slot 3 LED	Other
jnxContentsTable(3)	7	5	0	0	**jnxT320FPC.0**	FPC slot 4 LED	Other
jnxContentsTable(3)	7	6	0	0	**jnxT320FPC.0**	FPC slot 5 LED	Other
jnxContentsTable(3)	7	7	0	0	**jnxT320FPC.0**	FPC slot 6 LED	Other
jnxContentsTable(3)	7	8	0	0	**jnxT320FPC.0**	FPC slot 7 LED	Other
jnxContentsTable(3)	9	1	0	0	**jnxT320HM.0**	Host 0 LED	Blue
jnxContentsTable(3)	9	2	0	0	**jnxT320HM.0**	Host 1 LED	Green

jnxFilledLastChange

The object identifier for the **jnxFilledLastChange** object is **{jnxBoxAnatomy 11}**. This object indicates when the box filled status last changed. This variable is 0 if the **sysUpTime** value is unknown or it already existed when the agent was active.

jnxFilledTable

The object identifier for the **jnxFilledTable** object is **{jnxBoxAnatomy 12}**. This object indicates whether a specific container in the router is used (filled) or empty. This table is used for inventory and capacity planning.

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Entries in the **jnxFilledTable** are represented by the **jnxFilledEntry** object, whose object identifier is **{jnxFilledTable 1}**.

The **jnxFilledTable** describes the status of specific containers whose component objects are described as follows:

- **jnxFilledContainerIndex**—The associated **jnxContainersIndex** in the **jnxContainersTable**, whose object identifier is **{jnxFilledEntry 1}**.
- **jnxFilledL1Index**—The level-one index of the container housing the entry, whose object identifier is **{jnxFilledEntry 2}**.
- **jnxFilledL2Index**—The level-two index of the container housing the entry, whose object identifier is **{jnxFilledEntry 3}**.
- **jnxFilledL3Index**—The level-three index of the container housing the entry, whose object identifier is **{jnxFilledEntry 4}**.
- **jnxFilledDescr**—The entry's name or detailed description of the entry, whose object identifier is **{jnxFilledEntry 5}**.
- **jnxFilledState**—The entry's state (filled or empty), whose object identifier is **{jnxFilledEntry 6}**.

Table 22 on page 96 through Table 24 on page 103 provide examples of **jnxFilledEntry** objects in the **jnxFilledTable**. The following column headings for each table are abbreviated to correspond to the parts of the **jnxFilledEntry** objects:

- Container index—**jnxFilledContainerIndex**
- L1—**jnxFilledL1Index**
- L2—**jnxFilledL2Index**
- L3—**jnxFilledL3Index**
- Description—**jnxFilledDescr**
- State—**jnxFilledState**

Table 22 on page 96 provides an example of **jnxFilledEntry** objects in the **jnxFilledTable** of an M20 router.

Table 22: jnxFilledEntry Objects in the jnxFilledTable of an M20 Router

Container Index	L1	L2	L3	Description	State
1	1	0	0	Chassis frame compartment	Filled
1	1	1	0	Temperature sensor space 0	Filled
1	1	2	0	Temperature sensor space 1	Filled
2	1	0	0	Power supply compartment A	Filled

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Table 22: jnxFilledEntry Objects in the jnxFilledTable of an M20 Router *(continued)*

Container Index	L1	L2	L3	Description	State
2	2	0	0	Power supply compartment B	Empty
3	1	0	0	Rear top impeller compartment	Filled
3	2	0	0	Front bottom impeller compartment	Filled
4	1	0	0	Rear left fan compartment	Filled
4	2	0	0	Right center fan compartment	Filled
4	3	0	0	Rear right fan compartment	Filled
5	1	0	0	Host controller compartment	Filled
6	1	0	0	SCB slot	Filled
7	1	0	0	FPC slot 0	Empty
7	2	0	0	FPC slot 1	Empty
7	3	0	0	FPC slot 2	Filled
7	4	0	0	FPC slot 3	Filled
7	5	0	0	FPC slot 4	Empty
7	6	0	0	FPC slot 5	Filled
7	7	0	0	FPC slot 6	Empty
7	8	0	0	FPC slot 7	Empty
8	1	1	0	PIC space @ 0/0/*	Empty
8	1	2	0	PIC space @ 0/1/*	Empty
8	1	3	0	PIC space @ 0/2/*	Empty
8	1	4	0	PIC space @ 0/3/*	Empty
8	2	1	0	PIC space @ 1/0/*	Empty
8	2	2	0	PIC space @ 1/1/*	Empty
8	2	3	0	PIC space @ 1/2/*	Empty

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Table 22: jnxFilledEntry Objects in the jnxFilledTable of an M20 Router *(continued)*

Container Index	L1	L2	L3	Description	State
8	2	4	0	PIC space @ 1/3/*	Empty
8	3	1	0	PIC space @ 2/0/*	Filled
8	3	2	0	PIC space @ 2/1/*	Filled
8	3	3	0	PIC space @ 2/2/*	Filled
8	3	4	0	PIC space @ 2/3/*	Filled
8	4	1	0	PIC space @ 3/0/*	Filled
8	4	2	0	PIC space @ 3/1/*	Filled
8	4	3	0	PIC space @ 3/2/*	Filled
8	4	4	0	PIC space @ 3/3/*	Filled
8	5	1	0	PIC space @ 4/0/*	Empty
8	5	2	0	PIC space @ 4/1/*	Empty
8	5	3	0	PIC space @ 4/2/*	Empty
8	5	4	0	PIC space @ 4/3/*	Empty
8	6	1	0	PIC space @ 5/0/*	Filled
8	6	2	0	PIC space @ 5/1/*	Filled
8	6	3	0	PIC space @ 5/2/*	Filled
8	6	4	0	PIC space @ 5/3/*	Filled
8	7	1	0	PIC space @ 6/0/*	Empty
8	7	2	0	PIC space @ 6/1/*	Empty
8	7	3	0	PIC space @ 6/2/*	Empty
8	7	4	0	PIC space @ 6/3/*	Empty
8	8	1	0	PIC space @ 7/0/*	Empty
8	8	2	0	PIC space @ 7/1/*	Empty

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Table 22: jnxFilledEntry Objects in the jnxFilledTable of an M20 Router *(continued)*

Container Index	L1	L2	L3	Description	State
8	8	3	0	PIC space @ 7/2/*	Empty
8	8	4	0	PIC space @ 7/3/*	Empty
9	1	0	0	Routing Engine compartment	Filled

Table 23 on page 99 provides an example of **jnxFilledEntry** objects in the **jnxFilledTable** of a T640 router.

Table 23: jnxFilledEntry Objects in the jnxFilledTable of a T640 Router

Container Index	L1	L2	L3	Description	State
1	1	0	0	Chassis frame	Filled
2	1	0	0	PEM slot 0	Empty
2	2	0	0	PEM slot 1	Filled
4	1	1	0	Top left front fan slot	Filled
4	1	2	0	Top left middle fan slot	Filled
4	1	3	0	Top left rear fan slot	Filled
4	1	4	0	Top right front fan slot	Filled
4	1	5	0	Top right middle fan slot	Filled
4	1	6	0	Top right rear fan slot	Filled
4	2	1	0	Bottom left front fan slot	Filled
4	2	2	0	Bottom left middle fan slot	Filled
4	2	3	0	Bottom left rear fan slot	Filled
4	2	4	0	Bottom right front fan slot	Filled
4	2	5	0	Bottom right middle fan slot	Filled
4	2	6	0	Bottom right rear fan slot	Filled
4	3	1	0	Fourth blower from top slot	Filled

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Table 23: jnxFilledEntry Objects in the jnxFilledTable of a T640 Router *(continued)*

Container Index	L1	L2	L3	Description	State
4	3	2	0	Bottom blower slot	Filled
4	3	3	0	Middle blower slot	Filled
4	3	4	0	Top blower slot	Filled
4	3	5	0	Second blower from top slot	Filled
7	3	2	0	FPC slot 0	Empty
7	3	3	0	FPC slot 0 top temperature sensor	Empty
7	3	4	0	FPC slot 0 bottom temperature sensor	Empty
7	3	5	0	FPC slot 1	Filled
7	3	6	0	FPC slot 1 top temperature sensor	Filled
7	1	0	0	FPC slot 1 bottom temperature sensor	Filled
7	1	1	0	FPC slot 2	Empty
7	1	2	0	FPC slot 2 top temperature sensor	Empty
7	2	0	0	FPC slot 2 bottom temperature sensor	Empty
7	2	1	0	FPC slot 3	Empty
7	2	2	0	FPC slot 3 top temperature sensor	Empty
7	3	0	0	FPC slot 3 bottom temperature sensor	Empty
7	3	1	0	FPC slot 4	Empty
7	3	2	0	FPC slot 4 top temperature sensor	Empty
7	4	0	0	FPC slot 4 bottom temperature sensor	Empty
7	4	1	0	FPC slot 5	Filled
7	4	2	0	FPC slot 5 top temperature sensor	Filled

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Table 23: jnxFilledEntry Objects in the jnxFilledTable of a T640 Router *(continued)*

Container Index	L1	L2	L3	Description	State
7	5	0	0	FPC slot 5 bottom temperature sensor	Filled
7	5	1	0	FPC slot 6	Empty
7	5	2	0	FPC slot 6 top temperature sensor	Empty
7	6	0	0	FPC slot 6 bottom temperature sensor	Empty
7	6	1	0	FPC slot 7	Filled
7	6	2	0	FPC slot 7 top temperature sensor	Filled
7	7	0	0	FPC slot 7 bottom temperature sensor	Filled
8	1	1	0	PIC slot @ 0/0/*	Empty
8	1	2	0	PIC slot @ 0/1/*	Empty
8	1	3	0	PIC slot @ 0/2/*	Empty
8	1	4	0	PIC slot @ 0/3/*	Empty
8	2	1	0	PIC slot @ 1/0/*	Filled
8	2	2	0	PIC slot @ 1/1/*	Filled
8	2	3	0	PIC slot @ 1/2/*	Filled
8	2	4	0	PIC slot @ 1/3/*	Empty
8	3	1	0	PIC slot @ 2/0/*	Empty
8	3	2	0	PIC slot @ 2/1/*	Empty
8	3	3	0	PIC slot @ 2/2/*	Empty
8	3	4	0	PIC slot @ 2/3/*	Empty
8	4	1	0	PIC slot @ 3/0/*	Empty
8	4	2	0	PIC slot @ 3/1/*	Empty
8	4	3	0	PIC slot @ 3/2/*	Empty

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Table 23: jnxFilledEntry Objects in the jnxFilledTable of a T640 Router *(continued)*

Container Index	L1	L2	L3	Description	State
8	4	4	0	PIC slot @ 3/3/*	Empty
8	5	1	0	PIC slot @ 4/0/*	Empty
8	5	2	0	PIC slot @ 4/1/*	Empty
8	5	3	0	PIC slot @ 4/2/*	Empty
8	5	4	0	PIC slot @ 4/3/*	Empty
8	6	1	0	PIC slot @ 5/0/*	Filled
8	6	2	0	PIC slot @ 5/1/*	Filled
8	6	3	0	PIC slot @ 5/2/*	Filled
8	6	4	0	PIC slot @ 5/3/*	Empty
8	7	1	0	PIC slot @ 6/0/*	Empty
8	7	2	0	PIC slot @ 6/1/*	Empty
8	7	3	0	PIC slot @ 6/2/*	Empty
8	7	4	0	PIC slot @ 6/3/*	Empty
8	8	1	0	PIC slot @ 7/0/*	Filled
8	8	2	0	PIC slot @ 7/1/*	Filled
8	8	3	0	PIC slot @ 7/2/*	Filled
8	8	4	0	PIC slot @ 7/3/*	Filled
9	1	0	0	Host 0 slot	Filled
9	2	0	0	Host 1 slot	Filled
10	1	0	0	FPM slot	Filled
11	1	0	0	SCG slot 0	Filled
11	2	0	0	SCG slot 1	Filled
12	1	0	0	CB slot 0	Filled

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Table 23: jnxFilledEntry Objects in the jnxFilledTable of a T640 Router *(continued)*

Container Index	L1	L2	L3	Description	State
12	2	0	0	CB slot 1	Filled
13	1	0	0	CIP slot	Filled
14	1	0	0	SPMB slot 0	Filled
14	2	0	0	SPMB slot 1	Filled
15	1	0	0	SIB slot 0	Filled
15	2	0	0	SIB slot 1	Filled
15	3	0	0	SIB slot 2	Filled
15	4	0	0	SIB slot 3	Filled
15	5	0	0	SIB slot 4	Filled

Table 24 on page 103 provides an example of **jnxFilledEntry** objects in the **jnxFilledTable** of a T320 router.

Table 24: jnxFilledEntry Objects in the jnxFilledTable of a T320 Router

Container Index	L1	L2	L3	Description	State
1	1	0	0	Chassis frame	Filled
2	1	0	0	PEM slot 0	Filled
2	2	0	0	PEM slot 1	Empty
4	1	1	0	Top left front fan slot	Filled
4	1	2	0	Top left middle fan slot	Filled
4	1	3	0	Top left rear fan slot	Filled
4	1	4	0	Top right front fan slot	Filled
4	1	5	0	Top right middle fan slot	Filled
4	1	6	0	Top right rear fan slot	Filled
4	2	1	0	Bottom left front fan slot	Filled

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Table 24: jnxFilledEntry Objects in the jnxFilledTable of a T320 Router *(continued)*

Container Index	L1	L2	L3	Description	State
4	2	2	0	Bottom left middle fan slot	Filled
4	2	3	0	Bottom left rear fan slot	Filled
4	2	4	0	Bottom right front fan slot	Filled
4	2	5	0	Bottom right middle fan slot	Filled
4	2	6	0	Bottom right rear fan slot	Filled
4	3	1	0	Rear tray top fan slot	Filled
4	3	2	0	Rear tray second fan slot	Filled
4	3	3	0	Rear tray middle fan slot	Filled
4	3	4	0	Rear tray fourth fan slot	Filled
4	3	5	0	Rear tray bottom fan slot	Filled
7	1	0	0	FPC slot 0	Empty
7	1	1	0	FPC slot top temperature sensor	Empty
7	1	2	0	FPC slot 0 bottom temperature sensor	Empty
7	2	0	0	FPC slot 1	Empty
7	2	1	0	FPC slot 1 top temperature sensor	Empty
7	2	2	0	FPC slot 1 bottom temperature sensor	Empty
7	3	0	0	FPC slot 2	Empty
7	3	1	0	FPC slot 2 top temperature sensor	Empty
7	3	2	0	FPC slot 2 bottom temperature sensor	Empty
7	4	0	0	FPC slot 3	Filled
7	4	1	0	FPC slot 3 top temperature sensor	Filled
7	4	2	0	FPC slot 3 bottom temperature sensor	Filled

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Table 24: jnxFilledEntry Objects in the jnxFilledTable of a T320 Router *(continued)*

Container Index	L1	L2	L3	Description	State
7	5	1	0	FPC slot 4	Empty
7	5	2	0	FPC slot 4 top temperature sensor	Empty
7	5	0	0	FPC slot 4 bottom temperature sensor	Empty
7	6	1	0	FPC slot 5	Empty
7	6	2	0	FPC slot 5 top temperature sensor	Empty
7	6	0	0	FPC slot 5 bottom temperature sensor	Empty
7	7	1	0	FPC slot 6	Empty
7	7	2	0	FPC slot 6 top temperature sensor	Empty
7	7	0	0	FPC slot 6 bottom temperature sensor	Empty
7	8	1	0	FPC slot 7	Empty
7	8	2	0	FPC slot 7 top temperature sensor	Empty
7	8	0	0	FPC slot 7 bottom temperature sensor	Empty
8	1	1	0	PIC slot @ 0/0/*	Empty
8	1	2	0	PIC slot @ 0/1/*	Empty
8	2	1	0	PIC slot @ 1/0/*	Empty
8	2	2	0	PIC slot @ 1/1/*	Empty
8	3	1	0	PIC slot @ 2/0/*	Empty
8	3	2	0	PIC slot @ 2/1/*	Empty
8	4	1	0	PIC slot @ 3/0/*	Filled
8	4	2	0	PIC slot @ 3/1/*	Filled
8	5	1	0	PIC slot @ 4/0/*	Empty

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Table 24: jnxFilledEntry Objects in the jnxFilledTable of a T320 Router *(continued)*

Container Index	L1	L2	L3	Description	State
8	5	2	0	PIC slot @ 4/1/*	Empty
8	6	1	0	PIC slot @ 5/0/*	Empty
8	6	2	0	PIC slot @ 5/1/*	Empty
8	7	1	0	PIC slot @ 6/0/*	Empty
8	7	2	0	PIC slot @ 6/1/*	Empty
8	8	1	0	PIC slot @ 7/0/*	Empty
8	8	2	0	PIC slot @ 7/1/*	Empty
9	1	0	0	Host 0 slot	Filled
9	2	0	0	Host 1 slot	Filled
10	1	0	0	FPM slot	Filled
11	1	0	0	SCG slot 0	Filled
11	2	0	0	SCG slot 1	Filled
12	1	0	0	CB slot 0	Filled
12	2	0	0	CB slot 1	Filled
13	1	0	0	CIP slot	Filled
14	1	0	0	SPMB slot 0	Filled
14	2	0	0	SPMB slot 1	Filled
15	1	0	0	SIB slot 0	Filled
15	2	0	0	SIB slot 1	Filled
15	3	0	0	SIB slot 2	Filled

jnxOperatingTable

The object identifier for the **jnxOperatingTable** is **{jnxBoxAnatomy 13}**. This object reports the operating status of various components such as CPU, buffers, and memory.

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Juniper Networks devices implement packet forwarding and routing functions with two separate components, the Packet Forwarding Engine and the Routing Engine, to ensure stability. The clean separation of these two functions permits superior forwarding performance and a highly reliable operating system. Therefore, it is not necessary to monitor CPU, memory, and buffer utilization, as is the case with traditional, monolithic code base routers. The Routing Engine has its own CPU, memory, and buffers—separate from those of the Packet Forwarding Engine. The ASIC-based Packet Forwarding Engine forwards packets on all interfaces at wire speed, eliminating the need to monitor packet buffers being exhausted. As a result, CPU utilization under 2 percent is normal.

Entries in the **jnxOperatingTable** are represented by the **jnxOperatingEntry** object, whose object identifier is **{jnxOperatingTable 1}**.

The **jnxOperatingTable** describes the status of specific objects, which are described as follows:

- **jnxOperatingContents**—The associated **jnxContentsIndex** in the **jnxContentsTable**, whose object identifier is **{jnxOperatingEntry 1}**.
- **jnxOperatingL1Index**—The level-one index of the container housing the entry, whose object identifier is **{jnxOperatingEntry 2}**.
- **jnxOperatingL2Index**—The level-two index of the container housing the entry, whose object identifier is **{jnxOperatingEntry 3}**.
- **jnxOperatingL3Index**—The level-three index of the container housing the entry, whose object identifier is **{jnxOperatingEntry 4}**.
- **jnxOperatingDescr**—The name or detailed description of the entry, whose object identifier is **{jnxOperatingEntry 5}**.
- **jnxOperatingState**—The operating state of the entry, whose object identifier is **{jnxOperatingEntry 6}**. The state can be any of the following:
  - **Unknown(1)**—State of the component is unknown or unavailable
  - **Running(2)**—Up and running as an active primary
  - **Ready(3)**—Ready to run; not running yet
  - **Reset(4)**—Held in reset; not ready yet
  - **RunningAtFullSpeed(5)**—Valid for fans only
  - **Down(6)**—Power supply is down or off
  - **Standby(7)**—Running as a standby backup
- **jnxOperatingTemp**—The entry's temperature, in degrees Celsius (ºC), whose object identifier is **{jnxOperatingEntry 7}**.
- **jnxOperatingCPU**—The CPU utilization percentage of the entry, whose object identifier is **{jnxOperatingEntry 8}**. It is valid for the Control Board, the FPC, and the Routing Engine. It is a 5-second rolling-weighted average calculated every second for each of the CPUs. The value is sent to the Routing Engine every 10 seconds. The value for the

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Routing Engine is calculated as the average CPU usage over a 5-second period. The chassis process (**chassisd**) computes this value for every 5-second period and returns the latest value from the cache when probed for the value of **jnxOperatingCPU.9.1.0.0**, which maps to the Routing Engine CPU. The Routing Engine is the only object of interest; the rest of the objects are most likely zero because CPUs on those cards are only used for management purposes.

The following three MIB objects provide for better CPU utilization reporting over SNMP:

- **jnxOperating1MinAvgCPU**—Indicates the average utilization of CPU during the last minute.
- **jnxOperating5MinAvgCPU**—Indicates the average utilization of CPU during the last 5-minute period.
- **jnxOperating15MinAvgCPU**—Indicates the average utilization of CPU during the last 15-minute period.

  All these objects return a zero value if the data is not available or is not applicable.

- **jnxOperatingISR**—The CPU utilization percentage of the entry in relation to the interrupt service routing (ISR), whose object identifier is **{jnxOperatingEntry 9}**.
- **jnxOperatingDRAMSize**—The DRAM size of the entry, in bytes, whose object identifier is **{jnxOperatingEntry 10}**. It is valid for the FPC, Routing Engine, and Control Board.
- **jnxOperatingBuffer**—The buffer pool utilization of the entry (a percentage), whose object identifier is **{jnxOperatingEntry 11}**. It is valid for the FPC and Control Board as a percentage of utilization. Buffers are normally fixed-length memory preallocated for read/write, input/output, or reception/transmission. A measurement against these buffers gives some indication of how busy the system is. The larger the percentage utilization, the busier the system. In terms of absolute numbers, the bigger the buffer size, the better the system can handle bursty traffic patterns.
- **jnxOperatingHeap**—The heap utilization of the entry, whose object identifier is **{jnxOperatingEntry 12}**.
- **jnxOperatingUpTime**—The time interval, in 10-millisecond periods, that the entry has been up and running, whose object identifier is **{jnxOperatingEntry 13}**.
- **jnxOperatingLastRestart**—The value of **sysUpTime** when the entry was last restarted, whose object identifier is **{jnxOperatingEntry 14}**.
- **jnxOperatingMemory**—The entry's installed memory size, in megabytes (MB), whose object identifier is **{jnxOperatingEntry 15}**.
- **jnxOperatingStateOrdered**—The operating state of the entry, whose object identifier is **{jnxOperatingEntry 16}**. The state is represented by one of the following values:
  - **Running(1)**—Up and running as an active primary
  - **Standby(2)**—Running as a standby backup
  - **Ready(3)**—Ready to run; not running yet
  - **RunningAtFullSpeed(4)**—Valid for fans only

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- **Reset(5)**—Held in reset; not ready yet
- **Down(6)**—Power supply is down or off
- **Unknown(7)**—State of the component is unknown or unavailable

Table 25 on page 109 through Table 27 on page 113 provide examples of **jnxOperatingEntry** objects. The following column headings for each table are abbreviated to correspond to the parts of the **jnxOperatingEntry** objects:

- Contents index—**jnxOperatingContents**
- L1—**jnxOperatingL1Index**
- L2—**jnxOperatingL2Index**
- L3—**jnxOperatingL3Index**
- Description—**jnxOperatingDescr**
- State—**jnxOperatingState**
- Temp—**jnxOperatingTemp**
- CPU—**jnxOperatingCPU**
- ISR—**jnxOperatingISR**
- DRAM—**jnxOperatingDRAMSize**
- Buffer—**jnxOperatingBuffer**
- Heap—**jnxOperatingHeap**
- UpTime—**jnxOperatingUpTime**
- Last Restart—**jnxOperatingLastRestart**
- Memory—**jnxOperatingMemory**

Table 25 on page 109 provides an example of **jnxOperatingEntry** objects in the **jnxOperatingTable** of an M20 router.

Table 25: jnxOperatingEntry Objects in the jnxOperatingTable of an M20 Router

Index	L1	L2	L3	Description	State	Temp	CPU	ISR	DRAM	Buffer	Heap	UpTime	LastRestart	Memory
1	1	0	0	Midplane	Running	26	0	0	0	0	0	0	0:0:00:00.0	0
2	1	0	0	Power supply A	Running	28	0	0	0	0	0	0	0:0:00:00.0	0
2	2	0	0	Power supply B	Running	29	0	0	0	0	0	0	0:0:00:00.0	0
4	1	0	0	Front top fan	Running	0	0	0	0	0	0	0	0:0:00:00.0	0

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Table 25: jnxOperatingEntry Objects in the jnxOperatingTable of an M20 Router *(continued)*

Index	L1	L2	L3	Description	State	Temp	CPU	ISR	DRAM	Buffer	Heap	UpTime	LastRestart	Memory
4	2	0	0	Front middle fan	Running	0	0	0	0	0	0	0	0:0:00:00.0	0
4	3	0	0	Front bottom fan	Running	0	0	0	0	0	0	0	0:0:00:00.0	0
4	4	0	0	Rear fan	Running	0	0	0	0	0	0	0	0:0:00:00.0	0
6	1	0	0	SSB 0	Running	30	0	0	671088	6	0	67038195	0:0:00:35.41	64
7	1	0	0	FPC @ 0/*/*	Running	31	0	0	83886	3	0	67035034	0:0:01:06.91	8
7	2	0	0	FPC @ 1/*/*	Running	33	0	0	83886	4	0	67034422	0:0:01:13.04	8
7	3	0	0	FPC @ 2/*/*	Running	31	0	0	83886	3	0	67033809	0:0:01:19.18	8
9	1	0	0	Routing Engine 0	Running	29	4	0	802738	0	0	67046146	0:0:00:00.00	765

To verify the size of the memory, use the **show chassis fpc**, **show chassis routing-engine**, and **show chassis ssb** commands. For more information about the output of these commands, see the *Junos OS System Basics and Services Command Reference*.

Table 26 on page 110 provides an example of **jnxOperatingEntry** objects in the **jnxOperatingTable** of a T640 router.

Table 26: jnxOperatingEntry Objects in the jnxOperatingTable of a T640 Router

Index	L1	L2	L3	Description	State	Temp	CPU	ISR	DRAM	Buffer	Heap	UpTime	LastRestart	Memory
1	1	0	0	Midplane	Running	0	–	–	–	–	–	–	–	–
2	2	0	0	PEM 1	Running	29	–	–	–	–	–	–	–	–
4	1	1	0	Top left front fan	Running	0	–	–	–	–	–	–	–	–
4	1	2	0	Top left middle fan	Running	0	–	–	–	–	–	–	–	–
4	1	3	0	Top left rear fan	Running	0	0	0	0	0	0	0	0:0:00:00.00	0

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Table 26: jnxOperatingEntry Objects in the jnxOperatingTable of a T640 Router *(continued)*

Index	L1	L2	B	Description	State	Temp	CPU	ISR	DRAM	Buffer	Heap	UpTime	LastRestart	Memory
4	1	4	0	Top right front fan	Running	0	0	0	0	0	0	0	0:0:00:00.00	0
4	1	5	0	Top right middle fan	Running	0	0	0	0	0	0	0	0:0:00:00.00	0
4	1	6	0	Top right rear fan	Running	0	0	0	0	0	0	0	0:0:00:00.00	0
4	2	1	0	Bottom left front fan	Running	0	0	0	0	0	0	0	0:0:00:00.00	0
4	2	2	0	Bottom left middle fan	Running	0	0	0	0	0	0	0	0:0:00:00.00	0
4	2	3	0	Bottom left rear fan	Running	0	0	0	0	0	0	0	0:0:00:00.00	0
4	2	4	0	Bottom right front fan	Running	0	0	0	0	0	0	0	0:0:00:00.00	0
4	2	5	0	Bottom right middle fan	Running	0	0	0	0	0	0	0	0:0:00:00.00	0
4	3	1	0	Bottom right rear fan	Running	0	0	0	0	0	0	0	0:0:00:00.00	0
4	3	1	0	Bottom blower	Running	0	0	0	0	0	0	0	0:0:00:00.00	0
4	3	2	0	Bottom blower	Running	0	0	0	0	0	0	0	0:0:00:00.00	0
4	3	3	0	Middle blower	Running	0	0	0	0	0	0	0	0:0:00:00.00	0
4	3	4	0	Top blower	Running	0	0	0	0	0	0	0	0:0:00:00.00	0
4	3	5	0	Second blower from top	Running	0	0	0	0	0	0	0	0:0:00:00.00	0
7	2	0	0	FPC @ 1/*/*	Running	0	1	0	512	41	3	138367	0:18:56:48.81	512

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Table 26: jnxOperatingEntry Objects in the jnxOperatingTable of a T640 Router *(continued)*

Index	L1	L2	B	Description	State	Temp	CPU	ISR	DRAM	Buffer	Heap	UpTime	LastRestart	Memory
7	2	1	0	FPC @ 1/0/* top temperature sensor	Running	35	0	0	0	0	0	0	0:18:56:48.81	0
7	2	2	0	FPC @ 1/1/* bottom temperature sensor	Running	32	0	0	0	0	0	0	0:18:56:48.81	0
7	6	0	0	FPC @ 5/*/*	Running	0	3	0	256	41	14	136976	0:18:57:02.71	256
7	6	1	0	FPC @ 5/0/* top temperature sensor	Running	44	0	0	0	0	0	0	0:18:57:02.71	0
7	6	2	0	FPC @ 5/1/* bottom temperature sensor	Running	33	0	0	0	0	0	0	0:18:57:02.71	0
7	8	0	0	FPC @ 7/*/*	Running	0	2	0	256	41	7	137963	0:18:56:52.85	256
7	8	1	0	FPC @ 7/0/* top temperature sensor	Running	38	0	0	0	0	0	0	0:18:56:52.85	0
7	8	2	0	FPC @ 7/1/* bottom temperature sensor	Running	33	0	0	0	0	0	0	0:18:56:52.85	0
9	1	0	0	Host 0	Running	35	0	0	2048	0	0	6963005	0:19:20:30.07	2048
9	2	0	0	Host 1	Standby	32	2	0	2048	0	0	24401100	2:19:46:51.00	2048
10	1	0	0	FPM	Running	30	0	0	0	0	0	0	0:0:00:00.00	0
11	1	0	0	SCG 0	Running	36	0	0	0	0	0	0	0:0:00:00.00	0
11	2	0	0	SCG 1	Standby	35	0	0	0	0	0	0	0:0:00:00.00	0
12	1	0	0	CB 0	Running	36	0	0	0	0	0	0	0:0:00:00.00	0

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Table 26: jnxOperatingEntry Objects in the jnxOperatingTable of a T640 Router *(continued)*

Index	L1	L2	L3	Description	State	Temp	CPU	ISR	DRAM	Buffer	Heap	UpTime	LastRestart	Memory
12	2	0	0	CB 1	Standby	39	0	0	0	0	0	0	0:0:00:00.00	0
14	1	0	0	SPMB 0	Running	36	1	0	128	40	0	142576	0:18:56:06.72	128
14	2	0	0	SPMB 1	Standby	39	0	0	128	40	0	142447	0:18:56:08.01	128
15	1	0	0	SIB 0	Unknown	40	0	0	0	0	0	0	0:0:00:00.00	0
15	2	0	0	SIB 1	Unknown	39	0	0	0	0	0	0	0:0:00:00.00	0
15	3	0	0	SIB 2	Unknown	39	0	0	0	0	0	0	0:0:00:00.00	0
15	4	0	0	SIB 3	Unknown	40	0	0	0	0	0	0	0:0:00:00.00	0
15	5	0	0	SIB 4	Unknown	40	0	0	0	0	0	0	0:0:00:00.00	0

Table 27 on page 113 provides an example of **jnxOperatingEntry** objects in the **jnxOperatingTable** of a T320 router.

Table 27: jnxOperatingEntry Objects in the jnxOperatingTable of a T320 Router

Index	L1	L2	L3	Description	State	Temp	CPU	ISR	DRAM	Buffer	Heap	UpTime	LastRestart	Memory
1	1	0	0	Midplane	Running	0	0	0	0	0	0	0	(0) 0:00:00.00	0
2	1	0	0	PEM 0	Running	0	0	0	0	0	0	0	(0) 0:00:00.00	0
4	1	1	0	Top left front fan	Running	0	0	0	0	0	0	0	(0) 0:00:00.00	0
4	1	2	0	Top left middle fan	Running	0	0	0	0	0	0	0	(0) 0:00:00.00	0
4	1	3	0	Top left rear fan	Running	0	0	0	0	0	0	0	(0) 0:00:00.00	0
4	1	4	0	Top right front fan	Running	0	0	0	0	0	0	0	(0) 0:00:00.00	0
4	1	5	0	Top right middle fan	Running	0	0	0	0	0	0	0	(0) 0:00:00.00	0
4	2	6	0	Top right rear fan	Running	0	0	0	0	0	0	0	(0) 0:00:00.00	0

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Table 27: jnxOperatingEntry Objects in the jnxOperatingTable of a T320 Router *(continued)*

Index	L1	L2	L3	Description	State	Temp	CPU	ISR	DRAM	Buffer	Heap	UpTime	LastRestart	Memory
4	2	1	0	Bottom left front fan	Running	0	0	0	0	0	0	0	(0) 0:00:00.00	0
4	2	2	0	Bottom left middle fan	Running	0	0	0	0	0	0	0	(0) 0:00:00.00	0
4	2	3	0	Bottom left rear fan	Running	0	0	0	0	0	0	0	(0) 0:00:00.00	0
4	2	4	0	Bottom right front fan	Running	0	0	0	0	0	0	0	(0) 0:00:00.00	0
4	2	5	0	Bottom right middle fan	Running	0	0	0	0	0	0	0	(0) 0:00:00.00	0
4	2	6	0	Bottom right rear fan	Running	0	0	0	0	0	0	0	(0) 0:00:00.00	0
4	3	1	0	Rear tray top fan	Running	0	0	0	0	0	0	0	(0) 0:00:00.00	0
4	3	2	0	Rear tray second fan	Running	0	0	0	0	0	0	0	(0) 0:00:00.00	0
4	3	3	0	Rear tray middle fan	Running	0	0	0	0	0	0	0	(0) 0:00:00.00	0
4	3	4	0	Rear tray fourth fan	Running	0	0	0	0	0	0	0	(0) 0:00:00.00	0
4	3	5	0	Rear tray bottom fan	Running	0	0	0	0	0	0	0	(0) 0:00:00.00	0
7	4	0	0	FPC @ 3/*/*	Running	0	1	0	256	41	7	6568428	(26190949) 3 days, 0:45:09.49	256
7	4	1	0	FPC @ 3/0/* top temperature sensor	Running	41	0	0	0	0	0	0	(26190949) 3 days, 0:45:09.49	0
7	4	2	0	FPC @ 3/1/* bottom temperature sensor	Running	37	0	0	0	0	0	0	(26190949) 3 days, 0:45:09.49	0

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Table 27: jnxOperatingEntry Objects in the jnxOperatingTable of a T320 Router *(continued)*

Index	L1	L2	L3	Description	State	Temp	CPU	ISR	DRAM	Buffer	Heap	UpTime	LastRestart	Memory
9	1	0	0	Host 0	Running	34	1	0	2048	0	0	32763001	(32763004) 3 days, 19:00:30.04	2048
9	2	0	0	Host 1	Standby	32	1	0	2048	0	0	110271900	(110271900) 12 days, 18:18:39.00	2048
10	1	0	0	FPM	Running	30	0	0	0	0	0	0	(0) 0:00:00.00	0
11	1	0	0	SCG 0	Running	33	0	0	0	0	0	0	(0) 0:00:00.00	0
11	2	0	0	SCG 1	Standby	31	0	0	0	0	0	0	(0) 0:00:00.00	0
12	1	0	0	CB 0	Running	37	0	0	0	0	0	0	(0) 0:00:00.00	0
12	2	0	0	CB 1	Standby	34	0	0	0	0	0	0	(0) 0:00:00.00	0
14	1	0	0	SPMB 0	Running	36	0	0	128	40	0	6572381	(26186997) 3 days, 0:44:29.97	128
14	2	0	0	SPMB 1	Standby	36	1	0	128	40	0	6572465	(26186913) 3 days, 0:44:29.13	128
15	1	0	0	SIB 0	Standby	36	0	0	0	0	0	0	(0) 0:00:00.00	0
15	2	0	0	SIB 1	Running	36	0	0	0	0	0	0	(0) 0:00:00.00	0
15	3	0	0	SIB 2	Running	38	0	0	0	0	0	0	(0) 0:00:00.00	0

jnxRedundancyTable

The object identifier for the **jnxRedundancyTable** is **{jnxBoxAnatomy 14}**. This object shows the internal configuration settings for the redundant subsystems or components in the chassis.

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Entries within the **jnxRedundancyTable** are represented by the **jnxRedundancyEntry** object, whose object identifier is **{jnxRedundancyEntry 1}**. This **jnxRedundancyEntry** contains the following objects, which describe the internal configuration settings for the redundant subsystems or components in the chassis:

- **jnxRedundancyContentsIndex**—The index value of an entry in **jnxRedundancyEntry**, whose object identifier is **{jnxContainersEntry 1}**.
- **jnxRedundancyL1Index**—The level-one index associated with the redundant component, whose object identifier is **{jnxContainersEntry 2}**.
- **jnxRedundancyL2Index**—The level-two index associated with the redundant component, whose object identifier is **{jnxContainersEntry 3}**.
- **jnxRedundancyL3Index**—The level-three index associated with the redundant component, whose object identifier is **{jnxContainersEntry 4}**.
- **jnxRedundancyDescr**—The description of the redundant component, whose object identifier is **{jnxContainersEntry 5}**.
- **jnxRedundancyConfig**—The election priority of redundancy configuration, whose object identifier is **{jnxContainersEntry 6}**.
- **jnxRedundancyState**—The current running state of the redundant component, whose object identifier is **{jnxContainersEntry 7}**.
- **jnxRedundancySwitchoverCount**—The total number of switchovers, defined as a change in the **jnxRedundancyState** from master to backup or vice versa, as perceived by the redundant component since the Routing Engine is up and running, whose object identifier is **{jnxContainersEntry 8}**.
- **jnxRedundancySwitchoverTime**—The value of **sysUpTime** when the **jnxRedundancyState** was last switched over from master to backup or vice versa, whose object identifier is **{jnxContainersEntry 9}**.
- **jnxRedundancySwitchoverReason**—The reason for the last switchover to the redundant component, whose object identifier is **{jnxContainersEntry 10}**.
- **jnxKeepaliveHeartbeat**—The period of sending keepalive messages between the master and the backup subsystem, which is a system-wide preset value in seconds used by internal mastership resolution, whose object identifier is **{jnxContainersEntry 11}**.
- **jnxRedundancyKeepaliveTimeout**—The timeout period in seconds used by the watchdog timer before it initiates a switchover to the backup subsystem, whose object identifier is **{jnxContainersEntry 12}**.
- **jnxRedundancyKeepaliveElapsed**—The elapsed time since the redundant component received the last keepalive message from the outer subsystems, whose object identifier is **{jnxContainersEntry 13}**.
- **jnxRedundancyKeepaliveLoss**—The total number of keepalive messages lost between the master and the backup subsystems as perceived by the redundant component since the Routing Engine is up and running, whose object identifier is **{jnxContainersEntry 14}**.

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Table 28 on page 117 through Table 30 on page 120 provide examples of **jnxRedundancyEntry** objects. The following column headings for each table are abbreviated to correspond to the parts of the **jnxOperatingTable** objects:

- Contents index—**jnxRedundancyContentsIndex**
- L1—**jnxRedundancyL1Index**
- L2—**jnxRedundancyL2Index**
- L3—**jnxRedundancyL3Index**
- Description—**jnxRedundancyDescr**
- Config—**jnxRedundancyConfig**
- State—**jnxRedundancyState**
- Count—**jnxRedundancySwitchoverCount**
- Time—**jnxRedundancySwitchoverTime**
- Reason—**jnxRedundancySwitchoverReason**
- Heartbeat—**jnxKeepaliveHeartbeat**
- Timeout—**jnxRedundancyKeepaliveTimeout**
- Elapsed—**jnxRedundancyKeepaliveElapsed**
- Loss—**jnxRedundancyKeepaliveLoss**

Table 28 on page 117 provides an example of **jnxRedundancyEntry** objects in the **jnxRedundancyTable** of an M20 router.

Table 28: jnxRedundancyEntry Objects in the jnxRedundancyTable of an M20 Router

Index	L1	L2	L3	Description	Config	State	Count	Time	Reason	Heart beat	Time out	Elapsed	Loss
6	1	0	0	SSB 0 Internet Processor II	Master	Master	0	3383	Never switched	0	0	0	0
6	2	0	0	SSB 1	Disabled	Disabled	0	0	Never switched	0	0	0	0
9	1	0	0	Routing Engine 0	Master	Master	1	421	User switched	3	300	1	0
9	2	0	0	Routing Engine 1	Backup	Backup	0	0	Other	0	0	0	0

To verify Routing Engine status, use the **show chassis routing-engine** command. Sample command output from an M20 router is as follows:

```
user@host> show chassis routing-engine
```

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```
Routing Engine status:
 Slot 0:
  Current state      Master
  Election priority      Master (default)
  Temperature      26 degrees C / 78 degrees F
  DRAM                 768 Mbytes
  CPU utilization:
   User                  2 percent
   Background      0 percent
   Kernel            0 percent
   Interrupt         0 percent
   Idle                98 percent
  Model              teknor
  Serial ID      32000004f8ff1201
  Start time      2002-01-29 12:30:42 PST
  Uptime             21 hours, 17 minutes, 14 seconds
  Load averages:      1 minute  5 minute 15 minute
    0.03    0.02    0.00
Routing Engine status:
 Slot 1:
  Current state      Backup
  Election priority  Backup (default)
  DRAM                805306368 Mbytes
  CPU utilization:
   User                 0 percent
   Background      0 percent
   Kernel            1 percent
   Interrupt         0 percent
   Idle                99 percent
  Model              teknor
  Serial ID      100000078c10df01
  Start time      2002-01-24 16:47:39 PST
  Uptime             5 days, 17 hours, 14 seconds
```

To verify SSB status, use the **show chassis ssb** command. Sample command output from an M20 router is as follows:

```
user@host> show chassis ssb
SSB status:
Slot 0 information:
 State                    Master
 Temperature                24 degrees C / 75 degrees F
 CPU utilization              2 percent
 Interrupt utilization           0 percent
 Heap utilization             16 percent
 Buffer utilization           43 percent
 Total CPU DRAM              64 Mbytes
 Internet Processor II            Version 1, Foundry IBM, Part number 9
 Start time:                 2002-01-29 12:32:24 PST
 Uptime:                    21 hours, 30 minutes, 53 seconds
Slot 1 information:
 State                    Backup
```

Table 29 on page 119 provides an example of **jnxRedundancyEntry** objects in the **jnxRedundancyTable** of a T640 router.

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Table 29: jnxRedundancyEntry Objects in the jnxRedundancyTable of a T640 Router

Index	L1	L2	L3	Description	Config	State	Count	Time	Reason	Heart beat	Time out	Elapsed	Loss
9	1	0	0	Host 0	Master	Master	3	0:18:55:49.42	User switched	20	300	1	0
9	2	0	0	Host 1	Backup	Backup	0	0:0:00:00.00	Other	0	0	0	0
15	1	0	0	SIB 0	Unknown	Backup	1	0:0:00:00.00	0	0	0	0	0
15	2	0	0	SIB 1	Unknown	Master	1	0:0:00:00.00	0	0	0	0	0
15	3	0	0	SIB 2	Unknown	Master	1	0:0:00:00.00	0	0	0	0	0
15	4	0	0	SIB 3	Unknown	Master	1	0:0:00:00.00	0	0	0	0	0
15	5	0	0	SIB 4	Unknown	Master	1	0:0:00:00.00	0	0	0	0	0

To verify Routing Engine status, use the **show chassis routing-engine** command. Sample command output from a T640 router is as follows:

```
user@host> show chassis routing-engine
Routing Engine status:
  Slot 0:
    Current state        Master
    Election priority       Master (default)
    Temperature       35 degrees C / 95 degrees F
    DRAM            2048 MB
    CPU utilization:
      User             1 percent
      Background       0 percent
      Kernel           5 percent
      Interrupt        0 percent
      Idle            94 percent
    Model               unknown
    Start time     2002-03-31 14:26:49 PST
    Uptime         19 hours, 22 minutes, 13 seconds
    Load averages: 1 minute   5 minute  15 minute
       0.00       0.00       0.00
Routing Engine status:
  Slot 1:
    Current state  Backup
    Election priority              Backup (default)
    Temperature 32 degrees C / 89 degrees F
    DRAM      2048 MB
    CPU utilization:
      User       0 percent
      Background 0 percent
      Kernel     0 percent
      Interrupt  0 percent
      Idle     100 percent
    Model          RE-3.0
```

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```
    Start time     2002-03-29 14:00:18 PST
    Uptime         2 days, 19 hours, 48 minutes, 32 seconds
```

Table 30 on page 120 provides an example of **jnxRedundancyEntry** objects in the **jnxRedundancyTable** of a T320 router.

Table 30: jnxRedundancyEntry Objects in the jnxRedundancyTable of a T320 Router

Index	L1	L2	L3	Description	Config	State	Count	Time	Reason	Heart beat	Timeout	Elapsed	Loss
9	1	0	0	Host 0	Master	Master	6	(26185188)3 days, 0:44:11.88	User switched	20	300	1	0
9	2	0	0	Host 1	Backup	Backup	0	(0) 0:00:00.00	Other	0	0	0	0
15	1	0	0	SIB 0	Backup	Backup	1	(0) 0:00:00.00	0	0	0	0	0
15	2	0	0	SIB 1	Master	Master	1	(0) 0:00:00.00	0	0	0	0	0
15	3	0	0	SIB 2	Master	Master	1	(0) 0:00:00.00	0	0	0	0	0

To verify Routing Engine status, use the **show chassis routing-engine** command. Sample command output from a T320 router is as follows:

```
user@host> show chassis routing-engine
Routing Engine status:
  Slot 0:
    Current state  Master
    Election priority             Master (default)
    Temperature 34 degrees C / 93 degrees F
    DRAM      2048 MB
    CPU utilization:
      User       0 percent
      Background 0 percent
      Kernel     1 percent
      Interrupt  0 percent
      Idle      98 percent
    Model          RE-3.0
    Start time     2002-04-05 14:43:16 PST
    Uptime         17 days, 23 hours, 3 minutes, 47
seconds
    Load averages: 1 minute   5 minute  15 minute
       0.00       0.00       0.00
Routing Engine status:
  Slot 1:
    Current state  Backup
    Election priority             Backup (default)
    Temperature 32 degrees C / 89 degrees F
    DRAM      2048 MB
    CPU utilization:
      User       0 percent
      Background 0 percent
```


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```
  Kernel     0 percent
  Interrupt  0 percent
  Idle     100 percent
Model          RE-3.0
Start time     2002-03-27 15:25:07 PST
Uptime         26 days, 22 hours, 21 minutes, 44 seconds
```

jnxFruTable

The object identifier for the **jnxFruTable** is **{jnxBoxAnatomy 15}**. This object shows the status of field-replaceable units (FRUs) in the chassis.

Entries within the **jnxFruTable** are represented by the **jnxFruEntry** object, whose object identifier is **{jnxFruEntry 1}**. This **jnxFruEntry** object contains the following objects, which describe the FRUs in the chassis:

- **jnxFruContentsIndex**—The index value of an entry in **jnxFruEntry**, whose object identifier is **{jnxFruEntry 1}**.
- **jnxFruL1Index**—The level-one index associated with the FRU, whose object identifier is **{jnxFruEntry 2}**.
- **jnxFruL2Index**—The level-two index associated with the FRU, whose object identifier is **{jnxFruEntry 3}**.
- **jnxFruL3Index**—The level-three index associated with the FRU, whose object identifier is **{jnxFruEntry 4}**.
- **jnxFruName**—The name or detailed description of the FRU, whose object identifier is **{jnxFruEntry 5}**.
- **jnxFruType**—The FRU type, whose object identifier is **{jnxFruEntry 6}**. The FRU type is represented by one of the following values:
  - **other(1)**
  - **clockGenerator(2)**
  - **flexiblePicConcentrator(3)**
  - **switchingAndForwardingModule(4)**
  - **controlBoard(5)**
  - **routingEngine(6)**
  - **powerEntryModule(7)**
  - **frontPanelModule(8)**
  - **switchInterfaceBoard(9)**
  - **processorMezzanineBoardForSIB(10)**
  - **portInterfaceCard(11)**
  - **craftInterfacePanel(12)**

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- **fan(13)**
- **lineCardChassis(14)**
- **forwardingEngineBoard(13)**
- **protectedSystemDomain(13)**

- **jnxFruSlot**—The slot number of the FRU, whose object identifier is **{jnxFruEntry 7}**. This is equivalent to **jnxFruL1Index**. The slot number is zero if unavailable or inapplicable.
- **jnxFruState**—The current state of the FRU, whose object identifier is **{jnxFruEntry 8}**. The FRU state can be any of the following:
  - **unknown(1)**
  - **empty(2)**
  - **present(3)**
  - **ready(4)**
  - **announceOnline(5)**
  - **online(6)**
  - **announceOffline(7)**
  - **offline(8)**
  - **diagnostic(9)**
  - **standby(10)**
- **jnxFruTemp**—The temperature of the FRU, in degrees Celsius, whose object identifier is **{jnxFruEntry 9}**. The value is zero if unavailable or inapplicable.
- **jnxFruOfflineReason**—The reason the FRU is offline, whose object identifier is **{jnxFruEntry 10}**. The reason is represented by the following values:
  - **unknown(1)**—Unknown or other
  - **none(2)**—None
  - **error(3)**—Error
  - **noPower(4)**—No power
  - **configPowerOff(5)**—Configured to power off
  - **configHoldInReset(6)**—Configured to hold in reset
  - **cliCommand(7)**—Brought offline by a command-line interface (CLI) command
  - **buttonPress(8)**—Brought offline by button press
  - **cliRestart(9)**—Restarted by CLI command

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  - **overtempShutdown(10)**—Overtemperature shutdown
  - **masterClockDown(11)**—Master clock down
  - **singleSfmModeChange(12)**—Single SFM mode change
  - **packetSchedulingModeChange(13)**—Packet scheduling mode change
  - **physicalRemoval(14)**—Physical removal
  - **unresponsiveRestart(15)**—Restarting unresponsive board
  - **sonetClockAbsent(16)**—SONET out clock absent

- **jnxFruLastPowerOff**—The value of **sysUpTime** when this subject was last powered off, whose object identifier is **{jnxFruEntry 11}**. The value is zero if unavailable or inapplicable.
- **jnxFruLastPowerOn**—The value of **sysUpTime** when this subject was last powered on, whose object identifier is **{jnxFruEntry 12}**. The value is zero if unavailable or inapplicable.
- **jnxFruPowerUpTime**—The time interval in 10-millisecond periods that this subject has been up and running since the last power-on time, whose object identifier is **{jnxFruEntry 13}**. The value is zero if unavailable or inapplicable.
- **jnxFruChassisId**—The chassis type of this subject. The object identifier for this object is **{jnxFruEntry 14}**.
- **jnxFruChassisDescr**—The textual description for the chassis type of this subject. The object identifier is **{jnxFruEntry 15}**.
- **jnxFruPsdAssignment**—The protected system domain (PSD) assignment for this subject. The object identifier is **{jnxFruEntry 16}**.

Table 31 on page 124 through Table 36 on page 148 provide examples of **jnxFruEntry** objects. The following column headings for each table are abbreviated to correspond to the parts of the **jnxFruEntry** objects:

- Contents index—**jnxFruContentsIndex**
- L1—**jnxFruL1Index**
- L2—**jnxFruL2Index**
- L3—**jnxFruL3Index**
- Name—**jnxFruName**
- Type—**jnxFruType**
- Slot—**jnxFruSlot**
- State—**jnxFruState**
- Temp—**jnxFruTemp**
- Offline—**jnxFruOffline**
- PowerOff—**jnxFruPowerOff**

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- PowerOn—**jnxFruPowerOn**
- Uptime—**jnxFruPowerUpTime**

Table 31 on page 124 provides an example of **jnxFruEntry** objects in the **jnxFruTable** of an M10 router.

Table 31: jnxFruContents Objects in the jnxFruTable of an M10 Router

Index	L1	L2	B	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
2	1	0	0	Power supply A	powerEntryModule	1	Online	0	None	0:0:00:00.00	0:0:11:08.73	264319
2	2	0	0	Power supply B	powerEntryModule	2	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
4	1	1	0	Left fan 1	fan	1	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	1	2	0	Left fan 2	fan	1	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	1	3	0	Left fan 3	fan	1	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	1	4	0	Left fan 4	fan	1	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
6	1	0	0	FEB Internet Processor II	controlBoard	1	Online	24	None	0:0:00:00.00	0:0:00:00.00	0
7	1	0	0	FPC @ 0/*/*	flexiblePicConcentrator	1	Online	24	None	0:0:00:00.00	0:0:00:00.00	0
7	2	0	0	FPC @ 1/*/*	flexiblePicConcentrator	2	Online	24	None	0:0:00:00.00	0:0:00:00.00	0
8	1	1	0	PIC: @ 0/0/*	portInterfaceCard	1	Ready	24	None	0:0:00:00.00	0:0:00:00.00	0
8	1	2	0	PIC: 1x Monitor @ 0/1/*	portInterfaceCard	1	Ready	24	None	0:0:00:00.00	0:0:00:00.00	0
8	1	3	0	PIC: 1x OC-12 ATM, MM @ 0/2/*	portInterfaceCard	1	Ready	24	None	0:0:00:00.00	0:0:00:00.00	0
8	1	4	0	PIC: 4x T3 @ 0/3/*	portInterfaceCard	1	Ready	24	None	0:0:00:00.00	0:0:00:00.00	0

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Table 31: jnxFruContents Objects in the jnxFruTable of an M10 Router *(continued)*

Index	L1	L2	B	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
8	2	1	0	PIC: 4x OC-3 SONET, SMIR @ 1/0/*	portInterfaceCard	2	Ready	24	None	0:0:00:00.00	0:0:00:00.00	0
8	2	2	0	PIC: 4x OC-3 SONET, MM @ 1/1/*	portInterfaceCard	2	Ready	24	None	0:0:00:00.00	0:0:00:00.00	0
8	2	3	0	PIC: 2x OC-3 ATM, MM @ 1/2/*	portInterfaceCard	2	Ready	24	None	0:0:00:00.00	0:0:00:00.00	0
8	2	4	0	PIC: 2x OC-3 ATM, MM @ 1/3/*	portInterfaceCard	2	Ready	24	None	0:0:00:00.00	0:0:00:00.00	0
9	1	0	0	Routing Engine	routingEngine	1	Online	27	None	0:0:00:00.00	0:0:00:00.00	0

To verify the L1, L2, and L3 indexes, use the **show chassis hardware** command. Sample command output from an M10 router is as follows:.

```
user@host> show chassis hardware
Hardware inventory:
Item             Version  Part number  Serial number     Description
Chassis                                58974             M10
Midplane         REV 03   710-001950   HB1590
Power Supply A   Rev 03   740-002498   LK33505           DC
Display          REV 04   710-001995   HE8442
Routing Engine   REV 01   740-003239   9001025728        RE-2.0
FEB              REV 12   710-001948   HA4221            Internet Processor II
FPC 0
  PIC 1          REV 01   750-004188   AR2912            1x Monitor
  PIC 2          REV 04   750-001551   AN7869            1x OC-12 ATM, MM
  PIC 3          REV 02   750-002485   AN2803            4x T3
FPC 1
  PIC 0          REV 03   750-002970   HF2293            4x OC-3 SONET, SMIR
  PIC 1          REV 03   750-002971   HA8094            4x OC-3 SONET, MM
  PIC 2          REV 03   750-002977   HD9352            2x OC-3 ATM, MM
  PIC 3          REV 03   750-002977   HD9393            2x OC-3 ATM, MM
```

To verify FPC status, use the **show chassis fpc** command. Sample command output from an M10 router is as follows:

```
user@host> show chassis fpc
Temp  CPU Utilization (%)   Memory    Utilization (%)
Slot State            (C)  Total  Interrupt      DRAM (MB) Heap     Buffer
```

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```
  0  Online            24      3         1        64        44        17
  1  Online            24      3         1        64        44        17
```

To verify Routing Engine status, use the **show chassis routing-engine** command. Sample command output from an M10 router is as follows:

```
user@host> show chassis routing-engine
Routing Engine status:
    Temperature                 26 degrees C / 78 degrees F
    DRAM                       768 MB
    Memory utilization           9 percent
    CPU utilization:
      User                       0 percent
      Background                 0 percent
      Kernel                     0 percent
      Interrupt                  0 percent
      Idle                     100 percent
    Model                        RE-2.0
    Serial ID                    b7000007c81ce801
    Start time                   2002-06-21 09:33:45 PDT
    Uptime                       3 days, 1 hour, 23 minutes, 27 seconds
    Load averages:               1 minute   5 minute  15 minute
                                     0.07       0.03       0.01
```

To verify FEB status, use the **show chassis feb** command. Sample command output from an M10 router is as follows:

```
user@host> show chassis feb
FEB status:
  Temperature                24 degrees C / 75 degrees F
  CPU utilization             3 percent
  Interrupt utilization       1 percent
  Heap utilization           17 percent
  Buffer utilization         44 percent
  Total CPU DRAM             64 MB
  Internet Processor II       Version 1, Foundry IBM, Part number 9
  Start time:       2002-06-21 09:45:46 PDT
  Uptime:       3 days, 1 hour, 11 minutes, 33 seconds
```

Table 32 on page 126 provides an example of **jnxFruEntry** objects in the **jnxFruTable** of an M20 router.

Table 32: JnxFruContents Objects in the jnxFruTable of an M20 Router

Index	l1	l2	l3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
2	1	0	0	Power supply A	**powerEntryModule**	1	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
2	2	0	0	Power supply B	**powerEntryModule**	2	Online	25	None	0:0:00:00.00	0:0:00:43.45	24993357
4	1	0	0	Rear fan	**fan**	1	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	2	0	0	Front upper fan	**fan**	2	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	3	0	0	Front middle fan	**fan**	3	Present	0	None	0:0:00:00.00	0:0:00:00.00	0

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Table 32: JnxFruContents Objects in the jnxFruTable of an M20 Router *(continued)*

Index	I1	I2	I3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
4	4	0	0	Front bottom fan	**fan**	4	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
6	1	0	0	SSB 0	**controlBoard**	1	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
6	2	0	0	SSB 1 Internet Processor I	**controlBoard**	2	Online	29	None	0:0:00:00.00	0:0:00:00.00	0
7	1	0	0	FPC @ 0/*/*	**flexible PicConcentrator**	1	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
7	2	0	0	FPC @ 1/*/*	**flexible PicConcentrator-**	2	Online	27	None	0:0:00:00.00	0:0:00:00.00	0
7	3	0	0	FPC @ 2/*/*	**flexible PicConcentrator**	3	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
7	4	0	0	FPC @ 3/*/*	**flexible PicConcentrator**	4	Online	27	None	0:0:00:00.00	0:0:00:00.00	0
8	1	1	0	PIC: @ 0/0/*	**portInterfaceCard**	1	Offline	0	None	0:0:00:00.00	0:0:00:00.00	0
8	1	2	0	PIC: @ 0/1/*	**portInterfaceCard**	1	Offline	28	None	0:0:00:00.00	0:0:00:00.00	0
8	1	3	0	PIC: @ 0/2/*	**portInterfaceCard**	1	Offline	0	None	0:0:00:00.00	0:0:00:00.00	0
8	1	4	0	PIC: @ 0/3/*	**portInterfaceCard**	1	Offline	0	None	0:0:00:00.00	0:0:00:00.00	0
8	2	1	0	PIC: 1x Tunnel @ 1/0/*	**portInterfaceCard**	2	Ready	0	None	0:0:00:00.00	0:0:00:00.00	0
8	2	2	0	PIC: 4x T3 @ 1/1/*	**portInterfaceCard**	2	Ready	0	None	0:0:00:00.00	0:0:00:00.00	0
8	2	3	0	PIC: 2x OC-3 ATM, MM @ 1/2/*	**portInterfaceCard**	2	Ready	27	None	0:0:00:00.00	0:0:00:00.00	0
8	2	4	0	PIC: 1x G/E, 1000Base-SX @ 1/3/*	**portInterfaceCard**	2	Ready	27	None	0:0:00:00.00	0:0:00:00.00	0

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Table 32: JnxFruContents Objects in the jnxFruTable of an M20 Router *(continued)*

Index	L1	L2	L3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
8	3	1	0	PIC: @ 2/0/*	**portInterfaceCard**	3	Offline	27	None	0:0:00:00.00	0:0:00:00.00	0
8	3	2	0	PIC: @ 2/1/*	**portInterfaceCard**	3	Offline	0	None	0:0:00:00.00	0:0:00:00.00	0
8	3	3	0	PIC: @ 2/2/*	**portInterfaceCard**	3	Offline	0	None	0:0:00:00.00	0:0:00:00.00	0
8	3	4	0	PIC: @ 2/3/*	**portInterfaceCard**	3	Offline	0	None	0:0:00:00.00	0:0:00:00.00	0
8	4	1	0	PIC: @ 3/0/*	**portInterfaceCard**	4	Ready	0	None	0:0:00:00.00	0:0:00:00.00	0
8	4	2	0	PIC: @ 3/1/*	**portInterfaceCard**	4	Ready	28	None	0:0:00:00.00	0:0:00:00.00	0
8	4	3	0	PIC: 2x OC-3 SONET, SMIR @ 3/2/*	**portInterfaceCard**	4	Ready	28	None	0:0:00:00.00	0:0:00:00.00	0
8	4	4	0	PIC: @ 3/3/*	**portInterfaceCard**	4	Ready	28	None	0:0:00:00.00	0:0:00:00.00	0
9	1	0	0	Routing Engine 0	**routingEngine**	1	Online	25	None	0:0:00:00.00	0:0:00:00.00	0
9	2	0	0	Routing Engine 1	**routingEngine**	2	Online	24	None	0:0:00:00.00	0:0:00:00.00	0
10	1	0	0	Front panel display	**frontPanelModule**	1	Online	0	None	0:0:00:00.00	0:0:00:00.00	0

To verify the L1, L2, and L3 indexes, use the **show chassis hardware** command. Sample command output from an M20 router is as follows:

```
user@host> show chassis hardware
Hardware inventory:
Item             Version  Part number  Serial number     Description
Chassis                                20200             M20
Backplane        REV 07   710-001517   AB5911
Power Supply B   Rev 02   7            000240            AC
Display          REV 04   710-001519   AD1903
Routing Engine 0 REV01    740          umeshk            RE-2.0
Routing Engine 1                       270000078ba48501  RE-2.0
SSB slot 0       N/A      N/A          N/A               backup
SSB slot 1       REV 04   710-001411   AD0281            Internet Processor I
FPC 1            REV 01   710-001292   AC9230
```


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```
  PIC 0          REV 01   750-001323   AA2812           1x Tunnel
  PIC 1          REV 01   750-002963   AK8586           4x T3
  PIC 2          REV 03   750-000612   AM8116           2x OC-3 ATM, MM
  PIC 3          REV 08   750-001072   AB9884           1x G/E, 1000 BASE-SX
FPC 3            REV 01   710-001197   AA8661
  PIC 2          REV 01   750-003748   HE9734           2x OC-3 SONET, SMIR
```

```
user@host> show chassis environment
Class Item                 Status     Measurement
Power Power Supply A       Absent
      Power Supply B       OK         25 degrees C / 77 degrees F
Temp  FPC 1                OK         27 degrees C / 80 degrees F
      FPC 3                OK         28 degrees C / 82 degrees F
      SSB 1                OK         29 degrees C / 84 degrees F
      Backplane            OK         23 degrees C / 73 degrees F
      Routing Engine 0     OK         25 degrees C / 77 degrees F
      Routing Engine 1     OK         24 degrees C / 75 degrees F
Fans  Rear Fan             OK         Spinning at normal speed
      Front Upper Fan      OK         Spinning at normal speed
      Front Middle Fan     OK         Spinning at normal speed
      Front Bottom Fan     OK         Spinning at normal speed
Misc  Craft Interface      OK
```

```
user@host> show chassis fpc
                    Temp  CPU Utilization (%)   Memory    Utilization (%)
Slot State           (C)  Total  Interrupt      DRAM (MB) Heap     Buffer
  0  Empty             0      0          0         0         0          0
  1  Online           27      8          7         8         9         14
  2  Empty             0      0          0         0         0          0
  3  Online           28      0          0         8         8         14
```

To verify Routing Engine status, use the **show chassis routing-engine** command. Sample command output from an M10 router is as follows:

```
user@host> show chassis routing-engine
Routing Engine status:
  Slot 0:
    Current state                Master
    Election priority            Master (default)
    Temperature                 25 degrees C / 77 degrees F
    DRAM                       768 MB
    Memory utilization           8 percent
    CPU utilization:
      User                       0 percent
      Background                 0 percent
      Kernel                     1 percent
      Interrupt                  0 percent
      Idle                      99 percent
    Model                        RE-2.0
    Serial ID                    ba0000061779d601
    Start time                   2002-06-21 15:37:36 PDT
    Uptime                       2 days, 21 hours, 27 minutes, 25 seconds
    Load averages:               1 minute   5 minute  15 minute
                                     0.00       0.00       0.00
Routing Engine status:
  Slot 1:
    Current state                Backup
    Election priority            Backup (default)
    Temperature                 24 degrees C / 75 degrees F
    DRAM                       768 MB
    Memory utilization           9 percent
    CPU utilization:
```

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```
    User                     0 percent
    Background               0 percent
    Kernel                   0 percent
    Interrupt                0 percent
    Idle                    99 percent
  Model                        RE-2.0
  Serial ID                    270000078ba48501
  Start time                   2002-06-17 14:30:21 PDT
  Uptime                       6 days, 22 hours, 34 minutes, 28 seconds
```

To verify SSB status, use the **show chassis ssb** command. Sample command output from an M10 router is as follows:

```
user@host> show chassis ssb
SSB status:
Slot 0 information:
  State                        Backup
Slot 1 information:
  State                        Master
  Temperature                29 degrees C / 84 degrees F
  CPU utilization             1 percent
  Interrupt utilization       0 percent
  Heap utilization            8 percent
  Buffer utilization         43 percent
  Total CPU DRAM             64 MB
  Internet Processor I         Version 1, Foundry IBM, Part number 3
  Start time:                  2002-06-21 15:38:53 PDT
  Uptime:                      2 days, 21 hours, 26 minutes, 26 seconds
```

Table 33 on page 130 provides an example of **jnxFruEntry** objects in the **jnxFruTable** of an M160 router.

Table 33: jnxFruContents Objects in the jnxFruTable of an M160 Router

Index	L1	L2	B	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
2	1	0	0	PEM 0	**powerEntryModule**	1	Online	0	None	0:00:00.00	0:00:12.83	6906955
2	2	0	0	PEM 1	**powerEntryModule**	2	Online	0	None	0:00:00.00	0:00:12.83	6906955
4	1	0	0	Front top blower	**fan**	1	Present	0	None	0:00:00.00	0:00:00.00	0
4	2	1	0	Fan tray front left	**fan**	2	Present	0	None	0:00:00.00	0:00:00.00	0
4	2	2	0	Fan tray front right	**fan**	2	Present	0	None	0:00:00.00	0:00:00.00	0
4	2	3	0	Fan tray rear left	**fan**	2	Present	0	None	0:00:00.00	0:00:00.00	0
4	2	4	0	Fan tray rear right	**fan**	2	Present	0	None	0:00:00.00	0:00:00.00	0
4	3	0	0	Rear top blower	**fan**	3	Present	0	None	0:00:00.00	0:00:00.00	0

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Table 33: jnxFruContents Objects in the jnxFruTable of an M160 Router *(continued)*

Index	L1	L2	L3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
4	4	0	0	Rear bottom blower	**fan**	4	Present	0	None	0:00:00.00	0:00:00.00	0
6	1	1	0	SFM 0 SPP	**switchingAndForwardingMode**	1	Online	35	None	0:00:03.13	0:00:00.00	0
6	1	2	0	SFM 0 SPR Internet Processor II	**switchingAndForwardingMode**	1	Online	35	None	0:00:03.13	0:00:00.00	0
6	2	1	0	SFM 1 SPP	**switchingAndForwardingMode**	2	Empty	0	None	0:00:00.00	0:00:00.00	0
6	2	2	0	SFM 1 SPR	**switchingAndForwardingMode**	2	Empty	0	None	0:00:00.00	0:00:00.00	0
6	3	1	0	SFM 2 SPP	**switchingAndForwardingMode**	3	Online	44	None	0:00:03.20	0:00:00.00	0
6	3	2	0	SFM 2 SPR Internet Processor II	**switchingAndForwardingMode**	3	Online	44	None	0:00:03.20	0:00:00.00	0
6	4	1	0	SFM 3 SPP	**switchingAndForwardingMode**	4	Offline	0	Config-ured to power off	0:00:03.22	0:00:00.00	0
6	4	2	0	SFM 3 SPR	**switchingAndForwardingMode**	4	Offline	0	Config-ured to power off	0:00:03.22	0:00:00.00	0
7	1	0	0	FPC @ 0/*/*	**flexiblePic Concentrator**	1	Offline	0	Config-ured to power off	0:00:02.28	0:00:00.00	0
7	2	0	0	FPC @ 1/*/*	**flexiblePic Concentrator**	2	Offline	0	Error	0:13:08.12	0:00:00.00	0

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Table 33: jnxFruContents Objects in the jnxFruTable of an M160 Router *(continued)*

Index	L1	L2	L3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
7	3	0	0	FPC @ 2/*/*	**flexiblePic Concentrator**	3	Online	30	None	0:00:02.32	0:00:00.00	0
7	4	0	0	FPC: 1x OC-192 SM LR @ 3/*/*	**flexiblePic Concentrator**	4	Online	41	None	0:00:02.34	0:00:00.00	0
7	5	0	0	FPC @ 4/*/*	**flexiblePic Concentrator**	5	Empty	0	None	0:00:00.00	0:00:00.00	0
7	6	0	0	FPC @ 5/*/*	**flexiblePic Concentrator**	6	Offline	0	Configured to power off	0:00:02.37	0:00:00.00	0
7	7	0	0	FPC @ 6/*/*	**flexiblePic Concentrator**	7	Empty	0	None	0:00:00.00	0:00:00.00	0
7	8	0	0	FPC @ 7/*/*	**flexiblePic Concentrator**	8	Online	41	None	0:00:03.11	0:00:00.00	0
8	1	1	0	PIC: @ 0/0/*	**portInterfaceCard**	1	Online	40	None	0:00:00.00	0:00:00.00	0
8	1	2	0	PIC: @ 0/1/*	**portInterface Card**	1	Online	40	None	0:00:00.00	0:00:00.00	0
8	1	3	0	PIC: @ 0/2/*	**portInterfaceCard**	1	Online	40	None	0:00:00.00	0:00:00.00	0
8	1	4	0	PIC: @ 0/3/*	**portInterfaceCard**	1	Online	40	None	0:00:00.00	0:00:00.00	0
8	2	1	0	PIC: @ 1/0/*	**portInterfaceCard**	2	Online	46	None	0:00:00.00	0:00:00.00	0
8	2	2	0	PIC: @ 1/1/*	**portInterfaceCard**	2	Online	46	None	0:00:00.00	0:00:00.00	0
8	2	3	0	PIC: @ 1/2/*	**portInterfaceCard**	2	Online	46	None	0:00:00.00	0:00:00.00	0
8	2	4	0	PIC: @ 1/3/*	**portInterfaceCard**	2	Online	46	None	0:00:00.00	0:00:00.00	0

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Table 33: jnxFruContents Objects in the jnxFruTable of an M160 Router *(continued)*

Index	L1	L2	L3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
8	3	1	0	PIC: @ 2/0/*	**portInterfaceCard**	3	Offline	0	Config-ured to power off	0:00:02.28	0:00:00.00	0
8	3	2	0	PIC: @ 2/1/*	**portInterfaceCard**	3	Offline	0	Config-ured to power off	0:00:02.28	0:00:00.00	0
8	3	3	0	PIC: @ 2/2/*	**portInterfaceCard**	3	Offline	0	Config ured to power off	0:00:02.28	0:00:00.00	0
8	3	4	0	PIC: @ 2/3/*	**portInterfaceCard**	3	Offline	0	Config-ured to power off	0:00:02.28	0:00:00.00	0
8	4	1	0	PIC: 1x OC-192 SM LR @ 3/0/*	**portInterfaceCard**	4	Offline	0	Error	0:13:08.12	0:00:00.00	0
8	4	2	0	PIC continued	**portInterfaceCard**	4	Offline	0	Error	0:13:08.12	0:00:00.00	0
8	4	3	0	PIC continued	**portInterfaceCard**	4	Offline	0	Error	0:13:08.12	0:00:00.00	0
8	4	4	0	PIC continued	**portInterfaceCard**	4	Offline	0	Error	0:13:08.12	0:00:00.00	0
8	5	1	0	PIC: @ 4/0/*	**portInterfaceCard**	5	Online	30	None	0:00:02.32	0:00:00.00	0
8	5	2	0	PIC: @ 4/1/*	**portInterfaceCard**	5	Online	30	None	0:00:02.32	0:00:00.00	0
8	5	3	0	PIC: @ 4/2/*	**portInterfaceCard**	5	Online	30	None	0:00:02.32	0:00:00.00	0
8	5	4	0	PIC: @ 4/3/*	**portInterfaceCard**	5	Online	30	None	0:00:02.32	0:00:00.00	0

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Table 33: jnxFruContents Objects in the jnxFruTable of an M160 Router *(continued)*

Index	L1	L2	L3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
8	6	1	0	PIC: @ 5/0/*	**portInterfaceCard**	6	Online	41	None	0:00:02.34	0:00:00.00	0
8	6	2	0	PIC: @ 5/1/*	**portInterfaceCard**	6	Online	41	None	0:00:02.34	0:00:00.00	0
8	6	3	0	PIC: @ 5/2/*	**portInterfaceCard**	6	Online	41	None	0:00:02.34	0:00:00.00	0
8	6	4	0	PIC: @ 5/3/*	**portInterfaceCard**	6	Online	41	None	0:00:02.34	0:00:00.00	0
8	7	1	0	PIC: @ 6/0/*	**portInterfaceCard**	7	Empty	0	None	0:00:00.00	0:00:00.00	0
8	7	2	0	PIC: @ 6/1/*	**portInterfaceCard**	7	Empty	0	None	0:00:00.00	0:00:00.00	0
8	7	3	0	PIC: @ 6/2/*	**portInterfaceCard (11)**	7	Empty	0	None	0:00:00.00	0:00:00.00	0
8	7	4	0	PIC: @ 6/3/*	**portInterfaceCard (11)**	7	Empty	0	None	0:00:00.00	0:00:00.00	0
8	8	1	0	PIC: 1x OC-12 SONET, SMIR @ 7/0/*	**portInterfaceCard**	8	Offline	0	Config-ured to power off	0:00:02.37	0:00:00.00	0
8	8	2	0	PIC: 4x E3 @ 7/1/*	**portInterfaceCard**	8	Offline	0	Config-ured to power off	0:00:02.37	0:00:00.00	0
8	8	3	0	PIC: 1x OC-12 SONET, MM @ 7/2/* jnxFruName	**portInterfaceCard**	8	Offline	0	Config-ured to power off	0:00:02.37	0:00:00.00	0
8	8	4	0	PIC: @ 7/3/*	**portInterfaceCard**	8	Offline	0	Config-ured to power off	0:00:02.37	0:00:00.00	0

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Table 33: jnxFruContents Objects in the jnxFruTable of an M160 Router *(continued)*

Index	L1	L2	L3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
9	1	0	0	Routing Engine 0	**routingEngine**	1	Online	31	None	0:00:00.00	0:00:00.00	0
9	2	0	0	Routing Engine 1	**routingEngine**	2	Present	0	None	0:00:00.00	0:00:00.00	0
10	1	1	0	FPM CMB	**frontPanelModule**	1	Online	28	None	0:00:00.00	0:00:00.00	0
10	1	2	0	FPM Display	**frontPanelModule**	1	Online	28	None	0:00:00.00	0:00:00.00	0
11	1	0	0	PCG 0	**clockGenerator**	1	Online	40	None	0:00:00.00	0:00:00.00	0
11	2	0	0	PCG 1	**clockGenerator**	2	Online	46	None	0:00:00.00	0:00:00.00	0
12	1	0	0	MCS 0	**controlBoard**	1	Online	47	None	0:00:00.00	0:00:00.00	0
12	2	0	0	MCS 1	**controlBoard**	2	Empty	0	None	0:00:00.00	0:00:00.00	0
13	1	0	0	CIP	**craftInterfacePanel-**	1	Present	0	None	0:00:00.00	0:00:00.00	0

To verify the L1, L2, and L3 indexes, use the **show chassis hardware** command. Sample command output from an M160 router is as follows:

```
user@host> show chassis hardware
Hardware inventory:
Item             Version  Part number  Serial number     Description
Chassis                                47                M160
Midplane         REV 02   710-001245   AB4113
FPM CMB          REV 01   710-001642   AA9721
FPM Display      REV 01   710-001647   AA2995
CIP              REV 02   710-001593   AA9886
PEM 0            Rev 01   740-001243   KJ35782           DC
PEM 1            Rev 01   740-001243   kj35756           DC
PCG 0            REV 01   710-001568   AA9796
PCG 1            REV 01   710-001568   AA9895
Routing Engine 0 REV01    740-003239   AARCH00           RE-2.0
Routing Engine 1
MCS 0            REV 03   710-001226   AA9779
SFM 0 SPP        REV 07   710-001228   AE5504
SFM 0 SPR        REV 03   710-002189   AE4707            Internet Processor II
SFM 2 SPP        REV 06   710-001228   AB3133
SFM 2 SPR        REV 01   710-002189   AB2941            Internet Processor II
SFM 3 SPP        REV 07   710-001228   AV3167
SFM 3 SPR        REV 04   710-002189   AV3439            Internet Processor II
FPC 0            REV 02   710-001611   AA9518            FPC Type 2
  CPU            REV 02   710-001217   AA9572
FPC 1            REV 03   710-001255   AA9812            FPC Type 1
  CPU
FPC 2            REV 02   710-001611   AA9527            FPC Type 2
```

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```
  CPU            REV 02   710-001217   AA9592
FPC 3            REV 01   710-003061   HB2029           FPC Type OC192
  CPU            REV 05   710-001217   AF5950
  PIC 0          REV 01   750-003063   HB2029           1x OC-192 SM LR
FPC 5            REV 01   710-001255   AA2914           FPC Type 1
  CPU            REV 02   710-001217   AA2893
FPC 7            REV 03   710-001255   AA9809           FPC Type 1
  CPU            REV 02   710-001217   AA9573
  PIC 0          REV 04   750-000613   AA0374           1x OC-12 SONET, SMIR
  PIC 1          REV 02   750-E3-PIC   AC1903           4x E3
  PIC 2          REV 02   750-001020   AA8944           1x OC-12 SONET, MM
```

To verify FPC status, use the **show chassis fpc** command. Sample command output from an M160 router is as follows:

```
user@host> show chassis fpc
Temp  CPU Utilization (%)   Memory    Utilization (%)
Slot State            (C)  Total  Interrupt      DRAM (MB) Heap     Buffer
  0  Announce offline   0      0          0         0         0          0
  1  Present            0      0          0         0         0          0
  2  Online            32      4          0        32         1         39
  3  Online            44      1          0        32         1         40
  4  Empty              0      0          0         0         0          0
  5  Offline         --- Chassis connection dropped ---
  6  Empty              0      0          0         0         0          0
  7  Online            42      4          0        32         1         40
```

To verify Routing Engine status, use the **show chassis routing-engine** command. Sample command output from an M160 router is as follows:

```
user@host> show chassis routing-engine
Routing Engine status:
  Slot 0:
    Current state                  Master
    Election priority              Master (default)
    Temperature                 35 degrees C / 95 degrees F
    DRAM                       768 MB
    Memory utilization          10 percent
    CPU utilization:
      User                       1 percent
      Background                 0 percent
      Kernel                    10 percent
      Interrupt                  3 percent
      Idle                      87 percent
    Model                          RE-2.0
    Serial ID                      0c000004f8d26401
    Start time                     2002-06-14 14:39:03 PDT
    Uptime                         11 minutes, 46 seconds
    Load averages:                 1 minute   5 minute  15 minute
                                       0.18       0.19       0.14
Routing Engine status:
  Slot 1:
    Current state                  Present
```

To verify SFM status, use the **show chassis sfm** command. Sample command output from an M160 router is as follows:

```
user@host> show chassis sfm
Temp  CPU Utilization (%)   Memory    Utilization (%)
Slot State            (C)  Total  Interrupt      DRAM (MB) Heap     Buffer
  0  Online            35      1          0        64        16         46
```


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```
  1  Empty              0      0          0         0         0         0
  2  Online            47      1          0        64        16        45
  3  Online            50      1          0        64        16        45
Packet scheduling mode : Disabled
```

Table 34 on page 137 provides an example of **jnxFruEntry** objects in the **jnxFruTable** of an M40 router.

Table 34: jnxFruContents Objects in the jnxFruTable of an M40 Router

Index	L1	L2	L3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
2	1	0	0	Power supply A	**powerEntry Module**	1	Online	0	None	0:0:00:00.00	0:0:00:00.00	101974
2	2	0	0	Power supply B	**powerEntry Module**	2	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
3	1	0	0	Top impeller	**fan**	1	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
3	2	0	0	Bottom impeller	**fan**	2	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	1	0	0	Rear left fan	**fan**	1	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	2	0	0	Rear center fan	**fan**	2	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	3	0	0	Rear right fan	**fan**	3	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
5	1	0	0	Host controller	**routingEngine**	1	Online	37	None	0:0:00:00.00	0:0:00:00.00	0
6	1	0	0	SCB Internet Processor I	**controlBoard**	1	Online	27	None	0:0:00:00.00	0:0:00:00.00	0
7	1	0	0	FPC @ 0/*/*	**flexiblePic Concentrator**	1	Online	28	None	0:0:00:00.00	0:0:00:00.00	0
7	2	0	0	FPC @ 1/*/*	**flexible PicConcentrator**	2	Online	29	None	0:0:00:00.00	0:0:00:00.00	0
7	3	0	0	FPC @ 2/*/*	**flexible PicConcentrator**	3	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
7	4	0	0	FPC @ 3/*/*	**flexiblePic Concentrator**	4	Online	24	None	0:0:00:00.00	0:0:00:00.00	0

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Table 34: jnxFruContents Objects in the jnxFruTable of an M40 Router *(continued)*

Index	L1	L2	L3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
7	5	0	0	FPC @ 4/*/*	**flexiblePic Concentrator**	5	Online	27	None	0:0:00:00.00	0:0:00:00.00	0
7	6	0	0	FPC @ 5/*/*	**flexiblePic Concentrator**	6	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
7	7	0	0	FPC: 1x OC-48 SONET, SMIR @ 6/*/*	**flexiblePic Concentrator**	7	Online	28	None	0:0:00:00.00	0:0:00:00.00	0
7	8	0	0	FPC @ 7/*/*	**flexible PicConcentrator**	8	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
8	1	1	0	PIC: 1x G/E, 1000Base-SX @ 0/0/*	**portInterfaceCard**	1	Ready	24	None	0:0:00:00.00	0:0:00:00.00	0
8	1	2	0	PIC: 1x Tunnel @ 0/1/*	**portInterfaceCard**	1	Ready	24	None	0:0:00:00.00	0:0:00:00.00	0
8	1	3	0	PIC: 4x T1, RJ48 @ 0/2/*	**portInterfaceCard**	1	Ready	24	None	0:0:00:00.00	0:0:00:00.00	0
8	1	4	0	PIC: 1x COC12, SMIR @ 0/3/*	**portIntefaceCard**	1	Ready	24	None	0:0:00:00.00	0:0:00:00.00	0
8	2	1	0	PIC: 2x OC-3 ATM, MM @ 1/0/*	**portInterfaceCard**	2	Ready	27	None	0:0:00:00.00	0:0:00:00.00	0
8	2	2	0	PIC: 4x OC-3 SONET, MM @ 1/1/*	**portInterfaceCard**	2	Ready	27	None	0:0:00:00.00	0:0:00:00.00	0
8	2	3	0	PIC: 2x T3 @ 1/2/*	**portInterfaceCard**	2	Ready	27	None	0:0:00:00.00	0:0:00:00.00	0
8	2	4	0	PIC: 1x CSTM1, SMIR @ 1/3/*	**portInterfaceCard**	2	Ready	27	None	0:0:00:00.00	0:0:00:00.00	0

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Table 34: jnxFruContents Objects in the jnxFruTable of an M40 Router *(continued)*

Index	L1	L2	L3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
8	3	1	0	PIC: @ 2/0/*	**portInterfaceCard**	3	Offline	0	None	0:0:00:00.00	0:0:00:00.00	0
8	3	2	0	PIC: @ 2/1/*	**portInterfaceCard**	3	Offline	0	None	0:0:00:00.00	0:0:00:00.00	0
8	3	3	0	PIC: @ 2/2/*	**portInterfaceCard**	3	Offline	0	None	0:0:00:00.00	0:0:00:00.00	0
8	3	4	0	PIC: @ 2/3/*	**portInterfaceCard**	3	Offline	0	None	0:0:00:00.00	0:0:00:00.00	0
8	4	1	0	PIC: @ 3/0/*	**portInterfaceCard**	4	Ready	24	None	0:0:00:00.00	0:0:00:00.00	0
8	4	2	0	PIC: 4x F/E, 100 BASE-TX @ 3/1/*	**portInterfaceCard**	4	Ready	24	None	0:0:00:00.00	0:0:00:00.00	0
8	4	3	0	PIC: 1x 800M Crypto @ 3/2/*	**portInterfaceCard**	4	Ready	24	None	0:0:00:00.00	0:0:00:00.00	0
8	4	4	0	PIC: 1x CT3-NxDS0 @ 3/3/*	**portInterfaceCard**	4	Ready	24	None	0:0:00:00.00	0:0:00:00.00	0
8	5	1	0	PIC: @ 4/0/*	**portInterfaceCard**	5	Ready	27	None	0:0:00:00.00	0:0:00:00.00	0
8	5	2	0	PIC: @ 4/1/*	**portInterfaceCard**	5	Ready	27	None	0:0:00:00.00	0:0:00:00.00	0
8	5	3	0	PIC: @ 4/2/*	**portInterfaceCard**	5	Ready	27	None	0:0:00:00.00	0:0:00:00.00	0
8	5	4	0	PIC: @ 4/3/*	**flexible PicConcentrator**	5	Ready	27	None	0:0:00:00.00	0:0:00:00.00	0
8	6	1	0	PIC: @ 5/0/*	**portInterfaceCard**	6	Offline	0	None	0:0:00:00.00	0:0:00:00.00	0
8	6	2	0	PIC: @ 5/1/*	**portInterfaceCard**	6	Offline	0	None	0:0:00:00.00	0:0:00:00.00	0
8	6	3	0	PIC: @ 5/2/*	**portInterfaceCard**	6	Offline	0	None	0:0:00:00.00	0:0:00:00.00	0

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Table 34: jnxFruContents Objects in the jnxFruTable of an M40 Router *(continued)*

Index	L1	L2	L3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
8	6	4	0	PIC: @ 5/3/*	**portInterfaceCard**	6	Offline	0	None	0:0:00:00.00	0:0:00:00.00	0
8	7	1	0	PIC: 1x OC-48 SONET, SMIR @ 6/0/*	**portInterfaceCard**	7	Ready	28	None	0:0:00:00.00	0:0:00:00.00	0
8	7	2	0	PIC continued	**portInterfaceCard**	7	Ready	28	None	0:0:00:00.00	0:0:00:00.00	0
8	7	3	0	PIC continued	**portInterfaceCard**	7	Ready	28	None	0:0:00:00.00	0:0:00:00.00	0
8	7	4	0	PIC continued	**portInterfaceCard**	7	Ready	28	None	0:0:00:00.00	0:0:00:00.00	0
8	8	1	0	PIC: @ 7/0/*	**portInterfaceCard**	8	Offline	0	None	0:0:00:00.00	0:0:00:00.00	0
8	8	2	0	PIC: @ 7/1/*	**portInterfaceCard**	8	Offline	0	None	0:0:00:00.00	0:0:00:00.00	0
8	8	3	0	PIC: @ 7/2/*	**portInterfaceCard**	8	Offline	0	None	0:0:00:00.00	0:0:00:00.00	0
8	8	4	0	PIC: @ 7/3/*	**portInterfaceCard**	8	Offline	0	None	0:0:00:00.00	0:0:00:00.00	0
9	1	0	0	Routing Engine	**routingEngine**	1	Online	0	None	0:0:00:00.00	0:0:00:00.00	0

To verify the L1, L2, and L3 indexes, use the **show chassis hardware** command. Sample command output from an M40 router is as follows:

```
user@host> show chassis hardware
Hardware inventory:
Item             Version  Part number  Serial number     Description
Chassis
Backplane        REV 03   710-000073   AA2005
Power Supply A   Rev A    740-000235   000119            DC
Maxicab          REV 04   710-000229   AA0691
Minicab          REV 02   710-000482   AA0270
Display          REV 06   710-000150   AA1042
Routing Engine                                          RE-1.0
SCB              REV 07   710-000075   AA1033            Internet Processor I
FPC 0            REV 01   710-001292   AB8159
  PIC 0          REV 08   750-001072   AP5525            1x G/E, 1000 BASE-SX
  PIC 1          REV 01   750-001323   AB1645            1x Tunnel
  PIC 2          REV 01   750-002953   AD9083            4x T1, RJ48
```


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```
  PIC 3          REV 03   750-001190   AE2907           1x COC12, SMIR
FPC 1            REV 10   710-000175   AA7219
  PIC 0          REV 03   750-002977   HD9331           2x OC-3 ATM, MM
  PIC 1          REV 04   750-002971   HC8020           4x OC-3 SONET, MM
  PIC 2          REV 02.1 710-000608   AA1592           2x T3
  PIC 3          REV 05   750-003248   AD9648           1x CSTM1, SMIR
FPC 3            REV 10   710-000175   AA4782
  PIC 1          REV 04   750-002992   HC3974           4x F/E, 100 BASE-TX
  PIC 2          REV 03   750-003844   AY4806           1x 800M Crypto
  PIC 3          REV 03   750-004743   BD9433           1x CT3-NxDS0
FPC 4            REV 01   710-001292   AC5265
FPC 6            REV 01   710-001292   AB7485
  PIC 0          REV 03   750-000617   AA4566           1x OC-48 SONET, SMIR
```

```
user@host> show chassis environment
Class Item                 Status     Measurement
Power Power Supply A       OK
      Power Supply B       Absent
Temp  FPC 0                OK         28 degrees C / 82 degrees F
      FPC 1                OK         29 degrees C / 84 degrees F
      FPC 3                OK         24 degrees C / 75 degrees F
      FPC 4                OK         27 degrees C / 80 degrees F
      FPC 6                OK         28 degrees C / 82 degrees F
      SCB                  OK         27 degrees C / 80 degrees F
      Backplane @ A1       OK         30 degrees C / 86 degrees F
      Backplane @ A2       OK         26 degrees C / 78 degrees F
      Routing Engine       OK         37 degrees C / 98 degrees F
Fans  Top Impeller         OK         Spinning at normal speed
      Bottom impeller      OK         Spinning at normal speed
      Rear Left Fan        OK         Spinning at normal speed
      Rear Center Fan      OK         Spinning at normal speed
      Rear Right Fan       OK         Spinning at normal speed
Misc  Craft Interface      OK
```

To verify FPC status, use the **show chassis fpc** command. Sample command output from an M40 router is as follows:

```
user@host> show chassis fpc
Temp  CPU Utilization (%)   Memory    Utilization (%)
Slot State            (C)  Total  Interrupt      DRAM (MB) Heap     Buffer
  0  Online            28      2          0         8        11         14
  1  Online            29      7          0         8        21         14
  2  Empty              0      0          0         0         0          0
  3  Online            24     17          0         8        22         15
  4  Online            27      1          0         8         6         13
  5  Empty              0      0          0         0         0          0
  6  Online            28      1          0         8         7         15
  7  Empty              0      0          0         0         0          0
```

To verify Routing Engine status, use the **show chassis routing-engine** command. Sample command output from an M40 router is as follows:

```
user@host> show chassis routing-engine
Routing Engine status:
    Temperature                37 degrees C / 98 degrees F
    DRAM                              256 MB
    Memory utilization         19 percent
    CPU utilization:
      User                      1 percent
      Background                0 percent
      Kernel                    3 percent
      Interrupt                 1 percent
```

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```
    Idle                          96 percent
  Model          RE-1.0
  Start time                      2002-06-24 17:28:30 UTC
  Uptime                          20 minutes, 30 seconds
  Load averages:                  1 minute   5 minute  15 minute
                                      0.00       0.04       0.11
```

To verify SCB status, use the **show chassis scb** command. Sample command output from an M40 router is as follows:

```
user@host> show chassis scb
SCB status:
  Temperature                          27 degrees C / 80 degrees F
  CPU utilization                      3 percent
  Interrupt utilization                  0 percent
  Heap utilization           9 percent
  Buffer utilization           44 percent
  Total CPU DRAM                            64 MB
  Internet Processor I             Version 1, Foundry IBM, Part number 3
  Start time:                      2002-06-24 17:30:10 UTC
  Uptime:                          19 minutes, 8 seconds
```

Table 35 on page 142 provides an example of **jnxFruEntry** objects in the **jnxFruTable** of an M40e router.

Table 35: jnxFruContents Objects in the jnxFruTable of an M40e Router

Index	L1	L2	L3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
2	1	0	0	PEM 0	**powerEntryModule**	1	Present	0	None	0:0:00:00.00	0:0:00:25.99	208927
2	2		0	PEM 1	**powerEntryModule**	2	Online	0	None	0:0:00:00.00	0:0:00:25.99	208928
4	1	0	0	Front top blower	**fan**	1	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	2	1	0	Fan tray front left	**fan**	2	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	2	2	0	Fan tray front right	**fan**	2	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	2	3	0	Fan tray rear left	**fan**	2	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	2	4	0	Fan tray rear right	**fan**	2	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	3	0	0	Rear top blower	**fan**	3	Present		None	0:0:00:00.00	0:0:00:00.00	0
4	4	0	0	Rear bottom blower	**fan**	4	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
6	1	1	0	SFM 0 SPP	**switchingAnd ForwardingModule**	1	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0

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Table 35: jnxFruContents Objects in the jnxFruTable of an M40e Router *(continued)*

Index	L1	L2	L3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
6	1	2	0	SFM 0 SPR	**switchingAnd ForwardingModule**	1	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
6	2	1	0	SFM 1 SPP	**switchingAnd ForwardingModule**	2	Online	42	None	0:0:00:21.95	0:0:00:00.00	0
6	2	2	0	SFM 1 SPR Internet Processor II	**switchingAnd ForwardingModule**	2	Online	42	None	0:0:00:21.95	0:0:00:00.00	0
7	1	0	0	FPC @ 0/*/*	**flexiblePic Concentrator**	1	Online	41	None	0:0:00:21.85	0:0:00:00.00	0
7	2	0	0	FPC @ 1/*/*	**flexible PicConcentrator**	2	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
7	3	0	0	FPC @ 2/*/*	**flexible PicConcentrator**	3	Online	43	None	0:0:00:21.87	0:0:00:00.00	0
7	4	0	0	FPC @ 3/*/*	**flexible PicConcentrator**	4	Online	38	None	0:0:00:21.89	0:0:00:00.00	0
7	5	0	0	FPC @ 4/*/*	**flexiblePic Concentrator**	5	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
7	6	0	0	FPC @ 5/*/*	**flexiblePic Concentrator**	6	Online	46	None	0:0:00:21.91	0:0:00:00.00	0
7	7	0	0	FPC @ 6/*/*	**flexiblePic Concentrator**	7	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
7	8	0	0	FPC @ 7/*/*	**flexiblePic Concentrator**	8	Online	44	None	0:0:00:21.93	0:0:00:00.00	0
8	1	1	0	PIC: @ 0/0/*	**portInterfaceCard**	1	Online	45	None	0:0:00:00.00	0:0:00:00.00	0
8	1	2	0	PIC: 1x OC-12 SONET, MM @ 0/1/*	**portInterfaceCard**	1	Online	45	None	0:0:00:00.00	0:0:00:00.00	0
8	1	3	0	PIC: 4x CT3 @ 0/2/*	**portInterfaceCard**	1	Online	45	None	0:0:00:00.00	0:0:00:00.00	0

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Table 35: jnxFruContents Objects in the jnxFruTable of an M40e Router *(continued)*

Index	L1	L2	L3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
8	1	4	0	PIC: 1x Multi Link(32) @ 0/3/*	**portInterfaceCard**	1	Online	45	None	0:0:00:00.00	0:0:00:00.00	0
8	2	1	0	PIC: @ 1/0/*	**portInterfaceCard**	2	Online	50	None	0:0:00:00.00	0:0:00:00.00	0
8	2	2	0	PIC: @ 1/1/*	**portInterfaceCard**	2	Online	50	None	0:0:00:00.00	0:0:00:00.00	0
8	2	3	0	PIC: @ 1/2/*	**portInterfaceCard**	2	Online	50	None	0:0:00:00.00	0:0:00:00.00	0
8	2	4	0	PIC: @ 1/3/*	**portInterfaceCard**	2	Online	50	None	0:0:00:00.00	0:0:00:00.00	0
8	3	1	0	PIC: 1x OC-12 SONET, MM @ 2/0/*	**portInterfaceCard**	3	Online	41	None	0:0:00:00.00	0:0:00:00.00	0
8	3	2	0	PIC: 1x OC-12 SONET, MM @ 2/1/*	**portInterfaceCard**	3	Online	41	None	0:0:00:21.85	0:0:00:00.00	0
8	3	3	0	PIC: 1x OC-12 SONET, MM @ 2/2/*	**portInterfaceCard**	3	Online	41	–	0:0:00:21.85	0:0:00:00.00	–
8	3	4	0	PIC: @ 2/3/*	**portInterfaceCard**	3	Online	41	–	0:0:00:21.85	0:0:00:00.00	–
8	4	1	0	PIC: 1x OC-48 SONET, SMIR @ 3/0/*	**portInterfaceCard**	4	Empty	0	–	0:0:00:00.00	0:0:00:00.00	0
8	4	2	0	PIC: @ 3/1/*	**portInterfaceCard**	4	Empty	0	–	0:0:00:00.00	0:0:00:00.00	0
8	4	3	0	PIC: @ 3/2/*	**portInterfaceCard**	4	Empty	0	–	0:0:00:00.00	0:0:00:00.00	0

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Table 35: jnxFruContents Objects in the jnxFruTable of an M40e Router *(continued)*

Index	L1	L2	L3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
8	4	4	0	PIC: @ 3/3/*	**portInterfaceCard**	4	Empty	0	–	0:0:00:00.00	0:0:00:00.00	0
8	5	1	0	PIC: @ 4/0/*	**portInterfaceCard**	5	Online	43	–	0:0:00:21.87	0:0:00:00.00	0
8	5	2	0	PIC: @ 4/1/*	**portInterfaceCard**	5	Online	43	–	0:0:00:21.87	0:0:00:00.00	0
8	5	3	0	PIC: @ 4/2/*	**portInterfaceCard**	5	Online	43	–	0:0:00:21.87	0:0:00:00.00	0
8	5	4	0	PIC: @ 4/3/*	**portInterfaceCard**	5	Online	43	–	0:0:00:21.87	0:0:00:00.00	0
8	6	1	0	PIC: @ 5/0/*	**portInterfaceCard**	6	Online	38	–	0:0:00:21.89	0:0:00:00.00	0
8	6	2	0	PIC: @ 5/1/*	**portInterfaceCard**	6	Online	38	–	0:0:00:21.89	0:0:00:00.00	0
8	6	3	0	PIC: 1x OC-12 SONET, SMIR @ 5/2/*	**portInterfaceCard**	6	Online	38	–	0:0:00:21.89	0:0:00:00.00	0
8	6	4	0	PIC: 1x OC-12 SONET, MM @ 5/3/*	**portInterfaceCard**	6	Online	38	–	0:0:00:21.89	0:0:00:00.00	0
8	7	1	0	PIC: @ 6/0/*	**portInterfaceCard**	7	Empty	0	–	0:0:00:00.00	0:0:00:00.00	0
8	7	2	0	PIC: @ 6/1/*	**portInterfaceCard**	7	Empty	0	–	0:0:00:00.00	0:0:00:00.00	0
8	7	3	0	PIC: @ 6/2/*	**portInterfaceCard**	7	Empty	0	–	0:0:00:00.00	0:0:00:00.00	0
8	7	4	0	PIC: @ 6/3/*	**portInterfaceCard**	7	Empty	0	–	0:0:00:00.00	0:0:00:00.00	0
8	8	1	0	PIC: 8x FE-FX, 100 BASE-FX @ 7/0/*	**portInterfaceCard**	8	Online	46	–	0:0:00:21.91	0:0:00:00.00	0

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Table 35: jnxFruContents Objects in the jnxFruTable of an M40e Router *(continued)*

Index	L1	L2	L3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
8	8	2	0	PIC: @ 7/1/*	**portInterfaceCard**	8	Online	46	–	0:0:00:21.91	0:0:00:00.00	0
8	8	3	0	PIC: 1x Link Service(4) @ 7/2/*	**portInterfaceCard**	8	Online	46	–	0:0:00:21.91	0:0:00:00.00	0
8	8	4	0	PIC: @ 7/3/*	**portInterfaceCard**	1	Online	46	–	0:0:00:00.00	0:0:00:00.00	0
9	1	0	0	Routing Engine 0	**routingEngine**	2	Online	46	–	0:0:00:00.00	0:0:00:00.00	0
9	2	0	0	Routing Engine 1	**routingEngine**	1	Present	34	–	0:0:00:00.00	0:0:00:00.00	0
10	1	1	0	FPM CMB	**frontPanelModule**	1	Online	28	–	0:0:00:00.00	0:0:00:00.00	0
10	1	2	0	FPM Display	**frontPanelModule**	1	Online	28	–	0:0:00:00.00	0:0:00:00.00	0
11	1	0	0	PCG 0	**clockGenerator**	1	Online	45	–	0:0:00:00.00	0:0:00:00.00	0
11	2	0	0	PCG 1	**clockGenerator**	2	Online	50	–	0:0:00:00.00	0:0:00:00.00	0
12	1	0	0	MCS 0	**controlBoard**	1	Online	46	–	0:0:00:00.00	0:0:00:00.00	0
12	2	0	0	MCS 1	**controlBoard**	2	Online	0	–	0:0:00:00.00	0:0:00:00.00	0
13	1	0	0	CIP	**craftInterfacePanel**	1	Present	0	–	0:0:00:00.00	0:0:00:00.00	0

To verify L1, L2, and L3 indexes, use the **show chassis hardware** command. The sample output from an M40e router is as follows:

```
user@host> show chassis hardware
Item             Version  Part number  Serial number     Description
Chassis                               19084             M40e
Midplane         REV 01   710-005071   AX3654
FPM CMB          REV 03   710-001642   AR9037
FPM Display      REV 03   710-001647   AP1334
CIP              REV 08   710-001593   AE8486
PEM 0            Rev 01   740-003787   ME13120           Power Entry Module
PEM 1            Rev 01   740-003787   MC25354           Power Entry Module
PCG 0            REV 07   710-001568   AG1377
PCG 1            REV 07   710-001568   AR3806
Routing Engine 0 REV 04   740-003239   9001026568        RE-2.0
Routing Engine 1
MCS 0            REV 11   710-001226   AN5810
MCS 1            REV 11   710-001226   AR0109
```


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```
SFM 1 SPP        REV 07   710-001228   BE0106
SFM 1 SPR        REV 05   710-002189   BE0062           Internet Processor II
FPC 0            REV 01   710-005078   BE0642           M40e-FPC Type 1
  CPU            REV 01   710-004600   BD2496
  PIC 1          REV 04   750-001895   HE0885           1x OC-12 SONET, MM
  PIC 2          REV 06   750-003009   HE1422           4x CT3
  PIC 3          REV 03   750-003837   AP7134           1x Multi Link(32)
FPC 2            REV 01   710-005078   BE0647           M40e-FPC Type 1
  CPU            REV 01   710-004600   AN4299
  PIC 0          REV 04   750-001895   HD2623           1x OC-12 SONET, MM
  PIC 1          REV 04   750-001895   HE0609           1x OC-12 SONET, MM
  PIC 2          REV 04   750-001895   HE0871           1x OC-12 SONET, MM
FPC 3            REV 01   710-005197   BD9846           M40e-FPC Type 2
  CPU            REV 01   710-004600   BD2364
  PIC 0          REV 01   750-001900   AA9649           1x OC-48 SONET, SMIR
FPC 5            REV 01   710-005078   BE0639           M40e-FPC Type 1
  CPU            REV 01   710-004600   BD2587
  PIC 2          REV 04   750-001896   AV4480           1x OC-12 SONET, SMIR
  PIC 3          REV 04   750-001895   HE1000           1x OC-12 SONET, MM
FPC 7            REV 01   710-005196   BD9456           M40e-FPC
  CPU            REV 01   710-004600   AN4323
  PIC 0          REV 01   750-004944   AY4645           8x FE-FX, 100 BASE-FX
  PIC 2          REV 01   750-007927   AP1919           1x Link Service(4)
```

To verify Routing Engine status, use the **show chassis routing-engine** command. Sample command output from an M40e router is as follows:

```
user@host> show chassis routing-engine
Routing Engine status:
  Slot 0:
    Current state          Master
    Election priority          Master (default)
    Temperature                      34 degrees C / 93 degrees F
    DRAM                            768 MB
    Memory utilization               9 percent
    CPU utilization:
      User                        0 percent
      Background                         0 percent
      Kernel                      2 percent
      Interrupt                   0 percent
      Idle                             97 percent
    Model                        RE-2.0
    Serial ID                    9c000007c8644701
    Start time                   2002-06-24 10:33:41 PDT
    Uptime                       31 minutes, 7 seconds
    Load averages:               1 minute   5 minute  15 minute
                                     0.01       0.02       0.00
Routing Engine status:
  Slot 1:
    Current state                Present
```

To verify FPC status, use the **show chassis fpc** command. Sample command output from an M40e router is as follows:

```
user@host> show chassis fpc
Temp  CPU Utilization (%)   Memory    Utilization (%)
Slot State            (C)  Total  Interrupt      DRAM (MB) Heap     Buffer
  0  Online            41      4          0        32         3         40
  1  Empty              0      0          0         0         0          0
  2  Online            43      4          0        32         1         40
  3  Online            38      1          0        32         1         40
```

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```
4  Empty          0    0    0    0    0    0
5  Online        46    4    0   32    1   40
6  Empty          0    0    0    0    0    0
7  Online        44    4    0   32    2   39
```

Table 36 on page 148 provides an example of **jnxFruEntry** objects in the **jnxFruTable** of a T640 router.

Table 36: jnxFruEntry Objects in the jnxFruTable of a T640 Router

Index	L1	L2	L3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
2	1	0	0	PEM 0	**power EntryModule**	1	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
2	2	0	0	PEM 1	**power EntryModule**	2	Online	27	None	0:0:00:00.00	0:0:00:00.00	217044
4	1	1	0	Top left front fan	**fan**	1	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	1	2	0	Top left middle fan	**fan**	1	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	1	3	0	Top left rear fan	**fan**	1	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	1	4	0	Top right front fan	**fan**	1	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	1	5	0	Top right middle fan	**fan**	1	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	1	6	0	Top right rear fan	**fan**	1	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	2	1	0	Bottom left front fan	**fan**	2	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	2	2	0	Bottom left middle fan	**fan**	2	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	2	3	0	Bottom left rear fan	**fan**	2	Present	0	None	0:0:00:00.00	0:0:00:00.00	0

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Table 36: jnxFruEntry Objects in the jnxFruTable of a T640 Router *(continued)*

Index	L1	L2	B	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
4	2	4	0	Bottom right front fan	**fan**	2	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	2	5	0	Bottom right middle fan	**fan**	2	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	2	6	0	Bottom right rear fan	**fan**	2	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	3	1	0	Fourth blower from top	**fan**	3	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	3	2	0	Bottom blower	**fan**	3	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	3	3	0	Middle blower	**fan**	3	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	3	4	0	Top blower	**fan**	3	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
4	3	5	0	Second blower from top	**fan**	3	Present	0	None	0:0:00:00.00	0:0:00:00.00	0
7	1	0	0	FPC @ 0/*/*	**flexiblePic Concentrator**	1	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
7	1	1	0	FPC @ 0/0/* top temper-ature sensor	**flexiblePic Concentrator**	1	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
7	1	2	0	FPC @ 0/1/* bottom temper-ature sensor	**flexiblePic Concentrator**	1	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0

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Table 36: jnxFruEntry Objects in the jnxFruTable of a T640 Router *(continued)*

Index	L1	L2	L3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
7	2	0	0	FPC @ 1/*/*	**flexiblePic Concentrator**	2	Online	30	None	0:0:00:01.94	0:0:00:00.00	0
7	2	1	0	FPC @ 1/0/* top temperature sensor	**flexiblePic Concentrator**	2	Online	30	None	0:0:00:01.94	0:0:00:00.00	0
7	2	2	0	FPC @ 1/1/* bottom temperature sensor	**flexiblePic Concentrator**	2	Online	30	None	0:0:00:01.94	0:0:00:00.00	0
7	3	0	0	FPC @ 2/*/*	**flexiblePic Concentrator**	3	Online	30	None	0:0:00:01.96	0:0:00:00.00	0
7	3	1	0	FPC @ 2/0/* top temperature sensor	**flexiblePic Concentrator**	3	Online	30	None	0:0:00:01.96	0:0:00:00.00	0
7	3	2	0	FPC @ 2/1/* bottom temperature sensor	**flexiblePic Concentrator**	3	Online	30	None	0:0:00:01.96	0:0:00:00.00	0
7	4	0	0	FPC @ 3/*/*	**flexiblePic Concentrator**	4	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
7	4	1	0	FPC @ 3/0/* top temperature sensor	**flexiblePic Concentrator**	4	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
7	4	2	0	FPC @ 3/1/* bottom temperature sensor	**flexiblePic Concentrator**	4	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0

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Table 36: jnxFruEntry Objects in the jnxFruTable of a T640 Router *(continued)*

Index	L1	L2	B	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
7	5	9	0	FPC @ 4/*/*	**flexiblePic Concentrator**	5	Online	36	None	0:0:00:01.98	0:0:00:00.00	0
7	5	1	0	FPC @ 4/0/* top temperature sensor	**flexiblePic Concentrator**	5	Online	36	None	0:0:00:01.98	0:0:00:00.00	0
7	5	2	0	FPC @ 4/1/* bottom temperature sensor	**flexiblePic Concentrator**	5	Online	36	None	0:0:00:01.98	0:0:00:00.00	0
7	6	0	0	FPC @ 5/*/*	**flexiblePic Concentrator**	6	Offline	0	Error	0:0:12:51.28	0:0:00:00.00	0
7	6	1	0	FPC @ 5/0/* top temperature sensor	**flexiblePic Concentrator**	6	Offline	0	Error	0:0:12:51.28	0:0:00:00.00	0
7	6	2	0	FPC @ 5/1/* bottom temperature sensor	**flexiblePic Concentrator**	6	Offline	0	Error	0:0:12:51.28	0:0:00:00.00	0
7	7	0	0	FPC @ 6/*/*	**flexiblePic Concentrator**	7	Online	30	None	0:0:00:02.05	0:0:00:00.00	0
7	7	1	0	FPC @ 6/0/* top temperature sensor	**flexiblePic Concentrator**	7	Online	30	None	0:0:00:02.05	0:0:00:00.00	0
7	7	2	0	FPC @ 6/1/* bottom temperature sensor	**flexiblePic Concentrator**	7	Online	30	None	0:0:00:02.05	0:0:00:00.00	0

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Table 36: jnxFruEntry Objects in the jnxFruTable of a T640 Router *(continued)*

Index	L1	L2	L3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
7	8	0	0	FPC @ 7/*/*	**flexiblePic Concentrator**	8	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
7	8	1	0	FPC @ 7/0/* top temperature sensor	**flexiblePic Concentrator**	8	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
7	8	2	0	FPC @ 7/1/* bottom temperature sensor	**flexiblePic Concentrator**	8	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
8	1	1	0	PIC: @ 0/0/*	**portInterface-Card**	1	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
8	1	2	0	PIC: @ 0/1/*	**portInterface-Card**	1	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
8	1	3	0	PIC: @ 0/2/*	**portInterface-Card**	1	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
8	1	4	0	PIC: @ 0/3/*	**portInterface-Card**	1	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
8	2	1	0	PIC: 1x OC-48 SONET, SMIR @ 1/0/*	**portInterface-Card**	2	Online		None	0:0:00:00.00	0:0:00:00.00	0
8	2	2	0	PIC: 1x OC-48 SONET, SMSR @ 1/1/*	**portInterface-Card**	2	Online	36	None	0:0:00:00.00	0:0:00:00.00	0
8	2	3	0	PIC: 1x OC-48 SONET, SMIR @ 1/2/*	**portInterface-Card**	2	Online	36	None	0:0:00:00.00	0:0:00:00.00	0

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Table 36: jnxFruEntry Objects in the jnxFruTable of a T640 Router *(continued)*

Index	L1	L2	L3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
8	2	4	0	PIC: 1x OC-48 SONET, SMIR @ 1/3/*	portInterface-Card	2	Online	36	None	0:0:00:00.00	0:0:00:00.00	0
8	3	1	0	PIC: @ 2/0/*	portInterface-Card	3	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
8	3	2	0	PIC: @ 2/1/*	portInterface-Card	3	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
8	3	3	0	PIC: @ 2/2/*	portInterface-Card	3	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
8	3	4	0	PIC: @ 2/3/*	portInterface-Card	3	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
8	4	1	0	PIC: @ 3/0/*	portInterface-Card	4	Online		None	0:0:00:01.00	0:0:00:00.00	0
8	4	2	0	PIC: @ 3/1/*	portInterface-Card	4	Online	30	None	0:0:00:01.94	0:0:00:00.00	0
8	4	3	0	PIC: @ 3/2/*	portInterface-Card	4	Online	30	None	0:0:00:01.94	0:0:00:00.00	0
8	4	4	0	PIC: @ 3/3/*	portInterface-Card	4	Online	30	None	0:0:00:01.94	0:0:00:00.00	0
8	5	1	0	PIC: 1x Tunnel @ 4/0/*	portInterface-Card	5	Online	30	None	0:0:00:01.94	0:0:00:00.00	0
8	5	2	0	PIC: 1x OC-192 SM SR2 @ 4/1/*	portInterface-Card	5	Online	30	None	0:0:00:01.96	0:0:00:00.00	0
8	5	3	0	PIC: 4x OC-48 SONET, SMSR @ 4/2/*	portInterface-Card	5	Online	30	None	0:0:00:01.96	0:0:00:00.00	0
8	5	4	0	PIC: 1x OC-192 SM SR1 @ 4/3/*	portInterface-Card	5	Online	30	None	0:0:00:01.96	0:0:00:00.00	0

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Table 36: jnxFruEntry Objects in the jnxFruTable of a T640 Router *(continued)*

Index	L1	L2	L3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
8	6	1	0	PIC: @ 5/0/*	portInterfaceCard	6	Empty	0	None	0:0:00:01.00	0:0:00:00.00	0
8	6	2	0	PIC: @ 5/1/*	portInterfaceCard	6	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
8	6	3	0	PIC: @ 5/2/*	portInterfaceCard	6	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
8	6	4	0	PIC: @ 5/3/*	portInterfaceCard	6	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
8	7	1	0	PIC: @ 6/0/*	portInterfaceCard	7	Online	30	None	0:0:00:00.00	0:0:00:00.00	0
8	7	2	0	PIC: @ 6/1/*	portInterfaceCard	7	Online	30	None	0:0:00:01.98	0:0:00:00.00	0
8	7	3	0	PIC: @ 6/2/*	portInterfaceCard	7	Online	30	None	0:0:00:01.98	0:0:00:00.00	0
8	7	4	0	PIC: @ 6/3/*	portInterfaceCard	7	Online	30	None	0:0:00:01.98	0:0:00:00.00	0
8	8	1	0	PIC: @ 7/0/*	portInterfaceCard	8	Offline	0	Error	0:0:12:51.28	0:0:00:00.00	0
8	8	2	0	PIC: @ 7/1/*	portInterfaceCard	8	Offline	0	Error	0:0:12:51.28	0:0:00:00.00	0
8	8	3	0	PIC: @ 7/2/*	portInterfaceCard	8	Offline	0	Error	0:0:12:51.28	0:0:00:00.00	0
8	8	4	0	PIC: @ 7/3/*	portInterfaceCard	8	Offline	0	Error	0:0:12:51.28	0:0:00:00.00	0
9	1	0	0	Routing Engine 0	routingEngine	1	Online	34	None	0:0:00:00.00	0:0:00:00.00	0
9	2	0	0	Routing Engine 1	routingEngine	2	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
10	1	1	0	FPM GBUS	frontPanelModule	1	Online	27	None	0:0:00:00.00	0:0:00:00.00	0
10	1	2	0	FPM Display	frontPanelModule	1	Online	27	None	0:0:00:00.00	0:0:00:00.00	0

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Table 36: jnxFruEntry Objects in the jnxFruTable of a T640 Router *(continued)*

Index	L1	L2	L3	Name	Type	Slot	State	Temp	Offline	PowerOff	PowerOn	Uptime
11	1	0	0	SCG 0	clockGenerator	1	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
11	2	0	0	SCG 1	clockGenerator	2	Online	27	None	0:0:00:00.00	0:0:00:00.00	0
12	1	0	0	CB 0	controlBoard	1	Online	27	None	0:0:00:01.94	0:0:00:00.00	0
12	2	0	0	CB 1	controlBoard	2	Unknown	0	None	0:0:00:01.96	0:0:00:00.00	0
13	1	0	0	CIP	craftInterfacePanel	1	Present	36	None	0:0:00:00.00	0:0:00:00.00	0
14	1	0	0	SPMB 0	processorMezzanineBoardForSIB	1	Online	34	None	0:0:00:00.00	0:0:00:00.00	0
14	2	0	0	SPMB 1	processorMezzanineBoardForSIB	2	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
15	1	0	0	SIB 0	switchInterfaceBoard	1	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
15	2	0	0	SIB 1	switchInterfaceBoard	2	Online	36	None	0:0:00:00.00	0:0:00:00.00	0
15	3	0	0	SIB 2	switchInterfaceBoard	3	Empty	0	None	0:0:00:00.00	0:0:00:00.00	0
15	4	0	0	SIB 3	switchInterfaceBoard	4	Online	30	None	0:0:00:01.94	0:0:00:00.00	0
15	5	0	0	SIB 4	switchInterfaceBoard	5	Online	30	None	0:0:00:01.96	0:0:00:00.00	0

To verify the L1, L2, and L3 indexes, use the **show chassis hardware** command. Sample command output from a T640 router is as follows:

```
user@host> show chassis hardware
Hardware inventory:
Item             Version  Part number  Serial number     Description
Chassis                                1234              T640
Midplane         REV 04   710-002726   AX5603
FPM GBUS         REV 02   710-002901   HE3062
FPM Display      REV 01   710-002897   HD3033
CIP              REV 05   710-002895   HA5022
PEM 1            RevX02   740-002595   MD21812           Power Entry Module
```

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```
SCG 1            REV 01   710-003423   HD3025
Routing Engine 0 REV 01   740-005022   210865700336      RE-3.0
CB 0             REV 02   710-002728   HE3025
CB 1
FPC 1            REV 01   710-002385   HE3173            FPC Type 2
  CPU            REV 06   710-001726   HC0042
  PIC 0          REV 03   750-001900   AD5737            1x OC-48 SONET, SMIR
  PIC 1          REV 07   750-001900   AR3613            1x OC-48 SONET, SMSR
  PIC 2          REV 01   750-001900   AA9604            1x OC-48 SONET, SMIR
  PIC 3          REV 01   750-001900   AA9602            1x OC-48 SONET, SMIR
  MMB 1          REV 03   710-001723   HC0111            MMB-144mbit
  ICBM           REV 04   710-003384   HA4497
  PPB 0          REV 02   710-003758   HA4543            PPB Type 2
  PPB 1          REV 02   710-003758   HA4540            PPB Type 2
FPC 2            REV 01   710-002385   HE3180            FPC Type 2
  CPU            REV 06   710-001726   HE7904
  MMB 1          REV 03   710-001723   HC0120            MMB-144mbit
  ICBM           REV 01   710-003384   HE3046
  PPB 0          REV 02   710-003758   HA4564            PPB Type 2
  PPB 1          REV 02   710-003758   HA4554            PPB Type 2
FPC 4            REV 04   710-001721   HE3145            FPC Type 3
  CPU            REV 06   710-001726   HC0034
  PIC 0                                                  1x Tunnel
  PIC 1          REV 01   750-003824   HE7803            1x OC-192 SM SR2
  PIC 2          REV 01   750-003336   HE3420            4x OC-48 SONET, SMSR
  PIC 3          REV 01   750-003824   HE7802            1x OC-192 SM SR1
  MMB 0          REV 03   710-001723   HE7230            MMB-144mbit
  MMB 1          REV 03   710-001723   HE7267            MMB-144mbit
  ICBM           REV 04   710-003384   HA4485
  PPB 0          REV 02   710-002845   HA4550            PPB Type 3
  PPB 1          REV 02   710-002845   HA4525            PPB Type 3
FPC 5            REV 04   710-001721   HE3175            FPC Type 3
  CPU
FPC 6            REV 01   710-002385   HD5027            FPC Type 2
  CPU            REV 06   710-001726   HC0033
  MMB 1          REV 03   710-001723   HC0080            MMB-144mbit
  ICBM           REV 04   710-003384   HA4486
  PPB 0          REV 02   710-003758   HA4541            PPB Type 2
  PPB 1          REV 02   710-003758   HA4539            PPB Type 2
SPMB 0           REV 01   710-003229   HA5999
SIB 0            REV 01   710-003980   HD5054            SIB-I8
SIB 2            REV 01   710-003980   HC0035            SIB-I8
SIB 3            REV 01   710-003980   HA5065            SIB-I8
SIB 4            REV 01   710-003980   HE3016            SIB-I8
```

To verify FPC status, use the **show chassis fpc** command. Sample command output from a T640 router is as follows:

```
user@host> show chassis fpc
Temp  CPU Utilization (%)   Memory    Utilization (%)
Slot State            (C)  Total  Interrupt      DRAM (MB) Heap     Buffer
  0  Empty              0      0          0         0         0          0
  1  Online            30      2          0       512         3         41
  2  Online            30      2          0       256         7         41
  3  Empty              0      0          0         0         0          0
  4  Online            30      4          0       512         6         41
  5  Offline         --- Unresponsive ---
  6  Online            30      2          0       256         7         41
  7  Empty              0      0          0         0         0          0
```


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To verify Routing Engine status, use the **show chassis routing-engine** command. Sample command output from a T640 router is as follows:

```
user@host> show chassis routing-engine
Routing Engine status:
  Slot 0:
    Current state                  Master
    Election priority              Master (default)
    Temperature                 35 degrees C / 95 degrees F
    DRAM                      2048 MB
    Memory utilization           4 percent
    CPU utilization:
      User                       0 percent
      Background                 0 percent
      Kernel                     2 percent
      Interrupt                  0 percent
      Idle                      97 percent
    Model                        RE-3.0
    Start time                   2002-06-24 10:33:34 PDT
    Uptime                       33 minutes, 38 seconds
    Load averages:               1 minute   5 minute  15 minute
                                     0.08       0.03       0.01
```

To verify SPMB status, use the **show chassis spmb** command. Sample command output from a T640 router is as follows:

```
user@host> show chassis spmb
Slot 0 information:
  State                          Online
  Total CPU Utilization             2%
  Interrupt CPU Utilization         0%
  Memory Heap Utilization           0%
  Buffer Utilization               40%
  Start time:                   2002-06-24 10:34:22 PDT
  Uptime:                        33 minutes, 3 seconds
```

jnxBoxKernelMemoryUsedPercent

The object identifier for **jnxBoxKernelMemoryUsedPercent** is **{jnxBoxAnatomy 16}**. This object represents the amount of the kernel memory used, expressed as a percentage of the total available memory. The object shows **0** if the kernel memory usage is unavailable or inapplicable. When the kernel memory usage exceeds 80 percent, a system log message is logged and an RMON rising threshold trap is generated if RMON health monitoring is enabled for the device.

jnxBoxSystemDomainType

The object identifier for **jnxBoxSystemDomainType** is **{jnxBoxAnatomy 17}**. This object indicates the domain type of the device, that is, whether it is a root system domain (RSD; represented by integer 2) or a protected system domain (PSD; represented by integer 3). This object returns an integer value of **1**, denoting not applicable, if the system domain type feature is not supported on the device.

Related Documentation

- Chassis MIBs on page 70
- Chassis MIB Textual Conventions on page 71
- Chassis Traps on page 158

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Chassis Traps

The chassis-related traps are defined under the **jnxTraps** and **jnxChassisOKtraps** branches. For the system logging severity levels for these traps, see ."Juniper Networks Enterprise-Specific SNMP Traps" on page 633.

These traps are defined as follows:

- Power failure (**jnxPowerSupplyFailure**)—Sent when the power supply, router circuit breaker, or power circuit fails, or when there is a power outage. When only one of the power supplies has failed, the service impact is minimal. One power supply can provide the necessary power for a fully loaded router. To determine the source of the failure, you must physically inspect the router. This trap is repeated every hour until the power supply is restored.
- Fan failure (**jnxFanFailure**)—Sent when the fan fuse blows or when the fan wiring shorts out. When only one fan has failed, there is no service impact. The remaining fans increase speed to compensate. However, you must resolve the problem before another fan fails. This trap is repeated every hour until the fan failure is fixed. To determine the source of the failure, you must physically inspect the router, taking care to check the fuses. See the hardware installation guide for your router model for more information.
- Overtemperature (**jnxOverTemperature**)—Sent when several fans fail or the room temperature increases significantly. The service impact of this trap depends on the temperature of the router. In general, the router increases the speed of the fans when any component exceeds a temperature of 55º C. The fans remain at the higher speed until the temperature decreases below the threshold. In this case, there is no service impact. However, if the temperature exceeds 75º C, the router transmits a warning and automatically shuts down. This scenario creates a significant service impact because the shutdown affects additional routers and equipment. This trap is repeated every minute until the temperature is brought down to normal. To determine the source of the overtemperature problem, you must physically inspect the router to determine whether any fans have failed in the router.
- Power Supply OK (**jnxPowerSupplyOK**)—Sent when a power supply recovers from failure.
- Fan OK (**jnxFanOK**)—Sent when a fan recovers from failure.
- Temperature OK (**jnxTemperatureOK**)—Sent when a chassis component recovers from an overtemperature condition.
- Redundancy Switchover (**jnxRedundancySwitchover**)—For certain platforms, such as the M20 or M160, some subsystems, such as the Routing Engine, have a redundant backup unit that can be brought online, manually or automatically, if the main unit malfunctions. The redundancy switchover trap indicates such a change.
- Field Replaceable Unit Removal (**jnxFruRemoval**)—Sent when the specified FRU has been removed from the chassis.
- Field Replaceable Unit Insertion (**jnxFruInsertion**)—Sent when the specified FRU has been inserted into the chassis.

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- Field Replaceable Unit Power Off (**jnxFruPowerOff**)—Sent when the specified FRU has been powered off in the chassis.

  The **jnxFruPowerOff** trap is also sent in the following scenarios:

  - When an FRU that is controlled using inter-process communication (IPC) goes offline or is removed from the chassis. For example, a switch interface board (SIB).
  - When an FRU that does not have a backup unit goes offline or is removed from the chassis. For example, a Flexible PIC Concentrator (FPC).

  

  NOTE: When a SONET Clock Generator (SCG) is taken offline, the unit is not powered down. Therefore, jnxFruPowerOff or jnxFruPowerOn traps are not sent when the unit is taken online or offline.

- Field Replaceable Unit Power On (**jnxFruPowerOn**)—Sent when the specified FRU has been powered on in the chassis.
- Field Replaceable Unit Failed (**jnxFruFailed**)—Sent when the specified FRU has failed in the chassis. Typically, this is due to the FRU not powering up or being unable to load software. FRU replacement might be required.
- Field Replaceable Unit Offline (**jnxFruOffline**)—Sent when the specified FRU goes offline. However, when an FRU that does not have a backup unit goes offline, the Junos OS generates the **jnxFruPowerOff** trap instead of the **jnxFruOffline** trap. Typically, a **jnxFruOffline** trap is generated to inform the backup FRU about the status of the primary FRU so that the backup FRU can take over when the primary FRU goes offline.

  The following are some scenarios when **jnxFruOffline** traps are generated:

  - When a Packet Forwarding Engine Clock Generator (PCG) goes offline (M40e).
  - When a SONET Clock Generator goes offline (T Series).
  - When a Line Card Chassis goes offline (TX4 internet router).
  - When a Routing Engine goes offline
- Field Replaceable Unit Online (**jnxFruOnline)**—Sent when the specified FRU goes online.
- Field Replaceable Unit Check (**jnxFruCheck)**—Sent when the specified FRU has encountered operational errors. On M120 and M320 routers, this trap is sent if the revision number for the ATM2 PIC FPGA is less than 8B44(4).
- Forwarding Engine Board (FEB) Switch Over (**jnxFEBSwitchover**)—Sent when the specified FEB undergoes a switchover.
- Hard disk Failure (**jnxHardDiskFailed**)—Sent when the hard disk in the specified Routing Engine is not operational.
- Hard disk Missing (**jnxHardDiskMissing**)—Sent when the hard disk in the specified Routing Engine is missing from the boot device list.

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This section contains the following topics:

- SNMPv1 Trap Format on page 160
- SNMPv2 Trap Format on page 161

SNMPv1 Trap Format

The SNMPv1 trap format for the chassis-related traps is described in Table 37 on page 160. To view the SNMPv1 chassis-related traps, see "Juniper Networks Enterprise-Specific SNMP Version 1 Traps" on page 634 .

The column headings describe the SNMPv1 traps format:

- Trap Name—The name of the trap.
- Enterprise ID—The identification number of the enterprise-specific trap.
- Generic Trap Number—The generic trap number field of the SNMP trap PDU. This field is **enterpriseSpecific(6)** for enterprise-specific traps, and other predefined values for standard traps.
- Specific Trap Number— The specific trap number field of the SNMP trap PDU. For standard traps, this field is zero; for enterprise-specific traps, this field is nonzero as defined in the enterprise-specific MIBs.

Table 37: SNMP Version 1 Trap Format

Trap Name	Enterprise ID	Generic Trap Number	Specific Trap Number
jnxFanFailure	1.3.6.1.4.1.2636.4.1	6	2
jnxFanOK	1.3.6.1.4.1.2636.4.2	6	2
jnxFruCheck	1.3.6.1.4.1.2636.4.1	6	12
jnxFruFailed	1.3.6.1.4.1.2636.4.1	6	9
jnxFruInsertion	1.3.6.1.4.1.2636.4.1	6	6
jnxFruOffline	1.3.6.1.4.1.2636.4.1	6	10
jnxFruOnline	1.3.6.1.4.1.2636.4.1	6	11
jnxFruPowerOff	1.3.6.1.4.1.2636.4.1	6	7
jnxFruPowerOn	1.3.6.1.4.1.2636.4.1	6	8
jnxFruRemoval	1.3.6.1.4.1.2636.4.1	6	5
jnxOverTemperature	1.3.6.1.4.1.2636.4.1	6	3
jnxPowerSupplyFailure	1.3.6.1.4.1.2636.4.1	6	1

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Table 37: SNMP Version 1 Trap Format *(continued)*

Trap Name	Enterprise ID	Generic Trap Number	Specific Trap Number
jnxPowerSupplyOK	1.3.6.1.4.1.2636.4.2	6	1
jnxRedundancySwitchover	1.3.6.1.4.1.2636.4.1	6	4
jnxTemperatureOK	1.3.6.1.4.1.2636.4.2	6	3

SNMPv2 Trap Format

The SNMPv2 trap format for the Chassis MIB traps is described in Table 38 on page 161.

The column headings describe the SNMPv2 traps format:

- Trap Name—The name of the trap.
- **snmpTrapOID**—The authoritative identification of the notification currently being sent. This variable occurs as the second varbind in every SNMPv2 trap PDU and InformRequest PDU.
- Description—The Junos OS enterprise-specific name of the trap.

Table 38: SNMP Version 2 Trap Format

Trap Name	snmpTrapOID	Description
jnxFanFailure	1.3.6.1.4.1.2636.4.1.2	The fan fuse has blown or the fan wiring has shorted out. This trap is generated every hour until the fan failure is fixed.
jnxFanOK	1.3.6.1.4.1.2636.4.2.2	The fan has recovered from a failure state.
jnxFruCheck	1.3.6.1.4.1.2636.4.1.12	The FRU has operational errors and has gone into a self-check diagnostic state. The revision number for the ATM2 PIC FPGA on an M120 or M320 router is less than 8B44(4).
jnxFruInsertion	1.3.6.1.4.1.2636.4.1.6	The FRU has been inserted into the chassis.
jnxFruFailed	1.3.6.1.4.1.2636.4.1.9	The FRU has failed in the chassis.
jnxFruOffline	1.3.6.1.4.1.2636.4.1.10	The FRU has gone offline.
jnxFruOnline	1.3.6.1.4.1.2636.4.1.11	The FRU has gone back online.
jnxFruPowerOff	1.3.6.1.4.1.2636.4.1.7	The FRU has been powered off in the chassis.

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Table 38: SNMP Version 2 Trap Format *(continued)*

Trap Name	snmpTrapOID	Description
jnxFruPowerOn	1.3.6.1.4.1.2636.4.1.8	The FRU has been powered on in the chassis.
jnxFruRemoval	1.3.6.1.4.1.2636.4.1.5	The FRU has been removed from the chassis.
jnxOverTemperature	1.3.6.1.4.1.2636.4.1.3	Several fans have failed or the room temperature has increased significantly. This trap is repeated every minute until the temperature is brought down to normal.
jnxPowerSupplyFailure	1.3.6.1.4.1.2636.4.1.1	The power supply, router circuit breaker, or power circuit failed, or there has been a power outage. This trap is generated every hour until the power supply is restored.
jnxPowerSupplyOK	1.3.6.1.4.1.2636.4.2.1	The power supply has recovered from a failure.
jnxRedundancySwitchover	1.3.6.1.4.1.2636.4.1.4	A redundant backup unit that can be brought online, manually or automatically, if the main unit malfunctions.
jnxTemperatureOK	1.3.6.1.4.1.2636.4.2.3	The component sensor has detected an overtemperature condition.

Related Documentation

- Chassis MIBs on page 70
- Chassis MIB Textual Conventions on page 71
- Configuring SNMP Trap Options and Groups on a Device Running Junos OS
- Managing Traps and Informs

Chassis Definitions for Router Model MIB

The enterprise-specific Chassis Definitions for Router Model MIB contain the OIDs that are used by the Chassis MIB to identify platform and chassis components. The Chassis MIB provides information that changes often. The Chassis Definitions for Router Model MIB provide information that changes less often.

The last number in each **sysObjectId**, shown in Table 39 on page 163, corresponds to the router model and therefore does not change.

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Table 39: Router Models and Their sysObjectIds

Model	SysObjectID	jnxProductName
J2300	1.3.6.1.4.1.2636.1.1.1.2.13	jnxProductNameJ2300
J4300	1.3.6.1.4.1.2636.1.1.1.2.14	jnxProductNameJ4300
J6300	1.3.6.1.4.1.2636.1.1.1.2.15	jnxProductNameJ6300
M5	1.3.6.1.4.1.2636.1.1.1.2.5	jnxProductNameM5
M7i	1.3.6.1.4.1.2636.1.1.1.2.10	jnxProductNameM7i
M10	1.3.6.1.4.1.2636.1.1.1.2.4	jnxProductNameM10
M10i	1.3.6.1.4.1.2636.1.1.1.2.11	jnxProductNameM10i
M20	1.3.6.1.4.1.2636.1.1.1.2.2	jnxProductNameM20
M40	1.3.6.1.4.1.2636.1.1.1.2.1	jnxProductNameM40
M40e	1.3.6.1.4.1.2636.1.1.1.2.8	jnxProductNameM40e
M120	1.3.6.1.4.1.2636.1.1.1.2.18	jnxProductNameM120
M160	1.3.6.1.4.1.2636.1.1.1.2.3	jnxProductNameM160
M320	1.3.6.1.4.1.2636.1.1.1.2.9	jnxProductNameM320
MX960	1.3.6.1.4.1.2636.1.1.1.2.21	jnxProductNameMX960
MX480	1.3.6.1.4.1.2636.1.1.1.1.25	jnxProductNameMX480
MX240	1.3.6.1.4.1.2636.1.1.1.1.29	jnxProductNameMX240
MX80	1.3.6.1.4.1.2636.1.1.1.1.57	jnxProductNameMX80
MX40	1.3.6.1.4.1.2636.1.1.1.1.88	jnxProductNameMX40
MX10	1.3.6.1.4.1.2636.1.1.1.1.89	jnxProductNameMX10
MX5	1.3.6.1.4.1.2636.1.1.1.1.90	jnxProductNameMX5
EX2200	1.3.6.1.4.1.2636.1.1.1.1.43	jnxProductNameEX2200
EX3200	1.3.6.1.4.1.2636.1.1.1.1.30	jnxProductNameEX3200
EX4200	1.3.6.1.4.1.2636.1.1.1.1.31	jnxProductNameEX4200
EX4500	1.3.6.1.4.1.2636.1.1.1.1.44	jnxProductNameEX4500

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Table 39: Router Models and Their sysObjectIds *(continued)*

Model	SysObjectID	jnxProductName
EX8208	1.3.6.1.4.1.2636.1.1.1.1.32	jnxProductNameEX8208
EX8216	1.3.6.1.4.1.2636.1.1.1.1.33	jnxProductNameEX8216
SRX100	1.3.6.1.4.1.2636.1.1.1.2.41	jnxProductNameSRX100
SRX210	1.3.6.1.4.1.2636.1.1.1.2.36	jnxProductNameSRX210
SRX220	1.3.6.1.4.1.2636.1.1.1.2.58	jnxProductNameSRX220
SRX240	1.3.6.1.4.1.2636.1.1.1.2.39	jnxProductNameSRX240
SRX650	1.3.6.1.4.1.2636.1.1.1.2.40	jnxProductNameSRX650
SRX1400	1.3.6.1.4.1.2636.1.1.1.2.49	jnxProductNameSRX1400
SRX3400	1.3.6.1.4.1.2636.1.1.1.2.35	jnxProductNameSRX3400
SRX3600	1.3.6.1.4.1.2636.1.1.1.2.34	jnxProductNameSRX3600
SRX5600	1.3.6.1.4.1.2636.1.1.1.2.28	jnxProductNameSRX5600
SRX5800	1.3.6.1.4.1.2636.1.1.1.2.26	jnxProductNameSRX5800
TX	1.3.6.1.4.1.2636.1.1.1.2.17	jnxProductNameTX
T320	1.3.6.1.4.1.2636.1.1.1.2.7	jnxProductNameT320
T640	1.3.6.1.4.1.2636.1.1.1.2.6	jnxProductNameT640
T1600	1.3.6.1.4.1.2636.1.1.1.2.27	jnxProductNameT1600
TX Plus	1.3.6.1.4.1.2636.1.1.1.2.37	jnxProductNameTXP

For a downloadable version of the Chassis Definitions for Router Model MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-chas-defines.txt .

Related Documentation

- Chassis MIBs on page 70
- Chassis MIB Textual Conventions on page 71
- Chassis Traps on page 158
- MIB Objects for the M120 Router on page 165

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MIB Objects for the M120 Router

New Chassis MIB objects for the M120 router include:

```
  jnxProductLineM120      OBJECT IDENTIFIER ::= { jnxProductLine      18 }
  jnxProductNameM120      OBJECT IDENTIFIER ::= { jnxProductName      18 }
  jnxProductModelM120     OBJECT IDENTIFIER ::= { jnxProductModel     18 }
  jnxProductVariationM120 OBJECT IDENTIFIER ::= { jnxProductVariation 18 }
  jnxChassisM120          OBJECT IDENTIFIER ::= { jnxChassis          18 }
  jnxSlotM120             OBJECT IDENTIFIER ::= { jnxSlot             18 }
    jnxM120SlotFPC        OBJECT IDENTIFIER ::= { jnxSlotM120 1  }
    jnxM120SlotFEB        OBJECT IDENTIFIER ::= { jnxSlotM120 2  }
    jnxM120SlotHM         OBJECT IDENTIFIER ::= { jnxSlotM120 3  }
    jnxM120SlotPower      OBJECT IDENTIFIER ::= { jnxSlotM120 4  }
    jnxM120SlotFan        OBJECT IDENTIFIER ::= { jnxSlotM120 5  }
    jnxM120SlotCB         OBJECT IDENTIFIER ::= { jnxSlotM120 6  }
    jnxM120SlotFPB        OBJECT IDENTIFIER ::= { jnxSlotM120 7  }

  jnxMediaCardSpaceM120      OBJECT IDENTIFIER ::= { jnxMediaCardSpace    18 }
    jnxM120MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceM120 1 }

  jnxMidplaneM120         OBJECT IDENTIFIER ::= { jnxBackplane        18 }
  jnxModuleM120           OBJECT IDENTIFIER ::= { jnxModule           18 }
    jnxM120FEB            OBJECT IDENTIFIER ::= { jnxModuleM120   1 }
```



NOTE: The M120 router does not support the enterprise-specific Dynamic Flow Capture MIB.

Sample command output from the **show chassis hardware** command for the M120 router is as follows:

```
user@host> show chassis hardware
Hardware inventory:
Item             Version  Part number  Serial number     Description
Chassis                               JN000019AC        M120
Midplane         REV 01   710-011382   RB3003            M120 Midplane
FPM Board        REV 01   710-011407   CK9165            M120 FPM Board
FPM Display      REV 01   710-011405   CE0032            M120 FPM Display
FPM CIP          REV 01   710-011410   CE0058            M120 FPM CIP
PEM 1            Rev 01   740-011935   RG10165           DC Power Entry Module
Routing Engine 0 REV 00   740-014082   1000604605        RE-A-2000
Routing Engine 1 REV 00   740-014082   1000604601        RE-A-2000
CB 0             REV 03   710-011403   CM8335            M120 Control Board
CB 1             REV 03   710-011403   CM8340            M120 Control Board
FPC 0            REV 01   710-012879   CH1622            M120 CFPC OC192
  PIC 0                   BUILTIN      BUILTIN           1x OC-192 SONET XFP
    Xcvr 0                NON-JNPR     T05J32698         XFP-OC192-SR
FPC 1            REV 01   710-012882   CE0062            M120 CFPC 10GE
  PIC 0                   BUILTIN      BUILTIN           1x 10GE(LAN/WAN) XFP
    Xcvr 0                NON-JNPR     T05A02227         XFP-10G-ER
FPC 2            REV 01   710-011388   CJ9092            M120 FPC Type 1
  PIC 0          REV 16   750-007444   HS1526            1x CHOC3 IQ SONET, SMIR

  PIC 1          REV 12   750-005637   HT0533             4x CHDS3 IQ
  PIC 2         REV 15   750-005634   HN1903            1x CHOC12 IQ SONET, SMIR

  PIC 3          REV 15   750-007631   NB5006             10x CHE1 IQ
```

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```
  Board B        REV 01   710-011390   CJ9109           M120 FPC Mezz Board
FPC 3            REV 03   710-011393   CJ9231           M120 FPC Type 2
  PIC 0          REV 05   750-010472   JE3146           1x OC-48 ATM-II IQ
    Xcvr 0       REV 01   740-009028   P5F05WU          SFP-SR
  PIC 1          REV 13   750-001901   HB4231           4x OC-12 SONET, SMIR
  PIC 2          REV 15   750-008155   HX5442           2x G/E IQ, 1000 BASE
    Xcvr 0       REV      740-007326   P11E5RR          SFP-SX
    Xcvr 1       REV 01   740-009029   4C81050          UNKNOWN
  PIC 3          REV 16   750-008155   HZ8871           2x G/E IQ, 1000 BASE
    Xcvr 0       REV 01   740-011613   P8E2KGF          SFP-SX
    Xcvr 1       REV 01   740-011782   P6M1E5X          SFP-SX
  Board B        REV 02   710-011395   CN3750           M120 FPC Mezz
FPC 4            REV 01   710-011388   CJ9089           M120 FPC Type 1
  PIC 0          REV 03   750-002911   AJ2279           4x F/E, 100 BASE-TX
  PIC 1         REV 15   750-005634   HN0435          1x CHOC12 IQ SONET, SMIR

  PIC 2          REV 02   750-003064   HD4548           4x T1, RJ48
  PIC 3          REV 04   750-011209   JC8254           Adaptive Services-II
  Board B        REV 01   710-011390   CJ9111           M120 FPC Mezz Board
FPC 5            REV 01   710-011388   CJ9360           M120 FPC Type 1
  PIC 0          REV 08   750-007631   HK0212           10x CHE1 IQ
  PIC 1          REV 05   750-003034   BD8705           4x OC-3 SONET, SMIR
  PIC 2          REV 11   750-007643   NA5967           1x G/E IQ, 1000 BASE
    Xcvr 0       REV 01   740-007326   P4R0PNZ          SFP-SX
  PIC 3          REV 16   750-007444   HS1501           1x CHSTM1 IQ SDH, SMIR
  Board B        REV 01   710-011390   CJ9099           M120 FPC Mezz Board
FEB 0            REV 04   710-011663   CJ9364           M120 FEB
FEB 1            REV 04   710-011663   CJ9385           M120 FEB
FEB 2            REV 02   710-015795   CP6830           M120 FEB
FEB 3            REV 01   710-011663   CM2585           M120 FEB
FEB 4            REV 04   710-011663   CJ9416           M120 FEB
FEB 5            REV 01   710-011663   CM2600           M120 FEB
```

Related Documentation

- Chassis MIBs on page 70
- Chassis MIB Textual Conventions on page 71
- Chassis Traps on page 158
- Chassis Definitions for Router Model MIB on page 162

MIB Objects for the MX960 3D Universal Edge Router

The Chassis MIB objects for the MX960 3D Universal Edge Router include:

```
  jnxProductLineX960      OBJECT IDENTIFIER ::= { jnxProductLine      21 }
  jnxProductNameX960      OBJECT IDENTIFIER ::= { jnxProductName      21 }
  jnxProductModelX960     OBJECT IDENTIFIER ::= { jnxProductModel     21 }
  jnxProductVariationX960 OBJECT IDENTIFIER ::= { jnxProductVariation 21 }
  jnxChassisX960          OBJECT IDENTIFIER ::= { jnxChassis          21 }
  jnxSlotX960             OBJECT IDENTIFIER ::= { jnxSlot             21 }
    jnxX960SlotFPC        OBJECT IDENTIFIER ::= { jnxSlotX960 1  }
    jnxX960SlotHM         OBJECT IDENTIFIER ::= { jnxSlotX960 2  }
    jnxX960SlotPower      OBJECT IDENTIFIER ::= { jnxSlotX960 3  }
    jnxX960SlotFan        OBJECT IDENTIFIER ::= { jnxSlotX960 4  }
    jnxX960SlotCB         OBJECT IDENTIFIER ::= { jnxSlotX960 5  }
    jnxX960SlotFPB        OBJECT IDENTIFIER ::= { jnxSlotX960 6  }
  jnxMediaCardSpaceX960      OBJECT IDENTIFIER ::= { jnxMediaCardSpace 21 }
```


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```
  jnxX960MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceX960 1 }
jnxMidplaneX960         OBJECT IDENTIFIER ::= { jnxBackplane        21 }
```

Related Documentation

- Chassis MIBs on page 70
- Chassis MIB Textual Conventions on page 71
- MIB Objects for the MX480 3D Universal Edge Router on page 167
- MIB Objects for the MX240 3D Universal Edge Router on page 167

MIB Objects for the MX480 3D Universal Edge Router

The Chassis MIB objects for the MX480 3D Universal Edge Router include:

```
jnxProductLineMX480      OBJECT IDENTIFIER ::= { jnxProductLine      25 }
  jnxProductNameMX480      OBJECT IDENTIFIER ::= { jnxProductName      25 }
  jnxProductModelMX480     OBJECT IDENTIFIER ::= { jnxProductModel     25 }
  jnxProductVariationMX480 OBJECT IDENTIFIER ::= { jnxProductVariation 25 }
  jnxChassisMX480          OBJECT IDENTIFIER ::= { jnxChassis          25 }

  jnxSlotMX480             OBJECT IDENTIFIER ::= { jnxSlot             25 }
    jnxMX480SlotFPC        OBJECT IDENTIFIER ::= { jnxSlotMX480 1  }
    jnxMX480SlotHM         OBJECT IDENTIFIER ::= { jnxSlotMX480 2  }
    jnxMX480SlotPower      OBJECT IDENTIFIER ::= { jnxSlotMX480 3  }
    jnxMX480SlotFan        OBJECT IDENTIFIER ::= { jnxSlotMX480 4  }
    jnxMX480SlotCB         OBJECT IDENTIFIER ::= { jnxSlotMX480 5  }
    jnxMX480SlotFPB        OBJECT IDENTIFIER ::= { jnxSlotMX480 6  }

  jnxMediaCardSpaceMX480      OBJECT IDENTIFIER ::= { jnxMediaCardSpace    25 }
  jnxMX480MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceMX480 1 }


  jnxMidplaneMX480         OBJECT IDENTIFIER ::= { jnxBackplane        25 }
```

Related Documentation

- Chassis MIBs on page 70
- Chassis MIB Textual Conventions on page 71
- MIB Objects for the MX960 3D Universal Edge Router on page 166
- MIB Objects for the MX240 3D Universal Edge Router on page 167

MIB Objects for the MX240 3D Universal Edge Router

The Chassis MIB objects for the MX240 3D Universal Edge Router include:

```
jnxProductLineMX240      OBJECT IDENTIFIER ::= { jnxProductLine      29 }
  jnxProductNameMX240      OBJECT IDENTIFIER ::= { jnxProductName      29 }
  jnxProductModelMX240     OBJECT IDENTIFIER ::= { jnxProductModel     29 }
  jnxProductVariationMX240 OBJECT IDENTIFIER ::= { jnxProductVariation 29 }
  jnxChassisMX240          OBJECT IDENTIFIER ::= { jnxChassis          29 }

  jnxSlotMX240             OBJECT IDENTIFIER ::= { jnxSlot             29 }
  jnxMX240SlotFPC          OBJECT IDENTIFIER ::= { jnxSlotMX240 1  }
  jnxMX240SlotHM           OBJECT IDENTIFIER ::= { jnxSlotMX240 2  }
```

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```
jnxMX240SlotPower        OBJECT IDENTIFIER ::= { jnxSlotMX240 3  }
jnxMX240SlotFan          OBJECT IDENTIFIER ::= { jnxSlotMX240 4  }
jnxMX240SlotCB           OBJECT IDENTIFIER ::= { jnxSlotMX240 5  }
jnxMX240SlotFPB          OBJECT IDENTIFIER ::= { jnxSlotMX240 6  }

jnxMediaCardSpaceMX240   OBJECT IDENTIFIER ::= { jnxMediaCardSpace    29 }
jnxMX240MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceMX240 1 }

jnxMidplaneMX240         OBJECT IDENTIFIER ::= { jnxBackplane         29 }
```

Related Documentation

- Chassis MIBs on page 70
- Chassis MIB Textual Conventions on page 71
- MIB Objects for the MX960 3D Universal Edge Router on page 166
- MIB Objects for the MX480 3D Universal Edge Router on page 167

MIB Objects for the MX80 3D Universal Edge Router

The Chassis MIB objects for the MX80 3D Universal Edge Router include:

```
jnxProductLineMX80      OBJECT IDENTIFIER ::= { jnxProductLine      57 }
jnxProductNameMX80      OBJECT IDENTIFIER ::= { jnxProductName      57 }
jnxProductModelMX80     OBJECT IDENTIFIER ::= { jnxProductModel     57 }
jnxProductVariationMX80 OBJECT IDENTIFIER ::= { jnxProductVariation 57 }
  jnxProductMX80        OBJECT IDENTIFIER ::= { jnxProductVariationMX80 1 }
  jnxProductMX80-48T    OBJECT IDENTIFIER ::= { jnxProductVariationMX80 2 }
  jnxProductMX80-T      OBJECT IDENTIFIER ::= { jnxProductVariationMX80 3 }
jnxChassisMX80          OBJECT IDENTIFIER ::= { jnxChassis          57 }

jnxSlotMX80             OBJECT IDENTIFIER ::= { jnxSlot     57 }
  jnxMX80SlotFPC        OBJECT IDENTIFIER ::= { jnxSlotMX80 1  }
  jnxMX80SlotCFEB       OBJECT IDENTIFIER ::= { jnxSlotMX80 2  }
  jnxMX80SlotRE         OBJECT IDENTIFIER ::= { jnxSlotMX80 3  }
  jnxMX80SlotPower      OBJECT IDENTIFIER ::= { jnxSlotMX80 4  }
  jnxMX80SlotFan        OBJECT IDENTIFIER ::= { jnxSlotMX80 5  }

jnxMediaCardSpaceMX80      OBJECT IDENTIFIER ::= { jnxMediaCardSpace    57 }
  jnxMX80MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceMX80 1 }
  jnxMX80MediaCardSpaceMIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceMX80 2 }

jnxMidplaneMX80         OBJECT IDENTIFIER ::= { jnxBackplane        57 }

jnxModuleMX80           OBJECT IDENTIFIER ::= { jnxModule     57 }
  jnxMX80FPC            OBJECT IDENTIFIER ::= { jnxModuleMX80 1  }
  jnxMX80CFEB           OBJECT IDENTIFIER ::= { jnxModuleMX80 2  }
  jnxMX80RE             OBJECT IDENTIFIER ::= { jnxModuleMX80 3  }
  jnxMX80Power          OBJECT IDENTIFIER ::= { jnxModuleMX80 4  }
  jnxMX80PowerAC        OBJECT IDENTIFIER ::= { jnxModuleMX80 5  }
  jnxMX80Fan            OBJECT IDENTIFIER ::= { jnxModuleMX80 6  }
```

Related Documentation

- Chassis MIBs on page 70
- Chassis MIB Textual Conventions on page 71
- MIB Objects for the MX5 3D Universal Edge Router on page 170

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- MIB Objects for the MX10 3D Universal Edge Router on page 169
- MIB Objects for the MX40 3D Universal Edge Router on page 169

MIB Objects for the MX40 3D Universal Edge Router

The Chassis MIB objects for the MX40 3D Universal Edge Router include:

```
jnxProductLineMX40      OBJECT IDENTIFIER ::= { jnxProductLine      88 }
jnxProductNameMX40      OBJECT IDENTIFIER ::= { jnxProductName      88 }
jnxProductModelMX40     OBJECT IDENTIFIER ::= { jnxProductModel     88 }
jnxProductVariationMX40 OBJECT IDENTIFIER ::= { jnxProductVariation 88 }
  jnxProductMX40        OBJECT IDENTIFIER ::= { jnxProductVariationMX40 1 }
jnxChassisMX40          OBJECT IDENTIFIER ::= { jnxChassis          88 }

jnxSlotMX40             OBJECT IDENTIFIER ::= { jnxSlot     88 }
  jnxMX40SlotFPC        OBJECT IDENTIFIER ::= { jnxSlotMX40 1  }
  jnxMX40SlotCFEB       OBJECT IDENTIFIER ::= { jnxSlotMX40 2  }
  jnxMX40SlotRE         OBJECT IDENTIFIER ::= { jnxSlotMX40 3  }
  jnxMX40SlotPower      OBJECT IDENTIFIER ::= { jnxSlotMX40 4  }
  jnxMX40SlotFan        OBJECT IDENTIFIER ::= { jnxSlotMX40 5  }

jnxMediaCardSpaceMX40      OBJECT IDENTIFIER ::= { jnxMediaCardSpace    88 }
  jnxMX40MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceMX40 1 }
  jnxMX40MediaCardSpaceMIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceMX40 2 }

jnxMidplaneMX40         OBJECT IDENTIFIER ::= { jnxBackplane        88 }

jnxModuleMX40           OBJECT IDENTIFIER ::= { jnxModule     88 }
  jnxMX40FPC            OBJECT IDENTIFIER ::= { jnxModuleMX40 1  }
  jnxMX40CFEB           OBJECT IDENTIFIER ::= { jnxModuleMX40 2  }
  jnxMX40RE             OBJECT IDENTIFIER ::= { jnxModuleMX40 3  }
  jnxMX40Power          OBJECT IDENTIFIER ::= { jnxModuleMX40 4  }
  jnxMX40PowerAC        OBJECT IDENTIFIER ::= { jnxModuleMX40 5  }
  jnxMX40Fan            OBJECT IDENTIFIER ::= { jnxModuleMX40 6  }
```

Related Documentation

- Chassis MIBs on page 70
- Chassis MIB Textual Conventions on page 71
- MIB Objects for the MX5 3D Universal Edge Router on page 170
- MIB Objects for the MX10 3D Universal Edge Router on page 169
- MIB Objects for the MX80 3D Universal Edge Router on page 168

MIB Objects for the MX10 3D Universal Edge Router

The Chassis MIB objects for the MX10 3D Universal Edge Router include:

```
jnxProductLineMX10      OBJECT IDENTIFIER ::= { jnxProductLine      89 }
jnxProductNameMX10      OBJECT IDENTIFIER ::= { jnxProductName      89 }
jnxProductModelMX10     OBJECT IDENTIFIER ::= { jnxProductModel     89 }
jnxProductVariationMX10 OBJECT IDENTIFIER ::= { jnxProductVariation 89 }
  jnxProductMX10        OBJECT IDENTIFIER ::= { jnxProductVariationMX10 1 }
```

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```
jnxChassisMX10          OBJECT IDENTIFIER ::= { jnxChassis          89 }

jnxSlotMX10             OBJECT IDENTIFIER ::= { jnxSlot     89 }
  jnxMX10SlotFPC        OBJECT IDENTIFIER ::= { jnxSlotMX10 1  }
  jnxMX10SlotCFEB       OBJECT IDENTIFIER ::= { jnxSlotMX10 2  }
  jnxMX10SlotRE         OBJECT IDENTIFIER ::= { jnxSlotMX10 3  }
  jnxMX10SlotPower      OBJECT IDENTIFIER ::= { jnxSlotMX10 4  }
  jnxMX10SlotFan        OBJECT IDENTIFIER ::= { jnxSlotMX10 5  }

jnxMediaCardSpaceMX10      OBJECT IDENTIFIER ::= { jnxMediaCardSpace    89 }
  jnxMX10MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceMX10 1 }
  jnxMX10MediaCardSpaceMIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceMX10 2 }

jnxMidplaneMX10         OBJECT IDENTIFIER ::= { jnxBackplane        89 }

jnxModuleMX10           OBJECT IDENTIFIER ::= { jnxModule     89 }
  jnxMX10FPC            OBJECT IDENTIFIER ::= { jnxModuleMX10 1  }
  jnxMX10CFEB           OBJECT IDENTIFIER ::= { jnxModuleMX10 2  }
  jnxMX10RE             OBJECT IDENTIFIER ::= { jnxModuleMX10 3  }
  jnxMX10Power          OBJECT IDENTIFIER ::= { jnxModuleMX10 4  }
  jnxMX10PowerAC        OBJECT IDENTIFIER ::= { jnxModuleMX10 5  }
  jnxMX10Fan            OBJECT IDENTIFIER ::= { jnxModuleMX10 6  }
```

Related Documentation

- Chassis MIBs on page 70
- Chassis MIB Textual Conventions on page 71
- MIB Objects for the MX5 3D Universal Edge Router on page 170
- MIB Objects for the MX40 3D Universal Edge Router on page 169
- MIB Objects for the MX80 3D Universal Edge Router on page 168

MIB Objects for the MX5 3D Universal Edge Router

The Chassis MIB objects for the MX5 3D Universal Edge Router include:

```
jnxProductLineMX5       OBJECT IDENTIFIER ::= { jnxProductLine      90 }
jnxProductNameMX5       OBJECT IDENTIFIER ::= { jnxProductName      90 }
jnxProductModelMX5      OBJECT IDENTIFIER ::= { jnxProductModel     90 }
jnxProductVariationMX5 OBJECT IDENTIFIER ::= { jnxProductVariation 90 }
  jnxProductMX5         OBJECT IDENTIFIER ::= { jnxProductVariationMX5 1 }
jnxChassisMX5           OBJECT IDENTIFIER ::= { jnxChassis          90 }

jnxSlotMX5              OBJECT IDENTIFIER ::= { jnxSlot     90 }
  jnxMX5SlotFPC         OBJECT IDENTIFIER ::= { jnxSlotMX5 1  }
  jnxMX5SlotCFEB        OBJECT IDENTIFIER ::= { jnxSlotMX5 2  }
  jnxMX5SlotRE          OBJECT IDENTIFIER ::= { jnxSlotMX5 3  }
  jnxMX5SlotPower       OBJECT IDENTIFIER ::= { jnxSlotMX5 4  }
  jnxMX5SlotFan         OBJECT IDENTIFIER ::= { jnxSlotMX5 5  }

jnxMediaCardSpaceMX5       OBJECT IDENTIFIER ::= { jnxMediaCardSpace    90 }
  jnxMX5MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceMX5 1 }
  jnxMX5MediaCardSpaceMIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceMX5 2 }

jnxMidplaneMX5          OBJECT IDENTIFIER ::= { jnxBackplane        90 }

jnxModuleMX5            OBJECT IDENTIFIER ::= { jnxModule     90 }
```


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```
   jnxMX5FPC            OBJECT IDENTIFIER ::= { jnxModuleMX5 1  }
   jnxMX5CFEB           OBJECT IDENTIFIER ::= { jnxModuleMX5 2  }
   jnxMX5RE             OBJECT IDENTIFIER ::= { jnxModuleMX5 3  }
   jnxMX5Power          OBJECT IDENTIFIER ::= { jnxModuleMX5 4  }
   jnxMX5PowerAC        OBJECT IDENTIFIER ::= { jnxModuleMX5 5  }
   jnxMX5Fan            OBJECT IDENTIFIER ::= { jnxModuleMX5 6  }
```

Related Documentation

- Chassis MIBs on page 70
- Chassis MIB Textual Conventions on page 71
- MIB Objects for the MX10 3D Universal Edge Router on page 169
- MIB Objects for the MX40 3D Universal Edge Router on page 169
- MIB Objects for the MX80 3D Universal Edge Router on page 168

MIB Objects for the EX Series Ethernet Switches

This topic contains chassis definition objects for the following EX Series switches:

- EX2200 on page 171
- EX3200 on page 172
- EX4200 on page 172
- EX8208 on page 173
- EX8216 on page 173
- EX4500 on page 174

EX2200

```
jnxProductLineEX2200      OBJECT IDENTIFIER ::= { jnxProductLine      43 }
  jnxProductNameEX2200      OBJECT IDENTIFIER ::= { jnxProductName      43 }
  jnxProductModelEX2200     OBJECT IDENTIFIER ::= { jnxProductModel     43 }
  jnxProductVariationEX2200 OBJECT IDENTIFIER ::= { jnxProductVariation 43 }
    jnxProductEX2200port24T    OBJECT IDENTIFIER ::= { jnxProductVariationEX2200
 1 }
    jnxProductEX2200port24P    OBJECT IDENTIFIER ::= { jnxProductVariationEX2200
 2 }
    jnxProductEX2200port48T    OBJECT IDENTIFIER ::= { jnxProductVariationEX2200
 3 }
    jnxProductEX2200port48P    OBJECT IDENTIFIER ::= { jnxProductVariationEX2200
 4 }

  jnxChassisEX2200          OBJECT IDENTIFIER ::= { jnxChassis          43 }

  jnxSlotEX2200             OBJECT IDENTIFIER ::= { jnxSlot             43 }
    jnxEX2200SlotFPC        OBJECT IDENTIFIER ::= { jnxSlotEX2200 1 }
      jnxEX2200SlotPower    OBJECT IDENTIFIER ::= { jnxEX2200SlotFPC 1 }
      jnxEX2200SlotFan      OBJECT IDENTIFIER ::= { jnxEX2200SlotFPC 2 }
      jnxEX2200SlotRE       OBJECT IDENTIFIER ::= { jnxEX2200SlotFPC 3 }

  jnxMediaCardSpaceEX2200      OBJECT IDENTIFIER ::= { jnxMediaCardSpace    43 }

    jnxEX2200MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceEX2200 1
}
```

```
  jnxModuleEX2200            OBJECT IDENTIFIER ::= { jnxModule    43 }
    jnxEX2200FPC             OBJECT IDENTIFIER ::= { jnxModuleEX2200 1 }
      jnxEX2200Power         OBJECT IDENTIFIER ::= { jnxEX2200FPC 1 }
      jnxEX2200Fan           OBJECT IDENTIFIER ::= { jnxEX2200FPC 2 }
      jnxEX2200RE            OBJECT IDENTIFIER ::= { jnxEX2200FPC 3 }
```

EX3200

```
jnxProductLineEX3200      OBJECT IDENTIFIER ::= { jnxProductLine      30 }
jnxProductNameEX3200    OBJECT IDENTIFIER ::= { jnxProductName      30 }
jnxProductModelEX3200   OBJECT IDENTIFIER ::= { jnxProductModel     30 }
jnxProductVariationEX3200 OBJECT IDENTIFIER ::= { jnxProductVariation 30 }
  jnxProductEX3200port24T  OBJECT IDENTIFIER ::= { jnxProductVariationEX3200 1 }

  jnxProductEX3200port24P  OBJECT IDENTIFIER ::= { jnxProductVariationEX3200 2 }

  jnxProductEX3200port48T  OBJECT IDENTIFIER ::= { jnxProductVariationEX3200 3 }

  jnxProductEX3200port48P  OBJECT IDENTIFIER ::= { jnxProductVariationEX3200 4 }


  jnxChassisEX3200          OBJECT IDENTIFIER ::= { jnxChassis          30 }

  jnxSlotEX3200             OBJECT IDENTIFIER ::= { jnxSlot             30 }
    jnxEX3200SlotFPC        OBJECT IDENTIFIER ::= { jnxSlotEX3200 1 }
      jnxEX3200SlotPower    OBJECT IDENTIFIER ::= { jnxEX3200SlotFPC 1 }
      jnxEX3200SlotFan      OBJECT IDENTIFIER ::= { jnxEX3200SlotFPC 2 }
      jnxEX3200SlotRE       OBJECT IDENTIFIER ::= { jnxEX3200SlotFPC 3 }

  jnxMediaCardSpaceEX3200      OBJECT IDENTIFIER ::= { jnxMediaCardSpace    30 }

    jnxEX3200MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceEX3200 1
 }

  jnxModuleEX3200            OBJECT IDENTIFIER ::= { jnxModule    30 }
    jnxEX3200FPC             OBJECT IDENTIFIER ::= { jnxModuleEX3200 1 }
      jnxEX3200Power         OBJECT IDENTIFIER ::= { jnxEX3200FPC 1 }
      jnxEX3200Fan           OBJECT IDENTIFIER ::= { jnxEX3200FPC 2 }
      jnxEX3200RE            OBJECT IDENTIFIER ::= { jnxEX3200FPC 3 }
```

EX4200

```
jnxProductLineEX4200      OBJECT IDENTIFIER ::= { jnxProductLine      31 }
jnxProductNameEX4200      OBJECT IDENTIFIER ::= { jnxProductName      31 }
jnxProductModelEX4200     OBJECT IDENTIFIER ::= { jnxProductModel     31 }
jnxProductVariationEX4200 OBJECT IDENTIFIER ::= { jnxProductVariation 31 }
  jnxProductEX4200port24T  OBJECT IDENTIFIER ::= { jnxProductVariationEX4200 1 }

  jnxProductEX4200port24P  OBJECT IDENTIFIER ::= { jnxProductVariationEX4200 2 }

  jnxProductEX4200port48T  OBJECT IDENTIFIER ::= { jnxProductVariationEX4200 3 }

  jnxProductEX4200port48P  OBJECT IDENTIFIER ::= { jnxProductVariationEX4200 4 }

  jnxProductEX4200port24F  OBJECT IDENTIFIER ::= { jnxProductVariationEX4200 5 }


  jnxChassisEX4200          OBJECT IDENTIFIER ::= { jnxChassis          31 }
    jnxEX4200RE0            OBJECT IDENTIFIER ::= { jnxChassisEX4200 1  }
```


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```
    jnxEX4200RE1            OBJECT IDENTIFIER ::= { jnxChassisEX4200 2  }
  jnxSlotEX4200             OBJECT IDENTIFIER ::= { jnxSlot             31 }
    jnxEX4200SlotFPC        OBJECT IDENTIFIER ::= { jnxSlotEX4200 1 }
      jnxEX4200SlotPower    OBJECT IDENTIFIER ::= { jnxEX4200SlotFPC 1 }
      jnxEX4200SlotFan      OBJECT IDENTIFIER ::= { jnxEX4200SlotFPC 2 }

  jnxMediaCardSpaceEX4200      OBJECT IDENTIFIER ::= { jnxMediaCardSpace    31 }

    jnxEX4200MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceEX4200 1
}

  jnxModuleEX4200           OBJECT IDENTIFIER ::= { jnxModule    31 }
    jnxEX4200FPC            OBJECT IDENTIFIER ::= { jnxModuleEX4200 1 }
      jnxEX4200Power        OBJECT IDENTIFIER ::= { jnxEX4200FPC 1 }
      jnxEX4200Fan          OBJECT IDENTIFIER ::= { jnxEX4200FPC 2 }
```

EX8208

```
jnxProductLineEX8208       OBJECT IDENTIFIER ::= { jnxProductLine      32 }
  jnxProductNameEX8208       OBJECT IDENTIFIER ::= { jnxProductName      32 }
  jnxProductModelEX8208      OBJECT IDENTIFIER ::= { jnxProductModel     32 }
  jnxProductVariationEX8208  OBJECT IDENTIFIER ::= { jnxProductVariation 32 }
  jnxChassisEX8208           OBJECT IDENTIFIER ::= { jnxChassis          32 }

  jnxSlotEX8208              OBJECT IDENTIFIER ::= { jnxSlot             32 }
    jnxEX8208SlotFPC         OBJECT IDENTIFIER ::= { jnxSlotEX8208  1 }
                            -- Flexible Port Concentrator
      jnxEX8208Slot48S       OBJECT IDENTIFIER ::= { jnxEX8208SlotFPC  1 }
      jnxEX8208Slot48T       OBJECT IDENTIFIER ::= { jnxEX8208SlotFPC  2 }
      jnxEX8208Slot8XS       OBJECT IDENTIFIER ::= { jnxEX8208SlotFPC  3 }
    jnxEX8208HM              OBJECT IDENTIFIER ::= { jnxSlotEX8208  3 }
                            -- Host Module (also called Routing Engine)
    jnxEX8208SlotPower       OBJECT IDENTIFIER ::= { jnxSlotEX8208  4 }
    jnxEX8208SlotFan         OBJECT IDENTIFIER ::= { jnxSlotEX8208  5 }
      jnxEX8208SlotFT        OBJECT IDENTIFIER ::= { jnxEX8208SlotFan  1 }
    jnxEX8208SlotCBD         OBJECT IDENTIFIER ::= { jnxSlotEX8208  6 }
                            -- Control Board

  jnxMediaCardSpaceEX8208    OBJECT IDENTIFIER ::= { jnxMediaCardSpace   32 }
    jnxEX8208MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceEX8208 1
}

  jnxBackplaneEX8208         OBJECT IDENTIFIER ::= { jnxBackplane        32 }
```

EX8216

```
jnxProductLineEX8216       OBJECT IDENTIFIER ::= { jnxProductLine      33 }
  jnxProductNameEX8216       OBJECT IDENTIFIER ::= { jnxProductName      33 }
  jnxProductModelEX8216      OBJECT IDENTIFIER ::= { jnxProductModel     33 }
  jnxProductVariationEX8216  OBJECT IDENTIFIER ::= { jnxProductVariation 33 }
  jnxChassisEX8216           OBJECT IDENTIFIER ::= { jnxChassis          33 }

  jnxSlotEX8216              OBJECT IDENTIFIER ::= { jnxSlot             33 }
    jnxEX8216SlotFPC         OBJECT IDENTIFIER ::= { jnxSlotEX8216   1 }
                            -- Flexible Port Concentrator
      jnxEX8216Slot48S       OBJECT IDENTIFIER ::= { jnxEX8216SlotFPC   1 }
      jnxEX8216Slot48T       OBJECT IDENTIFIER ::= { jnxEX8216SlotFPC   2 }
      jnxEX8216Slot8XS       OBJECT IDENTIFIER ::= { jnxEX8216SlotFPC   3 }
    jnxEX8216SIB             OBJECT IDENTIFIER ::= { jnxSlotEX8216   2 }
```

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```
                          -- Swtich Interface Board
   jnxEX8216HM             OBJECT IDENTIFIER ::= { jnxSlotEX8216   3 }
                          -- Host Module (also called Routing Engine)
   jnxEX8216SlotPower      OBJECT IDENTIFIER ::= { jnxSlotEX8216   4 }
   jnxEX8216SlotFan        OBJECT IDENTIFIER ::= { jnxSlotEX8216   5 }
     jnxEX8216SlotFT       OBJECT IDENTIFIER ::= { jnxEX8216SlotFan   1 }
     jnxEX8216SlotRFT      OBJECT IDENTIFIER ::= { jnxEX8216SlotFan   2 }
   jnxEX8216SlotCBD        OBJECT IDENTIFIER ::= { jnxSlotEX8216   6 }
                          -- Control Board

 jnxMediaCardSpaceEX8216    OBJECT IDENTIFIER ::= { jnxMediaCardSpace   33 }
   jnxEX8216MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceEX8216 1
}

 jnxMidplaneEX8216          OBJECT IDENTIFIER ::= { jnxBackplane         33 }
```

EX4500

```
jnxProductLineEX4500      OBJECT IDENTIFIER ::= { jnxProductLine      44 }
 jnxProductNameEX4500      OBJECT IDENTIFIER ::= { jnxProductName      44 }
 jnxProductModelEX4500     OBJECT IDENTIFIER ::= { jnxProductModel     44 }
 jnxProductVariationEX4500 OBJECT IDENTIFIER ::= { jnxProductVariation 44 }
   jnxProductEX4500port40F    OBJECT IDENTIFIER ::= { jnxProductVariationEX4500
1 }
   jnxProductEX4500port20F    OBJECT IDENTIFIER ::= { jnxProductVariationEX4500
2 }

 jnxChassisEX4500          OBJECT IDENTIFIER ::= { jnxChassis          44 }
   jnxEX4500RE0            OBJECT IDENTIFIER ::= { jnxChassisEX4500 1  }
   jnxEX4500RE1            OBJECT IDENTIFIER ::= { jnxChassisEX4500 2  }
 jnxSlotEX4500             OBJECT IDENTIFIER ::= { jnxSlot             44 }
   jnxEX4500SlotFPC        OBJECT IDENTIFIER ::= { jnxSlotEX4500 1 }
     jnxEX4500SlotPower    OBJECT IDENTIFIER ::= { jnxEX4500SlotFPC 1 }
     jnxEX4500SlotFan      OBJECT IDENTIFIER ::= { jnxEX4500SlotFPC 2 }
     jnxEX4500SlotRE       OBJECT IDENTIFIER ::= { jnxEX4500SlotFPC 3 }

 jnxMediaCardSpaceEX4500      OBJECT IDENTIFIER ::= { jnxMediaCardSpace    44 }

   jnxEX4500MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceEX4500 1
}

 jnxModuleEX4500            OBJECT IDENTIFIER ::= { jnxModule    44 }
   jnxEX4500FPC             OBJECT IDENTIFIER ::= { jnxModuleEX4500 1 }
     jnxEX4500Power         OBJECT IDENTIFIER ::= { jnxEX4500FPC 1 }
     jnxEX4500Fan           OBJECT IDENTIFIER ::= { jnxEX4500FPC 2 }
     jnxEX4500RE            OBJECT IDENTIFIER ::= { jnxEX4500FPC 3 }
```

Related Documentation

- Chassis MIBs on page 70
- Chassis MIB Textual Conventions on page 71
- Chassis Traps on page 158

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MIB Objects for the QFX Series

This topic contains the SNMP Juniper Networks enterprise-specific Chassis MIB definition objects for the QFX Series:

- QFX3500 Switch (Standalone Mode) on page 175
- QFX3500 Node Device on page 175
- QFX3008 Interconnect Device on page 175
- Virtual Chassis (Control Plane Switch) on page 176

QFX3500 Switch (Standalone Mode)

```
jnxProductLineQFXSwitch       OBJECT IDENTIFIER ::= { jnxProductLine      82 }
  jnxProductNameQFXSwitch       OBJECT IDENTIFIER ::= { jnxProductName      82 }
  jnxProductModelQFXSwitch      OBJECT IDENTIFIER ::= { jnxProductModel     82 }
  jnxProductVariationQFXSwitch  OBJECT IDENTIFIER ::= { jnxProductVariation 82 }
    jnxProductQFX3500s          OBJECT IDENTIFIER ::= { jnxProductVariationQFXSwitch 1 }

  jnxChassisQFXSwitch           OBJECT IDENTIFIER ::= { jnxChassis          82 }

  jnxSlotQFXSwitch              OBJECT IDENTIFIER ::= { jnxSlot             82 }
    jnxQFXSwitchSlotFPC         OBJECT IDENTIFIER ::= { jnxSlotQFXSwitch   1 }
    jnxQFXSwitchSlotHM          OBJECT IDENTIFIER ::= { jnxSlotQFXSwitch   2 }
    jnxQFXSwitchSlotPower       OBJECT IDENTIFIER ::= { jnxSlotQFXSwitch   3 }
    jnxQFXSwitchSlotFan         OBJECT IDENTIFIER ::= { jnxSlotQFXSwitch   4 }
    jnxQFXSwitchSlotFPB         OBJECT IDENTIFIER ::= { jnxSlotQFXSwitch   5 }

  jnxMediaCardSpaceQFXSwitch    OBJECT IDENTIFIER ::= { jnxMediaCardSpace   82 }
    jnxQFXSwitchMediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceQFXSwitch 1 }
```

QFX3500 Node Device

```
jnxProductLineQFXNode       OBJECT IDENTIFIER ::= { jnxProductLine      61 }
  jnxProductNameQFXNode       OBJECT IDENTIFIER ::= { jnxProductName      61 }
  jnxProductModelQFXNode      OBJECT IDENTIFIER ::= { jnxProductModel     61 }
  jnxProductVariationQFXNode  OBJECT IDENTIFIER ::= { jnxProductVariation 61 }
    jnxProductQFX3500         OBJECT IDENTIFIER ::= { jnxProductVariationQFXNode 1 }

  jnxChassisQFXNode           OBJECT IDENTIFIER ::= { jnxChassis          61 }

  jnxSlotQFXNode              OBJECT IDENTIFIER ::= { jnxSlot             61 }
    jnxQFXNodeSlotFPC         OBJECT IDENTIFIER ::= { jnxSlotQFXNode   1 }
    jnxQFXNodeSlotHM          OBJECT IDENTIFIER ::= { jnxSlotQFXNode   2 }
    jnxQFXNodeSlotPower       OBJECT IDENTIFIER ::= { jnxSlotQFXNode   3 }
    jnxQFXNodeSlotFan         OBJECT IDENTIFIER ::= { jnxSlotQFXNode   4 }
    jnxQFXNodeSlotFPB         OBJECT IDENTIFIER ::= { jnxSlotQFXNode   5 }

  jnxMediaCardSpaceQFXNode    OBJECT IDENTIFIER ::= { jnxMediaCardSpace   61 }
    jnxQFXNodeMediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceQFXNode 1 }
```

QFX3008 Interconnect Device

```
  jnxProductLineQFXInterconnect OBJECT IDENTIFIER ::= { jnxProductLine      60 }
  jnxProductNameQFXInterconnect OBJECT IDENTIFIER ::= { jnxProductName      60 }
  jnxProductModelQFXInterconnect OBJECT IDENTIFIER ::= { jnxProductModel     60 }
  jnxProductVariationQFXInterconnect  OBJECT IDENTIFIER ::= { jnxProductVariation 60 }
```

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```
    jnxProductQFX3008           OBJECT IDENTIFIER ::= { jnxProductVariationQFXInterconnect 1 }

  jnxChassisQFXInterconnect     OBJECT IDENTIFIER ::= { jnxChassis          60 }

  jnxSlotQFXInterconnect        OBJECT IDENTIFIER ::= { jnxSlot             60 }
    jnxQFXInterconnectSlotFPC   OBJECT IDENTIFIER ::= { jnxSlotQFXInterconnect   1 }
    jnxQFXInterconnectSlotHM    OBJECT IDENTIFIER ::= { jnxSlotQFXInterconnect   2 }
    jnxQFXInterconnectSlotPower OBJECT IDENTIFIER ::= { jnxSlotQFXInterconnect   3 }
    jnxQFXInterconnectSlotFan   OBJECT IDENTIFIER ::= { jnxSlotQFXInterconnect   4 }
    jnxQFXInterconnectSlotCBD   OBJECT IDENTIFIER ::= { jnxSlotQFXInterconnect   5 }
    jnxQFXInterconnectSlotFPB   OBJECT IDENTIFIER ::= { jnxSlotQFXInterconnect   6 }

  jnxMediaCardSpaceQFXInterconnect    OBJECT IDENTIFIER ::= { jnxMediaCardSpace   60 }
    jnxQFXInterconnectMediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceQFXInterconnect 1 }

  jnxMidplaneQFXInterconnect    OBJECT IDENTIFIER ::= { jnxBackplane        60 }
```

Virtual Chassis (Control Plane Switch)

```
jnxProductLineQFXJVRE      OBJECT IDENTIFIER ::= { jnxProductLine      62 }
   jnxProductNameQFXJVRE      OBJECT IDENTIFIER ::= { jnxProductName      62 }
   jnxProductModelQFXJVRE     OBJECT IDENTIFIER ::= { jnxProductModel     62 }
   jnxChassisQFXJVRE          OBJECT IDENTIFIER ::= { jnxChassis          62 }

   jnxSlotQFXJVRE             OBJECT IDENTIFIER ::= { jnxSlot             62 }
    jnxQFXJVRESlotFPC         OBJECT IDENTIFIER ::= { jnxSlotQFXJVRE   1 }
    jnxQFXJVRESlotHM          OBJECT IDENTIFIER ::= { jnxSlotQFXJVRE   2 }
    jnxQFXJVRESlotPower       OBJECT IDENTIFIER ::= { jnxSlotQFXJVRE   3 }
    jnxQFXJVRESlotFan         OBJECT IDENTIFIER ::= { jnxSlotQFXJVRE   4 }
    jnxQFXJVRESlotFPB         OBJECT IDENTIFIER ::= { jnxSlotQFXJVRE   5 }

  jnxMediaCardSpaceQFXJVRE    OBJECT IDENTIFIER ::= { jnxMediaCardSpace   62 }
    jnxQFXJVREMediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceQFXJVRE 1 }
```

MIB Objects for the SRX1400 Services Gateway

The Chassis MIB objects for the SRX1400 Services Gateway include:

```
  jnxProductLineSRX1400      OBJECT IDENTIFIER ::= { jnxProductLine  49 }
  jnxProductNameSRX1400      OBJECT IDENTIFIER ::= { jnxProductName  49 }
  jnxProductModelSRX1400     OBJECT IDENTIFIER ::= { jnxProductModel  49 }
  jnxProductVariationSRX1400 OBJECT IDENTIFIER ::= { jnxProductVariation 49 }
  jnxChassisSRX1400          OBJECT IDENTIFIER ::= { jnxChassis  49 }

  jnxSlotSRX1400             OBJECT IDENTIFIER ::= { jnxSlot 49 }

    jnxSRX1400SlotFPC        OBJECT IDENTIFIER ::= { jnxSlotSRX1400 1  }
    jnxSRX1400SlotHM         OBJECT IDENTIFIER ::= { jnxSlotSRX1400 2  }
    jnxSRX1400SlotPower      OBJECT IDENTIFIER ::= { jnxSlotSRX1400 3  }
    jnxSRX1400SlotFan        OBJECT IDENTIFIER ::= { jnxSlotSRX1400 4  }
    jnxSRX1400SlotCB         OBJECT IDENTIFIER ::= { jnxSlotSRX1400 5  }
    jnxSRX1400SlotFPB        OBJECT IDENTIFIER ::= { jnxSlotSRX1400 6  }

 jnxMediaCardSpaceSRX1400    OBJECT IDENTIFIER ::= { jnxMediaCardSpace  49 }
 jnxSRX1400MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceSRX1400 1}
```


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```
jnxMidplaneSRX1400          OBJECT IDENTIFIER ::= { jnxBackplane 49 }
```

Related Documentation

- Chassis MIBs on page 70
- Chassis MIB Textual Conventions on page 71
- MIB Objects for the SRX3400 Services Gateway on page 177
- MIB Objects for the SRX3600 Services Gateway on page 178
- MIB Objects for the SRX5600 Services Gateway on page 178
- MIB Objects for the SRX5800 Services Gateway on page 179
- MIB Objects for the SRX100 Services Gateway on page 180
- MIB Objects for the SRX210 Services Gateway on page 181
- MIB Objects for the SRX240 Services Gateway on page 182
- MIB Objects for the SRX650 Services Gateway on page 183

MIB Objects for the SRX3400 Services Gateway

The Chassis MIB objects for the SRX3400 Services Gateway include:

```
 jnxProductLineSRX3400      OBJECT IDENTIFIER ::= { jnxProductLine  35 }
 jnxProductNameSRX3400      OBJECT IDENTIFIER ::= { jnxProductName  35 }
 jnxProductModelSRX3400     OBJECT IDENTIFIER ::= { jnxProductModel  35 }
 jnxProductVariationSRX3400 OBJECT IDENTIFIER ::= { jnxProductVariation 35 }
 jnxChassisSRX3400          OBJECT IDENTIFIER ::= { jnxChassis  35 }

 jnxSlotSRX3400             OBJECT IDENTIFIER ::= { jnxSlot 35 }

   jnxSRX3400SlotFPC        OBJECT IDENTIFIER ::= { jnxSlotSRX3400 1  }
   jnxSRX3400SlotHM         OBJECT IDENTIFIER ::= { jnxSlotSRX3400 2  }
   jnxSRX3400SlotPower      OBJECT IDENTIFIER ::= { jnxSlotSRX3400 3  }
   jnxSRX3400SlotFan        OBJECT IDENTIFIER ::= { jnxSlotSRX3400 4  }
   jnxSRX3400SlotCB         OBJECT IDENTIFIER ::= { jnxSlotSRX3400 5  }
   jnxSRX3400SlotFPB        OBJECT IDENTIFIER ::= { jnxSlotSRX3400 6  }

jnxMediaCardSpaceSRX3400    OBJECT IDENTIFIER ::= { jnxMediaCardSpace  35 }
jnxSRX3400MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceSRX3400 1}


jnxMidplaneSRX3400          OBJECT IDENTIFIER ::= { jnxBackplane 35 }
```

Related Documentation

- Chassis MIBs on page 70
- Chassis MIB Textual Conventions on page 71
- MIB Objects for the SRX1400 Services Gateway on page 176
- MIB Objects for the SRX3600 Services Gateway on page 178
- MIB Objects for the SRX5600 Services Gateway on page 178

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- MIB Objects for the SRX5800 Services Gateway on page 179
- MIB Objects for the SRX100 Services Gateway on page 180
- MIB Objects for the SRX210 Services Gateway on page 181
- MIB Objects for the SRX240 Services Gateway on page 182
- MIB Objects for the SRX650 Services Gateway on page 183

MIB Objects for the SRX3600 Services Gateway

The Chassis MIB objects for the SRX3600 Services Gateway include:

```
  jnxProductLineSRX3600       OBJECT IDENTIFIER ::= { jnxProductLine  34 }
  jnxProductNameSRX3600       OBJECT IDENTIFIER ::= { jnxProductName  34 }
  jnxProductModelSRX3600      OBJECT IDENTIFIER ::= { jnxProductModel  34 }
  jnxProductVariationSRX3600  OBJECT IDENTIFIER ::= { jnxProductVariation 34 }
  jnxChassisSRX3600           OBJECT IDENTIFIER ::= { jnxChassis  34 }

   jnxSlotSRX3600             OBJECT IDENTIFIER ::= { jnxSlot 34 }
    jnxSRX3600SlotFPC         OBJECT IDENTIFIER ::= { jnxSlotSRX3600 1 }
    jnxSRX3600SlotHM          OBJECT IDENTIFIER ::= { jnxSlotSRX3600 2 }
    jnxSRX3600SlotPower       OBJECT IDENTIFIER ::= { jnxSlotSRX3600 3 }
    jnxSRX3600SlotFan         OBJECT IDENTIFIER ::= { jnxSlotSRX3600 4 }
    jnxSRX3600SlotCB          OBJECT IDENTIFIER ::= { jnxSlotSRX3600 5 }
    jnxSRX3600SlotFPB         OBJECT IDENTIFIER ::= { jnxSlotSRX3600 6 }

  jnxMediaCardSpaceSRX3600    OBJECT IDENTIFIER ::= { jnxMediaCardSpace  34 }
  jnxSRX3600MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceSRX3600
1}

  jnxMidplaneSRX3600          OBJECT IDENTIFIER ::= { jnxBackplane  34 }
```

Related Documentation

- Chassis MIBs on page 70
- Chassis MIB Textual Conventions on page 71
- MIB Objects for the SRX3400 Services Gateway on page 177
- MIB Objects for the SRX5600 Services Gateway on page 178
- MIB Objects for the SRX5800 Services Gateway on page 179
- MIB Objects for the SRX100 Services Gateway on page 180
- MIB Objects for the SRX210 Services Gateway on page 181
- MIB Objects for the SRX240 Services Gateway on page 182
- MIB Objects for the SRX650 Services Gateway on page 183

MIB Objects for the SRX5600 Services Gateway

The Chassis MIB objects for the SRX5600 Services Gateway include:

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```
jnxProductLineSRX5600      OBJECT IDENTIFIER ::= { jnxProductLine      28 }
  jnxProductNameSRX5600      OBJECT IDENTIFIER ::= { jnxProductName      28 }
  jnxProductModelSRX5600     OBJECT IDENTIFIER ::= { jnxProductModel     28 }
  jnxProductVariationSRX5600 OBJECT IDENTIFIER ::= { jnxProductVariation 28 }
  jnxChassisSRX5600          OBJECT IDENTIFIER ::= { jnxChassis          28 }

  jnxSlotSRX5600             OBJECT IDENTIFIER ::= { jnxSlot             28 }
    jnxSRX5600SlotFPC        OBJECT IDENTIFIER ::= { jnxSlotSRX5600 1  }
    jnxSRX5600SlotHM         OBJECT IDENTIFIER ::= { jnxSlotSRX5600 2  }
    jnxSRX5600SlotPower      OBJECT IDENTIFIER ::= { jnxSlotSRX5600 3  }
    jnxSRX5600SlotFan        OBJECT IDENTIFIER ::= { jnxSlotSRX5600 4  }
    jnxSRX5600SlotCB         OBJECT IDENTIFIER ::= { jnxSlotSRX5600 5  }
    jnxSRX5600SlotFPB        OBJECT IDENTIFIER ::= { jnxSlotSRX5600 6  }

  jnxMediaCardSpaceSRX5600      OBJECT IDENTIFIER ::= { jnxMediaCardSpace    28
}
  jnxSRX5600MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceSRX5600 1
 }


  jnxMidplaneSRX5600         OBJECT IDENTIFIER ::= { jnxBackplane        28 }
```

Related Documentation

- Chassis MIBs on page 70
- Chassis MIB Textual Conventions on page 71
- MIB Objects for the SRX1400 Services Gateway on page 176
- MIB Objects for the SRX3400 Services Gateway on page 177
- MIB Objects for the SRX3600 Services Gateway on page 178
- MIB Objects for the SRX5800 Services Gateway on page 179
- MIB Objects for the SRX100 Services Gateway on page 180
- MIB Objects for the SRX210 Services Gateway on page 181
- MIB Objects for the SRX240 Services Gateway on page 182
- MIB Objects for the SRX650 Services Gateway on page 183

MIB Objects for the SRX5800 Services Gateway

The Chassis MIB objects for the SRX5800 Services Gateway include:

```
jnxProductLineSRX5800      OBJECT IDENTIFIER ::= { jnxProductLine      26 }
  jnxProductNameSRX5800      OBJECT IDENTIFIER ::= { jnxProductName      26 }
  jnxProductModelSRX5800     OBJECT IDENTIFIER ::= { jnxProductModel     26 }
  jnxProductVariationSRX5800 OBJECT IDENTIFIER ::= { jnxProductVariation 26 }
  jnxChassisSRX5800          OBJECT IDENTIFIER ::= { jnxChassis          26 }

  jnxSlotSRX5800             OBJECT IDENTIFIER ::= { jnxSlot             26 }
    jnxSRX5800SlotFPC        OBJECT IDENTIFIER ::= { jnxSlotSRX5800 1  }
    jnxSRX5800SlotHM         OBJECT IDENTIFIER ::= { jnxSlotSRX5800 2  }
    jnxSRX5800SlotPower      OBJECT IDENTIFIER ::= { jnxSlotSRX5800 3  }
    jnxSRX5800SlotFan        OBJECT IDENTIFIER ::= { jnxSlotSRX5800 4  }
    jnxSRX5800SlotCB         OBJECT IDENTIFIER ::= { jnxSlotSRX5800 5  }
    jnxSRX5800SlotFPB        OBJECT IDENTIFIER ::= { jnxSlotSRX5800 6  }
```

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```
  jnxMediaCardSpaceSRX5800      OBJECT IDENTIFIER ::= { jnxMediaCardSpace    26
}
    jnxSRX5800MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceSRX5800
 1 }

  jnxMidplaneSRX5800         OBJECT IDENTIFIER ::= { jnxBackplane        26 }
```

Related Documentation

- Chassis MIBs on page 70
- Chassis MIB Textual Conventions on page 71
- MIB Objects for the SRX1400 Services Gateway on page 176
- MIB Objects for the SRX3400 Services Gateway on page 177
- MIB Objects for the SRX3600 Services Gateway on page 178
- MIB Objects for the SRX5600 Services Gateway on page 178
- MIB Objects for the SRX100 Services Gateway on page 180
- MIB Objects for the SRX210 Services Gateway on page 181
- MIB Objects for the SRX240 Services Gateway on page 182
- MIB Objects for the SRX650 Services Gateway on page 183

MIB Objects for the SRX100 Services Gateway

The Chassis MIB objects for the SRX100 Services Gateway include:

```
  jnxProductLineSRX100   OBJECT IDENTIFIER ::= { jnxProductLine 41 }
  jnxProductNameSRX100   OBJECT IDENTIFIER ::= { jnxProductName 41 }
  jnxChassisSRX100       OBJECT IDENTIFIER ::= { jnxChassis     41 }

  jnxSlotSRX100          OBJECT IDENTIFIER ::= { jnxSlot    41 }
  jnxSRX100SlotFPC       OBJECT IDENTIFIER ::= { jnxSlotSRX100  1 }
  jnxSRX100SlotRE        OBJECT IDENTIFIER ::= { jnxSlotSRX100  2 }
  jnxSRX100SlotPower     OBJECT IDENTIFIER ::= { jnxSlotSRX100  3 }
  jnxSRX100SlotFan       OBJECT IDENTIFIER ::= { jnxSlotSRX100  4 }

  jnxMediaCardSpaceSRX100    OBJECT IDENTIFIER ::= { jnxMediaCardSpace       41
}
  jnxSRX100MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceSRX100  1
}

  jnxMidplaneSRX100      OBJECT IDENTIFIER ::= { jnxBackplane 41 }

  jnxModuleSRX100        OBJECT IDENTIFIER ::= { jnxModule    41 }
  jnxSRX100FPC           OBJECT IDENTIFIER ::= { jnxModuleSRX100 1 }
  jnxSRX100RE            OBJECT IDENTIFIER ::= { jnxModuleSRX100 2 }
  jnxSRX100Power         OBJECT IDENTIFIER ::= { jnxModuleSRX100 3 }
  jnxSRX100Fan           OBJECT IDENTIFIER ::= { jnxModuleSRX100 4 }
```

Related Documentation

- Chassis MIBs on page 70
- Chassis MIB Textual Conventions on page 71
- MIB Objects for the SRX3400 Services Gateway on page 177

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- MIB Objects for the SRX3600 Services Gateway on page 178
- MIB Objects for the SRX5600 Services Gateway on page 178
- MIB Objects for the SRX5800 Services Gateway on page 179
- MIB Objects for the SRX210 Services Gateway on page 181
- MIB Objects for the SRX240 Services Gateway on page 182
- MIB Objects for the SRX650 Services Gateway on page 183

MIB Objects for the SRX210 Services Gateway

The Chassis MIB objects for the SRX210 Services Gateway include:

```
   jnxProductLineSRX210    OBJECT IDENTIFIER ::= { jnxProductLine      36 }
   jnxProductNameSRX210    OBJECT IDENTIFIER ::= { jnxProductName      36 }
   jnxChassisSRX210        OBJECT IDENTIFIER ::= { jnxChassis          36 }

   jnxSlotSRX210           OBJECT IDENTIFIER ::= { jnxSlot    36 }
   jnxSRX210SlotFPC        OBJECT IDENTIFIER ::= { jnxSlotSRX210  1 }
   jnxSRX210SlotRE         OBJECT IDENTIFIER ::= { jnxSlotSRX210  2 }
   jnxSRX210SlotPower      OBJECT IDENTIFIER ::= { jnxSlotSRX210  3 }
   jnxSRX210SlotFan        OBJECT IDENTIFIER ::= { jnxSlotSRX210  4 }

   jnxMediaCardSpaceSRX210    OBJECT IDENTIFIER ::= { jnxMediaCardSpace       36
 }
   jnxSRX210MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceSRX210  1
 }

   jnxMidplaneSRX210       OBJECT IDENTIFIER ::= { jnxBackplane 36 }

   jnxModuleSRX210         OBJECT IDENTIFIER ::= { jnxModule    36 }
   jnxSRX210FPC            OBJECT IDENTIFIER ::= { jnxModuleSRX210 1 }
   jnxSRX210RE             OBJECT IDENTIFIER ::= { jnxModuleSRX210 2 }
   jnxSRX210Power          OBJECT IDENTIFIER ::= { jnxModuleSRX210 3 }
   jnxSRX210Fan            OBJECT IDENTIFIER ::= { jnxModuleSRX210 4 }
```

Related Documentation

- Chassis MIBs on page 70
- Chassis MIB Textual Conventions on page 71
- MIB Objects for the SRX3400 Services Gateway on page 177
- MIB Objects for the SRX3600 Services Gateway on page 178
- MIB Objects for the SRX5600 Services Gateway on page 178
- MIB Objects for the SRX5800 Services Gateway on page 179
- MIB Objects for the SRX100 Services Gateway on page 180
- MIB Objects for the SRX220 Services Gateway on page 182
- MIB Objects for the SRX240 Services Gateway on page 182
- MIB Objects for the SRX650 Services Gateway on page 183

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MIB Objects for the SRX220 Services Gateway

The Chassis MIB objects for the SRX220 Services Gateway include:

```
  jnxProductLineSRX220       OBJECT IDENTIFIER ::= { jnxProductLine      58 }
  jnxProductNameSRX220       OBJECT IDENTIFIER ::= { jnxProductName      58 }
  jnxChassisSRX220           OBJECT IDENTIFIER ::= { jnxChassis          58 }

    jnxSlotSRX220              OBJECT IDENTIFIER ::= { jnxSlot    58 }
    jnxSRX220SlotFPC         OBJECT IDENTIFIER ::= { jnxSlotSRX220  1  }
    jnxSRX220SlotRE          OBJECT IDENTIFIER ::= { jnxSlotSRX220  2  }
    jnxSRX220SlotPower       OBJECT IDENTIFIER ::= { jnxSlotSRX220  3  }
    jnxSRX220SlotFan         OBJECT IDENTIFIER ::= { jnxSlotSRX220  4  }

  jnxMediaCardSpaceSRX220    OBJECT IDENTIFIER ::= { jnxMediaCardSpace       58
}
    jnxSRX220MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceSRX220
1 }

  jnxMidplaneSRX220         OBJECT IDENTIFIER ::= { jnxBackplane 58 }

  jnxModuleSRX220            OBJECT IDENTIFIER ::= { jnxModule    58 }
    jnxSRX220FPC             OBJECT IDENTIFIER ::= { jnxModuleSRX220 1  }
    jnxSRX220RE              OBJECT IDENTIFIER ::= { jnxModuleSRX220 2  }
    jnxSRX220SlotPower       OBJECT IDENTIFIER ::= { jnxSlotSRX220  3  }
        jnxSRX220SlotFan         OBJECT IDENTIFIER ::= { jnxSlotSRX220  4  }
```

Related Documentation

- Chassis MIB Textual Conventions on page 71
- MIB Objects for the SRX100 Services Gateway on page 180
- MIB Objects for the SRX240 Services Gateway on page 182
- MIB Objects for the SRX650 Services Gateway on page 183

MIB Objects for the SRX240 Services Gateway

The Chassis MIB objects for the SRX240 Services Gateway include:

```
jnxProductLineJSRX240       OBJECT IDENTIFIER ::= { jnxProductLine   37 }
 jnxProductNameJSRX240       OBJECT IDENTIFIER ::= { jnxProductName  37 }
 jnxChassisJSRX240           OBJECT IDENTIFIER ::= { jnxChassis  37 }
 jnxSlotJSRX240              OBJECT IDENTIFIER ::= { jnxSlot    37 }
 jnxJSRX240SlotFPC           OBJECT IDENTIFIER ::= { jnxSlotJSRX240  1 }
 jnxJSRX240SlotRE            OBJECT IDENTIFIER ::= { jnxSlotJSRX240  2 }
 jnxJSRX240SlotPower         OBJECT IDENTIFIER ::= { jnxSlotJSRX240  3 }
 jnxJSRX240SlotFan           OBJECT IDENTIFIER ::= { jnxSlotJSRX240  4 }
 jnxMediaCardSpaceJSRX240    OBJECT IDENTIFIER ::= { jnxMediaCardSpace   37 }
 jnxJSRX240MediaCardSpacePIC OBJECT IDENTIFIER ::= { jnxMediaCardSpaceJSRX240 1
}
 jnxMidplaneJSRX240         OBJECT IDENTIFIER ::= { jnxBackplane 37 }
 jnxModuleJSRX240            OBJECT IDENTIFIER ::= { jnxModule    37 }
 jnxJSRX240FPC             OBJECT IDENTIFIER ::= { jnxModuleJSRX240 1 }
 jnxJSRX240RE              OBJECT IDENTIFIER ::= { jnxModuleJSRX240 2 }
 jnxJSRX240Power           OBJECT IDENTIFIER ::= { jnxModuleJSRX240 3 }
 jnxJSRX240Fan             OBJECT IDENTIFIER ::= { jnxModuleJSRX240 4 }
```


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Related Documentation

- Chassis MIBs on page 70
- Chassis MIB Textual Conventions on page 71
- MIB Objects for the SRX3400 Services Gateway on page 177
- MIB Objects for the SRX3600 Services Gateway on page 178
- MIB Objects for the SRX5600 Services Gateway on page 178
- MIB Objects for the SRX5800 Services Gateway on page 179
- MIB Objects for the SRX100 Services Gateway on page 180
- MIB Objects for the SRX210 Services Gateway on page 181
- MIB Objects for the SRX220 Services Gateway on page 182
- MIB Objects for the SRX650 Services Gateway on page 183

MIB Objects for the SRX650 Services Gateway

The Chassis MIB objects for the SRX650 Services Gateway include:

```
jnxProductLineJSRX650         OBJECT IDENTIFIER ::= { jnxProductLine   38 }
 jnxProductNameJSRX650         OBJECT IDENTIFIER ::= { jnxProductName   38 }
 jnxChassisJSRX650             OBJECT IDENTIFIER ::= { jnxChassis     38 }
 jnxSlotJSRX650                OBJECT IDENTIFIER ::= { jnxSlot    38 }
 jnxJSRX650SlotFPC             OBJECT IDENTIFIER ::= { jnxSlotJSRX650  1 }
 jnxJSRX650SlotRE              OBJECT IDENTIFIER ::= { jnxSlotJSRX650  2 }
 jnxJSRX650SlotPower           OBJECT IDENTIFIER ::= { jnxSlotJSRX650  3 }
 jnxJSRX650SlotFan             OBJECT IDENTIFIER ::= { jnxSlotJSRX650  4 }
 jnxMediaCardSpaceJSRX650      OBJECT IDENTIFIER ::= { jnxMediaCardSpace   38}
 jnxJSRX650MediaCardSpacePIC   OBJECT IDENTIFIER ::= { jnxMediaCardSpaceJSRX650
1}
 jnxMidplaneJSRX650            OBJECT IDENTIFIER ::= { jnxBackplane 38 }
 jnxModuleJSRX650              OBJECT IDENTIFIER ::= { jnxModule    38 }
 jnxJSRX650FPC                 OBJECT IDENTIFIER ::= { jnxModuleJSRX650 1 }
 jnxJSRX650RE                  OBJECT IDENTIFIER ::= { jnxModuleJSRX650 2 }
 jnxJSRX650Power               OBJECT IDENTIFIER ::= { jnxModuleJSRX650 3 }
 jnxJSRX650Fan                 OBJECT IDENTIFIER ::= { jnxModuleJSRX650 4 }
```

Related Documentation

- Chassis MIBs on page 70
- Chassis MIB Textual Conventions on page 71
- MIB Objects for the SRX3400 Services Gateway on page 177
- MIB Objects for the SRX3600 Services Gateway on page 178
- MIB Objects for the SRX5600 Services Gateway on page 178
- MIB Objects for the SRX5800 Services Gateway on page 179
- MIB Objects for the SRX100 Services Gateway on page 180
- MIB Objects for the SRX220 Services Gateway on page 182
- MIB Objects for the SRX210 Services Gateway on page 181

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- MIB Objects for the SRX240 Services Gateway on page 182

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CHAPTER 5

Interpreting the Enterprise-Specific Destination Class Usage MIB

- Destination Class Usage MIB on page 185
- jnxDCUsTable on page 186
- jnxDcuStatsTable on page 186

Destination Class Usage MIB

The Juniper Networks enterprise-specific Destination Class Usage (DCU) MIB counts packets from customers by performing a lookup of the IP destination address. DCU makes it possible to track traffic originating from the customer edge and destined for specific prefixes on the provider core router.

The DCU MIB is a subbranch of the **jnxMibs** branch of the enterprise-specific MIB **{enterprise 2636}** and has an object identifier of **{jnxMIB 6}**. The DCU MIB has one branch, **jnxDCUs**, which contains two tables: **jnxDCUsTable** and **jnxDcuStatsTable**. For information about configuring source and destination class usage, see the *Junos OS Routing Policy Configuration Guide* and *Junos OS Network Interfaces Configuration Guide* . For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-dcu.txt .



NOTE: **Class-based filter match conditions are not supported on J Series Services Routers.**

For information about the enterprise-specific DCU MIB objects, see the following topics:

- jnxDCUsTable on page 186
- jnxDcuStatsTable on page 186

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7
- Juniper Networks Enterprise-Specific SNMP Traps on page 633
- Standard SNMP Traps Supported on Devices Running Junos OS on page 653

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jnxDCUsTable

The entries in the **jnxDCUsTable**, whose object identifier is **{jnxDCUTable 1}**, are represented by **jnxDCUsEntry** and are listed in Table 40 on page 186.

Table 40: jnxDCUsEntry

Object	Object Identifier	Description
jnxDCUSrcIfIndex	**jnxDUCsEntry 1**	The interface index of the ingress interface.
jnxDCUDstClassName	**jnxDCUsEntry 2**	The destination class name specified in a routing policy and applied to the forwarding table.
jnxDCUPackets	**jnxDCUsEntry 3**	The number of packets passing through the network.
jnxDCUBytes	**jnxDCUsEntry 4**	The number of bytes passing through the network.

Related Documentation

- Destination Class Usage MIB on page 185
- jnxDcuStatsTable on page 186

jnxDcuStatsTable

jnxDcuStatsTable contains statistics for traffic that satisfies the rules in each configured destination class. A separate set of statistics is kept for each destination class on each interface and address family on which this feature is enabled. This is essentially a replacement for **jnxDCUsTable**.

The entries in the **jnxDcuStatsTable**, whose object identifier is **{jnxDCUs 2}**, are represented by **jnxDCUsStatsEntry** and are listed in Table 41 on page 186.

Table 41: jnxDCUsStatsEntry

Ojbect	Object Identifier	Description
jnxDcuStatsSrcIfIndex	**jnxDcuStatsEntry** 1	The interface index of the ingress interface for traffic counted in each entry.
jnxDcuStatsAddrFamily	**jnxDcuStatsEntry 2**	The address family of the entry's traffic.
nxDcuStatsClassName	**jnxDcuStatsEntry 3**	The name of the destination class that applies to the entry's traffic.
jnxDcuStatsPackets	**jnxDcuStatsEntry 4**	The number of packets received on this interface and belonging to this address family that match this destination class.

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Table 41: jnxDCUsStatsEntry *(continued)*

Ojbect	Object Identifier	Description
jnxDcuStatsBytes	**jnxDcuStatsEntry 5**	The number of bytes received on this interface and belonging to this address family that match this destination class.
jnxDcuStatsClName	**jnxDcuStatsEntry 6**	The name of the destination class. This object is a duplicate of **jnxDcuStatsClassName** and is included to satisfy those network management applications that cannot extract the destination class name from the instance portion of the OID.

Related Documentation

- Destination Class Usage MIB on page 185
- jnxDCUsTable on page 186

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CHAPTER 6

Interpreting the Enterprise-Specific BGP4 V2 MIB

- BGP4 V2 MIB on page 189
- jnxBgpM2PrefixCountersTable on page 190
- Enterprise-Specific BGP Traps on page 190

BGP4 V2 MIB

The Juniper Networks enterprise-specific BGP version 4 (BGP4) V2 MIB, whose object identifier is **{jnxBgpM2Experiment 1}**, contains objects used to monitor BGP peer-received prefix counters. It is based upon similar objects in the MIB documented in Internet draft draft-ietf-idr-bgp4-mibv2-03.txt, *Definitions of Managed Objects for the Fourth Version of Border Gateway Protocol (BGP-4), Second Version*. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-bgpmib2.txt .



NOTE:

For the BGP4 V2 MIB, the Junos OS supports only the following objects:

- **jnxBgpM2PrefixInPrefixes**
- **jnxBgpM2PrefixInPrefixesAccepted**
- **jnxBgpM2PrefixInPrefixesRejected**

For information about the enterprise-specific BGP4 V2 MIB objects, see the following topics:

- jnxBgpM2PrefixCountersTable on page 190
- Juniper Networks Enterprise-Specific BGP Traps on page 648

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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jnxBgpM2PrefixCountersTable

jnxBgpM2PrefixCountersTable contains counters associated with a BGP peer.

jnxBgpM2PrefixCountersEntry contains information about the prefix counters of a BGP peer, and the objects listed in Table 42 on page 190.

Table 42: jnxBgpM2PrefixCountersEntry

Object	Object Identifier	Description
jnxBgpM2PrefixInPrefixes	jnxBgpM2PrefixCountersEntry 7	The total number of prefixes received from a peer.
jnxBgpM2PrefixInPrefixesAccepted	jnxBgpM2PrefixCountersEntry 8	The total number of prefixes received from a peer that are eligible to be active in the routing table.
jnxBgpM2PrefixInPrefixesRejected	jnxBgpM2PrefixCountersEntry 9	The total number of prefixes received from a peer that are not eligible to be active in the routing table.

Related Documentation

- BGP4 V2 MIB on page 189
- Juniper Networks Enterprise-Specific BGP Traps on page 648

Enterprise-Specific BGP Traps

For information about the enterprise-specific BGP traps, see "Juniper Networks Enterprise-Specific BGP Traps" on page 648.

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CHAPTER 7

Interpreting the Enterprise-Specific DHCP MIB

- DHCP MIB on page 191
- DHCP Local Server on page 191
- DHCP Relay on page 197

DHCP MIB

The Juniper Networks enterprise-specific DHCP MIB, whose object identifier is **{jnxMibs 61}**, stores both local server and relay objects. This MIB provides support for Junos OS DHCP for bindings and lease tables, and for statistics.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/ topics/reference/mibs/mib-jnx-jdhcp.txt .

The DHCP MIB contains two managed object groups:

- **jnxJdhcp** local server objects **{jnxJdhcpMib 1}**
- **jnxJdhcp** relay objects **{jnxJdhcpMib 2}**

For information about using the Traceroute MIB and enterprise-specific DHCP MIB, see the *Junos OS Network Management Configuration Guide*.

For information about DHCP MIB objects, see the following topics:

- DHCP Local Server on page 191
- DHCP Relay on page 197

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

DHCP Local Server

The **jnxJdhcpLocalServerObjects** contains objects that describe the current statistics and bindings of a DHCP local server.

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Object IDs for the DHCP local server are as follows:

- **jnxJdhcpLocalServerStatistics—{jnxJdhcpLocalServerObjects 1}**
- **jnxJdhcpLocalServerBindings—{jnxJdhcpLocalServerObjects 2}**
- **jnxJdhcpLocalServerTraps—{jnxJdhcpLocalServerObjects 3}**
- **jnxJdhcpLocalServerTrapVars—{jnxJdhcpLocalServerObjects 4}**
- **jnxJdhcpLocalServerIfcStats—{jnxJdhcpLocalServerObjects 5}**

The **jnxJdhcpLocalServerStatistics**, whose object ID is **{jnxJdhcpLocalServerObjects 1}**, contains objects that describe the current status and statistics of a DHCP local server.

The **jnxJdhcpLocalServerStatistics** objects are listed in Table 43 on page 192.

Table 43: jnxJdhcpLocalServerStatistics

Object	Object Identifier	Description
jnxJdhcpLocalServerTotalDropped	jnxJdhcpLocalServerStatistics 1	Total number of packets dropped.
jnxJdhcpLocalServerBadHardwareDropped	jnxJdhcpLocalServerStatistics 2	Total number of packets dropped due to bad hardware address.
jnxJdhcpLocalServerBadBootpOpcodeDropped	jnxJdhcpLocalServerStatistics 3	Number of packets dropped due to no BOOTP message.
jnxJdhcpLocalServerBadOptionsDropped	jnxJdhcpLocalServerStatistics 4	Number of packets dropped due to bad options in the packet.
jnxJdhcpLocalServerBadAddressDropped	jnxJdhcpLocalServerStatistics 5	Number of packets dropped due to invalid server address.
jnxJdhcpLocalServerNoAddressDropped	jnxJdhcpLocalServerStatistics 6	Number of packets dropped due to no available addresses.
jnxJdhcpLocalServerNoInterfaceDropped	jnxJdhcpLocalServerStatistics 7	Number of packets dropped due to no interface match.
jnxJdhcpLocalServerNoRoutingInstanceDropped	jnxJdhcpLocalServerStatistics 8	Number of packets dropped due to no routing instance match.
jnxJdhcpLocalServerNoLocalAddressDropped	jnxJdhcpLocalServerStatistics 9	Number of packets dropped due to no valid local address.
jnxJdhcpLocalServerShortPacketDropped	jnxJdhcpLocalServerStatistics 10	Number of packets dropped due to packet being too short.
jnxJdhcpLocalServerBadReadDropped	jnxJdhcpLocalServerStatistics 11	Number of packets dropped due to read error.
jnxJdhcpLocalServerBadSendDropped	jnxJdhcpLocalServerStatistics 12	Number of packets dropped due to send error.

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Table 43: jnxJdhcpLocalServerStatistics *(continued)*

Object	Object Identifier	Description
jnxJdhcpLocalServerAuthenticationDropped	jnxJdhcpLocalServerStatistics 13	Number of packets dropped due to authentication failure.
jnxJdhcpLocalServerDynamicProfileDropped	jnxJdhcpLocalServerStatistics 14	Number of packets dropped due to dynamic profile error.
jnxJdhcpLocalServerLicenseDropped	jnxJdhcpLocalServerStatistics 15	Number of packets dropped due to license error.
jnxJdhcpLocalServerBootRequestReceived	jnxJdhcpLocalServerStatistics 16	Number of Boot Request packets received.
jnxJdhcpLocalServerDhcpDeclineReceived	jnxJdhcpLocalServerStatistics 17	Number of DHCP Decline packets received.
jnxJdhcpLocalServerDhcpDiscoverReceived	jnxJdhcpLocalServerStatistics 18	Number of DHCP Discover packets received.
jnxJdhcpLocalServerDhcpInformReceived	jnxJdhcpLocalServerStatistics 19	Number of DHCP inform packets received.
jnxJdhcpLocalServerDhcpReleaseReceived	jnxJdhcpLocalServerStatistics 20	Number of DHCP release packets received.
jnxJdhcpLocalServerDhcpRequestReceived	jnxJdhcpLocalServerStatistics 21	Number of DHCP request packets received.
jnxJdhcpLocalServerDhcpBootReplySent	jnxJdhcpLocalServerStatistics 22	Number of DHCP Boot Reply packets sent.
jnxJdhcpLocalServerDhcpOfferSent	jnxJdhcpLocalServerStatistics 23	Number of DHCP Offer packets sent.
jnxJdhcpLocalServerDhcpAckSent	jnxJdhcpLocalServerStatistics 24	Number of DHCP Ack packets sent.
jnxJdhcpLocalServerDhcpNakSent	jnxJdhcpLocalServerStatistics 25	Number of DHCP Nak packets sent.
jnxJdhcpLocalServerForceRenewSent	jnxJdhcpLocalServerStatistics 26	Number of DHCP Force Renew packets sent.

The **jnxJdhcpLocalServerBindings**, whose object ID is **{jnxJdhcpLocalServerObjects 2}**, contains information about bindings managed by a DHCP local server.

The **jnxJdhcpLocalServerBindings** objects are listed in Table 44 on page 194.

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Table 44: jnxJdhcpLocalServerBindings

Object	Object Identifier	Description
jnxJdhcpLocalServerBindingsIpAddress	jnxJdhcpLocalServerBindingsEntry 1	IP address associated with this entry in the bindings table.
jnxJdhcpLocalServerBindingsMacAddress	jnxJdhcpLocalServerBindingsEntry 2	MAC address associated with this entry in the bindings table and corresponding to the IP address denoted by the table index.
jnxJdhcpLocalServerBindingsState	jnxJdhcpLocalServerBindingsEntry 3	State associated with this entry in the bindings table.
jnxJdhcpLocalServerBindingsLeaseEndTime	jnxJdhcpLocalServerBindingsEntry 4	Time the lease expires on this binding.
jnxJdhcpLocalServerBindingsLeaseExpireTime	jnxJdhcpLocalServerBindingsEntry 5	Ttime remaining until the lease expires for this binding.
jnxJdhcpLocalServerBindingsLeaseStartTime	jnxJdhcpLocalServerBindingsEntry 6	Ttime the lease was started for this binding.
jnxJdhcpLocalServerBindingsIncomingClientInterface	jnxJdhcpLocalServerBindingsEntry 7	Incoming interface for this binding.
jnxJdhcpLocalServerBindingsClientInterfaceVlanId	jnxJdhcpLocalServerBindingsEntry 8	VLAN ID for this binding.
jnxJdhcpLocalServerBindingsDemuxInterfaceName	jnxJdhcpLocalServerBindingsEntry 9	Demux interface for this binding.
jnxJdhcpLocalServerBindingsServerIpAddress	jnxJdhcpLocalServerBindingsEntry 10	IP address associated with the server for this entry in the bindings table.
jnxJdhcpLocalServerBindingsBootpRelayAddress	jnxJdhcpLocalServerBindingsEntry 11	BOOTP relay address associated with the server for this entry in the bindings table.
jnxJdhcpLocalServerBindingsPreviousBootpRelayAddress	jnxJdhcpLocalServerBindingsEntry 12	Previous BOOTP relay address associated with the server for this entry in the bindings table.
jnxJdhcpLocalServerBindingsClientPoolName	jnxJdhcpLocalServerBindingsEntry 13	Client pool name for this binding.
jnxJdhcpLocalServerBindingsClientProfileName	jnxJdhcpLocalServerBindingsEntry 14	Client profile name for this binding.

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The **jnxJdhcpLocalServerTraps**, whose object ID is **{jnxJdhcpLocalServerObjects 3}** contains objects that describe the DHCP local server notifications.

The **jnxJdhcpLocalServerTraps** objects are listed in "Juniper Networks Enterprise-Specific SNMP Traps" on page 633.

jnxJdhcpLocalServerTrapVars, whose object ID is **{jnxJdhcpLocalServerObjects 4}**, contains objects that describe the contents of a table of interface statistics maintained by the DHCP local server.

jnxJdhcpLocalServerTrapVars objects are listed in Table 45 on page 195.

Table 45: jnxJdhcpLocalServerTrapVars

Object	Object Identifier	Description
jnxJdhcpLocalServerLastDetected	jnxJdhcpLocalServerTrapVars 1	Time the duplicate client was last detected.
jnxJdhcpLocalServerRouterName	jnxJdhcpLocalServerTrapVars 2	VRF ID in Junos OS. Represented as the logical router (LR) name followed by the router instance (RI) name.
jnxJdhcpLocalServerMacAddress	jnxJdhcpLocalServerTrapVars 3	MAC address of the client that changed interfaces."
jnxJdhcpLocalServerInterfaceName	jnxJdhcpLocalServerTrapVars 4	Interface where the DHCP client was detected.
jnxJdhcpLocalServerInterfaceLimit	jnxJdhcpLocalServerTrapVars 5	Number of clients supported on this interface."

jnxJdhcpLocalServerIfcStats, whose object ID is **{jnxJdhcpLocalServerObjects 5}**, contains objects that describe the contents of a table of interface statistics maintained by the DHCP local server.

jnxJdhcpLocalServerIfcStats objects are listed in Table 46 on page 195.

Table 46: jnxJdhcpLocalServerIfcStats

Object	Object Identifier	Description
jnxJdhcpLocalServerIfcStatsIfIndex	jnxJdhcpLocalServerIfcStatsEntry 1	IfIndex value of the interface for which this entry contains information.
jnxJdhcpLocalServerIfcStatsTotalDropped	jnxJdhcpLocalServerIfcStatsEntry 2	Total number of packets dropped.
jnxJdhcpLocalServerIfcStatsBadHardwareDropped	jnxJdhcpLocalServerIfcStatsEntry 3	Number of packets dropped due to bad hardware address.
jnxJdhcpLocalServerIfcStatsBadBootpOpcodeDropped	jnxJdhcpLocalServerIfcStatsEntry 4	Number of packets dropped due to no BOOTP message.
jnxJdhcpLocalServerIfcStatsBadOptionsDropped	jnxJdhcpLocalServerIfcStatsEntry 5	Number of packets dropped due to bad options in the packet.

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Table 46: jnxJdhcpLocalServerIfcStats *(continued)*

Object	Object Identifier	Description
jnxJdhcpLocalServerIfcStatsBadAddressDropped	jnxJdhcpLocalServerIfcStatsEntry 6	Number of packets dropped due to invalid server address.
jnxJdhcpLocalServerIfcStatsNoAddressDropped	jnxJdhcpLocalServerIfcStatsEntry 7	Number of packets dropped due to no available addresses.
jnxJdhcpLocalServerIfcStatsNoInterfaceCfgDropped	jnxJdhcpLocalServerIfcStatsEntry 8	Number of packets dropped due to no interface match.
jnxJdhcpLocalServerIfcStatsNoLocalAddressDropped	jnxJdhcpLocalServerIfcStatsEntry 9	Number of packets dropped due to no valid local address.
jnxJdhcpLocalServerIfcStatsShortPacketDropped	jnxJdhcpLocalServerIfcStatsEntry 10	Number of packets dropped due to packet too short.
jnxJdhcpLocalServerIfcStatsBadSendDropped	jnxJdhcpLocalServerIfcStatsEntry 11	Number of packets dropped due to send error.
jnxJdhcpLocalServerIfcStatsAuthenticationDropped	jnxJdhcpLocalServerIfcStatsEntry 12	Number of packets dropped due to authentication failure.
jnxJdhcpLocalServerIfcStatsDynamicProfileDropped	jnxJdhcpLocalServerIfcStatsEntry 13	Number of packets dropped due dynamic profile error.
jnxJdhcpLocalServerIfcStatsLicenseDropped	jnxJdhcpLocalServerIfcStatsEntry 14	Number of packets dropped due to license error.
jnxJdhcpLocalServerIfcStatsBootRequestReceived	jnxJdhcpLocalServerIfcStatsEntry 15	Number of Boot Request packets received.
jnxJdhcpLocalServerIfcStatsDhcpDeclineReceived	jnxJdhcpLocalServerIfcStatsEntry 16	Number of DHCP Decline packets received.
jnxJdhcpLocalServerIfcStatsDhcpDiscoverReceived	jnxJdhcpLocalServerIfcStatsEntry 17	Number of DHCP Discover packets received.
jnxJdhcpLocalServerIfcStatsDhcpInformReceived	jnxJdhcpLocalServerIfcStatsEntry 18	Number of DHCP Inform packets received.
jnxJdhcpLocalServerIfcStatsDhcpReleaseReceived	jnxJdhcpLocalServerIfcStatsEntry 19	Number of DHCP Release packets received.
jnxJdhcpLocalServerIfcStatsDhcpRequestReceived	jnxJdhcpLocalServerIfcStatsEntry 20	Number of DHCP Request packets received.
jnxJdhcpLocalServerIfcStatsDhcpBootReplySent	jnxJdhcpLocalServerIfcStatsEntry 21	Number of DHCP Boot Reply packets sent.

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Table 46: jnxJdhcpLocalServerIfcStats *(continued)*

Object	Object Identifier	Description
jnxJdhcpLocalServerIfcStatsDhcpOfferSent	jnxJdhcpLocalServerIfcStatsEntry 22	Number of DHCP Offer packets sent.
jnxJdhcpLocalServerIfcStatsDhcpAckSent	jnxJdhcpLocalServerIfcStatsEntry 23	Number of DHCP Ack packets sent.
jnxJdhcpLocalServerIfcStatsDhcpNakSent	jnxJdhcpLocalServerIfcStatsEntry 24	Number of DHCP Nak packets sent.
jnxJdhcpLocalServerIfcStatsForceRenewSent	jnxJdhcpLocalServerIfcStatsEntry 25	Number of DHCP Force Renew packets sent.
jnxJdhcpLocalServerIfcStatsTotalLeaseCount	jnxJdhcpLocalServerIfcStatsEntry 26	Number of Bound DHCP clients.
jnxJdhcpLocalServerIfcStatsBadDhcpOpcodeDropped	jnxJdhcpLocalServerIfcStatsEntry 27	Number of packets dropped with bad DHCP opcode.
jnxJdhcpLocalServerIfcStatsNoOptionsDropped	jnxJdhcpLocalServerIfcStatsEntry 28	Number of packets dropped with no options.
jnxJdhcpLocalServerIfcStatsHopLimitDropped	jnxJdhcpLocalServerIfcStatsEntry 29	Number of packets dropped due to hop limit violation.
jnxJdhcpLocalServerIfcStatsTtlExpiredDropped	jnxJdhcpLocalServerIfcStatsEntry 30	Number of packets dropped due to TTL expiration.
jnxJdhcpLocalServerIfcStatsBadUdpCksumDropped	jnxJdhcpLocalServerIfcStatsEntry 31	Number of packets dropped due to bad UDP checksum.
jnxJdhcpLocalServerIfcStatsOption60Dropped	jnxJdhcpLocalServerIfcStatsEntry 32	Number of packets dropped due to bad option 60.

Related Documentation

- DHCP MIB on page 191
- DHCP Relay on page 197

DHCP Relay

The **jnxJdhcpRelayServerObjects** contains objects that describe the current statistics and bindings of a DHCP relay server.

Object IDs for DHCP relay server are as follows:

- **jnxJdhcpRelayStatistics—{jnxJdhcpRelayObjects 1}**
- **jnxJdhcpRelayBindings—{jnxJdhcpRelayObjects 2}**
- **jnxJdhcpRelayIfcStats—{jnxJdhcpRelayObjects 3}**

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The **jnxJdhcpRelayStatistics**, whose object ID is **{jnxJdhcpRelayObjects 1}**, contains objects that describe the current status and statistics of a DHCP relay.

The **jnxJdhcpRelayStatistics** objects are listed in Table 47 on page 198.

Table 47: jnxJdhcpRelayStatistics

Object	Object Identifier	Description
jnxJdhcpRelayTotalDropped	jnxJdhcpRelayStatistics 1	Total number of packets dropped.
jnxJdhcpRelayBadHardwareDropped	jnxJdhcpRelayStatistics 2	Total number of packets dropped due to bad hardware address.
jnxJdhcpRelayBadBootpOpcodeDropped	jnxJdhcpRelayStatistics 3	Number of packets dropped due to no BOOTP message.
jnxJdhcpRelayBadOptionsDropped	jnxJdhcpRelayStatistics 4	Number of packets dropped due to bad options in the packet.
jnxJdhcpRelayBadAddressDropped	jnxJdhcpRelayStatistics 5	Number of packets dropped due to invalid server address.
jnxJdhcpRelayNoAddressDropped	jnxJdhcpRelayStatistics 6	Number of packets dropped due to no available addresses.
jnxJdhcpRelayNoInterfaceDropped	jnxJdhcpRelayStatistics 7	Number of packets dropped due to no interface match.
jnxJdhcpRelayNoRoutingInstanceDropped	jnxJdhcpRelayStatistics 8	Number of packets dropped due to no routing instance match.
jnxJdhcpRelayNoLocalAddressDropped	jnxJdhcpRelayStatistics 9	Number of packets dropped due to no valid local address.
jnxJdhcpRelayShortPacketDropped	jnxJdhcpRelayStatistics 10	Number of packets dropped due to packet being too short.
jnxJdhcpRelayBadReadDropped	jnxJdhcpRelayStatistics 11	Number of packets dropped due to read error.
jnxJdhcpRelayBadSendDropped	jnxJdhcpRelayStatistics 12	Number of packets dropped due to send error.
jnxJdhcpRelayOption82Dropped	jnxJdhcpRelayStatistics 13	Number of packets dropped due to failure to add Option 82.
jnxJdhcpRelayOption60Dropped	jnxJdhcpRelayStatistics 14	Number of packets dropped due to failure to add Option 60.
jnxJdhcpRelayAuthenticationDropped	jnxJdhcpRelayStatistics 15	Number of packets dropped due to authentication failure.
jnxJdhcpRelayDynamicProfileDropped	jnxJdhcpRelayStatistics 16	Number of packets dropped due to dynamic profile error.

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Table 47: jnxJdhcpRelayStatistics *(continued)*

Object	Object Identifier	Description
jnxJdhcpRelayLicenseDropped	jnxJdhcpRelayStatistics 17	Number of packets dropped due to license error.
jnxJdhcpRelayBootRequestReceived	jnxJdhcpRelayStatistics 18	Number of Boot Request packets received.
jnxJdhcpRelayDhcpDeclineReceived	jnxJdhcpRelayStatistics 19	Number of DHCP Decline packets received.
jnxJdhcpRelayDhcpDiscoverReceived	jnxJdhcpRelayStatistics 20	Number of DHCP Discover packets received.
jnxJdhcpRelayDhcpInformReceived	jnxJdhcpRelayStatistics 21	Number of DHCP Inform packets received.
jnxJdhcpRelayDhcpReleaseReceived	jnxJdhcpRelayStatistics 22	Number of DHCP Release packets received.
jnxJdhcpRelayDhcpRequestReceived	jnxJdhcpRelayStatistics 23	Number of DHCP Request packets received.
jnxJdhcpRelayDhcpBootReplySent	jnxJdhcpRelayStatistics 24	Number of DHCP Boot Reply packets sent.
jnxJdhcpRelayDhcpOfferSent	jnxJdhcpRelayStatistics 25	Number of DHCP Offer packets sent.
jnxJdhcpRelayDhcpAckSent	jnxJdhcpRelayStatistics 26	Number of DHCP Ack packets sent.
jnxJdhcpRelayDhcpNakSent	jnxJdhcpRelayStatistics 27	Number of DHCP Nak packets sent.
jnxJdhcpRelayForceRenewSent	jnxJdhcpRelayStatistics 28	Number of DHCP Force Renew packets sent.

The **jnxJdhcpRelayBindings**, whose object ID is **{jnxJdhcpRelayObjects 2}**, contains objects that describe the current statistics and bindings of a DHCP relay server.

The **jnxJdhcpRelayBindings** objects are listed in Table 48 on page 199.

Table 48: jnxJdhcpRelayBindings

Object	Object Identifier	Description
jnxJdhcpRelayBindingsIpAddress	jnxJdhcpRelayBindingsEntry 1	IP address associated with this entry in the bindings table.
jnxJdhcpRelayBindingsLeaseState	jnxJdhcpRelayBindingsEntry 2	State associated with this entry in the bindings table.
jnxJdhcpRelayBindingsLeaseEndTime	jnxJdhcpRelayBindingsEntry 3	Time the lease expires on this binding.
jnxJdhcpRelayBindingsLeaseExpireTime	jnxJdhcpRelayBindingsEntry 4	Time remaining until the lease expires for this binding.
jnxJdhcpRelayBindingsLeaseStartTime	jnxJdhcpRelayBindingsEntry 5	Time the lease started for this binding.
jnxJdhcpRelayBindingsIncomingClientInterface	jnxJdhcpRelayBindingsEntry 6	Iincoming interface for this binding.

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Table 48: jnxJdhcpRelayBindings *(continued)*

Object	Object Identifier	Description
jnxJdhcpRelayBindingsClientInterfaceVlanId	jnxJdhcpRelayBindingsEntry 7	VLAN ID for this binding.
jnxJdhcpRelayBindingsDemuxInterfaceName	jnxJdhcpRelayBindingsEntry 8	Demux interface for this binding.
jnxJdhcpRelayBindingsServerIpAddress	jnxJdhcpRelayBindingsEntry 9	IP address associated with the server for this entry in the bindings table.
jnxJdhcpRelayBindingsServerInterface	jnxJdhcpRelayBindingsEntry 10	Demux server for this binding.
jnxJdhcpRelayBindingsBootpRelayAddress	jnxJdhcpRelayBindingsEntry 11	IP address associated with the BOOTP relay for this entry in the bindings table.
jnxJdhcpRelayBindingsPreviousBootpRelayAddress	jnxJdhcpRelayBindingsEntry 12	IP address associated with the previous BOOTP relay for this entry in the bindings table.
jnxJdhcpRelayBindingsClientProfileName	jnxJdhcpRelayBindingsEntry 13	Client profile name.

jnxJdhcpRelayIfcStats, whose object ID is **{jnxJdhcpRelayObjects 3}**, contains objects that describe the contents of a table of interface statistics maintained by the DHCP relay.

jnxJdhcpRelayIfcStats objects are listed in Table 49 on page 200.

Table 49: jnxJdhcpRelayIfcStats

Object	Object Identifier	Description
jnxJdhcpRelayIfcStatsIfIndex	jnxJdhcpRelayIfcStatsEntry 1	IfIndex value of the interface for which this entry contains information.
jnxJdhcpRelayIfcStatsTotalDropped	jnxJdhcpRelayIfcStatsEntry 2	Total number of packets dropped.
jnxJdhcpRelayIfcStatsBadHardwareDropped	jnxJdhcpRelayIfcStatsEntry 3	Total number of packets dropped due to bad hardware address.
jnxJdhcpRelayIfcStatsBadBootpOpcodeDropped	jnxJdhcpRelayIfcStatsEntry 4	Number of packets dropped due to no BOOTP message.
jnxJdhcpRelayIfcStatsBadOptionsDropped	jnxJdhcpRelayIfcStatsEntry 5	Number of packets dropped due to bad options in the packet.
jnxJdhcpRelayIfcStatsBadAddressDropped	jnxJdhcpRelayIfcStatsEntry 6	Number of packets dropped due to invalid server address.
jnxJdhcpRelayIfcStatsNoAddressDropped	jnxJdhcpRelayIfcStatsEntry 7	Number of packets dropped due to no available addresses.
jnxJdhcpRelayIfcStatsNoInterfaceCfgDropped	jnxJdhcpRelayIfcStatsEntry 8	Number of packets dropped due to no interface match.

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Table 49: jnxJdhcpRelayIfcStats *(continued)*

Object	Object Identifier	Description
jnxJdhcpRelayIfcStatsNoLocalAddressDropped	jnxJdhcpRelayIfcStatsEntry 9	The number of packets dropped due to no valid local address.
jnxJdhcpRelayIfcStatsShortPacketDropped	jnxJdhcpRelayIfcStatsEntry 10	Number of packets dropped due to packet too short.
jnxJdhcpRelayIfcStatsBadSendDropped	jnxJdhcpRelayIfcStatsEntry 11	Number of packets dropped due to send error.
jnxJdhcpRelayIfcStatsAuthenticationDropped	jnxJdhcpRelayIfcStatsEntry 12	Number of packets dropped due to authentication failure.
jnxJdhcpRelayIfcStatsDynamicProfileDropped	jnxJdhcpRelayIfcStatsEntry 13	Number of packets dropped due to dynamic profile error.
jnxJdhcpRelayIfcStatsLicenseDropped	jnxJdhcpRelayIfcStatsEntry 14	Number of packets dropped due to license error.
jnxJdhcpRelayIfcStatsBootRequestReceived	jnxJdhcpRelayIfcStatsEntry 15	Number of Boot Request packets received.
jnxJdhcpRelayIfcStatsDhcpDeclineReceived	jnxJdhcpRelayIfcStatsEntry 16	Number of DHCP Decline packets received.
jnxJdhcpRelayIfcStatsDhcpDiscoverReceived	jnxJdhcpRelayIfcStatsEntry 17	Number of DHCP Discover packets received.
jnxJdhcpRelayIfcStatsDhcpInformReceived	jnxJdhcpRelayIfcStatsEntry 18	Number of DHCP Inform packets received.
jnxJdhcpRelayIfcStatsDhcpReleaseReceived	jnxJdhcpRelayIfcStatsEntry 19	Number of DHCP Release packets received.
jnxJdhcpRelayIfcStatsDhcpRequestReceived	jnxJdhcpRelayIfcStatsEntry 20	Number of DHCP Request packets received.
jnxJdhcpRelayIfcStatsDhcpBootReplySent	jnxJdhcpRelayIfcStatsEntry 21	Number of DHCP Boot Reply packets sent.
jnxJdhcpRelayIfcStatsDhcpOfferSent	jnxJdhcpRelayIfcStatsEntry 22	Number of DHCP Offer packets sent.
jnxJdhcpRelayIfcStatsDhcpAckSent	jnxJdhcpRelayIfcStatsEntry 23	Number of DHCP Ack packets sent.
jnxJdhcpRelayIfcStatsDhcpNakSent	jnxJdhcpRelayIfcStatsEntry 24	Number of DHCP Nak packets sent.
jnxJdhcpRelayIfcStatsForceRenewSent	jnxJdhcpRelayIfcStatsEntry 25	Number of DHCP Force Renew packets sent.
jnxJdhcpRelayIfcStatsTotalLeaseCount	jnxJdhcpRelayIfcStatsEntry 26	Number of Bound DHCP clients.
jnxJdhcpRelayIfcStatsBadDhcpOpcodeDropped	jnxJdhcpRelayIfcStatsEntry 27	Number of packets dropped with bad DHCP opcode.

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Table 49: jnxJdhcpRelayIfcStats *(continued)*

Object	Object Identifier	Description
jnxJdhcpRelayIfcStatsNoOptionsDropped	**jnxJdhcpRelayIfcStatsEntry 28**	Number of packets dropped with no options.
jnxJdhcpRelayIfcStatsHopLimitDropped	**jnxJdhcpRelayIfcStatsEntry 29**	Number of packets dropped due to hop limit violation.
jnxJdhcpRelayIfcStatsTtlExpiredDropped	**jnxJdhcpRelayIfcStatsEntry 30**	Number of packets dropped due to TTL expiration.
jnxJdhcpRelayIfcStatsBadUdpCksumDropped	**jnxJdhcpRelayIfcStatsEntry 31**	Number of packets dropped due to bad UDP checksum.
jnxJdhcpRelayIfcStatsOption82Dropped	**jnxJdhcpRelayIfcStatsEntry 32**	Number of packets dropped due to failure to add option 82.

Related Documentation

- DHCP MIB on page 191
- DHCP Local Server on page 191

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CHAPTER 8

Interpreting the Enterprise-Specific DHCPv6 MIB

- DHCPv6 MIB on page 203
- DHCPv6 Local Server on page 203

DHCPv6 MIB

The Juniper Networks enterprise-specific DHCPv6 MIB, whose object identifier is **{jnxMibs 62}**, stores local server objects. This MIB provides support for Junos OS DHCPv6 for bindings and lease tables, and for statistics.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-jdhcpv6.txt .

The DHCPv6 MIB contains the following object group:

- **jnxJdhcpv6** local server objects **{jnxJdhcpv6Mib 1}**

For information about DHCPv6 MIB objects, see the following topic:

- DHCPv6 Local Server on page 203

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

DHCPv6 Local Server

The **jnxJdhcpv6LocalServerObjects** contains objects that describe the current statistics and bindings of a DHCPv6 local server.

Object IDs for the DHCPv6 local server are as follows:

- **jnxJdhcpv6LocalServerStatistics—{jnxJdhcpv6LocalServerObjects 1}**
- **jnxJdhcpv6LocalServerBindings—{jnxJdhcpv6LocalServerObjects 2}**
- **jnxJdhcpv6LocalServerTraps—{jnxJdhcpv6LocalServerObjects 3}**

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- jnxJdhcpv6LocalServerTrapVars—{jnxJdhcpv6LocalServerObjects 4}
- jnxJdhcpv6LocalServerIfcStats—{jnxJdhcpv6LocalServerObjects 5}

The **jnxJdhcpv6LocalServerStatistics**, whose object ID is **{jnxJdhcpv6LocalServerObjects 1}**, contains objects that describe the current status and statistics of a DHCPv6 local server.

The **jnxJdhcpv6LocalServerStatistics** objects are listed in Table 50 on page 204.

Table 50: jnxJdhcpv6LocalServerStatistics

Object	Object Identifier	Description
jnxJdhcpv6LocalServerTotalDropped	jnxJdhcpv6LocalServerStatistics 1	Number of DHCPv6 packets dropped.
jnxJdhcpv6LocalServerNoSafdDropped	jnxJdhcpv6LocalServerStatistics 2	Number of DHCPv6 packets dropped due to no safd match.
jnxJdhcpv6LocalServerBadSendDropped	jnxJdhcpv6LocalServerStatistics 3	Number of DHCPv6 packets dropped due to send error.
jnxJdhcpv6LocalServerShortPacketDropped	jnxJdhcpv6LocalServerStatistics 4	Number of DHCPv6 packets dropped due to packet being too short.
jnxJdhcpv6LocalServerBadMsgtypeDropped	jnxJdhcpv6LocalServerStatistics 5	Number of DHCPv6 packets dropped due to bad opcode in the packet.
jnxJdhcpv6LocalServerBadOptionsDropped	jnxJdhcpv6LocalServerStatistics 6	Number of DHCPv6 packets dropped due to bad options in the packet.
jnxJdhcpv6LocalServerBadSrcAddressDropped	jnxJdhcpv6LocalServerStatistics 7	Number of DHCPv6 packets dropped due to invalid addr family.
jnxJdhcpv6LocalServerRelayHopCountDropped	jnxJdhcpv6LocalServerStatistics 8	Number of DHCPv6 packets dropped due to max relays supported.
jnxJdhcpv6LocalServerNoClientIdDropped	jnxJdhcpv6LocalServerStatistics 9	Number of DHCPv6 packets dropped due to missing client ID.
jnxJdhcpv6LocalServerDeclineReceived	jnxJdhcpv6LocalServerStatistics 10	Number of DHCPv6 Decline packets received.
jnxJdhcpv6LocalServerSolicitReceived	jnxJdhcpv6LocalServerStatistics 11	Number of DHCPv6 Solicit packets received.
jnxJdhcpv6LocalServerInformationRequestReceived	jnxJdhcpv6LocalServerStatistics 12	The number of DHCPv6 Information Request packets received.
jnxJdhcpv6LocalServerReleaseReceived	jnxJdhcpv6LocalServerStatistics 13	Number of DHCPv6 Release packets received.

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Table 50: jnxJdhcpv6LocalServerStatistics *(continued)*

Object	Object Identifier	Description
jnxJdhcpv6LocalServerRequestReceived	jnxJdhcpv6LocalServerStatistics 14	Number of DHCPv6 Request packets received.
jnxJdhcpv6LocalServerConfirmReceived	jnxJdhcpv6LocalServerStatistics 15	Number of DHCPv6 Confirm packets received.
jnxJdhcpv6LocalServerRenewReceived	jnxJdhcpv6LocalServerStatistics 16	Number of DHCPv6 Renew packets received.
jnxJdhcpv6LocalServerRebindReceived	jnxJdhcpv6LocalServerStatistics 17	Number of DHCPv6 Rebind packets received.
jnxJdhcpv6LocalServerRelayForwReceived	jnxJdhcpv6LocalServerStatistics 18	Number of DHCPv6 Relay Forw packets received.
jnxJdhcpv6LocalServerRelayReplReceived	jnxJdhcpv6LocalServerStatistics 19	Number of DHCPv6 Relay Repl packets received.
jnxJdhcpv6LocalServerAdvertiseSent	jnxJdhcpv6LocalServerStatistics 20	Number of DHCPv6 Advertise packets sent.
jnxJdhcpv6LocalServerReplySent	jnxJdhcpv6LocalServerStatistics 21	Number of DHCPv6 Reply packets sent.
jnxJdhcpv6LocalServerReconfigureSent	jnxJdhcpv6LocalServerStatistics 22	Number of DHCPv6 Reconfigure packets sent.

The **jnxJdhcpv6LocalServerBindings**, whose object ID is **{jnxJdhcpv6LocalServerObjects 2}**, contains objects that describe the current statistics and bindings of a DHCPv6 local server.

The **jnxJdhcpv6LocalServerBindings** objects are listed in Table 51 on page 205.

Table 51: jnxJdhcpv6LocalServerBindings

Object	Object Identifier	Description
jnxJdhcpv6LocalServerBindingsPrefix	jnxJdhcpv6LocalServerBindingsEntry 1	Prefix associated with this entry in the bindings table.
jnxJdhcpv6LocalServerBindingsLength	jnxJdhcpv6LocalServerBindingsEntry 2	Length of the prefix in bits.
jnxJdhcpv6LocalServerBindingsState	jnxJdhcpv6LocalServerBindingsEntry 3	State associated with this entry in the bindings table.
jnxJdhcpv6LocalServerBindingsLeaseEndTime	jnxJdhcpv6LocalServerBindingsEntry 4	Time the lease expires on this binding.

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Table 51: jnxJdhcpv6LocalServerBindings *(continued)*

Object	Object Identifier	Description
jnxJdhcpv6LocalServerBindingsLeaseExpireTime	jnxJdhcpv6LocalServerBindingsEntry 5	Time remaining until the lease expires for this binding.
jnxJdhcpv6LocalServerBindingsLeaseStartTime	jnxJdhcpv6LocalServerBindingsEntry 6	Time the lease started for this binding.
jnxJdhcpv6LocalServerBindingsIncomingClientInterface	jnxJdhcpv6LocalServerBindingsEntry 7	Incoming interface for this binding.
jnxJdhcpv6LocalServerBindingsClientInterfaceVlanId	jnxJdhcpv6LocalServerBindingsEntry 8	VLAN ID for this binding.
jnxJdhcpv6LocalServerBindingsDemuxInterfaceName	jnxJdhcpv6LocalServerBindingsEntry 9	Demux interface for this binding.
jnxJdhcpv6LocalServerBindingsServerIpAddress	jnxJdhcpv6LocalServerBindingsEntry 10	IP Address associated with the server for this entry in the bindings table.
jnxJdhcpv6LocalServerBindingsBootpRelayAddress	jnxJdhcpv6LocalServerBindingsEntry 11	BOOTP relay address associated with the server for this entry in the bindings table.
jnxJdhcpv6LocalServerBindingsPreviousBootpRelayAddress	jnxJdhcpv6LocalServerBindingsEntry 12	Previous BOOTP relay address associated with the server for this entry in the bindings table.
jnxJdhcpv6LocalServerBindingsClientPoolName	jnxJdhcpv6LocalServerBindingsEntry 13	Dlient pool name.
jnxJdhcpv6LocalServerBindingsClientProfileName	jnxJdhcpv6LocalServerBindingsEntry 14	Client profile name.

The **jnxJdhcpv6LocalServerTraps**, whose object ID is **{jnxJdhcpv6LocalServerObjects 3}** contains objects that describe the DHCPv6 local server notifications.

The **jnxJdhcpv6LocalServerTraps** objects are listed in "Juniper Networks Enterprise-Specific SNMP Traps" on page 633.

The **jnxJdhcpv6LocalServerTrapVars**, whose object ID is **{jnxJdhcpv6LocalServerObjects 4}**, contains objects that describe the current status and statistics of a DHCPv6 relay server.

The **jnxJdhcpv6LocalServerTrapVars** objects are listed in Table 52 on page 207.

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Table 52: jnxJdhcpv6LocalServerTrapVars

Object	Object Identifier	Description
jnxJdhcpv6LocalServerRouterName	jnxJdhcpv6LocalServerTrapVars 1	VRF ID in Junos OS. Represented as the Logical Router (LR) Name followed by the Router Instance (RI) Name.
jnxJdhcpv6LocalServerInterfaceName	jnxJdhcpv6LocalServerTrapVars 2	Interface where the DHCP client was detected.
jnxJdhcpv6LocalServerInterfaceLimit	jnxJdhcpv6LocalServerTrapVars 3	Number of clients supported on this interface.

jnxJdhcpv6LocalServerIfcStats, whose object ID is **{jnxJdhcpv6LocalServerObjects 5}**, contains objects that describe the contents of a table of interface statistics maintained by the DHCPv6 local server.

jnxJdhcpv6LocalServerIfcStats objects are listed in Table 53 on page 207.

Table 53: jnxJdhcpv6LocalServerIfcStats

Object	Object Identifier	Description
jnxJdhcpv6LocalServerIfcStatsIfIndex	jnxJdhcpv6LocalServerIfcStatsEntry 1	IfIndex value of the interface for which this entry contains information.
jnxJdhcpv6LocalServerIfcStatsTotalDropped	jnxJdhcpv6LocalServerIfcStatsEntry 2	Total number of DHCPv6 packets dropped.
jnxJdhcpv6LocalServerIfcStatsNoSafdDropped	jnxJdhcpv6LocalServerIfcStatsEntry 3	Number of DHCPv6 packets dropped due to no safd match.
jnxJdhcpv6LocalServerIfcStatsBadSendDropped	jnxJdhcpv6LocalServerIfcStatsEntry 4	Number of DHCPv6 packets dropped due to send error.
jnxJdhcpv6LocalServerIfcStatsShortPacketDropped	jnxJdhcpv6LocalServerIfcStatsEntry 5	Number of DHCPv6 packets dropped due to packet being too short.
jnxJdhcpv6LocalServerIfcStatsBadMsgtypeDropped	jnxJdhcpv6LocalServerIfcStatsEntry 6	Number of DHCPv6 packets dropped due to bad options code in the packet.
jnxJdhcpv6LocalServerIfcStatsBadOptionsDropped	jnxJdhcpv6LocalServerIfcStatsEntry 7	Number of DHCPv6 packets dropped due to bad options in the packet.
jnxJdhcpv6LocalServerIfcStatsBadSrcAddressDropped	jnxJdhcpv6LocalServerIfcStatsEntry 8	Number of DHCPv6 packets dropped due to invalid address family.

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Table 53: jnxJdhcpv6LocalServerIfcStats *(continued)*

Object	Object Identifier	Description
jnxJdhcpv6LocalServerIfcStatsRelayCountDropped	jnxJdhcpv6LocalServerIfcStatsEntry 9	Number of DHCPv6 packets dropped due to maximum relays supported.
jnxJdhcpv6LocalServerIfcStatsNoClientIdDropped	jnxJdhcpv6LocalServerIfcStatsEntry 10	Number of DHCPv6 packets dropped due to missing client ID.
jnxJdhcpv6LocalServerIfcStatsDeclineReceived	jnxJdhcpv6LocalServerIfcStatsEntry 11	Number of DHCPv6 Decline packets received.
jnxJdhcpv6LocalServerIfcStatsSolicitReceived	jnxJdhcpv6LocalServerIfcStatsEntry 12	Number of DHCPv6 Solicit packets received.
jnxJdhcpv6LocalServerIfcStatsInformationRequestReceived	jnxJdhcpv6LocalServerIfcStatsEntry 13	Number of DHCPv6 Information Request packets received.
jnxJdhcpv6LocalServerIfcStatsReleaseReceived	jnxJdhcpv6LocalServerIfcStatsEntry 14	Number of DHCPv6 Release packets received.
jnxJdhcpv6LocalServerIfcStatsRequestReceived	jnxJdhcpv6LocalServerIfcStatsEntry 15	Number of DHCPv6 Request packets received.
jnxJdhcpv6LocalServerIfcStatsConfirmReceived	jnxJdhcpv6LocalServerIfcStatsEntry 16	Number of DHCPv6 Confirm packets received.
jnxJdhcpv6LocalServerIfcStatsRenewReceived	jnxJdhcpv6LocalServerIfcStatsEntry 17	Number of DHCPv6 Renew packets received.
jnxJdhcpv6LocalServerIfcStatsRebindReceived	jnxJdhcpv6LocalServerIfcStatsEntry 18	Number of DHCPv6 Rebind packets received.
jnxJdhcpv6LocalServerIfcStatsRelayForwReceived	jnxJdhcpv6LocalServerIfcStatsEntry 19	Number of DHCPv6 Relay Forward packets received.
jnxJdhcpv6LocalServerIfcStatsRelayReplReceived	jnxJdhcpv6LocalServerIfcStatsEntry 20	Number of DHCPv6 Relay Reply packets received.
jnxJdhcpv6LocalServerIfcStatsAdvertiseSent	jnxJdhcpv6LocalServerIfcStatsEntry 21	Number of DHCPv6 Advertise packets sent.
jnxJdhcpv6LocalServerIfcStatsReplySent	jnxJdhcpv6LocalServerIfcStatsEntry 22	Number of DHCPv6 Reply packets sent.

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Table 53: jnxJdhcpv6LocalServerIfcStats *(continued)*

Object	Object Identifier	Description
jnxJdhcpv6LocalServerIfcStatsReconfigureSent	jnxJdhcpv6LocalServerIfcStatsEntry 23	Number of DHCPv6 Reconfigure packets sent.
jnxJdhcpv6LocalServerIfcStatsTotalLeaseCount	jnxJdhcpv6LocalServerIfcStatsEntry 24	Number of Bound DHCP clients.
jnxJdhcpv6LocalServerIfcStatsStrictReconfigDropped	jnxJdhcpv6LocalServerIfcStatsEntry 25	Number of packets dropped due to strict reconfigure.
jnxJdhcpv6LocalServerIfcStatsAuthenticationDropped	jnxJdhcpv6LocalServerIfcStatsEntry 26	Number of packets dropped due to authentication failure.
jnxJdhcpv6LocalServerIfcStatsDynamicProfileDropped	jnxJdhcpv6LocalServerIfcStatsEntry 27	Number of packets dropped due to dynamic profile error.
jnxJdhcpv6LocalServerIfcStatsLicenseDropped	jnxJdhcpv6LocalServerIfcStatsEntry 28	Number of packets dropped due to license error.

Related Documentation

- DHCPv6 MIB on page 203

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CHAPTER 9

Interpreting the Enterprise-Specific SNMP IDP MIB

- SNMP IDP MIB on page 211
- jnxJsIdpObjects on page 212
- jnxJsIdpAttackTable on page 212
- IDP Signature Update and Attack Traps on page 213

SNMP IDP MIB

The Juniper Networks enterprise-specific SNMP IDP MIB, **jnxJsIdpMIB**, whose object ID is **{jnxJsIdpRoot 1}**, extends SNMP support to the following features on Juniper Networks SRX100, SRX210, SRX220, SRX240, and SRX650 Services Gateways:

- Key monitoring and threshold-crossing traps
- Attack-related monitoring and traps
- IDP database update status and traps

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-js-idp.txt .

For information about the enterprise-specific SNMP IDP MIB objects, see the following topics:

- jnxJsIdpObjects on page 212
- jnxJsIdpAttackTable on page 212
- IDP Signature Update and Attack Traps on page 213

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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jnxJsIdpObjects

You use **jnxJsIdpObjects**, whose object ID is **{jnxJsIdpMIB 1}**, to display the IDP query-related statistics listed in Table 54 on page 212.

Table 54: jnxJsIdpObjects

Object	Object ID	Description
jnxJsIdpDataPlaneMemoryUsage	jnxJsIdpObjects 1	The percentage of memory used by IDP in the data plane.
jnxJsIdpSessionsUsage	jnxJsIdpObjects 2	The percentage of currently allocated sessions by IDP.
jnxJsIdpSessionsMaximum	jnxJsIdpObjects 3	The maximum number of sessions IDP can support.
jnxJsIdpPoliciesSupported	jnxJsIdpObjects 4	The number of policies that can be simultaneously loaded on the IDP device.
jnxJsIdpPoliciesLoaded	jnxJsIdpObjects 5	The number of policies that are currently loaded on the IDP device.
jnxJsIdpActivePolicyName	jnxJsIdpObjects 6	The name of the policy that is currently active on the device.
jnxJsIdpRunningDetectorVersion	jnxJsIdpObjects 8	The version of the detector that is currently loaded.
jnxJsIdpSecurityPackageVersion	jnxJsIdpObjects 9	The version of the security that is package currently loaded.
jnxJsIdpLastSignatureUpdateTime	jnxJsIdpObjects 10	The value of sysUpTime when the signature was last updated successfully. Zero if unknown.
jnxJsIdpSignatureUpdateStatus	jnxJsIdpObjects 11	The status of the signature update: success or failure.

Related Documentation

- SNMP IDP MIB on page 211
- jnxJsIdpAttackTable on page 212
- IDP Signature Update and Attack Traps on page 213

jnxJsIdpAttackTable

The **jnxJsIdpAttackTable**, whose object ID is **{jnxJsIdpObjects 7}**, contains the details of each attack and the number of hits involved. The objects are listed in Table 55 on page 213.

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Table 55: jnxJsIdpAttackTable

Object	Object ID	Description
jnxJsIdpAttackEntry	**jnxJsIdpAttackTable 1**	The attributes of an attack Sequence of parameters: • **jnxJsIdpAttackIndex** • **jnxJsIdpAttackName** • **jnxJsIdpAttackHits**
jnxJsIdpAttackIndex	**jnxJsIdpAttackEntry 1**	The index for the attack table.
jnxJsIdpAttackName	**jnxJsIdpAttackEntry 2**	The name of an attack identified by IDP.
jnxJsIdpAttackHits	**jnxJsIdpAttackEntry 3**	The number of times an attack has been launched.

Related Documentation

- SNMP IDP MIB on page 211
- IDP Signature Update and Attack Traps on page 213

IDP Signature Update and Attack Traps

The IDP MIB generates an SNMP trap whenever you install or update a signature, and provides information about the last updated version and a timestamp. It also generates SNMP traps to notify users about attacks that are detected on the device.

See Table 56 on page 213 for information about the signature update and attack traps.

Table 56: IDP Signature Update and Attack Traps

Object	Object ID	Description
jnxJsIdpSignatureUpdateNotification	**jnxJsIdpNotificationPrefix 1**	Signifies that a signature update has occurred.
jnxJsIdpAttackLogNotification	**jnxJsIdpNotificationPrefix 2**	Signifies that attacks have been detected. **jnxJsIdpAttackName** lists the names of the attacks that have been detected so far, and **jnxJsIdpAttackHits** is the number of times each attack has been detected.

Related Documentation

- SNMP IDP MIB on page 211
- jnxJsIdpObjects on page 212
- jnxJsIdpAttackTable on page 212
- Configuring SNMP Trap Options and Groups on a Device Running Junos OS
- Managing Traps and Informs

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CHAPTER 10

Interpreting the Enterprise-Specific Ping MIB

- PING MIB on page 215
- jnxPingCtlTable on page 216
- jnxPingResultsTable on page 220
- jnxPingProbeHistoryTable on page 224
- jnxPingLastTestResultTable on page 226

PING MIB

The Juniper Networks enterprise-specific Ping MIB extends the standard Ping MIB control table (RFC 2925). The Ping MIB, whose object identifier is **{jnxMIbs 7}**, allows you to monitor network delay (latency), packet loss, network delay variation (jitter), one-way latency, and other network statistics.

Items in this MIB are created when entries are created in the **pingCtlTable** of the Ping MIB. Each item is indexed exactly as in the Ping MIB.

To view a complete copy of the enterprise-specific extensions to the Ping MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-ping.txt .

For information about using the Ping MIB and enterprise-specific Ping MIB, see the *Junos OS Network Management Configuration Guide*.

For information about the enterprise-specific PING MIB objects, see the following topics:

- jnxPingCtlTable on page 216
- jnxPingResultsTable on page 220
- jnxPingProbeHistoryTable on page 224
- jnxPingLastTestResultTable on page 226

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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jnxPingCtlTable

The enterprise-specific Ping MIB structure includes one main object, **jnxPingCtlTable**, whose object identifier is **{jnxPingObjects 2}**, and defines the **jnxPing** control table for providing enterprise-specific options to the corresponding **pingCtlEntry**. **jnxPingCtlTable** monitors thresholds; for example, the maximum allowed jitter in the trip time during a text.

Each **jnxPingCtlEntry** has two indexes identical to those of the corresponding **pingCtlEntry**. Entries created in **pingCtlTable** are mirrored here. **jnxPingCtlTable** objects are listed in Table 57 on page 216.

Table 57: jnxPingCtlTable

Object	Object Identifier	Description
jnxCtlOwnerIndex	**jnxPingCtlEntry 1**	The first index. It is identical to the **pingCtlOwnerIndex** of the corresponding **pingCtlEntry** in the **pingCtlTable**.
jnxPingCtlTestName	**jnxPingCtlEntry 2**	The other index and is identical to the **pingCtlTestName** of the corresponding **pingCtlEntry** in the **pingCtlTable**.
jnxPingCtlIfName	**jnxPingCtlEntry 3**	Specifies the name of the outgoing interface for ping probes. This is the name-based complement to **pingCtlIfIndex**. A zero-length string value for this object means that this option is not enabled. The following values can be set simultaneously, but only one value is used. The precedence order is as follows: • **pingCtlIfIndex** (see **pingCtlTable** in "PING MIB" on page 215) • **jnxPingCtlIfName** • **jnxPingCtlRoutingInstanceName**
jnxPingCtlRoutingInstanceName	**jnxPingCtlEntry 6**	Specifies the name of the routing instance used when directing outgoing ping packets. The instance name specified must be configured at the **[edit routing-instances]** hierarchy level of the Junos configuration. The instance-type must be **vrf**.
jnxPingCtlRttThreshold	**jnxPingCtlEntry 7**	The maximum round-trip time allowed. If this threshold is crossed by any probe, a **jnxPingRttThresholdExceeded** trap is sent.

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Table 57: jnxPingCtlTable *(continued)*

Object	Object Identifier	Description
jnxPingCtlRttStdDevThreshold	**jnxPingCtlEntry 8**	The maximum round-trip time standard deviation allowed over the course of any test. If the calculated standard deviation of the round-trip time at the end of any test exceeds this threshold, a **jnxPingRttStdDevThresholdExceeded** trap is sent.
jnxPingCtlRttJitterThreshold	**jnxPingCtlEntry 9**	The maximum allowed jitter in the round-trip time over the course of any test. Jitter is the difference between the maximum and minimum round-trip times measured over the course of a single test (**jnxPingResultsMaxRttUs** minus **jnxPingResultsMinRttUs**). If the measured jitter exceeds this threshold, a **jnxPingRttJitterThresholdExceeded** trap is sent.
jnxPingCtlEgressTimeThreshold	**jnxPingCtlEntry 10**	Maximum egress trip time allowed. If this threshold is crossed by any probe, a **jnxPingEgressThresholdExceeded** trap is sent. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. Currently, **jnxPingIcmpTimeStamp** is the only supported probe type with this property.
jnxPingCtlEgressStdDevThreshold	**jnxPingCtlEntry 11**	The maximum egress trip time standard deviation allowed over the course of any test. If the calculated standard deviation of the egress trip time at the end of any test exceeds this threshold, a **jnxPingEgressStdDevThresholdExceeded** trap is sent. This applies only if the probe type (**pingCtlType**) provides one way delay measurements. The **jnxPingIcmpTimeStamp** is the only supported probe type with this property.
jnxPingCtlEgressJitterThreshold	**jnxPingCtlEntry 12**	The maximum allowed jitter in the egress trip time over the course of any test. Jitter is defined as the difference between the maximum and minimum egress trip times measured over the course of a single test (**jnxPingResultsMaxSrcDstt** minus **jnxPingResultsMinSrcDstt**). If the measured jitter exceeds this threshold, a **jnxPingEgressJitterThresholdExceeded** trap is sent. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. The **jnxPingIcmpTimeStamp** is the only supported probe type with this property.

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Table 57: jnxPingCtlTable *(continued)*

Object	Object Identifier	Description
jnxPingCtlIngressTimeThreshold	**jnxPingCtlEntry 13**	The maximum ingress trip time allowed. If this threshold is crossed by any probe, a **jnxPingIngressThresholdExceeded** trap is sent. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. The **jnxPingIcmpTimeStamp** is the only supported probe type with this property.
jnxPingCtlIngressStddevThreshold	**jnxPingCtlEntry 14**	The maximum ingress trip time standard deviation allowed over the course of any test. If the calculated standard deviation of the ingress trip time at the end of any test exceeds this threshold, a **jnxPingIngressStddevThresholdExceeded** trap is sent. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. Currently, **jnxPingIcmpTimeStamp** is the only supported probe type with this property.
jnxPingCtlIngressJitterThreshold	**jnxPingCtlEntry 15**	The maximum allowed jitter in the ingress trip time over the course of any test. Jitter is defined as the difference between the maximum and minimum ingress trip times measured over the course of a single test (**jnxPingResultsMaxDstSrct** minus **jnxPingResultsMinDstSrct**). If the measured jitter exceeds this threshold, a **jnxPingIngressJitterThresholdExceeded** trap is sent. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. The **jnxPingIcmpTimeStamp** is the only supported probe type with this property.

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Table 57: jnxPingCtlTable *(continued)*

Object	Object Identifier	Description
jnxPingCtlTrapGeneration	**jnxPingCtlEntry 16**	The value of this object determines when and if to generate a notification for this entry. **rttThreshold(0)**—Generate a **jnxPingRttThresholdExceeded** notification when the configured RTT threshold is exceeded. **rttStdDevThreshold(1)**—Generate a **jnxPingRttStdDevThresholdExceeded** notification when the configured RTT standard deviation threshold is exceeded. **rttJitterThreshold(2)**—Generate a **jnxPingRttJitterThresholdExceeded** notification when the configured RTT jitter threshold is exceeded. **egressThreshold(3)**—Generate a **jnxPingEgressThresholdExceeded** notification when the configured egress threshold is exceeded. This applies only if the probe type supports one-way measurements.

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Table 57: jnxPingCtlTable *(continued)*

Object	Object Identifier	Description
		egressStdDevThreshold(4)—Generate a **jnxPingEgressStdDevThresholdExceeded** notification when the configured egress standard deviation threshold is exceeded. This applies only if the probe type supports one-way measurements. **egressJitterThreshold(5)**—Generate a **jnxPingEgressJitterThresholdExceeded** notification when the configured egress jitter threshold is exceeded. This applies only if the probe type supports one-way measurements. **ingressThreshold(6)**—Generate a **jnxPingIngressThresholdExceeded** notification when the configured ingress threshold is exceeded. This applies only if the probe type supports one-way measurements. **ingressStdDevThreshold(7)**—Generate a **jnxPingIngressStdDevThresholdExceeded** notification when the configured ingress standard deviation threshold is exceeded. This applies only if the probe type supports one way measurements. **ingressJitterThreshold(8)**—Generate a **jnxPingIngressJitterThresholdExceeded** notification when the configured ingress jitter threshold is exceeded. This applies only if the probe type supports one-way measurements. The value of this object defaults to zero, indicating that none of the above options have been selected.

Related Documentation

- PING MIB on page 215
- jnxPingResultsTable on page 220
- jnxPingProbeHistoryTable on page 224
- jnxPingLastTestResultTable on page 226

jnxPingResultsTable

jnxPingResultsTable, whose object identifier is **{jnxPingObjects 3}**, gathers ping test results on traffic on round-trip, ingress, and egress trip delays. This is useful when you want to measure the performance of your network and verify service-level agreements with your vendors.

The **jnxPingResultsEntry** objects are listed in Table 58 on page 221.

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Table 58: jnxPingsResultsEntry

Object	Object Identifier	Description
jnxPingResultsRttUs	**jnxPingResultsEntry 1**	The round-trip delays measured for the most recent successful probe during this test, in microseconds.
jnxPingResultsSumRttUs	**jnxPingResultsEntry 2**	The sum of the round-trip delays measured for all the probes during this test, in microseconds.
jnxPingResultsMinRttUs	**jnxPingResultsEntry 3**	The minimum of the round-trip delays measured for all the probes during this test, in microseconds.
jnxPingResultsMaxRttUs	**jnxPingResultsEntry 4**	The maximum of the round-trip delays measured for all the probes during this test, in microseconds.
jnxPingResultsAvgRttUs	**jnxPingResultsEntry 5**	The average of the round-trip delays measured for all the probes during this test, in microseconds.
jnxPingResultsStdDevRttUs	**jnxPingResultsEntry 6**	The standard deviation of the round-trip delays measured during this test, in microseconds.
jnxPingResultsEgressUs	**jnxPingResultsEntry 7**	The egress trip delays measured for the most recent successful probe during this test, in microseconds. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, their values are irrelevant and this object returns 0.
jnxPingResultsMinEgressUs	**jnxPingResultsEntry 8**	The minimum of the egress trip delays measured over all probes during this test, in microseconds. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, their values are irrelevant and this object returns 0.
jnxPingResultsMaxEgressUs	**jnxPingResultsEntry 9**	The maximum of the egress trip delays measured over all probes during this test, in microseconds. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, their values are irrelevant and this object returns 0.

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Table 58: jnxPingsResultsEntry *(continued)*

Object	Object Identifier	Description
jnxPingResultsAvgEgressUs	jnxPingResultsEntry 10	The average of the egress trip delays measured over all probes during this test, in microseconds. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, their values are irrelevant and this object returns 0.
jnxPingResultsStddevEgressUs	jnxPingResultsEntry 11	The standard deviation of the egress trip delays measured over all probes during this test, in microseconds. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, their values are irrelevant and this object returns 0.
jnxPingResultsIngressUs	jnxPingResultsEntry 12	The ingress trip delays measured for the most recent successful probe during this test, in microseconds. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, their values are irrelevant and this object returns 0.
jnxPingResultsMinIngressUs	jnxPingResultsEntry 13	The minimum of the ingress trip delays measured over all probes during this test, in microseconds. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, their values are irrelevant and this object returns 0.
jnxPingResultsMaxIngressUs	jnxPingResultsEntry 14	The maximum of the ingress trip delays measured over all probes during this test, in microseconds. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, their values are irrelevant and this object returns 0.
jnxPingResultsAvgIngressUs	jnxPingResultsEntry 15	The average of the ingress trip delays measured over all probes during this test, in microseconds. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, their values are irrelevant and this object returns 0.

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Table 58: jnxPingsResultsEntry *(continued)*

Object	Object Identifier	Description
jnxPingResultsStddevIngressUs	jnxPingResultsEntry 16	The standard deviation of the ingress trip delays measured over all probes during this test, in microseconds. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, their values are irrelevant and this object returns 0.
jnxPingResultsJitterRttUs	jnxPingResultsEntry 17	The jitter of the round-trip delays measured for all probes during this test, in microseconds.
jnxPingResultsJitterEngressUs	jnxPingResultsEntry 18	The jitter of the engress trip delays measured for all probes during this test, in microseconds. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, their values are irrelevant and this object returns 0.
jnxPingResultsJitterIngressUs	jnxPingResultsEntry 19	The jitter of the ingress trip delays measured for all probes during this test, in microseconds. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, their values are irrelevant and this object returns 0.
jnxPingResultsStatus	jnxPingResultsEntry 20	The result of the most recent probe.
jnxPingResultsTime	jnxPingResultsEntry 21	The date and time of the most recent probe result.
jnxPingResultsOwnerIndex	jnxPingResultsEntry 22	The first index. It has the same value as **pingCtlOwnerIndex** and is provided for applications that are unable to obtain the value of **pingCtlOwnerIndex** from the instance portion of the object identifiers belonging to this table.
jnxPingResultsTestName	jnxPingResultsEntry 23	The other index. It has the same value as **pingCtlTestName** and is provided for applications that are unable to obtain the value of **pingCtlTestName** from the instance portion of the object identifiers belonging to this table.

Related Documentation

- PING MIB on page 215
- jnxPingCtlTable on page 216
- jnxPingProbeHistoryTable on page 224

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- jnxPingLastTestResultTable on page 226

jnxPingProbeHistoryTable

jnxpingProbeHistoryTable, whose object identifier is **{jnxPingObjects 4}**, contains the history of all ping tests.

The **jnxPingProbeHistoryEntry** objects are listed in Table 59 on page 224.

Table 59: jnxPingProbeHistoryEntry

Object	Object Identifier	Description
jnxPingProbeHistoryResponseUs	**jnxPingProbeHistoryEntry 1**	The amount of time, in microseconds, from when a probe was sent to when its response was received or when it timed out. The value of this object is reported as 0 when it is not possible to transmit a probe.
jnxPingProbeHistoryJitterUs	**jnxPingProbeHistoryEntry 2**	The time difference, in microseconds, between the maximum and minimum round-trip times. Each history entry provides a running calculation of the jitter (calculated over the current test) at the time a probe was completed.
jnxPingProbeHistoryResponseEgressUs	**jnxPingProbeHistoryEntry 3**	The amount of time, in microseconds, from when a probe was sent to when it was received by destination. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, the value is irrelevant and this object returns 0.

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Table 59: jnxPingProbeHistoryEntry *(continued)*

Object	Object Identifier	Description
jnxPingProbeHistoryResponseIngressUs	**jnxPingProbeHistoryEntry 4**	The amount of time, in microseconds, from when a probe was sent from the destination to when it was received. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, the value is irrelevant and this object returns 0.
jnxPingProbeHistoryEgressJitterUs	**jnxPingProbeHistoryEntry 5**	The time difference, in microseconds, between the maximum and minimum egress trip times. Each history entry provides a running calculation of the jitter (calculated over the current test) at the time a probe was completed. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, the value is irrelevant and this object returns 0.
jnxPingProbeHistoryIngressJitterUs	**jnxPingProbeHistoryEntry 6**	The time difference, in microseconds, between the maximum and minimum ingress trip times. Each history entry provides a running calculation of the jitter (calculated over the current test) at the time a probe was completed. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, the value is irrelevant and this object returns 0.

Related Documentation

- PING MIB on page 215
- jnxPingCtlTable on page 216
- jnxPingResultsTable on page 220

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- jnxPingLastTestResultTable on page 226

jnxPingLastTestResultTable

jnxPingLastTestResultTable, whose object identifier is **{jnxPingObjects 5}**, contains the results of the last completed ping tests. Entries corresponding to a test are created only after completion of the first test. This is useful when you want to ensure that a test has been completed before collecting test results.

The **jnxPingLastTestResultEntry** objects are listed in Table 60 on page 226.

Table 60: jnxPingLastTestResultEntry

Object	Object Identifier	Description
jnxPingLastTestResultProbeResponses	jnxPingLastTestResultEntry 1	The number of responses received in the most recently completed test.
jnxPingLastTestResultSentProbes	jnxPingLastTestResultEntry 2	The number of probes sent in the most recently completed test.
jnxPingLastTestResultSumRttUs	jnxPingLastTestResultEntry 3	The sum of the round-trip delays measured for all the probes during the most recently completed test, in microseconds.
jnxPingLastTestResultMinRttUs	jnxPingLastTestResultEntry 4	The minimum of the round-trip delays measured for all the probes during the most recently completed test, in microseconds.
jnxPingLastTestResultMaxRttUs	jnxPingLastTestResultEntry 5	The maximum of the round-trip delays measured for all the probes during the most recently completed test, in microseconds.
jnxPingLastTestResultAvgRttUs	jnxPingLastTestResultEntry 6	The average of the round-trip delays measured for all the probes during the most recently completed test, in microseconds.

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Table 60: jnxPingLastTestResultEntry *(continued)*

Object	Object Identifier	Description
jnxPingLastTestResultStdDevRttUs	**jnxPingLastTestResultEntry 7**	The standard deviation of the round-trip delays measured for all the probes during the most recently completed test, in microseconds.
jnxPingLastTestResultMinEgressUs	**jnxPingLastTestResultEntry 8**	The minimum of the egress trip delays measured over all probes during the most recently completed test, in microseconds. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, their values are irrelevant and this object returns 0.
jnxPingLastTestResultMaxEgressUs	**jnxPingLastTestResultEntry 9**	The maximum of the egress trip delays measured over all probes during the most recently completed test, in microseconds. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, their values are irrelevant and this object returns 0.
jnxPingLastTestResultAvgEgressUs	**jnxPingLastTestResultEntry 10**	The average of the egress trip delays measured over all probes during the most recently completed test, in microseconds. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, their values are irrelevant and this object returns 0.

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Table 60: jnxPingLastTestResultEntry *(continued)*

Object	Object Identifier	Description
jnxPingLastTestResultStddevEgressUs	jnxPingLastTestResultEntry 11	The standard deviation of the egress trip delays measured over all probes during the most recently completed test, in microseconds. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, their values are irrelevant and this object returns 0.
jnxPingLastTestResultMinIngressUs	jnxPingLastTestResultEntry 12	The minimum of the ingress trip delays measured over all probes during the most recently completed test, in microseconds. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, their values are irrelevant and this object returns 0.
jnxPingLastTestResultMaxIngressUs	jnxPingLastTestResultEntry 13	The maximum of the ingress trip delays measured over all probes during the most recently completed test, in microseconds. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, their values are irrelevant and this object returns 0.
jnxPingLastTestResultAvgIngressUs	jnxPingLastTestResultEntry 14	The average of the ingress trip delays measured over all probes during the most recently completed test, in microseconds. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, their values are irrelevant and this object returns 0.

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Table 60: jnxPingLastTestResultEntry *(continued)*

Object	Object Identifier	Description
jnxPingLastTestResultStddevIngressUs	**jnxPingLastTestResultEntry 15**	The standard deviation of the ingress trip delays measured over all probes during the most recently completed test, in microseconds. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, their values are irrelevant and this object returns 0.
jnxPingLastTestResult PeakToPeakJitterRttUs	**jnxPingLastTestResultEntry 16**	The difference between the minimum and maximum delays over the course of the last completed test, in microseconds.
jnxPingLastTestResult PeakToPeakJitterEgressUs	**jnxPingLastTestResultEntry 17**	The difference between the minimum and maximum egress trip delays over the course of the last completed test, in microseconds. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, their values are irrelevant and this object returns 0.
jnxPingLastTestResult PeakToPeakJitterIngressUs	**jnxPingLastTestResultEntry 18**	The difference between the minimum and maximum ingress trip delays over the course of the last completed test, in microseconds. This applies only if the probe type (**pingCtlType**) provides one-way delay measurements. For all other probe types, their values are irrelevant and this object returns 0.
jnxPingLastTestResultTime	**jnxPingLastTestResultEntry 19**	The time the last test was completed.

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Related Documentation

- PING MIB on page 215
- jnxPingCtlTable on page 216
- jnxPingResultsTable on page 220
- jnxPingProbeHistoryTable on page 224

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CHAPTER 11

Interpreting the Enterprise-Specific Traceroute MIB

- Traceroute MIB on page 231
- jnxTraceRouteCtlTable on page 231

Traceroute MIB

The Juniper Networks enterprise-specific Traceroute MIB supports Junos OS extensions of traceroutes and remote operations. Items in this MIB are created when entries are created in the **traceRouteCtlTable** of the Traceroute MIB. Each item is indexed exactly the same way as it is in the enterprise-specific Traceroute MIB. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-traceroute.txt .

For more information about using the Traceroute MIB and enterprise-specific Traceroute MIB, see the *Junos OS Network Management Configuration Guide*.

For information about the enterprise-specific Traceroute MIB objects, see the following topic:

- jnxTraceRouteCtlTable on page 231

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxTraceRouteCtlTable

The **jnxTraceRouteCtlTable**, whose object identifier is **{jnxTraceRouteObjects 2}**, defines the **jnxTraceRoute** control table for providing enterprise-specific options to the corresponding **traceRouteCtlEntry**.

Each **jnxTraceRouteCtlEntry** has two indexes that are identical to those of the corresponding **TraceRouteCtlEntry**. Entries created in **TraceRouteCtlTable** are mirrored here and are listed in Table 61 on page 232.

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Table 61: jnxTraceRouteCtlTable

Object	Object Identifier	Description
jnxTRCtlOwnerIndex	**jnxTraceRouteCtlEntry 1**	Specifies the first index. It is identical to **jnxTraceRouteCtlOwnerIndex** of the corresponding **jnxTraceRouteCtlEntry** in the **jnxTraceRouteCtlTable**.
jnxTRCtlTestName	**jnxTraceRouteCtlEntry 2**	Specifies the other index. It is identical to **jnxTraceRouteCtlTestName** of the corresponding **jnxTraceRouteCtlEntry** in the **jnxTraceRouteCtlTable**.
jnxTRCtlIfName	**jnxTraceRouteCtlEntry 3**	Specifies the name of the outgoing interface for traceroute probes. This is the name-based complement to **traceRouteCtlIfIndex**. A zero-length string value for this object means that this option is not enabled. The following values can be set simultaneously, but only one value is used. The precedence order is as follows: • **traceRouteCtlIfIndex** (see **traceRouteCtlTable** in the "Traceroute MIB" on page 231) • **jnxTRCtlIfName** • **jnxTRCRoutingInstanceName**
jnxTRCtlRoutingInstanceName	**jnxTraceRouteCtlEntry 4**	Specifies the name of the routing instance used when directing outgoing traceroute packets. The instance name specified must be configured at the **[edit routing-instances]** hierarchy level of the Junos OS configuration.

Related Documentation

- Traceroute MIB on page 231

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CHAPTER 12

Interpreting the Enterprise-Specific RMON Events and Alarms MIB

- RMON Events and Alarms MIB on page 233
- jnxRmonAlarmTable on page 234
- RMON Event and Alarm Traps on page 235

RMON Events and Alarms MIB

The Juniper Networks enterprise-specific Remote Monitoring (RMON) Events and Alarms MIB monitors objects on a device and warns the network system administrator if one of those values exceeds the defined range. The alarm monitors objects in this MIB and triggers an event when the condition (falling or rising threshold) is reached.

The Juniper Networks enterprise-specific extension to the standard RMON MIB augments the **alarmTable** with additional information about each alarm. Two new traps, **jnxRmonAlarmGetFailure** and **jnxRmonGetOk**, are also defined to indicate when problems are encountered with an alarm.

To view a complete copy of the enterprise-specific extensions to the RMON MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-rmon.txt .

For more information about RMON alarms and events, see the *Junos OS Network Management Configuration Guide*.

For information about the enterprise-specific RMON Events and Alarms MIB objects, see the following topics:

- jnxRmonAlarmTable on page 234
- RMON Event and Alarm Traps on page 235

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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jnxRmonAlarmTable

The entries in the **jnxRmonAlarmTable**, whose object identifier is **{jnxMibs 13}**, are represented by **jnxRmonAlarmEntry**, whose object identifier is **{jnxRmonAlarmTable 1}** and are listed in Table 62 on page 234.

Table 62: jnxRmonAlarmEntry

Object	Object Identifier	Description
jnxRmonAlarmGetFailCnt	**jnxRmonAlarmEntry 1**	Represents the number of times the internal **Get** request for the variable monitored by this entry has failed.
jnxRmonAlarmGetFailTime	**jnxRmonAlarmEntry 2**	Represents the value of **sysUpTime** when an internal **Get** request for the variable monitored by this entry last failed.
jnxRmonAlarmGetFailReason	**jnxRmonAlarmEntry 3**	Represents the reason an internal **Get** request for the variable monitored by this entry last failed. This object contains the following values: • **other (1)**—An error was encountered that does not fit into one of the currently defined categories. • **noError (2)**—**Get** request processed successfully. • **noSuchObject (3)**—Requested object not available. • **outOfView (4)**—Requested object instance out of MIB view. • **noSuchInstance (5)**—Requested object instance not available. • **badReqId (6)**—Unexpected request ID encountered while processing **Get** request. • **oidMatchErr (7)**—Unexpected object ID encountered while processing **Get** request. • **oidBindErr (8)**—Unable to bind object ID to **Get** request PDU. • **createPktErr (9)**—Unable to create **Get** request PDU. • **badObjType (10)**—Unexpected object type encountered while processing **Get** request.
jnxRmonAlarmGetOkTime	**jnxRmonAlarmEntry 4**	Represents the value of **sysUpTime** when an internal **Get** request for the variable monitored by this entry succeeded and the entry left the **getFailure** state.

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Table 62: jnxRmonAlarmEntry *(continued)*

Object	Object Identifier	Description
jnxRmonAlarmState	**jnxRmonAlarmEntry 5**	Represents the current state of this RMON alarm entry. This object contains the following values: • **unknown (1)**—Alarm entry unknown • **underCreation (2)**—Alarm entry not activated • **active (3)**—Alarm entry active and within thresholds • **startup (4)**—Alarm entry still waiting for first value • **risingThreshold (5)**—Alarm entry has crossed the rising threshold. • **fallingThreshold (6)**—Alarm entry has crossed the falling threshold • **getFailure (7)**—Alarm entry internal **Get** request failed.

Related Documentation

- RMON Events and Alarms MIB on page 233
- RMON Event and Alarm Traps on page 235

RMON Event and Alarm Traps

The following traps send notifications when there is a problem with RMON alarm processing and are listed in Table 63 on page 235.

Table 63: RMON Event and Alarm Traps

Trap	Object Identifier	Description
jnxRmonAlarmGetFailure	**jnxRmonTrapPrefix 1**	Generated when the **Get** request for an alarm variable returns an error. The specific error is identified by **jnxRmonAlarmGetFailReason**.
jnxRmonGetOk	**jnxRmonTrapPrefix 2**	Generated when the **Get** request for an alarm variable is successful. This trap is sent only if the previous **Get** requests were unsuccessful.

Related Documentation

- RMON Events and Alarms MIB on page 233
- jnxRmonAlarmTable on page 234
- Configuring SNMP Trap Options and Groups on a Device Running Junos OS
- Managing Traps and Informs

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CHAPTER 13

Interpreting the Enterprise-Specific Reverse-Path-Forwarding MIB

- Reverse-Path-Forwarding MIB on page 237
- jnxRpfStatsTable on page 237

Reverse-Path-Forwarding MIB

The Juniper Networks enterprise-specific Reverse-Path-Forwarding (RPF) MIB monitors statistics for traffic that is rejected because of RPF processing. The RPF MIB includes one main object, **jnxRpfStats**, with an object identifier of **{jnxRpf 1}**. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-rpf.txt .

For information about the enterprise-specific Reverse Path Forwarding MIB objects, see the following topic:

- jnxRpfStatsTable on page 237

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxRpfStatsTable

The **jnxRpfStatsTable**, whose object identifier is **{jnxRpfStats 1}**, provides a list of RPF entries in table format.

The **jnxRpfStatsEntry**, whose object identifier is **{jnxRpfStatsTable 1}**, has four objects, which are listed in Table 64 on page 237.

Table 64: jnxRpfStatsEntry

Object	Object Identifier	Description
jnxRpfStatsIfIndex	**jnxRpfStatsEntry 1**	The ingress interface for traffic that is counted in an **RpfStats** entry.

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Table 64: jnxRpfStatsEntry *(continued)*

Object	Object Identifier	Description
jnxRpfStatsAddrFamily	**jnxRpfStatsEntry 2**	The address family of an entry's traffic, which can be in IPv4 or IPv6 format.
jnxRpfStatsPackets	**jnxRpfStatsEntry 3**	The number of packets received on this interface, belonging to this address family, that have been rejected due to RPF processing.
jnxRpfStatsBytes	**jnxRpfStatsEntry 4**	The number of bytes received on this interface, belonging to this address family, that have been rejected due to RPF processing.

Related Documentation

- Reverse-Path-Forwarding MIB on page 237

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CHAPTER 14

Interpreting the Enterprise-Specific Source Class Usage MIB

- Source Class Usage MIB on page 239
- jnxScuStatsTable on page 239

Source Class Usage MIB

The Juniper Networks enterprise-specific Source Class Usage (SCU) MIB counts packets sent to customers by performing a lookup on the IP source address and the IP destination address. SCU makes it possible to track traffic originating from specific prefixes on the provider core and destined for specific prefixes on the customer edge.

The enterprise-specific SCU MIB is an object of the **jnxMibs** branch of the enterprise-specific MIB **{enterprise 2636}** and has an object identifier of **{jnxMIB 16}**. The enterprise-specific SCU MIB includes one object, **jnxScuStats**, which has an object identifier of **{jnxScu 1}**. For information about configuring source and destination class usage, see the *Junos OS Routing Policy Configuration Guide* and the *Junos OS Network Interfaces Configuration Guide*. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-scu.txt .



NOTE: **Class-based filter match conditions are not supported on J Series Services Routers.**

For information about the enterprise-specific SCU MIB objects, see the following topic:

- jnxScuStatsTable on page 239

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxScuStatsTable

The **jnxRpfStatsTable**, whose object identifier is **{jnxRpfStats 1}**, provides a list of RPF entries in table format.

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The **jnxRpfStatsEntry**, whose object identifier is **{jnxRpfStatsTable 1}**, has four objects, which are listed in Table 65 on page 240.

Table 65: jnxRpfStatsEntry

Object	Object Identifier	Description
jnxRpfStatsIfIndex	**jnxRpfStatsEntry 1**	The ingress interface for traffic that is counted in an **RpfStats** entry.
jnxRpfStatsAddrFamily	**jnxRpfStatsEntry 2**	The address family of an entry's traffic, which can be in IPv4 or IPv6 format.
jnxRpfStatsPackets	**jnxRpfStatsEntry 3**	The number of packets received on this interface, belonging to this address family, that have been rejected due to RPF processing.
jnxRpfStatsBytes	**jnxRpfStatsEntry 4**	The number of bytes received on this interface, belonging to this address family, that have been rejected due to RPF processing.

Related Documentation

- Source Class Usage MIB on page 239

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CHAPTER 15

Interpreting the Enterprise-Specific Passive Monitoring MIB

- Passive Monitoring MIB on page 241
- jnxPMonFlowTable on page 242

Passive Monitoring MIB

The Juniper Networks enterprise-specific Passive Monitoring MIB, whose object identifier is **{jnxMibs 19}**, performs traffic-flow monitoring and lawful interception of packets transiting between two routers. This MIB enables you to do the following:

- Gather and export detailed information about IPv4 and IPv6 traffic flows between source and destination nodes in your network.
- Sample all incoming IPv4 and IPv6 traffic on the monitoring interface and present the data in **cflowd** record format.
- Encrypt or tunnel outgoing **cflowd** records, intercepted IPv4 and IPv6 traffic, or both.
- Direct filtered traffic to different packet analyzers and present the data in its original format.

The Passive Monitoring MIB has three tables: **jnxPMonFlowTable**, **JnxPMonErrorTable**, and **jnxPMonMemoryTable**. **jnxPMonFlowTable** monitors and collects statistics on the flow of traffic on a Passive Monitoring PIC. **jnxPMonErrorTable** monitors and collects statistics on packet and memory errors on a Passive Monitoring PIC. **jnxPMonMemoryTable** monitors and collects statistics on memory usage on a Passive Monitoring PIC. For information about system requirements, see the *Junos OS Feature Guide*. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-pmon.txt .

For information about the Passive Monitoring MIB objects, see the following topic:

- jnxPMonFlowTable on page 242

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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- Passive Monitoring on Ethernet Interfaces Overview

jnxPMonFlowTable

jnxPMonFlowTable has an object identifier of **{jnxPMon 1}**. Its entries are represented by **jnxPMonFlowEntry**, which contains the objects listed in Table 66 on page 242.

Table 66: jnxPMFlowEntry

Object	Object Identifier	Description
jnxPMonCurrentActiveFlows	jnxPMonFlowEntry 1	Monitors the number of currently active flows on a Passive Monitoring PIC.
jnxPMonTotalFlows	jnxPMonFlowEntry 2	Monitors the total flows on a Passive Monitoring PIC.
jnxPMonTotalFlowsPackets	jnxPMonFlowEntry 3	Monitors the total packet flows on a Passive Monitoring PIC.
jnxPMonTenSecondAverageFlowsPackes	jnxPMonFlowEntry 4	Monitors the number of packets in all flows in a 10-second average on a Passive Monitoring PIC.
jnxPMonTotalFlowsBytes	jnxPMonFlowEntry 5	Monitors the number of total of bytes in all flows on a Passive Monitoring PIC.
jnxPMonTenSecondAverageFlowBytes	jnxPMonFlowEntry 6	Monitors the number of bytes in all flows in a 10-second average on a Passive Monitoring PIC.
jnxPMonTotalFlowsExpired	jnxPMonFlowEntry 7	Monitors the number of total flows expired on a Passive Monitoring PIC.
jnxPMonTotalFlowsAged	jnxPMonFlowEntry 8	Monitors the number of total flows aged on a Passive Monitoring PIC.
jnxPMonTotalFlowsExported	jnxPMonFlowEntry 9	Monitors the number of total flows exported on a Passive Monitoring PIC.
jnxPMonTotalFlowsPacketsExported	jnxPMonFlowEntry10	Monitors the number of total flow packets exported on a Passive Monitoring PIC.

Related Documentation

- Passive Monitoring MIB on page 241

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CHAPTER 16

Interpreting the Enterprise-Specific SONET/SDH Interface Management MIB

- SONET/SDH Interface Management MIB on page 243
- jnxSonetAlarmsTable on page 243

SONET/SDH Interface Management MIB

The Juniper Networks enterprise-specific SONET/SDH Interface Management MIB sends the current alarm state for each SONET/SDH interface. When the alarm state changes on an interface, the MIB updates its alarm status. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-sonet.txt .

For information about the enterprise-specific SONET/SDH Interface Management MIB objects, see the following topic:

- jnxSonetAlarmsTable on page 243

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxSonetAlarmsTable

The **jnxSonetAlarmsTable**, whose object identifier is **{jnxSonetAlarm 1}**, provides information about alarm status on SONET/SDH physical interfaces.

The **jnxSonetAlarmEntry**, whose object identifier is **{jnxSonetAlarmTable 1}**, has five objects, which are listed in Table 67 on page 243.

Table 67: jnxSonetAlarmTable

Object	Object Identifier	Description
jnxSonetCurrentAlarms	**jnxSonetAlarmEntry 1**	Identifies all the active SONET/SDH alarms on this interface.

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Table 67: jnxSonetAlarmTable *(continued)*

Object	Object Identifier	Description
jnxSonetLastAlarmId	**jnxSonetAlarmEntry 2**	Identifies the SONET/SDH alarm that most recently was set or cleared.
jnxSonetLastAlarmTime	**jnxSonetAlarmEntry 3**	The value of **sysUpTime** when the management subsystem learned of the last alarm event.
jnxSonetLastAlarmDate	**jnxSonetAlarmEntry 4**	The system date and time when the management subsystem learned of the last alarm event.
jnxSonetLastAlarmEvent	**jnxSonetAlarmEntry 5**	Indicates whether the last alarm event set a new alarm or cleared an existing alarm.

Table 68 on page 244 provides an example of **jnxSonetAlarmInterface** objects of an M20 router.

Table 68: jnxSonetAlarmInterface Objects in the jnxSonetAlarmTable of an M20 Router

Alarm Interface	CurrentAlarms	Last Alarm ID	Last Alarm Time (System Up Time)	Last Alarm Date and Time	Last Alarm Event
14	**sonetLolAlarm(0) sonetLosAlarm(3)**	**sonetLosAlarm(3)**	0:01:37.15	2002-10-15, 10:21:14.0,-7:0	**set(2)**
15	**sonetLosAlarm(3)**	**sonetLosAlarm(3)**	8 days, 4:09:46.22	2002-10-23,14:29:23.0,-7:0	**set(2)**
16	**sonetLolAlarm(0) sonetLosAlarm(3)**	**sonetBerrSdAlarm(8)**	8 days, 4:09:46.21	2002-10-23,14:29:23.0,-7:0	**cleared(3)**
17	**sonetLofAlarm(2)**	**sonetLaisAlarm(5)**	8 days, 4:09:47.21	2002-10-23,14:29:24.0,-7:0	**cleared(3)**
18	–	**sonetLosAlarm(3)**	7 days, 4:31:27.53	2002-10-22,14:51:4.0,-7:0	**cleared(3)**
19	**sonetLolAlarm(0) sonetLosAlarm(3)**	**sonetLosAlarm(3)**	0:01:37.16	2002-10-15,10:21:14.0,-7:0	**set(2)**
20	**sonetLolAlarm(0) sonetLosAlarm(3)**	**sonetLosAlarm(3)**	0:01:37.17	2002-10-15,10:21:14.0,-7:0	**set(2)**
21	–	**sonetLofAlarm(2)**	7 days, 11:15:00.15	2002-10-22,21:34:37.0,-7:0	**cleared(3)**
22	**sonetLolAlarm(0) sonetLosAlarm(3)**	**sonetLolAlarm(0)**	7 days, 6:33:32.02	2002-10-22,16:53:8.0,-7:0	**set(2)**
23	–	**sonetLosAlarm(3)**	7 days, 6:33:45.02	2002-10-22,16:53:21.0,-7:0	**cleared(3)**

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Table 68: jnxSonetAlarmInterface Objects in the jnxSonetAlarmTable of an M20 Router *(continued)*

Alarm Interface	CurrentAlarms	Last Alarm ID	Last Alarm Time (System Up Time)	Last Alarm Date and Time	Last Alarm Event
24	**sonetLolAlarm(0)** **sonetLosAlarm(3)**	**sonetLosAlarm(3)**	0:01:37.07	2002-10-15,10:21:14.0,-7:0	**set(2)**
25	**sonetLolAlarm(0)** **sonetLosAlarm(3)**	**sonetLosAlarm(3)**	0:01:37.08	2002-10-15,10:21:14.0,-7:0	**set(2)**
26	–	–	0:00:00.00	0-0-0,0:0:0.0,	**none(1)**
27	**sonetLolAlarm(0)** **sonetLosAlarm(3)**	**sonetLosAlarm(3)**	0:01:38.04	2002-10-15,10:21:14.0,-7:0	**set(2)**
28	**sonetLolAlarm(0)** **sonetLosAlarm(3)**	**sonetLosAlarm(3)**	0:01:38.04	2002-10-15,10:21:14.0,-7:0	**set(2)**
29	**sonetLolAlarm(0)** **sonetLosAlarm(3)**	**sonetLosAlarm(3)**	0:01:38.04	2002-10-15,10:21:14.0,-7:0	**set(2)**

Related Documentation

- SONET/SDH Interface Management MIB on page 243

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CHAPTER 17

Interpreting the Enterprise-Specific SONET APS MIB

- SONET APS MIB on page 247
- apsConfigTable on page 248
- apsStatusTable on page 249
- apsChanConfigTable on page 252
- apsChanStatusTable on page 254

SONET APS MIB

The Juniper Networks enterprise-specific SONET Automatic Protection Switching (APS) MIB monitors any SONET interface that participates in APS. APS is used by SONET add/drop multiplexers (ADMs) to protect against circuit failures. The Junos OS implementation of APS allows you to protect against circuit failures between an ADM and one or more routers, and between multiple interfaces in the same router. When a circuit or router fails, a backup immediately takes over. For more information about APS, see the *Junos OS Network Interfaces Configuration Guide*.



NOTE: **The Junos OS supports only read access, 1+1 architecture, bidirectional, revertive, and nonrevertive mode.**

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-sonetaps.txt .

For information about the enterprise-specific SONET APS MIB objects, see the following topics:

- apsConfigTable on page 248
- apsStatusTable on page 249
- apsChanConfigTable on page 252
- apsChanStatusTable on page 254

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Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

apsConfigTable

apsConfigTable lists the APS groups that are configured on the system.

apsConfigEntry objects have read access only and are listed in Table 69 on page 248.

Table 69: apsConfigTable

Object	Object Identifier	Description
apsConfigName	apsConfigEntry 1	A text name for the APS group. An entry cannot exist in the active state unless all objects in the entry have an appropriate value. Also, all associated **apsChanConfigEntry** rows must represent a set of consecutive channel numbers beginning with 0 or 1, depending on the selected architecture.
apsConfigRowStatus	apsConfigEntry 2	The status of a APS group entry.
apsConfigMode	apsConfigEntry 3	The architecture of the APS group. Junos OS supports only the 1+1 architecture.
apsConfigRevert	apsConfigEntry 4	The revertive mode of the APS group. • Revertive mode—When the condition that caused a switch to the protection line has been cleared, the signal is switched back to the working line. Switching can optionally be revertive with 1+1 architecture. • Nonrevertive mode—Traffic remains on the protection line until another switch request is received.
apsConfigDirection	apsConfigEntry 5	The directional mode of the APS group. Junos OS supports only bidirectional mode. Bidirectional mode provides protection in both directions.
apsConfigExtraTraffic	apsConfigEntry 6	This object always returns the value disabled.
apsConfigSdBerThreshold	apsConfigEntry 7	The signal degrade bit error rate (BER). The negative value of this number is used as the exponent of 10 for computing the threshold value for the BER. For example, a value of 5 indicates a BER threshold of 10^-5.

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Table 69: apsConfigTable *(continued)*

Object	Object Identifier	Description
apsConfigSfBerThreshold	**apsConfigEntry 8**	The signal failure bit error rate (BER). The negative value of this number is used as the exponent of 10 for computing the threshold value for the BER. For example, a value of 5 indicates a BER threshold of 10^-5.
apsConfigWaitToRestore	**apsConfigEntry 9**	The wait to restore period, in seconds. After a condition that necessitated an automatic switch is cleared, the wait to restore period must elapse before reverting. This avoids rapid switch oscillations. GR-253-CORE specifies a range of 5 to 12 minutes. G.783 defines a range of 5 to 12 minutes in section 5.4.1.1.3, but also allows a shorter period in Table 2-1, **WaitToRestore** value (MI_WTRtime: 0..(5)..12 minutes).
apsConfigCreationTime	**apsConfigEntry 10**	The value of **sysUpTime** at the time the row was created.
apsConfigStorageType	**apsConfigEntry 11**	The storage type for this conceptual row. For information about conceptual rows, see RFC 2579, *Textual Conventions for SMIv2*.

Related Documentation

- SONET APS MIB on page 247
- apsStatusTable on page 249
- apsChanConfigTable on page 252
- apsChanStatusTable on page 254

apsStatusTable

apsStatusTable provides status information about configured APS groups.

apsStatusEntry objects have read access only and are listed in Table 70 on page 249.

Table 70: apsStatusTable

Object	Object Identifier	Description
apsStatusK1K2Rcv	**apsStatusEntry 1**	The current value of the K1 and K2 bytes received on the protection channel.
apsStatusK1K2Trans	**apsStatusEntry 2**	The current value of the K1 and K2 bytes transmitted on the protection channel.

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Table 70: apsStatusTable *(continued)*

Object	Object Identifier	Description
apsStatusCurrent	**apsStatusEntry 3**	The current status of the APS group. This object has the following values: • **modeMismatch**—Modes other than 1+1 unidirectional monitor protection line K2 bit 5, which indicates the architecture, and K2 bits 6 through 8, which indicate whether the mode is unidirectional or bidirectional. A conflict between the current local mode and the received K2 mode information constitutes a mode mismatch. Junos OS supports only bidirectional mode. • **channelMismatch**—A mismatch between the transmitted K1 channel and the received K2 channel has been detected. • **psbf**—A protection switch byte failure (PSBF) is in effect. This condition occurs when either an inconsistent APS byte or an invalid code is detected. An inconsistent APS byte occurs when no 3 consecutive K1 bytes of the last 12 successive frames are identical, starting with the last frame containing a previously consistent byte. An invalid code occurs when the incoming K1 byte contains an unused code or a code irrelevant for the specific switching operation (for example, reverse request while no switching request is outstanding) in three consecutive frames. An invalid code also occurs when the incoming K1 byte contains an invalid channel number in three consecutive frames. • **feplf**—Modes other than 1+1 unidirectional monitor the K1 byte for far-end protection-line failures. A far-end protection-line defect is declared based on receiving a signal failure (SF) on the protection line. • **extraTraffic**—Indicates whether extra traffic is currently being accepted on the protection line. • **extraTraffic**—Indicates whether extra traffic is currently being accepted on the protection line.
apsStatusModeMismatches	**apsStatusEntry 4**	Counts mode mismatch conditions. Discontinuities in the value of this counter can occur when the management system is reinitialized, and at other times as indicated by the value of **apsStatusDiscontinuityTime**.

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Table 70: apsStatusTable *(continued)*

Object	Object Identifier	Description
apsStatusChannelMis-matches	apsStatusEntry 5	Counts channel mismatch conditions. Discontinuities in the value of this counter can occur when the management system is reinitialized, and at other times as indicated by the value of **apsStatusDiscontinuityTime**.
apsStatusPSBFs	apsStatusEntry 6	Counts protection switch byte failure conditions. This condition occurs when either an inconsistent APS byte or an invalid code is detected. An inconsistent APS byte occurs when no 3 consecutive K1 bytes of the last 12 successive frames are identical, starting with the last frame containing a previously consistent byte. An invalid code occurs when the incoming K1 byte contains an unused code or a code irrelevant for the specific switching operation (for example, reverse request while no switching request is outstanding) in three consecutive frames. An invalid code also occurs when the incoming K1 byte contains an invalid channel number in three consecutive frames. Discontinuities in the value of this counter can occur when the management system is reinitialized, and at other times as indicated by the value of **apsStatusDiscontinuityTime**.
apsStatusFEPLFs	apsStatusEntry 7	Counts far-end protection-line failure conditions. This condition is declared based on receiving a signal failure (SF) on the protection line in the K1 byte. Discontinuities in the value of this counter can occur when the management system is reinitialized, and at other times as indicated by the value of **apsStatusDiscontinuityTime**.
apsStatusSwitchedChannel	apsStatusEntry 8	This field is set to the number of the channel that is currently switched to protection. The value 0 indicates that no channel is switched to protection. The values 1 through 14 indicate that the working channel is switched to protection.

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Table 70: apsStatusTable *(continued)*

Object	Object Identifier	Description
apsStatusDiscontinuity-Time	**apsStatusEntry 9**	The value of **sysUpTime** when the last one or more of this APS group's counters experienced a discontinuity. The relevant counters are the specific instances associated with this APS group of any **Counter32** object contained in **apsStatusTable**. If no such discontinuities have occurred since the last reinitialization of the local management subsystem, then this object contains a zero value.

Related Documentation

- SONET APS MIB on page 247
- apsConfigTable on page 248
- apsChanConfigTable on page 252
- apsChanStatusTable on page 254

apsChanConfigTable

apsChanConfigTable lists the APS channels that have been configured in APS groups.

apsChanConfigEntry objects have read access only and are listed in Table 71 on page 252.

Table 71: apsChanConfigEntry

Object	Object Identifier	Description
apsChanConfigGroupName	**apsChanConfigEntry 1**	A text name for the APS group in which this channel is included.
apsChanConfigNumber	**apsChanConfigEntry 2**	A unique channel number within an APS group. The value 0 indicates the null channel. The values 1 through 14 define a working channel.

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Table 71: apsChanConfigEntry *(continued)*

Object	Object Identifier	Description
apsChanConfigRowStatus	**apsChanConfigEntry 3**	The status of this APS channel entry. An entry cannot exist in the active state unless all objects in the entry have an appropriate value. Junos OS supports only 1+1 architecture. The values 1 through 14 define a working channel. When an attempt is made to set the corresponding **apsConfigRowStatus** field to active, the **apsChanConfigNumber** values of all entries with equal **apsChanConfigGroupName** fields must be a set of consecutive integer values beginning with 0 or 1, depending on the architecture of the group, and ending with *n*, where *n* is greater than or equal to 1 and less than or equal to 14. Otherwise, the error **inconsistentValue** is returned to the **apsConfigRowStatus** set attempt.
apsChanConfigIfIndex	**apsChanConfigEntry 4**	The interface index assigned to a SONET LTE. This is an interface with **ifType sonet(39)**. The value of this object must be unique among all instances of **apsChanConfigIfIndex**. In other words, a particular SONET LTE can only be configured in one APS group. This object cannot be set if the **apsChanConfigGroupName** instance associated with this row is equal to an instance of **apsConfigName** and the corresponding **apsConfigRowStatus** object is set to active. In other words, this value cannot be changed if the APS group is active. However, this value can be changed if the **apsConfigRowStatus** value is equal to **notInService**. Junos OS supports only read access.
apsChanConfigPriority	**apsChanConfigEntry 5**	The priority of the channel. This field returns the value low priority. Junos OS supports only 1+1 architecture.
apsChanConfigStorageType	**apsChanConfigEntry 6**	The storage type for this conceptual row. Conceptual rows having the value permanent need not allow write access to any columnar objects in the row. For information about conceptual rows, see RFC 2579, *Textual Conventions for SMIv2*.

Related Documentation

- SONET APS MIB on page 247
- apsConfigTable on page 248

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- apsStatusTable on page 249
- apsChanStatusTable on page 254

apsChanStatusTable

apasChanStatusTable provides APS channel statistics.

apsChanStatusEntry objects have read access only and are listed in Table 72 on page 254.

Table 72: apsChanStatusTable

Object	Object Identifier	Description
apsChanStatusCurrent	**apsChanStatusEntry** 1	The current state of the port. This object has the following values: **lockedOut**—This bit, when applied to a working channel, indicates that the channel is prevented from switching to the protection line. When applied to the null channel, this bit indicates that no working channel can switch to the protection line. **sd**—A signal degrade condition is in effect. **sf**—A signal failure condition is in effect switched. The switched bit is applied to a working channel if that channel is currently switched to the protection line. **wtr**—A wait-to-restore state is in effect.
apsChanStatusSignalDegrades	**apsChanStatusEntry** 2	A count of signal degrade conditions. A signal degrade condition occurs when the line bit error rate (BER) exceeds the currently configured value of the relevant instance of **apsConfigSdBerThreshold**. Discontinuities in the value of this counter can occur when the management system is reinitialized, and at other times as indicated by the value of **apsChanStatusDiscontinuityTime**.

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Table 72: apsChanStatusTable *(continued)*

Object	Object Identifier	Description
apsChanStatusSignalFailures	**apsChanStatusEntry 3**	A count of signal failure conditions that have been detected on the incoming signal. A signal failure condition occurs when a loss of signal, loss of frame, AIS-L or line bit error rate exceeds the currently configured value of the relevant instance of **apsConfigSfBerThreshold**. Discontinuities in the value of this counter can occur when the management system is reinitialized, and at other times as indicated by the value of **apsChanStatusDiscontinuityTime**.
apsChanStatusSwitchovers	**apsChanStatusEntry 4**	When queried with index value **apsChanConfigNumber** other than 0, this object returns the number of times this channel has switched to the protection line. When queried with index value **s** set to 0, which is the protection line, this object returns the number of times that any working channel has switched back to the working line from this protection line. Discontinuities in the value of this counter can occur when the management system is reinitialized, and at other times as indicated by the value of **apsChanStatusDiscontinuityTime**.
apsChanStatusLastSwitchover	**apsChanStatusEntry 5**	When queried with index value **apsChanConfigNumber** other than 0, this object returns the value of **sysUpTime** when this channel last completed a switch to the protection line. If this channel has never switched to the protection line, the value 0 is returned. When queried with index value **apsChanConfigNumber** set to 0, which is the protection line, this object returns the value of **sysUpTime** the last time that a working channel was switched back to the working line from this protection line. If no working channel has ever switched back to the working line from this protection line, the value 0 is returned.

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Table 72: apsChanStatusTable *(continued)*

Object	Object Identifier	Description
apsChanStatusSwitchoverSeconds	**apsChanStatusEntry 6**	The cumulative Protection Switching Duration (PSD) time, in seconds. For a working channel, this is the cumulative number of seconds that service was carried on the protection line. For the protection line, this is the cumulative number of seconds that the protection line has been used to carry any working channel traffic. This information is only valid if revertive switching is enabled. The value 0 is returned. Otherwise, discontinuities in the value of this counter can occur at reinitialization of the management system, and at other times as indicated by the value of **apsChanStatusDiscontinuityTime**. For example, if the value of an instance of **apsChanStatusSwitchoverSeconds** changes from a nonzero value to zero due to revertive switching being disabled. It is expected that the corresponding value of **apsChanStatusDiscontinuityTime** is updated to reflect the time of the configuration change.
apsChanStatusDiscontinuityTime	**apsChanStatusEntry 7**	The value of **sysUpTime** on the most recent occasion at which any one or more of this channel's counters suffered a discontinuity. The relevant counters are the specific instances associated with this channel of any **Counter32** object contained in **apsChanStatusTable**. If no such discontinuities have occurred since the last reinitialization of the local management subsystem, then this object contains a zero value for **apsChanStatusEntry**.

Related Documentation

- SONET APS MIB on page 247
- apsConfigTable on page 248
- apsStatusTable on page 249
- apsChanConfigTable on page 252

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CHAPTER 18

Interpreting the Enterprise-Specific IPsec Monitoring MIB

- IPSec Monitoring MIB on page 257
- jnxIkeTunnelTable on page 257
- jnxIPSecTunnelTable on page 260
- jnxIPSecSaTable on page 263

IPSec Monitoring MIB

The Juniper Networks enterprise-specific IPsec Monitoring MIB, whose object identifier is **{jnxMibs 22}**, provides operational and statistical information related to the IPsec and Internet Key Exchange (IKE) tunnels on Juniper Networks routers. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ipsec-monitor-asp.txt .

For information about IPsec Monitoring MIB objects, see the following topics:

- jnxIkeTunnelTable on page 257
- jnxIPSecTunnelTable on page 260
- jnxIPSecSaTable on page 263

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxIkeTunnelTable

The IKE tunnel table (**jnxIkeTunnelTable**), whose object identifier is **{jnxIPSecPhaseOne 1}**, is used to monitor the IKE security associations established with the remote peers. The MIB variables in this table are used to display the IKE SA attributes and the SA statistics. There is one entry for each IKE SA present.

The key for this table is the combination of a service set name, remote gateway address, and the IKE tunnel index. The service set name is used from the **jnxSpSvcSetTable** which is implemented as part of the Services PIC MIB. The SNMP manager uses the

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jnxSpSvcSetTable to get the service set name, and this information can then be used to query the **jnxIkeTunnelTable** for the given service set.

To get only IKE tunnels specific to a particular remote gateway in a service set, the SNMP manager can specify the corresponding service set name and the remote gateway address in the query.

The **jnxIkeTunnelEntry**, whose object identifier is **{jnxIkeTunnelTable 1}**, has 25 objects, which are listed in Table 73 on page 258. Each entry contains attributes associated with an active IPsec phase 1 IKE tunnel.

Table 73: jnxIkeTunnelTable

Object	Object Identifier	Description
jnxIkeTunIndex	**jnxIkeTunnelEntry 1**	Index for the table. The value of the index is a number that begins at 1 and is incremented with each tunnel that is created. When the index number reaches 2,147,483,647, the value wraps back to 1.
jnxIkeTunLocalRole	**jnxIkeTunnelEntry 2**	The role of the local peer identity. The role can be **initiator** or **responder**.
jnxIkeTunNegState	**jnxIkeTunnelEntry 3**	The state of the current negotiation. The state can be **matured** or **non matured**.
jnxIkeTunInitiatorCookie	**jnxIkeTunnelEntry 4**	Cookie generated by the peer that initiated the IKE phase 1 negotiation. This cookie is carried in the ISAKMP header.
jnxIkeTunResponderCookie	**jnxIkeTunnelEntry 5**	Cookie generated by the peer responding to the IKE phase 1 negotiation. This cookie is carried in the ISAKMP header.
jnxIkeTunLocalIdType	**jnxIkeTunnelEntry 6**	The type of local peer identity. A local peer can be identified by an IP address, a fully qualified domain name (FQDN), or a distinguished name.
jnxIkeTunLocalIdValue	**jnxIkeTunnelEntry 7**	The value of the local peer identity. • If the local peer type is an IP address, then this is the IP address used to identify the local peer. • If the local peer type is an FQDN(**if_fqdn**), then this is the FQDN of the remote peer. • If the local peer type is a distinguished name (**id_dn**), then this is the distinguished name of the local peer.
jnxIkeTunLocalGwAddrType	**jnxIkeTunnelEntry 8**	The IP address type of the local endpoint (gateway) for the IPsec phase 1 IKE tunnel.

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Table 73: jnxIkeTunnelTable *(continued)*

Object	Object Identifier	Description
jnxIkeTunLocalGwAddr	**jnxIkeTunnelEntry 9**	The IP address of the local endpoint (gateway) for the IPsec phase 1 IKE tunnel.
jnxIkeTunLocalCertName	**jnxIkeTunnelEntry 10**	The name of the certificate used for authentication of the local tunnel endpoint. This object has a valid value only if the negotiated IKE authentication method is something other than a preshared key. If the IKE negotiation does not use certificates for authentication, the value is **NULL**.
jnxIkeTunRemoteIdType	**jnxIkeTunnelEntry 11**	The type of remote peer identity. A remote peer can be identified by an IP address, an FQDN, or a distinguished name.
jnxIkeTunRemoteIdValue	**jnxIkeTunnelEntry 12**	The value of the remote peer identity. • If the remote peer type is an IP address, then this is the IP address used to identify the remote peer. • If the remote peer type is an FQDN (**if_fqdn**), then this is the FQDN of the remote peer. • If the remote peer type is a distinguished name (**id_dn**), then this is the distinguished name of the remote peer.
jnxIkeTunRemoteGwAddrType	**jnxIkeTunnelEntry 13**	The IP address type of the remote gateway (endpoint) for the IPsec phase 1 IKE tunnel.
jnxIkeTunRemoteGwAddr	**jnxIkeTunnelEntry 14**	The IP address of the remote gateway (endpoint) for the IPsec phase 1 IKE tunnel.
jnxIkeTunNegoMode	**jnxIkeTunnelEntry 15**	The negotiation mode of the IPsec phase 1 IKE tunnel.
jnxIkeTunDiffHellmanGrp	**jnxIkeTunnelEntry 16**	The Diffie Hellman Group used in IPsec phase 1 IKE negotiations.
jnxIkeTunEncryptAlgo	**jnxIkeTunnelEntry 17**	The encryption algorithm used in IPsec phase 1 IKE negotiations.
jnxIkeTunHashAlgo	**jnxIkeTunnelEntry 18**	The hash algorithm used in IPsec phase 1 IKE negotiations.
jnxIkeTunAuthMethod	**jnxIkeTunnelEntry 19**	The authentication method used in IPsec phase 1 IKE negotiations.
jnxIkeTunLifeTime	**jnxIkeTunnelEntry 20**	The negotiated lifetime (in seconds) of the IPsec phase 1 IKE tunnel.

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Table 73: jnxIkeTunnelTable *(continued)*

Object	Object Identifier	Description
jnxIkeTunActiveTime	**jnxIkeTunnelEntry 21**	The length of time (in hundredths of seconds) that the IPsec phase 1 IKE tunnel has been active.
jnxIkeTunInOctets	**jnxIkeTunnelEntry 22**	The total number of octets received by this IPsec phase 1 IKE security association.
jnxIkeTunInPkts	**jnxIkeTunnelEntry 23**	The total number of packets received by this IPsec phase 1 IKE security association.
jnxIkeTunOutOcets	**jnxIkeTunnelEntry 24**	The total number of octets sent by this IPsec phase 1 IKE security association.
jnxIkeTunOutPkts	**jnxIkeTunnelEntry 25**	The total number of octets sent by this IPsec phase 1 IKE security association.

Related Documentation

- IPSec Monitoring MIB on page 257
- jnxIPSecTunnelTable on page 260
- jnxIPSecSaTable on page 263

jnxIPSecTunnelTable

The IPsec phase 2 tunnel table **jnxIPSecTunnelTable)**, whose object identifier is **{jnxIPSecPhaseTwo 1}**, is used to monitor the IPsec phase 2 tunnel attributes along with the statistics for the tunnel. There is one entry for each tunnel to the peer security gateway. This table does not contain information about IPsec security associations (SAs) because multiple SAs can be present for each tunnel.

Similar to the IKE tunnel table (**jnxIkeTunnelTable**), the key of this table is a combination of the service set name, remote gateway address, and the IPsec tunnel index. This table can be queried just like the IKE tunnel table.

To get only IPsec tunnels specific to a particular remote gateway in a service set, the SNMP manager can specify the corresponding service set name and the remote gateway address in the query.

The **jnxIPSecTunnelEntry**, whose object identifier is **{jnxIPSecTunnelTable 1}**, has 27 objects, which are listed in Table 74 on page 261. Each entry contains attributes associated with an active IPsec phase 2 tunnel.

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Table 74: jnxIPSecTunnelTable

Object	Object Identifier	Description
jnxIPSecTunIndex	jnxIPSecTunnelEntry 1	Index for the table. The value of the index is a number that begins at 1 and is incremented with each tunnel that is created. When the index number reaches 2,147,483,647, the value wraps back to 1.
jnxIPSecRuleName	jnxIPSecTunnelEntry 2	The name of the rule defined in the IPsec configuration.
jnxIPSecTermName	jnxIPSecTunnelEntry 3	The name of the term configured under the IPsec rule.
jnxIPSecTunLocalGwAddrType	jnxIPSecTunnelEntry 4	The IP address type of the local gateway (endpoint) for the IPsec phase 2 tunnel.
jnxIPSecTunLocalGwAddr	jnxIPSecTunnelEntry 5	The IP address of the local gateway (endpoint) for the IPsec phase 2 tunnel.
jnxIPSecTunRemoteGwAddrType	jnxIPSecTunnelEntry 6	The IP address type of the remote gateway (endpoint) for the IPsec phase 2 tunnel.
jnxIPSecTunRemoteGwAddr	jnxIPSecTunnelEntry 7	The IP address of the remote gateway (endpoint) for the IPsec phase 2 tunnel.
jnxIPSecTunLocalProxyId	jnxIPSecTunnelEntry 8	The identifier for the local endpoint.
jnxIPSecTunRemoteProxyId	jnxIPSecTunnelEntry 9	The identifier for the remote endpoint.
jnxIPSecTunKeyType	jnxIPSecTunnelEntry 10	The type of key used by the IPsec phase 2 tunnel. The key type can be **IKE negotiated** or **Manually installed**.
jnxIPSecRemotePeerType	jnxIPSecTunnelEntry 11	The type of the remote peer gateway (endpoint). If the remote peer's IP address is known beforehand, the type is **static**. If the IP address is not known beforehand, the type is **dynamic**.
jnxIPSecTunMtu	jnxIPSecTunnelEntry 12	The maximum transmission unit (MTU) value of the IPsec phase 2 tunnel.
jnxIPSecTunOutEncryptedBytes	jnxIPSecTunnelEntry 13	The number of bytes encrypted by the IPsec phase 2 tunnel.
jnxIPSecTunOutEncryptedPkts	jnxIPSecTunnelEntry 14	The number of packets encrypted by the IPsec phase 2 tunnel.

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Table 74: jnxIPSecTunnelTable *(continued)*

Object	Object Identifier	Description
jnxIPSecTunInDecryptedBytes	jnxIPSecTunnelEntry 15	The number of bytes decrypted by the IPsec phase 2 tunnel.
jnxIPSecTunInDecryptedPkts	jnxIPSecTunnelEntry 16	The number of packets decrypted by the IPsec phase 2 tunnel.
jnxIPSecTunAHInBytes	jnxIPSecTunnelEntry 17	The number of incoming bytes authenticated using the authentication header (AH) by the IPsec phase 2 tunnel.
jnxIPSecTunAHInPkts	jnxIPSecTunnelEntry 18	The number of incoming packets authenticated using the authentication header (AH) by the IPsec phase 2 tunnel.
jnxIPSecTunAHOutBytes	jnxIPSecTunnelEntry 19	The number of outgoing bytes on the IPsec phase 2 tunnel where the AH is applied.
jnxIPSecTunHAOutPkts	jnxIPSecTunnelEntry 20	The number of outgoing packets on the IPsec phase 2 tunnel where the AH is applied.
jnxIPSecTunReplayDropPkts	jnxIPSecTunnelEntry 21	The number of packets dropped by the IPsec phase 2 tunnel because of an anti-replay check failure.
jnxIPSecTunAhAuthFails	jnxIPSecTunnelEntry 22	The number of packets received by the IPsec phase 2 tunnel that failed AH authentication.
jnxIPSecTunEspAuthFails	jnxIPSecTunnelEntry 23	The number of packets received by this IPsec phase 2 tunnel that failed ESP authentication.
jnxIPSecTunDecryptFails	jnxIPSecTunnelEntry 24	The number of packets received by this IPsec phase 2 tunnel that failed decryption.
jnxIPSecTunBadHeaders	jnxIPSecTunnelEntry 25	The number of packets received by this IPsec phase 2 tunnel that failed because of bad headers.
jnxIPSecTunBadTrailers	jnxIPSecTunnelEntry 26	The number of packets received by this IPsec phase 2 tunnel that failed because of bad ESP trailers.
jnxIPSecTunDroppedPkts	jnxIPSecTunnelEntry 27	The total number of packets dropped from this IPsec phase 2 tunnel.

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Related Documentation

- IPSec Monitoring MIB on page 257
- jnxIkeTunnelTable on page 257
- jnxIPSecSaTable on page 263

jnxIPSecSaTable

The IPsec phase 2 security association table (**jnxIPSecSaTable)**, whose object identifier is **{jnxIPSecPhaseTwo 2}**, is used to monitor the IPsec SAs present for each tunnel in the IPsec tunnel table (**jnxIPSecTunnelTable**). More than one pair of SAs can be present for each of the IPsec tunnels.

The key for this table is a combination of a service set name, remote gateway address, IPsec tunnel index, and the SA index. While the IPsec tunnel table is queried using the service set name, the SA table can be queried for the IPsec tunnel using the service set name, remote gateway address, and the IPsec tunnel index.

The **jnxIPSecSaEntry**, whose object identifier is **{jnxIPSecSaTable 1}**, has 16 objects, which are listed in Table 75 on page 263. Each entry contains SA components for an active IPsec phase 2 tunnel.

Table 75: jnxIPSecSaTable

Object	Object Identifier	Description
jnxIpSecSaProtocol	**jnxIpSecSaEntry** 1	The index represents the security protocol (AH, ESP, or IPComp) for which the SA was created.
jnxIpSecSaIndex	**jnxIpSecSaEntry 2**	The index (in the context of the IPsec tunnel) for the SA. The value of the index is a number that begins at 1 and is incremented with each security parameter index (SPI) associated with an IPsec phase 2 tunnel. When the index number reaches 2,147,483,647, the value wraps back to 1.
jnxIpSecSaInSpi	**jnxIpSecSaEntry 3**	The value of the incoming SPI.
jnxIpSecSaOutSpi	**jnxIpSecSaEntry 4**	The value of the outgoing SPI.
jnxIpSecSaInAuxSpi	**jnxIpSecSaEntry 5**	The value of the incoming auxiliary SPI. This object is valid for AH and ESP bundles.
jnxIpSecSaOutAuxSpi	**jnxIpSecSaEntry 6**	The value of the outgoing auxiliary SPI. This object is valid for AH and ESP bundles.
jnxIpSecSaType	**jnxIpSecSaEntry 7**	The type of SA (**manual** or **dynamic**).
jnxIpSecSaEncapMode	**jnxIpSecSaEntry 8**	The encapsulation mode used by the IPsec phase 2 tunnel.

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Table 75: jnxIPSecSaTable *(continued)*

Object	Object Identifier	Description
jnxIpSecSaLifeSize	**jnxIpSecSaEntry 9**	The negotiated size (in kilobytes) of the IPsec phase 2 tunnel.
jnxIpSecSaLifeTime	**jnxIpSecSaEntry 10**	The negotiated **lifetime** (in seconds) of the IPsec phase 2 tunnel.
jnxIpSecSaActiveTime	**jnxIpSecSaEntry 11**	The number of seconds the IPsec phase 2 tunnel has been active.
jnxIpSecSaLifeSizeThreshold	**jnxIpSecSaEntry 12**	The refresh threshold (in kilobytes) of the SA size.
jnxIpSecSaLifeTimeThreshold	**jnxIpSecSaEntry 13**	The refresh threshold (in seconds) of the SA lifetime.
jnxIpSecSaEncryptAlgo	**jnxIpSecSaEntry 14**	The algorithm used to encrypt the packets (**es-cbc** or **3des-cbc**).
jnxIpSecSaAuthAlgo	**jnxIpSecSaEntry 15**	The algorithm used to authenticate the packets (**hmac-md5-96** or **hmac-sha1-96**).
jnxIpSecSaState	**jnxIpSecSaEntry 16**	The status of the SA. Status can be **active** (ready for active use) or **expiring** (any state an SA goes through before being purged).

Related Documentation

- IPSec Monitoring MIB on page 257
- jnxIkeTunnelTable on page 257
- jnxIPSecTunnelTable on page 260

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CHAPTER 19

Interpreting the Enterprise-Specific Ethernet MAC MIB

- Ethernet MAC MIB on page 265
- jnxMacStatsTable on page 265

Ethernet MAC MIB

The Juniper Networks enterprise-specific Ethernet Media Access Control (MAC) MIB, whose object identifier is **{jnxMibs 23}**, monitors media access control statistics on Gigabit Ethernet intelligent queuing (IQ) interfaces. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-mac.txt .

For information about the enterprise-specific Ethernet MAC MIB objects, see the following topic:

- jnxMacStatsTable on page 265

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxMacStatsTable

The **jnxMacStatsTable** contains a list of MAC statistics for Gigabit Ethernet interfaces.

jnxMacStatsEntry objects are listed in Table 76 on page 265.

Table 76: jnxMacStatsTable

Object	Object Identifier	Description
jnxVlanIndex	**jnxMacStatsEntry 1**	The virtual LAN (VLAN) ID of a VLAN.
jnxSourceMacAddress	**jnxMacStatsEntry 2**	The source MAC address.
jnxMacHCInOctets	**jnxMacStatsEntry 3**	The number of total octets received in this VLAN/MAC address.

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Table 76: jnxMacStatsTable *(continued)*

Object	Object Identifier	Description
jnxMacHCInFrames	**jnxMacStatsEntry 4**	The number of total frames received in this VLAN/MAC address
jnxMacHCOutOctets	**jnxMacStatsEntry 5**	The number of total octets transmitted in this VLAN/MAC address.
jnxMacHCOutFrames	**jnxMacStatsEntry 6**	The number of total frames transmitted in this VLAN/MAC address.

Related Documentation

- Ethernet MAC MIB on page 265

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CHAPTER 20

Interpreting the Enterprise-Specific Interface MIB

- Interface MIB on page 267
- jnxIfTable on page 267
- ifChassisTable on page 270

Interface MIB

The Juniper Networks enterprise-specific Interface MIB extends the standard **ifTable** (RFC 2863) with additional statistics and Juniper Networks enterprise-specific chassis information. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-if-extensions.txt .

For information about the enterprise-specific Interface MIB objects, see the following topics:

- jnxIfTable on page 267
- ifChassisTable on page 270

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxIfTable

jnxIfTable lists traffic statistics, input errors, and output errors for an interface.

jnxIfEntry objects are listed in Table 77 on page 267.

Table 77: jnxIfTable

Object	Object Identifier	Description
ifIn1SecRates	**ifJnxEntry 1**	The number of bits per second (bps) delivered by this sublayer to its next higher sublayer.

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Table 77: jnxIfTable *(continued)*

Object	Object Identifier	Description
ifIn1SecOctets	**ifJnxEntry 2**	The number of octets per second delivered by this sublayer to its next higher sublayer.
ifIn1SecPkts	**ifJnxEntry 3**	The number of packets per second (pps) delivered by this sublayer to its next higher sublayer.
ifOut1SecRate	**ifJnxEntry 4**	The number of bps delivered by this sublayer to its next lower sublayer.
ifOut1SecOctets	**ifJnxEntry 5**	The number of octets per second delivered by this sublayer to its next lower sublayer.
ifOut1SecPkts	**ifJnxEntry 6**	The number of pps delivered by this sublayer to its next lower sublayer.
ifHCIn1SecRate	**ifJnxEntry 7**	The number of bps delivered by this sublayer to its next higher sublayer. This object is a 64-bit version of **ifIn1SecRate**.
ifHCOut1SecRate	**ifJnxEntry 8**	The number of bps delivered by this sublayer to its next lower sublayer. This object is a 64-bit version of **ifOut1SecRate**.
ifJnxInErrors	**ifJnxEntry 9**	Errors: The sum of the incoming frame aborts and FCS errors.
ifJnxInFrameErrors	**ifJnxEntry 10**	Framing Errors: The number of input packets that were misaligned.
ifJnxInQDrops	**ifJnxEntry 11**	Drops: The number of packets dropped by the input queue of the I/O Manager ASIC.
ifJnxInRunts	**ifJnxEntry 12**	Runts: Frames received that are smaller than the runt threshold.
ifJnxInGiants	**ifJnxEntry 13**	Giants: Frames received that are larger than the giant threshold.
ifJnxInDiscards	**ifJnxEntry 14**	Policed discards: Frames that the incoming packet match code discarded because they were not recognized or of interest.
ifJnxInHslCrcErrors	**ifJnxEntry 15**	HS link CRC errors: The number of CRC errors on the high-speed links between the ASICs responsible for handling the router interfaces while receiving packets.
ifJnxInHslFifoOverFlows	**ifJnxEntry 16**	HS link FIFO overflows: The number of FIFO overflows on the high-speed links between the ASICs responsible for handling the router interfaces.

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Table 77: jnxIfTable *(continued)*

Object	Object Identifier	Description
ifJnxInL3Incompletes	**ifJnxEntry 17**	L3 incompletes: The number of incoming packets that fail Layer 3 sanity checks of the header.
ifJnxInL2ChanErrors	**ifJnxEntry 18**	L2 channel errors: The number of incoming packets for which the software could not find a valid logical interface.
ifJnxInL2MismatchTimeouts	**ifJnxEntry 19**	L2 mismatch timeouts: The count of malformed or short packets that cause the incoming packet handler to discard the frame as unreadable.
ifJnxInInvalidVCs	**ifJnxEntry 20**	Invalid virtual circuits (VCs): The number of cells that arrived for a nonexistent VC.
ifJnxInFifoErrors	**ifJnxEntry 21**	FIFO errors: The number of FIFO errors in the received direction as reported by the ASIC on the PIC.
ifJnxBucketDrops	**ifJnxEntry 22**	Bucket drops: Drops because traffic load exceeded the interface transmit and receive leaky bucket configuration.
ifJnxSramErrors	**ifJnxEntry 23**	SRAM errors: This counter increments when a hardware error has occurred in the SRAM on the PIC.
ifJnxOutErrors	**ifJnxEntry 24**	Errors: The sum of the outgoing frame aborts and FCS errors.
ifJnxCollisions	**ifJnxEntry 25**	Collisions: The number of output collisions detected on this interface.
ifJnxCarrierTrans	**ifJnxEntry 26**	Carrier transitions: The number of times the interface saw the carrier signal transition.
ifJnxOutQDrops	**ifJnxEntry 27**	Drops: The number of packets dropped by the output queue of the I/O Manager ASIC.
ifJnxOutAgedErrors	**ifJnxEntry 28**	Aged packets: The number of packets that remained in shared packet SDRAM for so long that the system automatically purged them.
ifJnxOutFifoErrors	**ifJnxEntry 29**	FIFO errors: The number of FIFO errors in the transmit direction as reported by the ASIC on the PIC.
ifJnxOutHslFifoUnderFlows	**ifJnxEntry 30**	HS link FIFO underflows: The number of FIFO underflows on the high-speed links between the ASICs responsible for handling the router interfaces.

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Table 77: jnxIfTable *(continued)*

Object	Object Identifier	Description
ifJnxOutHslCrcErrors	**ifJnxEntry 31**	HS link CRC errors: The number of CRC errors on the high-speed links between the ASICs responsible for handling the router interfaces while transmitting packets.

Related Documentation

- Interface MIB on page 267
- ifChassisTable on page 270

ifChassisTable

ifChassisTable provides additional interface and chassis information.

ifChassisEntry objects are listed in Table 78 on page 270.

Table 78: ifChassisTable

Object	Object Identifier	Description
ifChassisFpc	**ifChassisEntry 1**	The number of the FPC card on which the interface is located in the chassis. It is the chassis slot in which the FPC card is installed for the specified interface. Although the number is labeled from 0 and up in the chassis, the return value for this object always starts from 1 according to network management convention. Therefore, a value of zero means there is no real or physical FPC associated with the specified interface.
ifChassisPic	**ifChassisEntry 2**	The number of the PIC card on which the interface is located in the chassis. It is the PIC location on the FPC card for the specified interface. Although the number is labeled from 0 and up in the chassis, the return value for this object always starts from 1 according to network management convention. Therefore, a value of zero means there is no real or physical PIC associated with the specified interface.
ifChassisPort	**ifChassisEntry 3**	The number of the port on the PIC card on which the interface is located in the chassis. It is the port number on the PIC card for the specified interface. Although the number is labeled from 0 and up in the chassis, the return value for this object always starts from 1 according to network management convention. Therefore, a value of zero means there is no real or physical port associated with the specified interface.

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Table 78: ifChassisTable *(continued)*

Object	Object Identifier	Description
ifChassisChannel	**ifChassisEntry 4**	The channel identifier for the specified interface if it is part of a channelized interface. Although the channel is numbered from 0 and up in the interface naming, the return value for this object always starts from 1 according to network management convention. For an interface that could not be channelized, this object returns zero.
ifChassisLogicalUnit	**ifChassisEntry 5**	The logical unit number of the specified interface. It is the logical part of the interface that is configured on the physical or channel part, if any. Although the logical unit number is numbered from 0 and up in the interface naming, the return value for this object always starts from 1 according to network management convention. For an interface that is really a physical device, this value returns zero.
ifChassisPicIndex	**ifChassisEntry 6**	The indexes for the Chassis MIB tables. This is the instance index that keys into **jnxContentsTable** in the Chassis MIB. For example, the octet string of 8.1.2.0 means a PIC (“ 8&” first digit) at FPC slot 0 (“1–1” , second digit minus one if nonzero) PIC number 1 (“2–1” , third digit) minus one if nonzero port number, whatever (fourth digit currently unused). In turn, this PIC index can be plugged in by the NMS directly after any MIB objects in the **jnxContentsTable** obtain that PIC object for the specified interface. This object is valid only for interfaces having real and physical PIC cards. Otherwise, it returns an octet string “0.0.0.0.”

Related Documentation

- Interface MIB on page 267
- jnxIfTable on page 267

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CHAPTER 21

Interpreting the Enterprise-Specific VPN MIB

- VPN MIB on page 273
- jnxVpnInfo on page 274
- jnxVpnTable on page 275
- jnxVpnIfTable on page 276
- jnxVpnPwTable on page 279
- jnxVpnRTTable on page 283
- VPN Traps on page 284

VPN MIB

The Juniper Networks enterprise-specific Virtual Private Network (VPN) MIB, whose object identifier is **{jnxMibs 26}**, provides monitoring for the following type of VPNs:

- Layer 2 based on Internet draft draft-kompella-l2ppvpn-version.txt, *MPLS-based Layer 2 VPNs*.
- Layer 3 based on Internet draft draft-ietf-l3vpn-rfc2547bis-03.txt, *BGP and MPLS IP VPNs.*
- VPLS based on Internet draft draft-ietf-ppvpn-vpls-bgp-00.txt, *Virtual Private LAN Service*.



NOTE: **SNMP cannot be associated with any routing instances other than the master routing instance.**

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-vpn.txt .

For information about the enterprise-specific VPN MIB objects, see the following topics:

- jnxVpnInfo on page 274
- jnxVpnTable on page 275

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- jnxVpnIfTable on page 276
- jnxVpnPwTable on page 279
- jnxVpnRTTable on page 283
- VPN Traps on page 284

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxVpnInfo

jnxVpnInfo, whose object identifier is **{jnxVpnMibObjects 1}**, contains information about the number of configured VPNs and active VPNs.

Table 79 on page 274 lists the supported **jnxVpnInfo** objects, VPNs, and circuit connection services.

Table 79: Supported jnxVpnInfo Objects, VPNs, and Circuit Connection Services

Object	Object Identifier	Layer 3 VPN	Layer 2 VPN	VPLS VPN	Circuit Cross-Connect	L2 Circuit	Optical VPN	Description
jnxVpnConfiguredVpns	**jnxVpnInfo 1**	Yes	Yes	Yes	No	Yes	–	Number of configured VPNs.
jnxVpnActiveVpns	**jnxVpnInfo 2**	Yes	Yes	Yes	No	Yes	–	Number of active VPNs.
jnxVpnNextIfIndex	**jnxVpnInfo 3**	–	–	–	–	–	–	Next free VPN interface index.
jnxVpnNextPwIndex	**jnxVpnInfo 4**	–	–	–	–	–	–	Next free pseudowire index.
jnxVpnNextRTIndex	**jnxVpnInfo 5**	–	–	–	–	–	–	Next free route target index.

Related Documentation

- VPN MIB on page 273
- jnxVpnTable on page 275
- jnxVpnIfTable on page 276
- jnxVpnPwTable on page 279
- jnxVpnRTTable on page 283
- VPN Traps on page 284

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jnxVpnTable

jnxVpnTable, whose object identifier is **{jnxVpnMibobjects 2}**, lists configured VPNs.

jnxVpnEntry contains information about a configured VPN with the objects listed in Table 80 on page 275 and their supported VPNs and circuit connection services. The first two objects in **jnxVpnEntry** (**jnxVpnType** and **jnxVpnname**) are indexes and are not included in this table.

Table 80: Supported jnxVpnEntry Objects, VPNs, and Circuit Connection Services

Object	Object Identifier	Layer 3 VPN	Layer 2 VPN	VPLS VPN	L2 Circuit	Circuit Cross-Connect	Optical VPN	Description
jnxVpnRowStatus	**jnxVpnEntry 3**	–	–	–	–	–	–	Creates, modifies, or deletes a row in this table.
jnxVpnStorageType	**jnxVpnEntry 4**	–	–	–	–	–	–	The storage type.
jnxVpnDescription	**jnxVpnEntry 5**	Yes	Yes	Yes	Yes	No	–	VPN description.
jnxVpnIdentifierType	**jnxVpnEntry 6**	Yes	Yes	Yes	Yes	No	–	Type of **jnxVpnIdentifer**.
jnxVpnIdentifier	**jnxVpnEntry 7**	Yes	Yes	Yes	Yes	No	–	For BGP VPNs, the route distinguisher for the VPN. For LDP VPNs, the virtual circuit (VC) ID for the circuit. A value of all zeros indicates that a route distinguisher and a VC ID are not configured for the VPN.
jnxVpnConfiguredSites	**jnxVpnEntry 8**	–	No	No	No	No	–	The number of sites configured in the VPN.
jnxVpnActiveSites	**jnxVpnEntry 9**	–	No	No	No	No	–	The number of active sites in the VPN.
jnxVpnLocalAddresses	**jnxVpnEntry 10**	No	No	No	No	No	–	The number of addresses learned from the CE device.
jnxVpnTotalAddresses	**jnxVpnEntry 11**	No	No	No	No	No	–	The total number of addresses in the VPN routing table.

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Table 80: Supported jnxVpnEntry Objects, VPNs, and Circuit Connection Services *(continued)*

Object	Object Identifier	Layer 3 VPN	Layer 2 VPN	VPLS VPN	L2 Circuit	Circuit Cross-Connect	Optical VPN	Description
jnxVpnVpnAge	**jnxVpnEntry 12**	Yes	Yes	Yes	Yes	No	–	The age of the VPN, in hundredths of a second.

Related Documentation

- VPN MIB on page 273
- jnxVpnInfo on page 274
- jnxVpnIfTable on page 276
- jnxVpnPwTable on page 279
- jnxVpnRTTable on page 283
- VPN Traps on page 284

jnxVpnIfTable

The **jnxVpnIfTable**, whose object identifier is **{jnxVpnMibObjects 3}**, lists VPN interfaces.

jnxVpnIfEntry contains information about VPN interfaces, and has the objects listed in Table 81 on page 276. The first three objects (**jnxVpnIfVpnType**, **jnxVpnIfVpnName**, and **jnxVpnIfIndex**) are indexes and are not included in this table.

Table 81: Supported jnxVpnIfEntry Objects, VPNs, and Circuit Connection Services

Object	Object Identifier	Layer 3 VPN	Layer 2 VPN	VPLS VPN	L2 Circuit	Circuit Cross-Connect	Optical VPN	Description
jnxVpnIfRowStatus	**jnxVpnifEntry 4**	–	–	–	–	–	–	Creates, modifies, or deletes a row in this table.
jnxVpnIfStorageType	**jnxVpnifEntry 5**	–	–	–	–	–	–	Identifies the storage type for an object.

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Table 81: Supported jnxVpnIfEntry Objects, VPNs, and Circuit Connection Services *(continued)*

Object	Object Identifier	Layer 3 VPN	Layer 2 VPN	VPLS VPN	L2 Circuit	Circuit Cross-Connect	Optical VPN	Description
jnxVpnIfAssociationPw	**jnxVpnifEntry 6**	–	Yes	Yes	Yes	No	–	The index of the associated pseudowire. If no index is associated with a pseudowire, the index is 0. A pseudowire is a mechanism that carries essential elements of an emulated circuit from one provider edge (PE) device to one or more other PEs over a PSN.
jnxVpnIfProtocol	**jnxVpnifEntry 7**	No	Yes	Yes	Yes	No	–	Indicates the protocol running over a VPN interface. This object contains the following values: • **other(0)** • **frameRelay(1)** • **atmAal5(2)** • **atmCell(3)** • **ethernetVlan(4)** • **ethernet(5)** • **ciscoHdlc(6)** • **ppp(7)** • **cem(8)** • **atmVcc(9)** • **atmVpc(10)** • **vpls(11)** • **ipInter-working(12)** • **snapInter-working(13)** • **static(20)** • **rip(21)** • **ospf(22)** • **bgp(23)** • **atmTrunkNNI (129)** • **atmTrunkUNI (130)**

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Table 81: Supported jnxVpnIfEntry Objects, VPNs, and Circuit Connection Services *(continued)*

Object	Object Identifier	Layer 3 VPN	Layer 2 VPN	VPLS VPN	L2 Circuit	Circuit Cross-Connect	Optical VPN	Description
jnxVpnIfInBandwidth	**jnxVpnifEntry 8**	No	No	No	No	No	–	The maximum bandwidth that the customer edge (CE) device connected over a VPN can send to the PE device, in Kbps. A value of 0 indicates that there is no configured maximum.
jnxVpnIfOutBandwidth	**jnxVpnifEntry 9**	No	No	No	No	No	–	The maximum bandwidth that the PE device can send to the CE device over a VPN interface, in Kbps. A value of 0 indicates that there is no configured maximum.
jnxVpnIfStatus	**jnxVpnifEntry 10**	Yes	Yes	Yes	Yes	No	–	Status of a monitored VPN interface. This object contains the following values: • **unknown(0)** • **noLocal-Interface(1)** • **disabled(2)** • **encapsulation-Mismatch(3)** • **down(4)** • **up(5)**

Related Documentation

- VPN MIB on page 273
- jnxVpnInfo on page 274
- jnxVpnTable on page 275
- jnxVpnPwTable on page 279
- jnxVpnRTTable on page 283
- VPN Traps on page 284

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jnxVpnPwTable

jnxVpnPwTable, whose object identifier is **{jnxVpnMibObjects 4}**, lists pseudowire connections.

jnxVpnPwEntry contains pseudowire information about a VPN that is being monitored, and has the objects listed in Table 82 on page 279. The first three objects (**jnxVpnPwVpnType**, **jnxVpnPwVpnName**, and **jnxVpnPwIndex)** are indexes and are not listed in this table.

Table 82: Supported jnxVpnPwEntry Objects, VPNs, and Circuit Connection Services

Object	Object Identifier	Layer 3 VPN	Layer 2 VPN	VPLS VPN	L2 Circuit	Circuit Cross-Connect	Optical VPN	Description
jnxVpnPwRowStatus	jnxVpnPwEntry 4	–	–	–	–	–	–	Creates, modifies, and deletes a row in this table.
jnxVpnPwStorageType	jnxVpnPwEntry 5	–	–	–	–	–	–	The storage type.
jnxVpnPwAssociatedInterface	jnxVpnPwEntry 6	–	Yes	Yes	Yes	No	–	The VPN index of the interface associated with a pseudowire. If no interface is associated with a pseudowire, 0 is returned.
jnxVpnPwLocalSiteId	jnxVpnPwEntry 7	–	Yes	Yes	Yes	No	–	The local site identifier for a pseudowire. When there is no local site identifier, 0 is returned.
jnxVpnPwRemoteSiteId	jnxVpnPwEntry 8	–	Yes	Yes	Yes	No	–	The remote site identifier. For example, the site at the end of the pseudowire. When there is no remote site identifier, 0 is returned.

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Table 82: Supported jnxVpnPwEntry Objects, VPNs, and Circuit Connection Services *(continued)*

Object	Object Identifier	Layer 3 VPN	Layer 2 VPN	VPLS VPN	L2 Circuit	Circuit Cross-Connect	Optical VPN	Description
jnxVpnRemotetPeIdAddrType	**jnxVpnPwEntry 9**	–	Yes	Yes	Yes	No	–	The remote PE address. For example, the router at the end of the pseudowire.
jnxVpnRemotePeIdAddress	**jnxVpnPwEntry 10**	–	Yes	Yes	Yes	No	–	The type of tunnel over which the pseudowire is carried. If several pseudowires can be carried in one tunnel, each pseudowire is identified by the multiplexer or demultiplexer within a tunnel. This object can contain the following values: • **static(1)** • **gre(2)** • **l2tpv3(3)** • **ipSec(4)** • **ldp(5)** • **rsvpTe(6)** • **crLdp(7)**
jnxVpnPwTunnelType	**jnxVpnPwEntry 11**	–	Yes	Yes	Yes	No	–	The type of tunnel over which the pseudowire is carried.
jnxVpnPwTunnelName	**jnxVpnPwEntry 12**	–	Yes	Yes	Yes	No	–	The name of the tunnel over which a pseudowire is carried.

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Table 82: Supported jnxVpnPwEntry Objects, VPNs, and Circuit Connection Services *(continued)*

Object	Object Identifier	Layer 3 VPN	Layer 2 VPN	VPLS VPN	L2 Circuit	Circuit Cross-Connect	Optical VPN	Description
jnxVpnPwReceiveDemux	**jnxVpnPwEntry 13**	–	Yes	Yes	Yes	No	–	The demultiplexer value that identifies received packets associated with this pseudowire.
jnxVpnPwTransmitDemux	**jnxVpnPwEntry 14**	–	Yes	Yes	Yes	No	–	The demultiplexer value that identifies the transmitted packets associated with this pseudowire.
jnxVpnPwStatus	**jnxVpnPwEntry 15**	–	Yes	Yes	Yes	No	–	The status of the pseudowire. This object can have the following values: • **unknown(0)** • **down(1)** • **up(2)**
jnxVpnPwTunnelStatus	**jnxVpnPwEntry 16**	–	No	No	No	No	–	The status of the PE-to-PE tunnel over which the pseudowire is carried.

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Table 82: Supported jnxVpnPwEntry Objects, VPNs, and Circuit Connection Services *(continued)*

Object	Object Identifier	Layer 3 VPN	Layer 2 VPN	VPLS VPN	L2 Circuit	Circuit Cross-Connect	Optical VPN	Description
jnxVpnPwRemoteSiteStatus	**jnxVpnPwEntry 17**	–	No	No	No	No	–	The interface status at the remote end of the pseudowire. This object can have the following values: • **unknown(0)** • **outOf - Range(1)** • **down(2)** • **up(3)**
jnxVpnPwTimeUp	**jnxVpnPwEntry 18**	–	Yes	Yes	Yes	No	–	The time, in hundredths of a second, that a pseudowire has been operational.
jnxVpnPwTransitions	**jnxVpnPwEntry 19**	–	Yes	Yes	Yes	No	–	The number of state transitions (up to down and down to up) that a tunnel has undergone.
jnxVpnPwLastTransition	**jnxVpnPwEntry 20**	–	Yes	Yes	Yes	No	–	The time, in hundredths of a second, since the last transition occurred in a tunnel.
jnxVpnPwPacketsSent	**jnxVpnPwEntry 21**	–	No	No	No	No	–	The number of packets sent over a pseudowire.
jnxVpnPwOctetsSent	**jnxVpnPwEntry 22**	–	No	No	No	No	–	The number of octets sent over a pseudowire.

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Table 82: Supported jnxVpnPwEntry Objects, VPNs, and Circuit Connection Services *(continued)*

Object	Object Identifier	Layer 3 VPN	Layer 2 VPN	VPLS VPN	L2 Circuit	Circuit Cross-Connect	Optical VPN	Description
jnxVpnPwPacketsReceived	**jnxVpnPwEntry 23**	No	No		No	No	–	The number of packets received over a pseudowire.
jnxVpnPwOctetsReceived	**jnxVpnPwEntry 24**	No	No		No	No	–	The number of octets received over a pseudowire.
jnxVpnPwLRPacketsSent	**jnxVpnPwEntry 25**	No	No		No	No	–	The number of packets sent over a pseudowire.
jnxVpnPwLROctetsSent	**jnxVpnPwEntry 26**	No	No		No	No	–	The number of octets sent over a pseudowire.
jnxVpnPwLRPacketsReceived	**jnxVpnPwEntry 27**	No	No		No	No	–	The number of packets received over a pseudowire.
jnxVpnPwLROctetsReceived	**jnxVpnPwEntry 28**	No	No		No	No	–	The number of octets received over a pseudowire.

Related Documentation

- VPN MIB on page 273
- jnxVpnInfo on page 274
- jnxVpnTable on page 275
- jnxVpnIfTable on page 276
- jnxVpnRTTable on page 283
- VPN Traps on page 284

jnxVpnRTTable

The **jnxVpnRTTable**, whose object identifier is **{jnxVpnMibObjects 4}**, contains route targets for a VPN.

jnxVpnRTEntry lists route targets for a given VPN, and has the objects listed in Table 83 on page 284. The first three objects (**jnxVpnRTVpnType**, **jnxVpnRTVpnName**, and **jnxVpnRTIndex**) are indexes and are not listed in this table.

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Table 83: Supported jnxVpnRTEntry Objects, VPNs, and Circuit Connection Services

Object	ObjectIdentifier	Layer 3 VPN	Layer 2 VPN	VPLS	L2 Circuit	Circuit Cross-Connect	Optical VPN	Description
jnxVpnRTRowStatus	**jnxVpnRTEntry 4**	–	–	–	–	–	–	Creates, modifies, or deletes a row in this table.
jnxVpnRTStorageType	**jnxVpnRTEntry 5**	–	–	–	–	–	–	Identifies the storage type for an object.
jnxVpnRTType	**jnxVpnRTEntry 6**	Yes	Yes	Yes	–	No	–	The type of the following route target. The type can be **routeTarget[012]** or **none**.
jnxVpnRT	**jnxVpnRTEntry 7**	Yes	Yes	Yes	–	No	–	The VPN route target. If **jnxVpnRTType** is **none**, the value must be all zeros.
jnxVpnRTFunction	**jnxVpnRTEntry 8**	Yes	Yes	Yes	–	No	–	The route target export distribution type.

Related Documentation

- VPN MIB on page 273
- jnxVpnInfo on page 274
- jnxVpnTable on page 275
- jnxVpnIfTable on page 276
- jnxVpnPwTable on page 279
- VPN Traps on page 284

VPN Traps

The enterprise-specific VPN MIB provides traps for monitoring VPNs. Table 84 on page 285 lists supported VPN traps, VPNs, and circuit connection services.

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Table 84: Supported VPN Traps, VPNs, and Circuit Connection Services

Object	Object Identifier	Layer 3 VPN	Layer 2 VPN	VPLS VPN	L2 Circuit	Circuit Cross-Connect	Optical VPN	Description
jnxVpnIfUp	**jnxVpnMIBnotificatios 1**	Yes	Yes	–	Yes	No	–	Indicates that the interface with the index **jnxVpnIfIndex** belonging to the **jnxVpnIfVpnName** of type **jnxVpnIfVpnType** went up.
jnxVpnIfDown	**jnxVpnMIBnotificatios 2**	Yes	Yes	–	Yes	No	–	Indicates that the interface with index **jnxVpnIfIndex** belonging to **jnxVpnIfVpnName** of type **jnxVpnIfVpnType** went down.
jnxVpnPwUp	**jnxVpnMIBnotificatios 3**	No	Yes	Yes	Yes	No	–	Indicates that the pseudowire with the index **jnxVpnPwIndex** belonging to **jnxVpnPwVpnName** of type **jnxVpnPwVpnType** went up.
jnxVpnPwDown	**jnxVpnMIBnotificatios 4**	No	Yes	Yes	Yes	No	–	Indicates that the pseudowire with index **jnxVpnPwIndex** belonging to **jnxVpnPwVpnName** of type **jnxVpnPwVpnType** went down.

Related Documentation

- VPN MIB on page 273
- jnxVpnInfo on page 274
- jnxVpnTable on page 275
- jnxVpnIfTable on page 276
- jnxVpnPwTable on page 279
- jnxVpnRTTable on page 283
- Configuring SNMP Trap Options and Groups on a Device Running Junos OS
- Managing Traps and Informs

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CHAPTER 22

Interpreting the Enterprise-Specific Flow Collection Services MIB

- Flow Collection Services MIB on page 287
- jnxCollGlobalStats on page 287
- jnxCollPicIfTable on page 288
- jnxCollFileTable on page 290

Flow Collection Services MIB

The Juniper Networks enterprise-specific Flow Collection Services MIB, whose object identifier is **{jnxMibs 28}**, provides statistics on files, records, memory, FTP, and error states of flow collection services on a Monitoring Services PIC. It also provides SNMP traps for unavailable destinations, unsuccessful file transfers, flow overloading, and memory overloading. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-coll.txt .

For information about how to configure the flow collection services interface, see the *Junos OS Network Interfaces Configuration Guide* and the *Junos OS Feature Guide*.

For information about the enterprise-specific Flow Collection Services MIB objects, see the following topics:

- jnxCollGlobalStats on page 287
- jnxCollPicIfTable on page 288
- jnxCollFileTable on page 290

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxCollGlobalStats

jnxCollGlobalStats provides statistics on all the router's Monitoring Services PICs and has the objects listed in Table 85 on page 288.

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Table 85: jnxCollGlobalStats

Object	Object Identifier	Description
jnxCollGlobalCreatedFiles	jnxCollGlobalStats 1	The number of files created by all the Monitoring Services PICs on the router since the last time the router was reset.
jnxCollGlobalOpenFiles	jnxCollGlobalStats 2	The number of open flow collection services files currently on the router.

Related Documentation

- Flow Collection Services MIB on page 287
- jnxCollPicIfTable on page 288
- jnxCollFileTable on page 290

jnxCollPicIfTable

jnxCollPicIfTable contains statistics about each Monitoring Services PIC.

JnxCollPicIfEntry contains information about a Monitoring Services PIC. Each Monitoring Services PIC contains one interface and is identified by **IfIndex**. It has objects listed in Table 86 on page 288.

Table 86: jnxCollPicIfEntry

Object	Object Identifier	Description
File Statistics		
jnxCollPicIfCreatedFiles	jnxCollPicIfEntry 1	The number of files created by a Monitoring Services PIC since the last time the PIC was reset.
jnxCollPicIfCreatedFileRate	jnxCollPicIfEntry 2	The number of files created per second during the current 10-second interval.
jnxCollPicIfPeakCreatedFileRate	jnxCollPicIfEntry 3	The peak number of files created per second.
jnxCollPicIfExportedFiles	jnxCollPicIfEntry 4	The number of files exported by a Monitoring Services PIC.
jnxCollPicIfExportedFileRate	jnxCollPicIfEntry 5	The number of files exported per second during the current 10-second interval.
jnxCollPicIfPeakExportedFileRate	jnxCollPicIfEntry 6	The peak number of files exported per second.
jnxCollPicIfDestroyedFiles	jnxCollPicIfEntry 7	The number of files successfully exported and files dropped by the Monitoring Services PIC. Files are destroyed after they are transferred to the FTP server or when there is not enough memory.
jnxCollPicIfDestroyedFileRate	jnxCollPicIfEntry 8	The number of files dropped per second during the current 10-second interval. Files are dropped after they are transferred to the FTP server or when there is not enough memory.

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Table 86: jnxCollPicIfEntry *(continued)*

Object	Object Identifier	Description
jnxCollPicIfPeakDestroyedFileRate	**jnxCollPicIfEntry 9**	The peak number of files dropped, per second. Files are dropped after they are transferred to the FTP server or when there is not enough memory.
Record Statistics		
jnxCollPicIfProcRecords	**jnxCollPicIfEntry 10**	The number of flow records processed by a Monitoring Services PIC.
jnxCollPicIfProcRecordsRate	**jnxCollPicIfEntry 11**	The number of flow records processed per second during the current 10-second interval.
jnxCollPicIfPeakProcRecordsRate	**jnxCollPicIfEntry 12**	The peak number of flow records processed.
Memory Statistics		
jnxCollPicIfMemoryUsed	**jnxCollPicIfEntry 13**	The amount of memory used, in bytes, by a Monitoring Services PIC.
jnxCollPicIfMemoryFree	**jnxCollPicIfEntry 14**	The amount of free memory, in bytes, on a Monitoring Services PIC.
FTP Statistics		
jnxCollPicIfFtpBytes	**jnxCollPicIfEntry 15**	The number of bytes transferred by a Monitoring Services PIC using FTP.
jnxCollPicIfFtpByteRate	**jnxCollPicIfEntry 16**	The number of bytes per second transferred using FTP, measured during the current 10-second interval.
jnxCollPicIfPeakFtpByteRate	**jnxCollPicIfEntry 17**	The peak number of bytes per second transferred using FTP.
jnxCollPicIfFtpFiles	**jnxCollPicIfEntry 18**	The number of files transferred by a Monitoring Services PIC using FTP.
jnxCollPicIfFtpFileRate	**jnxCollPicIfEntry 19**	The number of files per second transferred using FTP.
jnxCollPicIfPeakFtpFileRate	**jnxCollPicIfEntry 20**	The peak number of files per second transferred using FTP.
jnxCollPicIfFtpFailures	**jnxCollPicIfEntry 21**	The number of FTP transfer failures transferred by a Monitoring Services PIC.
Error State Statistics		
jnxCollPicIfCurrentState	**jnxCollPicIfEntry 22**	The current state of various error conditions on a Monitoring Services PIC.
jnxCollPicIfLastStateChange	**jnxCollPicIfEntry 23**	The error condition of the last changed state.
jnxCollPicIfStateChangeTime	**jnxCollPicIfEntry 24**	The value of **sysUpTime** when the management subsystem last learned of a change to the **jnxCollPicIfCurrentState** for a Monitoring Services PIC.

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Table 86: jnxCollPicIfEntry *(continued)*

Object	Object Identifier	Description
jnxCollPicIfStateChangeDate	**jnxCollPicIfEntry 25**	The system date and time when the management subsystem last learned of a change to the **jnxCollPicIfCurrentState** on a Monitoring Services PIC.
jnxCollPicIfStateChangeType	**jnxCollPicIfEntry 26**	Indicates whether the last state change set a new error condition or cleared an existing one. This object contains the following values: • **none(1)** • **set(2)** • **cleared(3)**

Related Documentation

- Flow Collection Services MIB on page 287
- jnxCollGlobalStats on page 287
- jnxCollFileTable on page 290

jnxCollFileTable

jnxCollFileTable contains information about each flow collection services file on the router.

jnxCollFileEntry contains information about a single file open on a Monitoring Services PIC, and has the objects listed in Table 87 on page 290.

Table 87: jnxCollFileTable

Object	Object Identifier	Description
jnxCollFileName	**jnxCollFileEntry 1**	The name of a flow collection services file on a Monitoring Services PIC.
jnxCollFileFname	**jnxCollFileEntry 2**	The name of a flow collection services file on this Monitoring Services PIC. This object is included for those network management applications that cannot parse the filename from the instance portion of the OIDs and provides the value of **jnxCollFileName**.
jnxCollFileRecords	**jnxCollFileEntry 3**	The number of flow records in this file.
jnxCollFileRecordRate	**jnxCollFileEntry 4**	The number of flow records per second added to this file, measured during the current 10-second interval.
jnxCollFilePeakRecordRate	**jnxCollFileEntry 5**	The peak number of flow records per second added to this file.
jnxCollFileUncompBytes	**jnxCollFileEntry 6**	The number of uncompressed bytes in this file.
jnxCollFileUncompByteRate	**jnxCollFileEntry 7**	The number of uncompressed bytes per second added to this file.
jnxCollFilePeakUncompByteRate	**jnxCollFileEntry 8**	The peak number of uncompressed bytes per second added to this file.

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Table 87: jnxCollFileTable *(continued)*

Object	Object Identifier	Description
jnxCollFileCompBytes	**jnxCollFileEntry 9**	The number of compressed bytes in this file.
jnxCollFileCompByteRate	**jnxCollFileEntry 10**	The number of compressed bytes per second added to this file during the current 10-second interval.
jnxCollFilePeakCompByteRate	**jnxCollFileEntry 11**	The peak number of compressed bytes per second added to this file.
jnxCollFileBlocks	**jnxCollFileEntry 12**	The number of blocks in this file.
jnxCollFileCompBlocks	**jnxCollFileEntry 14**	The number of compressed blocks in this file.
jnxCollFileTransferAttempts	**jnxCollFileEntry 15**	The number of FTP transfer attempts in this file.
jnxCollFileState	**jnxCollFileEntry 16**	The current state of this file. This object contains the following values: • **unknown(1)** • **active(2)**—The file is actively receiving flow records. • **wait(3)**—The file is waiting for export. • **export1(4)**—The file is being exported to the primary server. • **export2(5)**—The file is being exported to the secondary server.

Related Documentation

- Flow Collection Services MIB on page 287
- jnxCollGlobalStats on page 287
- jnxCollPicIfTable on page 288

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CHAPTER 23

Interpreting the Enterprise-Specific Services PIC MIB

- Services PIC MIB on page 293
- jnxSpSvcSetTable on page 294
- jnxSpSvcSetSvcTypeTable on page 295
- jnxSpSvcSetIfTable on page 296
- Service Traps on page 297
- Redundant Interfaces on page 298

Services PIC MIB

The Adaptive Services (AS) PIC allows you to provide multiple services on a single PIC by configuring a set of services and applications. The AS PIC offers a special range of services you configure in one or more service sets: stateful firewalls, Network Address Translation (NAT), and intrusion detection system (IDS).

The Juniper Networks enterprise-specific Services PIC MIB, whose object identifier is **{jnxMibs 32}**, sends the current operational status for each AS PIC. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-sp.txt .



NOTE: **The Services PIC MIB is not supported on J Series Services Routers unless the appropriate services license is enabled.**

For information about the enterprise-specific Services PIC MIB objects, see the following topics:

- jnxSpSvcSetTable on page 294
- jnxSpSvcSetSvcTypeTable on page 295
- jnxSpSvcSetIfTable on page 296
- Service Traps on page 297
- Redundant Interfaces on page 298

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Related Documentation
- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxSpSvcSetTable

The **jnxSpSvcSetTable**, whose object identifier is **{jnxSPSvcSet 1}**, provides information about each service set on each Adaptive Services (AS) PIC on the router.

The **jnxSpSvcSetEntry**, whose object identifier is **{jnxSpSvcSetTable 1}**, has 11 objects, which are listed in Table 88 on page 294. Each entry provides information about a single service set. The service set is identified by the name of the service set. The AS PIC on which the service set is configured is identified by **JnxSpSvcSetIFName**.

Table 88: jnxSpSvcSetTable

Object	Object Identifier	Description
jnxSpSvcSetName	**jnxSpSvcSetEntry 1**	A text name for the service set.
jnxSpSvcSetSvcType	**jnxSpSvcSetEntry 2**	The name of the service type associated with the service set.
jnxSpSvcSetTypeIndex	**jnxSpSvcSetEntry 3**	An integer used to identify the service type for the service set.
jnxSpSvcSetIfName	**jnxSpSvcSetEntry 4**	The name of the interface identifying the AS PIC. If more than one interface is associated with the AS PIC, the name associated with the lower layer interface is used.
jnxSpSvcSetIfIndex	**jnxSpSvcSetEntry 5**	An index number associated with the interface name.
jnxSpSvcSetMemoryUsage	**jnxSpSvcSetEntry 6**	Amount of memory used by the service set, in bytes.
jnxSpSvcSetCpuUtil	**jnxSpSvcSetEntry 7**	Amount of CPU processing used by the service set, expressed as a percentage of total CPU usage. J Series Services Routers do not have a dedicated CPU for services. CPU usage on these routers appears as **0**.

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Table 88: jnxSpSvcSetTable *(continued)*

Object	Object Identifier	Description
jnxSpSvcSetSvcStyle	**jnxSpSvcSetEntry 8**	Type of service for the service set. Service types include: • **Unknown**—The service type is not known. • **Interface-service**—The service is interface based. • **Next-hop-service**—The service is next-hop based.
jnxSpSvcSetMemLimitPktDrops	**jnxSpSvcSetEntry 9**	Number of packets dropped because the service set exceeded its memory limits (operating in the **Red** zone).
jnxSpSvcSetCpuLimitPktDrops	**jnxSpSvcSetEntry 10**	Number of packets dropped because the service set exceeded the average CPU limits (when total CPU usage exceeds 85%).
jnxSpSvcSetFlowLimitPktDrops	**jnxSpSvcSetEntry 11**	Number of packets dropped because the service set exceeded the flow limit.

Related Documentation

- Services PIC MIB on page 293
- jnxSpSvcSetSvcTypeTable on page 295
- jnxSpSvcSetIfTable on page 296
- Redundant Interfaces on page 298
- Service Traps on page 297

jnxSpSvcSetSvcTypeTable

The **jnxSpSvcSetSvcTypeTable**, whose object identifier is **{jnxSPSvcSet 2}**, provides information about each service on each Adaptive Services (AS) PIC on the router. The stateful firewall, NAT, or IDS service sets are categorized as one **SvcType** (SFW/NAT/IDS).

The **jnxSpSvcSetSvcTypeEntry**, whose object identifier is **{jnxSpSvcSetSvcTypeTable 1}**, has seven objects, which are listed in Table 89 on page 295. Each entry provides information about a single service on each AS PIC. Each AS PIC is identified by its corresponding index number, while each service is identified by **jnxSpSvcSetSvcTypeIndex**. The service type associated with this index is provided by **jnxSpSvcSetSvcTypeName**.

Table 89: jnxSpSvcSetSvcTypeTable

Object	Object Identifier	Description
jnxSpSvcSetSvcTypeIndex	**jnxSpSvcSetSvcTypeEntry 1**	An integer used to identify the service type.

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Table 89: jnxSpSvcSetSvcTypeTable *(continued)*

Object	Object Identifier	Description
jnxSpSvcSetSvcTypeIfName	jnxSpSvcSetSvcTypeEntry 2	The name of the interface identifying the AS PIC. If more than one interface is associated with the AS PIC, the name associated with the lower layer interface is used.
jnxSpSvcSetSvcTypeName	jnxSpSvcSetSvcTypeEntry 3	The name of the service type.
jnxSpSvcSetSvcTypeSvcSets	jnxSpSvcSetSvcTypeEntry 4	Number of service sets configured on the AS PIC that use this service type.
jnxSpSvcSetSvcTypeMemoryUsage	jnxSpSvcSetSvcTypeEntry 5	Amount of memory used by this service type, expressed in bytes.
jnxSpSvcSetSvcTypePctMemoryUsage	jnxSpSvcSetSvcTypeEntry 6	Amount of memory used by this service type, expressed as a percentage of total memory.
jnxSpSvcSetSvcTypeCpuUtil	jnxSpSvcSetSvcTypeEntry 7	Amount of CPU processing used by the service set, expressed as a percentage of total CPU usage. J Series Services Routers do not have a dedicated CPU for services. CPU usage on these routers appears as **0**.

Related Documentation

- Services PIC MIB on page 293
- jnxSpSvcSetTable on page 294
- jnxSpSvcSetIfTable on page 296
- Redundant Interfaces on page 298
- Service Traps on page 297

jnxSpSvcSetIfTable

The **jnxSpSvcSetIfTable**, whose object identifier is **{jnxSPSvcSet 3}**, provides service set information for each Adaptive Services (AS) PIC on the router.

The **jnxSpSvcSetIfEntry**, whose object identifier is **{jnxSpSvcSetIfTable 1}**, has eight objects, which are listed in Table 90 on page 296. Each entry provides service set information about a single AS PIC. Each AS PIC is identified by its corresponding index number.

Table 90: jnxSpSvcSetIfTable

Object	Object Identifier	Description
jnxSpSvcSetIfTableName	jnxSpSvcSetIfEntry 1	The name of the interface used to identify the AS PIC. If more than one interface is associated with the AS PIC, the name associated with the lower layer interface is used.

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Table 90: jnxSpSvcSetIfTable *(continued)*

Object	Object Identifier	Description
jnxSpSvcSetIfsvcSets	**jnxSpSvcSetIfEntry 2**	The number of service sets configured on the AS PIC.
jnxSpSvcSetIfMemoryUsage	**jnxSpSvcSetIfEntry 3**	Amount of memory used by the AS PIC, expressed in bytes.
jnxSpSvcSetIfPctMemoryUsage	**jnxSpSvcSetIfEntry 4**	Amount of memory used by the AS PIC, expressed as a percentage of total memory.
jnxSpSvcSetIfPolMemoryUsage	**jnxSpSvcSetIfEntry 5**	Amount of policy memory used by the AS PIC, expressed in bytes.
jnxSpSvcSetIfPctPolMemoryUsage	**jnxSpSvcSetIfEntry 6**	Amount of policy memory used by the AS PIC, expressed as a percentage of the total.
jnxSpSvcSetIfMemoryZone	**jnxSpSvcSetIfEntry 7**	The memory usage zone currently occupied by the AS PIC. The definitions of each zone are: • Green—All new flows are allowed. • Yellow—Unused memory is reclaimed. All new flows are allowed. • Orange—New flows are allowed only for service sets that use less than their equal share of memory. • Red—No new flows are allowed.
jnxSpSvcSetIfCpuUtil	**jnxSpSvcSetIfEntry 8**	Amount of CPU processing used by the AS PIC, expressed as a percentage of total CPU usage. J Series Services Routers do not have a dedicated CPU for services. CPU usage on these routers appears as **0**.

Related Documentation

- Services PIC MIB on page 293
- jnxSpSvcSetTable on page 294
- jnxSpSvcSetSvcTypeTable on page 295
- Redundant Interfaces on page 298
- Service Traps on page 297

Service Traps

The enterprise-specific Services PIC MIB provides traps for monitoring Adaptive Services (AS) PICs. Table 91 on page 298 lists the supported traps.

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Table 91: Supported Traps for Services PIC MIB

Object	Object Identifier	Description
jnxSpSvcSetZoneEntered	**jnxSPNotificationPrefix 1**	Indicates that an AS PIC has entered a more severe memory usage zone from a less severe memory usage zone. The zone entered is identified by **JnxSpSvcSetIfMemoryZone**.
jnxSpSvcSetZoneExited	**jnxSPNotificationPrefix 2**	Indicates that an AS PIC has exited a more severe memory usage zone to a less severe memory usage zone. The zone entered is identified by **JnxSpSvcSetIfMemoryZone**.
jnxSpSvcSetCpuExceeded	**jnxSPNotificationPrefix 3**	Indicates that an AS PIC has over 85% CPU usage. This trap is not supported on J Series Services Routers.
jnxSpSvcSetCpuOk	**jnxSPNotificationPrefix 4**	Indicates that an AS PIC has returned to less than 85% CPU usage. This trap is not supported on J Series Services Routers.

Related Documentation

- Services PIC MIB on page 293
- jnxSpSvcSetTable on page 294
- jnxSpSvcSetSvcTypeTable on page 295
- jnxSpSvcSetIfTable on page 296
- Configuring SNMP Trap Options and Groups on a Device Running Junos OS
- Managing Traps and Informs

Redundant Interfaces

On M Series routers and T Series routers, redundant adaptive services interfaces (**rsp**) appear in the **jnxSpSvcSetIfTable** just like any other adaptive services interface (**sp**). With the exception of the index, information presented for an **rsp** interface is similar to the underlying **sp** interface. In the **jnxSpSvcSetTable**, only the underlying **sp** interface is shown because the Adaptive Services (AS) PIC does not track the overlying **rsp** interface,

Related Documentation

- Services PIC MIB on page 293
- jnxSpSvcSetIfTable on page 296

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CHAPTER 24

Interpreting the Enterprise-Specific Dynamic Flow Capture MIB

- Dynamic Flow Capture MIB on page 299
- jnxDfcCSTable on page 300
- jnxDfcCDTable on page 303
- DFC Notification Variables on page 304
- DFC Notification Definitions on page 305

Dynamic Flow Capture MIB

The Dynamic Flow Capture (DFC) PIC forwards passively monitored packets matching a particular filter list to one or more destinations.

The DFC architecture consists of one or more control sources that send requests to a Juniper Networks router to monitor incoming data and then forward any packets that match specific filter criteria to a set of one or more content destinations.

The Juniper Networks enterprise-specific DFC MIB, whose object identifier is **{jnxMibs 33}**, sends the current operational status for each DFC PIC. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-dfc.txt .



NOTE: **The DFC PIC is supported on M320 and T320 routers and the T640 Core Router.**

For information about the enterprise-specific DFC MIB objects, see the following topics:

- jnxDfcCSTable on page 300
- jnxDfcCDTable on page 303
- DFC Notification Variables on page 304
- DFC Notification Definitions on page 305

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Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxDfcCSTable

The **jnxDfcCSTable**, whose object identifier is **{jnxDfc 1}**, provides information about the DFC control source on each DFC PIC on the router.

The **jnxDfcCSEntry**, whose object identifier is **{jnxDfcCSTable 1}**, has 44 objects, which are listed in Table 92 on page 300.

Table 92: jnxDfcCSTable

Object	Object Identifier	Description
jnxDfcCSGrpName	**jnxDfcCSEntry 1**	The name assigned to a DFC group. A DFC group defines a profile of dynamic flow capture configuration information.
jnxDfcCSId	**jnxDfcCSEntry 2**	Control source identifier. The control source is a client that monitors electronic data or voice transfer over the network. The control source sends filter requests to the Juniper Networks router using a control protocol. The control source has a unique identifier and an optional list of IP addresses.
jnxDfcCSControlProtocolAddRequests	**jnxDfcCSEntry 3**	The number of control protocol **Add** requests received. The **Add** request specifies new filter criteria to be included in the current filter configuration for a given control source and content destination.
jnxDfcCSCriteriaAdded	**jnxDfcCSEntry 4**	The number of filter criteria added successfully by the control source.
jnxDfcCSCriteriaAdditionFailed	**jnxDfcCSEntry 5**	The number of filter criteria **Add** requests that could not be processed successfully.
jnxDfcCSControlProtocolDeleteRequests	**jnxDfcCSEntry 6**	The number of control protocol **Delete** requests received. The **Delete** request specifies filter criteria to be removed from the current filter configuration for a given control source and content destination.
jnxDfcCSCriteriaDeleted	**jnxDfcCSEntry 7**	The number of filter criteria deleted successfully as requested by the control source.
jnxDfcCSCriteriaDeletionFailed	**jnxDfcCSEntry 8**	The number of filter criteria **Delete** requests that could not be processed successfully.
jnxDfcCSCriteriaDeletedTimeoutIdle	**jnxDfcCSEntry 9**	The number of criteria deleted by timeout idle.
jnxDfcCSCriteriaDeletedTimeoutTotal	**jnxDfcCSEntry 10**	The number of criteria deleted by timeout total.
jnxDfcCSCriteriaDeletedPackets	**jnxDfcCSEntry 11**	The number of criteria deleted by packets.

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Table 92: jnxDfcCSTable *(continued)*

Object	Object Identifier	Description
jnxDfcCSCriteriaDeletedBytes	jnxDfcCSEntry 12	The number of criteria deleted by bytes.
jnxDfcCSControlProtocolRefreshRequests	jnxDfcCSEntry 13	The number of control protocol **Refresh** requests received. The **Refresh** request updates the timeout for a particular filter criterion (or a set of filter criteria) for a given control source and content destination.
jnxDfcCSCriteriaRefreshed	jnxDfcCSEntry 14	The number of filter criteria **Refresh** requests processed successfully as requested by the control source.
jnxDfcCSCriteriaRefreshFailed	jnxDfcCSEntry 15	The number of filter criteria **Refresh** requests that could not be processed successfully.
jnxDfcCSControlProtocolListRequests	jnxDfcCSEntry 16	The number of control protocol **List** requests received. The **List** request returns a list of all criteria that a particular control source has added and are currently active.
jnxDfcCSListSuccess	jnxDfcCSEntry 17	The number of **List** requests processed successfully as requested by the control source.
jnxDfcCSListFailed	jnxDfcCSEntry 18	The number of **List** requests that could not be processed successfully.
jnxDfcCSControlProtocolNoopRequests	jnxDfcCSEntry 19	The number of control protocol **Noop** requests received. This request is used to verify the end-to-end connectivity between the control source and the DFC PIC.
jnxDfcCSNoopSuccess	jnxDfcCSEntry 20	The number of **Noop** requests processed successfully as requested by the control source.
jnxDfcCSNoopFailed	jnxDfcCSEntry 21	The number of **Noop** requests that could not be processed successfully.
jnxDfcCSDynamicCrtieriaActive	jnxDfcCSEntry 22	The number of active dynamic filter criteria.
jnxDfcCSStaticCrtieriaActive	jnxDfcCSEntry 23	The number of active static filter criteria.
jnxDfcCSBadRequest	jnxDfcCSEntry 24	The number of **Bad** requests received.
jnxDfcCSResponseSuccessful	jnxDfcCSEntry 25	The number of successful responses corresponding to the **Add**, **Delete**, **Refresh**, **List**, and **Noop** requests sent to the control source.
jnxDfcCSResponseImproperCriteria	jnxDfcCSEntry 26	The number of responses generated because of improper filter criteria included in an **Add** request.
jnxDfcCSResponseUnknownContentDest	jnxDfcCSEntry 27	The number of responses generated because of an unknown content destination included in an **Add**, **Delete**, **Refresh**, or **List** request.

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Table 92: jnxDfcCSTable *(continued)*

Object	Object Identifier	Description
jnxDfcCSResponseUnknownControlSrc	**jnxDfcCSEntry 28**	The number of responses generated because of an unknown control source included in an **Add**, **Delete**, **Refresh**, or **List** request.
jnxDfcCSResponseUnknownCriteriaId	**jnxDfcCSEntry 29**	The number of responses generated because of an unknown criteria identifier included in an **Add**, **Delete**, **Refresh**, or **List** request.
jnxDfcCSResponseImproperTimeout	**jnxDfcCSEntry 30**	The number of responses generated because of an improper timeout specified in an **Add** or **Refresh** request.
jnxDfcCSResponseInvalidAuthentication	**jnxDfcCSEntry 31**	The number of responses generated because of invalid authentication information included in an **Add**, **Delete**, **Refresh**, **List**, or **Noop** request.
jnxDfcCSResponseInvalidSequenceNumber	**jnxDfcCSEntry 32**	The number of responses generated because of an invalid sequence number included in an **Add**, **Delete**, **Refresh**, **List**, or **Noop** request.
jnxDfcCSResponseInternalError	**jnxDfcCSEntry 33**	The number of responses generated because an internal error occurred on the DFC PIC processing the request.
jnxDfcCSNotificationRestart	**jnxDfcCSEntry 34**	The number of **Restart** notifications sent to configured notification recipients. A notification is generated when a system failure occurs and all DFC filter criteria are lost.
jnxDfcCSNotificationRollover	**jnxDfcCSEntry 35**	The number of **Rollover** notifications sent to configured notification recipients. A notification is generated when a sequence number rollover occurs on the DFC PIC.
jnxDfcCSNotificationNoop	**jnxDfcCSEntry 36**	The number of **Noop** notifications sent to configured notification recipients. A notification is generated when the DFC PIC receives a **Noop** message that includes a **SendAsync** parameter.
jnxDfcCSNotificationTimeout	**jnxDfcCSEntry 37**	The number of **Timeout** notifications sent to configured notification recipients. This notification is generated when a DFC PIC times out a filter criterion (based on any one of its configured timeout parameters) and the criterion contains a **SendTimeoutAsync** parameter.
jnxDfcCSNotificationCongestion	**jnxDfcCSEntry 38**	A **Congestion** notification is generated when the total 10-second average packet forwarding rate (in bps) summed over all active filter criteria to a configured content destination exceeds the configured *soft* limit for the destination. The **jnxDfcCSNotificationCongestion** object contains the number of **Congestion** notifications sent to configured notification recipients.

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Table 92: jnxDfcCSTable *(continued)*

Object	Object Identifier	Description
jnxDfcCSNotificationCongestionDelete	**jnxDfcCSEntry 39**	A **Congestion Delete** notification is generated when the total 10-second average packet forwarding rate (in bps) summed over all active filter criteria to a configured content destination exceeds the configured *hard* limit for the destination. The **jnxDfcCSNotificationCongestionDelete** object contains the number of **Congestion Delete** notifications sent to configured notification recipients.
jnxDfcCSNotificationDuplicatesDropped	**jnxDfcCSEntry 40**	The number of **Duplicated Dropped** notifications sent to configured notification recipients. This notification is generated when the configurable **Maximum Duplicates** parameter has been exceeded and packets matching criteria added by the corresponding control source are dropped.
jnxDfcCSAddRequestRate	**jnxDfcCSEntry 41**	The request processing rate (in requests processed per second).
jnxDfcCSAddRequestPeakRate	**jnxDfcCSEntry 42**	The peak request processing rate (in requests processed per second.
jnxDfcCSAggrCriteriaBandwidth	**jnxDfcCSEntry 43**	Bandwidth (in bps).
jnxDfcCSSequenceNumber	**jnxDfcCSEntry 44**	Protocol sequence number.

Related Documentation

- Dynamic Flow Capture MIB on page 299
- jnxDfcCDTable on page 303
- DFC Notification Variables on page 304
- DFC Notification Definitions on page 305

jnxDfcCDTable

The **jnxDfcCDTable**, whose object identifier is **{jnxDfc 2}**, provides statistical information for content destinations.

The **jnxDfcCDEntry**, whose object identifier is **{jnxDfcCDTable 1}**, has seven objects, which are listed in Table 93 on page 303.

Table 93: jnxDfcCDTable

Object	Object Identifier	Description
jnxDfcCDGrpName	**jnxDfcCDEntry 1**	The name assigned to a DFC group. A DFC group defines a profile of dynamic flow capture configuration information.

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Table 93: jnxDfcCDTable *(continued)*

Object	Object Identifier	Description
jnxDfcCDId	jnxDfcCDEntry 2	Content destination identifier. The DFC router processes the requests from the control sources, creates the filters, monitors incoming data flows, and sends the matched packets to their respective content destinations. Content destinations receive the matched packets from the router.
jnxDfcCDCriteria	jnxDfcCDEntry 3	The number of filter criteria configured for the content destination.
jnxDfcCDByteRate	jnxDfcCDEntry 4	The average data rate (in bytes per second) summed over all active filter criteria configured for a given content destination.
jnxDfcCDMatchedPackets	jnxDfcCDEntry 5	The number of packets that match the filter criteria configured for a content destination.
jnxDfcCDMatchedBytes	jnxDfcCDEntry 6	The number of bytes that match the filter criteria configured for a content destination.
jnxDfcCDCongestionNotification	jnxDfcCDEntry 7	The number of **Congestion** notifications sent to a configured notification recipient.

Related Documentation

- Dynamic Flow Capture MIB on page 299
- jnxDfcCSTable on page 300
- DFC Notification Variables on page 304
- DFC Notification Definitions on page 305

DFC Notification Variables

The enterprise-specific DFC MIB provides notifications for monitoring dynamic flow capture. Table 94 on page 304 lists the supported notification variables.

Table 94: Supported Notification Variables for the DFC MIB

Object	Object Identifier	Description
jnxDfcInputPktRate	jnxDfcNotifyVars 1	Data packet rate (in pps).
jnxDfcPpsSoftOverloadLowWatermark	jnxDfcNotifyVars 2	Configured lowest value for the data packet rate (in pps).
jnxDfcPpsSoftOverloadHighWatermark	jnxDfcNotifyVars 3	Configured highest value for the data packet rate (in pps).
jnxDfcPpsHardOverloadLowWatermark	jnxDfcNotifyVars 4	Recommended lowest value for the data packet rate (in pps).
jnxDfcPpsHardOverloadHighWatermark	jnxDfcNotifyVars 5	Recommended highest value for the data packet rate (in pps).
jnxDfcFlowsUsage	jnxDfcNotifyVars 6	Percent (%) usage of the total number of flows.

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Table 94: Supported Notification Variables for the DFC MIB *(continued)*

Object	Object Identifier	Description
jnxDfcCriteriaUsage	jnxDfcNotifyVars 7	Percent (%) usage of matching criteria for all filters.
jnxDfcMemSoftOverloadLowWatermark	jnxDfcNotifyVars 8	Configured lowest watermark percent for memory load.
jnxDfcMemSoftOverloadHighWatermark	jnxDfcNotifyVars 9	Configured highest watermark percent for memory load.
jnxDfcFlowLowWatermark	jnxDfcNotifyVars 10	Recommended lowest value for the number of flows allowed.
jnxDfcFlowHighWatermark	jnxDfcNotifyVars 11	Recommended highest value for the number of flows allowed.
jnxDfcCriteriaLowWatermark	jnxDfcNotifyVars 12	Recommended lowest value for the number of criteria allowed.
jnxDfcCriteriaHighWatermark	jnxDfcNotifyVars 13	Recommended highest value for the number of criteria allowed.

Related Documentation

- Dynamic Flow Capture MIB on page 299
- jnxDfcCSTable on page 300
- jnxDfcCDTable on page 303
- DFC Notification Definitions on page 305

DFC Notification Definitions

Table 95 on page 305 lists the supported notification definitions.

Table 95: Supported Notification Definitions for the DFC MIB

Notification Type	Objects	Identifier	Description
jnxDfcSoftPpsThresholdExceeded	jnxDfcInputPktRate jnxDfcPpsSoftOverloadLowWatermark jnxDfcPpsSoftOverloadHighWatermark	jnxDfcNotificationPrefix 1	Notification that occurs when the input packet rate (in pps) exceeds the configured limit.
jnxDfcSoftPpsUnderThreshold	jnxDfcInputPktRate jnxDfcPpsSoftOverloadLowWatermark jnxDfcPpsSoftOverloadHighWatermark	jnxDfcNotificationPrefix 2	Notification that occurs when the input packet rate (in pps) returns to below the configured limit.
jnxDfcHardPpsThresholdExceeded	jnxDfcInputPktRate jnxDfcPpsHardOverloadLowWatermark jnxDfcPpsHardOverloadHighWatermark	jnxDfcNotificationPrefix 3	Notification that occurs when the input packet rate (in pps) exceeds the recommended limit.

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Table 95: Supported Notification Definitions for the DFC MIB *(continued)*

Notification Type	Objects	Identifier	Description
jnxDfcHardPpsUnderThreshold	jnxDfcInputPktRate jnxDfcPpsHardOverloadLowWatermark jnxDfcPpsHardOverloadHighWatermark	jnxDfcNotificationPrefix 4	Notification that occurs when the input packet rate (in pps) returns to below the recommended limit.
jnxDfcSoftMemThresholdExceeded	jnxDfcFlowUsage jnxDfcCriteriaUsage jnxDfcMemSoftOverloadLowWatermark jnxDfcMemSoftOverloadHighWatermark	jnxDfcNotificationPrefix 5	Notification that occurs when memory usage exceeds the configured limit.
jnxDfcSoftMemUnderThreshold	jnxDfcFlowUsage jnxDfcCriteriaUsage jnxDfcMemSoftOverloadLowWatermark jnxDfcMemSoftOverloadHighWatermark	jnxDfcNotificationPrefix 6	Notification that occurs when memory usage returns to below the configured limit.
jnxDfcHardMemThresholdExceeded	jnxDfcFlowUsage jnxDfcFlowLowWatermark jnxDfcFlowHighWatermark jnxDfcCriteriaUsage jnxDfcCriteriaLowWatermark jnxDfcCriteriaHighWatermark	jnxDfcNotificationPrefix 7	Notification that occurs when memory usage exceeds the recommended limit.
jnxDfcHardMemUnderThreshold	jnxDfcFlowUsage jnxDfcFlowLowWatermark jnxDfcFlowHighWatermarkjnxDfcCriteriaUsage jnxDfcCriteriaLowWatermark jnxDfcCriteriaHighWatermark	jnxDfcNotificationPrefix 8	Notification that occurs when memory usage returns to below the recommended limit.

Related Documentation

- Dynamic Flow Capture MIB on page 299
- jnxDfcCSTable on page 300
- jnxDfcCDTable on page 303
- DFC Notification Variables on page 304

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CHAPTER 25

Interpreting the Enterprise-Specific Chassis Forwarding MIB

- Chassis Forwarding MIB on page 307
- jnxFwddProcess on page 307

Chassis Forwarding MIB

The Juniper Networks enterprise-specific Chassis Forwarding MIB, whose object identifier is **{jnxMibs 34}**, enables J Series Services Routers to fully support the Junos OS health monitor. This MIB extends the scope of health monitoring to include Junos OS forwarding process (**fwdd**) components on J Series Services Routers. The forwarding process is responsible for most of the packet transmission through a J Series Services Router. The overall performance of the router is largely determined by the effectiveness of the forwarding process.

The Junos OS health monitor uses objects in the Chassis Forwarding MIB to access information about the forwarding process such as microkernel CPU usage and real-time thread CPU usage.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-chassis-fwdd.txt .

For information about the enterprise-specific Chassis Forwarding MIB objects, see the following topic:

- jnxFwddProcess on page 307

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxFwddProcess

The object identifier for **jnxFwddProcess** is **{jnxFwdd 1}**. This object shows health monitoring statistics for the forwarding process (**fwdd**). SeeTable 96 on page 308.

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Table 96: jnxFwddProcess

Object	Object Identifier	Description
jnxFwddMicroKernelCPUUsage	jnxFwddProcess 1	Percentage of the CPU being used by the forwarding process microkernel. If this information is unavailable or is not applicable, the value is 0 (zero).
jnxFwddRtThreadsCPUUsage	jnxFwddProcess 2	Percentage of the CPU being used by the forwarding process real-time threads. If this information is unavailable or is not applicable, the value is 0 (zero).
jnxFwddHeapUsage	jnxFwddProcess 3	Percentage of heap space being used by the forwarding process. If this information is unavailable or is not applicable, the value is 0 (zero).
jnxFwddDmaMemUsage	jnxFwddProcess 4	Percentage of DMA memory used by the forwarding process. If this information is unavailable or is not applicable, the value is 0 (zero).
jnxFwddUpTime	jnxFwddProcess 5	Forwarding process uptime expressed in terms of system uptime. If this information is unavailable or is not applicable, the value is 0 (zero).

Related Documentation

- Chassis Forwarding MIB on page 307

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CHAPTER 26

Interpreting the Enterprise-Specific System Log MIB

- System Log MIB on page 309
- jnxSyslogTable on page 310
- jnxSyslogAvTable on page 312

System Log MIB

Event policies can include an action that raises traps for events based on system log messages. This feature enables notification of an SNMP trap-based application when an important system log message occurs. You can convert any system log message (for which there are no corresponding traps) into a trap. This feature is valuable for customers who use network management system traps rather than system log messages to monitor their networks. For more information about converting system log messages into traps, see the *Junos OS Configuration and Operations Automation Guide*.

The Juniper Networks enterprise-specific System Log MIB, whose object identifier is **{jnxMibs 35}**, provides support for this feature.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-syslog.txt .

For information about the enterprise-specific System Log MIB objects, see the following topics:

- jnxSyslogTable on page 310
- jnxSyslogAvTable on page 312

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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jnxSyslogTable

The **jnxSyslogTable**, whose object identifier is **{jnxSyslog 1}**, provides information about each system log message generated by the router.

The **jnxSyslogEntry**, whose object identifier is **{jnxSyslogTable 1}**, has nine objects, which are listed in Table 97 on page 310. Each entry provides information about a single system log message.

Table 97: jnxSyslogTable

Object	Object Identifier	Description
jnxSyslogID	**jnxSyslogEntry 1**	System log message identifier. This identifier is a numerical value and cannot be consecutive. This identifier is also used as the primary index in **jnxSyslogAttrValTable**.
jnxSyslogEventName	**jnxSyslogEntry 2**	An octet string that contains the system log event name.
jnxSyslogTimestamp	**jnxSyslogEntry 3**	Time the message was generated. This value is expressed as **DateAndTime**.
jnxSyslogSeverity	**jnxSyslogEntry 4**	Severity of the system log message. The severity value is equal to the value that the system log uses + 1. For example, an emergency message (severity 0 in the system log) has a severity of 1. Severity values include: • **emergency (1)** • **alert (2)** • **critical (3)** • **error (4)** • **warning (5)** • **notice (6)** • **info (7)** • **debug (8)**
jnxSyslogFacility	**jnxSyslogEntry 5**	Identifies the facility used to generate the log message. (See Table 98 on page 311.)
jnxSyslogProcessID	**jnxSyslogEntry 6**	Process ID of the process that generated the system log message.
jnxSyslogProcessName	**jnxSyslogEntry 7**	Process that generated the system log message.
jnxSyslogHostName	**jnxSyslogEntry 8**	Hostname of the machine that generated the system log.
jnxSyslogMessage	**jnxSyslogEntry 9**	System log message that was generated.

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Table 98 on page 311 lists the facilities that generate system log messages.

Table 98: Facilities That Generate System Log Messages

Index	Item	Description
1	kernel	Kernel messages
2	user	User level messages
3	mail	Mail system
4	daemon	System processes
5	auth	Authorization messages
6	syslog	Messages generated by the system log process (**syslogd**)
7	lpr	Line printer subsystem
8	news	Network news subsystem
9	uucp	UUCP subsystem
10	cron	Clock process
11	authPriv	Authorization messages
12	ftp	FTP process
13	ntp	NTP subsystem
14	security	Security subsystems (for example, firewall)
15	console	**/dev/console** output
16	reserved	Reserved for system use
17	loca10	–
18	dfc	Junos names
19	loca12	–
20	firewall	Junos names
21	pfe	Junos names
22	conflict	Junos names
23	change	Junos names

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Table 98: Facilities That Generate System Log Messages *(continued)*

Index	Item	Description
24	interact	Junos names

Related Documentation

- System Log MIB on page 309
- jnxSyslogAvTable on page 312

jnxSyslogAvTable

The **jnxSyslogAvTable**, whose object identifier is **{jnxSyslogNotifyVars 2}**, provides information about each system log message generated by the router.

The **jnxSyslogAvEntry**, whose object identifier is **{jnxSyslogAvTable 1}**, has three objects, which are listed in Table 99 on page 312. Each entry provides information about attribute value pairs of system log messages generated by a device.

Table 99: jnxSyslogAvTable

Object	Object Identifier	Description
jnxSyslogAvIndex	**jnxSyslogAvEntry 1**	Index for the attribute value pair in the system log message.
jnxSyslogAvAttribute	**jnxSyslogAvEntry 2**	Attribute of the system log message (identified by **jnxSyslogID**).
jnxSyslogAvValue	**jnxSyslogAvEntry 3**	Value of the attribute (identified by **jnxSyslogAvAttribute**).

Related Documentation

- System Log MIB on page 309
- jnxSyslogTable on page 310

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CHAPTER 27

Interpreting the Enterprise-Specific MPLS LDP MIB

- MPLS LDP MIB on page 313

MPLS LDP MIB

The Juniper Networks enterprise-specific MPLS LDF MIB, whose object identifier is **{jnxMibs 36}**, contains object definitions as described in RFC 3815, *Definitions of Managed Objects for the Multiprotocol Label Switching (MPLS), Label Distribution Protocol (LDP)*.

This MIB was supported in earlier releases of Junos OS as a proprietary LDP MIB (**mib-ldpmib.txt**). Because the branch used by the proprietary LDP MIB (**mib-ldpmib.txt**) conflicts with RFC 3812, the proprietary LDP MIB (**mib-ldpmib.txt**) has been deprecated and replaced by the enterprise-specific MPLS LDP MIB (**mib-jnx-mpls-ldp.txt**).

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-mpls-ldp.txt .

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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CHAPTER 28

Interpreting the Enterprise-Specific Packet Forwarding Engine MIB

- Packet Forwarding Engine MIB on page 315
- jnxPfeNotifyGlTable on page 315
- jnxPfeNotifyTypeTable on page 317

Packet Forwarding Engine MIB

The Juniper Networks enterprise-specific Packet Forwarding Engine MIB, whose object identifier is **{jnxPfeMibRoot 1}**, provides notification statistics for Packet Forwarding Engines. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-pfe.txt .

NOTE: Information provided by this MIB is modeled after information provided by the **show pfe statistics notification** command-line interface (CLI) command.

For information about the enterprise-specific Packet Forwarding Engine MIB objects, see the following topics:

- jnxPfeNotifyGlTable on page 315
- jnxPfeNotifyTypeTable on page 317

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxPfeNotifyGlTable

jnxPfeNotifyGlTable contains global Packet Forwarding Engine notification statistics.

JnxPfeNotifyGlEntry contains notification statistics for each Packet Forwarding Engine slot. Each entry has objects listed in Table 100 on page 316.

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Table 100: jnxPfeNotifyGlEntry

Object	Object Identifier	Description
jnxPfeNotifyGISlot	jnxPfeNotifyGIEntry 1	The slot number for a set of Packet Forwarding Engine notification statistics.
jnxPfeNotifyGIParsed	jnxPfeNotifyGIEntry 2	The number of notifications reported by the Packet Forwarding Engine controller, which manages packet forwarding functions.
jnxPfeNotifyGIAged	jnxPfeNotifyGIEntry 3	The number of notifications dropped because they have expired.
jnxPfeNotifyGICorrupt	jnxPfeNotifyGIEntry 4	The number of notifications dropped because the notification result format is invalid. This counter is valid for Internet Processor-I and Internet Processor-II only.
jnxPfeNotifyGIIllegal	jnxPfeNotifyGIEntry 5	The number of notifications dropped because the notification has an illegal notification type.
jnxPfeNotifyGISample	jnxPfeNotifyGIEntry 6	The number of sample notifications reported by the Packet Forwarding Engine controller.
jnxPfeNotifyGIGiants	jnxPfeNotifyGIEntry 7	The number of notifications dropped because the notification is larger than the supported direct memory access (DMA) size.
jnxPfeNotifyGITtlExceeded	jnxPfeNotifyGIEntry 8	The number of options/TTL-expired notifications sent to service interfaces as transit packets. This counter is valid for Internet Processor-I and Internet Processor-II only.
jnxPfeNotifyGITtlExcErrors	jnxPfeNotifyGIEntry 9	The number of options/TTL-expired notifications that could not be sent to service interfaces as transit packets because the output interface could not be determined. This counter is valid for Internet Processor-I and Internet Processor-II only.
jnxPfeNotifyGISvcOptAsp	jnxPfeNotifyGIEntry 10	The number of IP options packets sent to a Services PIC.
jnxPfeNotifyGISvcOptRe	jnxPfeNotifyGIEntry 11	The number of IP options packets sent to the Routing Engine.
jnxPfeNotifyGIPostSvcOptOut	jnxPfeNotifyGIEntry 12	The number of notifications re-injected by a Services PIC after processing the associated packets. The Packet Forwarding Engine will forward these notifications to their actual destination. This counter is valid for Internet Processor-I and Internet Processor-II only.
jnxPfeNotifyGIOptTtlExp	jnxPfeNotifyGIEntry 13	The number of TTL-expired transit packets.
jnxPfeNotifyGIDiscSample	jnxPfeNotifyGIEntry 14	The number of sample notifications dropped because the notifications refer to discarded packets in the Packet Forwarding Engine.
jnxPfeNotifyGIRateLimited	jnxPfeNotifyGIEntry 15	The number of notification ignored because of Packet Forwarding Engine software throttling (delaying or refusing requests).
jnxPfeNotifyGIPktGetFails	jnxPfeNotifyGIEntry 16	The number of notifications ignored because DMA memory could not be allocated.

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Table 100: jnxPfeNotifyGlEntry *(continued)*

Object	Object Identifier	Description
jnxPfeNotifyGIDmaFails	**jnxPfeNotifyGIEntry 17**	The number of notifications where the DMA of associated packets failed for miscellaneous reasons. This counter is valid for T Series routers only.
jnxPfeNotifyGIDmaTotals	**jnxPfeNotifyGIEntry 18**	The number of notifications for which the packet DMA completed. This counter is valid for T Series routers only.
jnxPfeNotifyGIUnknowns	**jnxPfeNotifyGIEntry 19**	The number of notifications that could not be resolved to a known, next hop destination. This counter is valid for T Series routers only.

Related Documentation

- Packet Forwarding Engine MIB on page 315
- jnxPfeNotifyTypeTable on page 317

jnxPfeNotifyTypeTable

jnxPfeNotifyTypeTable contains information about type-specific Packet Forwarding Engine notifications for each Packet Forwarding Engine slot. Packet Forwarding Engine notification types are listed in Table 101 on page 317.

Table 101: Packet Forwarding Engine Notification Types

Index	Item	Description
1	Illegal	Packets with an invalid notification type.
2	Unclassified	Packets that did not have a key lookup performed on them.
3	Option	Packets that include L3 options.
4	Next Hop	Packets that are destined to the host.
5	Discard	Discarded packets sent to the route processor.
6	Sample	Unused.
7	Redirect	Packets sent back to the interfaces from which they arrived.
8	Do Not Fragment	Packets that need to be fragmented, but have a **don't fragment (DF)** value set.
9	CFDF	Packets that have a DF value set and have triggered a maximum transmission unit (MTU) exceeded indicator.
10	Poison	Packets that have a poisoned next-hop index.

For information, see **jnxPfeNotifyTypeEntry**.

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jnxPfeNotifyTypeEntry

jnxPfeNotifyTypeEntry contains information about type-specific Packet Forwarding Engine notifications, and has the objects listed in Table 102 on page 318.

Table 102: jnxPfeNotifyTypeEntry

Object	Object Identifier	Description
jnxPfeNotifyTypeId	jnxPfeNotifyTypeEntry 1	Identifies the Packet Forwarding Engine notification type. See Table 101 on page 317 for a list of notification types.
jnxPfeNotifyTypeDescr	jnxPfeNotifyTypeEntry 2	A description of the Packet Forwarding Engine notification type.
jnxPfeNotifyTypeParsed	jnxPfeNotifyTypeEntry 3	The number of notifications that are parsed successfully.
jnxPfeNotifyTypeInput	jnxPfeNotifyTypeEntry 4	The number of notifications whose associated packets are stored in router processor memory using direct memory access.
jnxPfeNotifyTypeFailed	jnxPfeNotifyTypeEntry 5	The number of notifications that are not parsed successfully.
jnxPfeNotifyTypeIgnored	jnxPfeNotifyTypeEntry 6	The number of notifications where the notification type in the message does not match any of the valid notification types.

Related Documentation

- Packet Forwarding Engine MIB on page 315
- jnxPfeNotifyGlTable on page 315

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CHAPTER 29

Interpreting the Enterprise-Specific Packet Mirror MIB

- Packet Mirror MIB Overview on page 319
- jnxJsPacketMirrorTrapVars on page 319
- Packet Mirroring-Related Traps on page 321

Packet Mirror MIB Overview

The Juniper Networks enterprise-specific Packet Mirror MIB, **jnxJsPacketMirrorMIB**, whose object ID is **{jnxJsPacketMirror 1}**, enables you to capture and view packet mirroring-related information. Packet mirroring traps are an extension of the standard SNMP implementation and are only available to SNMPv3 users.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-packet-mirror.txt .

For information about the enterprise-specific Packet Mirror MIB objects, see the following topics:

- jnxJsPacketMirrorTrapVars on page 319
- Packet Mirroring-Related Traps on page 321

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxJsPacketMirrorTrapVars

The **jnxJsPacketMirrorTrapVars** object, whose object identifier is **{jnxJsPacketMirrorObjects 1}**, contains the objects listed in Table 103 on page 320.

You can use the objects listed in Table 103 on page 320 to retrieve information about the subscriber and to track their traffic statistics.

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Table 103: jnxJsPacketMirrorTrapVars

Object	Object ID	Description
jnxJsPacketMirrorIdentifier	jnxJsPacketMirrorTrapVars 1	Unique identifier of the mirrored session. Consists of the version, mirror ID, and session ID.
jnxJsPacketMirrorSessionIdentifier	jnxJsPacketMirrorTrapVars 2	Accounting session ID.
jnxJsPacketMirrorTrigger	jnxJsPacketMirrorTrapVars 3	Event that triggered the packet mirroring trap.
jnxJsPacketMirrorTriggerType	jnxJsPacketMirrorTrapVars 4	Type of event that triggered the packet mirroring trap.
jnxJsPacketMirrorConfigurationSource	jnxJsPacketMirrorTrapVars 5	Source IP address from where the packet mirroring request was configured.
jnxJsPacketMirrorErrorCause	jnxJsPacketMirrorTrapVars 6	Reason for the error message generated.
jnxJsPacketMirrorErrorString	jnxJsPacketMirrorTrapVars 7	Error message as reported by the Junos OS application.
jnxJsPacketMirrorApplicationName	jnxJsPacketMirrorTrapVars 8	Name of the Junos OS application reporting the trap.
jnxJsPacketMirrorAnalyzerAddress	jnxJsPacketMirrorTrapVars 9	IP address of the analyzer.
jnxJsPacketMirrorUserName	jnxJsPacketMirrorTrapVars 10	Login ID of the user whose traffic is being mirrored.
jnxJsPacketMirrorDateAndTime	jnxJsPacketMirrorTrapVars 11	Date and time when the event occurred.
jnxJsPacketMirrorRouterId	jnxJsPacketMirrorTrapVars 12	VRF ID in Junos OS. Represented as the logical router (LR) name followed by the router instance (RI) name.
jnxJsPacketMirrorDirection	jnxJsPacketMirrorTrapVars 13	Mirroring direction. Possible values: • ingress (0) - Ingress traffic is mirrored. • egress (1) - Egress traffic is mirrored. • bidirectional (2) - Both ingress and egress traffic is mirrored.
jnxJsPacketMirrorTargetIpAddress	jnxJsPacketMirrorTrapVars 14	IP address of the mirrored interface.
jnxJsPacketMirrorTerminationReason	jnxJsPacketMirrorTrapVars 15	Reason for terminating a packet mirroring session.
jnxPacketMirrorCallingStationIdentifier	jnxJsPacketMirrorTrapVars 16	Unique calling station ID of the subscriber whose traffic is being monitored.
jnxPacketMirrorNasIdentifier	jnxJsPacketMirrorTrapVars 17	Unique NAS ID of the router on which the traffic is being monitored.

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Table 103: jnxJsPacketMirrorTrapVars *(continued)*

Object	Object ID	Description
jnxJsPacketMirrorOctetsReceived	jnxJsPacketMirrorTrapVars 18	Number of octets of combined IPv4 and IPv6 subscriber traffic received.
jnxJsPacketMirrorOctetsTransmitted	jnxJsPacketMirrorTrapVars 19	Number of octets of combined IPv4 and IPv6 subscriber traffic transmitted.
jnxJsPacketMirrorTargetIpv6Address	jnxJsPacketMirrorTrapVars 20	IPv6 address of the mirrored interface.

Related Documentation

- Packet Mirror MIB Overview on page 319
- Packet Mirroring-Related Traps on page 321

Packet Mirroring-Related Traps

The **jnxJsPacketMirrorNotifications** object, whose object identifier is **{jnxJsPacketMirrorMIB 0}**, contains the objects listed in Table 104 on page 321.

You can use the objects listed in Table 104 on page 321 to retrieve information about the packet mirroring traps and when they are triggered.

Table 104: Packet Mirroring-Related Traps

Object	Object ID	Description
jnxJsPacketMirrorMirroringFailure	jnxJsPacketMirrorNotifications 1	The packet mirroring request failed.
jnxJsPacketMirrorLiSubscriberLoggedIn	jnxJsPacketMirrorNotifications 2	Subscriber has logged in successfully.
jnxJsPacketMirrorLiSubscriberLogInFailed	jnxJsPacketMirrorNotifications 3	Subscriber was unable to log in.
jnxJsPacketMirrorLiSubscriberLoggedOut	jnxJsPacketMirrorNotifications 4	Subscriber has logged out.
jnxJsPacketMirrorLiServiceActivated	jnxJsPacketMirrorNotifications 5	Packet mirroring on an interface is now active.
jnxJsPacketMirrorLiServiceActivationFailed	jnxJsPacketMirrorNotifications 6	Packet mirroring on an interface has failed.
jnxJsPacketMirrorLiServiceDeactivated	jnxJsPacketMirrorNotifications 7	Packet mirroring on an interface is now deactivated.

Related Documentation

- Packet Mirror MIB Overview on page 319
- jnxJsPacketMirrorTrapVars on page 319
- Configuring SNMP Trap Options and Groups on a Device Running Junos OS
- Managing Traps and Informs

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CHAPTER 30

Interpreting the Enterprise-Specific Event MIB

- Event MIB on page 323
- jnxEventAvTable on page 323
- Notifications for the Event MIB on page 324

Event MIB

The Juniper Networks enterprise-specific Event MIB, whose object identifier is **{jnxMibs 37}**, defines a generic trap that can be generated using an op script or event policy. This MIB provides the ability to specify a system log string and raise a trap if that system log string is found.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-event.txt .

For information about the enterprise-specific Event MIB objects, see the following topics:

- jnxEventAvTable on page 323
- Notifications for the Event MIB on page 324

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxEventAvTable

The **jnxEventAvTable**, whose object identifier is **{jnxEventNotifyVars 2}**, provides information about traps generated by op scripts or event policies.

jnxEventAvEntry, whose object identifier is **{jnxEventAvTable 1}**, has three objects, which are listed in Table 105 on page 324.

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Table 105: jnxEventAvTable

Object	Object Identifier	Description
jnxEventAvIndex	**jnxEventAvEntry 1**	The sequence number of the attribute value pair in the trap generated by a op script or event policy.
jnxEventAvAttribute	**jnxEventAvEntry 2**	The attribute name in the trap generated by an op script or event policy.
jnxEventAvValue	**jnxEventAvEntry 3**	The value of the attribute identified by **jnxEventAvAttribute**.

Related Documentation

- Event MIB on page 323
- Notifications for the Event MIB on page 324

Notifications for the Event MIB

Table 106 on page 324 lists the supported notifications for the Event MIB.

Table 106: Supported Notifications for the Event MIB

Object	Object Identifier	Description
jnxEventTrapDescr	**jnxEventNotificationPrefix 1**	A notification generated by an op script or event policy. In addition to the **jnxEventTrap** objects, this notification can include one or more attribute value pairs (identified by **jnxEventAvAttribute** and **jnxEventAvValue**).

Related Documentation

- Event MIB on page 323
- jnxEventAvTable on page 323

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CHAPTER 31

Interpreting the Enterprise-Specific Bidirectional Forwarding Detection MIB

- Bidirectional Forwarding Detection MIB on page 325
- jnxBfdSessTable on page 325
- Notifications for the BFD MIB on page 326

Bidirectional Forwarding Detection MIB

The Juniper Networks enterprise-specific Bidirectional Forwarding Detection (BFD) MIB, whose object identifier is **{jnxBfdMibRoot 1}**, sends the current operational status for the transmit interval and detection time of BFD sessions. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-bfd.txt .

For information about the enterprise-specific BFD MIB objects, see the following topics:

- jnxBfdSessTable on page 325
- Notifications for the BFD MIB on page 326

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxBfdSessTable

The **jnxBfdSessTable**, whose object identifier is **{jnxBfdObjects 1}**, is an extension to the **jnxSessTable**. It provides information about threshold values for the transmit interval and detection time on BFD sessions.

The **jnxBfdSessEntry**, whose object identifier is **{jnxBfdSessTable 1}**, has four objects, which are listed in Table 107 on page 326.

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Table 107: jnxBfdSessTable

Object	Object Identifier	Description
jnxBfdSessThresTxInterval	**jnxBfdSessEntry 1**	The threshold value (in microseconds) for the transmit interval. If the current transmit interval value (**jnxBfdSessCurrTxInterval**) adapts to a value greater than the threshold value, a trap is raised (**jnxBfdSessTxIntervalHigh**).
jnxBfdSessCurrTxInterval	**jnxBfdSessEntry 2**	The current transmit interval for the session (in microseconds).
jnxBfdSessThreshDectTime	**jnxBfdSessEntry 3**	The threshold value (in microseconds) for the detection time. If the current detection time value (**jnxBfdSessCurrDectTime)** adapts to a value greater than the threshold value, a trap is raised (**jnxBfdSessDetectionTimeHigh**).
jnxBfdSessCurrDectTime	**jnxBfdSessEntry 4**	The current detection time for the session (in microseconds).

Related Documentation

- Bidirectional Forwarding Detection MIB on page 325
- Notifications for the BFD MIB on page 326

Notifications for the BFD MIB

Table 108 on page 326 lists the supported notifications for the BFD MIB.

Table 108: Supported Notifications for the BFD MIB

Object	Object Identifier	Description
jnxBfdSessTxIntervalHigh	**jnxBfdNotification 1**	A notification generated when the threshold value for the transmit interval is configured (**jnxBfdSessThresTxInterval**) and the BFD session transmit interval (**jnxBfdSessCurrTxInterval**) adapts to a value greater than the threshold value. This trap is sent only once, when the threshold is first exceeded. The transmit interval can continue to adapt beyond the threshold value.
jnxBfdSessDetectionTimeHigh	**jnxBfdNotification 2**	A notification generated when the threshold value for the detection time is configured (**jnxBfdSessThresDectTime**) and the BFD session detection time (**jnxBfdSessCurrDectTime**) adapts to a value greater than the threshold value. This trap is sent only once, when the threshold is first exceeded. The detection time can continue to adapt beyond the threshold value.

Related Documentation

- Bidirectional Forwarding Detection MIB on page 325
- jnxBfdSessTable on page 325

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CHAPTER 32

Interpreting the Enterprise-Specific L2TP MIB

- L2TP MIB on page 327
- L2TP Scalar Status and Statistics Group on page 328
- jnxL2tpTunnelGroupStatsTable on page 329
- jnxL2tpTunnelStatsTable on page 329
- jnxL2tpSessionStatsTable on page 332
- jnxL2tpMlpppBundleStatsTable on page 338

L2TP MIB

The Juniper Networks enterprise-specific Layer 2 Tunneling Protocol (L2TP) MIB enables you to monitor L2TP tunnels and sessions using SNMP on both M Series and MX Series routers. The MX Series router uses the Common Edge L2TP process, jlt2pd. Some L2TP MIB objects are not supported by jl2tpd. If an object is not supported, the object returns either zero or the default value. The L2TP MIB, whose object identifier is **{jnxMibs 49}**, provides information related to L2TP tunnels and sessions.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-l2tp.txt .

For information about the enterprise-specific L2TP MIB objects, see the following topics:

- L2TP Scalar Status and Statistics Group on page 328
- jnxL2tpTunnelGroupStatsTable on page 329
- jnxL2tpTunnelStatsTable on page 329
- jnxL2tpSessionStatsTable on page 332
- jnxL2tpMlpppBundleStatsTable on page 338

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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L2TP Scalar Status and Statistics Group

Table 109 on page 328 lists the objects in the L2TP scalar status and statistics group.

Table 109: L2TP Scalar Status and Statistics Group

Object	Object Identifier	Description
jnxL2tpStatsTotalTunnels	jnxL2tpStats 1	Returns the total number of tunnels that are in established state at the time of querying.
jnxL2tpStatsTotalSessions	jnxL2tpStats 2	Returns the total number of sessions that are in established state at the time of querying.
jnxL2tpStatsControlRxOctets	jnxL2tpStats 3	Returns the number of control channel octets received by the tunnels in established state at the time of querying.
jnxL2tpStatsControlRxPkts	jnxL2tpStats 4	Returns the number of control packets received by the tunnels in established state at the time of querying.
jnxL2tpStatsControlTxOctets	jnxL2tpStats 5	Returns the number of control channel octets that were transmitted to tunnel endpoints that are in established state at the time of querying.
jnxL2tpStatsControlTxPkts	jnxL2tpStats 6	Returns the number of control packets that were transmitted to the tunnel endpoints that are in established state at the time of querying.
jnxL2tpStatsPayloadRxOctets	jnxL2tpStats 7	Returns the number of payload channel octets that were received on the tunnels that are in established state at the time of querying.
jnxL2tpStatsPayloadRxPkts	jnxL2tpStats 8	Returns the number of payload packets that were received on the tunnels that are in established state at the time of querying.
jnxL2tpStatsPayloadTxOctets	jnxL2tpStats 9	Returns the number of payload channel octets that were transmitted to the tunnel peers that are in established state at the time of querying.
jnxL2tpStatsPayloadTxPkts	jnxL2tpStats 10	Returns the number of payload packets that were transmitted to existing tunnel peers that are in established state at the time of querying.
jnxL2tpStatsErrorTxPkts	jnxL2tpStats 11	Returns the number of packet transmission attempts with errors to the tunnel peers that are in established state at the time of querying.
jnxL2tpStatsErrorRxPkts	jnxL2tpStats 12	Returns the number of packets with errors that were received from the existing tunnel peers that are in established state at the time of querying.

Related Documentation

- L2TP MIB on page 327
- jnxL2tpTunnelGroupStatsTable on page 329
- jnxL2tpTunnelStatsTable on page 329
- jnxL2tpSessionStatsTable on page 332

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- jnxL2tpMlpppBundleStatsTable on page 338

jnxL2tpTunnelGroupStatsTable

The **jnxL2tpTunnelGroupStatsTable**, whose object ID is **{jnxL2tpObjects 2}**, contains objects that describe the current status and statistics of an L2TP tunnel group.

The **jnxL2tpTunnelGroupStatsEntry** objects are listed in Table 110 on page 329.

Table 110: jnxL2tpTunnelGroupStatsTable

Object	Object Identifier	Description
jnxL2tpTunnelGroupStatsTnlGrpName	jnxL2tpTunnelGroupStatsEntry 1	The name of the particular tunnel group.
jnxL2tpTunnelGroupStatsGatewayAddrType	jnxL2tpTunnelGroupStatsEntry 2	The type of local IP address for L2TP tunnels that are part of the group.
jnxL2tpTunnelGroupStatsGatewayAddr	jnxL2tpTunnelGroupStatsEntry 3	The local IP address for L2TP tunnels that are part of the group.
jnxL2tpTunnelGroupStatsSvcIntfName	jnxL2tpTunnelGroupStatsEntry 4	The name of the service interface that is hosting the tunnel group.
jnxL2tpTunnelGroupStatsTotalTunnels	jnxL2tpTunnelGroupStatsEntry 5	The total number of tunnels that are in the established state at the time of querying.
jnxL2tpTunnelGroupStatsTotalSessions	jnxL2tpTunnelGroupStatsEntry 6	The total number of established sessions in the tunnel group at the time of querying.

Related Documentation

- L2TP MIB on page 327
- L2TP Scalar Status and Statistics Group on page 328
- jnxL2tpTunnelStatsTable on page 329
- jnxL2tpSessionStatsTable on page 332
- jnxL2tpMlpppBundleStatsTable on page 338

jnxL2tpTunnelStatsTable

jnxL2tpTunnelStatsTable, whose object ID is **{jnxL2tpObjects 3}**, contains objects that describe the current status and statistics of an L2TP tunnel.

A **jnxL2tpTunnelStatsEntry** represents an L2TP tunnel interface statistics entry and has objects that are listed in Table 111 on page 330.

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Table 111: jnxL2tpTunnelStatsTable

Object	Object Identifier	Description
jnxL2tpTunnelStatsLocalTID	jnxL2tpTunnelStatsEntry 1	The local tunnel Identifier.
jnxL2tpTunnelStatsServiceInterface	jnxL2tpTunnelStatsEntry 2	The name of the service interface on which the tunnel is being hosted. NOTE: This attribute is not supported on jl2tpd LAC.
jnxL2tpTunnelStatsTunnelGroup	jnxL2tpTunnelStatsEntry 3	The name of the tunnel group to which the tunnel belongs. NOTE: This attribute is not supported on jl2tpd LAC.
jnxL2tpTunnelStatsRemoteTID	jnxL2tpTunnelStatsEntry 4	The remote tunnel identifier. See RFC 2661, Section 3.1.
jnxL2tpTunnelStatsRemoteIpAddrType	jnxL2tpTunnelStatsEntry 5	The type of the remote-end address of the tunnel.
jnxL2tpTunnelStatsRemoteIpAddress	jnxL2tpTunnelStatsEntry 6	The remote-end address of the tunnel.
jnxL2tpTunnelStatsRemoteUdpPort	jnxL2tpTunnelStatsEntry 7	The remote-end UDP port of the tunnel.
jnxL2tpTunnelStatsActiveSessions	jnxL2tpTunnelStatsEntry 8	The total number of sessions that are in established state for the tunnel.
jnxL2tpTunnelStatsStat	jnxL2tpTunnelStatsEntry 9	One of the following states for the control tunnel: • **cc_responder_accept_new**—Shows that the tunnel has received and accepted the start control connection request (SCCRQ). • **cc_responder_reject_new**—Shows that the tunnel has received and rejected the SCCRQ. • **cc_responder_idle**—Shows that the tunnel has just been created. • **cc_responder_wait_ctl_conn**—Shows that the tunnel has sent the start control connection response (SCCRP) and is waiting for the start control connection connected (SCCCN) message. • **Cleanup**—Shows that the tunnel is being cleaned up. • **Closed**—Shows that the tunnel is being closed. • **Destroyed**—Shows that the tunnel is being destroyed. • **Established**—Shows that the tunnel is operational. • **Terminate**—Shows that the tunnel is being terminated. • **Unknown**—Shows that the tunnel is not connected to the router.
jnxL2tpTunnelStatsLocalIpAddrType	jnxL2tpTunnelStatsEntry 10	The type of local-end address of the tunnel.
jnxL2tpTunnelStatsLocalIpAddress	jnxL2tpTunnelStatsEntry 11	The local-end address of the tunnel.
jnxL2tpTunnelStatsLocalUdpPort	jnxL2tpTunnelStatsEntry 12	The local-end UDP port number of the tunnel.

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Table 111: jnxL2tpTunnelStatsTable *(continued)*

Object	Object Identifier	Description
jnxL2tpTunnelStatsLocalHostName	jnxL2tpTunnelStatsEntry 13	The local hostname of the tunnel.
jnxL2tpTunnelStatsRemoteHostName	jnxL2tpTunnelStatsEntry 14	The hostname of the L2TP peer, as discovered during the tunnel establishment phase (using the Host Name AVP). If the tunnel is idle, this object shows the value based on the data from the last time it was connected.
jnxL2tpTunnelMaxSessions	jnxL2tpTunnelStatsEntry 15	The maximum number of sessions configured on the tunnel. Value could be a positive number or zero (unlimited).
jnxL2tpTunnelStatsWindowSize	jnxL2tpTunnelStatsEntry 16	The send window size for the tunnel.
jnxL2tpTunnelStatsHelloInterval	jnxL2tpTunnelStatsEntry 17	The hello interval for the tunnel.
jnxL2tpTunnelStatsCreationTime	jnxL2tpTunnelStatsEntry 18	The time when the tunnel was created.
jnxL2tpTunnelStatsUpTime	jnxL2tpTunnelStatsEntry 19	The time elapsed since the tunnel was established.
jnxL2tpTunnelStatsIdleTime	jnxL2tpTunnelStatsEntry 20	The time elapsed since the last data activity, transmission or reception, on the tunnel.
jnxL2tpTunnelStatsCollectionStart	jnxL2tpTunnelStatsEntry 21	The time at which the statistics gathering started for the tunnel.
jnxL2tpTunnelStatsControlTxPkts	jnxL2tpTunnelStatsEntry 22	The number of control packets transmitted to the tunnel peer.
jnxL2tpTunnelStatsControlTxBytes	jnxL2tpTunnelStatsEntry 23	The number of control bytes transmitted to the tunnel peer.
jnxL2tpTunnelStatsControlRxPkts	jnxL2tpTunnelStatsEntry 24	The number of control packets received on the tunnel.
jnxL2tpTunnelStatsControlRxByte	jnxL2tpTunnelStatsEntry 25	The number of control bytes received from the tunnel peer.
jnxL2tpTunnelStatsDataTxPkts	jnxL2tpTunnelStatsEntry 26	The number of data packets transmitted to the tunnel.
jnxL2tpTunnelStatsDataTxBytes	jnxL2tpTunnelStatsEntry 27	The number of data bytes transmitted to the tunnel peer.
JnxL2tpTunnelStatsDataRxPkts	jnxL2tpTunnelStatsEntry 28	The number of data packets received from the tunnel.
jnxL2tpTunnelStatsDataRxBytes	jnxL2tpTunnelStatsEntry 29	The number of data bytes received from the tunnel peer.
jnxL2tpTunnelStatsErrorTxPkts	jnxL2tpTunnelStatsEntry 30	The number of error transmit packets on the tunnel. NOTE: This attribute is not available on Trio MPC/MICs.

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Table 111: jnxL2tpTunnelStatsTable *(continued)*

Object	Object Identifier	Description
jnxL2tpTunnelStatsErrorRxPkts	**jnxL2tpTunnelStatsEntry 31**	The number of error receive packets on the tunnel. NOTE: This attribute is not available on Trio MPC/MICs.

Related Documentation

- L2TP MIB on page 327
- L2TP Scalar Status and Statistics Group on page 328
- jnxL2tpTunnelGroupStatsTable on page 329
- jnxL2tpSessionStatsTable on page 332
- jnxL2tpMlpppBundleStatsTable on page 338

jnxL2tpSessionStatsTable

jnxL2tpSessionStatsTable, whose object ID is **{jnxL2tpObjects 4}**, contains the objects that describe the current status and statistics of a single L2TP tunneled session.

A **jnxL2tpSessionStatsEntry** represents an L2TP session interface status and has the objects that are listed in Table 112 on page 332.

Table 112: jnxL2tpSessionStatsTable

Object	Object Identifier	Description
jnxL2tpSessionStatsLocalTID	**jnxL2tpSessionStatsEntry 1**	The local tunnel Identifier.
jnxL2tpSessionStatsLocalSID	**jnxL2tpSessionStatsEntry 2**	The local session Identifier.
jnxL2tpSessionStatsServiceInterface	**jnxL2tpSessionStatsEntry 3**	The name of the service interface on which this session is being hosted. NOTE: This attribute is not supported on jl2tpd LAC.
jnxL2tpSessionStatsTunnelGroup	**jnxL2tpSessionStatsEntry 4**	The name of the tunnel group to which this session belongs. NOTE: This attribute is not supported on jl2tpd LAC.
jnxL2tpSessionStatsRemoteSID	**jnxL2tpSessionStatsEntry 5**	The remote-end assigned session identifier for this session. This value remains zero from the time of starting the session until the time the remote end point responds.

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Table 112: jnxL2tpSessionStatsTable *(continued)*

Object	Object Identifier	Description
jnxL2tpSessionStatsInterfaceUnit	**jnxL2tpSessionStatsEntry 6**	The interface unit number that corresponds to the logical service interface on which the session is being hosted.
jnxL2tpSessionStatsEncapType	**jnxL2tpSessionStatsEntry 7**	The tunnel encapsulation type.
jnxL2tpSessionStatsBundleID	**jnxL2tpSessionStatsEntry 8**	The ID of the bundle to which the session is linked. This field is valid only for tunnel encapsulation type **multilink-ppp**.
jnxL2tpSessionStatsStat	**jnxL2tpSessionStatsEntry 9**	One of the following status messages to show the state of the session at the time of querying: • Established— The session is operational. • Closed—The session has been closed. • Destroyed—The session has been destroyed. • Cleanup—The session has been cleaned up. • lns_ic_accept_new—The new session has been accepted. • lns_ic_idle—The session has been created but is in idle state. • lns_ic_reject_new—A new session has been rejected. • lns_ic_wait_connect—The session is waiting for the peer's incoming call connected (ICCN) message.
jnxL2tpSessionStatsUserName	**jnxL2tpSessionStatsEntry 10**	The peer session name on the interface. This is typically the login name of the remote user. This object contains a null string when the user name is unknown to the local tunnel peer. NOTE: This attribute is not supported on jl2tpd.
jnxL2tpSessionStatsMode	**jnxL2tpSessionStatsEntry 11**	The configured mode value for this session.
jnxL2tpSessionStatsLocalAddrType	**jnxL2tpSessionStatsEntry 12**	The type of the local-end address of the tunnel that hosts the session.
jnxL2tpSessionStatsLocalAddress	**jnxL2tpSessionStatsEntry 13**	The local end address of the tunnel that hosts the session.

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Table 112: jnxL2tpSessionStatsTable *(continued)*

Object	Object Identifier	Description
jnxL2tpSessionStatsLocalUdpPort	jnxL2tpSessionStatsEntry 14	The UDP port of the local end of the tunnel that hosts the session.
jnxL2tpSessionStatsRemoteAddrType	jnxL2tpSessionStatsEntry 15	The type of the remote end address of the tunnel that hosts the session.
jnxL2tpSessionStatsRemoteAddress	jnxL2tpSessionStatsEntry 16	The remote end address of the tunnel that hosts the session.
jnxL2tpSessionStatsRemoteUdpPort	jnxL2tpSessionStatsEntry 17	The UDP port of the remote-end of the tunnel that hosts the session.
jnxL2tpSessionStatsLocalHostName	jnxL2tpSessionStatsEntry 18	The local hostname of the tunnel that hosts the session.
jnxL2tpSessionStatsRemoteHostName	jnxL2tpSessionStatsEntry 19	The hostname as discovered during the tunnel establishment phase (using the Host Name AVP) of the L2TP peer.
jnxL2tpSessionAssignedIpAddrType	jnxL2tpSessionStatsEntry 20	The type of IP address of PPP client being tunneled as obtained from IPCP configuration while establishing the session. NOTE: This attribute is not supported on jl2tpd.
jnxL2tpSessionAssignedIpAddress	jnxL2tpSessionStatsEntry 21	The IP address of the PPP client being tunneled as obtained from IPCP configuration while establishing the session. NOTE: This attribute is not supported on jl2tpd.
jnxL2tpSessionLocalMRU	jnxL2tpSessionStatsEntry 22	The MRU for the local PPP entity. This value is the MRU that the remote entity uses when sending packets to the session. NOTE: This attribute is not supported on jl2tpd.
jnxL2tpSessionRemoteMRU	jnxL2tpSessionStatsEntry 23	The MRU for the remote PPP entity. This value is the MRU that the local entity uses when sending packets to the remote PPP client. NOTE: This attribute is not supported on jl2tpd.
jnxL2tpSessionStatsTxSpeed	jnxL2tpSessionStatsEntry 24	The last known transmit baud rate for the session.

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Table 112: jnxL2tpSessionStatsTable *(continued)*

Object	Object Identifier	Description
jnxL2tpSessionStatsRxSpeed	jnxL2tpSessionStatsEntry 25	The last known receive baud rate for the session.
jnxL2tpSessionStatsCallBearerType	jnxL2tpSessionStatsEntry 26	The bearer type of this session.
jnxL2tpSessionStatsFramingType	jnxL2tpSessionStatsEntry 27	The framing type of the session.
jnxL2tpSessionStatsLCPRenegotiation	jnxL2tpSessionStatsEntry 28	The ON/OFF state of the LCP renegotiation for the session.
jnxL2tpSessionStatsAuthMethod	jnxL2tpSessionStatsEntry 29	The proxy authentication method employed by the LAC for the session. NOTE: This attribute is not supported on jl2tpd.
nxL2tpSessionStatsNasIpAddrType	jnxL2tpSessionStatsEntry 30	The type of IP address of the RADIUS network address server to which the accounting records for this session are being sent. NOTE: This attribute is not supported on jl2tpd.
jnxL2tpSessionStatsNasIpAddress	jnxL2tpSessionStatsEntry 31	The IP address of the RADIUS network address server to which the accounting records for the session are being sent. NOTE: This attribute is not supported on jl2tpd.
jnxL2tpSessionStatsNasIpPort	jnxL2tpSessionStatsEntry 32	The port on which RADIUS network address server accounting messages are sent. NOTE: This attribute is not supported on jl2tpd.
jnxL2tpSessionStatsFramedProtocol	jnxL2tpSessionStatsEntry 33	The frame protocol attribute obtained from the RADIUS server for the session. NOTE: This attribute is not supported on jl2tpd.
jnxL2tpSessionStatsFramedIpAddrType	jnxL2tpSessionStatsEntry 34	The IP address to be configured for the user, as provided by the RADIUS server in response to authentication request. NOTE: This attribute is not supported on jl2tpd.

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Table 112: jnxL2tpSessionStatsTable *(continued)*

Object	Object Identifier	Description
jnxL2tpSessionStatsFramedIpAddress	**jnxL2tpSessionStatsEntry 35**	The IP address to be configured for the user, as provided by the RADIUS server in response to the authentication request. NOTE: This attribute is not supported on jl2tpd.
jnxL2tpSessionStatsCallingStationID	**jnxL2tpSessionStatsEntry 36**	The phone number that the call came from. The RADIUS NAS obtains the phone number that the call came from by using Automatic Number Identification (ANI) or similar technology. It is used only in access-request packets.
jnxL2tpSessionStatsCalledStationID	**jnxL2tpSessionStatsEntry 37**	The phone number that the user called. The RADIUS NAS obtains the phone number that the user called by using Dialed Number Identification Service (DNIS) or similar technology. It is used only in Access-Request packets.
jnxL2tpSessionStatsAcctDelayTime	**jnxL2tpSessionStatsEntry 38**	The duration (in seconds) for which the RADIUS accounting client has been trying to send a record for. This value can be subtracted from the time of arrival on the server to find the approximate time of the event generating this Accounting-Request. NOTE: This attribute is not supported on jl2tpd.
jnxL2tpSessionStatsAcctSessionID	**jnxL2tpSessionStatsEntry 39**	A unique Accounting ID to help match start and stop records in a log file. NOTE: This attribute is not supported on jl2tpd.
jnxL2tpSessionStatsAcctMethod	**jnxL2tpSessionStatsEntry 40**	The accounting method employed for this session. NOTE: This attribute is not supported on jl2tpd.
jnxL2tpSessionStatsAcctSessionTime	**jnxL2tpSessionStatsEntry 41**	The number of seconds for which the user has received service. NOTE: This attribute is not supported on jl2tpd.

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Table 112: jnxL2tpSessionStatsTable *(continued)*

Object	Object Identifier	Description
jnxL2tpSessionStatsAcctNasPortType	jnxL2tpSessionStatsEntry 42	The type of the physical port of the NAS that performs accounting for the user. NOTE: This attribute is not supported on jl2tpd.
jnxL2tpSessionStatsAcctTnlClientEndPoint	jnxL2tpSessionStatsEntry 43	This object contains the remote tunnel identifier of the tunnel that hosts the session.
jnxL2tpSessionStatsAcctTnlServerEndPoint	jnxL2tpSessionStatsEntry 44	The local tunnel identifier of the tunnel that hosts the session.
jnxL2tpSessionStatsAcctTnlClientAuthID	jnxL2tpSessionStatsEntry 45	The remote hostname of the tunnel that hosts the session as discovered during the tunnel establishment phase (using the Host Name AVP) of the L2TP peer. NOTE: This attribute is not supported on jl2tpd.
jnxL2tpSessionStatsAcctTnlServerAuthID	jnxL2tpSessionStatsEntry 46	The local hostname of the tunnel that hosts the session. NOTE: This attribute is not supported on jl2tpd.
jnxL2tpSessionStatsUserProfileName	jnxL2tpSessionStatsEntry 47	The configured access profile name that is being used for the session. NOTE: This attribute is not supported on jl2tpd.
jnxL2tpSessionStatsInterfaceID	jnxL2tpSessionStatsEntry 48	The interface identification (name) for the service interface that bears the session. NOTE: This attribute is not supported on jl2tpd LAC.
jnxL2tpSessionStatsCallSerialNumber	jnxL2tpSessionStatsEntry 49	The serial number assigned to the session.
jnxL2tpSessionStatsCreationTim	jnxL2tpSessionStatsEntry 50	Time when the session was created.
jnxL2tpSessionStatsUpTim	jnxL2tpSessionStatsEntry 51	The time elapsed since the session was established.
jnxL2tpSessionStatsIdleTime	jnxL2tpSessionStatsEntry 52	The time elapsed since the session had any data activity (transmission or reception).

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Table 112: jnxL2tpSessionStatsTable *(continued)*

Object	Object Identifier	Description
jnxL2tpSessionStatsCollectionStart	jnxL2tpSessionStatsEntry 53	The time at which the statistics gathering started for the session.
jnxL2tpSessionStatsControlTxPkts	jnxL2tpSessionStatsEntry 54	The number of control packets transmitted to the session peer.
jnxL2tpSessionStatsControlTxBytes	jnxL2tpSessionStatsEntry 55	The number of control bytes that were transmitted to the session peer.
jnxL2tpSessionStatsControlRxPkts	jnxL2tpSessionStatsEntry 56	The number of control packets received on the session.
jnxL2tpSessionStatsControlRxBytes	jnxL2tpSessionStatsEntry 57	The number of control bytes received from the session peer.
jnxL2tpSessionStatsDataTxPkts	jnxL2tpSessionStatsEntry 58	The number of data packets transmitted to the remote session peer.
jnxL2tpSessionStatsDataTxBytes	jnxL2tpSessionStatsEntry 59	The number of data bytes transmitted to the session peer.
jnxL2tpSessionStatsDataRxPkts	jnxL2tpSessionStatsEntry 60	The number of data packets received on this session.
jnxL2tpSessionStatsDataRxBytes	jnxL2tpSessionStatsEntry 61	The number of data bytes received from the session peer.
jnxL2tpSessionStatsErrorTxPkt	jnxL2tpSessionStatsEntry 62	The number of error transmit packets on the session.
jnxL2tpSessionStatsErrorRxPkts	jnxL2tpSessionStatsEntry 63	The number of error receive packets on the session.

Related Documentation

- L2TP MIB on page 327
- L2TP Scalar Status and Statistics Group on page 328
- jnxL2tpTunnelGroupStatsTable on page 329
- jnxL2tpTunnelStatsTable on page 329
- jnxL2tpMlpppBundleStatsTable on page 338

jnxL2tpMlpppBundleStatsTable

The **jnxL2tpMlpppBundleStatsTable**, whose object ID is **{jnxL2tpObjects 5}**, contains objects that describe the current status and statistics of a single L2TP tunneled multilink PPP bundle.

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A **jnxL2tpMlpppBundleStatsEntry** represents the L2TP MLPPP bundle statistics and has the objects listed in Table 113 on page 339.

Table 113: jnxL2tpMlpppBundleStatsTable

Object	Object Identifier	Description
jnxL2tpMlpppBundleStatsBundleID	**jnxL2tpMlpppBundleStatsEntry 1**	Identifies the session's associated bundle.
jnxL2tpMlpppBundleStatsNumLinks	**jnxL2tpMlpppBundleStatsEntry 2**	Shows the current number of links that have joined the bundle.
jnxL2tpMlpppBundleStatsEndpoint	**jnxL2tpMlpppBundleStatsEntry 3**	Shows the username of the MLPPP bundle.
jnxL2tpMlpppBundleStatsInputMrru	**jnxL2tpMlpppBundleStatsEntry 4**	Shows the maximum packet size that the input interface can process.
jnxL2tpMlpppBundleStatsOutputMrru	**jnxL2tpMlpppBundleStatsEntry 5**	Shows the maximum packet size that the output interface can process.

Related Documentation

- L2TP MIB on page 327
- L2TP Scalar Status and Statistics Group on page 328
- jnxL2tpTunnelGroupStatsTable on page 329
- jnxL2tpTunnelStatsTable on page 329
- jnxL2tpSessionStatsTable on page 332

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CHAPTER 33

Interpreting the Enterprise-Specific Real-Time Performance Monitoring MIB

- Real-Time Performance Monitoring MIB on page 341
- jnxRpmResultsSampleTable on page 342
- JnxRpmTimestampType on page 344
- jnxRpmResultsSummaryTable on page 344
- jnxRpmResultsCalculatedTable on page 346
- jnxRpmHistorySampleTable on page 347
- jnxRpmHistorySummaryTable on page 347
- jnxRpmHistoryCalculatedTable on page 348

Real-Time Performance Monitoring MIB

The Juniper Networks enterprise-specific Real-Time Performance Monitoring (RPM) MIB enables you to access real-time performance-related data over SNMP. Starting with Junos OS Release 8.4, you can access jitter measurements and calculations over SNMP.

The RPM MIB represents a restructuring of the standard Ping MIB and converts the flat structure of the Ping MIB into a hierarchical collection of data. For more information about Ping MIB, see Chapter 25, “Interpreting the Enterprise-Specific Ping MIB.” Similar to the Ping MIB, the RPM MIB also has two groups of tables: the Results group and the History group. The RPM MIB, however, groups its data into separate collection types and measurement sets.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-rpm.txt .

For information about the enterprise-specific RPM MIB objects, see the following topics:

- jnxRpmResultsSampleTable on page 342
- JnxRpmTimestampType on page 344
- jnxRpmResultsSummaryTable on page 344
- jnxRpmResultsCalculatedTable on page 346
- jnxRpmHistorySampleTable on page 347

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- jnxRpmHistorySummaryTable on page 347
- jnxRpmHistoryCalculatedTable on page 348

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxRpmResultsSampleTable

The **jnxRpmResultsSampleTable** provides you measurements from the latest individual RPM probe samples. Each **jnxRpmResultsSampleEntry** has the objects listed in Table 114 on page 342.

NOTE: **jnxRpmResultsSampleTable does not maintain entries for unsuccessful probes.**

Table 114: jnxRpmResultsSampleTable

Object	Object Identifier	Description
jnxRpmResSampleType	**jnxRpmResultsSampleEntry 1**	The measurement type for the particular **jnxRpmResultsSampleEntry**.
jnxRpmResSampleValue	**jnxRpmResultsSampleEntry 2**	The measurement for the entry.
jnxRpmResSampleTsType	**jnxRpmResultsSampleEntry 3**	The type of timestamp used to obtain the measurement.
jnxRpmResSampleDate	**jnxRpmResultsSampleEntry 4**	The date and time at which the measurement was obtained.

For information about **JnxRpmMeasurementType**, see the following section:

- JnxRpmMeasurementType on page 342

JnxRpmMeasurementType

Different types of measurements can be made for each probe. Table 115 on page 342 lists the measurement types used in **jnxRpm.mib**.

Table 115: jnxRpmMeasurementType

Measurement Type	Description
roundTripTime	The delay between the transmission of a probe and the arrival of its response.
rttJitter	The difference between the current round-trip time measurement and the previous one.

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Table 115: jnxRpmMeasurementType *(continued)*

Measurement Type	Description
rttInterarrivalJitter	An estimate of the statistical variance of a packet's interarrival time defined in RFC 1889 as: J=J+(|D(i-1,i)|-J)/16 where J is the interarrival jitter and D (i-1, i) is the egressJitter measurement.
egress	The delay between the transmission of a probe and its arrival at the destination.
egressJitter	The difference between the current egress delay and the previous measurement.
egressInterarrivalJitter	An estimate of the statistical variance of a packet's interarrival time. Defined in RFC 1889 as: J=J+(|D(i-1,i)|-J)/16 where J is the interarrival jitter and D (i-1, i) is the egressJitter measurement
ingress	The delay between the transmission of a probe response and its arrival at the destination.
ingressJitter	The difference between the current ingress delay and the previous measurement.
ingressInterarrivalJitter	An estimate of the statistical variance of a packet's interarrival time. Defined in RFC 1889 as: J=J+(|D(i-1,i)|-J)/16 where J is the interarrival jitter and D (i-1, i) is the current ingressJitter measurement.

Not all types of measurements are performed for every probe. For example, the jitter measurements are available only for those RPM entries that use hardware timestamps on both client and server. Similarly, the ingress and egress measurements are available only for those probe types that measure one-way delays or where hardware timestamps are used (for this, the one-way-hardware timestamp knob must be enabled). However, in these cases, if the one-way delay is greater than the round-trip time, the corresponding entries are not stored.



NOTE: To avoid possible variations in one-way jitter measurements and calculations due to clock synchronization issues, one-way jitter measurements are performed only on samples that are less than 10 seconds apart.

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Related Documentation
- Real-Time Performance Monitoring MIB on page 341
- jnxRpmResultsSummaryTable on page 344
- jnxRpmResultsCalculatedTable on page 346
- jnxRpmHistorySampleTable on page 347
- jnxRpmHistorySummaryTable on page 347
- jnxRpmHistoryCalculatedTable on page 348

JnxRpmTimestampType

The following three types of timestamps are used to obtain measurements:

- **software**—Indicates that software-based timestamps are used on both client and server.
- **clientHardware**—Indicates that hardware-based timestamps are used on the client.
- **clientAndServerHardware**—Indicates that hardware-based timestamps are used on the RPM client and the server.

Related Documentation
- Real-Time Performance Monitoring MIB on page 341
- jnxRpmResultsSampleTable on page 342
- jnxRpmResultsSummaryTable on page 344
- jnxRpmResultsCalculatedTable on page 346
- jnxRpmHistorySampleTable on page 347
- jnxRpmHistorySummaryTable on page 347
- jnxRpmHistoryCalculatedTable on page 348

jnxRpmResultsSummaryTable

The **jnxRpmResultsSummaryTable** provides a summary of the results for each RPM entry (identified by **pingCtlOwnerIndex**/**pingCtlTestName** in the Ping MIB) and for each data collection maintained by that entry. The RPM feature maintains several different collections of probe data, providing overall summaries as well as detailed calculations for each collection.

The **jnxRpmResultsSummaryTable** maintains the following collection types:

- **currentTest**—The test that is being executed currently.
- **lastCompletedTest**—The most recently completed test.

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- **movingAverage**—A list of most recent probes. You can configure the number of probes for this list using **jnxPingCtlMovAvgSize** or the **moving-average-size** command-line interface (CLI) command.
- **allTests**—All the probes that were sent. The value gets reset when the 64-bit value storing the square rolls over.

For each collection type, the table provides the following details:

- Number of probes sent
- Number of probes received
- Percentage of probes lost
- Timestamp for the latest sample in the collection

The **jnxRpmResultsSummaryEntry** has the objects listed in Table 116 on page 345.

Table 116: jnxRpmResultsSummaryTable

Object	Object Identifer	Description
jnxRpmResSumCollection	**jnxRpmResultsSummaryEntry 1**	The collection of probes to which the **jnxRpmResultsSummaryEntry** refers. NOTE: No entries are created for collection types that are not supported or not configured.
jnxRpmResSumSent	**jnxRpmResultsSummaryEntry 2**	The number of probes sent within the collection.
jnxRpmResSumReceived	**jnxRpmResultsSummaryEntry 3**	The number of probes received within the collection.
jnxRpmResSumPercentLost	**jnxRpmResultsSummaryEntry 4**	The percentage of probes that are lost within the collection.
jnxRpmResSumDate	**jnxRpmResultsSummaryEntry 5**	The timestamp for the most recent probe within the collection.

Related Documentation

- Real-Time Performance Monitoring MIB on page 341
- jnxRpmResultsSampleTable on page 342
- JnxRpmTimestampType on page 344
- jnxRpmResultsCalculatedTable on page 346
- jnxRpmHistorySampleTable on page 347
- jnxRpmHistorySummaryTable on page 347
- jnxRpmHistoryCalculatedTable on page 348

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jnxRpmResultsCalculatedTable

The **jnxRpmResultsCalculatedTable** provides a set of calculated values for each RPM entry, for each collection of probes maintained within that entry, and for each supported measurement set within that collection of probes.

The **jnxRpmResultsCalculatedEntry** has the objects listed in Table 117 on page 346.

Table 117: jnxRpmResultsCalculatedTable

Object	Object Identifier	Description
jnxRpmResCalcSet	jnxRpmResultsCalculatedEntry 1	The measurement set for the particular **jnxRpmResultsCalculatedEntry**.
jnxRpmResCalcSamples	jnxRpmResultsCalculatedEntry 2	The number of samples used in the calculations.
jnxRpmResCalcMin	jnxRpmResultsCalculatedEntry 3	The minimum (in microseconds) of all the samples in the collection and the measurement set associated with the entry.
jnxRpmResCalcMax	jnxRpmResultsCalculatedEntry 4	The maximum (in microseconds) of all the samples in the collection and the measurement set.
jnxRpmResCalcAverage	jnxRpmResultsCalculatedEntry 5	The average (in microseconds) of all the samples in the collection and the measurement set associated with the entry.
jnxRpmResCalcPkToPk	jnxRpmResultsCalculatedEntry 6	The difference (in microseconds) between the minimum and maximum of all the samples in the collection and the measurement set associated with the entry.
jnxRpmResCalcStdDev	jnxRpmResultsCalculatedEntry 7	The standard deviation (in microseconds) calculated over all the samples in the collection and the measurement set associated with the entry.
jnxRpmResCalcSum	jnxRpmResultsCalculatedEntry 8	The sum (in microseconds) of all the samples in the collection and the measurement set associated with the entry.

Related Documentation

- Real-Time Performance Monitoring MIB on page 341
- jnxRpmResultsSampleTable on page 342
- JnxRpmTimestampType on page 344
- jnxRpmResultsSummaryTable on page 344
- jnxRpmHistorySampleTable on page 347
- jnxRpmHistorySummaryTable on page 347
- jnxRpmHistoryCalculatedTable on page 348

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jnxRpmHistorySampleTable

The **jnxRpmHistorySampleTable** provides measurements for each sample stored in the history table of RPM probe entries. In addition to the last completed probe, the table also provides data for a configurable number of most recent probes (all the history tables in this MIB provide the same number of entries as the **pingProbeHistoryTable**). However, the table does not maintain entries for:

- Unsuccessful probes
- Invalid measurement types

The **jnxRpmHistorySampleEntry** has the objects listed in Table 118 on page 347.

Table 118: jnxRpmHistorySampleTable

Object	Object Identifier	Description
jnxRpmHistSampleType	**jnxRpmHistorySampleEntry 1**	The measurement type associated with the entry.
jnxRpmHistSampleValue	**jnxRpmHistorySampleEntry 2**	The measurement for the entry.
jnxRpmHistSampleTsType	**jnxRpmHistorySampleEntry 3**	The type of timestamp used to obtain the measurement.

Related Documentation

- Real-Time Performance Monitoring MIB on page 341
- jnxRpmResultsSampleTable on page 342
- JnxRpmTimestampType on page 344
- jnxRpmResultsSummaryTable on page 344
- jnxRpmResultsCalculatedTable on page 346
- jnxRpmHistorySummaryTable on page 347
- jnxRpmHistoryCalculatedTable on page 348

jnxRpmHistorySummaryTable

Similar to the **jnxRpmResultsSummaryTable**, the **jnxRpmHistorySummaryTable** provides you with summary data for each collection of probes within each RPM entry. In addition to summary data for the current probe, the table also provides summary information for a number of the most recent probes. You can configure the number of most recent probes that should be stored in the table.

The **jnxRpmHistorySummaryEntry** has the objects listed in Table 119 on page 348.

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Table 119: jnxRpmHistorySummaryTable

Object	Object Identifier	Description
jnxRpmHistSumCollection	**jnxRpmHistorySummaryEntry 1**	The collection of probes associated with the entry. NOTE: Historical summaries are available only for the current test (**currentTest**).
jnxRpmHistSumSent	**jnxRpmHistorySummaryEntry 2**	The number of probes sent within the collection.
jnxRpmHistSumReceived	**jnxRpmHistorySummaryEntry 3**	The number of probes received within the collection.
jnxRpmHistSumPercentLost	**jnxRpmHistorySummaryEntry 4**	The percentage of probes lost within the collection.

Related Documentation

- Real-Time Performance Monitoring MIB on page 341
- jnxRpmResultsSampleTable on page 342
- JnxRpmTimestampType on page 344
- jnxRpmResultsSummaryTable on page 344
- jnxRpmResultsCalculatedTable on page 346
- jnxRpmHistorySampleTable on page 347
- jnxRpmHistoryCalculatedTable on page 348

jnxRpmHistoryCalculatedTable

As with the **jnxRpmResultsCalculatedTable**, the **jnxRpmHistoryCalculatedTable** provides a set of calculated values for each RPM entry, for each collection of probes maintained within that entry, and for each supported calculated type within that collection of probes.

In addition to data from the current probe, this table also provides data from a configurable number of the most recent probes.



NOTE: **The only collection type that is stored in jnxRpmHistoryCalculatedTable is the currentTest.**

Each **jnxRpmHistoryCalculatedEntry** has the objects listed in Table 120 on page 348.

Table 120: jnxRpmHistoryCalculatedTable

Object	Object Identifier	Description
jnxRpmHistCalcSet	**jnxRpmHistoryCalculatedEntry 1**	The measurement set for the **jnxRpmHistoryCalculatedEntry**.
jnxRpmHistCalcSamples	**jnxRpmHistoryCalculatedEntry 2**	The number of samples used in the calculations for this entry.

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Table 120: jnxRpmHistoryCalculatedTable *(continued)*

Object	Object Identifier	Description
jnxRpmHistCalcMin	jnxRpmHistoryCalculatedEntry 3	The minimum (in microseconds) of all the samples in the collection and the measurement set associated with the entry.
jnxRpmHistCalcMax	jnxRpmHistoryCalculatedEntry 4	The maximum (in microseconds) of all the samples in the collection and the measurement set associated with the entry.
jnxRpmHistCalcAverage	jnxRpmHistoryCalculatedEntry 5	The average (in microseconds) of all the samples in the collection and the measurement set associated with the entry.
jnxRpmHistCalcPkToPk	jnxRpmHistoryCalculatedEntry 6	The difference (in microseconds) between the minimum and the maximum of all the samples in the collection and the measurement set associated with the row.
jnxRpmHistCalcStdDev	jnxRpmHistoryCalculatedEntry 7	The standard deviation (in microseconds) calculated over all the samples in the collection and the measurement set associated with the entry.
jnxRpmHistCalcSum	jnxRpmHistoryCalculatedEntry 8	The sum of all the samples in the collection and the measurement set associated with the entry.

Related Documentation

- Real-Time Performance Monitoring MIB on page 341
- jnxRpmResultsSampleTable on page 342
- JnxRpmTimestampType on page 344
- jnxRpmResultsSummaryTable on page 344
- jnxRpmResultsCalculatedTable on page 346
- jnxRpmHistorySampleTable on page 347
- jnxRpmHistorySummaryTable on page 347

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CHAPTER 34

Interpreting the Enterprise-Specific Class-of-Service MIB

- Class-of-Service MIB on page 351
- jnxCosInvQstatTable on page 351

Class-of-Service MIB

The Juniper Networks enterprise-specific Class-of-Service (CoS) MIB provides support for monitoring interface output queue statistics per interface and per forwarding class.

The CoS MIB is an object of the **jnxMibs** branch of the enterprise-specific MIB and has an object identifier of **{jnxMIB 15}**. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-cos.txt .

For information about the enterprise-specific CoS MIB objects, see the following topic:

- jnxCosInvQstatTable on page 351

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxCosInvQstatTable

This table provides output queue statistics for each interface. Only those physical interfaces that support output queue statistics appear in this table. This table provides the same information as the **jnxCosQstatTable**, but the information is ordered by queue number and interface index, instead of by interface index and queue number.

The entries in the **jnxCosInvQstatTable**, whose object identifier is **{jnxCos 6}**, are represented by **jnxCosInvQstatEntry**, whose object identifier is **{jnxCosIfqStatsTable 1}**, and are listed in Table 121 on page 351.

Table 121: jnxCosInvQstatEntry

Object	Object Identifier	Description
jnxCosInvQstatQedPkts	**jnxCosInvQstatEntry 1**	The number of packets queued at the output queue.

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Table 121: jnxCosInvQstatEntry *(continued)*

Object	Object Identifier	Description
jnxCosInvQstatQedPktRate	jnxCosInvQstatEntry 2	The rate (in pps) at which packets are queued at the output queue.
jnxCosInvQstatQedBytes	jnxCosInvQstatEntry 3	The number of bytes queued at the output queue.
jnxCosInvQstatQedByteRate	jnxCosInvQstatEntry 4	The rate (in bytes per second) at which bytes are queued at the output queue.
jnxCosInvQstatQedTxedPkts	jnxCosInvQstatEntry 5	The number of packets transmitted on the queue.
jnxCosInvQstatQedTxedPktRate	jnxCosInvQstatEntry 6	The packet transmission rate of the output queue (in pps).
jnxCosInvQstatQedTxedBytes	jnxCosInvQstatEntry 7	The number of bytes transmitted on the queue.
jnxCosInvQstatQedTxedByteRate	jnxCosInvQstatEntry 8	The byte transmission rate of the output queue (in bytes per second).
jnxCosInvQstatQedTailDropPkts	jnxCosInvQstatEntry 9	The number of packets tail dropped at the output queue.
jnxCosInvQstatQedTailDropPktRate	jnxCosInvQstatEntry 10	The tail drop packet rate (in pps) for the queue.
jnxCosInvQstatTotalRedDropPkts	jnxCosInvQstatEntry 11	The number of packets dropped on the interface due to random early detection (RED) at the output.
jnxCosInvQstatTotalRedDropPktRate	jnxCosInvQstatEntry 12	The most-recent estimate of the drop rate (in pps) for packets dropped on the interface due to RED at the output.
jnxCosInvQstatLpNonTcpRedDropPkts	jnxCosInvQstatEntry 13	The number of low PLP nonTCP packets dropped on the interface due to RED at the output.
jnxCosInvQstatLpNonTcpRedDropPktRate	jnxCosInvQstatEntry 14	The rate (in pps) at which low PLP nonTCP packets are dropped on the interface due to RED at the output.
jnxCosInvQstatLpTcpRedDropPkts	jnxCosInvQstatEntry 15	The number of low PLP TCP packets dropped on the interface due to RED at the output.
jnxCosInvQstatLpTcpRedDropPktRate	jnxCosInvQstatEntry 16	The rate (in pps) at which low PLP TCP packets are dropped on the interface due to RED at the output.
jnxCosInvQstatHpNonTcpRedDropPkts	jnxCosInvQstatEntry 17	The number of high PLP nonTCP packets dropped on the interface due to RED at the output.
jnxCosInvQstatHpNonTcpRedDropPktRate	jnxCosInvQstatEntry 18	The rate (in pps) at which high PLP nonTCP packets are dropped on the interface due to RED at the output.

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Table 121: jnxCosInvQstatEntry *(continued)*

Object	Object Identifier	Description
jnxCosInvQstatHpTcpRedDropPkts	jnxCosInvQstatEntry 19	The number of high PLP TCP packets dropped on the interface due to RED at the output.
jnxCosInvQstatHpTcpRedDropPktRate	jnxCosInvQstatEntry 20	The rate (in pps) at which high PLP TCP packets are dropped on the interface due to RED at the output.
jnxCosInvQstatTotalRedDropBytes	jnxCosInvQstatEntry 21	The number of bytes dropped on the interface due to RED at the output.
jnxCosInvQstatTotalRedDropByteRate	jnxCosInvQstatEntry 22	The rate (in bytes per second) at which bytes are dropped on the interface due to RED at the output.
jnxCosInvQstatLpNonTcpRedDropBytes	jnxCosInvQstatEntry 23	The number of low PLP nonTCP bytes dropped on the interface due to RED at the output.
jnxCosInvQstatLpNonTcpRedDropByteRate	jnxCosInvQstatEntry 24	The rate (in bytes per second) at which low PLP nonTCP bytes are dropped on the interface due to RED at the output.
jnxCosInvQstatLpTcpRedDropBytes	jnxCosInvQstatEntry 25	The number of low PLP TCP bytes dropped on the interface due to RED at the output.
jnxCosInvQstatLpTcpRedDropByteRate	jnxCosInvQstatEntry 26	The rate (in bytes per second) at which low PLP TCP bytes are dropped on the interface due to RED at the output.
jnxCosInvQstatHpNonTcpRedDropBytes	jnxCosInvQstatEntry 27	The number of high PLP nonTCP bytes dropped on the interface due to RED at the output.
jnxCosInvQstatHpNonTcpRedDropByteRate	jnxCosInvQstatEntry 28	The rate (in bytes per second) at which high PLP nonTCP bytes are dropped on the interface due to RED at the output.
jnxCosInvQstatHpTcpRedDropBytes	jnxCosInvQstatEntry 29	The number of high PLP TCP bytes dropped on the interface due to RED at the output.
jnxCosInvQstatHpTcpRedDropByteRate	jnxCosInvQstatEntry 30	The rate (in bytes per second) at which high PLP TCP bytes are dropped on the interface due to RED at the output.
jnxCosInvQstatLpRedDropPkts	jnxCosInvQstatEntry 31	The number of low PLP packets dropped on the interface due to RED at the output.
jnxCosInvQstatLpRedDropPktRate	jnxCosInvQstatEntry 32	The rate (in pps) at which low PLP packets are dropped on the interface due to RED at the output.
jnxCosInvQstatMLpRedDropPkts	jnxCosInvQstatEntry 33	The number of medium-low PLP packets dropped on the interface due to RED at the output.
jnxCosInvQstatMLpRedDropPktRate	jnxCosInvQstatEntry 34	The rate (in pps) at which medium-low PLP packets are dropped on the interface due to RED at the output.

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Table 121: jnxCosInvQstatEntry *(continued)*

Object	Object Identifier	Description
jnxCosInvQstatMHpRedDropPkts	jnxCosInvQstatEntry 35	The number of medium-high PLP packets dropped on the interface due to RED at the output.
jnxCosInvQstatMHpRedDropPktRate	jnxCosInvQstatEntry 36	The rate (in pps) at which medium-high PLP packets are dropped on the interface due to RED at the output.
jnxCosInvQstatHpRedDropPkts	jnxCosInvQstatEntry 37	The number of high PLP packets dropped on the interface due to RED at the output.
jnxCosInvQstatHpRedDropPktRate	jnxCosInvQstatEntry 38	The rate (in pps) at which high PLP packets are dropped on the interface due to RED at the output.
jnxCosInvQstatLpRedDropBytes	jnxCosInvQstatEntry 39	The number of low PLP bytes dropped on the interface due to RED at the output.
jnxCosInvQstatLpRedDropByteRate	jnxCosInvQstatEntry 40	The rate (in bytes per second) at which low PLP bytes are dropped on the interface due to RED at the output.
jnxCosInvQstatMLpRedDropBytes	jnxCosInvQstatEntry 41	The number of medium-low PLP bytes dropped on the interface due to RED at the output.
jnxCosInvQstatMLpRedDropByteRate	jnxCosInvQstatEntry 42	The rate (in bytes per second) at which medium-low PLP bytes are dropped on the interface due to RED at the output.
jnxCosInvQstatMHpRedDropBytes	jnxCosInvQstatEntry 43	The number of medium-high PLP bytes dropped on the interface due to RED at the output.
jnxCosInvQstatMHpRedDropByteRate	jnxCosInvQstatEntry 44	The rate (in bytes per second) at which medium-high PLP bytes are dropped on the interface due to RED at the output.
jnxCosInvQstatHpRedDropBytes	jnxCosInvQstatEntry 45	The number of high PLP bytes dropped on the interface due to RED at the output.
jnxCosInvQstatHpRedDropByteRate	jnxCosInvQstatEntry 46	The rate (in bytes per second) at which high PLP bytes are dropped on the interface due to RED at the output.

Related Documentation

- Class-of-Service MIB on page 351

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CHAPTER 35

Interpreting the Enterprise-Specific IP Forward MIB

- IP Forward MIB on page 355
- jnxIpCidrRouteTable on page 356

IP Forward MIB

The Juniper Networks enterprise-specific IP Forward MIB, whose object identifier is **{jnxMibs 38}**, extends the **ipCidrRouteTable** in the IP Forwarding Table MIB (as defined in RFC 2096) to include a tunnel name when the next hop is through an RSVP-signaled LSP.

This MIB adds an **jnxIpCidrRouteTunnelName** attribute to the **ipCidrRouteTable**. The attribute exists for each entry in the **ipCidrRouteTable**. (One entry in the **ipCidrRouteTable** represents each route in **inet.0**). If the route's next hop is an RSVP-signaled MPLS LSP, the new attribute contains the LSP name. If the route's next hop is not an RSVP-signaled MPLS LSP, the new attribute is defined as null.

The attribute's name is **jnxIpCidrRouteTunnelName**. Its object identifier (OID) is 1.3.6.1.4.1.2636.3.38.1.1.1. As with any SNMP attribute, an index is appended to the OID to form the instance identifier. Because this attribute augments the **ipCidrRouteTable**, the index is identical to that used in the **ipCidrRouteTable**. The index is formed by concatenating the destination address, subnet mask, ToS byte, and next hop.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ipforward.txt .

For more information about the enterprise-specific IP Forward MIB objects, see the following topic:

- jnxIpCidrRouteTable on page 356

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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jnxIpCidrRouteTable

The **jnxIpCidrRouteTable**, whose object identifier is **{jnxIpForwardMIB 1}**, extends the **ipCidrRouteTable** with additional data.

jnxIpCidrRouteEntry, whose object identifier is **{jnxIpCidrRouteTable 1}**, has one object, which is listed in Table 122 on page 356.

Table 122: jnxIpCidrRouteTable

Object	Object Identifier	Description
jnxIpCidrRouteTunnelName	**nxIpCidrRouteEntry 1**	The canonical name assigned to the tunnel. The router forwards traffic bound for the destination through this tunnel.

Related Documentation

- IP Forward MIB on page 355

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CHAPTER 36

Interpreting the Enterprise-Specific IPv6 MIB

- IPv6 MIB on page 357
- jnxIpv6GlobalStats on page 357
- jnxIcmpv6GlobalStats on page 360

IPv6 MIB

The Juniper Networks enterprise-specific IPv6 MIB, whose object identifier is **{jnxMibs 11}**, functions as an extension of the **ifTable** defined in RFC 2465, *IF MIB*, and defines the branches of IPv6 configuration.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-ipv6.txt .

For more information about the enterprise-specific IPv6 MIB objects, see the following topics:

- jnxIpv6GlobalStats on page 357
- jnxIcmpv6GlobalStats on page 360

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxIpv6GlobalStats

The **jnxIpv6GlobalStats**, whose object ID is **{jnxIpv6Stats 1}**, represents the **jnxIpv6GlobalStats** branch of IPv6 stats. Table 123 on page 357 identifies objects in the **jnxIpv6GlobalStats** branch.

Table 123: jnxIpv6GlobalStats

Object	Object ID	Description
jnxIpv6StatsReceives	**jnxIpv6GlobalStats 1**	The total number of IPv6 packets received.

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Table 123: jnxIpv6GlobalStats *(continued)*

Object	Object ID	Description
jnxIpv6StatsTooShorts	jnxIpv6GlobalStats 2	The total number of IPv6 packets received with size smaller than minimum.
jnxIpv6StatsTooSmalls	jnxIpv6GlobalStats 3	The total number of IPv6 packets with data size less than data length.
jnxIpv6StatsBadOptions	jnxIpv6GlobalStats 4	The total number of IPv6 packets encountering errors in option processing.
jnxIpv6StatsBadVersions	jnxIpv6GlobalStats 5	The total number of IPv6 packets with incorrect version number.
jnxIpv6StatsFragments	jnxIpv6GlobalStats 6	The total number of packet fragments received.
jnxIpv6StatsFragDrops	jnxIpv6GlobalStats 7	The total number of fragments dropped (duplicates or out of place).
jnxIpv6StatsFragTimeOuts	jnxIpv6GlobalStats 8	The total number of fragments dropped due to timeout.
jnxIpv6StatsFragOverFlows	jnxIpv6GlobalStats 9	The total number of fragments that exceeded limit.
jnxIpv6StatsReasmOKs	jnxIpv6GlobalStats 10	The total number of IPv6 packets reassembled successfully.
jnxIpv6StatsDelivers	jnxIpv6GlobalStats 11	The total number of IPv6 packets delivered to the upper layers.
jnxIpv6StatsForwards	jnxIpv6GlobalStats 12	The total number of IPv6 packets forwarded.
jnxIpv6StatsUnreachables	jnxIpv6GlobalStats 13	The total number of IPv6 packets received for unreachable destinations.
jnxIpv6StatsRedirects	jnxIpv6GlobalStats 14	The total number of IPv6 packets forwarded on the same net as received.
jnxIpv6StatsOutRequests	jnxIpv6GlobalStats 15	The total number of IPv6 packets generated by the current host.
jnxIpv6StatsRawOuts	jnxIpv6GlobalStats 16	The total number of IPv6 packets sent with fabricated IPv6 headers.
jnxIpv6StatsOutDiscards	jnxIpv6GlobalStats 17	The total number of output IPv6 packets dropped due to no buffers and so on.
jnxIpv6StatsOutNoRoutes	jnxIpv6GlobalStats 18	The total number of output IPv6 packets discarded because no route was found.
jnxIpv6StatsOutFragOKs	jnxIpv6GlobalStats 19	The total number of output IPv6 packets fragmented.
jnxIpv6StatsOutFragCreates	jnxIpv6GlobalStats 20	The total number of output IPv6 packets that have been generated as a result of fragmentation at the output interface.

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Table 123: jnxIpv6GlobalStats *(continued)*

Object	Object ID	Description
jnxIpv6StatsOutFragFails	jnxIpv6GlobalStats 21	The total number of output IPv6 packets that cannot be fragmented.
jnxIpv6StatsBadScopes	jnxIpv6GlobalStats 22	The total number of IPv6 packets that violate scope rules.
jnxIpv6StatsNotMcastMembers	jnxIpv6GlobalStats 23	The total number of IPv6 multicast packets that the users do not join.
jnxIpv6StatsHdrNotContinuous	jnxIpv6GlobalStats 24	The total number of IPv6 packets whose headers are not continuous.
jnxIpv6StatsNoGifs	jnxIpv6GlobalStats 25	The total number of tunneling IPv6 packets that canot find graphic interchange format (GIF).
jnxIpv6StatsTooManyHdrs	jnxIpv6GlobalStats 26	The total number of IPv6 packets discarded due to numerous headers.
jnxIpv6StatsForwCacheHits	jnxIpv6GlobalStats 27	The total number of forward cache hits.
jnxIpv6StatsForwCacheMisses	jnxIpv6GlobalStats 28	The total number of forward cache misses.
jnxIpv6StatsOutDeadNextHops	jnxIpv6GlobalStats 29	The total number of output IPv6 packets destined to dead next hops.
jnxIpv6StatsOptRateDrops	jnxIpv6GlobalStats 30	The total number of IPv6 option packets dropped due to rate limits.
jnxIpv6StatsMCNoDests	jnxIpv6GlobalStats 31	The total number of IPv6 Multicast packets dropped due to no destination.

Table 124 on page 359 identifies the count of all the other IPv6 next header types received.

Table 124: Count of All Other IPv6 Next Header Types Received

Object	Object ID	Description
jnxIpv6StatsInHopByHops	jnxIpv6GlobalStats 32	The total number of IPv6 packets received with a Hop by Hop next header.
jnxIpv6StatsInIcmps	jnxIpv6GlobalStats 33	The total number of IPv6 packets received with an ICMP(v4) next header.
jnxIpv6StatsInIgmps	jnxIpv6GlobalStats 34	The total number of IPv6 packets received with an IGMP next header.
jnxIpv6StatsInIps	jnxIpv6GlobalStats 35	The total number of IPv6 packets received with an IPv4 next header.
jnxIpv6StatsInTcps	jnxIpv6GlobalStats 36	The total number of IPv6 packets received with a TCP next header.

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Table 124: Count of All Other IPv6 Next Header Types Received *(continued)*

Object	Object ID	Description
jnxIpv6StatsInUdps	jnxIpv6GlobalStats 37	The total number of IPv6 packets received with a UDP next header.
jnxIpv6StatsInIdps	jnxIpv6GlobalStats 38	The total number of IPv6 packets received with an xns IDP next header.
jnxIpv6StatsInTps	jnxIpv6GlobalStats 39	The total number of IPv6 packets received with a TP4 next header.
jnxIpv6StatsInIpv6s	jnxIpv6GlobalStats 40	The total number of IPv6 packets with an IPv6 next header.
jnxIpv6StatsInRoutings	jnxIpv6GlobalStats 41	The total number of IPv6 packets with an IPv6 Routing next header.
jnxIpv6StatsInFrags	jnxIpv6GlobalStats 42	The total number of IPv6 packets with an IPv6 Fragment next header.
jnxIpv6StatsInEsps	jnxIpv6GlobalStats 43	The total number of IPv6 packets with an IPv6 ESP next header.
jnxIpv6StatsInAhs	jnxIpv6GlobalStats 44	The total number of IPv6 packets with an IPv6 AH next header.
jnxIpv6StatsInIcmpv6s	jnxIpv6GlobalStats 45	The total number of IPv6 packets with an ICMPv6 next header.
jnxIpv6StatsInNoNhs	jnxIpv6GlobalStats 46	The total number of IPv6 packets with no next header.
jnxIpv6StatsInDestOpts	jnxIpv6GlobalStats 47	The total number of IPv6 packets with an IPv6 Destination Option next header.
jnxIpv6StatsInIsoIps	jnxIpv6GlobalStats 48	The total number of IPv6 packets with an ISO CLNP next header.
jnxIpv6StatsInOspfs	jnxIpv6GlobalStats 49	The total number of IPv6 packets with an OSPF next header.
jnxIpv6StatsInEths	jnxIpv6GlobalStats 50	The total number of IPv6 packets with an Ethernet next header.
jnxIpv6StatsInPims	jnxIpv6GlobalStats 51	The total number of IPv6 packets with a PIM next header.

Related Documentation

- IPv6 MIB on page 357
- jnxIcmpv6GlobalStats on page 360

jnxIcmpv6GlobalStats

The **jnxIcmpv6GlobalStats**, whose object ID is **{jnxIpv6Stats 2}**, represents the jnxIcmpv6GlobalStats branch of IPv6 stats. Table 125 on page 361 identifies objects in the jnxIcmpv6GlobalStats branch.

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Table 125: jnxIcmpv6GlobalStats

Object	Object ID	Description
jnxIcmpv6StatsErrors	jnxIcmpv6GlobalStats 1	The total number of calls to **icmp_error**.
jnxIcmpv6StatsCantErrors	jnxIcmpv6GlobalStats 2	The total number of errors that are not generated if the old message was an ICMP error.
jnxIcmpv6StatsTooFreqs	jnxIcmpv6GlobalStats 3	The total number of errors that are not generated due to rate limitations.
jnxIcmpv6StatsBadCodes	jnxIcmpv6GlobalStats 4	The total number of messages with bad code fields.
jnxIcmpv6StatsTooShorts	jnxIcmpv6GlobalStats 5	The total number of messages less than the minimum length.
jnxIcmpv6StatsBadChecksums	jnxIcmpv6GlobalStats 6	The total number of messages with bad checksums.
jnxIcmpv6StatsBadLenths	jnxIcmpv6GlobalStats 7	The total number of messages with bad length.
jnxIcmpv6StatsNoRoutes	jnxIcmpv6GlobalStats 8	The total number of “no route” messages generated.
jnxIcmpv6StatsAdminProhibits	jnxIcmpv6GlobalStats 9	The total number of “administratively prohibited” messages generated.
jnxIcmpv6StatsBeyondScopes	jnxIcmpv6GlobalStats 10	The total number of “beyond scope” messages generated.
jnxIcmpv6StatsAddrUnreachs	jnxIcmpv6GlobalStats 11	The total number of “address unreachable” messages generated.
jnxIcmpv6StatsPortUnreachs	jnxIcmpv6GlobalStats 12	The total number of “port unreachable” messages generated.
jnxIcmpv6StatsTooBigs	jnxIcmpv6GlobalStats 13	The total number of “packet too big” messages generated.
jnxIcmpv6StatsExceedTrans	jnxIcmpv6GlobalStats 14	The total number of “time exceed transit” messages generated.
jnxIcmpv6StatsExceedReasms	jnxIcmpv6GlobalStats 15	The total number of “time exceed reassembly” messages generated.
jnxIcmpv6StatsBadHdrFields	jnxIcmpv6GlobalStats 16	The total number of “erroneous header field” messages generated.
jnxIcmpv6StatsBadNextHdrs	jnxIcmpv6GlobalStats 17	The total number of “unrecognized next header” messages generated.
jnxIcmpv6StatsBadOptions	jnxIcmpv6GlobalStats 18	The total number of “unrecognized option” messages generated.
jnxIcmpv6StatsRedirects	jnxIcmpv6GlobalStats 19	The total number of “redirect” messages generated.

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Table 125: jnxIcmpv6GlobalStats *(continued)*

Object	Object ID	Description
jnxIcmpv6StatsOthers	jnxIcmpv6GlobalStats 20	The total number of generated message types other than the ones previously listed.
jnxIcmpv6StatsResponses	jnxIcmpv6GlobalStats 21	The total number of message responses generated.
jnxIcmpv6StatsExcessNDOptions	jnxIcmpv6GlobalStats 22	The total number of messages processed with too many ND options.

Table 126 on page 362 identifies the count of all the other ICMPv6 message types received and transmitted.

Table 126: Count of All Other ICMPv6 Message Types Received and Transmitted

Object	Object ID	Description
jnxIcmpv6StatsInUnreachables	jnxIcmpv6GlobalStats 23	The total number of unreachable messages received.
jnxIcmpv6StatsInPktTooBigs	jnxIcmpv6GlobalStats 24	The total number of “packet too big” messages received.
jnxIcmpv6StatsInTimeExceeds	jnxIcmpv6GlobalStats 25	The total number of “time exceeded” messages received.
jnxIcmpv6StatsInParamProbs	jnxIcmpv6GlobalStats 26	The total number of “parameter problem” messages received.
jnxIcmpv6StatsInEchoReqs	jnxIcmpv6GlobalStats 27	The total number of “echo request” messages received.
jnxIcmpv6StatsInEchoReplies	jnxIcmpv6GlobalStats 28	The total number of “echo reply” messages received.
jnxIcmpv6StatsInMLQueries	jnxIcmpv6GlobalStats 29	The total number of “multicast listener query” messages received.
jnxIcmpv6StatsInMLReports	jnxIcmpv6GlobalStats 30	The total number of “multicast listener report” messages received.
jnxIcmpv6StatsInMLDones	jnxIcmpv6GlobalStats 31	The total number of “multicast listener done” messages received.
jnxIcmpv6StatsInRtrSolicits	jnxIcmpv6GlobalStats 32	The total number of “router solicitation” messages received.
jnxIcmpv6StatsInRtrAdvs	jnxIcmpv6GlobalStats 33	The total number of “router advertisement” messages received.
jnxIcmpv6StatsInNbrSolicits	jnxIcmpv6GlobalStats 34	The total number of “neighbor solicitation” messages received.
jnxIcmpv6StatsInNbrAdvs	jnxIcmpv6GlobalStats 35	The total number of “neighbor advertisement” messages received.
jnxIcmpv6StatsInRedirects	jnxIcmpv6GlobalStats 36	The total number of “redirect” messages received.

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Table 126: Count of All Other ICMPv6 Message Types Received and Transmitted *(continued)*

Object	Object ID	Description
jnxIcmpv6StatsInRtrRenumbers	jnxIcmpv6GlobalStats 37	The total number of "router renumber" messages received.
jnxIcmpv6StatsInNIReqs	jnxIcmpv6GlobalStats 38	The total number of "node information request" messages received.
jnxIcmpv6StatsInNIReplies	jnxIcmpv6GlobalStats 39	The total number of "node information report" messages received.
jnxIcmpv6StatsOutUnreachables	jnxIcmpv6GlobalStats 40	The total number of "unreachable" messages transmitted.
jnxIcmpv6StatsOutPktTooBigs	jnxIcmpv6GlobalStats 41	The total number of "packet too big" messages to be transmitted. Some of the icmp messages are not sent out (problems with rate limits, source addresses, and so on). **jnxIcmpv6StatsTooBigs** represents the number of 'too big packet' messages that actually goes out.
jnxIcmpv6StatsOutTimeExceeds	jnxIcmpv6GlobalStats 42	The total number of "time exceeded" messages transmitted.
jnxIcmpv6StatsOutParamProbs	jnxIcmpv6GlobalStats 43	The total number of "parameter problem" messages transmitted.
jnxIcmpv6StatsOutEchoReqs	jnxIcmpv6GlobalStats 44	The total number of "echo request" messages transmitted.
jnxIcmpv6StatsOutEchoReplies	jnxIcmpv6GlobalStats 45	The total number of "echo reply" messages transmitted.
jnxIcmpv6StatsOutMLQueries	jnxIcmpv6GlobalStats 46	The total number of "multicast listener query" messages transmitted.
jnxIcmpv6StatsOutMLReports	jnxIcmpv6GlobalStats 47	The total number of "multicast listener report" messages transmitted.
jnxIcmpv6StatsOutMLDones	jnxIcmpv6GlobalStats 48	The total number of "multicast listener done" messages transmitted.
jnxIcmpv6StatsOutRtrSolicits	jnxIcmpv6GlobalStats 49	The total number of "router solicitation" messages transmitted.
jnxIcmpv6StatsOutRtrAdvs	jnxIcmpv6GlobalStats 50	The total number of "router advertisement" messages transmitted.
jnxIcmpv6StatsOutNbrSolicits	jnxIcmpv6GlobalStats 51	The total number of "neighbor solicitation" messages transmitted.
jnxIcmpv6StatsOutNbrAdvs	jnxIcmpv6GlobalStats 52	The total number of "neighbor advertisement" messages transmitted.
jnxIcmpv6StatsOutRedirects	jnxIcmpv6GlobalStats 53	The total number of "redirect" messages transmitted.

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Table 126: Count of All Other ICMPv6 Message Types Received and Transmitted *(continued)*

Object	Object ID	Description
jnxIcmpv6StatsOutRtrRenumbers	jnxIcmpv6GlobalStats 54	The total number of "router renumber" messages transmitted.
jnxIcmpv6StatsOutNIReqs	jnxIcmpv6GlobalStats 55	The total number of "node information request" messages transmitted.
jnxIcmpv6StatsOutNIReplies	jnxIcmpv6GlobalStats 56	The total number of "node information report" messages transmitted.

Related Documentation

- IPv6 MIB on page 357
- jnxIpv6GlobalStats on page 357

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CHAPTER 37

Interpreting the Enterprise-Specific ATM Class-of-Service MIB

- ATM Class-of-Service MIB on page 365
- jnxCosAtmVcTable on page 366
- jnxCosAtmVcScTable on page 366
- jnxCosAtmVcQstatsTable on page 368
- jnxCosAtmTrunkTable on page 369

ATM Class-of-Service MIB

The Juniper Networks enterprise-specific ATM Class-of-Service (CoS) MIB provides information about the ATM CoS infrastructure.

The Juniper Networks enterprise-specific ATM CoS MIB uses the following objects and definitions as per the RFCs and MIBs:

- **ifIndex** (RFC 2233, *IF MIB*)
- **atmVclvpi** and **atmVclVci** (RFC 2515, *ATM MIB*)
- **jnxMibs** (*Juniper Networks enterprise-specific SMI MIB*) and **jnxCoSFcIf** (*Juniper Networks enterprise-specific CoS MIB*)

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-atm-cos.txt .

For information about the enterprise-specific ATM CoS MIB objects, see the following topics:

- jnxCosAtmVcTable on page 366
- jnxCosAtmVcScTable on page 366
- jnxCosAtmVcQstatsTable on page 368
- jnxCosAtmTrunkTable on page 369

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Related Documentation
- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxCosAtmVcTable

The **jnxCosAtmVcTable**, whose object identifier is **{jnxAtmCos 1}**, contains information about virtual circuits (VC) that have CoS configured.

Each **jnxCosAtmVcEntry**, whose object identifier is **{jnxCosAtmVcTable 1}**, is indexed using **ifIndex**, **atmVclVpi**, and **atmVclVci**, and contains the **jnxCosAtmVcCosMode** object. The **jnxCosAtmVcCosMode** object represents the mode of CoS queue priority for the VC.

There are two modes, **strict** mode and **alternate** mode:

- **Strict** mode (represented by integer value **0**): One of the four queues has high priority, and is always serviced before the other three queues. The remaining queues are serviced in a round robin fashion.
- **Alternate** mode (represented by integer value **1**): Even though one of the four queues has high priority, the servicing of the queues alternates between the high priority queue and the other queues.

Related Documentation
- ATM Class-of-Service MIB on page 365
- jnxCosAtmVcScTable on page 366
- jnxCosAtmVcQstatsTable on page 368
- jnxCosAtmTrunkTable on page 369

jnxCosAtmVcScTable

The **jnxCosAtmVcScTable**, whose object identifier is **{jnxAtmCos 2}**, contains ATM-scheduler configuration parameters for each forwarding class within a specified VC.

NOTE: **The schedulers mentioned in this MIB are specific to an ATM interface, and are different from the typical schedulers specified using the CoS configuration command-line interface (CLI).**

Each **jnxCosAtmVcScEntry**, whose object identifier is **{jnxCosAtmVcScTable 1}**, represents ATM-scheduler configuration parameters per forwarding class and per VC, and is indexed using **ifIndex**, **atmVclVpi**, and **atmVclVci**. Table 127 on page 367 lists the ATM scheduler parameters as represented by a **jnxCosAtmVcScEntry**.

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Table 127: jnxCosAtmVcScTable

Object	Object ID	Description
jnxCosAtmVcScPriority	**jnxCosAtmVcScEntry 1**	Represents the ATM scheduler priority for the queue associated with the specified forwarding class within the VC.
jnxCosAtmVcScTxWeightType	**jnxCosAtmVcScEntry 2**	Represents the ATM scheduler transmit-weight-type for the queue associated with the specified forwarding class inside the VC. The transmit-weight-type is represented either as the number of cells or as a percentage of the queue size.
jnxCosAtmVcScTxWeight	**jnxCosAtmVcScEntry 3**	Represents the transmit weight of the ATM scheduler for the queue associated with the specified forwarding class and the VC. **jnxCosAtmVcScTxWeight** is expressed either as the number of cells or as a percentage of the total VC bandwidth. The value of **jnxCosAtmVcScTxWeightType**.determines the unit used.
jnxCosAtmVcScDpType	**jnxCosAtmVcScEntry 4**	Shows the type of RED drop profile configured for the specified forwarding class within the VC. A scheduler can specify either linear or constant drop profile. A constant type drop profile (also known as EPD) specifies that all the cells should be dropped when the number of queued cells exceeds a threshold. A linear type drop profile specifies that only a percentage of cells be dropped based on the number of queued cells at any time.
jnxCosAtmVcScLrdpQueueDepth	**jnxCosAtmVcScEntry 5**	Represents the maximum queue size in cells, as specified by the linear RED drop profile associated with the specified forwarding class within the VC. This object is valid only when the value of the object **jnxCosAtmVcScDpType** is **linearRed(0).**
jnxCosAtmVcScLrdpLowPlpThresh	**jnxCosAtmVcScEntry 6**	Represents the threshold percentage of fill-level beyond which the low packet loss priority (PLP) packets belonging to the specified forwarding class within the VC are randomly dropped. This value is specified by the linear RED drop profile configuration. This object is valid only when the object **jnxCosAtmVcScDpType** is set to **linearRed(0)**.

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Table 127: jnxCosAtmVcScTable *(continued)*

Object	Object ID	Description
jnxCosAtmVcScLrdpHighPlpThresh	**jnxCosAtmVcScEntry 7**	Represents the threshold percentage of the fill level beyond which high PLP packets belonging to the specified forwarding class within the VC are randomly dropped. This value is specified by the linear RED drop profile configuration. This object is valid only when the object **jnxCosAtmVcScDpType** is set to **linearRed(0)**.
jnxCosAtmVcEpdThreshold	**jnxCosAtmVcScEntry 8**	Shows the EPD drop threshold configured for the specified forwarding class within the VC. When the number of cells queued exceeds the value specified for this object, all the cells in the queue are dropped. This object is valid only when the **jnxCosAtmVcScDpType** object is set to **epd(1)**.

Related Documentation

- ATM Class-of-Service MIB on page 365
- jnxCosAtmVcTable on page 366
- jnxCosAtmVcQstatsTable on page 368
- jnxCosAtmTrunkTable on page 369

jnxCosAtmVcQstatsTable

The **jnxCosAtmVcQstatsTable**, whose object identifier is **{jnxAtmCos 3}**, contains queue statistics for VCs and forwarding classes. Each **jnxCosAtmVcQstatsEntry** in the **jnxCosAtmVcQstatsTable** contains the queue status information for a particular forwarding class and VC. The **jnxCosAtmVcQstatsEntry** object uses **ifIndex**, **atmVclVpi**, **atmVclVci**, and **jnxCosFcId** for indexing.

Each **jnxCosAtmVcQstatsEntry** contains the objects listed in Table 128 on page 368.

Table 128: jnxCosAtmVcQstatsTable

Object	Object ID	Description
jnxCosAtmVcQstatsOutPackets	**jnxCosAtmVcQstatsEntry 1**	Represents the number of packets belonging to a particular forwarding class that is transmitted on a specific VC.
jnxCosAtmVcQstatsOutBytes	**jnxCosAtmVcQstatsEntry 2**	Represents the number of bytes of a particular forwarding class that are transmitted on a specific VC.
jnxCosAtmVcQstatsOutRedDropPkts	**jnxCosAtmVcQstatsEntry 3**	Represents the number of RED-dropped outgoing packets of a particular forwarding class that are transmitted on a specific VC.

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Table 128: jnxCosAtmVcQstatsTable *(continued)*

Object	Object ID	Description
jnxCosAtmVcQstatsOutNonRedDrops	**jnxCosAtmVcQstatsEntry 4**	Represents the number of outgoing packets, of a particular forwarding class and transmitted on a specific VC, that are dropped because of errors in packets.
jnxCosAtmVcQstatsOutLpBytes	**jnxCosAtmVcQstatsEntry 5**	Represents the number of low PLP (PLP0) bytes transmitted.
jnxCosAtmVcQstatsOutLpPkts	**jnxCosAtmVcQstatsEntry 6**	Represents the number of low PLP(PLP0) packets that are transmitted.
jnxCosAtmVcQstatsOutLpDropBytes	**jnxCosAtmVcQstatsEntry 7**	Represents the number of low PLP (PLP0) bytes dropped at the output queue.
jnxCosAtmVcQstatsOutHpDropBytes	**jnxCosAtmVcQstatsEntry 8**	Represents the number of high PLP (PLP1) bytes dropped at the output queue.
jnxCosAtmVcQstatsOutLpDropPkts	**jnxCosAtmVcQstatsEntry 9**	Represents the number of low PLP (PLP0) packets dropped at the output queue.
jnxCosAtmVcQstatsOutHpDropPkts	**jnxCosAtmVcQstatsEntry 10**	Represents the number of high PLP (PLP1) packets dropped at the output queue.

Related Documentation

- ATM Class-of-Service MIB on page 365
- jnxCosAtmVcTable on page 366
- jnxCosAtmVcScTable on page 366
- jnxCosAtmTrunkTable on page 369

jnxCosAtmTrunkTable

The **jnxCosAtmTrunkTable**, whose object identifier is **{jnxAtmCos 4}**, contains statistics and configuration information related to the ATM trunk CoS interface.

The **jnxCosAtmTrunkEntry**, whose object identifier is **{jnxCosAtmTrunkTable 1}**, uses **ifIndex** and **jnxCosFcId**, and contains the objects listed in Table 129 on page 370.

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Table 129: jnxCosAtmTrunkTable

Object	Object ID	Description
jnxCosAtmTrunkMode	**jnxCosAtmTrunkEntry 1**	Represents the mode of CoS queue priority for the trunk: • **Strict** mode (represented by integer value **0**): One of the four queues has high priority, and is always serviced before the other three queues. The remaining queues are serviced in a round-robin fashion. • **Alternate** mode (represented by integer value **1**): Even though one of the four queues has high priority, the servicing of the queues alternates between the high priority queue and the other queues.
jnxCosAtmTrunkScPriority	**jnxCosAtmTrunkEntry 2**	Represents the ATM scheduler priority for the queue associated with a particular forwarding class within the trunk.
jnxCosAtmTrunkScTxWeightType	**jnxCosAtmTrunkEntry 3**	Represents the ATM scheduler transmit weight type for the queue associated with a particular forwarding class inside the trunk. The weight type can be expressed either as the number of cells or as a percentage of the queue size.
jnxCosAtmTrunkScTxWeight	**jnxCosAtmTrunkEntry 4**	Represents the transmit weight for the queue. The transmit weight can be expressed either as the number of cells or as a percentage of the total trunk bandwidth. The unit is determined by the value set for **jnxCosAtmTrunkScTxWeightType**.
jnxCosAtmTrunkQaType	**jnxCosAtmTrunkEntry 5**	Represents the ATM queue admission type used for the specified trunk. Available values for this object are: **red**(**1**), **singleEpd**(**2**), and **dualEpd**(**3**).
jnxCosAtmTrunkEpdThresholdPlp0	**jnxCosAtmTrunkEntry 6**	Represents the threshold value beyond which all PLP0 cells get dropped. This object has a valid value only when the value for **jnxCosAtmTrunkQaType** is set to **singleEpd** or **dualEpd**.
jnxCosAtmTrunkEpdThresholdPlp1	**jnxCosAtmTrunkEntry 7**	Represents the threshold value beyond which all PLP1 cells get dropped. This object has a valid value only when the **jnxCosAtmTrunkQaType** object is set to **dualEpd**.
jnxCosAtmTrunkQstatsOutPackets	**jnxCosAtmTrunkEntry 8**	Represents the number of packets that belong to a particular forwarding class, and are transmitted on the specific trunk.
jnxCosAtmTrunkQstatsOutBytes	**jnxCosAtmTrunkEntry 9**	Represents the number of bytes that belong to a particular forwarding class, and are transmitted on the specific trunk.
jnxCosAtmTrunkQstatsOutDrops	**jnxCosAtmTrunkEntry 10**	Represents the number of outgoing packets on the trunk that are dropped.
jnxCosAtmTrunkQstatsOutLpBytes	**jnxCosAtmTrunkEntry 11**	Represents the number of low PLP (PLP0) bytes that are transmitted on the trunk.

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Table 129: jnxCosAtmTrunkTable *(continued)*

Object	Object ID	Description
jnxCosAtmTrunkQstatsOutLpPkt	jnxCosAtmTrunkEntry 12	Represents the number of low PLP (PLP0) packets that are transmitted on the trunk.
jnxCosAtmTrunkQstatsOutLpDropBytes	jnxCosAtmTrunkEntry 13	Represents the number of low PLP (PLP0) bytes dropped at the output queue.
jnxCosAtmTrunkQstatsOutHpDropBytes	jnxCosAtmTrunkEntry 14	Represents the number of high PLP (PLP1) bytes that are dropped at the output queue.
jnxCosAtmTrunkQstatsOutLpDropPkts	jnxCosAtmTrunkEntry 15	Represents the number of low PLP (PLP0) packets that are dropped at the output queue.
jnxCosAtmTrunkQstatsOutHpDropPkts	jnxCosAtmTrunkEntry 16	Represents the number of high PLP (PLP1) packets dropped at the output queue.
jnxCosAtmTrunkQstatsOutHpBytes	jnxCosAtmTrunkEntry 17	Represents the number of high PLP (PLP1) bytes that are transmitted on the trunk.
jnxCosAtmTrunkQstatsOutHpPkts	jnxCosAtmTrunkEntry 18	Represents the number of high PLP (PLP1) packets that are transmitted on the trunk.

Related Documentation

- ATM Class-of-Service MIB on page 365
- jnxCosAtmVcTable on page 366
- jnxCosAtmVcScTable on page 366
- jnxCosAtmVcQstatsTable on page 368

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CHAPTER 38

Interpreting the Enterprise-Specific Firewall MIB

- Firewall MIB on page 373
- jnxFirewallsTable on page 374
- jnxFirewallCounterTable on page 374

Firewall MIB

The Juniper Networks enterprise-specific Firewall MIB, whose object identifier is **{jnxMibs 5}**, contains information about firewall filters and policies.

Firewall MIB contains two tables, **jnxFirewallsTable** and **jnxFirewallCounterTable**.

The **jnxFirewallsTable** does not support the following conditions:

- Counter and filter names that have more than 24 characters.
- Duplicate counter names, even if the counter types are different.

Because of the preceding limitations, the **jnxFirewallsTable** has been deprecated and replaced with **jnxFirewallCounterTable**. However, for backward compatibility, the **jnxFirewallsTable** is retained in the Firewall MIB.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-firewall.txt .

For information about the enterprise-specific Firewall MIB objects, see the following topics:

- jnxFirewallsTable on page 374
- jnxFirewallCounterTable on page 374

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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jnxFirewallsTable

The deprecated **jnxFirewallsTable** contains **jnxFirewallsEntry**, whose object ID is **{jnxFirewallsTable 1}**. Each **jnxFirewallsEntry** contains the objects listed in Table 130 on page 374.

Table 130: jnxFirewallsEntry

Object	Object ID	Description
jnxFWFilter	**jnxFirewallsEntry 1**	The name of the firewall filter. This object does not support names that have more than 24 characters.
jnxFWCounter	**jnxFirewallsEntry 2**	The name of the counter or the policer. This name is specific within the firewall filter.
jnxFWType	**jnxFirewallsEntry 3**	The type of the **jnxFWCounter** object. The value of **jnxFWType** can be **1** (other), **2** (counter), or **3** (policer).
jnxFWPackets	**jnxFirewallsEntry 4**	The number of packets that are associated with the specified counter or policer.
jnxFWBytes	**jnxFirewallsEntry 5**	The number of bytes that are associated with the counter. For policers, the value of **jnxFWBytes** is always zero because the policers do not count the number of bytes.

Related Documentation

- Firewall MIB on page 373
- jnxFirewallCounterTable on page 374

jnxFirewallCounterTable

The **jnxFirewallCounterTable**, whose object identifier is **{jnxFirewalls 2}**, replaces the deprecated **jnxFirewallsTable**. Each **jnxFirewallCounterEntry** contains the objects listed in Table 131 on page 374.

Table 131: jnxFirewallCounterEntry

Object	Object ID	Description
jnxFWCounterFilterName	**jnxFirewallCounterEntry 1**	The name of the firewall filter. The name can have up to 127 characters.
jnxFWCounterName	**jnxFirewallCounterEntry 2**	The name of the counter or the policer. The name can have up to 127 characters.
jnxFWCounterType	**jnxFirewallCounterEntry 3**	The type of the **jnxFWCounterName** object. The value of **jnxFWType** can be **1** (other), **2** (counter), or **3** (policer).

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Table 131: jnxFirewallCounterEntry *(continued)*

Object	Object ID	Description
jnxFWCounterPacketCount	jnxFirewallCounterEntry 4	The number of packets that are associated with the specified counter or policer.
jnxFWCounterByteCount	jnxFirewallCounterEntry 5	The number of bytes that are associated with the counter or policer.
jnxFWCounterDisplayFilterName	jnxFirewallCounterEntry 6	The name of the firewall filter. The name can have up to 127 characters.
jnxFWCounterDisplayName	jnxFirewallCounterEntry 7	The name of the counter or the policer.
jnxFWCounterDisplayType	jnxFirewallCounterEntry 8	The type of the **jnxFWCounterName** object. The value of **jnxFWType** can be **1** (other), **2** (counter), or **3** (policer).

Related Documentation

- Firewall MIB on page 373
- jnxFirewallsTable on page 374

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CHAPTER 39

Interpreting the Enterprise-Specific ATM MIB

- ATM MIB on page 377
- jnxAtmIfTable on page 377
- jnxAtmVCTable on page 380
- jnxAtmVpTable on page 383
- jnxAtmTrunkTable on page 385

ATM MIB

The Juniper Networks enterprise-specific ATM MIB, whose object identifier is **{jnxMibs 10}**, extends the standard ATM MIB, RFC 1695, and contains information about ATM interfaces and VCs.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-atm.txt .

For information about the enterprise-specific ATM MIB objects, see the following topics:

- jnxAtmIfTable on page 377
- jnxAtmVCTable on page 380
- jnxAtmVpTable on page 383
- jnxAtmTrunkTable on page 385

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxAtmIfTable

The **jnxAtmIfTable** augments the **atmInterfaceConfTable** in the standard ATM MIB. The object identifier for **jnxAtmIfTable** is **{jnxAtm 1}**. Each **jnxAtmIfEntry** in the **jnxAtmIfTable** contains the configuration and statistical information for a particular ATM port. The

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jnxAtmIfEntry, whose object identifier is **{jnxAtmIfTable 1}**, is an extension of the **atmInterfaceConfEntry** in the standard ATM MIB.

Table 132 on page 378 lists the objects associated with the **jnxAtmIfEntry**.

Table 132: jnxAtmIfTable

Object	Object ID	Description
jnxAtmIfPortType	**jnxAtmIfEntry 1**	Represents the type of the physical port. This object uses the following integer values to denote the port type: • **1** (other) • **2** (oc3) • **3** (oc1) • **4** (t3) • **5** (e3) • **6** (oc48)
jnxAtmIfEncaps	**jnxAtmIfEntry 2**	Represents the type of ATM physical link layer encapsulation. This object uses the following integer values to denote the type of encapsulation: • **1** (**other**) • **2** (**atmPvc**) • **3** (**atmCccCellRelay**)
jnxAtmIfLpBackInfo	**jnxAtmIfEntry 3**	Represents the loopback configuration and type. This object uses the following integer values to denote the loopback configuration type: • **1** (**noLoopBack**) • **2** (**localLoopBack**) • **3** (**remoteLoopBack**)
jnxAtmIfScrambleEnable	**jnxAtmIfEntry 4**	Indicates whether scrambling is enabled (object value **1**) or disabled (object value **2**). Default value for this object is **2**, disabled.
jnxAtmIfTxCellCount	**jnxAtmIfEntry 5**	Indicates the number of ATM cells, including the idle cells, transmitted by the interface.
jnxAtmIfRxCellCount	**jnxAtmIfEntry 6**	Indicates the number of ATM cells, excluding the idle cells, received by the interface.
jnxAtmIfTxIdleCellCount	**jnxAtmIfEntry 7**	Indicates the number of idle cells transmitted by the interface. When the interface does not have anything to send, it sends idle cells to fill the time slot.

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Table 132: jnxAtmIfTable *(continued)*

Object	Object ID	Description
jnxAtmIfUncorrHCSErrs	jnxAtmIfEntry 8	Indicates the number of uncorrectable cell Header Check Sequence (HCS) errors that occurred on the interface.
jnxAtmIfCorrHCSErrs	jnxAtmIfEntry 9	Indicates the number of correctable cell HCS errors.
jnxAtmIfTxCellFIFOOverRuns	jnxAtmIfEntry 10	Indicates the number of overruns in the Transmit FIFO.
jnxAtmIfRxCellFIFOOverRuns	jnxAtmIfEntry 11	Indicates the number of overruns in the Receive FIFO.
jnxAtmIfRxCellFIFOUnderRuns	jnxAtmIfEntry 12	Indicates the number of underruns in the Receive FIFO.
jnxAtmIfInInvalidVCCells	jnxAtmIfEntry 13	Indicates the number of cells that are arrived for a nonexistent VC.
jnxAtmIfInNoBufferOAMCells	jnxAtmIfEntry 14	Indicates the number of incoming OAM cells or raw cells that are dropped because of lack of buffer to handle them.
jnxAtmIfInNoBufDropPkts	jnxAtmIfEntry 15	Indicates the number of AAL5 packets that are dropped because of lack of buffer to handle them.
jnxAtmIfOutVCQueueDrops	jnxAtmIfEntry 16	Indicates the number of packets dropped because of queue limits on each VC.
jnxAtmIfInBadCrcs	jnxAtmIfEntry 17	Indicates the total number of incoming CRC errors.
jnxAtmIfInLenErrPkts	jnxAtmIfEntry 18	Indicates the number of AAL5 packets that were dropped because of incorrect length.
jnxAtmIfInTimeoutPkts	jnxAtmIfEntry 19	Indicates the number of AAL5 packets that were dropped because of reassembly timeout.

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Table 132: jnxAtmIfTable *(continued)*

Object	Object ID	Description
jnxAtmIfL2CircuitMode	**jnxAtmIfEntry 20**	Indicates the layer 2 circuit mode of the ATM interface (on an ATMII PIC). This object uses the following integer values to denote the circuit modes: • **1** (not applicable) • **2** (none) • **3** (aal5) • **4** (cell) • **5** (uniTrunk) • **6** (nniTrunk)

Related Documentation

- ATM MIB on page 377
- jnxAtmVCTable on page 380
- jnxAtmVpTable on page 383
- jnxAtmTrunkTable on page 385

jnxAtmVCTable

The **jnxAtmVCTable**, whose object identifier is **{jnxAtm 2}**, extends the **atmVclTable** in the standard ATM MIB, and contains information about peer-to-peer, point-to-multipoint, and multicast virtual circuit entries.

Each **jnxAtmVCEntry**, whose object identifier is **{jnxAtmVCTable 1}**, in the **jnxAtmVCTable** contains the objects listed in Table 133 on page 381.

However, the **jnxAtmVCEntry** does not store any statistic for multicast VCs. A value of zero indicates this. Similarly, a value of **0.0.0.0** indicates that the multipoint destination IP address for a P2P VC is invalid. If **atmCccCellRelay** is set as the encapsulation type for the logical interface (to which the VC belongs), only the generic VC info is valid. And, if OAM is disabled (indicated by zero value for **jnxAtmVCFlags**), then all the OAM config and f5 statistics information is invalid.

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Table 133: jnxAtmVCTable

Object	Object ID	Description
jnxAtmVCConnType	**jnxAtmVCEntry 1**	Indicates the type of connection. This object uses the following integer values to denote the connection types: • **1** (other; unknown connection type or a connection type that is not one of the other connection types explicitly represented) • **2** (peer-to-peer) • **3** (point to multipoint) • **4** (multicast)
jnxAtmVCEncapsulation	**jnxAtmVCEntry 2**	Indicates the ATM encapsulation type associated with the VC. This object uses the following integer values to denote the encapsulation types: • **1 other** • **2 atmCccCellRelay** (ATM cell relay for CCC) • **3 atmCccVcMux** (ATM VC for CCC) • **4 atmCiscoNlpid** (Cisco-compatible ATM NLPID encapsulation) • **5 atmNlpid** (ATM NLPID encapsulation) • **6 atmSnap** (ATM LLC/SNAP encapsulation) • **7 atmVcMux** (ATM VC multiplexing) • **8 atmTccVcmux** (Translational Cross Connection (TCC) over ATM VC MUX encapsulation) • **9 atmTccSnap** (TCC over ATM LLC/SNAP encapsulation)
jnxAtmVCMpDestIPv4Addr	**jnxAtmVCEntry 3**	Represents the multipoint destination IPv4 address for a point-to-multipoint connection. This object returns an all zero address in the following cases: • When the connection type is not P2MP. • When the multipoint destination address type is IPv6.
jnxAtmVCMpDestIPv6Addr	**jnxAtmVCEntry 4**	Represents the multipoint destination IPv6 address for a P2MP connection. This object returns an all zero address in the following cases: • When the connection type is not P2MP. • When the multipoint destination address type is IPv4.

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Table 133: jnxAtmVCTable *(continued)*

Object	Object ID	Description
jnxAtmVCFlags	jnxAtmVCEntry 5	Contains the flags related to the VC.
jnxAtmVCTotalDownTime	jnxAtmVCEntry 6	Shows the total downtime for the VC after the last reboot of the system.
jnxAtmVCInBytes	jnxAtmVCEntry 7	Represents the number of bytes received on the VC.
jnxAtmVCOutBytes	jnxAtmVCEntry 8	Represents the number of bytes transmitted from the VC.
jnxAtmVCInPkts	jnxAtmVCEntry 9	Represents the number of packets received on the VC.
jnxAtmVCOutPkts	jnxAtmVCEntry 10	Represents the number of packets transmitted from the VC.
jnxAtmVCTailQueuePktDrops	jnxAtmVCEntry 11	Represents the number of packets that were dropped because of bandwidth constraints.
jnxAtmVCOAMPeriod	jnxAtmVCEntry 12	Shows the frequency at which the F5 cells are transmitted to check the status of the VC.
jnxAtmVCOAMUpCellCount	jnxAtmVCEntry 13	Shows the minimum number of loopback cells that are required to confirm that a VC is up.
jnxAtmVCOAMDownCellCount	jnxAtmVCEntry 14	Shows the minimum number of loopback cells that are required to confirm that a VC is down. NOTE: This object returns a zero value if OAM is not enabled.
jnxAtmVCInOAMF5LoopCells	jnxAtmVCEntry 15	Shows the number of OAM F5 loopback cells received on a VC. NOTE: This object returns a zero value if OAM is not enabled.
jnxAtmVCOutOAMF5LoopCells	jnxAtmVCEntry 16	Shows the number of OAM F5 loopback cells transmitted from a VC. NOTE: This object returns a zero value if OAM is not enabled.

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Table 133: jnxAtmVCTable *(continued)*

Object	Object ID	Description
jnxAtmVCInOAMF5RDICells	**jnxAtmVCEntry 17**	Shows the number of OAM F5 cells that are received with RDI (Remote Defect Indication) bit set. NOTE: This object returns a zero value if OAM is not enabled.
jnxAtmVCOutOAMF5RDICells	**jnxAtmVCEntry 18**	Shows the number of OAM F5 cells that are transmitted with RDI (Remote Defect Indication) bit set. NOTE: This object returns a zero value if OAM is not enabled.
jnxAtmVCInOAMF5AISCells	**jnxAtmVCEntry 19**	Shows the number of OAM F5 cells that are received with alarm indication signal (AIS) bit set. NOTE: This object returns a zero value if OAM is not enabled.
jnxAtmVCOutOAMF5AISCells	**jnxAtmVCEntry 20**	Shows the number of OAM F5 cells that are transmitted with AIS bit set. NOTE: This object returns a zero value if OAM is not enabled.

Related Documentation

- ATM MIB on page 377
- jnxAtmIfTable on page 377
- jnxAtmTrunkTable on page 385

jnxAtmVpTable

The **jnxAtmVpTable** extends the **atmVplTable** defined in RFC 2515, *Definitions of Managed Objects for ATM Management*, and contains additional information about ATM virtual paths (VP).

The **jnxAtmVpTable**, whose object identifier is **{jnxAtm 3}**, contains **jnxAtmVpEntry**. Each **jnxAtmVpEntry**, whose object ID is **{jnxAtmVpTable 1}**, contains the objects listed in Table 134 on page 384.

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Table 134: jnxAtmVpTable

Object	Object ID	Description
jnxAtmVpEntry	**jnxAtmVpTable 1**	Represents configuration status and statistics information related to an ATM VP. However, traffic stats are available per VP tunnel only if shaping is configured on the VP. You can use the **jnxAtmVpFlags** to determine whether shaping is enabled. Similarly, the values for OAM config and OAM stat objects are invalid (default value: **0**) if no OAM is configured. You can use the **jnxAtmVpFlags** to determine whether OAM is configured. NOTE: For an ATM-1 VP, the only valid object is **jnxAtmVpFlags**.
jnxAtmVpFlags	**jnxAtmVpEntry 1**	Represents the flags associated with the VP. This object uses the following values: • **0 active** • **1 down** • **2 oamEnabled** • **3 shapingEnabled** • **4 passiveOam**
jnxAtmVpTotalDownTime	**jnxAtmVpEntry 2**	Represents the total downtime for the VP since the last reboot of the system.
jnxAtmVpOamPeriod	**jnxAtmVpEntry 3**	Indicates the frequency at which the OAM F4 cells are transmitted to find out the status of the VP. This object returns a value of zero if OAM is not enabled for the VP.
jnxAtmVpOamUpCellCount	**jnxAtmVpEntry 4**	Indicates the minimum number of consecutive loopback cells required to confirm that a VP is up.
jnxAtmVpOamDownCellCount	**jnxAtmVpEntry 5**	Indicates the minimum number of consecutive loopback cells required to confirm that a VP is down.
jnxAtmVpInBytes	**jnxAtmVpEntry 6**	Indicates the number of bytes received on the VP.
jnxAtmVpOutBytes	**jnxAtmVpEntry 7**	Indicates the number of bytes sent out of the VP.
jnxAtmVpInPkts	**jnxAtmVpEntry 8**	Indicates the number of packets received on the VP.
jnxAtmVpOutPkts	**jnxAtmVpEntry 9**	Indicates the number of packets sent out on the VP.
jnxAtmVpInOamF4Cells	**jnxAtmVpEntry 10**	Indicates the number of OAM F4 cells received on the VP.
jnxAtmVpOutOamF4Cells	**jnxAtmVpEntry 11**	Indicates the number of OAM F4 cells transmitted on the VP.
jnxAtmVpInOamF4LoopCells	**jnxAtmVpEntry 12**	Indicates the number of OAM F4 loopback cells received on the VP.

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Table 134: jnxAtmVpTable *(continued)*

Object	Object ID	Description
jnxAtmVpOutOamF4LoopCells	jnxAtmVpEntry 13	Indicates the number of OAM F4 cells transmitted on the VP.
jnxAtmVpInOamF4RdiCells	jnxAtmVpEntry 14	Indicates the number of OAM F4 RDI cells received on the VP.
jnxAtmVpOutOamF4RdiCells	jnxAtmVpEntry 15	Indicates the number of OAM F4 RDI cells transmitted on the VP.
jnxAtmVpInOamF4AisCells	jnxAtmVpEntry 16	Indicates the number of OAM F4 AIS cells received on the VP.

Related Documentation

- ATM MIB on page 377
- jnxAtmIfTable on page 377
- jnxAtmVCTable on page 380
- jnxAtmTrunkTable on page 385

jnxAtmTrunkTable

The **jnxAtmTrunkTable**, whose object identifier is **{jnxAtm 4}**, contains information related to ATM trunks. Each **jnxAtmTrunkEntry** in **jnxAtmTrunkTable** contains the objects listed in Table 135 on page 385.

NOTE: If the encapsulation type for the logical interface to which the trunk belongs is **atmCccCellRelay**, only the generic trunk information (**jnxAtmTrunkConnType**, **jnxAtmTrunkEncapsulation**, **nxAtmTrunkFlags**, and **jnxAtmTrunkTotalDownTime**) is valid.

Table 135: jnxAtmTrunkTable

Object	Object ID	Description
jnxAtmTrunkId	jnxAtmTrunkEntry 1	Represents the identifier of the ATM trunk.
jnxAtmTrunkConnType	jnxAtmTrunkEntry 2	Indicates the type of connection. This object uses the following integer values to denote the type of connection: • 1 other • 2 P2P
jnxAtmTrunkEncapsulation	jnxAtmTrunkEntry 3	Represents the ATM encapsulation type associated with the VC or trunk. This object uses the following integer values to denote the encapsulation type: • 1 other • 2 atmCccCellRelay
jnxAtmTrunkFlags	jnxAtmTrunkEntry 4	Represents the flags related to the trunk.

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Table 135: jnxAtmTrunkTable *(continued)*

Object	Object ID	Description
jnxAtmTrunkTotalDownTime	**jnxAtmTrunkEntry 5**	Indicates the total downtime (in seconds) for the trunk since the last reboot of the system.
jnxAtmTrunkInBytes	**jnxAtmTrunkEntry 6**	Indicates the number of bytes received on the trunk.
jnxAtmTrunkOutBytes	**jnxAtmTrunkEntry 7**	Indicates the number of bytes sent out on the trunk.
jnxAtmTrunkInPkts	**jnxAtmTrunkEntry 8**	Indicates the number of packets received on the trunk.
jnxAtmTrunkOutPkts	**jnxAtmTrunkEntry 9**	Indicates the number of packets sent out on the trunk.
jnxAtmTrunkTailQueuePktDrops	**jnxAtmTrunkEntry 10**	Represents the number of packets that were dropped because of bandwidth constraints. This object indicates that the packets were queued to be transmitted at a rate faster than allowed.
jnxAtmTrunkInOAMF4AISCells	**jnxAtmTrunkEntry 15**	Indicates the number of OAM F4 cells that are received with AIS (Alarm Indication Signal) bit set.
jnxAtmTrunkOutOAMF4AISCells	**jnxAtmTrunkEntry 16**	Indicates the number of OAM F4 cells that are sent out with AIS bit set.

Related Documentation

- ATM MIB on page 377
- jnxAtmIfTable on page 377
- jnxAtmVCTable on page 380
- jnxAtmVpTable on page 383

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CHAPTER 40

Interpreting the Enterprise-Specific Configuration Management MIB

- Configuration Management MIB on page 387
- Configuration Management MIB Textual Conventions on page 387
- Configuration Change Management Objects and jnxCmCfgChgEventTable on page 388
- Rescue Configuration Change Management Objects on page 390
- Configuration Management Notifications on page 391

Configuration Management MIB

The Juniper Networks enterprise-specific Configuration Management MIB, whose object identifier is **{jnxMIbs 18}**, defines the objects that are used for managing the configuration of Juniper Networks products.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-cfgmgmt.txt .

For information about the enterprise-specific Configuration Management MIB objects, see the following topics:

- Configuration Management MIB Textual Conventions on page 387
- Configuration Change Management Objects and jnxCmCfgChgEventTable on page 388
- Rescue Configuration Change Management Objects on page 390
- Configuration Management Notifications on page 391

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

Configuration Management MIB Textual Conventions

Table 136 on page 388 contains information about the textual conventions used in the enterprise-specific Configuration Management MIB.

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Table 136: Configuration Management MIB Textual Conventions

Object	Description	Syntax
jnxCmCfChgSource	Identifies the source of the configuration event.	This object uses the following integer values: • **1**—Other • **2**—CLI • **3**—Junos XML protocol • **4**—Synchronize • **5**—SNMP • **6**—Button • **7**—Autoinstall • **8**—Unknown
jnxCmRescueCfgState	Represents the state of the rescue configuration.	This object uses the following integer values: • **1**—Nonexistent • **2**—Updated

Related Documentation

- Configuration Management MIB on page 387
- Configuration Change Management Objects and jnxCmCfgChgEventTable on page 388
- Rescue Configuration Change Management Objects on page 390
- Configuration Management Notifications on page 391

Configuration Change Management Objects and jnxCmCfgChgEventTable

The configuration change management objects listed in Table 137 on page 388 along with the objects in the **jnxCmCfgChgEventTable** provide the configuration change history.

Each **jnxCmCfgChg**, whose object identifier is **{jnxCfgMgmt 1}**, contains the objects listed in Table 137 on page 388.

Table 137: Configuration Change Management Objects

Object	Object ID	Description
jnxCmCfgChgLatestIndex	**jnxCmCfgChg 1**	Contains the index used in **jnxCmCfgChgEventTable** to represent the latest configuration change event.
jnxCmCfgChgLatestTime	**jnxCmCfgChg 2**	Shows the value of **sysUpTime** at the time of the last configuration change. However, this object returns **0** if the management subsystem was reset after the last configuration change.
jnxCmCfgChgLatestDate	**jnxCmCfgChg 3**	Shows the date and time when the configuration was last modified.
jnxCmCfgChgLatestSource	**jnxCmCfgChg 4**	Shows the source from which the configuration event was triggered. This object uses **jnxCmCfChgSource** to represent the source of configuration event. For more information about **jnxCmCfChgSource**, see Table 136 on page 388.

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Table 137: Configuration Change Management Objects *(continued)*

Object	Object ID	Description
jnxCmCfgChgLatestUser	**jnxCmCfgChg 5**	Shows the login name of the current user. This object returns a zero-length string if the user name is not available or not applicable.
jnxCmCfgChgMaxEventEntries	**jnxCmCfgChg 6**	Shows the maximum number of entries that **jnxCmCfgChgEventTable** can contain. Allowable range is **0** though **2147483647**. When the number of entries in **jnxCmCfgChgEventTable** exceeds the maximum value set for **jnxCmCfgChgMaxEventEntries**, the latest entry displaces the oldest entry in the table.

For information about the **jnxCmCfgChgEventTable**, see the following section:

- jnxCmCfgChgEventTable on page 389

jnxCmCfgChgEventTable

The **jnxCmCfgChgEventTable**, whose object identifier is **{jnxCmCfgChg 7}**, contains **jnxCmCfgChgEventEntry** that maps to the most recent configuration change events on the router. The **jnxCmCfgChgMaxEventEntries** object discussed in Table 137 on page 388 controls the number of entries stored in **jnxCmCfgChgEventTable**.

Each **jnxCmCfgChgEventEntry**, whose object identifier is **{jnxCmCfgChgEventTable 1}**, contains the objects listed in Table 138 on page 389.

Table 138: jnxCmCfgChgEventTable

Object	Object ID	Description
jnxCmCfgChgEventIndex	**jnxCmCfgChgEventEntry 1**	Uniquely identifies a configuration change event. The SNMP process assigns monotonically increasing values to each event as it occurs. However, when the SNMP process is reset, the index values also are reset.
jnxCmCfgChgEventTime	**jnxCmCfgChgEventEntry 2**	Contains the value of **sysUpTime** when the event occurred.
jnxCmCfgChgEventDate	**jnxCmCfgChgEventEntry 3**	Contains the system date and time when the event occurred.
jnxCmCfgChgEventSource	**jnxCmCfgChgEventEntry 4**	Shows the source from which the configuration event was triggered. This object uses **JnxCmCfChgSource** to represent the source of configuration event. For more information about **JnxCmCfChgSource**, see Table 136 on page 388.
jnxCmCfgChgEventUser	**jnxCmCfgChgEventEntry 5**	Contains the name of the user who was logged in at the time of the event. Returns a zero-length string if the user name is not applicable or not available.
jnxCmCfgChgEventLog	**jnxCmCfgChgEventEntry 6**	Contains the log of the configuration event. Returns a zero-length string if no log is available.

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Related Documentation

- Configuration Management MIB on page 387
- Configuration Management MIB Textual Conventions on page 387
- Rescue Configuration Change Management Objects on page 390
- Configuration Management Notifications on page 391

Rescue Configuration Change Management Objects

The **jnxCmRescueChg**, whose object identifier is **{jnxCfgMgmt 2}**, contains information about changes to rescue configuration.

Table 139 on page 390 lists the objects associated with **jnxCmRescueChg**.

Table 139: Rescue Configuration Change Management Objects

Object	Object ID	Description
jnxCmRescueChgTime	**jnxCmRescueChg 1**	Contains the value of **sysUpTime** when the rescue configuration was last changed. If the management subsystem has been reset since the last configuration change, this object returns **0**.
jnxCmRescueChgDate	**jnxCmRescueChg 2**	Contains the date and time when the rescue configuration was last changed.
jnxCmRescueChgSource	**jnxCmRescueChg 3**	Shows the source from which the rescue configuration event was triggered. This object uses **jnxCmCfChgSource** to represent the source of configuration event. For more information about **jnxCmCfChgSource**, see Table 136 on page 388.
jnxCmRescueChgUser	**jnxCmRescueChg 4**	Contains the name of the user who was logged in at the time of the event. Returns a zero-length string if the user name is not applicable or not available.
jnxCmRescueChgState	**jnxCmRescueChg 5**	Shows the current state of the rescue configuration. For more information about the different states of rescue configuration, see Table 136 on page 388.

Related Documentation

- Configuration Management MIB on page 387
- Configuration Management MIB Textual Conventions on page 387
- Configuration Change Management Objects and jnxCmCfgChgEventTable on page 388
- Configuration Management Notifications on page 391

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Configuration Management Notifications

The Junos OS generates the following traps when a configuration or a rescue configuration event occurs:

- **jnxCmCfgChange**, whose object identifier is **{jnxCmNotificationsPrefix 1}**, contains **jnxCmCfgChgEventTime**, **jnxCmCfgChgEventDate**, **jnxCmCfgChgEventSource**, **jnxCmCfgChgEventUser**, and **jnxCmCfgChgEventLog**.

  NOTE: Because configuration rollback is handled by the master management process that uses the **root** user ID, the **jnxCmCfgChgEventUser** object in the **jnxCmCfgChange** trap always returns **root** as the user name for configuration rollback events.

- **jnxCmRescueChange**, whose object identifier is **{jnxCmNotificationsPrefix 2}**, contains **jnxCmRescueChgTime**, **jnxCmRescueChgDate**, **jnxCmRescueChgSource**, **jnxCmRescueChgUser**, and **jnxCmRescueChgState**.

Related Documentation

- Configuration Management MIB on page 387
- Configuration Management MIB Textual Conventions on page 387
- Configuration Change Management Objects and jnxCmCfgChgEventTable on page 388
- Rescue Configuration Change Management Objects on page 390

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CHAPTER 41

Interpreting the Enterprise-Specific IPv4 MIB

- IPv4 MIB on page 393
- jnxIpv4AddrTable on page 393

IPv4 MIB

The Juniper Networks enterprise-specific IPv4 MIB, whose object identifier is **{jnxMibs12}**, functions as an extension of the **ifTable** defined in RFC 1573, *IF MIB*, and defines the branches for IPV4 configuration.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-ipv4.txt .

For information about the enterprise-specific IPv4 MIB objects, see the following topic:

- jnxIpv4AddrTable on page 393

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxIpv4AddrTable

The **jnxIpv4AddrTable** defines the **jnxIpv4AddrEntry** and its attributes. Each **jnxIpv4AddrEntry** contains the objects listed in Table 140 on page 393.

Table 140: jnxIpv4AddrTable

Object	Object ID	Description
jnxIpv4AdEntIfIndex	**jnxIpv4AddrEntry 1**	A unique index value that identifies the interface with which a particular entry is associated. An interface identified by a particular value of **jnxIpv4AdEntIfIndex** is the same as the interface that is identified by the same value of **ifIndex** as defined in RFC 1573.
jnxIpv4AdEntAddr	**jnxIpv4AddrEntry 2**	The IP address of the interface with which the address information stored in this entry is associated.

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Table 140: jnxIpv4AddrTable *(continued)*

Object	Object ID	Description
jnxIpv4AdEntNetMask	**jnxIpv4AddrEntry 3**	The subnet mask associated with the IP address of this entry. The value of the mask is an IP address with all the network bits set to 1 and all the host bits set to 0.
jnxIpv4AdEntBcastAddr	**jnxIpv4AddrEntry 4**	The least significant bit in the IP broadcast address used for sending datagrams on the logical interface associated with the IP address of this entry. For example, when the Internet standard all-ones broadcast address is used, the value is 1. This value applies to both the subnet and network broadcast addresses used by the entry on the logical interface.
jnxIpv4AdEntReasmMaxSize	**jnxIpv4AddrEntry 5**	The size of the largest IP datagram that this entry can reassemble from the incoming fragmented IP datagrams received on the interface.

Related Documentation

- IPv4 MIB on page 393

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CHAPTER 42

Interpreting the Enterprise-Specific Alarm MIB

- Alarm MIB on page 395
- jnxAlarmRelayMode on page 395
- jnxYellowAlarms on page 396
- jnxRedAlarms on page 397

Alarm MIB

The Juniper Networks enterprise-specific Alarm MIB, whose object identifier is **{jnxMibs 4}**, contains information about alarms from the router chassis.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-chassis-alarm.txt .

For information about the enterprise-specific Alarm MIB objects, see the following topics:

- jnxAlarmRelayMode on page 395
- jnxYellowAlarms on page 396
- jnxRedAlarms on page 397

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxAlarmRelayMode

The **jnxAlarmRelayMode**, whose object identifier is **{jnxCraftAlarms 1}**, denotes the alarm relay mode of the craft interface panel for the yellow and red alarms. This object uses the following integer values:

- **1–Other**: Other or unknown state
- **2–passOn**: Alarms are passed on. The alarm relay is activated to pass on the yellow or red alarms to audible sirens or visual flashing devices.

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- **3–cutOff**: Alarms are turned off. Both the yellow and red alarms are cut off from the alarm relays and do not get passed on to audible sirens or visual flashing devices.



NOTE: Both the yellow and red alarms can be cut off from the alarm relay using a Alarm Cutoff/Lamp Test button on the front panel of the router chassis.

Related Documentation

- Alarm MIB on page 395
- jnxYellowAlarms on page 396
- jnxRedAlarms on page 397

jnxYellowAlarms

The **jnxYellowAlarms**, whose object identifier is **{jnxCraftAlarms 2}**, contains the objects listed in Table 141 on page 396.

Table 141: jnxYellowAlarms

Object	Object ID	Description
jnxYellowAlarmState	**jnxYellowAlarms 1**	Denotes the yellow alarm state on the craft interface panel of the router chassis. This object contains one of the following integer values: • **other–1**: The alarm state is unknown. • **off–2**: The yellow alarms are off. You can turn off the yellow alarms using the Alarm Cutoff/Lamp Test button on the craft interface panel of the router chassis. • **on–3**: The yellow alarms are on.
jnxYellowAlarmCount	**jnxYellowAlarms 2**	Shows the number of currently active and nonsilent yellow alarms. NOTE: The value of this object is independent of the state of the Alarm Cutoff/Lamp Test button.
jnxYellowAlarmLastChange	**jnxYellowAlarms 3**	Shows the value of the **sysUp** time when the state of the yellow alarm last changed from on to off or vice versa. This object returns **0** if the alarm state has not changed since the **sysUp** time was reset last time, or if the value is unknown.

Related Documentation

- Alarm MIB on page 395
- jnxAlarmRelayMode on page 395
- jnxRedAlarms on page 397

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jnxRedAlarms

The **jnxRedAlarms**, whose object identifier is **{jnxCraftAlarms 3}**, contains the objects listed in Table 142 on page 397.

Table 142: jnxRedAlarms

Object	Object ID	Description
jnxRedAlarmState	**jnxRedAlarms 1**	Denotes the state of red alarms on the craft interface panel of the router chassis. This object contains one of the following values: • **1–other**: The red alarm state is unknown. • **2–off**: The red alarm is turned off. • **3–on**: The red alarm is on. Typically, the red alarm is on when there is a system failure, power failure, or hardware malfunction, or when a threshold value is exceeded.
jnxRedAlarmCount	**jnxRedAlarms 2**	Shows the number of currently active and nonsilent red alarms. NOTE: The value of this object is independent of the state of the Alarm Cutoff/Lamp Test button.
jnxRedAlarmLastChange	**jnxRedAlarms 3**	Shows the value of the **sysUp** time when the red alarm last changed from on to off or vice versa. This object contains **0** value, if the alarm state has not changed since the **sysUp** time was reset last time, or if the value is unknown.

Related Documentation

- Alarm MIB on page 395
- jnxAlarmRelayMode on page 395
- jnxYellowAlarms on page 396

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CHAPTER 43

Interpreting the Enterprise-Specific RSVP MIB

- RSVP MIB on page 399
- jnxRsvpSessionTable on page 399

RSVP MIB

The Juniper Networks enterprise-specific RSVP MIB, whose object identifier is **{jnxMibs 30}**, contains information about RSVP-traffic engineering (TE) sessions that correspond to MPLS LSPs on transit routers in the service provider core network.

NOTE: To collect information about MPLS statistics on transit routers, use the enterprise-specific RSVP MIB (**mib-jnx-rsvp.txt**) instead of the enterprise-specific MPLS MIB (**mib-jnx-mpls.txt**).

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-rsvp.txt .

For information about the enterprise-specific RSVP MIB objects, see the following topic:

- jnxRsvpSessionTable on page 399

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxRsvpSessionTable

The **jnxRsvpSessionTable**, whose object identifier is **{jnxRsvpOperation 1}**, contains information about RSVP sessions. Each **jnxRsvpSessionEntry** (object identifier **{jnxRsvpSessionTable 1}**) is identified using a combination of two indexes, **jnxRsvpSessionName** and **jnxRsvpSessionIndex**. The **jnxRsvpSessionName** maps to the LSP name in MPLS entry, and can be used to correlate the **jnxRsvpSessionEntry** with **mplsLspEntry**. Because there can be multiple entries with the same RSVP session name,

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a secondary index, **jnxRsvpSessionIndex**, is used to uniquely identify each session in combination with the **jnxRsvpSessionName**.

Each **jnxRsvpSessionEntry** contains the objects listed in Table 143 on page 400.

Table 143: jnxRsvpSessionTable

Object	Object ID	Description
jnxRsvpSessionName	**jnxRsvpSessionEntry 1**	Contains the name of the RSVP session. This is the same as the LSP name in the **mplsLspEntry** and can contain up to 64 characters.
jnxRsvpSessionIndex	**jnxRsvpSessionEntry 2**	Uniquely identifies an RSVP session entry in combination with the **jnxRsvpSessionName**.
jnxRsvpSessionState	**jnxRsvpSessionEntry 3**	Shows the operational state of the RSVP session. This object contains one of the following integer values: • **1**–Up • **2**–Down
jnxRsvpSessionFrom	**jnxRsvpSessionEntry 4**	Contains the source IP address of the RSVP session.
jnxRsvpSessionTo	**jnxRsvpSessionEntry 5**	Contains the destination IP address of the RSVP session.
jnxRsvpSessionLspId	**jnxRsvpSessionEntry 6**	Contains the LSP ID of the sender for the RSVP session.
jnxRsvpSessionTunnelId	**jnxRsvpSessionEntry 7**	Contains the tunnel ID for the RSVP session.
jnxRsvpSessionPathType	**jnxRsvpSessionEntry 8**	Denotes the type of the path for the RSVP session. This object uses the following integer values to denote the path type: • **1**–Primary • **2**–Secondary • **3**–unknown
jnxRsvpSessionRole	**jnxRsvpSessionEntry 9**	Shows the role of an RSVP session with respect to the start and end points of the session. This object uses the following integer values to represent the role of the RSVP session: • **1**–Ingress (source) • **2**–Transit (intermediate nodes) • **3**–Egress (destination)
jnxRsvpSessionDiscontinuityTime	**jnxRsvpSessionEntry 10**	Shows the value of **sysUpTime** when either **jnxRsvpSessionMplsOctets** or **jnxRsvpSessionMplsPackets** counters experienced discontinuity. This object contains a zero value if no discontinuity occurred since the last initialization of the local management subsystem.

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Table 143: jnxRsvpSessionTable *(continued)*

Object	Object ID	Description
jnxRsvpSessionMplsOctets	jnxRsvpSessionEntry 11	Contains the number of MPLS octets that have been forwarded over the RSVP session. Because the MPLS statistics collection occurs at predefined intervals (default of 5 minutes), the value of this object might not reflect real-time statistics. This object is not updated if MPLS statistics collection is not enabled.
jnxRsvpSessionMplsPackets	jnxRsvpSessionEntry 12	Shows the number of MPLS packets that have been forwarded over the RSVP session. Because the MPLS statistics collection occurs at predefined intervals (default of 5 minutes), the value of this object might not reflect real-time statistics. This object is not updated if MPLS statistics collection is not enabled.

Related Documentation

- RSVP MIB on page 399

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CHAPTER 44

Interpreting the Enterprise-Specific MPLS MIB

- MPLS MIB on page 403
- mplsInfo on page 404
- mplsTEInfo on page 404
- mplsAdminGroup on page 405
- mplsLspInfoList on page 405
- Juniper Networks Enterprise-Specific MPLS Traps on page 409

MPLS MIB

The Juniper Networks enterprise-specific MPLS MIB, whose object identifier is **{jnxMibs 2}**, provides information about MPLS paths and defines MPLS notifications.

The table **mplsLspList** and the sequence for **mplsLspEntry** have been deprecated and replaced by a new table **mplsLspInfoList** to extend support for LSP names longer than 32 characters.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-mpls.txt .

For information about the enterprise-specific MPLS MIB objects, see the following topics:

- mplsInfo on page 404
- mplsTEInfo on page 404
- mplsAdminGroup on page 405
- mplsLspInfoList on page 405
- Juniper Networks Enterprise-Specific MPLS Traps on page 409

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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mplsInfo

The **mplsInfo** table, whose object identifier is **{mpls 1}**, contains the objects listed in Table 144 on page 404.

Table 144: mplsInfo

Object	Object ID	Description
mplsVersion	**mplsInfo 1**	Shows the MPLS version number.
mplsSignalingProto	**mplsInfo 2**	Indicates the MPLS signaling protocol. This object uses the following integer values to denote the MPLS signaling protocol: • **1**–None • **2**–Other • **3**–RSVP • **4**–LDP
mplsConfiguredLsps	**mplsInfo 3**	Indicates the number of LSPs configured on the router.
mplsActiveLsps	**mplsInfo 4**	Indicates the number of active LSPs on the router.

Related Documentation

- MPLS MIB on page 403
- mplsTEInfo on page 404
- mplsAdminGroup on page 405
- mplsLspInfoList on page 405
- Juniper Networks Enterprise-Specific MPLS Traps on page 409

mplsTEInfo

The **mplsTEInfo** table, whose object identifier is **{mpls 2}**, contains the objects listed in Table 145 on page 404.

Table 145: mplsTEInfo

Object	Object ID	Description
mplsTEDistProtocol	**mplsTEInfo 1**	Indicates the Interior Gateway Protocol (IGP) used to distribute Traffic Engineering information and topology to each Label Switch Router (LSR) for automatic path computation. This object uses the following integer values to denote the protocols: • **1**–None • **2**–IS-IS • **3**–OSPF • **4**–IS-IS – OSPF

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Table 145: mplsTEInfo *(continued)*

Object	Object ID	Description
mplsAdminGroupList	**mplsTEInfo 2**	Contains the list of administrative groups configured on the router. Administrative groups are used to label links in the Traffic Engineering topology for specifying constraints (include and exclude) on LSP paths.

Related Documentation

- MPLS MIB on page 403
- mplsInfo on page 404
- mplsAdminGroup on page 405
- mplsLspInfoList on page 405
- Juniper Networks Enterprise-Specific MPLS Traps on page 409

mplsAdminGroup

The **mplsAdminGroup**, whose object identifier is **{mplsAdminGroupList 1}**, contains **mplsAdminGroupNumber** and **mplsAdminGroupName** objects, and provides a mapping between the group number and name.

- **mplsAdminGroupNumber**–Whose object identifier is **{mplsAdminGroup 1}**, contains an integer value from 0 through 31. This object indexes the **mplsAdminGroup**.
- **mplsAdminGroupName**–Whose object identifier is **{mplsAdminGroup 2}**, contains the name of the **mplsAdminGroup**. This object can contain a string of not more than 16 characters.

Related Documentation

- MPLS MIB on page 403
- mplsInfo on page 404
- mplsTEInfo on page 404
- mplsLspInfoList on page 405
- Juniper Networks Enterprise-Specific MPLS Traps on page 409

mplsLspInfoList

The **mplsLspInfoList**, whose object identifier is **{mpls 5}**, contains a list of Label Switched Paths (LSPs) configured on the router. Each **mplsLspInfoEntry** contains information about a particular LSP, and has the objects listed in Table 146 on page 405.

Table 146: mplsLspInfoEntry

Object	Object ID	Description
mplsLspInfoName	**mplsLspInfoEntry 1**	Contains the name of the LSP. This object can contain a string of not more than 64 characters.

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Table 146: mplsLspInfoEntry *(continued)*

Object	Object ID	Description
mplsLspInfoState	**mplsLspInfoEntry 2**	Contains one of the following integer values to indicate the operational state of the LSP: • **1**–Unknown • **2**–Up • **3**–Down • **4**–notInService: Indicates that the LSP has been torn down or never been signaled due to the lack of demand for its protection. • **5**–backupActive: indicates that the LSP is up and carrying user traffic for at least one protected LSP due to the failure of the LSP, which has caused the creation of a backup LSP.
mplsLspInfoOctets	**mplsLspInfoEntry 3**	Indicates the number of octets that have been forwarded over the current LSP active path. Because the MPLS statistics are collected only at predefined intervals (default of 5 minutes), the value of this object might not reflect the real-time value. The value of the object is not updated if MPLS statistics collection is not enabled.
mplsLspInfoPackets	**mplsLspInfoEntry 4**	Indicates the number of packets that have been forwarded over the current LSP active path. Because the MPLS statistics are collected only at predefined intervals (default of 5 minutes), the value of this object might not reflect the real-time value. The value of the object is not updated if MPLS statistics collection is not enabled.
mplsLspInfoAge	**mplsLspInfoEntry 5**	Indicates the time duration (in 10-millisecond intervals) since the inception of the LSP.
mplsLspInfoTimeUp	**mplsLspInfoEntry 6**	Indicates the total time (in 10-millisecond intervals) that the LSP has been operational. The percentage of up time can be calculated using the following formula: **mplsLspInfoTimeUp**/**mplsLspInfoAge** x 100.
mplsLspInfoPrimaryTimeUp	**mplsLspInfoEntry 7**	Indicates the total time (in 10-millisecond intervals) that the primary path of the LSP has been operational.
mplsLspInfoTransitions	**mplsLspInfoEntry 8**	Indicates the number of state transitions, from up to down and down to up, that the LSP has undergone.
mplsLspInfoLastTransition	**mplsLspInfoEntry 9**	Shows the time (in 10-millisecond intervals) since the last state transition occurred on the LSP.
mplsLspInfoPathChanges	**mplsLspInfoEntry 10**	Shows the number of path changes that occurred on the LSP. Every path change (path down, path up, and path change) generates a syslog entry or trap or both if the corresponding configuration is enabled.
mplsLspInfoLastPathChange	**mplsLspInfoEntry 11**	Indicates the time (in 10-millisecond intervals) since the last path change occurred on the LSP.
mplsLspInfoConfiguredPaths	**mplsLspInfoEntry 12**	Indicates the number of paths configured for the LSP.
mplsLspInfoStandbyPaths	**mplsLspInfoEntry 13**	Indicates the number of standby paths configured on the LSP.

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Table 146: mplsLspInfoEntry *(continued)*

Object	Object ID	Description
mplsLspInfoOperationalPaths	**mplsLspInfoEntry 14**	Indicates the number of operational paths for the LSP. The value of this object includes the currently active path as well as the operational standby paths.
mplsLspInfoFrom	**mplsLspInfoEntry 15**	Contains the source IP address of the LSP.
mplsLspInfoTo	**mplsLspInfoEntry 16**	Contains the destination IP address of the LSP.
mplsPathInfoName	**mplsLspInfoEntry 17**	Shows the name of the active path for the LSP. If the path does not have a name, the **mplsLspInfoEntry** objects listed in this table are invalid.
mplsPathInfoType	**mplsLspInfoEntry 18**	Contains one of the following integer values to denote the type of the active path: • **1**–Other • **2**–Primary • **3**–Standby • **4**–Secondary • **5**–Bypass NOTE: The value of this object is invalid if **mplsPathInfoName** is blank.
mplsPathInfoExplicitRoute	**mplsLspInfoEntry 19**	Contains the explicit route used to set up the LSP. The explicit router can be one configured by the user or a generated route that satisfies the constraints set by the user. The value of this object is stored in the following format: **xxx.xxx.xxx.xxx S/L**, where **S/L** stands for Strict/Loose route. Each explicit route appears in a new line. NOTE: The value of this object is invalid if **mplsPathInfoName** is blank.
mplsPathInfoRecordRoute	**mplsLspInfoEntry 20**	Shows the route actually used for the LSP as recorded by the signaling protocol. NOTE: The value of this object is invalid if **mplsPathInfoName** is blank.
mplsPathInfoBandwidth	**mplsLspInfoEntry 21**	Indicates the configured bandwidth (in kbps) for the LSP. NOTE: The value of this object is invalid if **mplsPathInfoName** is blank.
mplsPathInfoCOS	**mplsLspInfoEntry 22**	Indicates the class of service (CoS) configured for the path. If the value of this object is from **0** through **7**, it goes in the 3-bit CoS field in the label. If the value is **255**, the value in the CoS field of the label depends on other factors. NOTE: The value of this object is invalid if **mplsPathInfoName** is blank.

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Table 146: mplsLspInfoEntry *(continued)*

Object	Object ID	Description
mplsPathInfoInclude	**mplsLspInfoEntry 23**	Contains a configured set of colors represented by bit vector. For each link this path goes through, the link must have colors associated with the path, and the intersection of the link's colors and the **include** set must be set to a value other than null. NOTE: The value of this object is invalid if **mplsPathInfoName** is blank.
mplsPathInfoExclude	**mplsLspInfoEntry 24**	Contains a configured set of colors represented by bit vector. For each link the path goes through, the link must have colors associated with the path, and the intersection of the link's colors and the **exclude** set must be set to null. NOTE: The value of this object is invalid if **mplsPathInfoName** is blank.
mplsPathInfoSetupPriority	**mplsLspInfoEntry 25**	Indicates the setup priority configured for the path. This object contains integer values from **0** through **7**. NOTE: The value of this object is invalid if **mplsPathInfoName** is blank.
mplsPathInfoHoldPriority	**mplsLspInfoEntry 26**	Indicates the hold priority configured for the path. This object contains integer values from **0** through **7**. NOTE: The value of this object is invalid if **mplsPathInfoName** is blank.
mplsPathInfoProperties	**mplsLspInfoEntry 27**	Denotes the properties configured for the path. This value is represented as a bit map. The possible values are: • **1**–Record-Route • **2**–Adaptive • **4**–CSPF • **8**–Mergeable • **16**–Preemptable • **32**–Preemptive • **64**–Fast-Reroute NOTE: The value of this object is invalid if **mplsPathInfoName** is blank.
mplsLspInfoAggrOctets	**mplsLspInfoEntry 28**	Indicates the total number of octets that have been forwarded over all instances of the current LSP. Because the MPLS statistics are collected only at predefined intervals (default of 5 minutes), the value of this object might not reflect the real-time value. The value of the object is not updated if MPLS statistics collection is not enabled.
mplsLspInfoAggrPackets	**mplsLspInfoEntry 29**	Indicates the total number of packets that have been forwarded over all instances of the current LSP. Because the MPLS statistics are collected only at predefined intervals (default of 5 minutes), the value of this object might not reflect the real-time value. The value of the object is not updated if MPLS statistics collection is not enabled.

Related Documentation

- MPLS MIB on page 403
- mplsInfo on page 404

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- mplsTEInfo on page 404
- mplsAdminGroup on page 405
- Juniper Networks Enterprise-Specific MPLS Traps on page 409

Juniper Networks Enterprise-Specific MPLS Traps

Table 147 on page 409 lists the enterprise-specific MPLS traps based on **mplsLspInfoName**.

Table 147: MPLS Traps

Object	Object ID	Description
mplsLspInfoUp	**mplsLspTraps 1**	Indicates that the LSP (**mplsLspInfoName**) is up. The current active path is represented by **mplsPathInfoName**.
mplsLspInfoDown	**mplsLspTraps 2**	Indicates that the LSP (**mplsLspInfoName**) is down because the current active path (**mplsPathInfoName**) has gone down.
mplsLspInfoChange	**mplsLspTraps 3**	Indicates that the LSP (**mplsLspInfoName**) has switched traffic to a new active path (**mplsPathInfoName**) without changing the state (up) before or after the switch.
mplsLspInfoPathDown	**mplsLspTraps 4**	Indicates that the specified path (**mplsPathInfoName**) for the LSP (**mplsLspInfoName**) has gone down.
mplsLspInfoPathUp	**mplsLspTraps 5**	Indicates that the specified path (**mplsPathInfoName**) for the LSP (**mplsLspInfoName**) has come up.

Related Documentation

- MPLS MIB on page 403
- mplsInfo on page 404
- mplsTEInfo on page 404
- mplsAdminGroup on page 405
- mplsLspInfoList on page 405
- Configuring SNMP Trap Options and Groups on a Device Running Junos OS
- Managing Traps and Informs

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CHAPTER 45

Interpreting the Enterprise-Specific Host Resources MIB

- Host Resources MIB on page 411
- jnxHrStorageTable on page 411

Host Resources MIB

The Juniper Networks enterprise-specific Host Resources MIB, whose object identifier is **{jnxMibs 31}**, extends the **hrStorageTable** defined in RFC 2790, the standard Host Resources MIB, to include the **jnxHrStoragePercentUsed** object.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-hostresources.txt .

For information about the enterprise-specific Host Resources MIB objects, see the following topic:

- jnxHrStorageTable on page 411

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxHrStorageTable

The **jnxHrStorageTable**, whose object identifier is **{jnxHrStorage 1}**, contains **jnxHrStorageEntry**. Each **jnxHrStorageEntry** augments the **hrStorageEntry** to provide additional file system data, and contains the following object:

- **jnxHrStoragePercentUsed**—Object identifier is **{jnxHrStorageEntry 1}**. —Shows what percentage of the total storage space has been used.

Related Documentation

- Host Resources MIB on page 411

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CHAPTER 46

Interpreting the Enterprise-Specific Layer 2 Control Protocol (L2CP) MIB

- L2CP MIB on page 413
- L2CP MIB Objects Supported by the Junos OS on page 413

L2CP MIB

The Juniper Networks enterprise-specific Layer 2 Control Protocol (L2CP) MIB, whose object identifier is **{jnxMibs 53}**, provides information about LC2P-based features on MX Series 3D Universal Edge Routers . Currently, the Junos OS supports only the **jnxDot1dStpPortRootProtectEnabled**, **jnxDot1dStpPortRootProtectState**, and **jnxPortRootProtectStateChangeTrap** objects.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-l2cp-features.txt .

For more information about the enterprise-specific L2CP MIB objects supported by the Junos OS, see the following topic:

- L2CP MIB Objects Supported by the Junos OS on page 413

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

L2CP MIB Objects Supported by the Junos OS

Table 148 on page 414 lists the L2CP MIB objects supported by the Junos OS.

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Table 148: L2CP MIB Objects Supported by the Junos OS

Object	Object ID	Description
jnxDot1dStpPortRootProtectEnabled	**jnxDot1dStpPortProtectEntry 1**	Indicates whether the root protect functionality is enabled on the port. If set to **true**, the port cannot be selected as the root port even if it has the best spanning-tree priority value. By default this object is set to **false**.
jnxDot1dStpPortRootProtectState	**jnxDot1dStpPortProtectEntry 2**	Returns one of the following integer values to indicate whether the port was ever prevented from being the root port or not: • **0** no-error—Indicates that the port was not prevented from being a root port. • **1** root-prevented—Indicates that the port was prevented from being a root port. This object always indicates a **0 no-error** state if the **jnxDot1dStpPortRootProtectEnabled** is set to **false**.
jnxPortRootProtectStateChangeTrap	**jnxL2cpProtectTraps 1**	Generated when there is a change in the **jnxDot1dStpPortRootProtectState** for a port.

Related Documentation

- L2CP MIB on page 413

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CHAPTER 47

Interpreting the Enterprise-Specific MIMSTP MIB

- MIMSTP MIB on page 415
- jnxMIDot1sJuniperMstTable on page 416
- jnxMIMstMstiBridgeTable on page 422
- jnxMIMstVlanInstanceMappingTable on page 424
- jnxMIMstCistPortTable on page 426
- jnxMIMstMstiPortTable on page 430
- Juniper Networks Enterprise-Specific MIMSTP Traps on page 431

MIMSTP MIB

The Junos OS provides SNMP support for spanning-tree protocols on MX Series 3D Universal Edge Routers .

The following standard and Juniper Networks enterprise-specific MIBs have been added to extend SNMP support to spanning-tree protocols:

- RFC 4188, *Definitions of Managed Objects for Bridges*—Supports 802.1d STP (1998) only.
- RFC 4318, *Definitions of Managed Objects for Bridges with Rapid Spanning Tree Protocol*—Supports 802.1w and 802.1t extensions for RSTP.
- Juniper Networks Enterprise-Specific Multiple Instance Virtual Switch MIB (**mib-jnx-mimstp.txt**)—Supports 802.1s (2002) for MSTP.

The Juniper Networks enterprise-specific Multiple Instance Multiple Spanning Tree Protocol (MIMSTP) MIB (**mib-jnx-mimstp.txt**) provides information about multiple spanning-tree instances, that is, routing instances of type Virtual Switch/Layer 2 control, also known as virtual contexts and associated VLANs.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-mimstp.txt .

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For information about the enterprise-specific MIMSTP MIB objects, see the following topics:

- jnxMIDot1sJuniperMstTable on page 416
- jnxMIMstMstiBridgeTable on page 422
- jnxMIMstVlanInstanceMappingTable on page 424
- jnxMIMstCistPortTable on page 426
- jnxMIMstMstiPortTable on page 430
- Juniper Networks Enterprise-Specific MIMSTP Traps on page 431

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxMIDot1sJuniperMstTable

The **jnxMIDot1sJuniperMstTable** provides MSTP module parameters for a given virtual context. Table 149 on page 416 lists the objects in the **jnxMIDot1sJuniperMstTable**.

Table 149: jnxMIDot1sJuniperMstTable

Object	Object Identifier	Description
jnxMIDot1sJuniperMstContextId	**jnxMIDot1sJuniperMstEntry 1**	Identifies the virtual context.
jnxMIMstSystemControl	**jnxMIDot1sJuniperMstEntry 2**	Indicates the status of MST on the ports of the device. The value **start (1)** indicates that MIMST is active on all ports of the device; the value **shutdown (2)** indicates that MIMST is shut down on all ports of the device.
jnxMIMstModuleStatus	**jnxMIDot1sJuniperMstEntry 3**	Indicates whether MST is enabled or disabled on the ports. When set to value 1, **enabled**, MST is enabled on all ports of the device; when set to value 2, **disabled**, MIMST is disabled on all ports. NOTE: The object can be set to **enabled(1)** only when **jnxMIMstSystemControl** is set to start.
jnxMIMstMaxMstInstanceNumber	**jnxMIDot1sJuniperMstEntry 4**	Indicates the maximum number of spanning-tree instances allowed on the bridge.
jnxMIMstNoOfMstiSupported	**jnxMIDot1sJuniperMstEntry 5**	Indicates the maximum number of spanning-tree instances that can be supported on the bridge.
jnxMIMstMaxHopCount	**jnxMIDot1sJuniperMstEntry 6**	Returns the Maximum Hop Count value.

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Table 149: jnxMIDot1sJuniperMstTable *(continued)*

Object	Object Identifier	Description
jnxMIMstBrgAddress	jnxMIDot1sJuniperMstEntry 7	Indicates the MAC address used by the bridge, which forms a unique bridge identifier when combined with **jnxMIMstCistBridgePriority** or **jnxMIMstMstiBridgePriority**.
jnxMIMstCistRoot	jnxMIDot1sJuniperMstEntry 8	Indicates the bridge identifier of the root of the common spanning-tree as determined by the Spanning Tree Protocol. This value is used as the common and internal spanning tree (CIST) Root Identifier parameter in all configuration bridge PDUs originating at this node.
jnxMIMstCistRegionalRoot	jnxMIDot1sJuniperMstEntry 9	Indicates the bridge identifier of the root of the multiple spanning-tree region as determined by the Spanning Tree Protocol. This value is used as the CIST Regional Root Identifier parameter in all configuration bridge PDUs originating at this node.
jnxMIMstCistRootCost	jnxMIDot1sJuniperMstEntry 10	Indicates the cost of the path to the CIST root from this bridge.
jnxMIMstCistRegionalRootCost	jnxMIDot1sJuniperMstEntry 11	Indicates the cost of the path to the CIST regional root from this bridge.
jnxMIMstCistRootPort	jnxMIDot1sJuniperMstEntry 12	Indicates the port number of the port that offers the lowest path cost from this bridge to the CIST Root Bridge.
jnxMIMstCistBridgePriority	jnxMIDot1sJuniperMstEntry 13	Indicates the value of the writable portion of the bridge identifier. The values set for Bridge Priority must be in multiples of 4096.
jnxMIMstCistBridgeMaxAge	jnxMIDot1sJuniperMstEntry 14	Indicates the value that a bridge uses for **MaxAge** when the bridge is acting as the root.
jnxMIMstCistBridgeForwardDelay	jnxMIDot1sJuniperMstEntry 15	Indicates the value that the bridge uses for **ForwardDelay** when this bridge is acting as the root. Note that 802.1D specifies that the range for this parameter is related to the value of **BridgeMaxAge**.
jnxMIMstCistHoldTime	jnxMIDot1sJuniperMstEntry 16	Sets the interval between transmitting two configuration bridge PDUs.

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Table 149: jnxMIDot1sJuniperMstTable *(continued)*

Object	Object Identifier	Description
jnxMIMstCistMaxAge	**jnxMIDot1sJuniperMstEntry 17**	Sets the maximum age of Spanning Tree Protocol (STP) information learned on the ports. The STP information is discarded when the age exceeds the set limit.
jnxMIMstCistForwardDelay	**jnxMIDot1sJuniperMstEntry 18**	Indicates the time period during which a port stays in a particular state before moving to the next state; for example, from spanning to forwarding state.
jnxMIMstMstpUpCount	**jnxMIDot1sJuniperMstEntry 19**	Shows the number of times the MSTP Module has been enabled on the bridge.
jnxMIMstMstpDownCount	**jnxMIDot1sJuniperMstEntry 20**	Shows the number of times the MSTP Module has been disabled on the bridge.
jnxMIMstPathCostDefaultType	**jnxMIDot1sJuniperMstEntry 21**	Shows the version of the spanning-tree default path costs that are to be used by the bridge. • A value of 8021d1998(1) uses the 16-bit default path costs from IEEE Std. 802.1D-1998. • A value of stp8021t2001(2) uses the 32-bit default path costs from IEEE Std. 802.1t.

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Table 149: jnxMIDot1sJuniperMstTable *(continued)*

Object	Object Identifier	Description
jnxMIMstTrace	**jnxMIDot1sJuniperMstEntry 22**	Enables trace statements in the MSTP module. A 4-byte integer can be used to represent the level of trace. The bit position maps to the following levels of trace: • **0**: Init and shutdown traces • **1**: Management traces • **2**: Data path traces • **3**: Control plane traces • **4**: Packet dump traces • **5**: Traces that are related to all resources except buffers • **6**: All failure traces • **7**: Buffer traces The rest of the bits remain unused. You can use a combination of trace levels to generate trace statements of multiple trace levels. NOTE: Trace options and debug options are mutually exclusive. When the trace option is set, the debug option is set to **0**.

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Table 149: jnxMIDot1sJuniperMstTable *(continued)*

Object	Object Identifier	Description
jnxMIMstDebug	**jnxMIDot1sJuniperMstEntry 23**	Enables debug statements in the MSTP module. A 4-byte integer can be used to set the level of debugging. The bit position maps to the following levels of debugging: • **0**: Init and shutdown debug statements • **1**: Management debug statements • **2**: Memory-related debug statements • **3**: BPDU-related debug statements • **4**: Event handling debug statements • **5**: Time module debug statements • **6**: Port information SEM debug statements • **7**: Port receive SEM debug statements (valid in the case of MSTP alone) • **8**: Role selection SEM debug statements • **9**: Role transition SEM debug statements • **10**: State transition SEM debug statements • **11**: Protocol migration SEM debug statements • **12**: Topology change SEM debug statements • **13**: Port transmit SEM debug statements • **14**: Bridge detection SEM debug statements • **15**: All failure debug statements • **16**: Redundancy code flow debug statements The rest of the bits remain unused. You can use a combination of debug levels to generate debug statements of multiple debug levels. NOTE: Debug options and trace options are mutually exclusive. When the debug option is set, the trace option is set to **0**.

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Table 149: jnxMIDot1sJuniperMstTable *(continued)*

Object	Object Identifier	Description
jnxMIMstForceProtocolVersion	jnxMIDot1sJuniperMstEntry 24	Indicates the version of the Spanning Tree Protocol that is running on the bridge. • **stpCompatible(0)**: Represents the Spanning Tree Protocol specified in IEEE 802.1D. • **rstp(2)**: Represents the Rapid Spanning Tree Protocol specified in IEEE 802.1w. • **mstp(3)**: Represents the Multiple Spanning Tree Protocol specified in IEEE 802.1s.
jnxMIMstTxHoldCount	jnxMIDot1sJuniperMstEntry 25	Indicates the value that the port transmit state machine uses to limit the maximum transmission rate.
jnxMIMstMstiConfigIdSel	jnxMIDot1sJuniperMstEntry 26	Indicates the Configuration Identifier Format Selector that the bridge uses.
jnxMIMstMstiRegionName	jnxMIDot1sJuniperMstEntry 27	Indicates the name of the region's configuration. By default, the region name and the MAC address of the bridge are the same.
jnxMIMstMstiRegionVersion	jnxMIDot1sJuniperMstEntry 28	Indicates the version of the multiple spanning-tree region.
jnxMIMstMstiConfigDigest	jnxMIDot1sJuniperMstEntry 29	Indicates the configuration digest value for the multiple spanning-tree region.
jnxMIMstBufferOverFlowCount	jnxMIDot1sJuniperMstEntry 30	Indicates the number of times buffer overflows or failures have occurred. This event generates a trap.
jnxMIMstMemAllocFailureCount	jnxMIDot1sJuniperMstEntry 31	Indicates the number of times memory allocation failures have occurred. This event generates a trap.
jnxMIMstRegionConfigChangeCount	jnxMIDot1sJuniperMstEntry 32	Indicates the number of times a Region Configuration Identifier Change was detected. This event generates a trap.
jnxMIMstCistBridgeRoleSelectionSemState	jnxMIDot1sJuniperMstEntry 33	Indicates the current state of the Port Role Selection State Machine of the bridge in a common spanning-tree context

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Table 149: jnxMIDot1sJuniperMstTable *(continued)*

Object	Object Identifier	Description
jnxMIMstCistTimeSinceTopologyChange	jnxMIDot1sJuniperMstEntry 34	Indicates the time since the TcWhile Timer for any port of the Bridge was nonzero for the common spanning-tree context.
jnxMIMstCistTopChanges	jnxMIDot1sJuniperMstEntry 35	Indicates the number of times when there was at least one nonzero TcWhile Timer on the bridge for the common spanning-tree context.
jnxMIMstCistNewRootBridgeCount	jnxMIDot1sJuniperMstEntry 36	Indicates how many times the bridge has detected a root bridge change for a common-spanning-tree context. This event generates a trap.
jnxMIMstCistHelloTime	jnxMIDot1sJuniperMstEntry 37	Specifies the interval between the transmission of configuration BPDUs by this node on any port when it is either the root of the spanning-tree or trying to become the root.
jnxMIMstCistBridgeHelloTime	jnxMIDot1sJuniperMstEntry 38	Specifies the interval between the transmission of configuration bridge PDUs by this node.
jnxMIMstCistDynamicPathcostCalculation	jnxMIDot1sJuniperMstEntry 39	Indicates whether the dynamic path cost calculation is allowed. If set to true, path cost is calculated dynamically from the port speed; otherwise, the link speed at the time of port creation is used for calculating the path cost. In both cases, the user has configured a path cost for the port that is used. By default, dynamic path cost calculation is set to false.

Related Documentation

- MIMSTP MIB on page 415
- jnxMIMstMstiBridgeTable on page 422
- jnxMIMstVlanInstanceMappingTable on page 424
- jnxMIMstCistPortTable on page 426
- jnxMIMstMstiPortTable on page 430
- Juniper Networks Enterprise-Specific MIMSTP Traps on page 431

jnxMIMstMstiBridgeTable

The **jnxMIMstMstiBridgeTable** provides information about a bridge that belongs to a given spanning-tree instance (MSTI). Each **jnxMIMstMstiBridgeEntry** points to a bridge in the spanning-tree instance and has the objects listed in Table 150 on page 423.

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Table 150: jnxMIMstMstiBridgeTable

Object	Object Identifier	Description
jnxMIMstMstiInstanceIndex	jnxMIMstMstiBridgeEntry 1	Identifies the spanning-tree instance to which the information belongs.
jnxMIMstMstiBridgeRegionalRoot	jnxMIMstMstiBridgeEntry 2	Indicates the MSTI Regional Root Identifier value for the Instance. This value is used as the Regional Root Identifier parameter in all the configuration bridge PDUs originated by this node.
jnxMIMstMstiBridgePriority	jnxMIMstMstiBridgeEntry 3	Indicates the writable portion of the MSTI Bridge Identifier that comprises the first two octets. The values that are set for Bridge Priority must be in multiples of 4096.
jnxMIMstMstiRootCost	jnxMIMstMstiBridgeEntry 4	Indicates the cost of the path to the MSTI Regional Root as calculated by the bridge.
jnxMIMstMstiRootPort	jnxMIMstMstiBridgeEntry 5	Indicates the port number of the port that offers the lowest path cost from the bridge to the MSTI Region Root Bridge.
jnxMIMstMstiTimeSinceTopologyChange	jnxMIMstMstiBridgeEntry 6	Indicates the time (in hundredths of a second) since the TcWhile Timer for any port on this bridge was nonzero for this spanning-tree instance.
jnxMIMstMstiTopChanges	jnxMIMstMstiBridgeEntry 7	Indicates the number of times when there was at least one nonzero TcWhile Timer on the bridge for the spanning-tree instance.
jnxMIMstMstiNewRootBridgeCount	jnxMIMstMstiBridgeEntry 8	Indicates the number of times the bridge has detected a root bridge change for the spanning-tree instance. This event generates a trap.
jnxMIMstMstiBridgeRoleSelectionSemState	jnxMIMstMstiBridgeEntry 9	Shows the current state of the Port Role Selection State Machine for the spanning-tree instance of this bridge.
jnxMIMstInstanceUpCount	jnxMIMstMstiBridgeEntry 10	Indicates the number of times a new spanning-tree instance has been created. This counter is incremented whenever a new spanning-tree instance is created and also whenever a VLAN is mapped to the instance. This event generates a trap.

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Table 150: jnxMIMstMstiBridgeTable *(continued)*

Object	Object Identifier	Description
jnxMIMstInstanceDownCount	jnxMIMstMstiBridgeEntry 11	Indicates the number of times a spanning-tree instance has been deleted. This counter is incremented whenever a spanning-tree instance is deleted and also whenever a VLAN is unmapped from the instance. This event generates a trap.
jnxMIMstOldDesignatedRoot	jnxMIMstMstiBridgeEntry 12	Indicates the bridge identifier of the old root of the spanning-tree instance as determined by the Spanning Tree Protocol.

Related Documentation

- MIMSTP MIB on page 415
- jnxMIDot1sJuniperMstTable on page 416
- jnxMIMstVlanInstanceMappingTable on page 424
- jnxMIMstCistPortTable on page 426
- jnxMIMstMstiPortTable on page 430
- Juniper Networks Enterprise-Specific MIMSTP Traps on page 431

jnxMIMstVlanInstanceMappingTable

The **jnxMIMstVlanInstanceMappingTable** contains information about the mapping between each instance of MSTP and associated VLANs. Each **jnxMIMstVlanInstanceMappingEntry** indicates the status and properties of a specific MSTP instance-VLAN mapping and has the objects listed in Table 151 on page 424.

Table 151: jnxMIMstVlanInstanceMappingTable

Object	Object Identifier	Description
jnxMIMstInstanceIndex	jnxMIMstVlanInstanceMappingEntry 1	Identifies a multiple spanning-tree instance using an arbitrary integer from 1 through the value of **Max Instance Number**.
jnxMIMstMapVlanIndex	jnxMIMstVlanInstanceMappingEntry 2	Indicates that the VLAN ID is mapped to the multiple spanning-tree instance specified.
jnxMIMstUnMapVlanIndex	jnxMIMstVlanInstanceMappingEntry 3	Indicates that the VLAN ID is unmapped from the spanning-tree instance to which it was mapped.

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Table 151: jnxMIMstVlanInstanceMappingTable *(continued)*

Object	Object Identifier	Description
jnxMIMstInstanceVlanMapped	jnxMIMstVlanInstanceMappingEntry 6	Represents a string of octets that contain one bit per VLAN. The first octet corresponds to VLANs with VLAN index values 1 through 8; the second octet to VLANs 9 through 16, and so on. The most significant bit of each octet corresponds to the lowest **VlanIndex** value in that octet. For each VLAN that is mapped to this MSTP instance, the bit corresponding to that VLAN is set to 1.
jnxMIMstInstanceVlanMapped2k	jnxMIMstVlanInstanceMappingEntry 7	Represents a string of octets that contain one bit per VLAN for VLANS with **VlanIndex** values from 1024 through 2047. The first octet corresponds to VLANs with **VlanIndex** values 1024 through 1031; the second octet to VLANs 1032 through 1039, and so on. The most significant bit of each octet corresponds to the lowest **VlanIndex** value in that octet. For each VLAN that is mapped to this MSTP instance, the bit corresponding to that VLAN is set to 1.
jnxMIMstInstanceVlanMapped3k	jnxMIMstVlanInstanceMappingEntry 8	Represents a string of octets that contain one bit per VLAN for VLANS with **VlanIndex** values from 2048 through 3071. The first octet corresponds to VLANs with **VlanIndex** values 2048 through 2055; the second octet to VLANs 2056 through 2063, and so on. The most significant bit of each octet corresponds to the lowest **VlanIndex** value in that octet. For each VLAN that is mapped to this MSTP instance, the bit corresponding to that VLAN is set to 1.
jnxMIMstInstanceVlanMapped4k	jnxMIMstVlanInstanceMappingEntry 9	Represents a string of octets that contain one bit per VLAN for VLANS with **VlanIndex** values from 3072 through 4095. The first octet corresponds to VLANs with **VlanIndex** values 3072 through 3079; the second octet to VLANs 3080 through 3087, and so on. The most significant bit of each octet corresponds to the lowest **VlanIndex** value in that octet. For each VLAN that is mapped to this MSTP instance, the bit corresponding to that VLAN is set to 1.

Related Documentation

- MIMSTP MIB on page 415
- jnxMIDot1sJuniperMstTable on page 416

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- jnxMIMstMstiBridgeTable on page 422
- jnxMIMstCistPortTable on page 426
- jnxMIMstMstiPortTable on page 430
- Juniper Networks Enterprise-Specific MIMSTP Traps on page 431

jnxMIMstCistPortTable

The **jnxMIMstCistPortTable** contains the information maintained by the ports of the Common and Internal Spanning Tree Protocol. Table 152 on page 426 lists the parameters maintained by each **jnxMIMstCistPortEntry**.

Table 152: jnxMIMstCistPortTable

Object	Object Identifier	Description
jnxMIMstCistPort	**jnxMIMstCistPortEntry 1**	Specifies the port number of the port to which this entry is mapped.
nxMIMstCistPortPathCost	**jnxMIMstCistPortEntry 2**	Indicates the contribution of this port to the path cost of paths towards the CIST root that includes this port.
jnxMIMstCistPortPriority	**jnxMIMstCistPortEntry 3**	Contains the four most significant bits of the Port Identifier of the spanning-tree instance that can be modified by setting the **CistPortPriority** value. The values that are set for Port Priority must be in multiples of 16.
jnxMIMstCistPortDesignatedRoot	**jnxMIMstCistPortEntry 4**	Specifies the unique Bridge Identifier that is recorded as the CIST root in the configuration bridge protocol data units (BPDUs).
jnxMIMstCistPortDesignatedBridge	**jnxMIMstCistPortEntry 5**	Specifies the unique Bridge Identifier of the bridge that is considered as the designated bridge for the port's segment.
jnxMIMstCistPortDesignatedPort	**jnxMIMstCistPortEntry 6**	Indicates the port identifier of the port on the designated bridge for this port's segment.

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Table 152: jnxMIMstCistPortTable *(continued)*

Object	Object Identifier	Description
jnxMIMstCistPortAdminP2P	jnxMIMstCistPortEntry 7	Indicates the administrative point-to-point status of the LAN segment attached to this port. • A value of **forceTrue(0)** indicates that this port must be treated as if it were connected to a point-to-point link. • A value of **forceFalse(1)** indicates that this port should be treated as having a shared media connection. • A value of **auto(2)** indicates that this port is considered to have a point-to-point link if it is an aggregator and all of its members are aggregatable, or if the MAC entity is configured for full duplex operation, either through autonegotiation or by management.
jnxMIMstCistPortOperP2P	jnxMIMstCistPortEntry 8	Indicates the operational point-to-point status of the LAN segment that is attached to this port. It indicates whether a port is considered to have a point-to-point connection or not. The value is determined by management or by autodetection, as described in the **jnxMIMstCistPortAdminP2P** object.
jnxMIMstCistPortAdminEdgeStatus	jnxMIMstCistPortEntry 9	Specifies the administrative value of the **EdgePort** parameter. A value of **TRUE(1)** indicates that this port must be treated as an edge port, and a value of **FALSE(2)** indicates that this port should be treated as a nonedge port.
jnxMIMstCistPortOperEdgeStatus	jnxMIMstCistPortEntry 10	Specifies the operational value of the **EdgePort** parameter. The object is initialized to the value of **jnxMIMstCistPortAdminEdgeStatus** and is set **FALSE** on reception of a BPDU.
jnxMIMstCistPortState	jnxMIMstCistPortEntry 12	Shows the current state of the port as defined by the Common Spanning Tree Protocol.
jnxMIMstCistPortForwardTransitions	jnxMIMstCistPortEntry 14	Indicates the number of times this port has transitioned to the forwarding state.
jnxMIMstCistPortRxMstBpduCount	jnxMIMstCistPortEntry 15	Indicates the number of multiple spanning-tree BPDUs that are received on this port.
jnxMIMstCistPortRxRstBpduCount	jnxMIMstCistPortEntry 16	Indicates the number of rapid spanning-tree BPDUs that are received on this port.
jnxMIMstCistPortRxConfigBpduCount	jnxMIMstCistPortEntry 17	Indicates the number of configuration BPDUs that are received on the port.
jnxMIMstCistPortRxTcnBpduCount	jnxMIMstCistPortEntry 18	Indicates the number of topology change notification BPDUs that are received on the port.

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Table 152: jnxMIMstCistPortTable *(continued)*

Object	Object Identifier	Description
jnxMIMstCistPortTxMstBpduCount	jnxMIMstCistPortEntry 19	Indicates the number of multiple spanning-tree BPDUs that are transmitted from the port.
jnxMIMstCistPortTxRstBpduCount	jnxMIMstCistPortEntry 20	Indicates the number of rapid spanning-tree BPDUs that are transmitted from the port.
jnxMIMstCistPortTxConfigBpduCount	jnxMIMstCistPortEntry 21	Indicates the number of configuration BPDUs that are transmitted from the port.
jnxMIMstCistPortTxTcnBpduCount	jnxMIMstCistPortEntry 22	Indicates the number of TCN BPDUs that are transmitted from the port.
jnxMIMstCistPortInvalidMstBpduRxCount	jnxMIMstCistPortEntry 23	Indicates the number of invalid MST BPDUs that are received on the port.
jnxMIMstCistPortInvalidRstBpduRxCount	jnxMIMstCistPortEntry 24	Indicates the number of invalid RST BPDUs that are received on the port.
jnxMIMstCistPortInvalidConfigBpduRxCount	jnxMIMstCistPortEntry 25	Indicates the number of invalid configuration BPDUs that are received on the port.
jnxMIMstCistPortInvalidTcnBpduRxCount	jnxMIMstCistPortEntry 26	Indicates the number of invalid TCN BPDUs that are received on the port.
jnxMIMstCistPortTransmitSemState	jnxMIMstCistPortEntry 27	Indicates the current state of the port transmit state machine.
jnxMIMstCistPortReceiveSemState	jnxMIMstCistPortEntry 28	Indicates the current state of the port receive state machine.
jnxMIMstCistPortProtMigrationSemState	jnxMIMstCistPortEntry 29	Indicates the current state of the port protocol migration state machine.
jnxMIMstCistProtocolMigrationCount	jnxMIMstCistPortEntry 30	Indicates the number of times the port has migrated from one Spanning Tree Protocol version to another. The relevant protocols are STP-COMPATIBLE and RSTP/MSTP. This event generates a trap.
jnxMIMstCistPortDesignatedCost	jnxMIMstCistPortEntry 31	Indicates the path cost of the designated port of the segment connected to this port.
jnxMIMstCistPortRegionalRoot	jnxMIMstCistPortEntry 32	Specifies the unique Bridge Identifier of the bridge recorded as the CIST Regional Root Identifier in the configuration BPDUs transmitted.
jnxMIMstCistPortRegionalPathCost	jnxMIMstCistPortEntry 33	Specifies the contribution of this port to the path cost of paths towards the CIST regional root that includes this port.

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Table 152: jnxMIMstCistPortTable *(continued)*

Object	Object Identifier	Description
jnxMIMstCistSelectedPortRole	jnxMIMstCistPortEntry 34	Indicates the selected port role of the port for this spanning-tree instance.
jnxMIMstCistCurrentPortRole	jnxMIMstCistPortEntry 35	Specifies the current port role of the port for this spanning-tree instance.
jnxMIMstCistPortInfoSemState	jnxMIMstCistPortEntry 36	Indicates the current state of the port information state machine for this port in this spanning-tree context.
jnxMIMstCistPortRoleTransitionSemState	jnxMIMstCistPortEntry 37	Indicates the current state of the port role transition state machine for this port in this spanning-tree context.
jnxMIMstCistPortStateTransitionSemState	jnxMIMstCistPortEntry 38	Indicates the current state of the port state transition state machine for this port in this spanning-tree context.
jnxMIMstCistPortTopologyChangeSemState	jnxMIMstCistPortEntry 39	Indicates the current state of the topology change state machine for this port in this spanning-tree context.
jnxMIMstCistPortHelloTime	jnxMIMstCistPortEntry 40	Indicates the interval between the transmission of configuration bridge PDUs on this port.
jnxMIMstCistPortOperVersion	jnxMIMstCistPortEntry 41	Indicates one of the following operational modes of the port: MSTP, RSTP, and STP-compatible.
jnxMIMstCistPortEffectivePortState	jnxMIMstCistPortEntry 42	Indicates the effective operational state of the port for CIST. This value is set to TRUE only when the port is operationally up in the Interface level and Protocol level for CIST. This value is set to FALSE for all other instances.
jnxMIMstCistPortAutoEdgeStatus	jnxMIMstCistPortEntry 43	Indicates one of the following states: • **TRUE(1)** when the detection of a port as edge post happens automatically • **FALSE(2)** when this feature is disabled.

Related Documentation

- MIMSTP MIB on page 415
- jnxMIDot1sJuniperMstTable on page 416
- jnxMIMstMstiBridgeTable on page 422
- jnxMIMstVlanInstanceMappingTable on page 424
- jnxMIMstMstiPortTable on page 430
- Juniper Networks Enterprise-Specific MIMSTP Traps on page 431

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jnxMIMstMstiPortTable

The **jnxMIMstMstiPortTable** contains information maintained by the nonCIST ports for each spanning-tree instance. Each **jnxMIMstMstiPortEntry** contains the parameters listed in Table 153 on page 430.

Table 153: jnxMIMstMstiPortTable

Object	Object Identifier	Description
jnxMIMstMstiPort	jnxMIMstMstiPortEntry 1	Specifies the port number of the port to which this entry maps.
jnxMIMstMstiPortPathCost	jnxMIMstMstiPortEntry 2	Shows the contribution of this port to the path cost of paths toward the MSTI root that includes this port.
jnxMIMstMstiPortPriority	jnxMIMstMstiPortEntry 3	Enables you to specify the four most significant bits of the Port Identifier for a given spanning-tree instance, which can be modified independently for each spanning-tree instance supported by the bridge. The values that are set for Port Priority must be in multiples of 16.
jnxMIMstMstiPortDesignatedRoot	jnxMIMstMstiPortEntry 4	Indicates the unique Bridge Identifier of the bridge recorded as the MSTI regional root in the configuration BPDUs that are transmitted.
jnxMIMstMstiPortDesignatedBridge	jnxMIMstMstiPortEntry 5	Indicates the unique Bridge Identifier of the bridge that this port considers to be the designated bridge for the port's segment.
jnxMIMstMstiPortDesignatedPort	jnxMIMstMstiPortEntry 6	Indicates the port identifier of the port on the designated bridge for this port's segment.
jnxMIMstMstiPortState	jnxMIMstMstiPortEntry 7	Indicates the current state of the port as defined by the Multiple Spanning Tree protocol. A port that is in forwarding state in one instance can be in discarding (blocking) state in another instance.
jnxMIMstMstiPortForwardTransitions	jnxMIMstMstiPortEntry 9	Indicates the number of times this port has transitioned to the forwarding state for a specific instance.
jnxMIMstMstiPortReceivedBPDUs	jnxMIMstMstiPortEntry 10	Indicates the number of BPDUs received by this port for this spanning-tree instance.
jnxMIMstMstiPortTransmittedBPDUs	jnxMIMstMstiPortEntry 11	Indicates the number of BPDUs transmitted on this port for this spanning-tree instance.
jnxMIMstMstiPortInvalidBPDUsRcvd	jnxMIMstMstiPortEntry 12	Indicates the number of invalid BPDUs received on this port for this spanning-tree instance.

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Table 153: jnxMIMstMstiPortTable *(continued)*

Object	Object Identifier	Description
jnxMIMstMstiPortDesignatedCost	jnxMIMstMstiPortEntry 13	Indicates the path cost of the designated port of the segment connected to this port.
jnxMIMstMstiSelectedPortRole	jnxMIMstMstiPortEntry 14	Indicates the selected Port Role of the port for this spanning-tree instance.
jnxMIMstMstiCurrentPortRole	jnxMIMstMstiPortEntry 15	Indicates the current Port Role of the port for this spanning-tree instance.
jnxMIMstMstiPortInfoSemState	jnxMIMstMstiPortEntry 16	Shows the current state of the port information state machine for this port in this spanning-tree context.
jnxMIMstMstiPortRoleTransitionSemState	jnxMIMstMstiPortEntry 17	Shows the current state of the port role transition state machine for this port in this spanning-tree context.
jnxMIMstMstiPortStateTransitionSemState	nxMIMstMstiPortEntry 18	Shows the current state of the port state transition state machine for this port in this spanning-tree context.
jnxMIMstMstiPortTopologyChangeSemState	nxMIMstMstiPortEntry 19	Shows the current state of the topology change state machine for this port in this spanning-tree context.
jnxMIMstMstiPortEffectivePortState	jnxMIMstMstiPortEntry 20	Shows the effective operational state of the port for the specific instance. The value is set to TRUE only when the port is operationally up in the interface level and protocol level for the specific instance. This is set to FALSE at all other times.

Related Documentation

- MIMSTP MIB on page 415
- jnxMIDot1sJuniperMstTable on page 416
- jnxMIMstMstiBridgeTable on page 422
- jnxMIMstVlanInstanceMappingTable on page 424
- jnxMIMstCistPortTable on page 426
- Juniper Networks Enterprise-Specific MIMSTP Traps on page 431

Juniper Networks Enterprise-Specific MIMSTP Traps

Table 154 on page 431 lists the enterprise-specific MIMSTP traps.

Table 154: Enterprise-Specific MIMSTP Traps

Object	Object Identifier	Description
jnxMIMstGenTrap	jnxMIMstTraps 1	Generated when any of the general events such as protocol up or protocol down occurs.

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Table 154: Enterprise-Specific MIMSTP Traps *(continued)*

Object	Object Identifier	Description
jnxMIMstErrTrap	**jnxMIMstTraps 2**	Generated when any of the error events such as a memory failure, buffer failure, protocol migration, or new root or topology change occurs.
jnxMIMstNewRootTrap	**jnxMIMstTraps 3**	Generated when a new root bridge is selected in the topology. The **jnxMIMstNewRootTrap** indicates that the sending agent has become the new root of the spanning-tree. The trap is sent by a bridge soon after its election as the new root
jnxMIMstTopologyChgTrap	**jnxMIMstTraps 4**	Generated when a topology change is detected.
jnxMIMstProtocolMigrationTrap	**jnxMIMstTraps 5**	Generated when a port protocol migration happens on the port.
jnxMIMstInvalidBpduRxdTrap	**jnxMIMstTraps 6**	Generated when an invalid packet is received for **bpdu/stp/rstp/maximum age/forward delay/hello time**.
jnxMIMstRegionConfigChangeTrap	**jnxMIMstTraps 7**	Generated when the multiple spanning-tree region's configuration identifier changes.

Related Documentation

- MIMSTP MIB on page 415
- jnxMIDot1sJuniperMstTable on page 416
- jnxMIMstMstiBridgeTable on page 422
- jnxMIMstVlanInstanceMappingTable on page 424
- jnxMIMstCistPortTable on page 426
- jnxMIMstMstiPortTable on page 430
- Configuring SNMP Trap Options and Groups on a Device Running Junos OS
- Managing Traps and Informs

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CHAPTER 48

Interpreting the Enterprise-Specific L2ALD MIB

- L2ALD MIB on page 433
- jnxl2aldInterfaceTable on page 433
- MAC Address Limit Traps on page 434

L2ALD MIB

The Juniper Networks enterprise-specific Layer 2 Address Learning Daemon (L2ALD) MIB, whose object identifier is **{jnxl2aldMibRoot 1}**, contains information about Layer 2 addresses and defines L2ALD traps.

The L2ALD MIB has the following two branches:

- **jnxl2aldNotification**, whose object identifier is **{jnxl2aldMib 0}**
- **jnxl2aldObjects**, whose object identifier is **{jnxl2aldMib 1}**

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-l2ald.txt .

For information about the enterprise-specific L2ALD MIB objects, see the following topics:

- jnxl2aldInterfaceTable on page 433
- MAC Address Limit Traps on page 434

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxl2aldInterfaceTable

The **jnxl2aldInterfaceTable**, whose object identifier is **{jnxl2aldObjects 1}**, contains objects that control the MAC address limit on each interface. Each **jnxl2aldEntry** (object identifier: **{jnxl2aldInterfaceTable 1}**) in the **jnxl2aldInterfaceTable** contains the objects listed in Table 155 on page 434.

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Table 155: jnxl2aldInterfaceTable

Object	Object ID	Description
jnxl2aldIntfLogicalRouter	**jnxl2aldEntry 1**	Identifies the logical system with which the **jnxl2aldEntry** is associated.
jnxl2aldIntfRoutingInst	**jnxl2aldEntry 2**	Identifies the routing instance with which the **jnxl2aldEntry** is associated.
jnxl2aldIntfBridgeDomain	**jnxl2aldEntry 3**	Identifies the bridge domain with which the **jnxl2aldEntry** is associated.
jnxl2aldIntfMacLimit	**jnxl2aldEntry 4**	Defines the MAC address limit for the physical interface associated with the **jnxl2aldEntry**.
Scalar Objects for Notifications		
jnxl2aldRoutingInst	**jnxl2aldObjects 2**	Specifies the routing instance for the **jnxl2aldRoutingInstMacLimit** trap.
jnxl2aldBridgeDomain	**jnxl2aldObjects 3**	Specifies the bridge domain for the **jnxl2aldRoutingInstMacLimit** trap.
jnxl2aldLogicalRouter	**jnxl2aldObjects 4**	Specifies the logical system for the **jnxl2aldRoutingInstMacLimit** trap.
jnxl2aldMacLimit	**jnxl2aldObjects 5**	Specifies the maximum number of MAC addresses that can be learned by the routing instance.
jnxl2aldGbMacLimit	**jnxl2aldObjects 6**	Specifies the maximum number of MAC addresses that can be learned by the router.

Related Documentation

For information about the enterprise-specific L2ALD MIB objects, see the following topics:

- L2ALD MIB on page 433
- MAC Address Limit Traps on page 434

MAC Address Limit Traps

The enterprise-specific L2ALD MIB defines the following traps:

- **jnxl2aldRoutingInstMacLimit**, whose object identifier is **{jnxl2aldNotification 1}**, is generated when the number of MAC addresses for the given routing instance, **jnxl2aldRoutingInst**, exceeds the set limit. This trap contains the following objects: **jnxl2aldLogicalRouter**, **jnxl2aldRoutingInst**, **jnxl2aldBridgeDomain**, and **jnxl2aldMacLimit**.
- **jnxl2aldInterfaceMacLimit**, whose object identifier is **{jnxl2aldNotification 2}**, is generated when the number of MAC addresses for the given physical interface exceeds the set limit. This trap contains the following objects: **jnxl2aldIntfLogicalRouter**, **jnxl2aldIntfRoutingInst**, **jnxl2aldIntfBridgeDomain**,**ifDescr**, and **jnxl2aldIntfMacLimit**.

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- **jnxl2aldGlobalMacLimit**, whose object identifier is **{jnxl2aldNotification 3}**, is generated when the MAC limit for the entire system exceeds the set limit.

Related Documentation

- L2ALD MIB on page 433
- jnxl2aldInterfaceTable on page 433
- Configuring SNMP Trap Options and Groups on a Device Running Junos OS
- Managing Traps and Informs

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CHAPTER 49

Interpreting the Enterprise-Specific Utility MIB

- Utility MIB on page 437
- jnxUtilCounter32Table on page 438
- jnxUtilCounter64Table on page 438
- jnxUtilIntegerTable on page 439
- jnxUtilUintTable on page 439
- jnxUtilStringTable on page 440

Utility MIB

The Juniper Networks enterprise-specific Utility MIB, whose object ID is **{jnxUtilMibRoot 1}**, defines objects for counters, intergers, and strings. The Utility MIB contains one table for each of the following five data types:

- 32-bit counters
- 64-bit counters
- Signed integers
- Unsigned integers
- Octet strings

Each data type has an arbitrary ASCII name, which is defined when the data is populated, and a timestamp that shows the last time when the data instance was modified. For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-util.txt .

For information about the enterprise-specific Utility MIB objects, see the following topics:

- jnxUtilCounter32Table on page 438
- jnxUtilCounter64Table on page 438
- jnxUtilIntegerTable on page 439

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- jnxUtilUintTable on page 439
- jnxUtilStringTable on page 440

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxUtilCounter32Table

jnxUtilCounter32Table, whose object ID is **{jnxUtilData 1}**,contains information about 32-bit counters.

Each **jnxUtilCounter32Entry** has the objects listed in Table 156 on page 438.

Table 156: jnxUtilCounter32Entry

Object	Object ID	Description
jnxUtilCounter32Name	**jnxUtilCounter32Entry 1**	Name assigned to the instance.
jnxUtilCounter32Value	**jnxUtilCounter32Entry 2**	Value of the instance.
jnxUtilCounter32Time	**jnxUtilCounter32Entry 3**	Time when the instance was last populated.

Related Documentation

- Utility MIB on page 437
- jnxUtilCounter64Table on page 438
- jnxUtilIntegerTable on page 439
- jnxUtilUintTable on page 439
- jnxUtilStringTable on page 440

jnxUtilCounter64Table

jnxUtilCounter64Table, whose object ID is **{jnxUtilData 2}**, contains information about 64-bit counters.

Each **jnxUtilCounter64Entry** has the objects listed in Table 157 on page 438.

Table 157: jnxUtilCounter64Entry

Object	Object ID	Description
jnxUtilCounter64Name	**jnxUtilCounter64Entry 1**	Name assigned to the instance.
jnxUtilCounter64Value	**jnxUtilCounter64Entry 2**	Value of the instance.
jnxUtilCounter64Time	**jnxUtilCounter64Entry 3**	Time when the instance was last populated.

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Related Documentation

- Utility MIB on page 437
- jnxUtilCounter32Table on page 438
- jnxUtilIntegerTable on page 439
- jnxUtilUintTable on page 439
- jnxUtilStringTable on page 440

jnxUtilIntegerTable

jnxUtilIntegerTable, whose object ID is **{jnxUtilData 3}**, contains information about signed integer values.

Each **jnxUtilIntegerEntry** contains the objects listed in Table 158 on page 439.

Table 158: jnxUtilIntegerEntry

Object	Object ID	Description
jnxUtilIntegerName	**jnxUtilIntegerEntry 1**	Name assigned to the instance.
jnxUtilIntegerValue	**jnxUtilIntegerEntry 2**	Value of the instance.
jnxUtilIntegerTime	**jnxUtilIntegerEntry 3**	Time when the instance was last populated.

Related Documentation

- Utility MIB on page 437
- jnxUtilCounter32Table on page 438
- jnxUtilCounter64Table on page 438
- jnxUtilUintTable on page 439
- jnxUtilStringTable on page 440

jnxUtilUintTable

jnxUtilUintTable, whose object ID is **{jnxUtilData 4}**, contains information about unsigned integer values.

Each **jnxUtilUintEntry** has the objects listed in Table 159 on page 439.

Table 159: jnxUtilUintEntry

Object	Object ID	Description
jnxUtilUintName	**jnxUtilUintEntry 1**	Name assigned to the instance.
jnxUtilUintValue	**jnxUtilUintEntry 2**	Value of the instance.
jnxUtilUintTime	**jnxUtilUintEntry 3**	Time when the instance was last populated.

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Related Documentation

- Utility MIB on page 437
- jnxUtilCounter32Table on page 438
- jnxUtilCounter64Table on page 438
- jnxUtilIntegerTable on page 439
- jnxUtilStringTable on page 440

jnxUtilStringTable

jnxUtilStringTable, whose object ID is **{jnxUtilData 5}**, contains information about octet strings.

Each **jnxUtilStringEntry** contains the objects listed in Table 160 on page 440.

Table 160: jnxUtilStringEntry

Object	Object ID	Description
jnxUtilStringName	**jnxUtilStringEntry 1**	Name assigned to the instance.
jnxUtilStringValue	**jnxUtilStringEntry 2**	Value of the instance.
jnxUtilStringTime	**jnxUtilStringEntry 31**	Time when the instance was last populated.

Related Documentation

- Utility MIB on page 437
- jnxUtilCounter32Table on page 438
- jnxUtilCounter64Table on page 438
- jnxUtilIntegerTable on page 439
- jnxUtilUintTable on page 439

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CHAPTER 50

Interpreting the Enterprise-Specific AAA Objects MIB

- AAA Objects MIB on page 441
- AAA Objects MIB Textual Conventions on page 442
- jnxUserAAAStatTable on page 444
- jnxUserAAAServerName on page 444
- Access Authentication-Related Traps on page 445
- jnxUserAAAAccessPool on page 445

AAA Objects MIB

The Juniper Networks enterprise-specific AAA Objects MIB, whose object ID is **{jnxUserAAAMibRoot 1}**, defines the objects pertaining to user authentication, authorization, and accounting.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-user-aaa.txt .

Object IDs for next branch nodes are as follows:

- jnxUserAAAGlobalStats—**{jnxUserAAAObjects 1}**
- jnxUserAAAAccessAuthStats—**{jnxUserAAAObjects 2}**
- jnxUserAAATrapVars—**{jnxUserAAAObjects 3}**
- jnxUserAAAAssignment—**{jnxUserAAAObjects 4}**
- jnxUserAAAAccessProfile—**{jnxUserAAAObjects 5}**

Object IDs for statistic counters related to access authentication are as follows:

- jnxTotalAuthenticationRequests—**{jnxUserAAAGlobalStats 1}**
- jnxTotalAuthenticationResponses—**{jjnxUserAAAGlobalStats 2}**

The object ID for address pools is as follows:

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- **jnxUserAAAAccessPool 1—{jnxUserAAAAccessPoolTable 1}**

For information about the enterprise-specific AAA Objects MIB, see the following topics

- AAA Objects MIB Textual Conventions on page 442
- jnxUserAAAStatTable on page 444
- jnxUserAAAServerName on page 444
- Access Authentication-Related Traps on page 445
- jnxUserAAAAccessPool on page 445

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

AAA Objects MIB Textual Conventions

jnxAuthenticateType—Defines the method to authenticate a user.

Table 161: jnxAuthenticateType

Method	Syntax Integer
none—no authentication	**(0)**
radius—authentication through a RADIUS server	**(1)**
local—local authenticaiton	**(2)**
ldap—authentication through an LDAP server	**(3)**
securid—authentication through an RSA SecurID authentication server	**(4)**
jsrc—authentication through JSRC	**(5)**

jnxaccountingType—Defines the accounting method to authenticate a user.

Table 162: jnxAccountingType

Method	Syntax Integer
none—no accounting	**(0)**
radius—accounting through a RADIUS server	**(1)**
local—local accounting	**(2)**
ldap—accounting through an LDAP server	**(3)**

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Table 162: jnxAccountingType *(continued)*

Method	Syntax Integer
securid—accounting through an RSA SecurID authentication server	(4)
jsrc—accounting through JSRC	(5)

jnxAuthorizationType—Defines the method to authorize a user.

Table 163: jnxAuthorizationType

Method	Syntax Integer
none—no authentication	(0)
radius—authorization through a RADIUS server	(1)
local—local authorization	(2)
ldap—authorization through an LDAP server	(3)
securid—authorization through an RSA SecurID authentication server	(4)
jsrc—authorization through JSRC	(5)

jnxProvisioningType—Defines the method to provision a user.

Table 164: jnxProvisioningType

Method	Syntax Integer
none—no provisioning	(0)
radius—provisioning through a RADIUS server	(1)
local—local provisioning	(2)
ldap—provisioning through an LDAP server	(3)
securid—provisioning through an RSA SecurID authentication server	(4)
jsrc—provisioning through JSRC	(5)

Related Documentation

- AAA Objects MIB on page 441
- jnxUserAAAStatTable on page 444
- jnxUserAAAServerName on page 444

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- jnxUserAAAAccessPool on page 445
- Access Authentication-Related Traps on page 445

jnxUserAAAStatTable

jnxUserAAAStatTable, whose object ID is **{jnxUserAAAAccessAuthStats 1}**, exposes the user authentication statistics listed in Table 165 on page 444.

Table 165: jnxUserAAAStatTable

Object	Object ID	Description
jnxUserAAAStatEntry	**jnxUserAAAStatTable 1**	Statistics entries collected for authentication. Sequence of parameters: • **jnxUserAAAStatAuthType** • **jnxUserAAAStatRequestReceived** • **jnxUserAAAStatAccessAccepted** • **jnxUserAAAStatAccessRejected**
jnxUserAAAStatAuthType	**jnxUserAAAStatEntry 1**	Indicates the authentication type. This entry uniquely identifies the statistics counters related to its authentication.
jnxUserAAAStatRequestReceived	**jnxUserAAAStatEntry 2**	The number of the request received.
jnxUserAAAStatAccessAccepted	**jnxUserAAAStatEntry 3**	The number of the access granted. This entry is an aggregated statistic for this type of authentication.
jnxUserAAAStatAccessRejected	**jnxUserAAAStatEntry 4**	This number of the access request rejected. This entry is an aggregated statistic for this type of authentication.

Related Documentation

- AAA Objects MIB on page 441
- AAA Objects MIB Textual Conventions on page 442
- jnxUserAAAServerName on page 444
- jnxUserAAAAccessPool on page 445
- Access Authentication-Related Traps on page 445

jnxUserAAAServerName

jnxUserAAAServerName, whose object ID is **{jnxUserAAAAccessAuthStats 1}**, specifies the server name that identifies the authentication server.

Related Documentation

- AAA Objects MIB on page 441
- AAA Objects MIB Textual Conventions on page 442
- jnxUserAAAStatTable on page 444

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- jnxUserAAAAccessPool on page 445
- Access Authentication-Related Traps on page 445

Access Authentication-Related Traps

Table 166 on page 445 identifies access authentication traps.

Table 166: Access Authentication-Related Traps

Object	Object ID	Description
jnxAccessAuthServiceUp	jnxUserAAANotifications 1	Access authentication trap to signify that the specified service has started.
jnxAccessAuthServiceDown	jnxUserAAANotifications 2	Access authentication trap to signify that the specified service has been stopped.
jnxAccessAuthServerDisabled	jnxUserAAANotifications 3	Access authentication trap to signify that the external authentication server is not responding.
jnxAccessAuthServerEnabled	jnxUserAAANotifications 4	Access authentication trap to signify that the external authentication server started responding again.

Related Documentation

- AAA Objects MIB on page 441
- AAA Objects MIB Textual Conventions on page 442
- jnxUserAAAStatTable on page 444
- jnxUserAAAServerName on page 444
- jnxUserAAAAccessPool on page 445
- Configuring SNMP Trap Options and Groups on a Device Running Junos OS
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jnxUserAAAAccessPool

jnxUserAAAAccessPoolTable, whose object ID is **{ jnxUserAAAAccessPoolTable 1 }**, provides the ability to track address resource usage off-chassis. **jnxUserAAAAccessPoolTable** objects are listed in Table 167 on page 445.

Table 167: jnxUserAAAAccessPoolTable

Object	Object ID	Description
jnxUserAAAAccessPoolRoutingInstance	jnxUserAAAAccessPoolEntry 1	Routing instance of the address pool.
jnxUserAAAAccessPoolName	jnxUserAAAAccessPoolEntry 2	Address pool name.

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Table 167: jnxUserAAAAccessPoolTable *(continued)*

Object	Object ID	Description
jnxUserAAAAccessPoolLinkName	jnxUserAAAAccessPoolEntry 3	Address pool link name.
jnxUserAAAAccessPoolFamilyType	jnxUserAAAAccessPoolEntry 4	Family type of this pool.
jnxUserAAAAccessPoolInetNetwork	jnxUserAAAAccessPoolEntry 5	Match criteria for this pool. Network or Prefix.
jnxUserAAAAccessPoolInetPrefixLength	jnxUserAAAAccessPoolEntry 6	Prefix length for an IPv6 pool.
jnxUserAAAAccessPoolOutOfMemory	jnxUserAAAAccessPoolEntry 7	Number of times this pool has flagged an Out of Memory condition.
jnxUserAAAAccessPoolOutOfAddresses	jnxUserAAAAccessPoolEntry 8	Number of times this pool has flagged an Out of Address condition.
jnxUserAAAAccessPoolAddressTotal	jnxUserAAAAccessPoolEntry 9	Total number of addresses or prefixes in this pool.
jnxUserAAAAccessPoolAddressesInUse	jnxUserAAAAccessPoolEntry 10	Total number of addresses or prefixes given out from this pool.
jnxUserAAAAccessPoolAddressUsage	jnxUserAAAAccessPoolEntry 11	Percentage of addresses used in this pool or linked pool. If this pool is the head of a linked chain of pools, this number reflects the usage for the entire chain. If this pool is part of a linked chain of pools, but not the head of the chain, the value is not used.
jnxUserAAAAccessPoolAddressUsageHigh	jnxUserAAAAccessPoolEntry 12	Configured high percentage threshold of addresses used in this pool or linked pool. An SNMP trap is generated when this threshold is exceeded. This trap is generated only for unlinked pools or pools that are the head of a linked chain of pools. If this pool is part of a linked chain of pools, but not the head of the chain, then no traps are generated.
jnxUserAAAAccessPoolAddressUsageAbate	jnxUserAAAAccessPoolEntry 13	Configured abate percentage threshold of addresses used in this pool or linked pool. An SNMP trap clear is generated when address use falls below this threshold percentage. This trap is generated only for unlinked pools or pools that are the head of a linked chain of pools. If this pool is part of a linked chain of pools, but not the head of the chain, then no traps are generated.

Related Documentation

- AAA Objects MIB on page 441

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- AAA Objects MIB Textual Conventions on page 442
- jnxUserAAAServerName on page 444
- Access Authentication-Related Traps on page 445

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CHAPTER 51

Interpreting the Enterprise-Specific Access Authentication Objects MIB

- Access Authentication Objects MIB on page 449
- jnxJsFwAuthStats on page 449
- jnxJsAuthTrapVars on page 450
- jnxJsAuthNotifications on page 451

Access Authentication Objects MIB

The Juniper Networks enterprise-specific Access Authentication Objects MIB, whose object ID is **{jnxJsAuth 1}**, defines the objects that pertain to access authentication. Firewall and security features restrict the accessing of protected resources (ideally on different zones) behind a firewall based on their source IP and other credentials.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-auth.txt .

For information about the enterprise-specific Access Authentication Objects MIB, see the following topics:

- jnxJsFwAuthStats on page 449
- jnxJsAuthTrapVars on page 450
- jnxJsAuthNotifications on page 451

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxJsFwAuthStats

jnxJsFwAuthStats, whose object ID is **{jnxJsAuthObjects 1}**, exposes the firewall authentication statistics listed in Table 168 on page 450.

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Table 168: jnxJsFwAuthStats

Object	Object ID	Description
jnxJsFwAuthNumPendingUsers	jnxJsFwAuthStats 1	Number of users currently waiting to be authenticated by the firewall user authentication mechanism.
jnxJsFwAuthNumSuccUsers	jnxJsFwAuthStats 2	Number of users currently allowed access by the firewall user authentication mechanism.
jnxJsFwAuthNumFailedUsers	jnxJsFwAuthStats 3	Number of users who currently failed to be authenticated by the firewall user authentication mechanism.
jnxJsFwAuthTotalUsers	jnxJsFwAuthStats 4	Total number of users that are accessing or attempting to access resources managed by the firewall user authentication mechanism.

Related Documentation

- Access Authentication Objects MIB on page 449
- jnxJsAuthTrapVars on page 450
- jnxJsAuthNotifications on page 451

jnxJsAuthTrapVars

jnxJsAuthTrapVars identifies access authentication traps variables listed in Table 169 on page 450.

Table 169: jnxJsAuthTrapVars

Object	Object ID	Description
jnxJsFwAuthUserName	jnxJsAuthTrapVars 1	Name of the user who is attempting to be authenticated or has been authenticated.
jnxJsFwAuthServiceDesc	jnxJsAuthTrapVars 2	Service or application name that the authentication is performed for: Telnet, FTP, HTTP.
jnxJsFwAuthReason	jnxJsAuthTrapVars 3	Reason for the trap being generated—authentication failure due to: timeout, invalid password, invalid username, and so on.
jnxJsFwAuthClientIpAddr	jnxJsAuthTrapVars 4	Authentication client's IP address.

Related Documentation

- Access Authentication Objects MIB on page 449
- jnxJsFwAuthStats on page 449
- jnxJsAuthNotifications on page 451

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jnxJsAuthNotifications

jnxJsAuthNotifications, whose object ID is **{jnxJsAuthMIB 0}**, identifies the user access authentication notifications listed in Table 170 on page 451 .

Table 170: jnxJsAuthNotifications

Object	Object ID	Description
jnxJsFwAuthFailure	**jnxJsAuthNotifications** 1	A firewall user authentication status trap to signify whether a user using the pass-through firewall authentication mechanism has been rejected due to the reason specified in the trap. • **jnxJsFwAuthUserName** is the user. • **jnxClientIPAddress** is the IP address the user came from. • **jnxJsFwAuthServiceDesc** specifies the application by which the authentication was performed. • **jnxJsFwAuthReason** indicates the reason for failure.
jnxJsFwAuthServiceUp	**jnxJsAuthNotifications** 2	Firewall user authentication service has started.
jnxJsFwAuthServiceDown	**jnxJsAuthNotifications** 3	Firewall user authentication service has stopped.
jnxJsFwAuthCapacityExceeded	**jnxJsAuthNotifications** 4	Firewall user authentication maximum capacity has been exceeded. **jnxJsFwAuthTotalUsers** indicates the total number of users being authenticated, and it has exceeded the maximum allowable users.

Related Documentation

- Access Authentication Objects MIB on page 449
- jnxJsFwAuthStats on page 449
- jnxJsAuthTrapVars on page 450

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CHAPTER 52

Interpreting the Enterprise-Specific DNS Objects MIB

- DNS Objects MIB on page 453
- jnxJsDnsProxyDataObjects on page 453

DNS Objects MIB

The Juniper Networks enterprise-specific DNS Objects MIB, **jnxJsDns**,whose object ID is **{jnxJsDnsRoot 1}**, provides collated statistics for the Domain Name System (DNS) proxy collected over all interfaces on which it is configured to serve.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-dns.txt .

For information about the enterprise-specific DNS Objects MIB, see the following topic:

- jnxJsDnsProxyDataObjects on page 453

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxJsDnsProxyDataObjects

jnxJsDnsProxyDataObjects, whose object ID is **{jnxJsDns 1}**, displays the DNS query-related statistics listed in Table 171 on page 453.

Table 171: jnxJsDnsProxyDataObjects

Object	Object ID	Description
jnxJsDNSProxyQueriesReceived	**jnxJsDnsProxyDataObjects 1**	Total number of DNS queries received by the DNS proxy.
jnxJsDnsProxyResponsesSent	**jnxJsDnsProxyDataObjects 2**	Number of DNS queries answered sent by the DNS proxy. This includes DNS cache hits and misses that were answered.

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Table 171: jnxJsDnsProxyDataObjects *(continued)*

Object	Object ID	Description
jnxJsDnsProxyQueriesForwarded	**jnxJsDnsProxyDataObjects 3**	Number of DNS queries forwarded to other DNS servers. This is the number of queries that have been proxied due to cache misses.
jnxJsDnsProxyNegativeResponses	**jnxJsDnsProxyDataObjects 4**	Number of negative DNS query responses. This is the count of DNS queries for which the proxy could not obtain answers.
jnxJsDnsProxyRetryRequests	**jnxJsDnsProxyDataObjects 5**	Number of DNS retry queries that this proxy received.
jnxJsDnsProxyPendingRequests	**jnxJsDnsProxyDataObjects 6**	Number of DNS requests yet to be answered.
jnxJsDnsProxyServerFailures	**jnxJsDnsProxyDataObjects 7**	Number of DNS proxy failures.

Related Documentation

- DNS Objects MIB on page 453

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CHAPTER 53

Interpreting the Enterprise-Specific IPsec Generic Flow Monitoring Object MIB

- IPsec Generic Flow Monitoring Object MIB on page 455
- Branch Tree Objects on page 456
- IPSec Generic Flow Monitoring Object MIB Textual Conventions on page 456
- Number of IKE Tunnels Currently Active on page 460
- IPsec Phase 1 IKE Tunnel Table on page 460
- IPsec Phase 2 IKE Tunnel Table on page 464
- IPsec Phase 2 Security Association Table on page 467

IPsec Generic Flow Monitoring Object MIB

The Juniper Networks enterprise-specific IPsec Generic Flow Monitoring Object MIB, whose object ID is **{jnxIpSecMibRoot 1}**, defines the objects used to monitor the entries pertaining to IPsec objects and the management of the IPsec VPN functionalities. This generic MIB models the standard, dynamic aspects of IPsec, including the counters and objects that are of management interest in a standard IPsec implementation.

This MIB module is based on the **jnxIpSecMonitorMib**. Building on the existing Internet Key Exchange (IKE) infrastructure, the security IKE implementation integrates the value-added features for the security products.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ipsec-flow-mon.txt .

For information about the enterprise-specific IPsec Generic Flow Monitoring Object MIB, see the following topics:

- Branch Tree Objects on page 456
- IPSec Generic Flow Monitoring Object MIB Textual Conventions on page 456
- Number of IKE Tunnels Currently Active on page 460
- IPsec Phase 1 IKE Tunnel Table on page 460

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- IPsec Phase 2 IKE Tunnel Table on page 464
- IPsec Phase 2 Security Association Table on page 467

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

Branch Tree Objects

The following branch tree objects are associated with the IPsec Generic Flow Monitoring Object MIB:

- **jnxIpSecFlowMonNotifications {jnxIpSecFlowMonMIB 0}**
- **jnxIpSecFlowMonPhaseOne {jnxIpSecFlowMonMIB 1}**
- **jnxIpSecFlowMonPhaseTwo {jnxIpSecFlowMonMIB 2}**

Related Documentation

- IPsec Generic Flow Monitoring Object MIB on page 455
- IPSec Generic Flow Monitoring Object MIB Textual Conventions on page 456
- Number of IKE Tunnels Currently Active on page 460
- IPsec Phase 1 IKE Tunnel Table on page 460
- IPsec Phase 2 IKE Tunnel Table on page 464
- IPsec Phase 2 Security Association Table on page 467

IPSec Generic Flow Monitoring Object MIB Textual Conventions

- **JnxIkePeerType**—The type of IPsec Phase 1 IKE peer identity. This identity is the local IKE identity to send in the exchange. The IKE peer can be identified by one of the ID types defined in IPsec DOI (see Table 172 on page 456).

Table 172: IKE Identity Type Textual Conventions

Type	Description	Syntax Integer
Unknown	Unknown type	**(0)**
idIpv4Addr	IPv4 address	**(1)**
idFqdn	Fully qualified domain name	**(2)**
idDn	Distinguished name	**(3)**
idUfqdn	User fully qualified domain name	**(4)**

- **JnxIkeNegoMode**—The IPsec Phase 1 IKE negotiation mode (see Table 173 on page 457).

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Table 173: IKE Negotiation Mode Text Conventions

Type	Description	Syntax Integer
Main mode	A six-message Phase 1 exchange that provides identity protection	(1)
Aggressive mode	A three-message Phase 1 exchange that does not provide identity protection	(2)

- **JnxIkeHashAlgo**—The hash algorithm used in IPsec Phase 1 IKE negotiations (see Table 174 on page 457).

Table 174: IKE Negotiations Hash Alogorithms

Hash Alogorithm	Syntax Integer
md5	(1)
sha	(2)

- **JnxIkeAuthMethod**—The authentication method used in IPsec Phase 1 IKE negotiations (see Table 175 on page 457).

Table 175: IKE Authentication Method

Method	Syntax Integer
preSharedKey	(1)
dssSignature	(2)
rsaSignature	(3)
rsaEncryption	(4)
revRsaEncryption	(5)
xauthPreSharedKey	(6)
xauthDssSignature	(7)
xauthRsaSignature	(8)
xauthRsaEncryption	(9)
xauthRevRsaEncryption	(10)

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- **JnxIkePeerRole**—The role of the local endpoint in negotiating the IPsec Phase 1 IKE security association (SA). It can be either initiator or responder (see Table 176 on page 458).

Table 176: Role of Local Endpoint in Negotiations

Role	Syntax Integer
Initiator	(1)
Responder	(2)

- **JnxIkeTunStateType**—State of the Phase 1 IKE negotiation (see Table 177 on page 458).

Table 177: State of Phase 1 IKE Negotiation

State	Syntax Integer
Up	(1)
Down	(2)

- **JnxDiffHellmanGrp**—The Diffie-Hellman Group used in negotiations (see Table 178 on page 458).

Table 178: Diffie-Hellman Group in Negotiations

Diffie Hellman Group	Description	Syntax Integer
Unknown	Unknown	(0)
modp768	768-bit MODP	(1)
modp1024	1024-bit MODP	(2)
modp1536	modp1536	(3)

- **JnxKeyType**—The type of key used by an IPsec Phase 2 Tunnel (see Table 179 on page 458).

Table 179: Key Used by IPsec Phase 2 Tunnel

Key	Syntax Integer
Unknown	(0)
keyIke	(1)
keyManual	(2)

- **JnxEncryptAlgo** —The encryption algorithm used in negotiations (see Table 180 on page 459).

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Table 180: Encryption Algorithm in Negotiations

Algorithm	Syntax Integer
espDes	**(1)**
esp3des	**(2)**
espNull	**(3)**
espAes128	**(4)**
espAes192	**(5)**
espAes256	**(6)**

- **JnxAuthAlgo**—The authentication algorithm used by an SA of an IPsec Phase 2 Tunnel (see Table 181 on page 459).

Table 181: Role of Local Endpoint in Negotiations

Algorithm	Syntax Integer
Unknown	**(0)**
hmacMd5	**(1)**
hmacSha	**(2)**

- **JnxRemotePeerType**—The type of the remote peer gateway (endpoint) (see Table 182 on page 459).

Table 182: Type of Remote Peer Gateway

Gateway Type	Syntax Integer and Description
Unknown	**(0)**
Static	**(1)** Static (remote peer whose IP address is known beforehand)
Dynamic	**(2)** Dynamic (remote peer whose IP address is not known beforehand)

- **JnxSpiType**—The type of the SPI associated with IPsec Phase 2 SAs. An unsigned 32-bit integer (256. . . 4294967295).
- **JnxSAType**—The SA type (see Table 183 on page 460).

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Table 183: Role of Local Endpoint in Negotiations

SA Type	Syntax Integer
Unknown	(0)
Manual	(1)
Dynamic	(2)

Related Documentation

- IPsec Generic Flow Monitoring Object MIB on page 455
- Branch Tree Objects on page 456
- Number of IKE Tunnels Currently Active on page 460
- IPsec Phase 1 IKE Tunnel Table on page 460
- IPsec Phase 2 IKE Tunnel Table on page 464
- IPsec Phase 2 Security Association Table on page 467

Number of IKE Tunnels Currently Active

NOTE: The **jnxIkeNumOfTunnels** object is not supported in this release.

Table 184 on page 460 reports the number of IKE Tunnels currently active.

Table 184: Number of IKE Tunnels Currently Active

Object	Object ID	Description
jnxIkeNumOfTunnels	**jnxIpSecFlowMonPhaseOne 1**	Number of IKE Tunnels (Phase 1) actively negotiating between peers. The SA can be in either the up or down state. This attribute details the number of IKE tunnels in **jnxIkeTunnelMonTable**.

Related Documentation

- IPsec Generic Flow Monitoring Object MIB on page 455
- Branch Tree Objects on page 456
- IPSec Generic Flow Monitoring Object MIB Textual Conventions on page 456
- IPsec Phase 1 IKE Tunnel Table on page 460
- IPsec Phase 2 IKE Tunnel Table on page 464
- IPsec Phase 2 Security Association Table on page 467

IPsec Phase 1 IKE Tunnel Table

Table 185 on page 461 identifies objects listed in the IPsec Phase 1 IKE Tunnel Table.

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Phase 1 is used to negotiate the parameter and key material required to establish an ISAKMP SA.

Phase 1 SA components include an encryption algorithm, authentication, Diffie-Hellman group values, and anthentication methods, such as preshared keys or certificates.

Table 185: IPsec Phase 1 IKE Tunnel Table

Object	Object ID	Description
jnxIkeTunnelMonTable	**jnxIpSecFlowMonPhaseOne 2**	The IPsec Phase 1 IKE Tunnel Table. There is one entry in this table for each active IPsec Phase 1 IKE tunnel.
jnxIkeTunnelMonEntry	**jnxIkeTunnelMonTable 1**	Attributes associated with an active IPsec Phase 1 IKE tunnel. Sequence of attributes: • **jnxIkeTunMonRemoteGwAddrType** • **jnxIkeTunMonRemoteGwAddr** • **jnxIkeTunMonIndex** • **jnxIkeTunMonLocalGwAddrType** • **jnxIkeTunMonLocalGwAddr** • **jnxIkeTunMonState** • **jnxIkeTunMonInitiatorCookie** • **jnxIkeTunMonResponderCookie** • **jnxIkeTunMonLocalRole** • **jnxIkeTunMonLocalIdType** • **jnxIkeTunMonLocalIdValue** • **jnxIkeTunMonLocalCertName** • **jnxIkeTunMonRemoteIdType** • **jnxIkeTunMonRemoteIdValue** • **jnxIkeTunMonNegoMode** • **jnxIkeTunMonDiffHellmanGrp** (not supported in this release) • **jnxIkeTunMonEncryptAlgo** • **jnxIkeTunMonHashAlgo** • **jnxIkeTunMonAuthMethod** • **jnxIkeTunMonLifeTime** • **jnxIkeTunMonActiveTime** • **jnxIkeTunMonInOctets** • **jnxIkeTunMonInPkts** • **jnxIkeTunMonOutOctets** • **jnxIkeTunMonOutPkts** • **jnxIkeTunMonXAuthUserId** • **jnxIkeTunMonDPDDownCount**
jnxIkeTunMonRemoteGwAddrType	**jnxIkeTunnelMonEntry 1**	IP address type of remote gateway (endpoint) for the IPsec Phase 1 IKE tunnel.

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Table 185: IPsec Phase 1 IKE Tunnel Table *(continued)*

Object	Object ID	Description
jnxJsFwAuthClientIpAddr	**jnxJsAuthTrapVars 4**	IP address of remote gateway (endpoint) for the IPsec Phase 1 IKE tunnel.
jnxIkeTunMonIndex	**jnxIkeTunnelMonEntry 3**	Index number of IPsec Phase 1 IKE Tunnel Table. The index number begins at 1 and is incremented with each tunnel that is created. The value of this object will wrap at 2,147,483,647.
jnxIkeTunMonLocalGwAddr	**jnxIkeTunnelMonEntry 4**	IP address of local endpoint (gateway) for the IPsec Phase 1 IKE tunnel.
jnxIkeTunMonLocalGwAddrType	**jnxIkeTunnelMonEntry 5**	IP address type of local endpoint (gateway) for the IPsec Phase 1 IKE tunnel.
jnxIkeTunMonState	**jnxIkeTunnelMonEntry 6**	State of IKE tunnel. It can be: • 1—up, negotiation completed. • 2—down, being negotiated.
jnxIkeTunMonInitiatorCookie	**jnxIkeTunnelMonEntry 7**	Cookie as generated by peer that initiated the IKE Phase 1 negotiation. This cookie is carried in the ISAKMP header.
jnxIkeTunMonResponderCookie	**jnxIkeTunnelMonEntry 8**	Cookie as generated by peer responding to the IKE Phase 1 negotiation initiated by the remote peer. This cookie is carried in the ISAKMP header.
jnxIkeTunMonLocalRole	**jnxIkeTunnelMonEntry 9**	Role of local peer identity. The role of the local peer can be: • Initiator • Responder
jnxIkeTunMonLocalIdType	**jnxIkeTunnelMonEntry 10**	Type of local peer identity. The local peer can be identified by: • IP address • Fully qualified domain name string • Distinguished name string
jnxIkeTunMonLocalIdValue	**jnxIkeTunnelMonEntry 11**	Value of local peer identity. If the local peer type is an IP address, then this is the IP address used to identify the local peer. If the local peer type is a fully qualified domain name string, then this is the fully qualified domain name string of the local peer. If the local peer type is a distinguished name string, then this is the distinguished name string of the local peer.

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Table 185: IPsec Phase 1 IKE Tunnel Table *(continued)*

Object	Object ID	Description
jnxIkeTunMonLocalCertName	**jnxIkeTunnelMonEntry 12**	Name of certificate used for authentication of the local tunnel endpoint. This object has some valid value only if the negotiated IKE authentication method is other than preshared key. If the IKE negotiation does not use a certificate-based authentication method, then the value of this object is a NULL string.
jnxIkeTunMonRemoteIdType	**jnxIkeTunnelMonEntry 13**	Type of remote peer identity. The remote peer can be identified by: • IP address • Fully qualified domain name string • Distinguished name string
jnxIkeTunMonRemoteIdValue	**jnxIkeTunnelMonEntry 14**	Value of remote peer identity. If the remote peer type is an IP address, then this is the IP address used to identify the remote peer. If the remote peer type is a fully qualified domain name string, then this is the fully qualified domain name string of the remote peer. If the remote peer type is a distinguished name string, then this is the distinguished name string of the remote peer.
jnxIkeTunMonNegoMode	**jnxIkeTunnelMonEntry 15**	Negotiation mode of IPsec Phase 1 IKE tunnel.
NOTE: The **jnxIkeTunMonDiffHellmanGrp** object is not supported in this release.		
jnxIkeTunMonDiffHellmanGrp	**jnxIkeTunnelMonEntry 16**	Diffie-Hellman Group used in IPsec Phase 1 IKE negotiations.
jnxIkeTunMonEncryptAlgo	**jnxIkeTunnelMonEntry 17**	Encryption algorithm used in IPsec Phase 1 IKE negotiations.
jnxIkeTunMonHashAlgo	**jnxIkeTunnelMonEntry 18**	Hash algorithm used in IPsec Phase 1 IKE negotiations.
jnxIkeTunMonAuthMethod	**jnxIkeTunnelMonEntry 19**	Authentication method used in IPsec Phase 1 IKE negotiations.
jnxIkeTunMonLifeTime	**jnxIkeTunnelMonEntry 20**	Negotiated lifetime of IPsec Phase 1 IKE tunnel in seconds.
jnxIkeTunMonActiveTime	**jnxIkeTunnelMonEntry 21**	Length of time IPsec Phase 1 IKE tunnel has been active in hundredths of seconds.
jnxIkeTunMonInOctets	**jnxIkeTunnelMonEntry 22**	Total number of octets received by this IPsec Phase 1 IKE SA.
jnxIkeTunMonInPkts	**jnxIkeTunnelMonEntry 23**	Total number of packets received by this IPsec Phase 1 IKE SA.

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Table 185: IPsec Phase 1 IKE Tunnel Table *(continued)*

Object	Object ID	Description
jnxIkeTunMonOutOctets	**jnxIkeTunnelMonEntry 24**	Total number of octets sent by this IPsec Phase 1 IKE SA.
jnxIkeTunMonOutPkts	**jnxIkeTunnelMonEntry 25**	Total number of packets sent by this IPsec Phase 1 IKE SA.
jnxIkeTunMonXAuthUserId	**jnxIkeTunnelMonEntry 26**	Extended Authentication (XAuth) User Identifier. Identifies the user associated with this IPsec Phase 1 negotiation.
jnxIkeTunMonDPDDownCount	**jnxIkeTunnelMonEntry 27**	Number of times that the remote peer is detected in a dead (or down) state.

Related Documentation

- IPsec Generic Flow Monitoring Object MIB on page 455
- Branch Tree Objects on page 456
- IPSec Generic Flow Monitoring Object MIB Textual Conventions on page 456
- Number of IKE Tunnels Currently Active on page 460
- IPsec Phase 2 IKE Tunnel Table on page 464
- IPsec Phase 2 Security Association Table on page 467

IPsec Phase 2 IKE Tunnel Table

Table 186 on page 464 identifies objects listed in the IPsec Phase 2 IKE Tunnel Table.

During this phase, IKE negotiates IPsec SA parameters and setup, matching IPsec SA in the peers.

Phase 2 VPN includes tunnel peer connection, associated with a specific policy or a tunnel interface. Phase 2 SA components include encryption and authentication algorithms, proxy-IDs, and optional DH group values.

Table 186: IPsec Phase 2 IKE Tunnel Table

Object	Object ID	Description
NOTE: The **jnxIpSecNumOfTunnels** object is not supported in this release.		
jnxIpSecNumOfTunnels	**jnxIpSecFlowMonPhaseTwo 1**	Number of IPsec VPN tunnels. This attribute should report the number of IPsec VPN tunnels in **jnxIpSecTunnelTable**.
jnxIpSecTunnelMonTable	**jnxIpSecFlowMonPhaseTwo 2**	The IPsec Phase 2 Tunnel Table. There is one entry in this table for each active IPsec Phase 2 tunnel. If the tunnel is terminated, then the entry is no longer available after the table has been refreshed.

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Table 186: IPsec Phase 2 IKE Tunnel Table *(continued)*

Object	Object ID	Description
jnxIpSecTunnelMonEntry	**jnxIpSecTunnelMonTable 1**	Each entry contains the attributes associated with an active IPsec Phase 2 tunnel. Sequence of attributes: • **jnxIpSecTunMonRemoteGwAddrType** • **jnxIpSecTunMonRemoteGwAddr** • **jnxIpSecTunMonIndex** • **jnxIpSecTunMonLocalGwAddrType** • **jnxIpSecTunMonLocalGwAddr** • **jnxIpSecTunMonLocalProxyId** • **jnxIpSecTunMonRemoteProxyId** • **jnxIpSecTunMonKeyType** • **jnxIpSecTunMonRemotePeerType** • **jnxIpSecTunMonOutEncryptedBytes** • **jnxIpSecTunMonOutEncryptedPkts** • **jnxIpSecTunMonInDecryptedBytes** • **jnxIpSecTunMonInDecryptedPkts** • **jnxIpSecTunMonAHInBytes** • **jnxIpSecTunMonAHInPkts** • **jnxIpSecTunMonAHOutBytes** • **jnxIpSecTunMonAHOutPkts** • **jnxIpSecTunMonReplayDropPkts** • **jnxIpSecTunMonAhAuthFails** • **jnxIpSecTunMonDecryptFails** • **jnxIpSecTunMonBadHeaders** • **jnxIpSecTunMonBadTrailers** • **jnxIkeTunMonOutOctets** • **jnxIpSecTunMonDroppedPkts** (not supported in this release)
jnxIpSecTunMonRemoteGwAddrType	**jnxIpSecTunnelMonEntry 1**	IP address type of remote gateway (endpoint) for the IPsec Phase 2 tunnel.
jnxIpSecTunMonRemoteGwAddr	**jnxIpSecTunnelMonEntry 2**	IP address of remote gateway (endpoint) for the IPsec Phase 2 tunnel.
jnxIpSecTunMonIndex	**jnxIpSecTunnelMonEntry 3**	Index number of IPsec Phase 2 Tunnel Table. The index number begins at 1 and is incremented with each tunnel that is created. The value of this object will wrap at 2,147,483,647.
jnxIpSecTunMonLocalGwAddrType	**jnxIpSecTunnelMonEntry 4**	IP address type of local gateway (endpoint) for the IPsec Phase 2 tunnel.
jnxIpSecTunMonLocalGwAddr	**jnxIpSecTunnelMonEntry 5**	IP address of local gateway (endpoint) for the IPsec Phase 2 tunnel.

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Table 186: IPsec Phase 2 IKE Tunnel Table *(continued)*

Object	Object ID	Description
jnxIpSecTunMonLocalProxyId	jnxIpSecTunnelMonEntry 6	Identifier for local end.
jnxIpSecTunMonRemoteProxyId	jnxIpSecTunnelMonEntry 7	Identifier for remote end.
jnxIpSecTunMonKeyType	jnxIpSecTunnelMonEntry 8	Type of key used by IPsec Phase 2 tunnel. It can be one of the following two types: • IKE-negotiated • Manually installed
jnxIpSecTunMonRemotePeerType	jnxIpSecTunnelMonEntry 9	Type of the remote peer gateway (endpoint). It can be one of the following two types: • Static (remote peer whose IP address is known beforehand) • Dynamic (remote peer whose IP address is not known beforehand)
jnxIpSecTunMonOutEncryptedBytes	jnxIpSecTunnelMonEntry 10	Number of bytes encrypted by this Phase 2 tunnel.
jnxIpSecTunMonOutEncryptedPkts	jnxIpSecTunnelMonEntry 11	Number of packets encrypted by this Phase 2 tunnel.
jnxIpSecTunMonInDecryptedBytes	jnxIpSecTunnelMonEntry 12	Number of bytes decrypted by this Phase 2 tunnel.
jnxIpSecTunMonInDecryptedPkts	jnxIpSecTunnelMonEntry 13	Number of packets decrypted by this Phase 2 tunnel.
jnxIpSecTunMonAHInBytes	jnxIpSecTunnelMonEntry 14	Number of incoming bytes authenticated using AH by this Phase 2 tunnel.
jnxIpSecTunMonAHInPkts	jnxIpSecTunnelMonEntry 15	Number of incoming packets authenticated using AH by this Phase 2 tunnel.
jnxIpSecTunMonAHOutBytes	jnxIpSecTunnelMonEntry 16	Number of outgoing bytes applied AH by this Phase 2 tunnel.
jnxIpSecTunMonAHOutPkts	jnxIpSecTunnelMonEntry 17	Number of outgoing packets applied AH by this Phase 2 tunnel.
jnxIpSecTunMonReplayDropPkts	jnxIpSecTunnelMonEntry 18	Number of packets dropped by this Phase 2 tunnel due to antireplay check failure.
jnxIpSecTunMonAhAuthFails	jnxIpSecTunnelMonEntry 19	Number of packets received by this Phase 2 tunnel that failed AH authentication.
jnxIpSecTunMonEspAuthFails	jnxIpSecTunnelMonEntry 20	Number of packets received by this Phase 2 tunnel that failed ESP authentication.
jnxIpSecTunMonDecryptFails	jnxIpSecTunnelMonEntry 21	Number of packets received by this Phase 2 tunnel that failed decryption.

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Table 186: IPsec Phase 2 IKE Tunnel Table *(continued)*

Object	Object ID	Description
jnxIpSecTunMonBadHeaders	**jnxIpSecTunnelMonEntry 22**	Number of packets received by this Phase 2 tunnel that failed due to bad headers.
jnxIpSecTunMonBadTrailers	**jnxIpSecTunnelMonEntry 23**	Number of packets received by this Phase 2 tunnel that failed due to bad ESP trailers.
NOTE: The **jnxIpSecTunMonDroppedPkts** object is not supported in this release.		
jnxIpSecTunMonDroppedPkts	**jnxIpSecTunnelMonEntry 26**	Total number of dropped packets for this Phase 2 tunnel.

Related Documentation

- IPsec Generic Flow Monitoring Object MIB on page 455
- Branch Tree Objects on page 456
- IPSec Generic Flow Monitoring Object MIB Textual Conventions on page 456
- Number of IKE Tunnels Currently Active on page 460
- IPsec Phase 1 IKE Tunnel Table on page 460
- IPsec Phase 2 Security Association Table on page 467

IPsec Phase 2 Security Association Table

jnxIpSecSaMonTable, whose object ID is **{jnxIpSecFlowMonPhaseTwo 3}**, identifies the objects listed in Table 187 on page 468. The IPsec Phase 2 Security Association table identifies the structure (in terms of component SAs) of each active Phase 2 IPsec tunnel. This table contains an entry for each active and expiring SA and maps each entry in the active Phase 2 tunnel table (**ipSecTunTable**) into a number of entries in this table.

The SA contains the information negotiated by IKE. The SA is like a contract laying out the rules of the VPN connection for the duration of the SA. An SA is assigned a 32-bit number that, when used in conjunction with the destination IP address, uniquely identifies the SA. This number is called the Security Parameters Index (SPI).

IPsec SAs are unidirectional and are unique in each security protocol. A set of SAs is needed for a protected data pipe, one per direction per protocol.

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Table 187: IPsec Phase 2 Security Association Table

Object	Object ID	Description
jnxIpSecSaMonEntry	**jnxIpSecSaMonTable 1**	Each entry contains the attributes associated with active and expiring IPsec Phase 2 SAs. Sequence of parameters: • **jnxIpSecSaMonIndex** • **jnxIpSecSaMonProtocol** • **jnxIpSecSaMonInSpi** • **jnxIpSecSaMonOutSpi** • **jnxIpSecSaMonType** • **jnxIpSecSaMonEncapMode** • **jnxIpSecSaMonLifeSize** • **jnxIpSecSaMonLifeTime** • **jnxIpSecSaMonActiveTime** • **jnxIpSecSaMonLifeSizeThreshold** (not supported in this release) • **jnxIpSecSaMonLifeTimeThreshold** • **jnxIpSecSaMonEncryptAlgo** • **jnxIpSecSaMonAuthAlgo** • **jnxIpSecSaMonState**
jnxIpSecSaMonIndex	**jnxIpSecSaMonEntry 1**	Index number, in the context of the IPsec tunnel **ipSecTunIndex**, of the SA represented by this table entry. The index number begins at 1 and is incremented with each SPI associated with an IPsec Phase 2 tunnel. The value of this object will wrap at 65535.
jnxIpSecSaMonProtocol	**jnxIpSecSaMonEntry 2**	Index number that represents the security protocol (AH, ESP or IPComp) for which this SA was set up.
jnxIpSecSaMonInSpi	**jnxIpSecSaMonEntry 3**	Value of the incoming SPI.
jnxIpSecSaMonOutSpi	**jnxIpSecSaMonEntry 4**	Value of the outgoing SPI.
jnxIpSecSaMonType	**jnxIpSecSaMonEntry 5**	Types of SAs that can be either manual or dynamic.
jnxIpSecSaMonEncapMode	**jnxIpSecSaMonEntry 6**	Encapsulation mode used by an IPsec Phase 2 tunnel.
jnxIpSecSaMonLifeSize	**jnxIpSecSaMonEntry 7**	Negotiated lifesize of the IPsec Phase 2 tunnel in kilobytes.
jnxIpSecSaMonLifeTime	**jnxIpSecSaMonEntry 8**	Negotiated lifetime of the IPsec Phase 2 tunnel in seconds.
jnxIpSecSaMonActiveTime	**jnxIpSecSaMonEntry 9**	Length of time the IPsec Phase 2 tunnel has been active in hundredths of seconds.
NOTE: The **jnxIpSecSaMonLifeSizeThreshold** object is not supported in this release.		
jnxIpSecSaMonLifeSizeThreshold	**jnxIpSecSaMonEntry 10**	SA lifesize refresh threshold in kilobytes.

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Table 187: IPsec Phase 2 Security Association Table *(continued)*

Object	Object ID	Description
jnxIpSecSaMonLifeTimeThreshold	**jnxIpSecSaMonEntry 11**	SA lifetime refresh threshold in seconds.
jnxIpSecSaMonEncryptAlgo	**jnxIpSecSaMonEntry 12**	Encryption algorithm used to encrypt the packets that can be either **es-cbc** or **3des-cbc**.
jnxIpSecSaMonAuthAlgo	**jnxIpSecSaMonEntry 13**	Algorithm used for authentication of packets that can be **hmac-md5-96** or **hmac-sha1-96**.
jnxIpSecSaMonState	**jnxIpSecSaMonEntry 14**	This column represents the status of the SA represented by this table entry. If the status of the SA is **active**, the SA is ready for active use. The status **expiring** represents any of the various states that the SA transitions through before being purged.

Related Documentation

- IPsec Generic Flow Monitoring Object MIB on page 455
- Branch Tree Objects on page 456
- IPSec Generic Flow Monitoring Object MIB Textual Conventions on page 456
- Number of IKE Tunnels Currently Active on page 460
- IPsec Phase 1 IKE Tunnel Table on page 460
- IPsec Phase 2 IKE Tunnel Table on page 464

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CHAPTER 54

Interpreting the Enterprise-Specific IPsec VPN Objects MIB

- IPsec VPN Objects MIB on page 471
- IPSec VPN Objects MIB Textual Conventions on page 472
- jnxJsIpSecTunnelTable on page 472

IPsec VPN Objects MIB

The Juniper Networks enterprise-specific IPsec VPN Objects MIB, **jnxJsIpSecVpnMib**, whose object ID is **{jnxJsIPSecVpn 1}**, defines the object used to monitor the entries pertaining to IPsec objects and the management of the IPsec VPN functionalities for Juniper Networks security product lines. This MIB models IPsec attributes specific to the appropriate Juniper Networks implementation.

This MIB module extends the Juniper Networks common IPsec flow monitoring MIB. Building on the existing common infrastructure, the security implementation integrates the value-added features for the security products.

Related IPsec VPN Objects MIBs include:

- **jnxJsIpSecVpnNotifications** {**jnxJsIpSecVpnMib 0**}
- **jnxJsIpSecVpnPhaseOne** {**jnxJsIpSecVpnMib 1**}
- **jnxJsIpSecVpnPhaseTwo** {**jnxJsIpSecVpnMib 2**}

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-ipsec-vpn.txt .

For information about the enterprise-specific IPsec VPN Objects MIB, see the following topics:

- IPSec VPN Objects MIB Textual Conventions on page 472
- jnxJsIpSecTunnelTable on page 472

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30

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- Standard SNMP MIBs Supported by Junos OS on page 7

IPSec VPN Objects MIB Textual Conventions

Table 188 on page 472 explains the possible values for **jnxJsIpSecVpnType**, which is the type of remote peer gateway (endpoint).

Table 188: jnxJsIpSecVpnType

Type	Description	Syntax Integer
policyBased	Tunnels require a policy with action **tunnel** to trigger IPsec VPN. The device receives traffic and matches it with a policy that has action **tunnel**, then performs the encryption/decryption and authentication options negotiated for this VPN Phase 2 negotiation.	**(1)**
routeBased	Requires a tunnel interface to a route directing traffic to protected networks to exit the system using that tunnel interface. The tunnel interface is bound to a Phase 2 VPN configuration that specifies all the tunnel parameters.	**(2)**

Related Documentation

- IPsec VPN Objects MIB on page 471
- jnxJsIpSecTunnelTable on page 472

jnxJsIpSecTunnelTable

jnxJsIpSecTunnelTable, whose object ID is **{jnxJsIpSecVpnPhaseTwo 1}**, is the IPsec Phase 2 Tunnel Table, with objects listed in Table 189 on page 472. There is one entry for each active IPsec Phase 2 tunnel. If the tunnel is terminated, then the entry is no longer available after the table has been refreshed.

This table augments **jnxIpSecTunnelMonTable** in Table 186 on page 464.

Table 189: jnxJsIpSecTunnelTable

Object	Object ID	Description
jnxJsIpSecTunnelEntry	**jnxJsIpSecTunnelTable 1**	Each entry contains the attributes associated with an active IPsec Phase 2 tunnel. Sequence of parameters: • **jnxJsIpSecTunPolicyName** • **jnxJsIpSecVpnTunType** • **jnxJsIpSecTunCfgMonState** • **jnxJsIpSecTunState**

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Table 189: jnxJsIpSecTunnelTable *(continued)*

Object	Object ID	Description
jnxJsIpSecTunPolicyName	**jnxJsIpSecTunnelEntry 1**	Policy name assoicated with this tunnel if the IPsec VPN is policy-based. If the IPsec VPN is not policy-based, this attribute is not applicable.
jnxJsIpSecVpnTunType	**jnxJsIpSecTunnelEntry 2**	Attribute to indicate whether the IPsec VPN tunnel is policy-based or route-based.
jnxJsIpSecTunCfgMonState	**jnxJsIpSecTunnelEntry 3**	According to user configuration, whether to monitor the IPsec tunnel to be enabled or not: • **disable—(1)** • **enable—(2)**
jnxJsIpSecTunState	**jnxJsIpSecTunnelEntry 4**	Attribute to indicate whether the IPsec tunnel is up or down, determined by ICMP **ping** if **jnxJsIpSecTunCfgMonState** is enabled: • **up—(1)**: VPN monitor detects the tunnel is up. • **down—(2)**: VPN monitor detects the tunnel is down. • **vpnMonitoringDisabled—(3)**: User has disabled VPN tunnel monitoring.

Related Documentation

- IPsec VPN Objects MIB on page 471
- IPSec VPN Objects MIB Textual Conventions on page 472

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CHAPTER 55

Interpreting the Enterprise-Specific Network Address Translation Objects MIB

- NAT Objects MIB on page 475
- Source NAT Table on page 476
- New Source NAT Table on page 477
- jnxJsNatIfSrcPoolPortTable on page 479
- NAT Rule Hit Table on page 480
- NAT Pool Hit Table on page 480
- NAT Trap Definitions on page 481

NAT Objects MIB

The Juniper Networks enterprise-specific Network Address Translation (NAT) Objects MIB, **jnxJsNatMIB**, whose object ID is **{jnxJsNAT 1}**, defines the objects that are used to monitor NAT attributes.

Related NAT Objects MIBs include:

- **jnxJsNatNotifications {jnxJsNatMIB 0}**
- **jnxJsNatObjects {jnxJsNatMIB 1}**
- **jnxJsNatTrapVars {jnxJsNatMIB 2}**

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-nat.txt .

For information about the enterprise-specific NAT Objects MIB, see the following topics:

- Source NAT Table on page 476
- New Source NAT Table on page 477
- jnxJsNatIfSrcPoolPortTable on page 479
- NAT Rule Hit Table on page 480

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- NAT Pool Hit Table on page 480
- NAT Trap Definitions on page 481

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

Source NAT Table

Table 190 on page 476 identifies objects in the Source NAT Table.

Table 190: Source NAT Table

Object	Object ID	Description
jnxJsSrcNatNumOfEntries	**jnxJsNatObjects 1**	Total number of dynamic addresses being translated.
jnxJsSrcNatTable	**jnxJsNatObjects 2**	Table that exposes the source NAT attributes of the translated addresses. When performing source IP address translation, the device translates the original source IP address or port number to a different one. The resource, address source pools, provides the security device with a supply of addresses from which to draw when performing source NAT. The security device has the following types of source pools: • Source pool with Port Address Translation (PAT) • Source pool without PAT • Static source pool This table contains information about source IP address translation only.
jnxJsSrcNatEntry	**jnxJsSrcNatTable 1**	Source NAT address entries. This object is indexed by the address pool table and the address allocated. Sequence of parameters: • **jnxJsNatSrcIpPoolName** • **jnxJsNatSrcGlobalAddr** • **jnxJsNatSrcPortPoolType** • **jnxJsNatSrcNumOfPortInuse** • **jnxJsNatSrcNumOfSessions** • **jnxJsNatSrcAssocatedIf**
jnxJsNatSrcIpPoolName	**jnxJsSrcNatEntry 1**	Address pool from which the translated address is allocated.
jnxJsNatSrcGlobalAddr	**jnxJsSrcNatEntry 2**	Name of dynamic source IP address allocated from the address pool used in the NAT translation.

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Table 190: Source NAT Table *(continued)*

Object	Object ID	Description
jnxJsNatSrcPortPoolType	**jnxJsSrcNatEntry 3**	Source NAT can do address translation with or without PAT. The source port pool type indicates whether the address translation is done with PAT, without PAT, or as a static translation: • **withPAT**—The security device translates both source IP address and port number of the packets. • **withoutPAT**—The device performs source NAT for the IP address without performing PAT for the source port number. • **static**—One range of IP addresses is statically mapped one-to-one to a shifted range of IP addresses.
jnxJsNatSrcNumOfPortInuse	**jnxJsSrcNatEntry 4**	Number of ports in use for this NAT address entry. This attribute is applicable to only NAT translation with PAT.
jnxJsNatSrcNumOfSessions	**jnxJsSrcNatEntry 5**	Number of sessions in use for this NAT address entry. This attribute is applicable to only NAT translation without PAT.
jnxJsNatSrcAssocatedIf	**jnxJsSrcNatEntry 6**	Index of interfaces associated with this NAT address entry. For each interface, the value is a unique value, greater than zero.

Related Documentation

- NAT Objects MIB on page 475
- New Source NAT Table on page 477
- jnxJsNatIfSrcPoolPortTable on page 479
- NAT Rule Hit Table on page 480
- NAT Pool Hit Table on page 480
- NAT Trap Definitions on page 481

New Source NAT Table

The **jnxJsSrcNatStatsTable**, whose object ID is **{jnxJsNatObjects 4}**, represents the source NAT attributes of the translated addresses. When performing source IP address translation, the security device translates the original source IP address, the port number, or both to a different address. The resource address source pool provides the security device with a supply of addresses from which to draw when performing source NAT. The new source NAT contains objects on source IP address translation only. Table 191 on page 478 identifies objects in the New Source NAT Table.

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Table 191: New Source NAT Table

Object	Object ID	Description
jnxJsSrcNatStatsEntry	jnxJsSrcNatStatsTable 1	Source NAT address entries. This object is indexed by the address pool table and the allocated address. Sequence of parameters: • jnxJsNatSrcPoolName • jnxJsNatSrcXlatedAddrType • jnxJsNatSrcXlatedAddr • jnxJsNatSrcPoolType • jnxJsNatSrcNumPortInuse • jnxJsNatSrcNumSessions
jnxJsNatSrcPoolName	jnxJsSrcNatStatsEntry 1	The name of the dynamic source IP address pool. This is the address pool from which the translated address is allocated.
jnxJsNatSrcXlatedAddrType	jnxJsSrcNatStatsEntry 2	The type of the dynamic source IP address allocated from the address pool used in the NAT translation. The MIB supports IPv4 and IPv6 addresses only.
jnxJsNatSrcXlatedAddr	jnxJsSrcNatStatsEntry 3	The name of the dynamic source IP address allocated from the address pool used in the NAT translation. The MIB supports IPv4 and IPv6 addresses only.
jnxJsNatSrcPoolType	jnxJsSrcNatStatsEntry 4	Source NAT can do address translation with or without Port Address Translation (PAT). The following source port pool type indicates whether the address translation is done with PAT, without PAT, or as a static translation: • **Source pool with PAT**—The security device translates both the source IP address and the port number of the packets. • **Source pool without PAT**—The device performs source NAT translation for the IP address without performing PAT for the source port number. • **Static**—One range of IP addresses is statically mapped one-to-one to a shifted range of IP addresses.
jnxJsNatSrcNumPortInuse	jnxJsSrcNatStatsEntry 5	The number of ports in use for this NAT address entry. This attribute is applicable only to NAT translation with PAT.
jnxJsNatSrcNumSessions	jnxJsSrcNatStatsEntry 6	The number of sessions in use based on this NAT address entry. This attribute is applicable only to NAT translation without PAT.

Related Documentation

- NAT Objects MIB on page 475
- Source NAT Table on page 476
- jnxJsNatIfSrcPoolPortTable on page 479
- NAT Rule Hit Table on page 480
- NAT Pool Hit Table on page 480

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- NAT Trap Definitions on page 481

jnxJsNatIfSrcPoolPortTable

jnxJsNatIfSrcPoolPortTable, whose object ID is **{jnxJsNatObjects 3}**, monitors the port usage of the NAT interface source IP address pool by displaying information about the objects listed in Table 192 on page 479.

The interface source pool is predefined. This source pool is referenced in a policy in which it is configured. The security device translates the source IP address to the address of the egress interface for the traffic, matching a policy that references the interface source pool. The security device always applies PAT for the interface source pool.

Table 192: jnxJsNatIfSrcPoolPortTable

Object	Object ID	Description
jnxJsNatIfSrcPoolPortEntry	**jnxJsNatIfSrcPoolPortTable 1**	Source NAT address entries. This object is indexed by the address pool table and the address. Sequence of parameters: • **jnxJsNatIfSrcPoolIndex** • **jnxJsNatIfSrcPoolTotalSinglePorts** • **jnxJsNatIfSrcPoolAllocSinglePorts** • **jnxJsNatIfSrcPoolTotalTwinPorts** • **jnxJsNatIfSrcPoolAllocTwinPorts**
jnxJsNatIfSrcPoolIndex	**jnxJsNatIfSrcPoolPortEntry 1**	Index number of the port pool of this address pool.
jnxJsNatIfSrcPoolTotalSinglePorts	**jnxJsNatIfSrcPoolPortEntry 2**	Total number of single ports in a port pool.
jnxJsNatIfSrcPoolAllocSinglePorts	**jnxJsNatIfSrcPoolPortEntry 3**	Number of single ports in a port pool allocated or in use.
jnxJsNatIfSrcPoolTotalTwinPorts	**jnxJsNatIfSrcPoolPortEntry 4**	Total number of twin ports in a port pool.
jnxJsNatIfSrcPoolAllocTwinPorts	**jnxJsNatIfSrcPoolPortEntry 5**	Number of twin ports in a port pool allocated or in use.

Related Documentation

- NAT Objects MIB on page 475
- Source NAT Table on page 476
- New Source NAT Table on page 477
- NAT Rule Hit Table on page 480
- NAT Pool Hit Table on page 480
- NAT Trap Definitions on page 481

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NAT Rule Hit Table

The **jnxJsNatRuleTable**, whose object ID is **{jnxJsNatObjects 5}**, monitors the NAT rule hits. Table 193 on page 480 identifies objects in the NAT Rule Hit Table.

Table 193: NAT Rule Hit Table

Object	Object ID	Description
jnxJsNatRuleEntry	**jnxJsNatRuleTable 1**	NAT rule hit entries. This object is indexed by the rule index. Sequence of parameters: • **jnxJsNatRuleName** • **jnxJsNatRuleType** • **jnxJsNatRuleTransHits**
jnxJsNatRuleName	**jnxJsNatRuleEntry 1**	NAT rule name.
jnxJsNatRuleType	**jnxJsNatRuleEntry 2**	NAT types such as source, destination, and static.
jnxJsNatRuleTransHits	**jnxJsNatRuleEntry 3**	Number of hits on this NAT rule.

Related Documentation

- NAT Objects MIB on page 475
- Source NAT Table on page 476
- New Source NAT Table on page 477
- jnxJsNatIfSrcPoolPortTable on page 479
- NAT Pool Hit Table on page 480
- NAT Trap Definitions on page 481

NAT Pool Hit Table

The **jnxJsNatPoolTable**, whose object ID is **{jnxJsNatObjects 6}**, monitors the NAT pool hits. Table 194 on page 480 identifies objects in the NAT Pool Hit Table.

Table 194: NAT Pool Hit Table

Object	Object ID	Description
jnxJsNatPoolEntry	**jnxJsNatPoolTable 1**	NAT pool hit entries. This object is indexed by the pool index. Sequence of parameters: • **jnxJsNatPoolName** • **jnxJsNatPoolType** • **jnxJsNatPoolTransHits**
jnxJsNatPoolName	**jnxJsNatPoolEntry 1**	NAT pool name.

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Table 194: NAT Pool Hit Table *(continued)*

Object	Object ID	Description
jnxJsNatPoolType	jnxJsNatPoolEntry 2	NAT types such as source, destination, and static.
jnxJsNatPoolTransHits	jnxJsNatPoolEntry 3	Number of hits on this NAT pool.

Related Documentation

- NAT Objects MIB on page 475
- Source NAT Table on page 476
- New Source NAT Table on page 477
- jnxJsNatIfSrcPoolPortTable on page 479
- NAT Rule Hit Table on page 480
- NAT Trap Definitions on page 481

NAT Trap Definitions

Table 195 on page 481 lists NAT trap definition objects.

Table 195: NAT Trap Definitions

Object	Object ID	Description
jnxJsNatAddrPoolThresholdStatus	jnxJsNatNotifications 1	NAT address pool utilization threshold status trap to signify that the address pool utilization either exceeds a certain percentage or is clear of that percentage. **jnxJsNatSrcIpPoolName** is the name of the resource pool. **jnxJsNatAddrPoolUtil** is the percentage of utilization of the address pool.
jnxJsNatAddrPoolUtil	jnxJsNatTrapVars 1	Dynamic address pool utilization expressed as a percentage.

Related Documentation

- NAT Objects MIB on page 475
- Source NAT Table on page 476
- New Source NAT Table on page 477
- jnxJsNatIfSrcPoolPortTable on page 479
- NAT Rule Hit Table on page 480
- NAT Pool Hit Table on page 480

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CHAPTER 56

Interpreting the Enterprise-Specific Network Address Translation Resources–Monitoring MIB

- Network Address Translation Resources–Monitoring MIB on page 483
- Source NAT Table on page 484
- NAT Pool Hit Table on page 485
- NAT Rule Hit Table on page 485
- Juniper Networks Enterprise-Specific NAT Traps on page 486

Network Address Translation Resources–Monitoring MIB

The Juniper Networks enterprise-specific SNMP MIB **jnxNatMIB** defines the objects that are used to monitor Network Address Translation (NAT) pools usage and NAT rules. The object ID of the MIB is **{jnxNAT 1}** and it is under the **JNX-SVCS-SMI** hierarchy.

The following MIB objects are supported by **JUNIPER-NAT-MIB** for implementing SNMP MIBs to monitor NAT resources on the Multiservices PIC and Multiservices DPC:

- **jnxNatNotifications {jnxNatMIB 0}**
- **jnxNatObjects {jnxNatMIB 1}**
- **jnxNatTrapVars {jnxNatMIB 2}**

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-sp-nat.txt.

For information about enterprise-specific SNMP NAT MIBs for monitoring NAT resources, see the following topics:

- NAT Pool Hit Table on page 485
- NAT Rule Hit Table on page 485
- Juniper Networks Enterprise-Specific NAT Traps on page 486
- Source NAT Table on page 484

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Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

Source NAT Table

The **jnxSrcNatStatsTable** table, whose object ID is **{jnxNatObjects 1}**, lists the source NAT attributes of the translated addresses.

When performing source IP address translation, the services PIC translates the original source IP address with or without the port number to a different address. The resource, that is, the address source pools provide the services PIC with a supply of addresses from which an address is selected to perform the source network address translation.

Table 196 on page 484 identifies objects in the **jnxNatPoolPortTable** branch. It contains information about source IP address translation only.

Table 196: Source NAT Table

Object	Object ID	Description
jnxSrcNatStatsEntry	**jnxSrcNatStatsTable 1**	The source NAT address entries. The source NAT address entries are indexed by the address pool table and the address allocated.
jnxNatSrcPoolName	**jnxSrcNatStatsEntry 1**	The name of dynamic source IP address pool. The translated address is allocated from this address pool.
jnxNatSrcXlatedAddrType	**jnxSrcNatStatsEntry 2**	The type of dynamic source IP address allocated from the address pool, which is used in the NAT translation. IPv4 and IPv6 addresses are supported for NAT MIB.
jnxNatSrcPoolType	**jnxSrcNatStatsEntry 3**	The source port pool type indicates whether the address translation is done with the port or without the port, or if it is a static translation. The source NAT can do an address translation with or without the port translation.
jnxNatSrcNumPortAvail	**jnxSrcNatStatsEntry 4**	Indicates the number of ports available with this pool.
jnxNatSrcNumPortInuse	**jnxSrcNatStatsEntry 5**	Indicates the number of ports in use for a given NAT address entry. This attribute is applicable only to the source NAT translation with port number translation.
jnxNatSrcNumAddressAvail	**jnxSrcNatStatsEntry 6**	Indicates the total number of addresses available in this pool.
jnxNatSrcNumAddressInUse	**jnxSrcNatStatsEntry 7**	Indicates the number of addresses in use from this pool. This attribute is applicable only to the pools used with source dynamic translations.

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Table 196: Source NAT Table *(continued)*

Object	Object ID	Description
jnxNatSrcNumSessions	**jnxSrcNatStatsEntry 8**	This NAT address entry indicates the number of sessions in use.

Related Documentation

- NAT Pool Hit Table on page 485
- NAT Rule Hit Table on page 485
- Juniper Networks Enterprise-Specific NAT Traps on page 486
- Network Address Translation Resources–Monitoring MIB on page 483

NAT Pool Hit Table

The **jnxNatPoolTable** table, whose object ID is **{jnxNatObjects 3}**, monitors NAT pool hits.

Table 197 on page 485 identifies objects in the **jnxNatPoolPortTable** branch.

Table 197: NAT Pool Hit Table

Object	Object ID	Description
jnxNatPoolEntry	**jnxNatPoolTable 1**	Specifies the NAT pool hit entries. The NAT pool hit entries are indexed by the pool index.
jnxNatPoolName	**jnxNatPoolEntry 1**	Specifies the NAT pool name.
jnxNatPoolType	**jnxNatPoolEntry 2**	Indicates the NAT types as one of the following: • Source • Destination • Static
jnxNatPoolTransHits	**jnxNatPoolEntry 3**	Indicates the number of hits in the specified NAT pool.

Related Documentation

- NAT Rule Hit Table on page 485
- Juniper Networks Enterprise-Specific NAT Traps on page 486
- Network Address Translation Resources–Monitoring MIB on page 483
- Source NAT Table on page 484

NAT Rule Hit Table

The **jnxNatRuleTable** table, whose object ID is **{jnxNatObjects 2}**, monitors the NAT rule hits.

Table 198 on page 486 identifies objects in the **jnxNatRuleTable** branch.

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Table 198: NAT Rule Hit Table

Object	Object ID	Description
jnxNatRuleEntry	**jnxNatRuleTable 1**	Specifies NAT rule hit entries. The NAT rule hit entries are indexed by the rule index.
jnxNatRuleName	**jnxNatRuleEntry 1**	Specifies the NAT rule name.
jnxNatRuleType	**jnxNatRuleEntry 2**	Indicates the NAT types as one of the following: • Source • Destination • Static
jnxNatRuleTransHits	**jnxNatRuleEntry 3**	Indicates the number of hits in this NAT rule.

Related Documentation

- NAT Pool Hit Table on page 485
- Juniper Networks Enterprise-Specific NAT Traps on page 486
- Network Address Translation Resources–Monitoring MIB on page 483

Juniper Networks Enterprise-Specific NAT Traps

The **jnxNatAddrPoolThresholdStatus** and **jnxNatAddrPoolUtil** objects notify the user about the usage of NAT resources in percentage.

Table 199 on page 486 lists the NAT trap definition objects.

Table 199: NAT Trap Definitions

Object	Object ID	Description
jnxNatAddrPoolUtil	**jnxNatTrapVars 1**	Displays the dynamic address pool utilization in percentage.
jnxNatTrapSrcPoolName	**jnxNatTrapVars 2**	Displays the source NAT pool name that issues the trap.
jnxNatAddrPoolThresholdStatus	**jnxNatNotifications 1**	The NAT address pool utilization threshold status trap indicates that the address pool utilization has either exceeded a certain set percentage or is lower than a set percentage. **jnxNatTrapPoolName** is the name of the resource pool. **jnxNatAddrPoolUtil** is the percentage of utilization of the address pool.

Related Documentation

- NAT Pool Hit Table on page 485
- NAT Rule Hit Table on page 485
- Network Address Translation Resources–Monitoring MIB on page 483

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- Source NAT Table on page 484
- Configuring SNMP Trap Options and Groups on a Device Running Junos OS
- Managing Traps and Informs

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CHAPTER 57

Interpreting the Enterprise-Specific Policy Objects MIB

- Policy Objects MIB on page 489
- Security Policy Table on page 490
- jnxJsPolicyStatsTable on page 491

Policy Objects MIB

The Juniper Networks enterprise-specific Policy Objects MIB, **jnxJsSecPolicyMIB**, whose object ID is **{jnxJsPolicies 1}**, defines the MIB for policy monitoring.

A security policy, which can be configured from the user interface, controls the traffic flow from one zone to another zone by defining one or more kinds of traffic permitted from specified IP sources to specified IP destinations at scheduled times.

The Juniper Networks security device enforces the security policies rules for the transit traffic in terms of which traffic can pass through the firewall and the actions taken on the traffic as it passes through the firewall.

Related MIB objects include the following:

- **jnxJsPolicyNotifications—{jnxJsSecPolicyMIB 0}**
- **jnxJsPolicyObjects—{jnxJsSecPolicyMIB 1}**
- **jnxJsPolicyTrapVars—{jnxJsSecPolicyMIB 2}**

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-policy.txt .

For information about the enterprise-specific Policy Objects MIB, see the following topics:

- Security Policy Table on page 490
- jnxJsPolicyStatsTable on page 491

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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Security Policy Table

Table 200 on page 490 describes the objects in the Security Policy Table.

Table 200: Security Policy Table

Object	Object ID	Description
jnxJsPolicyNumber	**jnxJsPolicyObjects 1**	Number of policies (regardless of their current state) present on this system.
jnxJsPolicyTable	**jnxJsPolicyObjects 2**	Exposes the security policy entries. Security devices and routers provide a network boundary with a single point of entry and exit, which allows the screening and directing of traffic through the implementation of access policies. The access policies can permit, deny, encrypt, authenticate, prioirtize, schedule, and monitor the traffic flow through the firewall. This table lists entries of policy. The number of policies are indicated by **jnxJsPolicyNumber**.
jnxJsPolicyEntry	**jnxJsPolicyTable 1**	An entry contains a security policy. Indexes: • **nxJsPolicyFromZone** • **jnxJsPolicyToZone** • **jnxJsPolicyName** Security policies are configured under FromZone and ToZone directions. Under a specific zone direction, each security policy contains a name, match-criteria, action, and other options. Sequence of parameters: • **jnxJsPolicyFromZone** • **jnxJsPolicyToZone** • **jnxJsPolicyName** • **jnxJsPolicySequenceNumber** • **jnxJsPolicyAction** • **jnxJsPolicyScheduler** • **jnxJsPolicyState** • **jnxJsPolicyStatsAvailability** • **jnxJsPolicyPerSecBytesThreshold** • **jnxJsPolicyPerMinKbytesThreshold**
jnxJsPolicyFromZone	**jnxJsPolicyEntry 1**	FromZone name.

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Table 200: Security Policy Table *(continued)*

Object	Object ID	Description
jnxJsPolicyToZone	jnxJsPolicyEntry 2	ToZone name.
jnxJsPolicyName	jnxJsPolicyEntry 3	Name of the policy defined. The name consists of up to 256 ASCII characters and uniquely identifies the policy entry.
jnxJsPolicySequenceNumber	jnxJsPolicyEntry 4	Indication of the policy sequence order of the policy within a specific FromZone and ToZone pair. Policies are matched in a sequence in which the ordering is specified by this number.
jnxJsPolicyAction	jnxJsPolicyEntry 5	Indication of the actions performed when the criteria are matched. The actions **permit**, **reject**, and **deny** are user-configured policies.
jnxJsPolicyScheduler	jnxJsPolicyEntry 6	Name of the schedule attached to this policy. Certain schedules have a specified duration that can affect the status of the policy.
jnxJsPolicyState	jnxJsPolicyEntry 7	State of this policy: active, inactive, or unavailable. The state can be affected by the scheduler if the scheduler has a specified duration.
jnxJsPolicyStatsAvailability	jnxJsPolicyEntry 8	Indication of whether the statistics counters are available and are actively updated. If available, a matching **jnxJsPolicyStatsEntry** exists for the policy.
jnxJsPolicyPerSecBytesThreshold	jnxJsPolicyEntry 9	Indication of the threshold value of bytes per second
jnxJsPolicyPerMinKbytesThreshold	jnxJsPolicyEntry 10	Indication of the threshold value of KB per minute

Related Documentation

- Policy Objects MIB on page 489
- jnxJsPolicyStatsTable on page 491

jnxJsPolicyStatsTable

jnxJsPolicyStatsTable, whose object ID is **{jnxJsPolicyObjects 3}**, exposes the security policy statistics entries listed in Table 201 on page 492. These statistics can be enabled and disabled by configuration on a per-policy basis.

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Table 201: jnxJsPolicyStatsTable

Object	Object ID	Description
jnxJsPolicyStatsEntry	**jnxJsPolicyStatsTable 1**	Contains security policy statistics. Indexes: • **jnxJsPolicyFromZone** • **jnxJsPolicyToZone** • **jnxJsPolicyName** Security policies are configured under FromZone and ToZone direction. Under a specific zone direction, each security policy contains name, match-criteria, action, and other options. Sequence of parameters: • **jnxJsPolicyStatsCreationTime** • **jnxJsPolicyStatsInputBytes** • **jnxJsPolicyStatsInputByteRate** • **jnxJsPolicyStatsOutputBytes** • **jnxJsPolicyStatsOutputByteRate** • **jnxJsPolicyStatsInputPackets** • **jnxJsPolicyStatsInputPacketRate** • **jnxJsPolicyStatsOutputPackets** • **jnxJsPolicyStatsOutputPacketRate** • **jnxJsPolicyStatsNumSessions** • **jnxJsPolicyStatsSessionRate** • **jnxJsPolicyStatsSessionDeleted** • **jnxJsPolicyStatsLookups** • **jnxJsPolicyStatsCountAlarm**
jnxJsPolicyStatsCreationTime	**jnxJsPolicyStatsEntry 1**	Creation timestamp of the policy statistics entry. The timestamp is modified during the creation and deletion of the policy statistics entry. When the timestamp changes, the policy entry statistics entry is assumed to be a new statistics entry and not associated with a previous statistic entry of the same indices.
jnxJsPolicyStatsInputBytes	**jnxJsPolicyStatsEntry 2**	Number of input bytes that enter the firewall through this policy.
jnxJsPolicyStatsInputByteRate	**jnxJsPolicyStatsEntry 3**	Number of input bytes per second or the rate that enters the firewall through this policy.
jnxJsPolicyStatsOutputBytes	**jnxJsPolicyStatsEntry 4**	Number of output bytes associated with this policy.

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Table 201: jnxJsPolicyStatsTable *(continued)*

Object	Object ID	Description
jnxJsPolicyStatsOutputByteRate	jnxJsPolicyStatsEntry 5	Number of output bytes per second or the rate associated with this policy.
jnxJsPolicyStatsInputPackets	jnxJsPolicyStatsEntry 6	Number of input packets that enter the firewall through this policy.
jnxJsPolicyStatsInputPacketRate	jnxJsPolicyStatsEntry 7	Number of input packets per second or the input packet rate of the firewall through this policy.
jnxJsPolicyStatsOutputPackets	jnxJsPolicyStatsEntry 8	Number of output packets associated with this policy.
jnxJsPolicyStatsOutputPacketRate	jnxJsPolicyStatsEntry 9	Number of output packets per second or the rate associated with this policy.
jnxJsPolicyStatsNumSessions	jnxJsPolicyStatsEntry 10	Number of sessions associated with this policy.
jnxJsPolicyStatsSessionRate	jnxJsPolicyStatsEntry 11	Rate of the sessions associated with this policy.
jnxJsPolicyStatsSessionDeleted	jnxJsPolicyStatsEntry 12	Number of sessions deleted from this policy.
jnxJsPolicyStatsLookups	jnxJsPolicyStatsEntry 13	Number of policy lookups performed.
jnxJsPolicyStatsCountAlarm	jnxJsPolicyStatsEntry 14	Number of alarms counted when the traffic exceeds a certain threshold configuration.

Related Documentation

- Policy Objects MIB on page 489
- Security Policy Table on page 490

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CHAPTER 58

Interpreting the Enterprise-Specific Security Interface Extension Objects MIB

- Security Interface Extension Objects MIB on page 495
- jnxJsIfMonTable on page 495

Security Interface Extension Objects MIB

The Juniper Networks enterprise-specific Security Interface Extension Objects MIB, **jnxJsIfMIB**, whose object ID is **{jnxJsIf 1}**, defines the objects that are used to monitor the entries in the interfaces that pertain to the security management of the interface.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-if-ext.txt .

For information about the enterprise-specific Security Interface Extension Objects MIB, see the following topic:

- jnxJsIfMonTable on page 495

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxJsIfMonTable

jnxJsIfMonTable, whose object ID is **{jnxJsIfExtension 1}**, extends the interface entries on a particular interface to support the security related-objects listed in Table 202 on page 496. The table is indexed by **ifIndex**.

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Table 202: jnxJsIfMonTable

Object	Object ID	Description
jnxJsIfMonEntry	**jnxJsIfMonTable 1**	Object related to interface monitoring. Sequence of parameters: • **jnxJsIfMonInIcmp** • **jnxJsIfMonInSelf** • **jnxJsIfMonInVpn** • **jnxJsIfMonInPolicyPermit** • **jnxJsIfMonOutPolicyPermit** • **jnxJsIfMonConn** • **jnxJsIfMonInMcast** • **jnxJsIfMonOutMcast** • **jnxJsIfMonPolicyDeny** • **jnxJsIfMonNoGateParent** • **jnxJsIfMonTcpProxyDrop** • **jnxJsIfMonNoDip** • **jnxJsIfMonNoNspTunnel** • **jnxJsIfMonNoNatCon** • **jnxJsIfMonInvalidZone** • **jnxJsIfMonIpClsFail** • **jnxJsIfMonAuthDrop** • **jnxJsIfMonMultiUserAuthDrop** • **jnxJsIfMonLoopMultiDipDrop** • **jnxJsIfMonAddrSpoof** • **jnxJsIfMonLpDrop** • **jnxJsIfMonNullZone** • **jnxJsIfMonNoGate** • **jnxJsIfMonNoMinorSess** • **jnxJsIfMonNvecErr** • **jnxJsIfMonTcpSeq** • **jnxJsIfMonIllegalPak** • **jnxJsIfMonNoRoute** • **jnxJsIfMonAuthFail** • **jnxJsIfMonSaInactive** • **jnxJsIfMonNoSa** • **jnxJsIfMonSelfPktDrop**
jnxJsIfMonInIcmp	**jnxJsIfMonEntry 1**	ICMP packets received.
jnxJsIfMonInSelf	**jnxJsIfMonEntry 2**	Self packets received.
jnxJsIfMonInVpn	**jnxJsIfMonEntry 3**	VPN packets received.
jnxJsIfMonInPolicyPermit	**jnxJsIfMonEntry 4**	Incoming bytes permitted by policy.

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Table 202: jnxJsIfMonTable *(continued)*

Object	Object ID	Description
jnxJsIfMonOutPolicyPermit	jnxJsIfMonEntry 5	Outgoing bytes permitted by policy.
jnxJsIfMonConn	jnxJsIfMonEntry 6	Incoming connections established.
jnxJsIfMonInMcast	jnxJsIfMonEntry 7	Multicast packets received.
jnxJsIfMonOutMcast	jnxJsIfMonEntry 8	Multicast packets sent.
jnxJsIfMonPolicyDeny	jnxJsIfMonEntry 9	Packets dropped due to policy denial.
jnxJsIfMonNoGateParent	jnxJsIfMonEntry 10	Packets dropped due to no parent for a gate.
jnxJsIfMonTcpProxyDrop	jnxJsIfMonEntry 11	Packets dropped due to syn-attack protection.
jnxJsIfMonNoDip	jnxJsIfMonEntry 12	Packets dropped due to DIP errors.
jnxJsIfMonNoNspTunnel	jnxJsIfMonEntry 13	Packets dropped because no NSP tunnel found.
jnxJsIfMonNoNatCon	jnxJsIfMonEntry 14	Packets dropped due to no more sessions.
jnxJsIfMonInvalidZone	jnxJsIfMonEntry 15	Packets dropped because an invalid zone received the packet.
jnxJsIfMonIpClsFail	jnxJsIfMonEntry 16	Packets dropped due to IP classification failure.
jnxJsIfMonAuthDrop	jnxJsIfMonEntry 17	Packets dropped due to user authentication errors.
jnxJsIfMonMultiUserAuthDrop	jnxJsIfMonEntry 18	Packets dropped due to multiple user authentications in loopback sessions.
jnxJsIfMonLoopMultiDipDrop	jnxJsIfMonEntry 19	Packets dropped due to multiple DIP in loopback sessions.
jnxJsIfMonAddrSpoof	jnxJsIfMonEntry 20	Packets dropped due to address spoofing.
jnxJsIfMonLpDrop	jnxJsIfMonEntry 21	Packets dropped due to no loopback.
jnxJsIfMonNullZone	jnxJsIfMonEntry 22	Packets dropped due to no zone or NULL zone binding.
jnxJsIfMonNoGate	jnxJsIfMonEntry 23	Packets dropped due to no NAT gateway.

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Table 202: jnxJsIfMonTable *(continued)*

Object	Object ID	Description
jnxJsIfMonNoMinorSess	jnxJsIfMonEntry 24	Packets dropped due to no minor session.
jnxJsIfMonNvecErr	jnxJsIfMonEntry 25	Packets dropped due to no session for gateway.
jnxJsIfMonTcpSeq	jnxJsIfMonEntry 26	Packets dropped because TCP sequence number out of window.
jnxJsIfMonIllegalPak	jnxJsIfMonEntry 27	Packets dropped because they did not make any sense.
jnxJsIfMonNoRoute	jnxJsIfMonEntry 28	Packets dropped because no route was present.
jnxJsIfMonAuthFail	jnxJsIfMonEntry 29	Packets dropped because authentication failed.
jnxJsIfMonSaInactive	jnxJsIfMonEntry 30	Packets dropped because security association (SA) is not active.
jnxJsIfMonNoSa	jnxJsIfMonEntry 31	Packets dropped because no SA found for incoming security parameter index (SPI).
jnxJsIfMonSelfPktDrop	jnxJsIfMonEntry 32	Packets dropped because there was no interest in self packets.

Related Documentation

- Security Interface Extension Objects MIB on page 495

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CHAPTER 59

Interpreting the Enterprise-Specific VPN Certificate Objects MIB

- VPN Certificate Objects MIB on page 499
- jnxJsLoadedCaCertTable on page 499
- jnxJsLoadedLocalCertTable on page 500

VPN Certificate Objects MIB

The Juniper Networks enterprise-specific VPN Certificate Objects MIB, **jnxJsCertificateMIB**, whose object ID is **{jnxJsCertificates 1}**, defines the objects that are used to monitor reference and attributes to the certificates.

A related VPN Certificate Object MIB is **jnxJsCertificateObjects {jnxJsCertificateMIB 1}**.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-cert.txt .

For information about the enterprise-specific VPN Certificate Objects MIB, see the following topics:

- jnxJsLoadedCaCertTable on page 499
- jnxJsLoadedLocalCertTable on page 500

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxJsLoadedCaCertTable

The **jnxJsLoadedCaCertTable**, whose object ID is **{jnxJsCertificateObjects 1}**, exposes the loaded Certification Authority (CA)-issued certificate objects listed in Table 203 on page 500. Certificates are used when establishing a secure connection in the device.

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Table 203: jnxJsLoadedCaCertTable

Object	Object ID	Description
jnxJsLoadedCaCertEntry	**jnxJsLoadedCaCertTable 1**	Loaded CA certificate entry. The loaded CA certificate entry is indexed by the CA certification name. Sequence of parameters: • **jnxJsLoadedCaCertName** • **jnxJsLoadedCaCertSubject** • **jnxJsLoadedCaCertExpire** • **jnxJsLoadedCaCertIssuer**
jnxJsLoadedCaCertName	**jnxJsLoadedCaCertEntry 1**	Loaded CA certificate name. This name is unique within the loaded CA certifcates.
jnxJsLoadedCaCertSubject	**jnxJsLoadedCaCertEntry 2**	CA certificate subject.
jnxJsLoadedCaCertExpire	**jnxJsLoadedCaCertEntry 3**	Expiration time and date of CA certificate.
jnxJsLoadedCaCertIssuer	**jnxJsLoadedCaCertEntry 4**	Issuer of CA certificate.

Related Documentation

- VPN Certificate Objects MIB on page 499
- jnxJsLoadedLocalCertTable on page 500

jnxJsLoadedLocalCertTable

The **jnxJsLoadedLocalCertTable**, whose object ID is **{jnxJsCertificateObjects 2}**, exposes the loaded local certificate objects listed in Table 204 on page 500. Certificates are used when establishing a secure connection in the device.

Table 204: jnxJsLoadedLocalCertTable

Object	Object ID	Description
jnxJsLoadedLocalCertEntry	**jnxJsLoadedLocalCertTable 1**	Default certificate entry. This entry is indexed by the certification name. Sequence of parameters: • **jnxJsLoadedLocalCertName** • **jnxJsLoadedLocalCertSubject** • **jnxJsLoadedLocalCertExpire** • **jnxJsLoadedLocalCertIssuer**
jnxJsLoadedLocalCertName	**jnxJsLoadedLocalCertEntry 1**	Name of the local certificate. The certificate name is unique within the loaded local certificates.
jnxJsLoadedLocalCertSubject	**jnxJsLoadedLocalCertEntry 2**	Certificate subject.
jnxJsLoadedLocalCertExpire	**jnxJsLoadedLocalCertEntry 3**	Expiration time and date of the local certificate.

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Table 204: jnxJsLoadedLocalCertTable *(continued)*

Object	Object ID	Description
jnxJsLoadedLocalCertIssuer	jnxJsLoadedLocalCertEntry 4	Issuer of the local certificate.

Related Documentation

- VPN Certificate Objects MIB on page 499
- jnxJsLoadedCaCertTable on page 499

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CHAPTER 60

Interpreting the Enterprise-Specific Security Screening Objects MIB

- Security Screening Objects MIB on page 503
- jnxJsScreenMonTable on page 504
- jnxJsScreenSweepTable on page 514
- jnxJsScreenNotifications on page 515
- jnxJsScreenTrapVars on page 515

Security Screening Objects MIB

The Juniper Networks enterprise-specific Security Screening Objects MIB, **jnxJsScreenMIB**, whose object ID is **{jnxJsScreening 1}**, defines the MIB for the Juniper Networks enterprise firewall screening functionality.

The Juniper Networks security firewall provides various detection methods and defense mechanisms to combat exploits at all stages of the path of execution, including:

- Screen option setting
- Firewall Denial-of-Service (DoS) attack
- Network DoS attack
- OS-specific DoS attack
- Fragment reassembly

Related Security Screening Objects MIBs include:

- **jnxJsScreenNotifications {jnxJsScreenMIB 0}**
- **jnxJsScreenObjects {jnxJsScreenMIB 1}**
- **jnxJsScreenTrapVars {jnxJsScreenMIB 2}**

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-screening.txt .

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For information about the enterprise-specific Security Screening Objects MIB, see the following topics:

- jnxJsScreenMonTable on page 504
- jnxJsScreenSweepTable on page 514
- jnxJsScreenNotifications on page 515
- jnxJsScreenTrapVars on page 515

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxJsScreenMonTable

The **jnxJsScreenMonTable**, whose object ID is **{jnxJsScreenObjects 1}**, collects the screen attributes that monitor the various attacks to enable the Juniper Networks Security Firewall to provide deep inspection (DI) protection on each of the security device's physical interfaces. These attributes are listed in Table 203 on page 500.

The screen options can be enabled at a security zone bounded to an interface or interfaces. When these options apply to traffic reaching the security device through interfaces (via a zone), they offer protection against a malicious information gathering probe or an attack to compromise, disable, or harm a network or network resources.

Table 205: jnxJsScreenMonTable

Object	Object ID	Description
jnxJsScreenMonEntry	**jnxJsScreenMonTable 1**	The screen option monitoring statistics entry. Each entry is uniquely identified by the zone name. The data is collected on a per zone basis. There can be multiple interfaces bound to a particular zone. Hence, the statistics are aggregated across the interfaces on a per zone basis. Sequence of parameters: • **jnxJsScreenZoneName** • **jnxJsScreenNumOfIf** • **jnxJsScreenMonSynAttk** • **jnxJsScreenMonTearDrop** • **jnxJsScreenMonSrcRoute** • **jnxJsScreenMonPingDeath** • **jnxJsScreenMonAddrSpoof** • **jnxJsScreenMonLand** • **jnxJsScreenMonIcmpFlood** • **jnxJsScreenMonUdpFlood** • **jnxJsScreenMonWinnuke** • **jnxJsScreenMonPortScan** • **jnxJsScreenMonIpSweep**

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Table 205: jnxJsScreenMonTable *(continued)*

Object	Object ID	Description
		• **jnxJsScreenMonSynFrag** • **jnxJsScreenMonTcpNoFlag** • **jnxJsScreenMonIpUnknownProt** • **jnxJsScreenMonIpOptBad** • **jnxJsScreenMonIpOptRecRt**—Record route option • **jnxJsScreenMonIpOptTimestamp**—Timestamp option • **jnxJsScreenMonIpOptSecurity** • **jnxJsScreenMonIpOptLSR**—Loose source route • **jnxJsScreenMonIpOptSSR**—Strict source route • **jnxJsScreenMonIpOptStream**—Stream options • **jnxJsScreenMonIcmpFrag** • **jnxJsScreenMonIcmpLarge** • **jnxJsScreenMonTcpSynFin** • **jnxJsScreenMonTcpFinNoAck** • **jnxJsScreenMonLimitSessSrc**—Session limit (source IP-based)
		• **jnxJsScreenMonLimitSessDest**—Session limit (destination IP-based) • **jnxJsScreenMonSynAckAck** • **jnxJsScreenMonIpFrag** • **jnxJsScreenSynAttackThresh**—Threshold data • **jnxJsScreenSynAttackTimeout**—Threshold data • **jnxJsScreenSynAttackAlmTh**—Threshold data • **jnxJsScreenSynAttackQueSize**—Threshold data • **jnxJsScreenSynAttackAgeTime**—Threshold data (obsolete in this release) • **jnxJsScreenIcmpFloodThresh**—Threshold data • **jnxJsScreenUdpFloodThresh**—Threshold data • **jnxJsScreenPortScanThresh**—Threshold data • **jnxJsScreenIpSweepThresh**—Threshold data • **jnxJsScreenSynAckAckThres**—Threshold data
jnxJsScreenZoneName	**jnxJsScreenMonEntry 1**	Name of the security zone under which the statistics are collected.
jnxJsScreenNumOfIf	**jnxJsScreenMonEntry 2**	Number of interfaces bound to this zone. Each counter contains the aggregated data of all the interfaces.
jnxJsScreenMonSynAttk	**jnxJsScreenMonEntry 3**	Number of SYN (TCP connection request) attacks. A SYN attack is a common denial of service (DoS) technique characterized by the following pattern: • Using a spoofed IP address not in use on the Internet, an attacker sends multiple SYN packets to the target machine. • For each SYN packet received, the target machine allocates resources and sends an acknowledgment (SYN-ACK) to the

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Table 205: jnxJsScreenMonTable *(continued)*

Object	Object ID	Description
		source IP address. This can cause the target machine to allocate resources for more than 3 minutes to respond to just one SYN attack, subsequently wasting resources.
jnxJsScreenMonTearDrop	jnxJsScreenMonEntry 4	Number of teardrop attacks. Teardrop attacks exploit the reassembly of fragmented IP packets. In the IP header, one of the fields is the fragment offset field, which indicates the starting position of the data contained in a fragmented packet relative to the data of the original unfragmented packet. When the sum of the offset and size of one fragmented packet differs from that of the next fragmented packet, the packets overlap. The server attempting to reassemble the packet can crash, especially if it is running an older operating system that has this vulnerability. When this option is enabled, the security device detects this discrepancy in a fragmented packet and drops it, and counts the number of packets dropped.
jnxJsScreenMonSrcRoute	jnxJsScreenMonEntry 5	Number of either loose source route option packets or strict source route attack packets. IP source route options can be used to hide their true address and access restricted areas of a network by specifying a different path. The security device should be able to either block any packets with loose or strict source route options set or detect such packets and then record the event for the ingress interface.
jnxJsScreenMonPingDeath	jnxJsScreenMonEntry 6	Number of ping-of-death attack packets. The maximum allowable IP packet size is 65,535 bytes, including the packet header (typically 20 bytes long). An ICMP echo request is an IP packet with a pseudo header, which is 8 bytes long. Therefore, the maximum allowable size of the data area of an ICMP echo request is 65,507 bytes. Many ping implementations, however, allow the user to specify a packet size larger than 65,507 bytes. A grossly oversized ICMP packet can trigger a range of adverse system reactions, such as DoS, crashing, freezing, and rebooting. When the ping-of-death option is enabled, the security device detects and rejects such oversized and irregular packet sizes, even when the attacker hides the total packet size by purposefully fragmenting it.
jnxJsScreenMonAddrSpoof	jnxJsScreenMonEntry 7	Number of address spoofing attack packets. One method to gain access to a restricted network is to insert a bogus source address in the packet header to make the packet appear to come from a trusted source. This technique is called IP spoofing. The mechanism to detect IP spoofing relies on route table entries.

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Table 205: jnxJsScreenMonTable *(continued)*

Object	Object ID	Description
		For example, if a packet with source IP address 10.1.1.6 arrives at port **eth3**, but the security device has a route to 10.1.1.0/24 through port **eth1**, IP spoofing checking notes that this address arrived at an invalid interface as defined in the route table. A valid packet from 10.1.1.6 can arrive only through **eth1**, not **eth3**. The security device concludes that the packet has a spoofed source IP address and discards it.
jnxJsScreenMonLand	**jnxJsScreenMonEntry 8**	Number of land attack packets. A SYN attack combined with an IP spoof is referred to as a land attack. A land attack occurs when an attacker sends spoofed SYN packets containing the IP address of the victim as both the destination and source IP address. The receiving victim responds by sending the SYN-ACK packet to itself, creating an empty connection that lasts until the idle timeout value is reached. Flooding a system with such empty connections can overwhelm the victim, causing a DoS.
jnxJsScreenMonIcmpFlood	**jnxJsScreenMonEntry 9**	Number of ICMP flood attack packets. An ICMP flood typically occurs when ICMP echo requests overload a victim with so many requests that the victim expends all its resources responding to the ICMP echo requests until it can no longer process valid network traffic. With ICMP flood protection enabled and a threshold set, if the threshold is exceeded, the victim invokes the flood attack protection feature. The default threshold value is 1000 pps. If the threshold is exceeded, the security device ignores further ICMP echo requests for the remainder of that second plus the next second as well.
jnxJsScreenMonUdpFlood	**jnxJsScreenMonEntry 10**	Number of UDP flood attack packets. UDP flooding occurs when an attacker sends IP packets containing UDP datagrams with the purpose of slowing down the victim to the point that it can no longer handle valid connections. With UDP flood protection enabled, a threshold can be set so that when the threshold is exceeded, the system invokes UDP flood attack protection. The default threshold value is 1000 pps. If the number of UDP datagrams from one or more sources to a single destination exceeds this threshold, the security device ignores further UDP datagrams to that destination for the remainder of that second plus the next second as well.
jnxJsScreenMonWinnuke	**jnxJsScreenMonEntry 11**	Number of NetBIOS attacks. WinNuke is a DoS attack targeting any computer on the Internet running Microsoft Windows. The attacker sends a TCP segment, usually to NetBIOS port 139 of a host with an established connection with segment's urgent (URG) flag set. This practice introduces a NetBIOS fragment overlap, which causes many machines running Microsoft Windows to crash.

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Table 205: jnxJsScreenMonTable *(continued)*

Object	Object ID	Description
jnxJsScreenMonPortScan	jnxJsScreenMonEntry 12	Number of port scan attempt attack packets. A port scan occurs when one source IP address sends IP packets containing TCP SYN segments to a defined number of different ports at the same destination IP address within a defined interval. The purpose of this attack is to scan the available services in the hope that at least one port will respond, thus identifying a service of the target. The security device should internally log the number of different ports scanned from one remote source.
jnxJsScreenMonIpSweep	jnxJsScreenMonEntry 13	Number of address sweep attempt attack packets. An address sweep occurs when one source IP address sends a defined number of ICMP packets to different hosts within a defined interval. The purpose of this attack is to send ICMP packets, typically echo requests, to various hosts in the hope that at least one replies, thus uncovering an address of the target. The security device internally logs the number of ICMP packets to different addresses from one remote source.
jnxJsScreenMonSynFrag	jnxJsScreenMonEntry 14	Number of SYN fragments. IP encapsulates a TCP SYN segment in the IP packet that initiates a TCP connection. The purpose is to initiate a connection and to invoke a SYN/ACK segment response. The SYN segment typically does not contain any data since the IP packet is small and there is no legitimate reason for it to be fragmented. A fragmented SYN packet is anomalous and is suspicious. To be cautious, it might be helpful to block such fragments from entering the protected network. When the SYN fragmentation check is enabled, the security device detects and drops the packets when the IP header indicates that the packet has been fragmented while the SYN flag is set in the TCP header.
jnxJsScreenMonTcpNoFlag	jnxJsScreenMonEntry 15	Number of TCP packets with no flag set. A normal TCP segment header has at least one flag control set. A TCP segment with no control flags set is an anomalous event. Operating systems respond to such anomalies in different ways. The response, or even lack of response, from the targeted device can provide a clue as to the target's OS type. When this option is enabled, if the security device discovers such a header with a missing or malformed flags field, it drops the packet.
jnxJsScreenMonIpUnknownProt	jnxJsScreenMonEntry 16	Number of of unknown protocol IP packets. According to RFC 1700, some protocol types in an IP header are reserved and unassigned at this time. Precisely because these protocols are undefined, there is no way to know in advance whether a particular unknown protocol is benign or malicious. Unless your network makes use of a nonstandard protocol with a reserved or unassigned protocol number, a cautious stance is to

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Table 205: jnxJsScreenMonTable *(continued)*

Object	Object ID	Description
		block such unknown elements from entering your protected network. When the Unknown Protocol Protection SCREEN option is enabled, the security device drops packets when the protocol field contains a protocol ID number of 137 or greater.
jnxJsScreenMonIpOptBad	jnxJsScreenMonEntry 17	Number of IP bad option packets. The IP protocol specifies a set of eight options that provide special routing controls, diagnostic tools, and security. These eight options can be used for malicious objectives. Either intentionally or accidentally, attackers sometimes configure IP options incorrectly, producing either incomplete or malformed fields. The incorrect formatting is anomalous and potentially harmful to the intended recipient. When the Bad IP Option Protection SCREEN option is enabled, the security device detects and blocks packets when any IP option in the IP packet header is incorrectly formatted.
jnxJsScreenMonIpOptRecRt	jnxJsScreenMonEntry 18	Number of IP record option packets. The IP standard RFC 791 specifies a set of options to provide special routing controls, diagnostic tools, and security. These options appear after the destination address in an IP packet header. When they do appear, they are frequently being put to some nefarious use. The record option is one of these options that an attacker can use for reconnaissance or for some unknown but suspicious purpose. When a record IP option is received, the security device flags it as a network reconnaissance attack and records the event for the ingress interface.
jnxJsScreenMonIpOptTimestamp	jnxJsScreenMonEntry 19	Number of IP timestamp option packets. The IP standard RFC 791 specifies a set of options to provide special routing controls, diagnostic tools, and security. These options appear after the destination address in an IP packet header. When they do appear, they are frequently being put to some nefarious use. Timestamp is one of these options that an attacker can use for reconnaissance or for some unknown but suspicious purpose. When a timestamp IP option is received, the security device flags this as a network reconnaissance attack and records the event for the ingress interface.
jnxJsScreenMonIpOptSecurity	jnxJsScreenMonEntry 20	Number of IP security option packets. The IP standard RFC 791 specifies a set of options to provide special routing controls, diagnostic tools, and security. These options appear after the destination address in an IP packet header. When they do appear, they are frequently being put to some nefarious

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Table 205: jnxJsScreenMonTable *(continued)*

Object	Object ID	Description
		use. Security is one of these options that an attacker can use for reconnaissance or for some unknown but suspicious purpose. When a security IP option is received, the security device flags this as a network reconnaissance attack and records the event for the ingress interface.
jnxJsScreenMonIpOptLSR	jnxJsScreenMonEntry 21	Number of strict source route packets. Attackers can use IP source route options to hide their true address and access restricted areas of a network by specifying a different path. The security device should be able to either block any packets with loose or strict source route options set or detect such packets and then record the event for the ingress interface.
jnxJsScreenMonIpOptStream	jnxJsScreenMonEntry 23	Number of IP stream option packets. The IP standard RFC 791 specifies a set of options to provide special routing controls, diagnostic tools, and security. These options appear after the destination address in an IP packet header. When they do appear, they are frequently being put to some nefarious use. Stream is one of these options that an attacker can use for reconnaissance or for some unknown but suspicious purpose. When a security IP option is received, the security device flags it as a network reconnaissance attack and records the event for the ingress interface.
jnxJsScreenMonIcmpFrag	jnxJsScreenMonEntry 24	Number of ICMP fragment packets. ICMP provides error reporting and network probe capabilities. Because ICMP packets contain very short messages, there is no legitimate reason for ICMP packets to be fragmented. If an ICMP packet is so large that it must be fragmented, something is wrong. With the ICMP Fragment Protection SCREEN option enabled, the security device should be able to block any ICMP packet with the More Fragments flag set or with an offset value indicated in the offset field.
jnxJsScreenMonIcmpLarge	jnxJsScreenMonEntry 25	Number of large ICMP packets. Because ICMP packets contain very short messages, there is no legitimate reason for ICMP packets to be fragmented. If an ICMP packet is unusually large, something is wrong. For example, the Loki program uses ICMP as a channel for transmitting covert messages. The presence of large ICMP packets might expose a compromised machine acting as a Loki agent. It might also indicate some other kind of malicious activity. When the the Large Size ICMP Packet Protection SCREEN option is enabled, the security device drops ICMP packets with a length greater than 1024 bytes.

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Table 205: jnxJsScreenMonTable *(continued)*

Object	Object ID	Description
jnxJsScreenMonTcpSynFin	jnxJsScreenMonEntry 26	Number of dropped TCP packets because SYN and FIN are both set. Both the SYN and FIN control flags are not normally set in the same TCP segment header. The SYN flag synchronizes sequence numbers to initiate a TCP connection. The FIN flag indicates the end of data transmission to finish a TCP connection. Their purposes are mutually exclusive. A TCP header with the SYN and FIN flags set is anomalous TCP behavior, causing various responses from the recipient, depending on the OS. When the blocking of TCP packets with both SYN and FIN is enabled, the security device drops the packet when it discovers such a header.
jnxJsScreenMonTcpFinNoAck	jnxJsScreenMonEntry 27	Number of TCP packets with FIN set, but without the ACK bit set. A FIN scan sends TCP segments with the FIN flag set in an attempt to provoke a response and thereby discover an active host or an active port on a host. The use of TCP segments with the FIN flag set might evade detection and thereby help attackers succeed in their reconnaissance efforts.
jnxJsScreenMonLimitSessSrc	jnxJsScreenMonEntry 28	Number of session connections for a source IP address that exceeds the specified limit. Because all the virus-generated traffic originates from the same IP address (generally from an infected server), a source-based session limit ensures that the firewall can curb such excessive amounts of traffic. This amount is based on a threshold value of the number of concurrent sessions required to fill up the session table of the particular firewall. The default maximum for a source-based session limit is 128 concurrent sessions, which can be adjusted accordingly.
jnxJsScreenMonLimitSessDest	jnxJsScreenMonEntry 29	Number of session connections for the destination source IP address that exceeds the specified limit. The user can limit the number of concurrent sessions to the same destination IP address. An attacker can launch a distributed denial of service (DDoS) attack using “zombie agents.” Setting a destination-based session limit can ensure that the security device allows only an acceptable number of concurrent connection requests, no matter what the source, to reach any one host. The default maximum for the destination-based session limit is 128 concurrent sessions.
jnxJsScreenMonSynAckAck	jnxJsScreenMonEntry 30	Number of SYN ACK ACK attacks. When an authentication user initiates a Telnet or FTP connection, the user sends a SYN segment to the Telnet or FTP server. The security device intercepts the SYN segment, creates an entry in its session table, and proxies a SYN-ACK segment to the user. The user then replies with an ACK segment. At that point, the initial

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Table 205: jnxJsScreenMonTable *(continued)*

Object	Object ID	Description
		three-way handshake is complete. The security device sends a login prompt to the user. When a malicious user does not log in, but instead continues initiating SYN-ACK-ACK sessions, the firewall session table can fill up to the point at which the security device begins rejecting legitimate connection requests. When the SYN-ACK-ACK proxy protection option is enabled, after the number of connections from the same IP address reaches the SYN-ACK-ACK proxy threshold, the security device rejects further connection requests from that IP address. By default, the threshold is 512 connections from any single IP address.
jnxJsScreenMonIpFrag	**jnxJsScreenMonEntry 31**	Number of block IP fragment packets. As a packets travels, it is sometimes necessary to break the packet into smaller fragments based on the maximum transmission unit (MTU) of each network. IP fragments might contain an attacker's attempt to exploit the vulnerabilities in the packet reassembly code of specific IP stack implementations. When the victim receives these packets, the results can range from processing the packets incorrectly to crashing the entire system. When the block IP framentation flag is enabled, the security device blocks all IP packet fragments that it receives at interfaces bound to that zone.
Threshold Values		
jnxJsScreenSynAttackThresh	**jnxJsScreenMonEntry 32**	SYN attack threshold value. The number of SYN segments to the same destination address and port number per second required to activate the SYN proxying mechanism. Setting the appropriate threshold value requires a thorough knowledge of the normal traffic patterns at the site. For example, if the security device normally gets 2000 SYN segments per second, the threshold value should be set at 3000 segments per second.
jnxJsScreenSynAttackTimeout	**jnxJsScreenMonEntry 33**	SYN attack timeout value. The maximum length of time before a half-completed connection is dropped from the queue. The default is 20 seconds.
jnxJsScreenSynAttackAlmTh	**jnxJsScreenMonEntry 34**	SYN attack alarm threshold value. The SYN attack alarm threshold causes an alarm to be generated when the number of proxied, half-completed TCP connection requests per second to the same destination address and port number exceeds its value.
jnxJsScreenSynAttackQueSize	**jnxJsScreenMonEntry 35**	SYN attack queue size. The number of proxied connection requests held in the proxied connection queue before the security device starts rejecting new connection requests.

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Table 205: jnxJsScreenMonTable *(continued)*

Object	Object ID	Description
NOTE: The **jnxJsScreenSynAttackAgeTime** object is obsolete in this release.		
jnxJsScreenSynAttackAgeTime	**jnxJsScreenMonEntry 36**	SYN flood age time.
jnxJsScreenIcmpFloodThresh	**jnxJsScreenMonEntry 37**	ICMP attack alarm threshold value. The security device can impose a limit on the number of SYN segments permitted to pass through the firewall per second. The default attack threshold value is 1000. The valid threshold range is 1 through 100,000. When the threshold value is exceeded, an alarm is triggered.
jnxJsScreenUdpFloodThresh	**jnxJsScreenMonEntry 38**	UDP attack alarm threshold value. UDP flooding occurs when an attacker sends IP packets containing UDP datagrams with the purpose of slowing down the victim to the point that it can no longer handle valid connections. The default threshold value is 1000 pps.
jnxJsScreenPortScanThresh	**jnxJsScreenMonEntry 39**	Port scan threshold value. The port scan threshold interval is in microseconds. The default threshold value is 5000. The valid threshold range is 1000 through 1,000,000. By using the default settings, if a remote host scans 10 ports in 0.005 seconds (5000 microseconds), the security device flags this occurrence as a port scan attack and rejects all further packets from the remote source for the remainder of the specified timeout period. The security device detects and drops the tenth packet that meets the port scan attack criterion.
jnxJsScreenIpSweepThresh	**jnxJsScreenMonEntry 40**	IP sweep threshold interval. The IP sweep threshold interval is in microseconds. The default threshold value is 5000. The valid threshold range is 1000 through 1,000,000. By using the default settings, if a remote host sends ICMP traffic to 10 addresses in 0.005 seconds (5000 microseconds), the security device flags this occurrence as an address sweep attack and rejects all further ICMP echo requests from that host for the remainder of the specified threshold time period. The security device detects and drops the tenth packet that meets the address sweep attack criterion.
jnxJsScreenSynAckAckThres	**jnxJsScreenMonEntry 41**	SYN-ACK-ACK alarm threshold value.

Related Documentation

- Security Screening Objects MIB on page 503
- jnxJsScreenSweepTable on page 514

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- jnxJsScreenNotifications on page 515
- jnxJsScreenTrapVars on page 515

jnxJsScreenSweepTable

The **jnxJsScreenSweepTable**, whose object ID is {**jnxJsScreenObjects 3**}, collects the screen attributes that monitor TCP/UDP sweep attacks. These attributes are listed in Table 206 on page 514.

Table 206: jnxJsScreenSweepTable

Object	Object ID	Description
jnxJsScreenSweepEntry	**jnxJsScreenSweepTable 1**	Maintains the TCP/UDP sweep thresholds and counters. Sequence of parameters: • **jnxJsScreenTcpSweepThresh** • **jnxJsScreenUdpSweepThresh** • **jnxJsScreenMonTcpSweep** • **jnxJsScreenMonUdpSweep**
jnxJsScreenTcpSweepThresh	**jnxJsScreenSweepEntry 1**	The TCP sweep threshold interval is in microseconds. The default threshold value is 5000. The valid threshold range is 1000-1,000,000. If a remote host sends TCP packets to 10 addresses in 0.005 seconds (5000 microseconds), the security device flags this as a TCP sweep, and rejects all further new TCP connections initiated from that host for the remainder of the specified threshold period.
jnxJsScreenUdpSweepThresh	**jnxJsScreenSweepEntry 2**	The UDP sweep threshold interval is in microseconds. The default threshold value is 5000. The valid threshold range is 1000-1,000,000. If a remote host sends UDP packets to 10 addresses in 0.005 seconds (5000 microseconds), the security device flags this as a UDP sweep, and rejects all further new UDP connections from that host for the remainder of the specified threshold period.
jnxJsScreenMonTcpSweep	**jnxJsScreenSweepEntry 3**	Number of TCP sessions dropped as a result of a TCP sweep.
jnxJsScreenMonUdpSweep	**jnxJsScreenSweepEntry 4**	Number of UDP packets dropped as a result of a UDP sweep.

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Related Documentation

- Security Screening Objects MIB on page 503
- jnxJsScreenMonTable on page 504
- jnxJsScreenNotifications on page 515
- jnxJsScreenTrapVars on page 515

jnxJsScreenNotifications

The **jnxJsScreenNotifications** object, whose object ID is **{jnxJsScreenNotifications 0}**, specifies the screen notifications listed in Table 207 on page 515.

Table 207: jnxJsScreenNotifications

Object	Object ID	Description
jnxJsScreenAttack	**jnxJsScreenNotifications 1**	Specifies that the number of bytes per minute has exceeded the specified threshold. • **jnxJsScreenZoneName** — Specifies the zone name in which the attack is occurring. • **ifName** — Specifies the interface at which the attack is occurring. • **jnxJsScreenAttackType** — Specifies the type of attack. • **jnxJsScreenAttackCounter** — Displays the number of attacks recorded. The value is the aggregated statistic of all the interfaces bound to that specific zone. • **jnxJsScreenAttackDescr** — Provides a general text description of this attack or the trap.
jnxJsScreenCfgChange	**jnxJsScreenNotifications 2**	Specifies the status of the screening option. • **jnxJsScreenCfgStatus** — Specifies if the screening option is enabled or disabled.

Related Documentation

- Security Screening Objects MIB on page 503
- jnxJsScreenMonTable on page 504
- jnxJsScreenSweepTable on page 514
- jnxJsScreenTrapVars on page 515

jnxJsScreenTrapVars

The **jnxJsScreenTrapVars** object identifies the screen trap variables listed in Table 208 on page 516.

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Table 208: jnxJsScreenTrapVars

Object	Object ID	Description
jnxJsScreenAttackType	**jnxJsScreenTrapVars 1**	Returns one of the following integer values to Identify the type of attack: • **32**—tcpsweeping • **33**—udpsweeping
jnxJsScreenAttackCounter	**jnxJsScreenTrapVars 2**	Returns an integer which defines the threshold that triggered the trap.
jnxJsScreenAttackDescr	**jnxJsScreenTrapVars 3**	Represents a string value that displays the description of the attack
jnxJsScreenCfgStatus	**jnxJsScreenTrapVars 4**	Returns one of the following integer values to indicate the state of the screening option: • **1**–disabled • **2**–enabled

Related Documentation

- Security Screening Objects MIB on page 503
- jnxJsScreenMonTable on page 504
- jnxJsScreenSweepTable on page 514
- jnxJsScreenNotifications on page 515

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CHAPTER 61

Interpreting the Enterprise-Specific LDP MIB

- LDP MIB on page 517
- LDP Notification Objects and Notification Types on page 518
- jnxLdpStatsTable on page 520

LDP MIB

The Juniper Networks enterprise-specific LDP MIB, whose object identifier is **{jnxMibs 14}**, contains LDP statistics, and defines LDP notification objects and types.

The enterprise-specific LDP MIB uses the following objects and definitions from standard MIBs and enterprise-specific MIB definitions:

- **IpAddress** from SNMPv2–SMI MIB
- **DisplayString** from SNMPv2–TC MIB
- **InterfaceIndex** and **InterfaceIndexOrzero** from IF MIB
- **jnxMibs** and **jnxLdpTraps** from Juniper Enterprise-Specific SMI MIB
- **jnxMplsLdpSesState** from Juniper Enterprise-Specific MPLS LDP MIB
- **MplsVpnName** from the standard MPLS VPN MIB
- **InetAddressType**, **InetAddress**, and **InetAddressPrefixLength** from the standard Inet Address MIB

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-ldp.txt .

For information about the enterprise-specific LDP MIB objects, see the following topics:

- LDP Notification Objects and Notification Types on page 518
- jnxLdpStatsTable on page 520

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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LDP Notification Objects and Notification Types

The **jnxLdpTrapVars** table, whose object identifier is **{jnxLdp 1}**, defines the objects used in the enterprise-specific LDP traps. For more information about LDP notification types, see Table 209 on page 518.

NOTE: The enterprise-specific LDP MIB requires jnxLdpTrapPrefix with 0 subidentifier for seamless translation of SNMPv2 notifications to SNMPv1 format.

Table 209: LDP Notification Objects

Object	Object ID	Description
jnxLdpLspFec	**jnxLdpTrapVars 1**	Contains the LSP forwarding equivalence class (FEC) in **IpAddress** format.
jnxLdpRtrid	**jnxLdpTrapVars 2**	Contains the router ID of the sending router in **IpAddress** format.
jnxLdpLspDownReason	**jnxLdpTrapVars 3**	Returns one of the following integer values to denote what might have caused the LSP to go down: • **1**–Change in topology • **2**–Withdrawal of label by the neighbor • **3**–Unavailability of the neighbor because the neighbor is down • **4**–Change in filter • **5**–Unknown reason
jnxLdpSesDownReason	**jnxLdpTrapVars 4**	Returns one of the following integer values to denote what might have caused the session transition into nonexistent state: • **0**–Unknown reason • **1**–Hold time expired • **2**–Connection time expired • **3**–All adjacencies are down • **4**–Received bad TLV (type, length, and value encoding scheme) • **5**–Received bad PDU (protocol data unit) • **6**–Connection error occurred • **7**–The connection was reset • **8**–Received notification from peer • **9**–Received unexpected end-of-file message • **10**–The authentication key was changed • **11**–Error occurred during initialization • **12**–Graceful restart was aborted • **13**– CLI command was issued to end the session
jnxLdpSesDownIf	**jnxLdpTrapVars 5**	Contains the SNMP index of the interface associated with the session-down event. If no interface is associated with the session-down event, then this object returns the SNMP index of any interface associated with one of the neighbors.

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Table 209: LDP Notification Objects *(continued)*

Object	Object ID	Description
jnxLdpLspFecLen	**jnxLdpTrapVars 6**	Represents the length of the LSP FEC prefix in bits. The allowable range is **0** through **32** bits.
jnxLdpSesUpIf	**jnxLdpTrapVars 7**	Contains the interface index of one of the neighbors associated with the session.
jnxLdpInstanceName	**jnxLdpTrapVars 8**	Contains the name of the VPN interface.

Table 210 on page 519 lists the enterprise-specific LDP notifications and the objects contained in each notification. The enterprise-specific LDP notifications use the objects listed in Table 209 on page 518.

Table 210: LDP Notification Types

Object	Object ID	Description
jnxLdpLspUp	**jnxLdpTrapPrefix 1**	Generated when an LSP comes back online. Typically, this trap is generated only when an LSP that has an active **jnxLdpLspDown** trap comes back online. • **jnxLdpLspFec** • **jnxLdpRtrid** • **jnxLdpLspFecLen** • **jnxLdpInstanceName**
jnxLdpLspDown	**jnxLdpTrapPrefix 2**	Generated when an LSP goes offline. This trap contains the following objects: • **jnxLdpLspFec** • **jnxLdpRtrid** • **jnxLdpLspDownReason** • **jnxLdpLspFecLen** • **jnxLdpInstanceName** NOTE: For every **jnxLdpLspDown** trap generated, Junos OS ensures that a **jnxLdpLspUp** trap is generated when the LSP comes back online.
jnxLdpSesUp	**jnxLdpTrapPrefix 3**	Generated when the **jnxMplsLdpSesState** object moves into the **operational (5)** state. This trap contains the following objects: • **jnxMplsLdpSesState** • **jnxLdpSesUpIf**

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Table 210: LDP Notification Types *(continued)*

Object	Object ID	Description
jnxLdpSesDown	**jnxLdpTrapPrefix 4**	Generated when the **jnxMplsLdpSesState** object moves out of the **operational (5)** state. This trap contains the following objects: • **jnxMplsLdpSesState** • **jnxLdpSesDownReason** • **jnxLdpSesDownIf** The **jnxLdpSesDownIf** object contains the address of the interface associated with the last neighbor when the value of **jnxLdpSesDownReason** was **allAdjacenciesDown (3)**.

Related Documentation
- LDP MIB on page 517
- jnxLdpStatsTable on page 520

jnxLdpStatsTable

The **jnxLdpStatsTable**, whose object identifier is **{jnxLdp 2}**, contains the statistics associated with a particular LDP FEC. Each **jnxLdpStatsEntry** in **jnxLdpStatsTable** contains the objects listed in Table 211 on page 520.

Table 211: jnxLdpStatsTable

Object	Object ID	Description
jnxLdpInstanceId	**jnxLdpStatsEntry 1**	Identifies the LDP instance.
jnxLdpFecType	**jnxLdpStatsEntry 2**	Denotes the type of the LDP instance.
jnxLdpFec	**jnxLdpStatsEntry 3**	Contains the **InetAddress** of the LDP FEC.
jnxLdpFecLength	**jnxLdpStatsEntry 4**	Shows the LDP FEC length in bits. The allowable range is **0** through **32** bits.
jnxLdpFecStatisticsStatus	**jnxLdpStatsEntry 5**	Contains one of the following integer values to indicate the status of traffic statistics for the FEC: • **1**–Enabled and available • **2**–Disabled • **3**–Unavailable The traffic statistics can be disabled for the penultimate hop FECs, and in such cases, the objects after **jnxLdpFecStatisticsStatus** in the **jnxLdpStatsEntry** return **0** value.

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Table 211: jnxLdpStatsTable *(continued)*

Object	Object ID	Description
jnxLdpIngressOctets	**jnxLdpStatsEntry 6**	Shows the number of octets of traffic originated from the router, and forwarded over the current LDP FEC. Because the LDP statistics are collected at preconfigured intervals and not in real time, this object might return a value that is different from the current value. The default interval for LDP statistics collection is **5** minutes.
jnxLdpIngressPackets	**jnxLdpStatsEntry 7**	Shows the number of packets originated from the router, and forwarded over the current LDP FEC. Because the LDP statistics are collected at preconfigured intervals and not in real time, this object might return a value that is different from the current value. The default interval for LDP statistics collection is **5** minutes.
jnxLdpTransitOctets	**jnxLdpStatsEntry 8**	Shows the number of octets of traffic originated from a different router, but destined for this FEC, and forwarded over the current LDP FEC. Because the LDP statistics are collected at preconfigured intervals and not in real time, this object might return a value that is different from the current value. The default interval for LDP statistics collection is **5** minutes.
jnxLdpTransitPackets	**jnxLdpStatsEntry 9**	Shows the number of packets of traffic originated from a different router, but destined for this FEC, and forwarded over the current LDP FEC. Because the LDP statistics are collected at preconfigured intervals and not in real time, this object might return a value that is different from the current value. The default interval for LDP statistics collection is **5** minutes.

Related Documentation

- LDP MIB on page 517
- LDP Notification Objects and Notification Types on page 518

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CHAPTER 62

Interpreting the Enterprise-Specific EX Series SMI MIB

- EX Series SMI MIB on page 523

EX Series SMI MIB

The Juniper Networks enterprise-specific EX Series Structure of Management Information (SMI) MIB leverages the **jnxExMibRoot** object from the enterprise-specific SMI MIB (http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-smi.txt), and defines a MIB branch for switching-related MIB definitions for the EX Series Ethernet Switches. MIB objects that are specific to EX Series are identified with a **jnxEx** prefix.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ex-smi.txt .

The **jnxExMibRoot** contains one branch, **jnxExSwitching**, whose object identifier is **{jnxExMibRoot 1}**.

The **jnxExSwitching** branch contains the objects listed in Table 212 on page 523.

Table 212: jnxExSwitching

Object	Object ID	Description
jnxExAnalyzer	**jnxExSwitching 1**	Defines the branch for the enterprise-specific Analyzer MIB. For more information about the Analyzer MIB, see "Analyzer MIB" on page 529.
jnxExSecureAccessPort	**jnxExSwitching 2**	Defines the branch for the enterprise-specific Secure Access Port MIB. NOTE: The Secure Access Port MIB is not currently supported on EX Series Switches.
jnxExPaeExtention	**jnxExSwitching 3**	Defines the branch for the enterprise-specific PAE Extensions MIB. For more information about the PAE Extensions MIB, see "PAE Extension MIB" on page 543.
jnxExVirtualChassis	**jnxExSwitching 4**	Defines the branch for the enterprise-specific Virtual Chassis MIB. For more information about the Virtual Chassis MIB, see "Virtual Chassis MIB" on page 541.

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Table 212: jnxExSwitching *(continued)*

Object	Object ID	Description
jnxExVlan	**jnxExSwitching 5**	Defines the branch for the enterprise-specific VLAN MIB. For more information about the VLAN MIB, see "VLAN MIB" on page 533.

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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CHAPTER 63

Interpreting the Enterprise-Specific EX MAC Notification MIB

- EX Series MAC Notification MIB on page 525
- MAC Notification Global Group on page 526
- MAC Notification MIB Global Objects on page 526
- MAC History Index on page 527
- MAC History Changes Message on page 527

EX Series MAC Notification MIB

The **jnxMacNotification** MIB module, whose object ID is **{jnxExSwitching 7}**, is for configuring the MAC notification feature on EX Series Ethernet Switches. MAC notification is a mechanism that is used to inform monitoring devices when there are MAC addresses learned or removed from the forwarding database of the monitored devices.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-ex-mac-notification.txt .

For information about the Juniper Networks enterprise-specific EX Series MAC Notification MIB objects, see the following topics:

- MAC Notification Global Group on page 526
- MAC Notification MIB Global Objects on page 526
- MAC History Index on page 527
- MAC History Changes Message on page 527

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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MAC Notification Global Group

The **jnxMacGlobalFeatureEnabled** object indicates whether the MAC notification feature is currently running in the device. To disable the MAC notification feature globally, set this object to false. The feature is disabled at each interface.

Set this object to true to start the MAC notification feature running on the device. Once the MAC notification is enabled, the **jnxIfConfigTable** controls the functioning of the feature.

Related Documentation

- EX Series MAC Notification MIB on page 525
- MAC Notification MIB Global Objects on page 526
- MAC History Index on page 527
- MAC History Changes Message on page 527

MAC Notification MIB Global Objects

jnxMacNotificationMIBGlobalObjects table, whose object identifier is **{jnxMacNotificationMIBGlobalObjects 1}**, contains details about MAC notifications. Each **jnxMacNotification** contains the objects listed in Table 213 on page 526.

Table 213: jnxMacNotificationGlobalObjects

Object	Object ID	Description
jnxMacNotificationInterval	**jnxMacNotificationMIBGlobalObjects 1**	The MAC notification interval object specifies the maximum interval of time between notifications that are generated by the device. If the value of **jnxNotificationsEnabled** is true, the device sends out the generated notifications and archives the MAC change notification events in the **jnxMacHistory** table. If the value of **jnxNotificationEnabled** is false, the device does not send out the generated notifications, but it will archive these events in the **jnxMacHistory** table.
jnxMacAddressesLearnt	**jnxMacNotificationMIBGlobalObjects 2**	Indicates the number of MAC addresses learned by the device.
jnxMacAddressesRemoved	**jnxMacNotificationMIBGlobalObjects 3**	Indicates the number of MAC addresses removed from the forwarding database.
jnxMacNotificationsEnabled	**jnxMacNotificationMIBGlobalObjects 4**	Indicates whether MAC notifications will or will not be sent when there are MAC addresses learned or removed from the device's forwarding database. Disabling notifications does not prevent the MAC address information from being added to the **jnxMacHistoryTable**.
jnxMacNotificationsSent	**jnxMacNotificationMIBGlobalObjects 5**	Indicates the number of MAC notifications sent out by the device.

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Table 213: jnxMacNotificationGlobalObjects *(continued)*

Object	Object ID	Description
jnxMacHistTableMaxLength	**jnxMacNotificationMIBGlobalObjects 6**	Indicates the upper limit on the number of entries that the **jnxMacHistoryTable** can contain. A value of 0 prevents any history from being retained. When the table is full, the oldest entry is deleted and a new entry is created.
jnxMacHistoryTable	**jnxMacNotificationMIBGlobalObjects 7**	Archives the MAC change notification events generated by the device. The MAC change notification events are archived here even if **jnxMacChangesNotifications** are not actually sent.
jnxMacHistoryEntry	**jnxMacNotificationMIBGlobalObjects 8**	A MAC change notification message that was previously generated by the device. Each entry is indexed by a message index.

Related Documentation

- EX Series MAC Notification MIB on page 525
- MAC Notification Global Group on page 526
- MAC History Index on page 527
- MAC History Changes Message on page 527

MAC History Index

jnxHistIndex is an index that uniquely identifies a MAC change notification event previously generated by the device. This index starts at 1 and increases by one when a MAC change notification is generated. When it reaches the maximum value (**4294967295**), the agent wraps the value back to 1.

Related Documentation

- EX Series MAC Notification MIB on page 525
- MAC Notification Global Group on page 526
- MAC Notification MIB Global Objects on page 526
- MAC History Changes Message on page 527

MAC History Changes Message

The **jnxHistMacChangedMsg** object contains details of a MAC change notification event. It consists of several tuples packed together in the format **tuple1, tuple2, ...**

Each tuple comprises 13 octets in the following format:
<operation><vlan><MAC><dot1dbaseport>.

<operation> is of size 1 octet and supports the following values:

- **0**–End of MIB object.
- **1**–MAC learned.

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- **2**–MAC removed.

<vlan> is the VLAN number of the VLAN whose MAC address is being used and has a size of 2 octets.

<Mac> is the Layer 2 MAC address and has a size of 6 octets.

<dot1dBasePort> is the value of **dot1dBasePort** for the interface from which the MAC address is learned and has a size of 4 octets.

Related Documentation

- EX Series MAC Notification MIB on page 525
- MAC Notification Global Group on page 526
- MAC Notification MIB Global Objects on page 526
- MAC History Index on page 527

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CHAPTER 64

Interpreting the Enterprise-Specific Analyzer MIB

- Analyzer MIB on page 529
- jnxAnalyzerTable on page 530
- jnxAnalyzerInputTable on page 530
- jnxAnalyzerOutputTable on page 531

Analyzer MIB

The Juniper Networks enterprise-specific Analyzer MIB, whose object identifier is **{jnxExAnalyzer 1}**, contains analyzer and remote analyzer data related to port mirroring on the EX Series Ethernet Switches. Port mirroring is a method used on enterprise switches to monitor and analyze traffic on the network.

When port mirroring is enabled, copies of all (or a sample set of) packets are forwarded from one port of the switch to another port on the same switch (analyzer) or on another switch (remote analyzer) where the packet can be analyzed and studied.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-analyzer.txt .

For information about the enterprise-specific Analyzer MIB objects, see the following topics:

- jnxAnalyzerTable on page 530
- jnxAnalyzerInputTable on page 530
- jnxAnalyzerOutputTable on page 531

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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jnxAnalyzerTable

The **jnxAnalyzerTable**, whose object identifier is **{jnxAnalyzerMIBObjects 1}**, contains information about analyzer parameters. Each **jnxAnalyzerEntry** contains the objects listed in Table 214 on page 530.

Table 214: jnxAnalyzerTable

Object	Object ID	Description
jnxAnalyzerName	**jnxAnalyzerEntry 1**	Uniquely identifies an analyzer configured on the switch.
jnxMirroringRatio	**jnxAnalyzerEntry 2**	Specifies the mirroring ratio. This object defines the sample size for mirroring. For example, **1** out of every ***x*** packets, where ***x*** is a number in the range of 1 through 2047.
jnxLossPriority	**jnxAnalyzerEntry 3**	Specifies the loss priority for a packet. When the number of packets at the analyzer port exceeds the bandwidth of the analyzer port, packets are dropped based on the loss priority value. When there is a bandwidth crunch, packets with high loss priority are dropped to clear the congestion. This object uses the following integer values to denote the loss priority: • **0**—Low loss priority • **1**—High loss priority

Related Documentation

- Analyzer MIB on page 529
- jnxAnalyzerOutputTable on page 531
- jnxAnalyzerInputTable on page 530

jnxAnalyzerInputTable

The **jnxAnalyzerInputTable** (obsolete in this release), whose object identifier is **{jnxAnalyzerMIBObjects 2}**, contains information about analyzer sessions. In a typical analyzer session, several source ports can be associated with a single destination port, and a range or series of ports can be mirrored.

Each **jnxAnalyzerInputEntry** provides information about input source ports, and contains the objects listed in Table 215 on page 530.

Table 215: jnxAnalyzerInputTable

Object	Object ID	Description
jnxAnalyzerInputValue	**jnxAnalyzerInputEntry 1**	Identifies an analyzer input source port. This object can contain a display string of not more than 255 characters. • If the value of **jnxAnalyzerInputType** is **1**, then the value of **jnxAnalyzerInputValue** denotes the interface name of the input source. • If the value of **jnxAnalyzerInputType** is **2**, then the value of **jnxAnalyzerInputValue** denotes the VLAN name of the input source.

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Table 215: jnxAnalyzerInputTable *(continued)*

Object	Object ID	Description
jnxAnalyzerInputOption	**jnxAnalyzerInputEntry 2**	Denotes the type of traffic to be mirrored from the source port; that is, whether it is ingress traffic or egress traffic. This object uses the following integer values: • **1**–Ingress traffic, where the analyzer monitors packets received by the source port. • **2**–Egress traffic, where the analyzer monitors packets transmitted by the source port. In both the cases, the number of packets mirrored to the destination port depends on the **jnxMirroringRatio**.
jnxAnalyzerInputType	**jnxAnalyzerInputEntry 3**	Denotes whether the mirroring source is an interface or a VLAN. This object uses integer values **1** (for interface) and **2** (for VLAN). For interfaces, you can configure either ingress or egress mirroring, whereas for VLANS, you can configure only ingress mirroring.

Related Documentation

- Analyzer MIB on page 529
- jnxAnalyzerTable on page 530
- jnxAnalyzerOutputTable on page 531

jnxAnalyzerOutputTable

The **jnxAnalyzerOutputTable**, whose object identifier is **{jnxAnalyzerMIBObjects 3}**, contains information about the destination port to which the packets are mirrored. Each **jnxAnalyzerOutputEntry** contains the objects listed in Table 216 on page 531, and provides information about the destination port or destination VLAN.

Table 216: jnxAnalyzerOutputTable

Object	Object ID	Description
jnxAnalyzerOutputValue	**jnxAnalyzerOutputEntry 1**	Uniquely identifies a destination port or VLAN. This object can contain a string of not more than 255 characters. If the value of **jnxAnalyzerOutputType** is **1**, then **jnxAnalyzerOutputValue** contains an interface name. If the value of **jnxAnalyzerOutputType** is **2**, then **jnxAnalyzerOutputValue** contains a VLAN name.
jnxAnalyzerOutputType	**jnxAnalyzerOutputEntry 2**	Denotes the type of the output destination port. This object uses integer values **1** (for a destination port that is on the same switch) and **2** (for remote analyzer, that is a dedicated VLAN on a different switch).

Related Documentation

- Analyzer MIB on page 529
- jnxAnalyzerTable on page 530

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- jnxAnalyzerInputTable on page 530

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CHAPTER 65

Interpreting the Enterprise-Specific VLAN MIB

- VLAN MIB on page 533
- VLAN Configuration Table on page 534
- VLAN Interface Table on page 535
- Port Group Table on page 537
- MAC List Table on page 539

VLAN MIB

The Juniper Networks enterprise-specific VLAN MIB for EX Series Ethernet Switches, whose object identifier is **{jnxExSwitching 5}**, contains information about prestandard IEEE 802.10 VLANS and their association with LAN Emulation Clients (LAC). Devices with prestandard implementation maintain port groupings and associated filters that are used to form a virtual bridge.

The enterprise-specific VLAN MIB leverages the following objects and data types from standard MIBs, RFCs, and Juniper Networks enterprise-specific MIBs:

- **Integer 32** and **IpAddress**–From SNMPv2–SMI
- **MacAddress**, **DisplayString**, and **TruthValue**–From SNMPv2–TC
- **InterfaceIndex**–From IF MIB
- **InetAddress** and **InetAddressType**–From Inet Address MIB
- **jnxExVlan**–From Juniper Networks enterprise-specific SMI MIB

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-vlan.txt .

NOTE: **jnxVlanTable, jnxVlanInterfaceTable,** and **jnxVlanPortGroupTable** have been deprecated and are replaced by **jnxExVlanTable, jnxExVlanInterfaceTable,** and **jnxExVlanPortGroupTable.**

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For information about the enterprise-specific VLAN MIB objects, see the following topics:

- VLAN Configuration Table on page 534
- VLAN Interface Table on page 535
- Port Group Table on page 537
- MAC List Table on page 539

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

VLAN Configuration Table

NOTE: **jnxVlanTable** has been deprecated and is replaced by **jnxExVlanTable**. See "jnxExVlanTable" on page 534.

The **jnxVlanTable**, whose object identifier is **{jnxVlanMIbObjects 1}**, contains VLAN names and properties. Each **jnxVlanEntry** contains the objects listed in Table 217 on page 534.

Table 217: jnxVlanTable

Object	Object ID	Description
jnxVlanName	**jnxVlanEntry 1**	Contains the name of the VLAN, which can be a string of not more than **255** characters.
jnxVlanID	**jnxVlanEntry 2**	Contains the identifier used internally by the device to reference the VLAN. This object can be an integer value in the range **1** through **4094**.
jnxVlanType	**jnxVlanEntry 3**	Denotes the type of the VLAN. This object uses the following integer values: • **1**–Static • **2**–Dynamic The default value for this object is **1**, static.
jnxVlanPortGroupInstance	**jnxVlanEntry 4**	Contains the index that identifies the subtree in the **jnxVlanPortGroupTable**.
jnxVlanMacListInstance	**jnxVlanEntry 5**	Contains an index that identifies the subtree to retrieve the list of MAC addresses to the **jnxVlanMacListTable** subtree to retrieve in.

jnxExVlanTable

jnxExVlanTable, whose object identifier is **{jnxVlanMIBObjects 5}**, replaces the deprecated **jnxVlanTable** and contains the objects listed in Table 218 on page 535.

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Table 218: jnxExVlanTable

Object	Object ID	Description
jnxExVlanID	**jnxExVlanEntry 1**	Contains the identifier used internally by the device to reference the VLAN. This object can be an integer value in the range **1** through **4094**.
jnxExVlanName	**jnxExVlanEntry 2**	Contains the name of the VLAN. VLAN name can be a string of not more than **255** characters.
jnxExVlanType	**jnxExVlanEntry 3**	Denotes the type of the VLAN. This object uses the following integer values: • **1**–Static • **2**–Dynamic The default value for this object is **1**, static.
jnxExVlanPortGroupInstance	**jnxExVlanEntry 4**	Contains the index that identifies the subtree in the **jnxExVlanPortGroupTable**.

Related Documentation

- VLAN MIB on page 533
- VLAN Interface Table on page 535
- Port Group Table on page 537
- MAC List Table on page 539

VLAN Interface Table



NOTE: **jnxVlanInterfaceTable** has been deprecated and is replaced by **jnxExVlanInterfaceTable**. See "jnxExVlanInterfaceTable" on page 536.

The **jnxVlanInterfaceTable**, whose object identifier is **{jnxVlanMIBObjects 2}**, contains information about the Layer 3 properties of VLANs. Each **jnxVlanInterfaceEntry**, indexed with **jnxVlanName**, contains the objects listed in Table 219 on page 535.

Table 219: jnxVlanInterfaceTable

Object	Object ID	Description
jnxVlanInterfaceIpAddress	**jnxVlanInterfaceEntry 1**	Contains the IP address of the interface.
jnxVlanInterfaceProtocol	**jnxVlanInterfaceEntry 2**	Specifies the protocol used.
jnxVlanInterfaceSubNetMask	**jnxVlanInterfaceEntry 3**	Specifies the subnet mask of the VLAN,
jnxVlanInterfaceBroadcastAddress	**jnxVlanInterfaceEntry 4**	Specifies the broadband address of the VLAN.

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Table 219: jnxVlanInterfaceTable *(continued)*

Object	Object ID	Description
jnxVlanInterfaceDescription	jnxVlanInterfaceEntry 5	Contains a description for the VLAN. This object can contain a string of not more than 255 characters.
jnxVlanInterfaceAdminStatus	jnxVlanInterfaceEntry 6	Denotes the administration status of the VLAN.
jnxVlanInterfaceOperStatus	jnxVlanInterfaceEntry 7	Denotes the operational status of the VLAN.
jnxVlanSnmpIfIndex	jnxVlanInterfaceEntry 8	Specifies the SNMP IF Index for the interface.

jnxExVlanInterfaceTable

jnxExVlanInterfaceTable, whose object identifier is **{jnxVlanMIBObjects 6}**, replaces the deprecated **jnxVlanInterfaceTable** and contains the objects listed in Table 220 on page 536.

Table 220: jnxExVlanInterfaceTable

Object	Object ID	Description
jnxExVlanInterfaceProtocol	jnxExVlanInterfaceEntry 1	Specifies the protocol used.
jnxExVlanInterfaceIpAddress	jnxExVlanInterfaceEntry 2	Contains the IP address of the interface.
jnxExVlanInterfacePrefixLength	jnxExVlanInterfaceEntry 3	Specifies the subnet mask of the VLAN,
jnxExVlanInterfaceBroadcastAddress	jnxExVlanInterfaceEntry 4	Specifies the broadband address of the VLAN.
jnxExVlanInterfaceDescription	jnxExVlanInterfaceEntry 5	Contains a description for the VLAN. This object can contain a string of not more than 255 characters.
jnxExVlanInterfaceAdminStatus	jnxExVlanInterfaceEntry 6	Denotes the administration status of the VLAN.
jnxExVlanInterfaceOperStatus	jnxExVlanInterfaceEntry 7	Denotes the operational status of the VLAN.
jnxExVlanSnmpIfIndex	jnxExVlanInterfaceEntry 8	Specifies the SNMP IF Index for the interface.

Related Documentation

- VLAN MIB on page 533
- VLAN Configuration Table on page 534
- Port Group Table on page 537
- MAC List Table on page 539

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Port Group Table

NOTE: **jnxVlanPortGroupTable** has been deprecated and is replaced by **jnxExVlanPortGroupTable**. See "jnxExVlanPortGroupTable" on page 537.

The **jnxVlanPortGroupTable** contains information about port groupings. Each **jnxVlanPortGroupEntry** contains the objects listed in Table 221 on page 537.

Table 221: jnxVlanPortGroupTable

Object	Object ID	Description
jnxVlanPortGroupIndex	**jnxVlanPortGroupEntry 1**	Uniquely identifies a port group.
jnxVlanPort	**jnxVlanPortGroupEntry 2**	Specifies the port on the VLAN with which this port group is associated.
jnxVlanPortStatus	**jnxVlanPortGroupEntry 3**	Shows the status of association between the port and the VLAN. This object uses the following integer values: • **1–autoActive**: The port is part of the VLAN because the switch has automatically added the port. • **2–allowed**: The port has been configured to be part of the VLAN, and is allowed to be part of the VLAN, if the port meets all other requirements. • **3–allowedActive**: The port has been configured to be part of the VLAN, and is allowed to be part of the VLAN, if the port meets all other requirements. However, unlike in the case of **allowed** ports, this port has a device that is participating in the VLAN associated with the port. • **4–allowedNotAvail**: The port is active on some other VLAN, and is not available currently. This value applies to devices that do not allow a port to be part of more than one VLAN at the same time. • **5–notAssociated**: The port is part of a port group that is not associated with the VLAN.

jnxExVlanPortGroupTable

jnxExVlanPortGroupTable replaces the deprecated **jnxVlanPortGroupTable** and contains the objects listed in Table 222 on page 538.

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Table 222: jnxExVlanPortGroupTable

Object	Object ID	Description
jnxExVlanPortGroupIndex	**jnxExVlanPortGroupEntry 1**	Uniquely identifies a port group.
jnxExVlanPort	**jnxExVlanPortGroupEntry 2**	Specifies the port on the VLAN with which this port group is associated.
jnxExVlanPortStatus	**jnxExVlanPortGroupEntry 3**	Shows the status of association between the port and the VLAN. This object uses the following integer values: • **1—autoActive**: The port is part of the VLAN because the switch has automatically added the port. • **2—allowed**: The port has been configured to be part of the VLAN, and is allowed to be part of the VLAN, if the port meets all other requirements. • **3—allowedActive**: The port has been configured to be part of the VLAN, and is allowed to be part of the VLAN, if the port meets all other requirements. However, unlike in the case of **allowed** ports, this port has a device that is participating in the VLAN associated with the port. • **4—allowedNotAvail**: The port is active on some other VLAN, and is not available currently. This value applies to devices that do not allow a port to be part of more than one VLAN at the same time. • **5—notAssociated**: The port is part of a port group that is not associated with the VLAN. Default value for this object is **allowed**.
jnxExVlanPortTagness	**jnxExVlanPortGroupEntry 4**	Specifies whether the port is tagged or untagged.
jnxExVlanPortAccessMode	**jnxExVlanPortGroupEntry 5**	Specifies whether the port access mode is Access or Trunk.

Related Documentation

- VLAN MIB on page 533
- VLAN Configuration Table on page 534
- VLAN Interface Table on page 535
- MAC List Table on page 539

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MAC List Table

The **jnxVlanMacListTable**, whose object identifier is **{jnxVlanMIBObjects 4}**, contains information about MAC address lists. Each **jnxVlanMacListEntry** contains the objects listed in Table 223 on page 539.

Table 223: jnxVlanMacListTable

Object	Object ID	Description
jnxVlanMacListIndex	**jnxVlanMacListEntry 1**	Uniquely identifies a MAC address list.
jnxVlanMacAddress	**jnxVlanMacListEntry 2**	Specifies a MAC address that belongs to the group.

Related Documentation

- VLAN MIB on page 533
- VLAN Configuration Table on page 534
- VLAN Interface Table on page 535
- Port Group Table on page 537

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CHAPTER 66

Interpreting the Enterprise-Specific Virtual Chassis MIB

- Virtual Chassis MIB on page 541
- jnxVirtualChassisMemberTable on page 541

Virtual Chassis MIB

The Juniper Networks enterprise-specific Virtual Chassis MIB, whose object identifier is **{jnxExSwitching 4}**, contains information about virtual chassis on EX Series Ethernet Switches. EX4200 Switches allow you to connect two or more switches (maximum ten) together to form a virtual chassis that can be managed as a single network element. The switches can be connected through dedicated 64 Gbps virtual chassis ports (VCPs) or through 10 Gbps fiber uplink ports.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-virtualchassis.txt .

For information about the enterprise-specific Virtual Chassis MIB objects, see the following topic:

- jnxVirtualChassisMemberTable on page 541

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

jnxVirtualChassisMemberTable

The **jnxVirtualChassisMemberTable**, whose object identifier is **{jnxVirtualChassisMemberMIB 1}**, contains information about the devices that form the virtual chassis. Each **jnxVirtualChassisMemberEntry** contains the objects listed in Table 224 on page 542.

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Table 224: jnxVirtualChassisMemberTable

Object	Object ID	Description
jnxVirtualChassisMemberId	jnxVirtualChassisMemberEntry 1	Uniquely identifies a virtual chassis member. This object contains integer values in the range **0** through **9**.
jnxVirtualChassisMemberSerialnumber	jnxVirtualChassisMemberEntry 2	Contains the serial number of the virtual chassis member.
jnxVirtualChassisMemberRole	jnxVirtualChassisMemberEntry 3	Specifies the type of virtual chassis member. This object uses the following integer values: • **1**–Master • **2**–Backup • **3**–Linecard
jnxVirtualChassisMemberMacAddBase	jnxVirtualChassisMemberEntry 4	Specifies the media access control (MAC) address base for the virtual-chassis member.
jnxVirtualChassisMemberSWVersion	jnxVirtualChassisMemberEntry 5	Identifies the Junos Base operating system software suite that is installed on the virtual chassis member.
jnxVirtualChassisMemberPriority	jnxVirtualChassisMemberEntry 6	Specifies the priority of the virtual-chassis member. This object contains integer values in the range **1** through **255**.
jnxVirtualChassisMemberUptime	jnxVirtualChassisMemberEntry 7	Specifies the virtual chassis member uptime.

Related Documentation

- Virtual Chassis MIB on page 541

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CHAPTER 67

Interpreting the Enterprise-Specific PAE Extension MIB

- PAE Extension MIB on page 543
- jnxAuthProfileName on page 544
- jnxPaeAuthConfigTable on page 544
- jnxStaticMacAuthBypassTable on page 545
- jnxStaticMacAuthBypassIfTable on page 545

PAE Extension MIB

The Juniper Networks enterprise-specific Port Access Entity (PAE) Extension MIB, whose object identifier is **{jnxExSwitching 3}**, is an extension of the standard IEEE802.1x PAE Extension MIB, and contains information for static MAC authentication. The enterprise-specific PAE Extension MIB has two branches, **jnxPaeExtensionMIBNotification** and **jnxPaeExtensionMIBObjects**.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-pae-extension.txt .

For information about the enterprise-specific PAE Extension MIB objects, see the following topics:

- jnxAuthProfileName on page 544
- jnxPaeAuthConfigTable on page 544
- jnxStaticMacAuthBypassTable on page 545
- jnxStaticMacAuthBypassIfTable on page 545

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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jnxAuthProfileName

The **jnxAuthProfileName** object, whose object identifier is **{jnxPaeExtensionMIBObjects 1}**, contains the authentication profile name. The authentication profile contains the RADIUS server IP address, the port number, and the secret key.

Related Documentation

- PAE Extension MIB on page 543
- jnxPaeAuthConfigTable on page 544
- jnxStaticMacAuthBypassTable on page 545
- jnxStaticMacAuthBypassIfTable on page 545

jnxPaeAuthConfigTable

The **jnxPaeAuthConfigTable**, whose object identifier is **{jnxPaeExtensionMIBObjects 2}**, contains the configuration objects for the Authenticator PAE associated with each port. Each **jnxPaeAuthConfigEntry**, indexed with **dot1xPaePortNumber** from the standard IEEE802.1x PAE Extension MIB, contains the objects listed in Table 225 on page 544.

Table 225: jnxPaeAuthConfigTable

Object	Object ID	Description
jnxPaeAuthConfigMacAuthStatus	jnxPaeAuthConfigEntry 1	Shows whether MAC authentication is enabled on the specified PAE port.
jnxPaeAuthConfigGuestVlan	jnxPaeAuthConfigEntry 2	Specifies the VLAN to which an unauthenticated client moves. This object can contain a string of not more than 255 characters.
jnxPaeAuthConfigNumberRetries	jnxPaeAuthConfigEntry 3	Specifies the maximum number of failed authentication retries allowed on an interface before the interface transitions into the quiet period. No authentication happens on the interface during the quiet period.

Related Documentation

- PAE Extension MIB on page 543
- jnxAuthProfileName on page 544
- jnxStaticMacAuthBypassTable on page 545
- jnxStaticMacAuthBypassIfTable on page 545

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jnxStaticMacAuthBypassTable

The **jnxStaticMacAuthBypassTable**, whose object identifier is **{jnxPaeExtensionMIBObjects 3}**, contains a static list of MAC addresses specified by a user. The static MAC address list contains the MAC addresses of clients associated with a port. The clients whose MAC addresses are in the MAC address list are allowed to connect to the port without authentication. The 802.1X or MAC authentication process is initiated for a connection request only when a matching entry is not available for the client in the **jnxStaticMacAuthBypassTable**.

The **jnxStaticMacAuthBypassTable** allows devices like printers that do not support 802.1X to connect to 802.1X-enabled ports.

Each **jnxStaticMacAuthBypassEntry**, whose object identifier is **{jnxStaticMacAuthBypassTable 1}**, contains the objects listed in Table 226 on page 545.

Table 226: jnxStaticMacAuthBypassTable

Object	Object ID	Description
jnxStaticMacAddress	**jnxStaticMacAuthBypassEntry 1**	Specifies the MAC address of the client connected to the PAE port.
jnxStaticMacVlanName	**jnxStaticMacAuthBypassEntry 2**	Specifies the VLAN to which the client is assigned.

Related Documentation

- PAE Extension MIB on page 543
- jnxAuthProfileName on page 544
- jnxPaeAuthConfigTable on page 544
- jnxStaticMacAuthBypassIfTable on page 545

jnxStaticMacAuthBypassIfTable

The **jnxStaticMacAuthBypassIfTable**, whose object identifier is **{jnxPaeExtensionMIBObjects 4}**, contains a list of interfaces associated with the MAC addresses in the **jnxStaticMacAuthBypassTable**.

Each **jnxStaticMacAuthBypassIfEntry**, whose object identifier is **{jnxStaticMacAuthBypassIfTable 1}**, is indexed with **jnxStaticMacAddress** (from **jnxStaticMacAuthBypassTable**) and **jnxStaticMacIfIndex**. The **jnxStaticMacIfIndex** contains a list of interfaces from which a MAC address is allowed. If the interface associated with a MAC address does not match the one stored in this entry, the authentication bypass does not occur.

Related Documentation

- PAE Extension MIB on page 543
- jnxAuthProfileName on page 544
- jnxPaeAuthConfigTable on page 544

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- jnxStaticMacAuthBypassTable on page 545

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CHAPTER 68

Interpreting the Enterprise-Specific Power Supply Unit MIB

- Power Supply Unit MIB on page 547
- Power Supply Unit Environment Group on page 548
- Power Supply Unit Device Power Group on page 548
- Power Supply Unit Outlets Group on page 549
- Power Source Equipment Objects Group on page 549
- Power Supply Unit FPC Power Allocated Information on page 550

Power Supply Unit MIB

The **jnxPsuMIB** MIB module is for enabling power monitoring and management of a switch. This is handled by two modules: **jnxPsuNotifications** module, whose object identifier is **{jnxPsuMIB 1}**, and **jnxPsuObjects** module, whose object identifier is **{jnxPsuMIB 2}**.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-power-supply-unit.txt .

For information about the Juniper Networks enterprise-specific Power Supply Unit (PSU) MIB objects, see the following topics:

- Power Supply Unit Environment Group on page 548
- Power Supply Unit Device Power Group on page 548
- Power Supply Unit Outlets Group on page 549
- Power Source Equipment Objects Group on page 549
- Power Supply Unit FPC Power Allocated Information on page 550

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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Power Supply Unit Environment Group

The **jnxPsuEnvironmentTable** lists all the environment entries for each PSU component. The **jnxPsuEnvironmentEntry** contains the objects listed in Table 227 on page 548.

Table 227: jnxPsuEnvironmentTable

Object	Object Identifier	Description
jnxPsuThermalValue	jnxPsuEnvironmentEntry 1	The temperature (in degrees Celsius) for each PSU component.
jnxPsuHumidityValue	jnxPsuEnvironmentEntry 2	The humidity (in percentage) for each PSU component.

Related Documentation

- Power Supply Unit MIB on page 547
- Power Source Equipment Objects Group on page 549
- Power Supply Unit Device Power Group on page 548
- Power Supply Unit Outlets Group on page 549
- Power Supply Unit FPC Power Allocated Information on page 550

Power Supply Unit Device Power Group

The **jnxPsuTable** lists all the power entries for each PSU component. The **jnxPsuEntry** contains the objects listed in Table 228 on page 548.

Table 228: jnxPsuTable

Object	Object Identifier	Description
jnxPsuAvgPower	jnxPsuEntry 1	The buffer that specifies the average power in watts used for each component.
jnxPsuMaxPower	jnxPsuEntry 2	The buffer that specifies the maximum power in watts available for each component.
jnxPsuMode	jnxPsuEntry 3	The mode for each PSU component.
jnxPsuOutletCount	jnxPsuEntry 4	The number of outlets present on a particular PSU component. The default number is 0.

Related Documentation

- Power Supply Unit MIB on page 547
- Power Source Equipment Objects Group on page 549
- Power Supply Unit Environment Group on page 548
- Power Supply Unit Outlets Group on page 549

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- Power Supply Unit FPC Power Allocated Information on page 550

Power Supply Unit Outlets Group

The **jnxPsuOutletTable** lists all the outlet entries. The number of entries is specified by the **jnxPsuOutletPsuIndex**. The **jnxPsuOutletEntry** contains the objects listed in Table 229 on page 549.

Table 229: jnxPsuOutletTable

Object	Object Identifier	Description
jnxPsuOutletName	jnxPsuOutletEntry 1	The outlet name associated with the power supply unit for each PSU Component.
jnxPsuOutletDescription	jnxPsuOutletEntry 2	The outlet description associated with the power supply unit for each PSU Component.
jnxPsuOutletAvgPower	jnxPsuOutletEntry 3	The buffer that specifies the average power in watts used for each component.
jnxPsuOutletMaxPower	jnxPsuOutletEntry 4	The buffer that specifies the maximum power in watts available for each component.
jnxPsuOutletCurrent	jnxPsuOutletEntry 5	The actual current in milliamps that flows through the given outlet.
jnxPsuOutletStatus	jnxPsuOutletEntry 8	The operational status for the given outlet. This can also be used to set the outlet state.

Related Documentation

- Power Supply Unit MIB on page 547
- Power Source Equipment Objects Group on page 549
- Power Supply Unit Device Power Group on page 548
- Power Supply Unit Environment Group on page 548

Power Source Equipment Objects Group

The **pethObjects** module consists of the following:

- The **pethPsePortTable** lists the objects that display and control the power characteristics of Power over Ethernet (POE) ports on a PSE device. The objects are identified by **pethPsePortEntry**.
- The **pethPsePortGroupIndex** identifies the group containing the port to which a power Ethernet PSE is connected.
- The **pethPsePortIndex** specifies the PSE port within the group **pethPsePortGroupIndex** to which a PSE entry is connected.

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- The **pethPsePortAdminEnable** specifies whether or not an interface can provide the PSE functions.
- The **pethPsePortPowerPairsControlAbility** specifies whether or not the device can control power pairs in order to switch pins for sourcing power.
- The **pethPsePortPowerPairs** specifies power controls for the power pairs in use.
  - **pethPsePortPowerPairsControl** value is true - Object is writable
  - **pethPsePortPowerPairsControl** value is signal(1) - Signal pairs only are in use
  - **pethPsePortPowerPairsControl** value is spare(2) - Spare pairs only are in use
- The **pethPsePortDetectionStatus** specifies the operational status of the port PD detection.
- The **pethPsePortPowerPriority** specifies the priority of the port to aid efficient power management.
- The **pethPsePortMPSAbsentCounter** displays the PSE state diagram transitions from **POWER_ON** to **IDLE** to **tmpdo_timer_done**.
- The **pethPsePortType** specifies the type of powered device that is connected to the port. The default value is a zero-length octet string.
- The **pethPsePortPowerClassifications** specifies how the different terminals on the Power over LAN network are tagged according to their power consumption.
- The **pethPsePortInvalidSignatureCounter** displays the invalid state of the port when the port enters the state **SIGNATURE_INVALID**.
- The **pethPsePortPowerDeniedCounter** indicates that the port is denied power when the port enters the state **POWER_DENIED**.
- The **pethPsePortOverLoadCounter** indicates that the port is overloaded with power when the port enters the state **ERROR_DELAY_OVER**.
- The **pethPsePortShortCounter** indicates that the port is not supplied with sufficient power when the port enters the state **ERROR_DELAY_OVER**.

Related Documentation

- Power Supply Unit MIB on page 547
- Power Supply Unit Device Power Group on page 548
- Power Supply Unit Environment Group on page 548
- Power Supply Unit Outlets Group on page 549
- Power Supply Unit FPC Power Allocated Information on page 550

Power Supply Unit FPC Power Allocated Information

The **jnxPsuFpcPowerTable** lists the entries for each FPC (Flexible PIC Concentrator), their assigned priority levels, and allocated power. The **jnxPsuFpcPowerEntry** contains the objects listed in Table 230 on page 551.

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Table 230: jnxPsuFpcPowerTable

Object	Object Identifier	Description
jnxPsuFpcPowerPriority	jnxPsuFpcPowerEntry 1	The power budget priority assigned to the FPC. A lower number indicates a higher priority.
jnxPsuFpcPowerAllocated	jnxPsuFpcPowerEntry 2	The power (in Watts) allocated to the FPC.

Related Documentation

- Power Supply Unit MIB on page 547
- Power Source Equipment Objects Group on page 549
- Power Supply Unit Device Power Group on page 548
- Power Supply Unit Environment Group on page 548
- Power Supply Unit Outlets Group on page 549

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CHAPTER 69

Interpreting the Enterprise-Specific SPU Monitoring MIB

- SPU Monitoring Objects MIB on page 553
- SPU Monitoring Objects Table on page 554

SPU Monitoring Objects MIB

The Juniper Networks enterprise-specific Services Processing Unit (SPU) Monitoring Objects MIB, **jnxJsSecPolicyMIB**, whose object ID is **{jnxJsSPUMonitoringMIB 1}**, defines the MIB for SPU monitoring for SRX5600 and SRX5800 services gateways.

Related MIB objects include the following:

- **jnxJsSPUMonitoringObjectsTable**, whose object identifier is **{jnxJsSPUMonitoringMIB 1}**, provides statistics on the utilization of SPUs.
- **jnxJsSPUMonitoringCurrentTotalSession**, whose object identifier is **{jnxJsSPUMonitoringMIB 2}**, provides the total number of sessions in use at the system level.
- **jnxJsSPUMonitoringMaxTotalSession**, whose object identifier is **{jnxJsSPUMonitoringMIB 3}**, provides the maximum number of sessions possible at the system level.
- **jnxSPUClusterObjectsTable**, whose object identifier is **{jnxJsSPUMonitoringMIB 4}**, identifies the SPU monitoring objects in a chassis cluster.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-js-spu-monitoring.txt

For information about the enterprise-specific SPU Monitoring Objects MIB or SPU Cluster MIB, see the following topics:

- SPU Monitoring Objects Table on page 554
- Cluster Objects Table

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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SPU Monitoring Objects Table

The object identifier for the **jnxJsMonitoringObjectsTable** is **{jnxJsSPUMonitoringMIB 1}**. SPU monitoring objects provide statistical information related to utilization of SPUs. Table 231 on page 554 describes the SPU monitoring objects.

Table 231: SPU Monitoring Objects Table

Object	Object ID	Description
jnxJsSPUMonitoringObjects	**jnxJsSPUMonitoringObjectsTable 1**	Each entry collects information about the memory utilization for an SPU.
JnxJsSPUMonitoringObjectsEntry		Indexes: • **jnxJsSPUMonitoringIndex** • **jnxJsSPUMonitoringFPCIndex** • **jnxJsSPUMonitoringSPUIndex** Sequence of parameters: • **jnxJsSPUMonitoringCPUUsage** • **jnxJsSPUMonitoringMemoryUsage** • **jnxJsSPUMonitoringCurrentFlowSession** • **jnxJsSPUMonitoringMaxFlowSession** • **jnxJsSPUMonitoringCurrentCPSession** • **jnxJsSPUMonitoringMaxCPSession** • **jnxJsSPUMonitoringMaxTotalSession** • **jnxJsSPUMonitoringNodeIndex** • **jnxJsSPUMonitoringNodeDescr**
jnxJsSPUMonitoringIndex	**jnxJsSPUMonitoringObjectsEntry 1**	Indicates an SPU's overall index in the system.
jnxJsSPUMonitoringFPCIndex	**jnxJsSPUMonitoringObjectsEntry 2**	Indicates which FPC the SPU is on .
jnxJsSPUMonitoringSPUIndex	**jnxJsSPUMonitoringObjectsEntry 3**	Indicates the index of an SPU inside the FPC.
jnxJsSPUMonitoringCPUUsage	**jnxJsSPUMonitoringObjectsEntry 4**	Indicates the current utilization percentage of an SPU.
jnxJsSPUMonitoringMemoryUsage	**jnxJsSPUMonitoringObjectsEntry 5**	Indicates the current percentage of memory usage of an SPU.
jnxJsSPUMonitoringCurrentFlowSession	**jnxJsSPUMonitoringObjectsEntry 6**	Indicates the current flow sessions of an SPU.
jnxJsSPUMonitoringMaxFlowSession	**jnxJsSPUMonitoringObjectsEntry 7**	Indicates the maximum flow sessions of an SPU.

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Table 231: SPU Monitoring Objects Table *(continued)*

Object	Object ID	Description
jnxJsSPUMonitoringCurrentCPSession	jnxJsSPUMonitoringObjectsEntry 8	Indicates the current number of central point (CP) sessions on an SPU.
jnxJsSPUMonitoringMaxCPSession	jnxJsSPUMonitoringObjectsEntry 9	Indicates the maximum number of CP sessions on an SPU.
jnxJsSPUMonitoringMaxTotalSession	jnxJsSPUMonitoringObjectsEntry 10	Indicates the maximum number of sessions on the device.
jnxJsSPUMonitoringNodeIndex	jnxJsSPUMonitoringObjectsEntry 11	Identifies a chassis in single or cluster mode.
jnxJsSPUMonitoringNodeDescr	jnxJsSPUMonitoringObjectsEntry 12	Describes the chassis cluster node information.

Related Documentation

- SPU Monitoring Objects MIB on page 553

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CHAPTER 70

Interpreting the Enterprise-Specific OTN Interface Management MIB

- OTN Interface Management MIB on page 557
- OTN Interface Management MIB Textual Conventions on page 557
- jnxOtnAlarmTable on page 558
- jnxOtnCurrentOdu15minTable on page 559

OTN Interface Management MIB

The Juniper Networks enterprise-specific Optical Transport Network (OTN) Interface Management MIB, whose object identifier is **{jnxMibs 56}**, extends the standard OTN Interface Management MIB and defines the objects for managing OTN interfaces for Juniper Networks devices.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-otn.txt .

For information about the enterprise-specific OTN Interface Management MIB objects, see the following topics:

- OTN Interface Management MIB Textual Conventions on page 557
- jnxOtnAlarmTable on page 558
- jnxOtnCurrentOdu15minTable on page 559

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

OTN Interface Management MIB Textual Conventions

JnxOtnAlarmId specifies the SONET/SDH alarms present on an interface. The possible values are:

- **otnLosAlarm (0)**—OTN loss of signal alarm
- **otnLofAlarm (1)**—OTN loss of frame alarm

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- **otnLomAlarm (2)**—OTN loss of multiframe alarm
- **otnWavelengthlockAlarm (3)**—OTN wavelength lock alarm
- **otnOtuAisAlarm (4)**—OTN OTU AIS alarm
- **otnOtuBdiAlarm (5)**—OTN OTU BDI alarm
- **otnOtuTtimAlarm (6)**—OTN OTU TTIM alarm
- **otnOtuIaeAlarm (7)**—OTN OTU IAE alarm
- **otnOtuSdAlarm (8)**—OTN OTU bit error rate defect alarm
- **otnOtuFecExcessiveErrsAlarm (10)**—OTN OTU FEC excessive errors alarm
- **otnOtuFecDegradedErrsAlarm (11)**—OTN OTU FEC degraded errors alarm
- **otnOtuBbeThreholdAlarm (12)**—OTN OTU BBE threshold alarm
- **otnOtuEsThreholdAlarm (13)**— OTN OTU ES threshold alarm
- **otnOtuSesThreholdAlarm (14)**—OTN OTU SES threshold alarm
- **otnOtuUasThreholdAlarm (15)**—OTN OTU UAS threshold alarm
- **otnOduAisAlarm (16)**—OTN ODU AIS alarm
- **otnOduOciAlarm (17)**—OTN ODU OCI alarm
- **otnOduLckAlarm (18)**—OTN ODU LCK alarm
- **otnOduBdiAlarm (19)**—OTN ODU BDI alarm
- **otnOduTtimAlarm (20)**—OTN ODU TTIM alarm
- **otnOduSdAlarm (21)**—OTN ODU bit error rate defect alarm
- **otnOduSfAlarm (22)**—OTN ODU bit error rate fault alarm
- **otnOduRxApsChange (23)**—OTN Rx APS change
- **otnOduBbeThreholdAlarm (24)**—OTN ODU BBE threshold alarm
- **otnOduEsThreholdAlarm (25)**—OTN OTU ES threshold alarm
- **otnOduSesThreholdAlarm (26)**—OTN OTU SES threshold alarm
- **otnOduUasThreholdAlarm (27)**—OTN ODU UAS threshold alarm
- **otnOpuPMTAlarm (28)**—OTN OPU payload mismatch alarm

Related Documentation

- OTN Interface Management MIB on page 557
- jnxOtnAlarmTable on page 558
- jnxOtnCurrentOdu15minTable on page 559

jnxOtnAlarmTable

jnxOtnAlarmTable, whose object ID is **{jnxOtnAlarms 1}**, contains information about alarms on the SONET/SDH physical interfaces on a device running the Junos OS. Alarm

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information is stored in **jnxOtnAlarmEntry**. Each **jnxOtnAlarmEntry**, whose object identifier is **{jnxOtnAlarmTable 1}**, contains the objects listed in Table 232 on page 559.

Table 232: jnxOtnAlarmEntry

Object	Object ID	Description
jnxOtnCurrentAlarms	jnxOtnAlarmEntry 1	Identifies the active OTN alarms on the interface.
jnxOtnLastAlarmId	jnxOtnAlarmEntry 2	Identifies the latest OTN alarm that was set or cleared on the interface.
jnxOtnLastAlarmTime	jnxOtnAlarmEntry 3	Shows the **sysUpTime** value when the management system was informed of the last alarm event.
jnxOtnLastAlarmDate	jnxOtnAlarmEntry 4	Shows the date and time when the management system was informed about the last alarm event.
jnxOtnLastAlarmEvent	jnxOtnAlarmEntry 5	Indicates whether the last alarm event was to set a new alarm or to clear a previously set alarm.

Related Documentation

- OTN Interface Management MIB on page 557
- OTN Interface Management MIB Textual Conventions on page 557
- jnxOtnCurrentOdu15minTable on page 559

jnxOtnCurrentOdu15minTable

jnxOtnCurrentOdu15minTable, whose object identifier is **{jnxOtnPerformanceMonitoring 1}**, stores information about the optical data unit (ODU) performance monitoring. Each **jnxOtnCurrentOdu15minEntry**, whose object identifier is **{jnxOtnCurrentOdu15minTable 1}**, contains the objects listed in Table 233 on page 559.

Table 233: jnxOtnCurrentOdu15minEntry

Object	Object ID	Description
jnxOtnCurrentOdu15minBIP	jnxOtnCurrentOdu15minEntry 1	Shows the current 15-minute BIP counter.
jnxOtnCurrentOdu15minBBE	jnxOtnCurrentOdu15minEntry 2	Shows the current 15-minute BBE counter.
jnxOtnCurrentOdu15minES	jnxOtnCurrentOdu15minEntry 3	Shows the current 15-minute ES counter.
jnxOtnCurrentOdu15minSES	jnxOtnCurrentOdu15minEntry 4	Shows the current 15-minute SES counter.
jnxOtnCurrentOdu15minUAS	jnxOtnCurrentOdu15minEntry 5	Shows the current 15-minute UAS counter.
jnxOtnCurrentOdu15minElapsedTime	jnxOtnCurrentOdu15minEntry 6	Shows the time elapsed so far in the 15-minute interval.

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Related Documentation

- OTN Interface Management MIB on page 557
- OTN Interface Management MIB Textual Conventions on page 557
- jnxOtnAlarmTable on page 558

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CHAPTER 71

Interpreting the Enterprise-Specific Digital Optical Monitoring MIB

- Digital Optical Monitoring MIB on page 561
- DOM MIB Textual Conventions on page 561
- jnxDomCurrentTable on page 563
- jnxDomNotifications on page 564

Digital Optical Monitoring MIB

The Juniper Networks enterprise-specific Digital Optical Monitoring (DOM) MIB, whose object identifier is **{jnxDomMib 1}**, under the existing **JUNIPER-SMI MIB** hierarchy, supports **SNMP Get** for statistics and **SNMP Trap** for alarms.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/ reference/mibs/mib-jnx-dom.txt .

For information about the enterprise-specific DOM MIB objects, see the following topics:

- DOM MIB Textual Conventions on page 561
- jnxDomCurrentTable on page 563
- jnxDomNotifications on page 564

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

DOM MIB Textual Conventions

jnxDomAlarmId specifies the DOM alarms present on an interface. The possible values are:

- **domRxLossSignalAlarm(0)**—Input loss of signal
- **domRxCDRLossLockAlarm(1)**—Input loss of lock
- **domRxNotReadyAlarm(2)**—Input rx path

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- **domRxLaserPowerHighAlarm(3)**—High input laser power
- **domRxLaserPowerLowAlarm(4)**—Low input laser power
- **domTxLaserBiasCurrentHighAlarm(5)**—High output laser bias current
- **domTxLaserBiasCurrentLowAlarm(6)**—Low output laser bias current
- **domTxLaserOutputPowerHighAlarm(7)**—High output laser power
- **domTxLaserOutputPowerLowAlarm(8)**—Low output laser power
- **domTxDataNotReadyAlarm(9)**—Output A/D data not ready
- **domTxNotReadyAlarm(10)**—Output tx path
- **domTxLaserFaultAlarm(11)**—Output laser safety
- **domTxCDRLossLockAlarm(12)**—Output clock data recovery
- **domModuleTemperatureHighAlarm(13)**—High module temperature
- **domModuleTemperatureLowAlarm(14)**—Low module temperature
- **domModuleNotReadyAlarm(15)**—Module not ready
- **domModulePowerDownAlarm(16)**—Module P_DOWN
- **domLinkDownAlarm(17)**—Wire unplugged or down
- **domModuleRemovedAlarm(18)**—Module unplugged or down

jnxDomWarningId specifies the DOM warnings present on an interface. The possible values are:

- **domRxLaserPowerHighWarning(0)**—High input laser power
- **domRxLaserPowerLowWarning(1)**—Low input laser power
- **domTxLaserBiasCurrentHighWarning(2)**—High output laser bias current
- **domTxLaserBiasCurrentLowWarning(3)**—Low output laser bias current
- **domTxLaserOutputPowerHighWarning(4)**—High output laser power
- **domTxLaserOutputPowerLowWarning(5)**—Low output laser power
- **domModuleTemperatureHighWarning(6)**—High module temperature
- **domModuleTemperatureLowWarning(7)**—Low module temperature

Related Documentation

- Digital Optical Monitoring MIB on page 561
- jnxDomCurrentTable on page 563
- jnxDomNotifications on page 564

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jnxDomCurrentTable

jnxDomCurrentTable is a table of **jnxDomCurrentEntry** objects that has DOM information about the router. **jnxDomCurrentEntry**, whose object ID is **{jnxDomCurrentTable 1}**, contains the objects listed in Table 234 on page 563. Objects in the **jnxDomCurrentEntry** are indexed by **ifIndex**.

Table 234: jnxDomCurrentEntry

Object	Object ID	Description
jnxDomCurrentAlarms	jnxDomCurrentEntry 1	Identifies all the active DOM alarms on an XFP physical interface on the device.
jnxDomCurrentAlarmDate	jnxDomCurrentEntry 2	Indicates the system date and time when the management subsystem learned of the current alarm.
jnxDomLastAlarms	jnxDomCurrentEntry 3	Identifies a copy of **jnxDomCurrentAlarms** before the last set or clear.
jnxDomCurrentWarnings	jnxDomCurrentEntry 4	Identifies all the active DOM warnings on an XFP physical interface on this device.
jnxDomCurrentRxLaserPower	jnxDomCurrentEntry 5	Indicates the receiver laser power in 0.01 dbm units.
jnxDomCurrentTxLaserBiasCurrent	jnxDomCurrentEntry 6	Indicates the receiver laser bias current in 0.001 mA units.
jnxDomCurrentTxLaserOutputPower	jnxDomCurrentEntry 7	Indicates the transmitter laser output power in 0.01 dbm units.
jnxDomCurrentModuleTemperature	jnxDomCurrentEntry 8	Indicates the module temperature in Celsius (degrees C).
jnxDomCurrentRxLaserPowerHighAlarmThreshold	jnxDomCurrentEntry 9	Indicates the receiver laser power high alarm threshold in 0.01 dbm units.
jnxDomCurrentRxLaserPowerLowAlarmThreshold	jnxDomCurrentEntry 10	Indicates the receiver laser power low alarm threshold in 0.01 dbm units.
jnxDomCurrentRxLaserPowerHighWarningThreshold	jnxDomCurrentEntry 11	Indicates the receiver laser power high warning threshold in 0.01 dbm units.
jnxDomCurrentRxLaserPowerLowWarningThreshold	jnxDomCurrentEntry 12	Indicates the receiver laser power low warning threshold in 0.01 dbm units.
jnxDomCurrentTxLaserBiasCurrentHighAlarmThreshold	jnxDomCurrentEntry 13	Indicates the transmitter laser bias current high alarm threshold in 0.001 mA units.

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Table 234: jnxDomCurrentEntry *(continued)*

Object	Object ID	Description
jnxDomCurrentTxLaserBiasCurrentLowAlarmThreshold	jnxDomCurrentEntry 14	Indicates the transmitter laser bias current low alarm threshold in 0.001 mA units.
jnxDomCurrentTxLaserBiasCurrentHighWarningThreshold	jnxDomCurrentEntry 15	Indicates the transmitter laser bias current high warning threshold in 0.001 mA units.
jnxDomCurrentTxLaserBiasCurrentLowWarningThreshold	jnxDomCurrentEntry 16	Indicates the transmitter laser bias current low warning threshold in 0.001 mA units.
jnxDomCurrentTxLaserOutputPowerHighAlarmThreshold	jnxDomCurrentEntry 17	Indicates the transmitter laser power high alarm threshold in 0.01 dbm units.
jnxDomCurrentTxLaserOutputPowerLowAlarmThreshold	jnxDomCurrentEntry 18	Indicates the transmitter laser power low alarm threshold in 0.01 dbm units.
jnxDomCurrentTxLaserOutputPowerHighWarningThreshold	jnxDomCurrentEntry 19	Indicates the transmitter laser power high warning threshold in 0.01 dbm units.
jnxDomCurrentTxLaserOutputPowerLowWarningThreshold	jnxDomCurrentEntry 20	Indicates the transmitter laser power low warning threshold in 0.01 dbm units.
jnxDomCurrentModuleTemperatureHighAlarmThreshold	jnxDomCurrentEntry 21	Indicates the module temperature high alarm threshold in degrees Celsius.
jnxDomCurrentModuleTemperatureLowAlarmThreshold	jnxDomCurrentEntry 22	Indicates the module temperature low alarm threshold in degrees Celsius.
jnxDomCurrentModuleTemperatureHighWarningThreshold	jnxDomCurrentEntry 23	Indicates the module temperature high warning threshold in degrees Celsius.
jnxDomCurrentModuleTemperatureLowWarningThreshold	jnxDomCurrentEntry 24	Indicates the module temperature low warning threshold in degrees Celsius.

Related Documentation

- Digital Optical Monitoring MIB on page 561
- DOM MIB Textual Conventions on page 561
- jnxDomNotifications on page 564

jnxDomNotifications

jnxDomNotifications contains information about notifications of recently set and cleared DOM alarms in the dense wavelength division multiplexing (DWDM) tunable 10-gigabit small form-factor pluggable (XFP) transceivers. **jnxDomAlarmSet** and **jnxDomAlarmCleared** are members of **jnxDomNotifications**.

jnxDomAlarmSet, whose object ID is **{jnxDomNotificationPrefix 1}**, contains information about a newly inserted DOM alarm condition on an interface, which is identified by **ifDescr**.

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There could be preexisting alarm conditions, which are indicated in **jnxDomLastAlarms**. The new alarm bit inserted is identified by comparing **jnxDomLastAlarms** and **jnxDomCurrentAlarms**. The time of this change is indicated in **jnxDomCurrentAlarmDate**. A change in the interface condition might trigger multiple **jnxDomAlarmSet** notifications. See Table 235 on page 565.

Table 235: jnxDomAlarmSet

Object	Object ID	Description
ifDescr	**ifEntry 2**	Contains information about the interface.
jnxDomLastAlarms	**jnxDomCurrentEntry 3**	Contains preexisting alarm conditions.
jnxDomCurrentAlarms	**jnxDomCurrentEntry 1**	Contains active alarm conditions at the time when the new alarm bit was set.
jnxDomCurrentAlarmDate	**jnxDomCurrentEntry 2**	Contains the current date and time when the management subsystem learned of the current set DOM alarm.

jnxDomAlarmCleared, whose object ID is **{jnxDomNotificationPrefix 2}**, contains information about notifications when an existing DOM alarm condition is removed from an interface, which is identified by **ifDescr**. The clearing alarm bit can be identified by comparing **jnxDomLastAlarms** and **jnxDomCurrentAlarms**. The time of this change is indicated in **jnxDomCurrentAlarmDate**. A change in the interface condition might trigger multiple **jnxDomAlarmCleared** notifications. See Table 236 on page 565.

Table 236: jnxDomAlarmCleared

Object	Object ID	Description
ifDescr	**ifEntry 2**	Contains information about the interface.
jnxDomLastAlarms	**jnxDomCurrentEntry 3**	Contains preexisting alarm conditions.
jnxDomCurrentAlarms	**jnxDomCurrentEntry 1**	Contains active alarm conditions at the time when a new alarm bit was cleared.
jnxDomCurrentAlarmDate	**jnxDomCurrentEntry 2**	Contains the current date and time when the management subsystem learned of the last cleared DOM alarm.

Related Documentation

- Digital Optical Monitoring MIB on page 561
- DOM MIB Textual Conventions on page 561
- jnxDomCurrentTable on page 563

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CHAPTER 72

Interpreting the Enterprise-Specific License MIB

- License MIB on page 567
- jnxLicenseInstallTable on page 568
- Supported Features and Configuration Parameters for Licensing on page 568
- Juniper Networks Enterprise-Specific License MIB Notifications on page 569

License MIB

The Juniper Networks enterprise-specific License MIB, whose object identifier is **{jnxMibs 63 }**, extends SNMP support to licensing information in the Junos OS.

The License MIB stores licensing-related information, such as licensed features installed on the device, licenses that are about to expire or have expired, licenses that are in use, the number of users accessing a licensed feature, and so on. It enables users to access licensing-related information using SNMP **get** and **get-next** commands. The License MIB also supports SNMP traps that alert users when a license is about to expire or has expired, when the number of users exceeds the maximum number specified in the license, and so on.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-license.txt .

For information about the enterprise-specific License MIB objects and traps, see the following topics:

- jnxLicenseInstallTable on page 568
- Supported Features and Configuration Parameters for Licensing on page 568
- Juniper Networks Enterprise-Specific License MIB Notifications on page 569

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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jnxLicenseInstallTable

The **jnxLicenseInstallTable**, whose object identifier is **{jnxLicenseInstallObjects 1}**, contains license-related information for the features installed on the device. The license-related information for installed features is stored in **jnxLicenseInstallEntry**.

Each **jnxLicenseInstallEntry**, whose object identifier is **{jnxLicenseInstallTable 1}**, contains the objects listed in Table 237 on page 568.

Table 237: jnxLicenseInstallEntry

Object	Object Identifier	Description
jnxLicenseId	**jnxLicenseInstallEntry 1**	Contains the license ID for the installed feature. It is also the index for **jnxLicenseInstallEntry**.
jnxLicenseVersion	**jnxLicenseInstallEntry 2**	Contains information about the license version.
jnxLicenseDeviceId	**jnxLicenseInstallEntry 3**	Contains the device ID for the license.
jnxLicenseType	**jnxLicenseInstallEntry 4**	Indicates the license type. Possible values are permanent, time-based, and invalid.
jnxLicenseKeys	**jnxLicenseInstallEntry 5**	Contains the license key.
jnxLicenseUsed	**jnxLicenseInstallEntry 6**	Indicates whether the license is in use (1) or not (0).
jnxLicenseInstalled	**jnxLicenseInstallEntry 7**	Contains details of the installed license.
jnxLicenseNeeded	**jnxLicenseInstallEntry 8**	Indicates whether the license needs to be installed.

Related Documentation

- License MIB on page 567
- Supported Features and Configuration Parameters for Licensing on page 568
- Juniper Networks Enterprise-Specific License MIB Notifications on page 569

Supported Features and Configuration Parameters for Licensing

This topic contains the following sections:

- jnxLicenseFeatureListTable on page 568
- License Configuration Parameters on page 569

jnxLicenseFeatureListTable

The **jnxLicenseFeatureListTable**, whose object identifier is **{jnxLicenseInstallObjects 2}**, contains a list of features that support licensing. Each **jnxLicenseFeatureListEntry**, whose object identifier is **{jnxLicenseFeatureListTable 1}**, contains the objects listed in Table 238 on page 569.

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Table 238: jnxLicenseFeatureListEntry

Object	Object Identifier	Description
jnxLicenseFeatureId	jnxLicenseFeatureListEntry 1	Contains the feature ID that is the index of the **jnxLicenseFeatureListEntry**.
jnxLicenseFeatureName	jnxLicenseFeatureListEntry 2	Contains the name of the feature.
jnxLicenseFeatureDescr	jnxLicenseFeatureListEntry 3	Contains a description of the feature.
jnxLicenseFeatureLicenseId	jnxLicenseFeatureListEntry 4	Contains the feature license ID (from the **jnxLicenseInstallTable**) that maps the feature to the license information.

License Configuration Parameters

The license MIB also contains the following objects that provide information about license configuration:

- **jnxLicenseRenewBeforExpiration**, whose object identifier is **{jnxLicenseSettings 1}**, indicates the number of days left before the license expires.
- **jnxLicenseRenewInterval**, whose object identifier is **{jnxLicenseSettings 2}**, indicates the license checking interval in hours.
- **jnxLicenseiAutoUpdate**, whose object identifier is **{jnxLicenseSettings 3}**, contains the auto-update URL of a license server.
- **jnxLicenseTraceOptions**, whose object identifier is **{jnxLicenseSettings 4}**, indicates the traceoption settings for the license. Possible values are all (1), events (2), and configuration (3).

Related Documentation

- License MIB on page 567
- jnxLicenseInstallTable on page 568
- Juniper Networks Enterprise-Specific License MIB Notifications on page 569

Juniper Networks Enterprise-Specific License MIB Notifications

Table 239 on page 569 lists the license MIB notifications.

Table 239: License MIB Notifications

Notification	Object Identifier	Description
jnxLicenseGraceExpired	jnxLicenseNotifications 1	Generated when the grace period for renewing the license for the feature (identified by **jnxLicenseFeatureName**) has expired.

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Table 239: License MIB Notifications *(continued)*

Notification	Object Identifier	Description
jnxLicenseGraceAboutToExpire	**jnxLicenseNotifications 2**	Generated when the grace period for renewing the license for the feature (identified by **jnxLicenseFeatureName**) is about to expire.
jnxLicenseAboutToExpire	**jnxLicenseNotifications 3**	Generated when the license for the feature (identified by **jnxLicenseFeatureName**) is about to expire.
jnxLicenseInfringeCumulative	**jnxLicenseNotifications 4**	Generated when the use of the feature (identified by **jnxLicenseFeatureName**) exceeds the maximum limit specified in the license.
jnxLicenseInfringeSingle	**jnxLicenseNotifications 5**	Generated when the license for the feature (identified by **jnxLicenseFeatureName**) is invalid; that is, either expired or not available.

Related Documentation

- License MIB on page 567
- jnxLicenseInstallTable on page 568
- Supported Features and Configuration Parameters for Licensing on page 568
- Standard SNMP Traps Supported on Devices Running Junos OS on page 653
- Juniper Networks Enterprise-Specific SNMP Traps on page 633

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CHAPTER 73

Interpreting the Enterprise-Specific Subscriber MIB

- Subscriber MIB on page 571
- Subscriber MIB Textual Conventions on page 572
- Managed Objects for General Subscriber Functions and jnxSubscriberTable on page 572
- Managed Objects for Logical System Subscriber Information and jnxSubscriberLogicalSystemTable on page 574
- Managed Objects for Routing Instance Subscriber Information and jnxSubscriberRoutingInstanceTable on page 575

Subscriber MIB

The Juniper Networks enterprise-specific Subscriber MIB, whose object ID is **{jnxSubscriberMibRoot 1}**, defines an object to enable retrieval of subscriber-related information. Subscriber state information is maintained in the session database. Using the object, you can view subscriber information using SNMP **get** and **get-next** requests.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-subscriber.txt .

For information about the enterprise-specific Subscriber MIB objects, see the following topics:

- Subscriber MIB Textual Conventions on page 572
- Managed Objects for General Subscriber Functions and jnxSubscriberTable on page 572
- Managed Objects for Logical System Subscriber Information and jnxSubscriberLogicalSystemTable on page 574
- Managed Objects for Routing Instance Subscriber Information and jnxSubscriberRoutingInstanceTable on page 575

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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Subscriber MIB Textual Conventions

Table 240 on page 572 contains information about the textual conventions used in the enterprise-specific Subscriber MIB.

Table 240: Subscriber MIB Textual Conventions

Object	Description	Syntax
jnxSubscriberState	State of the AAA subscriber.	This object uses the following integer values: • **0**—INIT state only • **1**—Configured state only • **2**—Active state only • **3**—Terminating state only • **4**—Terminated state only • **5**—Unknown state
jnxSubscriberClientType	Type of subscriber client.	This object uses the following integer values: • **0**—None • **1**—DHCP • **2**—VLAN • **3**—Generic • **4**—MobileIp • **5**—vplsPw • **6**—PPP • **7**—PPPoE • **8**—L2TP • **9**—Static • **10**—MLPPP

Related Documentation

- Subscriber MIB on page 571
- Managed Objects for General Subscriber Functions and jnxSubscriberTable on page 572
- Managed Objects for Logical System Subscriber Information and jnxSubscriberLogicalSystemTable on page 574
- Managed Objects for Routing Instance Subscriber Information and jnxSubscriberRoutingInstanceTable on page 575

Managed Objects for General Subscriber Functions and jnxSubscriberTable

The managed objects listed in Table 241 on page 573 along with the objects in **jnxSubscriberTable** provide general subscriber information such as the number of active subscribers and the total number of subscribers.

Each **jnxSubscriberGeneral**, whose object identifier is **{jnxSubscriberObjects 1}**, contains the objects listed in Table 241 on page 573.

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Table 241: Managed Objects for General Subscriber Functions

Object	Object ID	Description
jnxSubscriberTotalCount	jnxSubscriberGeneral 1	Total number of subscribers.
jnxSubscriberActiveCount	jnxSubscriberGeneral 2	Number of active subscribers.

For information about **jnxSubscriberTable**, see the following section:

- jnxSubscriberTable on page 573

jnxSubscriberTable

jnxSubscriberTable, whose object identifier is **{jnxSubscriberGeneral 3}**, contains **jnxSubscriberEntry** that maps to the specification of each subscriber.

Each **jnxSubscriberEntry**, whose object identifier is **{jnxSubscriberTable 1}**, contains the objects listed in Table 242 on page 573.

Table 242: jnxSubscriberTable

Object	Object ID	Description
jnxSubscriberHandleHiWord	jnxSubscriberEntry 1	Subscriber handle associated with each subscriber. Returns the most significant 32 bits of the 64-bit subscriber ID. The value of the subscriber handle is a monotonically increasing number.
jnxSubscriberHandleLoWord	jnxSubscriberEntry 2	Subscriber handle associated with each subscriber. Returns the least significant 32 bits of the 64-bit subscriber ID. The value of the subscriber handle is a monotonically increasing number.
jnxSubscriberUserName	jnxSubscriberEntry 3	Name of the user associated with the subscriber.
jnxSubscriberClientType	jnxSubscriberEntry 4	Type of the client associated with the subscriber.
jnxSubscriberIpAddress	jnxSubscriberEntry 5	Client IP address assigned to the subscriber.
jnxSubscriberIpAddressMask	jnxSubscriberEntry 6	Client IP address mask assigned to the subscriber.
jnxSubscriberLogicalSystem	jnxSubscriberEntry 7	Logical system associated with the subscriber.
jnxSubscriberRoutingInstance	jnxSubscriberEntry 8	Routing instance associated with the subscriber.
jnxSubscriberInterface	jnxSubscriberEntry 9	Textual name of the interface associated with the subscriber.

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Table 242: jnxSubscriberTable *(continued)*

Object	Object ID	Description
jnxSubscriberInterfaceType	**jnxSubscriberEntry 10**	Type of interface associated with the subscriber. Possible values include: • none (0) • static (1) • dynamic (2)
jnxSubscriberMacAddress	**jnxSubscriberEntry 11**	MAC address of the interface associated with the subscriber.
jnxSubscriberState	**jnxSubscriberEntry 12**	State of the subscriber.
jnxSubscriberLoginTime	**jnxSubscriberEntry 13**	System time when the subscriber logged in to the server.

Related Documentation

- Subscriber MIB on page 571
- Subscriber MIB Textual Conventions on page 572
- Managed Objects for Logical System Subscriber Information and jnxSubscriberLogicalSystemTable on page 574
- Managed Objects for Routing Instance Subscriber Information and jnxSubscriberRoutingInstanceTable on page 575

Managed Objects for Logical System Subscriber Information and jnxSubscriberLogicalSystemTable

The managed objects listed in Table 243 on page 574 along with the objects in **jnxSubscriberLogicalSystemTable** provide subscriber information based on the associated logical system, such as the number of active subscribers on the logical system and the total number of subscribers on the logical system.

Each **jnxSubscriberLogicalSystemObjects**, whose object identifier is **{jnxSubscriberObjects 2}**, contains the objects listed in Table 243 on page 574.

Table 243: Managed Objects for Logical System Subscriber Functions

Object	Object ID	Description
jnxSubscriberLogicalSystemTotalCount	**jnxSubscriberLogicalSystemObjects 1**	Total number of subscribers on the logical system.
jnxSubscriberLogicalSystemActiveCount	**jnxSubscriberLogicalSystemObjects 2**	Number of active subscribers on the logical system.

For information about **jnxSubscriberLogicalSystemTable**, see the following section:

- jnxSubscriberLogicalSystemTable on page 575

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jnxSubscriberLogicalSystemTable

jnxSubscriberLogicalSystemTable, whose object identifier is **{jnxSubscriberLogicalSystemObjects 3}**, contains **jnxSubscriberLogicalSystemEntry** that maps to the specification of each subscriber. You must provide the name of the logical system in the SNMP query. The logical system is identified either by the context field in V3 requests or by the community string in V1 and V2c requests. You can query for information specific to the logical system or routing instance defined within a logical system by using the prefix **<logical-system>/<routing-instance>** @ with the community string. In the prefix, @ acts as a separator between the community string and the name of the routing instance, and / acts as a separator between the logical system name and the routing instance. If you do not provide the name of the logical system, the default logical system data is displayed. The name of the routing instance is ignored for this table.

Each **jnxSubscriberLogicalSystemEntry**, whose object identifier is **{jnxSubscriberLogicalSystemTable 1}**, contains the objects listed in Table 244 on page 575.

Table 244: jnxSubscriberLogicalSystemTable

Object	Object ID	Description
jnxSubscriberLogicalSystemHandleHiWord	jnxSubscriberLogicalSystemEntry 1	Subscriber handle associated with each subscriber. Returns the most significant 32 bits of the 64-bit subscriber ID. The value of the subscriber handle is a monotonically increasing number.
jnxSubscriberLogicalSystemHandleLoWord	jnxSubscriberLogicalSystemEntry 2	Subscriber handle associated with each subscriber. Returns the least significant 32 bits of the 64-bit subscriber ID. The value of the subscriber handle is a monotonically increasing number.
jnxSubscriberLogicalSystemState	jnxSubscriberLogicalSystemEntry 3	State of the subscriber.

Related Documentation

- Subscriber MIB on page 571
- Subscriber MIB Textual Conventions on page 572
- Managed Objects for General Subscriber Functions and jnxSubscriberTable on page 572
- Managed Objects for Routing Instance Subscriber Information and jnxSubscriberRoutingInstanceTable on page 575

Managed Objects for Routing Instance Subscriber Information and jnxSubscriberRoutingInstanceTable

The managed objects listed in Table 245 on page 576 along with the objects in **jnxSubscriberRoutingInstanceTable** provide subscriber information based on the

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associated routing instance, such as the number of active subscribers on the routing instance and the total number of subscribers on the routing instance.

Each **jnxSubscriberRoutingInstanceObjects**, whose object identifier is **{jnxSubscriberObjects 3}**, contains the objects listed in Table 245 on page 576.

Table 245: Managed Objects for Routing Instance Subscriber Functions

Object	Object ID	Description
jnxSubscriberRoutingInstanceTotalCount	**jnxSubscriberRoutingInstanceObjects 1**	Total number of subscribers on the routing instance.
jnxSubscriberRoutingInstanceActiveCount	**jnxSubscriberRoutingInstanceObjects 2**	Number of active subscribers on the routing instance.

For information about **jnxSubscriberRoutingInstanceTable**, see the following section:

- jnxSubscriberRoutingInstanceTable on page 576

jnxSubscriberRoutingInstanceTable

jnxSubscriberRoutingInstanceTable, whose object identifier is **{jnxSubscriberRoutingInstanceObjects 3}**, contains **jnxSubscriberRoutingInstanceEntry** that maps to the specification of each subscriber. You must provide the name of the routing instance in the SNMP query. The routing instance is identified either by the context field in V3 requests or by the community string in V1 and V2c requests. You can query for information specific to the routing instance or routing instance defined within a logical system by using the prefix **<logical-system>/<routing-instance>** @ with the community string. In the prefix, @ acts as a separator between the community string and the name of the routing instance, and / acts as a separator between the logical system name and the routing instance. If you do not provide the name of the routing instance, the default routing instance data is displayed. The name of the logical system is ignored for this table.

Each **jnxSubscriberRoutingInstanceEntry**, whose object identifier is **{jnxSubscriberRoutingInstanceTable 1}**, contains the objects listed in Table 246 on page 576.

Table 246: jnxSubscriberRoutingInstanceTable

Object	Object ID	Description
jnxSubscriberRoutingInstanceHandleHiWord	**jnxSubscriberRoutingInstanceEntry 1**	Subscriber handle associated with each subscriber. Returns the most significant 32 bits of the 64-bit subscriber ID. The value of the subscriber handle is a monotonically increasing number.
jnxSubscriberRoutingInstanceHandleLoWord	**jnxSubscriberRoutingInstanceEntry 2**	Subscriber handle associated with each subscriber. Returns the least significant 32 bits of the 64-bit subscriber ID. The value of the subscriber handle is a monotonically increasing number.

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Table 246: jnxSubscriberRoutingInstanceTable *(continued)*

Object	Object ID	Description
jnxSubscriberRoutingInstanceState	**jnxSubscriberRoutingInstanceEntry 3**	State of the subscriber.

Related Documentation

- Subscriber MIB on page 571
- Subscriber MIB Textual Conventions on page 572
- Managed Objects for General Subscriber Functions and jnxSubscriberTable on page 572
- Managed Objects for Logical System Subscriber Information and jnxSubscriberLogicalSystemTable on page 574

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CHAPTER 74

Interpreting the Enterprise-Specific VPLS MIBs

- Juniper Networks Enterprise-Specific Implementation of VPLS MIBs on page 579
- Juniper Networks Enterprise-Specific VPLS Generic MIB on page 580
- Juniper Networks Enterprise-Specific VPLS LDP MIB on page 585
- Juniper Networks Enterprise-Specific VPLS BGP MIB on page 586

Juniper Networks Enterprise-Specific Implementation of VPLS MIBs

Starting with Release 11.4, Junos OS extends SNMP support to virtual private LAN service (VPLS) networks, and thus enables users to access VPLS-related data over SNMP.

The Junos OS SNMP support for VPLS is based on the enterprise-specific extension of the following IETF standard MIBs defined in the *Virtual Private LAN Service (VPLS) Management Information Base* Internet draft draft-ietf-l2vpn-vpls-mib-05.txt:

- **VPLS-Generic-Draft-01-MIB** implemented as **mib-jnx-vpls-generic.txt**
- **VPLS-BGP-Draft-01-MIB** implemented as **mib-jnx-vpls-bgp.txt**
- **VPLS-LDP-Draft-01-MIB** implemented as **mib-jnx-vpls-ldp.txt**

The enterprise-specific extensions are implemented as part of the **jnxExperiment** branch.

For downloadable versions of the enterprise-specific MIBs, see:

- http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-vpls-generic.txt
- http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-vpls-ldp.txt
- http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-vpls-bgp.txt

For more information about VPLS MIBs, see the following topics:

- Juniper Networks Enterprise-Specific VPLS Generic MIB on page 580
- Juniper Networks Enterprise-Specific LDP VPLS MIB on page 585

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- Juniper Networks Enterprise-Specific BGP VPLS MIB on page 586

Related Documentation

- Juniper Networks Enterprise-Specific VPLS Generic MIB on page 580
- Juniper Networks Enterprise-Specific LDP VPLS MIB on page 585
- Juniper Networks Enterprise-Specific BGP VPLS MIB on page 586
- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

Juniper Networks Enterprise-Specific VPLS Generic MIB

The enterprise-specific VPLS Generic MIB is a Juniper Networks implementation of the IETF standard **VPLS-Generic-Draft-01-MIB** defined in draft-ietf-l2vpn-vpls-mib-05.txt. The object identifier for the enterprise-specific version implemented under the **jnxExperiment** branch is **{jnxExperiment 8}**.

This MIB contains generic managed objects definitions for VPLS as defined in RFC 4761, *Virtual Private LAN Service (VPLS) Using BGP for Auto-Discovery and Signaling,* and RFC 4762, *Virtual Private LAN Service (VPLS) Using Label Distribution Protocol (LDP) Signaling*.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-vpls-generic.txt .

For information about the enterprise-specific VPLS Generic MIB objects, see the following topics:

- jnxVplsConfigTable on page 580
- jnxVplsStatusTable on page 582
- jnxVplsPwBindTable on page 582
- jnxVplsBgpADConfigTable on page 583
- jnxVplsBgpRteTargetTable on page 584

jnxVplsConfigTable

The **jnxVplsConfigTable**, whose object identifier is **{jnxVplsObjects 2}**, contains information related to configuring and monitoring VPLS. The **jnxVplsConfigTable** contains **jnxVplsConfigEntry**, whose object identifier is **{jnxVplsConfigTable 1}**.

Each **jnxVplsConfigEntry** is indexed with **jnxVplsConfigIndex** that uniquely identifies a VPLS, and contains the objects listed in Table 247 on page 580.

Table 247: jnxVplsConfigEntry

Object	Object Identifier	Description
jnxVplsConfigIndex	**jnxVplsConfigEntry 1**	Contains the unique index that identifies a row specific to a VPLS.

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Table 247: jnxVplsConfigEntry *(continued)*

Object	Object Identifier	Description
jnxVplsConfigName	**jnxVplsConfigEntry 2**	Contains the name of the VPLS.
jnxVplsConfigDescr	**jnxVplsConfigEntry 3**	Contains a brief description of the VPLS.
jnxVplsConfigAdminStatus	**jnxVplsConfigEntry 4**	Indicates the administrative state of the VPLS. Possible values are **1** (up), **2** (down), and **3** (testing). The testing (**3**) state indicates that no operational packets can be passed. The default value is **2** (down).
jnxVplsConfigMacLearning	**jnxVplsConfigEntry 6**	Indicates whether MAC learning is enabled (**true**) or not (**false**). The default value is **true**.
jnxVplsConfigDiscardUnknownDest	**jnxVplsConfigEntry 7**	Indicates whether frames received with an unknown destination MAC address are processed (**false**) or discarded (**true**). The default value is **false**.
jnxVplsConfigMacAging	**jnxVplsConfigEntry 8**	Indicates whether MAC aging is enabled (**true**) or not (**false**). The default value is **true**.
jnxVplsConfigRowStatus	**jnxVplsConfigEntry 12**	Indicates the row status.
jnxVplsConfigMtu	**jnxVplsConfigEntry 13**	Specifies the MTU of the VPLS. The range is **64** through **1518**. The default value is **1518**.
jnxVplsConfigVpnId	**jnxVplsConfigEntry 14**	Contains the IEEE 802-1990 VPN ID of the associated VPLS.
jnxVplsConfigServiceType	**jnxVplsConfigEntry 15**	Specifies the type of service—**1** (vlan) or **2** (ethernet)—emulated by the VPLS. The default value is **1**.

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Table 247: jnxVplsConfigEntry *(continued)*

Object	Object Identifier	Description
jnxVplsConfigStorageType	**jnxVplsConfigEntry 1**	Indicates the storage type for the row. The default value is **volatile**.

jnxVplsStatusTable

The **jnxVplsStatusTable**, whose object identifier is **{jnxVplsObjects 3}**, contains the **jnxVplsStatusEntry**. The **jnxVplsStatusEntry** represents a VPLS instance, and is indexed by **jnxVplsConfigIndex**. A **jnxVplsStatusEntry** is automatically created when a VPLS instance is configured.

Each **jnxVplsStatusEntry**, whose object identifier is **{jnxVplsStatusTable 1}**, contains the objects listed in Table 248 on page 582:

Table 248: jnxVplsStatusEntry

Object	Object Identifier	Description
jnxVplsStatusOperStatus	**jnxVplsStatusEntry 1**	Shows the current operational state of the VPLS.
jnxVplsStatusPeerCount	**jnxVplsStatusEntry 2**	Shows the number of peers for the VPLS.

jnxVplsPwBindTable

The **jnxVplsPwBindTable**, whose object identifier is **{jnxVplsObjects 4}**, provides information about associations between a VPLS instance and pseudowires. A VPLS can be associated with more than one pseudowire. Note that pseudowires are defined in the **pwTable**.

The **jnxVplsPwBindTable** contains the **jnxVplsPwBindEntry**, whose object identifier is **{jnxVplsPwBindTable 1}**. Each **jnxVplsPwBindEntry**, indexed with **jnxVplsConfigIndex** and **jnxVplsPwBindIndex**, contains the objects listed in Table 249 on page 582.

Table 249: jnxVplsPwBindEntry

Object	Object Identifier	Description
jnxVplsPwBindConfigType	**jnxVplsPwBindEntry 1**	Indicates whether the pseudowire binding was created manually or through autodiscovery. Possible values are **1** (manual) and **2** (autodiscovery).

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Table 249: jnxVplsPwBindEntry *(continued)*

Object	Object Identifier	Description
jnxVplsPwBindType	jnxVplsPwBindEntry 2	Indicates whether the pseudowire is of type **mesh** or **spoke**. A value of **1** denotes a mesh type pseudowire, whereas a value of **2** denotes spoke type.
jnxVplsPwBindRowStatus	jnxVplsPwBindEntry 3	Indicates the row status. No object in the table can be modified when the row status is **active**, denoted by **1**.
jnxVplsPwBindStorageType	jnxVplsPwBindEntry 4	Indicates the storage type for the row. The default value is **volatile**.
jnxVplsPwBindIndex	jnxVplsPwBindEntry 5	Contains the secondary index for the conceptual row to identify a pseudowire within the **pwEntry**. This must match with a **pwTable** entry that represents an already provisioned pseudowire.

jnxVplsBgpADConfigTable

The **jnxVplsBgpADConfigTable**, whose object identifier is **{jnxVplsObjects 5}**, contains information related to BGP autodiscovery parameters for a VPLS.

The **jnxVplsBgpADConfigEntry**, whose objected identifier is **{jnxVplsBgpADConfigTable 1}**, is indexed with **jnxVplsConfigIndex**. Each **jnxVplsBgpADConfigEntry** contains the objects listed in Table 250 on page 583.

Table 250: jnxVplsBgpADConfigEntry

Object	Object Identifier	Description
jnxVplsBgpADConfigRouteDistinguisher	jnxVplsBgpADConfigEntry 1	Contains the router distinguisher for the VPLS. For more information on router distinguisher and its use for a VPLS, see RFC 4364 and RFC 4761.

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Table 250: jnxVplsBgpADConfigEntry *(continued)*

Object	Object Identifier	Description
jnxVplsBgpADConfigPrefix	**jnxVplsBgpADConfigEntry 2**	Indicates whether the advertised prefix is the loopback IP address or a user-defined value. The default value is **0**, denoting that the advertised prefix is the loopback IP address.
jnxVplsBgpADConfigVplsId	**jnxVplsBgpADConfigEntry 3**	Contains a unique identifier that is applicable for all VSIs in a VPLS.
jnxVplsBgpADConfigRowStatus	**jnxVplsBgpADConfigEntry 4**	Indicates the row status.

jnxVplsBgpRteTargetTable

The **jnxVplsBgpRteTargetTable**, whose object identifier is **{jnxVplsObjects 6}**, contains the list of route targets imported or exported by BGP during the autodiscovery of the VPLS.

The **jnxVplsBgpRteTargetEntry**, whose object identifier is **{jnxVplsBgpRteTargetTable 1}**, is indexed with **jnxVplsConfigIndex** and **jnxVplsBgpRteTargetIndex**. Each **jnxVplsBgpRteTargetEntry** contains the objects listed in Table 251 on page 584.

Table 251: jnxVplsBgpRteTargetEntry

Object	Object Identifier	Description
jnxVplsBgpRteTargetIndex	**jnxVplsBgpRteTargetEntry 1**	In combination with the **jnxVplsConfigIndex**, this object uniquely identifies an entry in the **jnxVplsBgpRteTargetTable**. Users can configure multiple route targets for the same VPLS by using new values for this object while keeping the **jnxVplsConfigIndex** constant.
jnxVplsBgpRteTargetRTType	**jnxVplsBgpRteTargetEntry 2**	Indicates whether the route target is imported, exported, or both.
jnxVplsBgpRteTargetRT	**jnxVplsBgpRteTargetEntry 3**	Contains the route target associated with the VPLS.
jnxVplsBgpRteTargetRTRowStatus	**jnxVplsBgpRteTargetEntry 4**	Indicates the row status.

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Related Documentation

- Juniper Networks Enterprise-Specific Implementation of VPLS MIBs on page 579
- Juniper Networks Enterprise-Specific LDP VPLS MIB on page 585
- Juniper Networks Enterprise-Specific BGP VPLS MIB on page 586

Juniper Networks Enterprise-Specific VPLS LDP MIB

The enterprise-specific VPLS LDP MIB, **jnx-vpls-ldp-mib**, is a Juniper Networks implementation of the IETF standard VPLS LDP MIB. The enterprise-specific VPLS LDP MIB, whose object identifier is **{jnxExperiment 9}**, contains managed objects definitions for LDP-signaled VPLS (RFC 4762). This MIB also supports data related to pseudowires in the VPLS.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-vpls-ldp.txt .

The following topics contain information about VPLS LDP objects supported by Junos OS:

- jnxVplsLdpConfigTable on page 585
- jnxVplsLdpPwBindTable on page 585

jnxVplsLdpConfigTable

The **jnxVplsLdpConfigTable**, whose object identifier is **{jnxVplsLdpObjects 1}** contains the **jnxVplsLdpConfigEntry**. Each **jnxVplsLdpConfigEntry** is indexed by the **jnxVplsConfigIndex**, and contains the **jnxVplsLdpConfigMacAddrWithdraw** object. The object identifier for **jnxVplsLdpConfigEntry** is **{jnxVplsLdpConfigTable 1}**.

The **jnxVplsLdpConfigMacAddrWithdraw** object, whose object identifier is **{jnxVplsLdpConfigEntry 1}**, shows the MAC address withdrawal learning status for the VPLS. The value of this object is set to **true** when MAC address withdrawal learning is enabled for the VPLS. The value is set to **false** when MAC address withdrawal learning is disabled.

jnxVplsLdpPwBindTable

The **jnxVplsLdpPwBindTable**, whose object identifier is **{jnxVplsLdpObjects 2}**, contains the **jnxVplsLdpPwBindEntry**. Each **jnxVplsLdpPwBindEntry** is indexed with **jnxVplsConfigIndex** and **jnxVplsPwBindIndex**, and contains the **jnxVplsLdpPwBindMacAddressLimit** object.

The **jnxVplsLdpPwBindMacAddressLimit** object, whose object identifier is **{jnxVplsLdpPwBindEntry 1}**, shows the maximum number of learned and static MAC address entries allowed in the forwarding database for the pseudowire binding. The range is **0** through **4294967295**. The default value is **0**, which denotes that there is no maximum limit for the entries.

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Related Documentation

- Juniper Networks Enterprise-Specific Implementation of VPLS MIBs on page 579
- Juniper Networks Enterprise-Specific VPLS Generic MIB on page 580
- Juniper Networks Enterprise-Specific BGP VPLS MIB on page 586

Juniper Networks Enterprise-Specific VPLS BGP MIB

The enterprise-specific VPLS BGP MIB is a Juniper Networks implementation of the IETF standard BGP VPLS MIB. The enterprise-specific VPLS BGP MIB, whose object identifier is **{jnxExperiment 10}**, contains managed object definitions for the BGP-signaled VPLS (RFC 4761). This MIB also supports data related to pseudowires in the VPLS.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/mibs/mib-jnx-vpls-bgp.txt .

The following topics contain information about VPLS BGP objects supported by Junos OS:

- jnxVplsBgpConfigTable on page 586
- jnxVplsBgpVETable on page 586
- jnxVplsBgpPwBindTable on page 587

jnxVplsBgpConfigTable

The **jnxVplsBgpConfigTable**, whose object identifier is **{jnxVplsBgpObjects 1}**, contains the **jnxVplsBgpConfigEntry**. Each **jnxVplsBgpConfigEntry** is indexed with **jnxVplsConfigIndex**, and contains the **jnxVplsBgpConfigVERangeSize** object. The object identifier for **jnxVplsBgpConfigEntry** is **{jnxVplsBgpConfigTable 1}**.

The **jnxVplsBgpConfigVERangeSize** object, whose object identifier is **{jnxVplsBgpConfigEntry 1}**, specifies the range of the VPLS edge device (VE) identifiers in the VPLS. This number controls the size of the label block advertised for the VE by the provider edge (PE) device. The available range is **0** through **65535**. A value of **0** indicates that the range is not configured, and the PE derives the range value from received advertisements from other PE devices.

jnxVplsBgpVETable

The **jnxVplsBgpVETable**, whose object identifier is **{jnxVplsBgpObjects 2}**, contains information about the association of VE devices with a VPLS. This table contains an entry, **jnxVplsBgpVEEntry**, for each VE device identifier configured on a PE device for a particular VPLS instance.

Each **jnxVplsBgpVEEntry**, whose object identifier is **{jnxVplsBgpVETable 1}**, contains the objects listed in Table 252 on page 587.

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Table 252: jnxVplsBgpVEEntry

Object	Object Identifier	Description
jnxVplsBgpVEId	jnxVplsBgpVEEntry 1	Contains a secondary index that identifies a VE within a VPLS instance.
jnxVplsBgpVEName	jnxVplsBgpVEEntry 2	Contains the name of the site or the u-PE associated with the VE ID.
jnxVplsBgpVEPreference	jnxVplsBgpVEEntry 3	Specifies the preference for the VE ID on the PE device if the site is multihomed and the VE ID is reused.
jnxVplsBgpVERowStatus	jnxVplsBgpVEEntry 5	Indicates the status of the row. When this object is set to **active**, objects in this row cannot be modified.
jnxVplsBgpVEStorageType	jnxVplsBgpVEEntry 6	Indicates the storage type for this row.

jnxVplsBgpPwBindTable

The **jnxVplsBgpPwBindTable**, whose object identifier is **{jnxVplsBgpObjects 3}**, provides BGP-specific information about the association between a VPLS and the corresponding pseudowire. One VPLS can be associated with more than one pseudowire.

Each **jnxVplsBgpPwBindEntry**, whose object identifier is **{jnxVplsBgpPwBindTable 1}**, is indexed with **jnxVplsConfigIndex** and **jnxVplsPwBindIndex**, and contains the objects listed in Table 253 on page 587:

Table 253: jnxVplsBgpPwBindEntry

Object Name	Object Identifier	Description
jnxVplsBgpPwBindLocalVEId	jnxVplsBgpPwBindEntry 1	Identifies the local VE that the pseudowire is associated with.
jnxVplsBgpPwBindRemoteVEId	jnxVplsBgpPwBindEntry 2	Identifies the remote VE that the pseudowire is associated with

Related Documentation

- Juniper Networks Enterprise-Specific Implementation of VPLS MIBs on page 579
- Juniper Networks Enterprise-Specific VPLS Generic MIB on page 580
- Juniper Networks Enterprise-Specific LDP VPLS MIB on page 585

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CHAPTER 75

Interpreting the Enterprise-Specific PPP MIB

- PPP MIB on page 589
- PPP MIB Textual Conventions on page 590
- Managed Objects for PPP Link Control Protocol Layer on page 591
- Managed Objects for PPP IP Network Control Protocol Layer on page 596
- Managed Objects for PPP OSI Network Control Protocol Layer on page 599
- Managed Objects for PPP Session Attributes on page 602
- Managed Objects for Multilink PPP Layer on page 604
- Managed Objects for PPP Interface Summary Counts on page 609
- Managed Objects for PPP IPv6 Network Control Protocol Layer on page 615
- Managed Objects for PPP Global Attributes on page 617

PPP MIB

The Juniper Networks enterprise-specific PPP MIB, whose object ID is **{jnxPppMibRoot 1}**, extends SNMP support for the Point-to-Point Protocol (PPP) modules in Junos OS.

The PPP MIB stores PPP-related information, such as the authentication type, status of the operation of network protocols over each link, PPP session status, PPP interface characteristics, and statistics. You can view this information by using SNMP **get** and **get-next** requests.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/ mibs/mib-jnx-ppp.txt .

For information about the enterprise-specific PPP MIB objects, see the following topics:

- PPP MIB Textual Conventions on page 590
- Managed Objects for PPP Link Control Protocol Layer on page 591
- Managed Objects for PPP IP Network Control Protocol Layer on page 596
- Managed Objects for PPP OSI Network Control Protocol Layer on page 599

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- Managed Objects for PPP Session Attributes on page 602
- Managed Objects for Multilink PPP Layer on page 604
- Managed Objects for PPP Interface Summary Counts on page 609
- Managed Objects for PPP IPv6 Network Control Protocol Layer on page 615
- Managed Objects for PPP Global Attributes on page 617

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

PPP MIB Textual Conventions

Table 254 on page 590 contains information about the textual conventions used in the enterprise-specific PPP MIB.

Table 254: PPP MIB Textual Conventions

Object	Description	Syntax
jnxPppAuthentication	Specifies the types of PPP authentication used.	This object uses the following integer values: • **0**–None (no authentication is negotiated) • **1**–pap (Password Authentication Protocol negotiation only) • **2**–chap (Challenge Handshake Authentication Protocol negotiation only) • **3**–papChap (PAP negotiation is attempted first; if that fails, attempts CHAP negotiation) • **4**–chapPap (CHAP negotiation is attempted first; if that fails, attempts PAP negotiation)
jnxPppMlPppBundleName	Specifies the name of the MLPPP bundle.	An octet string that contains the name of the MLPPP bundle.
jnxPppAuthentication2	Specifies the type of PPP authentication used.	This object uses the following integer values: • **0**–None (no authentication is negotiated) • **1**–pap (PAP negotiation only) • **2**–chap (CHAP negotiation only) • **3**–eap (Extensible Authentication Protocol negotiation only)
jnxNibbleConfig	Specifies the configuration variable that consists of nibbles (4 bits).	-

Related Documentation

- PPP MIB Overview on page 589
- Managed Objects for PPP Link Control Protocol Layer on page 591
- Managed Objects for PPP IP Network Control Protocol Layer on page 596

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- Managed Objects for PPP OSI Network Control Protocol Layer on page 599
- Managed Objects for PPP Session Attributes on page 602
- Managed Objects for Multilink PPP Layer on page 604
- Managed Objects for PPP Interface Summary Counts on page 609
- Managed Objects for PPP IPv6 Network Control Protocol Layer on page 615
- Managed Objects for PPP Global Attributes on page 617

Managed Objects for PPP Link Control Protocol Layer

The **jnxPppLcp** object, whose object identifier is **{jnxPPPObjects 1}**, contains the objects listed in **jnxPppLinkStatusTable**, **jnxPppLinkConfigTable**, and **jnxPppNextifIndex**. You can use the managed objects listed in these tables to retrieve information about the PPP interfaces present in the system.

For information about **jnxPppLinkStatusTable**, **jnxPppLinkConfigTable**, and **jnxPppNextIfIndex**, see the following sections:

- jnxPppLinkStatusTable on page 591
- jnxPppLinkConfigTable on page 594
- jnxPppNextIfIndex on page 596

jnxPppLinkStatusTable

jnxPppLinkStatusTable, whose object identifier is **{jnxPppLcp 1}**, contains the objects listed in Table 255 on page 592 and provides information about the characteristics of the PPP interface.

Each **jnxPppLinkStatusEntry**, whose object identifier is **{jnxPppLinkStatusTable 1}**, maps to a specific characteristic of the PPP interface present in the system.

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Table 255: jnxPppLinkStatusTable

Object	Object ID	Description
jnxPppLinkStatusTerminateReason	**jnxPppLinkStatusEntry 1**	Reason the PPP link was terminated. Possible values: • none (0) - None. • other (1) - Not specified. • adminDisable (2) - Interface was administratively disabled. • lowerlayerDown (3) - Underlying interface is down. • noUpperInterface (4) - No interface above PPP. • authenticationFailure (5) - Authentication failed. • peerTerminated (6) - Peer initiated termination. • peerRenegotiated (7) - Peer initiated renegotiation. • maxretriesExceeded (8) - Maximum number of retries exceeded. • negotiationFailure (9) - Failed to negotiate LCP option. • keepaliveFailure (10) - Keepalive failed. • sessionTimeout (11) - Maximum session period expired. • inactivityTimeout (12) - Maximum inactivity period expired. • addressLeaseExpired (13) - Lease for network address expired. • adminLogout (14) - Session administratively terminated. • tunnelFailed (15) - Associated tunnel failed. • tunnelDisconnected (16) - Associated tunnel disconnected. • loopback (17) - Loopback detected.

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Table 255: jnxPppLinkStatusTable *(continued)*

Object	Object ID	Description
jnxPppLinkStatusTerminateNegFailOption	**jnxPppLinkStatusEntry 2**	PPP LCP option for which the negotiation failed when **jnxPppLinkStatusTerminateReason** has a value of negotiationFailure (9). Possible values: • none (0) - None. • other (1) - Not specified. • localMru (2) - MRU setting of the local device. • remoteMru (3) - MRU setting of the remote device. • localMagicNumber (4) - Configuration option that provides a method to detect looped-back links on the local device. • remoteMagicNumber (5) - Configuration option that provides a method to detect looped-back links on the remote device. • localAuthentication (6) - Authentication infromation about the local entity. • localToRemoteProtocolCompression (7) - Protocol compression is enabled or disabled. • localToRemoteACCompression (8) - AC compression is enabled or disabled.
jnxPppLinkStatusInKeepaliveRequests	**jnxPppLinkStatusEntry 3**	Number of keepalive requests received.
jnxPppLinkStatusOutKeepaliveRequests	**jnxPppLinkStatusEntry 4**	Number of keepalive requests transmitted.
jnxPppLinkStatusInKeepaliveReplies	**jnxPppLinkStatusEntry 5**	Number of keepalive replies received.
jnxPppLinkStatusOutKeepaliveReplies	**jnxPppLinkStatusEntry 6**	Number of keepalive replies transmitted.
jnxPppLinkStatusKeepaliveFailures	**jnxPppLinkStatusEntry 7**	Number of keepalive failures detected.
jnxPppLinkStatusLocalMagicNumber	**jnxPppLinkStatusEntry 8**	Magic number negotiated for the local side.
jnxPppLinkStatusRemoteMagicNumber	**jnxPppLinkStatusEntry 9**	Magic number negotiated for the remote side.
jnxPppLinkStatusLocalAuthentication	**jnxPppLinkStatusEntry 10**	Authentication protocol negotiated for the local side.

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Table 255: jnxPppLinkStatusTable *(continued)*

Object	Object ID	Description
jnxPppLinkStatusTunnelIfIndex	jnxPppLinkStatusEntry 11	IfIndex of an associated interface pertaining to a tunneling protocol. Zero indicates that no such interface exists.
jnxPppLinkStatusRenegoTerminates	jnxPppLinkStatusEntry 12	Number of times link was terminated because peer exceeded the maximum renegotiation attempts.

jnxPppLinkConfigTable

jnxPppLinkConfigTable, whose object identifier is **{jnxPppLcp 2}**, contains the objects listed in Table 256 on page 594 and provides information about the PPP interface.

Each **jnxPppLinkConfigEntry**, whose object identifier is **{jnxPppLinkConfigTable 1}**, maps to a specific characteristic of the PPP interface present in the system.

Table 256: jnxPppLinkConfigTable

Object	Object ID	Description
jnxPppLinkConfigIfIndex	jnxPppLinkConfigEntry 1	IfIndex of the PPP interface.
jnxPppLinkConfigRowStatus	jnxPppLinkConfigEntry 2	Row status for this entry. Possible values: • createAndGo • destroy
jnxPppLinkConfigLowerIfIndex	jnxPppLinkConfigEntry 3	IfIndex of the interface over which this PPP interface is layered. Zero indicates no layering.
jnxPppLinkConfigKeepalive	jnxPppLinkConfigEntry 4	Keepalive duration, in seconds. Value of zero disables keepalive.
jnxPppLinkConfigAuthentication	jnxPppLinkConfigEntry 5	Type of authentication, to be negotiated with a peer. Possible values: • none - No authentication is negotiated. • pap - PAP negotiation only. • chap - CHAP negotiation only. • papChap - PAP negotiation is attempted first; if that fails, CHAP negotiation is attempted. • chapPap - CHAP negotiation is attempted first; if that fails, PAP negotiation is attempted.
jnxPppLinkConfigMaxAuthenRetries	jnxPppLinkConfigEntry 6	Number of authentication retries permitted, in addition to a failed initial attempt.

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Table 256: jnxPppLinkConfigTable *(continued)*

Object	Object ID	Description
jnxPppLinkConfigStandardIfIndex	jnxPppLinkConfigEntry 7	IfIndex value for this interface in the standard PPP MIBs.
jnxPppLinkConfigChapMinChallengeLength	jnxPppLinkConfigEntry 8	Minimum value of the CHAP authenticator challenge length value.
jnxPppLinkConfigChapMaxChallengeLength	jnxPppLinkConfigEntry 9	Maximum value of the CHAP authenticator challenge length value.
jnxPppLinkConfigPassiveMode	jnxPppLinkConfigEntry 10	Status of the LCP state machine. Possible values: • enabled - Status of the LCP state machine is in passive mode. • disabled - Status of the LCP state machine is not in a passive mode.
jnxPppLinkConfigAuthenticatorLogicalSystem	jnxPppLinkConfigEntry 11	Name of the logical system used for authentication on the PPP interface.
jnxPppLinkConfigAuthenticatorRoutingInstance	jnxPppLinkConfigEntry 12	Name of the routing instance used for authentication on the PPP interface.
jnxPppLinkConfigAaaProfile	jnxPppLinkConfigEntry 13	Name of the access profile used for authentication on the PPP interface.
jnxPppLinkConfigAuthentication2	jnxPppLinkConfigEntry 14	Type of authentication. Possible values: • none (0) • pap (1) • chap (2) • eap (3)
jnxPppLinkConfigIgnoreMagicNumberMismatch	jnxPppLinkConfigEntry 15	Action to be taken when the peer has not negotiated yet or has sent either null or invalid magic number in the LCP packet. Possible values: • Ignore the mismatch and retain the connection (1) • Disallow the match and terminate the connection (2)
jnxPppLinkConfigMaxLcpRenegotiation	jnxPppLinkConfigEntry 16	Maximum number of allowed lcp renegotiation attempts from peer. NOTE: This attribute is not supported. The attribute returns the default value as mentioned in the description.

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jnxPppNextIfIndex

The **jnxPppNextIfIndex** object, whose object identifier is **{jnxPppLcp 3}**, contains the object listed in Table 257 on page 596. Successive **Get** requests usually return different values, to avoid collisions among clients seeking to create table entries simultaneously.

Table 257: jnxPppNextIfIndex

Object	Object ID	Description
jnxPppNextIfIndex	**jnxPppLcp 3**	Coordinates allocation of ifIndex value for entries in **jnxPppLinkConfigTable**.

Related Documentation

- PPP MIB Overview on page 589
- PPP MIB Textual Conventions on page 590
- Managed Objects for PPP IP Network Control Protocol Layer on page 596
- Managed Objects for PPP IPv6 Network Control Protocol Layer on page 615
- Managed Objects for Multilink PPP Layer on page 604
- Managed Objects for PPP Interface Summary Counts on page 609

Managed Objects for PPP IP Network Control Protocol Layer

jnxPppIpTable and **jnxPppIpConfigTable**, whose object identifier is **{jnxPPPObjects 3}**, contains the objects listed in Table 258 on page 596 and Table 259 on page 598, respectively. You can use these managed objects to retrieve information about the IPCP status information for a specific PPP link.

For information about **jnxPppIpTable** and **jnxPppIpConfigTable**, see the following sections:

- jnxPppIpTable on page 596
- jnxPppIpConfigTable on page 597

jnxPppIpTable

jnxPppIpTable, whose object identifier is **{jnxPppIp 1}**, contains the objects listed in Table 258 on page 596 and provides IPCP status information for a particular PPP link.

Each **jnxPppIpEntry**, whose object identifier is **{jnxPppIpTable 1}**, maps to a specific IP parameter for the local PPP entity.

Table 258: jnxPppIpTable

Object	Object ID	Description
jnxPppIpServiceStatus	**jnxPppIpEntry 1**	Indication as to whether IP protocol service is operating over this PPP link.

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Table 258: jnxPppIpTable *(continued)*

Object	Object ID	Description
jnxPppIpTerminateReason	**jnxPppIpEntry 2**	Reason the IPCP link was terminated. Possible values: • none (0) - None. • other (1) - Not specified. • noService (2) - No IP service was configured on this PPP link. • admin (3) - Administratively disabled. • linkDown (4) - Underlying link is down. • peerTerminated (5) - Peer initiated termination. • peerRenegotiated (6) - Peer initiated renegotiation. • maxRetriesExceeded (7) - Configured maximum number of retries exceeded. • negotiationFailure (8) - Failed to negotiate IPCP option.
jnxPppIpTerminateNegfailOption	**jnxPppIpEntry 3**	PPP IPCP option for which the negotiation failed when **jnxPppIpTerminateReason** has a value of negotiationFailure(8).
jnxPppIpLocalIpAddress	**jnxPppIpEntry 4**	IP address used by the local side.
jnxPppIpRemoteIpAddress	**jnxPppIpEntry 5**	IP address used by the remote side.
jnxPppIpRemotePrimaryDnsAddress	**jnxPppIpEntry 6**	Primary DNS server used by the remote side.
jnxPppIpRemoteSecondaryDnsAddress	**jnxPppIpEntry 7**	Secondary DNS server used by the remote side.
jnxPppIpRemotePrimaryWinsAddress	**jnxPppIpEntry 8**	Primary WINS server used by the remote side.
jnxPppIpRemoteSecondaryWinsAddress	**jnxPppIpEntry 9**	Secondary WINS server used by the remote side.
jnxPppIpNetworkStatusIpcpRenegoTerminates	**jnxPppIpEntry 10**	Number of times IPCP was terminated because peer exceeded the maximum renegotiation attempts.

jnxPppIpConfigTable

jnxPppIpConfigTable, whose object identifier is **{jnxPppIp 2}**, contains the objects listed in Table 259 on page 598 and provides IPCP configuration information for a particular PPP link.

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Each **jnxPppIpConfigEntry**, whose object identifier is **{jnxPppIpConfigTable 1}**, maps to a specific IP parameter for the local PPP entity.

Table 259: jnxPppIpConfigTable

Object	Object ID	Description
jnxPppIpConfigPeerDnsPriority	jnxPppIpConfigEntry 1	Determines whose DNS address prevails in the event of a negotiation conflict. Possible values: • enable - Peer's DNS address prevails. • disable - Local PPP interface DNS address prevails.
jnxPppIpConfigPeerWinsPriority	jnxPppIpConfigEntry 2	Determines whose WINS address prevails in the event of a negotiation conflict. Possible values: • enable - Peer's WINS address prevails. • disable - Local PPP interface WINS address prevails.
jnxPppIpConfigIpcpNetmask	jnxPppIpConfigEntry 3	Enables the negotiation of the IPCP option netmask during IPCP negotiation.
jnxPppIpConfigInitiateIp	jnxPppIpConfigEntry 4	Enables the initiation of negotiation of the IPCP.
jnxPppIpConfigMaxIpcpRenegotiation	jnxPppIpConfigEntry 5	Specifies the maximum number of allowed IPCP renegotiation attempts from peer. NOTE: This attribute is not supported. The attribute returns the default value.
jnxPppIpConfigPromptIpcpDnsOption	jnxPppIpConfigEntry 6	Controls prompting of IPCP DNS option to remote peer. NOTE: This attribute is not supported. The attribute returns the default value.
jnxPppIpConfigIpcpLockout	jnxPppIpConfigEntry 7	Enables IPCP lockout. IPCP negotiation is blocked after a different NCP service is up. NOTE: This attribute is not supported. The attribute returns the default value.

Related Documentation

- PPP MIB Overview on page 589
- PPP MIB Textual Conventions on page 590
- Managed Objects for PPP Link Control Protocol Layer on page 591
- Managed Objects for PPP IPv6 Network Control Protocol Layer on page 615

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- Managed Objects for PPP Interface Summary Counts on page 609

Managed Objects for PPP OSI Network Control Protocol Layer

jnxPppOsiTable and **jnxPppOsiConfigTable**, whose object identifier is **{jnxPPPObjects 4}**, contains the objects listed in Table 260 on page 599 and Table 261 on page 601, respectively. You can use these managed objects to retrieve information about the status of the NCP state machine as well as the outcome of the option parameter negotiation. The managed objects are used to manage the PPP Network Control Protocol for OSI protocol operation (OSICP).

For information about **jnxPppOsiTable** and **jnxPppOsiConfigTable**, see the following sections:

- jnxPppOsiTable on page 599
- jnxPppOsiConfigTable on page 601

jnxPppOsiTable

jnxPppOsiTable, whose object identifier is **{jnxPppOsi 1}**, contains the objects listed in Table 260 on page 599 and provides status information of the NCP state machine.

Each **jnxPppOsiEntry**, whose object identifier is **{jnxPppOsiTable 1}**, maps to a specific OSI parameter for the local PPP entity.

Table 260: jnxPppOsiTable

Object	Object ID	Description
jnxPppOsiServiceStatus	**jnxPppOsiEntry 1**	Indication as to whether OSI protocol service is operating over this PPP link.
jnxPppOsiOperStatus	**jnxPppOsiEntry 2**	Operational status of the OSI network protocol. Possible values: • Opened (1) - State machine status is open. • notOpened (2) - State machine status is not open.

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Table 260: jnxPppOsiTable *(continued)*

Object	Object ID	Description
jnxPppOsiTerminateReason	**jnxPppOsiEntry 3**	Reason the OSICP link was terminated. Possible values: • none (0) - None. • other (1) - Not specified. • noService (2) - No OSI service was configured on this PPP link. • admin (3) - Administratively disabled. • linkDown (4) - Underlying link is down. • peerTerminated (5) - Peer initiated termination. • peerRenegotiated (6) - Peer initiated renegotiation. • maxRetriesExceeded (7) - Configured maximum number of retries exceeded. • negotiationFailure (8) - Failed to negotiate IPCP option.
jnxPppOsiTerminateNegFailOption	**jnxPppOsiEntry 4**	PPP OSICP option for which negotiation failed when **jnxPppOsiTerminateReason** has a value of negotiationFailure (8).
jnxPppOsiLocalAlignNpdu	**jnxPppOsiEntry 5**	Local alignment of network PDU. Possible values: • none (0) - No alignment specified. • oneModulo4 (1) - Alignment on first octet. • twoModulo4 (2) - Alignment on second octet. • threeModulo4 (3) - Alignment on third octet. • fourModulo4 (4) - Alignment on fourth octet. • even (254) - Alignment on even-octet boundary. • odd (255) - Alignment on odd-octet boundary.

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Table 260: jnxPppOsiTable *(continued)*

Object	Object ID	Description
jnxPppOsiRemoteAlignNpdu	**jnxPppOsiEntry 6**	Remote Alignment of network PDU. Possible values: • none (0) - No alignment specified. • oneModulo4 (1) - Alignment on first octet. • twoModulo4 (2) - Alignment on second octet. • threeModulo4 (3) - Alignment on third octet. • fourModulo4 (4) - Alignment on fourth octet. • even (254) - Alignment on even-octet boundary. • odd (255) - Alignment on odd-octet boundary.

jnxPppOsiConfigTable

jnxPppOsiConfigTable, whose object identifier is **{jnxPppOsi 2}**, contains the objects listed in Table 261 on page 601, and provides administrative control over the NCP state machine and permits configuration of option parameters that can be used during NCP negotiation.

Each **jnxPppOsiConfigEntry**, whose object identifier is **{jnxPppOsiConfigTable 1}**, maps to a specific option parameter that can be used for NCP negotiation.

Table 261: jnxPppOsiConfigTable

Object	Object ID	Description
jnxPppOsiConfigAdminStatus	**jnxPppOsiConfigEntry 1**	Status of the OSI network protocol. Possible values: • open (1) - Administrative status of the NCP state machine changes to open. • close (2) - Administrative status of the NCP state machine changes to closed.

Related Documentation

- PPP MIB Overview on page 589
- PPP MIB Textual Conventions on page 590
- Managed Objects for PPP Link Control Protocol Layer on page 591
- Managed Objects for PPP IP Network Control Protocol Layer on page 596
- Managed Objects for PPP IPv6 Network Control Protocol Layer on page 615
- Managed Objects for PPP Global Attributes on page 617

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Managed Objects for PPP Session Attributes

jnxPppSessionTable, whose object identifier is **{jnxPppSession 1}**, contains the objects listed in Table 262 on page 602. You can use the managed objects to retrieve information about the status of each PPP session.

Each **jnxPppSessionEntry**, whose object identifier is **{jnxPppSessionTable 1}**, maps to a specific characteristic of the PPP interface.

Table 262: jnxPppSessionTable

Object	Object ID	Description
jnxPppSessionGrant	**jnxPppSessionEntry 1**	Indication whether the session was authenticated.
jnxPppSessionterminateReason	**jnxPppSessionEntry 2**	Reason the session was terminated. Possible values: • none (0) • unknown (1) • userRequest (2) • keepaliveFailure (3) • sessionTimeout (4) • inactivityTimeout (5) • adminDisable (6) • lowerLayerDown (7) • noUpperInterface (8) • deny (9) • noHardware (10) • noResources (11) • noInterface (12) • challengeTimeout (13) • requestTimeout (14) • authenticatorTimeout (15) • addressLeaseExpired (16) • adminLogout (17) • tunnelFailed (18)
jnxPppSessionStartTime	**jnxPppSessionEntry 3**	Value of the sysUpTime when this session last became active.
jnxPppSessionInOctets	**jnxPppSessionEntry 4**	Number of octets received since this session last became active.
jnxPppSessionOutOctets	**jnxPppSessionEntry 5**	Number of octets sent since this session last became active.
jnxPppSessionInPackets	**jnxPppSessionEntry 6**	Number of packets received since this session last became active.

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Table 262: jnxPppSessionTable *(continued)*

Object	Object ID	Description
jnxPppSessionOutPackets	jnxPppSessionEntry 7	Number of packets sent since this session last became active.
jnxPppSessionSessionTimeout	jnxPppSessionEntry 8	Maximum duration for the session, after which the session automatically terminates.
jnxPppSessionInactivityTimeout	jnxPppSessionEntry 9	Maximum inactivity duration for the session, after which the session automatically terminates.
jnxPppSessionAccountingInterval	jnxPppSessionEntry 10	Interval that must elapse between generation of accounting records for this session.
jnxPppSessionRemoteIpAddress	jnxPppSessionEntry 11	IP address of the remote PPP entity obtained from the authentication service for IPCP negotiation.
jnxPppSessionRemotePrimaryDnsAddress	jnxPppSessionEntry 12	IP address of the remote primary DNS server obtained from the authentication service for IPCP negotiation.
jnxPppSessionRemoteSecondaryDnsAddress	jnxPppSessionEntry 13	IP address of the remote secondary DNS server obtained from the authentication service for IPCP negotiation.
jnxPppSessionRemotePrimaryWinsAddress	jnxPppSessionEntry 14	IP address of the remote primary WINS server obtained from the authentication service for IPCP negotiation.
jnxPppSessionRemoteSecondaryWinsAddress	jnxPppSessionEntry 15	IP address of the remote primary WINS server obtained from the authentication service for IPCP negotiation.
jnxPppSessionRemoteIpv6AddressIfIdentifier	jnxPppSessionEntry 16	IPv6 address interface identifier obtained from the authentication service for IPCP negotiation.
jnxPppSessionInhibitIp	jnxPppSessionEntry 17	Indication whether the IP service is inhibited by the authentication service for this session.
jnxPppSessioninhibitIpv6	jnxPppSessionEntry 18	Indication whether the IPv6 service is inhibited by the authentication service for this session.

Related Documentation

- PPP MIB Overview on page 589
- PPP MIB Textual Conventions on page 590
- Managed Objects for PPP Link Control Protocol Layer on page 591
- Managed Objects for PPP Interface Summary Counts on page 609
- Managed Objects for PPP IP Network Control Protocol Layer on page 596
- Managed Objects for PPP OSI Network Control Protocol Layer on page 599

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Managed Objects for Multilink PPP Layer

The **jnxPppMlPpp** object, whose object identifier is **{jnxPPPObjects 6}**, defines the managed objects listed in Table 264 on page 605, along with the objects listed in **jnxPppMlPppBundleTable**, **jnxPppMlPppLinkConfigTable**, **jnxPppMlPppNetworkConfigTable**, **jnxPppMlPppNextNetworkIfIndex**, and **jnxPppMlPppLinkBindTable**. You can use these managed objects to retrieve information about the MLPPP bundles present in the system.

For information about **jnxPppMlPppBundleTable**, **jnxPppMlPppNextLinkIfIndex**,**jnxPppMlPppLinkConfigTable**, **jnxPppMlPppNetworkConfigTable**, **jnxPppMlPppNextNetworkIfIndex**, and **jnxPppMlPppLinkBindTable**, see the following sections:

- jnxPppMlPppBundleTable on page 604
- jnxPppMlPppNextLinkIfIndex on page 604
- jnxPppMlPppLinkConfigTable on page 605
- jnxPppMlPppNextNetworkIfIndex on page 607
- jnxPppMlPppNetworkConfigTable on page 608
- jnxPppMlPppLinkBindTable on page 608

jnxPppMlPppBundleTable

jnxPppMlPppBundleTable, whose object identifier is **{jnxPppMlPpp 1}**, contains the objects listed in Table 263 on page 604 and provides information about the characteristics of the MLPPP bundle.

Each **jnxPppMlPppBundleEntry**, whose object identifier is **{jnxPppMlPppBundleTable 1}**, maps to a specific characteristic of the MLPPP bundle in the system.

Table 263: jnxPppMLPppBundleTable

Object	Object ID	Description
jnxPppMlPppBundleName	**jnxPppMlPppBundleEntry 1**	Administrative name of the MLPPP bundle associated with this MLPPP network interfaces.
jnxPppMlPppBundleRowStatus	**jnxPppMlPppBundleEntry 2**	Row status for this entry. Possible values: • createAndGo (4) • destroy (6)
jnxPppMlPppBundleNetworkIfIndex	**jnxPppMlPppBundleEntry 3**	IfIndex of the MLPPP network interface.

jnxPppMlPppNextLinkIfIndex

The **jnxPppMlPppNextLinkIfIndex** object, whose object identifier is **{jnxPppMlPpp 2}**, contains the object listed in Table 264 on page 605. Successive **Get** requests usually return

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different values, to avoid collisions among clients seeking to create table entries simultaneously.

Table 264: jnxPppMlPppNextLinkIfIndex

Object	Object ID	Description
jnxPppMlPppNextLinkIfIndex	**jnxPppMlPpp 2**	Coordinates allocation of ifIndex value for entries in **jnxPppMlPppLinkConfigTable**.

jnxPppMlPppLinkConfigTable

jnxPppMlPppLinkConfigTable, whose object identifier is **{jnxPppMlPpp 3}**, contains the objects listed in Table 265 on page 605 and provides information about the MLPPP interfaces present in the system.

Each **jnxPppMlPppLinkConfigEntry**, whose object identifier is **{jnxPppMlPppLinkConfigTable 1}**, maps to a specific characteristic of the MLPPP interface present in the system.

Table 265: jnxPppMLPppLinkConfigTable

Object	Object ID	Description
jnxPppMlPppLinkConfigIfIndex	**jnxPppMlPppLinkConfigEntry 1**	IfIndex of the MLPPP interface.
jnxPppMlPppLinkConfigLowerIfIndex	**jnxPppMlPppLinkConfigEntry 2**	IfIndex of the interface over which this MLPPP interface is layered. Zero indicates no layering.
jnxPppMlPppLinkConfigKeepalive	**jnxPppMlPppLinkConfigEntry 3**	Keepalive duration, in seconds. Value of zero disables keepalive.
jnxPppMlPppLinkConfigAuthentication	**jnxPppMlPppLinkConfigEntry 4**	Type of authentication, to be negotiated with a peer. Possible values: • none - No authentication is negotiated. • pap - PAP negotiation only. • chap - CHAP negotiation only. • papChap - PAP negotiation is attempted first; if that fails, CHAP negotiation is attempted. • chapPap - CHAP negotiation is attempted first; if that fails, PAP negotiation is attempted.
jnxPppMlPppLinkConfigMaxAuthenRetries	**jnxPppMlPppLinkConfigEntry 5**	Number of authentication retries permitted, in addition to a failed initial attempt.

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Table 265: jnxPppMLPppLinkConfigTable *(continued)*

Object	Object ID	Description
jnxPppMlPppLinkConfigRowStatus	jnxPppMlPppLinkConfigEntry 6	Row status for this entry. Possible values: • createAndGo (4) • destroy (6)
jnxPppMlPppLinkConfigAaaProfile	jnxPppMlPppLinkConfigEntry 7	Name of the access profile used for authentication on the MLPPP interface.
jnxPppMlPppLinkConfigChapMinChallengeLength	jnxPppMlPppLinkConfigEntry 8	Minimum value of the CHAP authenticator challenge length value.
jnxPppMlPppLinkConfigChapMaxChallengeLength	jnxPppMlPppLinkConfigEntry 9	Maximum value of the CHAP authenticator challenge length value.
jnxPppMlPppLinkConfigPassiveMode	jnxPppMlPppLinkConfigEntry 10	Status of the MLPPP state machine. Possible values: • enabled - Status of the MLPPP state machine moves to passive mode. • disabled - Status of the MLPPP state machine does not move to passive mode.
jnxPppMlPppLinkConfigAuthenticatorLogicalSystem	jnxPppMlPppLinkConfigEntry 11	Name of the logical system used for authentication on the MLPPP interface.
jnxPppMlPppLinkConfigAuthenticatorRoutingInstance	jnxPppMlPppLinkConfigEntry 12	Name of the routing instance used for authentication on the MLPPP interface.
jnxPppMlPppLinkConfigFragmentation	jnxPppMlPppLinkConfigEntry 13	Status of MLPPP fragmentation. Possible values: • enabled - Fragmentation is enabled. • disabled - Fragmentation is disabled.
jnxPppMlPppLinkConfigReassembly	jnxPppMlPppLinkConfigEntry 14	Status of MLPPP reassembly. • enabled - Reassembly is enabled. • disabled - Reassembly is disabled.
jnxPppMlPppLinkConfigMaxReceiveReconstructedUnit	jnxPppMlPppLinkConfigEntry 15	Maximum receive reconstructed unit (MRRU) that the local MLPPP entity advertises to the remote MLPPP entity.

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Table 265: jnxPppMLPppLinkConfigTable *(continued)*

Object	Object ID	Description
jnxPppMlPppLinkConfigFragmentSize	jnxPppMlPppLinkConfigEntry 16	Size of the fragments transmittted by the local MLPPP entity.
jnxPppMlPppLinkConfigHashLinkSelection	jnxPppMlPppLinkConfigEntry 17	Status of MLPPP link selection for non-best effort traffic. • enabled - MLPPP hash-based link selection for non-best traffic. • disabled - MLPPP hash-based link selection is not selected.
jnxPppMlPppLinkConfigAuthentication2	jnxPppMlPppLinkConfigEntry 18	Type of authentication. Possible values: • none (0) • pap (1) • chap (2) • eap (3)
jnxPppMlPppLinkConfigIgnoreMagicNumberMismatch	jnxPppMlPppLinkConfigEntry 19	Action to be taken when the peer has not negotiated yet or has sent either null or invalid magic number in the LCP packet. Possible values: • Ignore the mismatch and retain the connection (1) • Disallow the match and terminate the connection (2)
jnxPppMlPppLinkConfigMultilinkMaxMultiClass	jnxPppMlPppLinkConfigEntry 20	Status of multiclass multilink PPP. Possible values: Enabled and disabled.
jnxPppMlPppLinkConfigMultilinkMaxMultiClasses	jnxPppMlPppLinkConfigEntry 21	Maximum number of MCML classes to be negotiated.

jnxPppMlPppNextNetworkIfIndex

The **jnxPppMlPppNextNetworkIfIndex** object, whose object identifier is **{jnxPppMlPpp 4}**, contains the object listed in Table 266 on page 607. Successive **Get** requests usually return different values, to avoid collisions among clients seeking to create table entries simultaneously.

Table 266: jnxPppMlPppNextNetworkIfIndex

Object	Object ID	Description
jnxPppMlPppNextNetworkIfIndex	jnxPppMlPpp 4	Coordinates allocation of ifIndex value for entries in **jnxPppMlPppNetworkConfigTable**.

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jnxPppMlPppNetworkConfigTable

jnxPppMlPppNetworkConfigTable, whose object identifier is **{jnxPppMlPpp 5}**, contains the objects listed in Table 267 on page 608 and provides information about the characteristics of the MLPPP network interfaces present in the system.

Each **jnxPppMlPppNetworkConfigEntry**, whose object identifier is **{jnxPppMlPppNetworkConfigTable 1}**, maps to a specific characteristic of the MLPPP network interface in the system.

Table 267: jnxPppMLPppNetworkConfigTable

Object	Object ID	Description
jnxPppMlPppNetworkConfigIfindex	jnxPppMlPppNetworkConfigEntry 1	IfIndex of the MLPPP network interface.
jnxPppMlPppNetworkConfigLowerIfIndex	jnxPppMlPppNetworkConfigEntry 2	IfIndex of a PPP link interface over which this MLPPP network interface is to be layered.
jnxPppMlPppNetworkBundleName	jnxPppMlPppNetworkConfigEntry 3	Administrative name of the MLPPP bundle.
jnxPppMlPppNetworkRowStatus	jnxPppMlPppNetworkConfigEntry 4	Row status for this entry. Possible values: • createAndGo • destroy

jnxPppMlPppLinkBindTable

jnxPppMlPppLinkBindTable, whose object identifier is **{jnxPppMlPpp 6}**, contains the objects listed in Table 268 on page 608 and provides information about the characteristics of the MLPPP link interface and MLPPP network interfaces bindings.

Each **jnxPppMlPppLinkBindEntry**, whose object identifier is **{jnxPppMlPppLinkBindTable 1}**, maps to a specific characteristic of the MLPPP link interface.

Table 268: jnxPppMLPppLinkBindTable

Object	Object ID	Description
jnxPppMlPppBindNetworkIfIndex	jnxPppMlPppLinkBindEntry 1	IfIndex of the MLPPP network interface.
jnxPppMlPppBindLinkIfIndex	jnxPppMlPppLinkBindEntry 2	IfIndex of the MLPPP link interface bound by the MLPPP network interface.
jnxPppMlPppBindRowStatus	jnxPppMlPppLinkBindEntry 3	Row status for this entry. Possible values: • createAndGo • destroy

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Related Documentation

- PPP MIB Overview on page 589
- PPP MIB Textual Conventions on page 590
- Managed Objects for PPP Link Control Protocol Layer on page 591
- Managed Objects for PPP IP Network Control Protocol Layer on page 596
- Managed Objects for PPP OSI Network Control Protocol Layer on page 599
- Managed Objects for PPP Session Attributes on page 602
- Managed Objects for PPP Interface Summary Counts on page 609

Managed Objects for PPP Interface Summary Counts

The PPP interface summary counts, whose object identifier is **{jnxPPPObjects 7}**, contains the managed objects listed in Table 269 on page 609. You can retrieve information about the number of PPP interfaces configured or created, as well as the status of the interfaces using the managed objects listed in Table 269 on page 609.

Each **jnxPppSummary** maps to a specific characteristic of the PPP interface configured on the system.

Table 269: jnxPppSummaryCounts

Object	Object ID	Description
jnxPppSummaryPppInterfaceCount	jnxPppSummary 1	Total number of PPP interfaces configured in the system.
jnxPppSummaryPppIpNCPs	jnxPppSummary 2	Total number of IP NCPs configured in the system.
jnxPppSummaryPppOsiNCPs	jnxPppSummary 3	Total number of OSI NCPs configured in the system.
jnxPppSummaryPppIfAdminUp	jnxPppSummary 4	Total number of PPP interfaces in the system with administrative status of up.
jnxPppSummaryPppIfAdminDown	jnxPppSummary 5	Total number of PPP interfaces in the system with administrative status of down.
jnxPppSummaryPppIfOperUp	jnxPppSummary 6	Total number of PPP interfaces in the system with operational status of up.
jnxPppSummaryPppIfOperDown	jnxPppSummary 7	Total number of PPP interfaces in the system with operational state of down.
jnxPppSummaryPppIfOperDormant	jnxPppSummary 8	Total number of PPP interfaces in the system with operational status of dormant.
jnxPppSummaryPppIfNotPresent	jnxPppSummary 9	Total number of PPP interfaces in the system with operational status of not present.

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Table 269: jnxPppSummaryCounts *(continued)*

Object	Object ID	Description
jnxPppSummaryPppIfLowerLayerDown	jnxPppSummary 10	Total number of PPP interfaces in the system with operational status of lower layer down.
jnxPppSummaryPppIpNcpOpened	jnxPppSummary 11	Total number of PPP IP NCPs in the system with an operational status of open.
jnxPppSummaryPppIpNcpClosed	jnxPppSummary 12	Total number of PPP interfaces in the system with an operational status of closed.
jnxPppSummaryPppOsiNcpOpened	jnxPppSummary 13	Total number of PPP OSI NCPs in the system with an operational status of open.
jnxPppSummaryPppOsiNcpClosed	jnxPppSummary 14	Total number of PPP OSI NCPs in the system with an operational status of closed.
jnxPppSummaryPppIfLastChangeTime	jnxPppSummary 15	Value of the sysUpTime since the last creation or deletion of a PPP interface in the system. Zero indicates that the number of PPP interfaces remains unchanged since the last reinitialization.
jnxPppSummaryPppLinkInterfaceCount	jnxPppSummary 16	Total number of PPP link interfaces configured in the system.
jnxPppSummaryPppLinkIfAdminUp	jnxPppSummary 17	Total number of PPP link interfaces in the system with administrative status of up.
jnxPppSummaryPppLinkIfAdminDown	jnxPppSummary 18	Total number of PPP link interfaces in the system with administrative status of down.
jnxPppSummaryPppLinkIfOperUp	jnxPppSummary 19	Total number of PPP link interfaces in the system with an operational status of up.
jnxPppSummaryPppLinkIfOperDown	jnxPppSummary 20	Total number of PPP link interfaces in the system with an operational status of down.
jnxPppSummaryPppLinkIfOperDormant	jnxPppSummary 21	Total number of PPP link interfaces in the system with an operational status of dormant.
jnxPppSummaryPppLinkIfNotPresent	jnxPppSummary 22	Total number of PPP link interfaces in the system with an operational status of not present.
jnxPppSummaryPppLinkIfLowerLayerDown	jnxPppSummary 23	Total number of PPP link interfaces in the system with an operational status of lower layer down.
jnxPppSummaryPppLinkIfLastChangeTime	jnxPppSummary 24	Value of the sysUpTime since the last creation or deletion of a PPP interface in the system. Zero indicates that the number of PPP interfaces remains unchanged since the last reinitialization.

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Table 269: jnxPppSummaryCounts *(continued)*

Object	Object ID	Description
jnxPppSummaryPppNetworkInterfaceCount	jnxPppSummary 25	Total number of PPP network interfaces configured in the system.
jnxPppSummaryPppNetworkIpNCPs	jnxPppSummary 26	Total number of IP NCPs in the system configured on PPP network interfaces.
jnxPppSummaryPppNetworkOsiNCPs	jnxPppSummary 27	Total number of OSI NCPs in the system configured on PPP network interfaces.
jnxPppSummaryPppNetworkIfAdminUp	jnxPppSummary 28	Total number of PPP network interfaces in the system with an administrative status of up.
jnxPppSummaryPppNetworkIfAdminDown	jnxPppSummary 29	Total number of PPP network interfaces in the system with an administrative status of down.
jnxPppSummaryPppNetworkIfOperUp	jnxPppSummary 30	Total number of PPP network interfaces in the system with an operational status of up.
jnxPppSummaryPppNetworkIfOperDown	jnxPppSummary 31	Total number of PPP network interfaces in the system with an operational status of down.
jnxPppSummaryPppNetworkIfOperDormant	jnxPppSummary 32	Total number of PPP network interfaces in the system with an operational status of dormant.
jnxPppSummaryPppNetworkIfNotPresent	jnxPppSummary 33	Total number of PPP network interfaces in the system with an operational status of not present.
jnxPppSummaryPppNetworkIfLowerLayerDown	jnxPppSummary 34	Total number of PPP network interfaces in the system with an operational status of lower layer down.
jnxPppSummaryPppNetworkIpNcpOpened	jnxPppSummary 35	Total number of PPP IP NCPs in the system with an operational status of open.
jnxPppSummaryPppNetworkIpNcpClosed	jnxPppSummary 36	Total number of PPP IP NCPs in the system with an operational status of closed.
jnxPppSummaryPppNetworkOsiNcpOpened	jnxPppSummary 37	Total number of PPP OSI NCPs in the system with an operational status of open.
jnxPppSummaryPppNetworkOsiNcpClosed	jnxPppSummary 38	Total number of PPP OSI NCPs in the system with an operational status of closed.
jnxPppSummaryPppNetworkIfLastChangeTime	jnxPppSummary 39	Value of the sysUpTime since the last creation or deletion of a PPP interface in the system. Zero indicates that the number of PPP interfaces remains unchanged since the last reinitialization.
jnxPppSummaryPppIpv6NCPs	jnxPppSummary 40	Total number of IPv6 NCPs configured in the system.

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Table 269: jnxPppSummaryCounts *(continued)*

Object	Object ID	Description
jnxPppSummaryPppIpv6NcpOpened	jnxPppSummary 41	Total number of IPv6 NCPs configured in the system with an operational status of open.
jnxPppSummaryPppIpv6NcpClosed	jnxPppSummary 42	Total number of IPv6 NCPs configured in the system with an operational status of closed.
jnxPppSummaryPppNetworkIpv6NCPs	jnxPppSummary 43	Total number of IPv6 NCPs configured in the system.
jnxPppSummaryPppNetworkIpv6NcpOpened	jnxPppSummary 44	Total number of PPP IPv6 NCPs configured in the system with an operational status of open.
jnxPppSummaryPppNetworkIpv6NcpClosed	jnxPppSummary 45	Total number of PPP IPv6 NCPs configured in the system with an operational status of closed.
jnxPppSummaryPppStaticInterfaceCount	jnxPppSummary 46	Total number of static PPP interfaces configured in the system.
jnxPppSummaryPppMplsNCPs	jnxPppSummary 47	Total number of MPLS NCPs configured in the system.
jnxPppSummaryPppIpAdminOpen	jnxPppSummary 48	Total number of IP NCPs in the system with an administrative status of open.
jnxPppSummaryPppIpAdminClose	jnxPppSummary 49	Total number of IP NCPs in the system with an administrative status of closed.
jnxPppSummaryPppIpv6AdminOpen	jnxPppSummary 50	Total number of IPv6 NCPs in the system with an administrative status of open.
jnxPppSummaryPppIpv6AdminClose	jnxPppSummary 51	Total number of IPv6 NCPs in the system with an administrative status of closed.
jnxPppSummaryPppOsiAdminOpen	jnxPppSummary 52	Total number of OSI NCPs in the system with an administrative status of open.
jnxPppSummaryPppOsiAdminClose	jnxPppSummary 53	Total number of OSI NCPs in the system with an administrative status of closed.
jnxPppSummaryPppMplsAdminOpen	jnxPppSummary 54	Total number of MPLS NCPs in the system with an administrative status of open.
jnxPppSummaryPppMplsAdminClose	jnxPppSummary 55	Total number of MPLS NCPs in the system with an administrative status of closed.
jnxPppSummaryPppIpNcpNotPresent	jnxPppSummary 56	Total number of PPP IP NCPs in the system with an operational status of not present.
jnxPppSummaryPppIpNcpNoResources	jnxPppSummary 57	Total number of PPP IP NCPs in the system with an operational status of no resources.

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Table 269: jnxPppSummaryCounts *(continued)*

Object	Object ID	Description
jnxPppSummaryPppIpv6NcpNotPresent	jnxPppSummary 58	Total number of PPP IPv6 NCPs in the system with an operational status of not present.
jnxPppSummaryPppIpv6NcpNoResources	jnxPppSummary 59	Total number of PPP IPv6 NCPs in the system with an operational status of no resources.
jnxPppSummaryPppOsiNcpNotPresent	jnxPppSummary 60	Total number of PPP OSI NCPs in the system with an operational status of not present.
jnxPppSummaryPppOsiNcpNoResources	jnxPppSummary 61	Total number of PPP OSI NCPs in the system with an operational status of no resources.
jnxPppSummaryPppMplsNcpOpened	jnxPppSummary 62	Total number of PPP MPLS NCPs in the system with an operational status of open.
jnxPppSummaryPppMplsNcpClosed	jnxPppSummary 63	Total number of PPP MPLS NCPs in the system with an operational status of closed.
jnxPppSummaryPppMplsNcpNotPresent	jnxPppSummary 64	Total number of PPP MPLS NCPs in the system with an operational status of not present.
jnxPppSummaryPppMplsNcpNoResources	jnxPppSummary 65	Total number of PPP MPLS NCPs in the system with an operational status of no resources.
jnxPppSummaryPppLinkStaticInterfaceCount	jnxPppSummary 66	Total number of static PPP link interfaces in the system.
jnxPppSummaryPppNetworkStaticInterfaceCount	jnxPppSummary 67	Total number of static PPP network interfaces in the system.
jnxPppSummaryPppNetworkMplsNCPs	jnxPppSummary 68	Total number of MPLS NCPs configured on PPP network interfaces in the system.
jnxPppSummaryPppNetworkIpAdminOpen	jnxPppSummary 69	Total number of IP NCPs configured on PPP network interfaces with an administrative status of open.
jnxPppSummaryPppNetworkIpAdminClose	jnxPppSummary 70	Total number of IP NCPs configured on PPP network interfaces with an administrative status of closed.
jnxPppSummaryPppNetworkIpv6AdminOpen	jnxPppSummary 71	Total number of IPv6 NCPs configured on PPP network interfaces with an administrative status of open.
jnxPppSummaryPppNetworkIpv6AdminClose	jnxPppSummary 72	Total number of IPv6 NCPs configured on PPP network interfaces with an administrative status of closed.

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Table 269: jnxPppSummaryCounts *(continued)*

Object	Object ID	Description
jnxPppSummaryPppNetworkOsiAdminOpen	jnxPppSummary 73	Total number of OSI NCPs configured on PPP network interfaces with an administrative status of open.
jnxPppSummaryPppNetworkOsiAdminClose	jnxPppSummary 74	Total number of OSI NCPs configured on PPP network interfaces with an administrative status of closed.
jnxPppSummaryPppNetworkMplsAdminOpen	jnxPppSummary 75	Total number of MPLS NCPs configured on PPP network interfaces with an administrative status of open.
jnxPppSummaryPppNetworkMplsAdminClose	jnxPppSummary 76	Total number of MPLS NCPs configured on PPP network interfaces with an administrative status of closed.
jnxPppSummaryPppNetworkIpNcpNotPresent	jnxPppSummary 77	Total number of IP NCPs configured on PPP network interfaces with an operational status of not present.
jnxPppSummaryPppNetworkIpNcpNoResources	jnxPppSummary 78	Total number of IP NCPs configured on PPP network interfaces with an operational status of no resources.
jnxPppSummaryPppNetworkIpv6NcpNotPresent	jnxPppSummary 79	Total number of IPv6 NCPs configured on PPP network interfaces with an operational status of not present.
jnxPppSummaryPppNetworkIpv6NcpNoResources	jnxPppSummary 80	Total number of IPv6 NCPs configured on PPP network interfaces with an operational status of no resources.
jnxPppSummaryPppNetworkOsiNcpNotPresent	jnxPppSummary 81	Total number of OSI NCPs configured on PPP network interfaces with an operational status of not present.
jnxPppSummaryPppNetworkOsiNcpNoresources	jnxPppSummary 82	Total number of OSI NCPs configured on PPP network interfaces with an operational status of no resources.
jnxPppSummaryPppNetworkMplsNcpOpened	jnxPppSummary 83	Total number of MPLS NCPs configured on PPP network interfaces with an operational status of open.
jnxPppSummaryPppNetworkMplsNcpClosed	jnxPppSummary 84	Total number of MPLS NCPs configured on PPP network interfaces with an operational status of closed.
jnxPppSummaryPppNetworkMplsNcpNotPresent	jnxPppSummary 85	Total number of MPLS NCPs configured on PPP network interfaces with an operational status of not present.

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Table 269: jnxPppSummaryCounts *(continued)*

Object	Object ID	Description
jnxPppSummaryPppNetworkMplsNcpNoResources	jnxPppSummary 86	Total number of MPLS NCPs configured on PPP network interfaces with an operational status of no resources.

Related Documentation

- PPP MIB Overview on page 589
- PPP MIB Textual Conventions on page 590
- Managed Objects for PPP IP Network Control Protocol Layer on page 596
- Managed Objects for PPP Link Control Protocol Layer on page 591
- Managed Objects for PPP OSI Network Control Protocol Layer on page 599
- Managed Objects for PPP Session Attributes on page 602
- Managed Objects for PPP IPv6 Network Control Protocol Layer on page 615
- Managed Objects for PPP Global Attributes on page 617

Managed Objects for PPP IPv6 Network Control Protocol Layer

jnxPppIpv6Table and **jnxPppIpv6ConfigTable**, whose object identifier is **{jnxPPPObjects 8}**, contains the objects listed in Table 270 on page 615 and Table 271 on page 617, respectively. You can use these managed objects to retrieve information about the IPv6CP status information for a specific PPP link.

For information about **jnxPppIpv6Table** and **jnxPppIpv6ConfigTable**, see the following sections:

- jnxPppIpv6Table on page 615
- jnxPppIpv6ConfigTable on page 616

jnxPppIpv6Table

jnxPppIpv6Table, whose object identifier is **{jnxPppIpv6 1}**, contains the objects listed in Table 270 on page 615 and provides IPv6CP status information for a particular PPP link.

Each **jnxPppIpv6Entry**, whose object identifier is **{jnxPppIpv6Table 1}**, maps to a specific IPv6 parameter for the local PPP entity.

Table 270: jnxPppIpv6Table

Object	Object ID	Description
jnxPppIpv6ServiceStatus	jnxPppIpv6Entry 1	Indication as to whether IPv6 protocol service is operating over this PPP link.
jnxPppIpv6OperStatus	jnxPppIpv6Entry 2	Operational status of the IPv6 network protocol.

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Table 270: jnxPppIpv6Table *(continued)*

Object	Object ID	Description
jnxPppIpv6TerminateReason	**jnxPppIpv6Entry 3**	Reason the IPv6CP link was terminated. Possible values: • none (0) - None. • other (1) - Not specified. • noService (2) - No IPv6 service was configured on this PPP link. • admin (3) - Administratively disabled. • linkDown (4) - Underlying link is down. • peerTerminated (5) - Peer initiated termination. • peerRenegotiated (6) - Peer initiated renegotiation. • maxRetriesExceeded (7) - Configured maximum number of retries exceeded. • negotiationFailure (8) - Failed to negotiate IPv6CP option.
jnxPppIpv6TerminateNegfailOption	**jnxPppIpv6Entry 4**	PPP IPv6CP option for which the negotiation failed when **jnxPppIpTerminateReason** has a value of negotiationFailure(8).
jnxPppIpv6LocalIpv6AddressIfIdentifier	**jnxPppIpv6Entry 5**	IPv6 address interface identifier used by the local side.
jnxPppIpv6RemoteIpv6AddressIfIdentifier	**jnxPppIpv6Entry 6**	IPv6 address interface identifier used by the remote side.
jnxPppIpv6NetworkStatusIpv6cpRenegoTerminates	**jnxPppIpv6Entry 7**	Number of times IPv6CP was terminated because peer exceeded the maximum renegotiation attempts.

jnxPppIpv6ConfigTable

jnxPppIpv6ConfigTable, whose object identifier is **{jnxPppIpv6 2}**, contains the objects listed in Table 271 on page 617 and provides IPv6CP configuration information for a particular PPP link.

Each **jnxPppIpv6ConfigEntry**, whose object identifier is **{jnxPppIpv6ConfigTable 1}**, maps to a specific IPv6 parameter for the local PPP entity.

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Table 271: jnxPppIpv6ConfigTable

Object	Object ID	Description
jnxPppIpv6ConfigAdminStatus	jnxPppIpv6ConfigEntry 1	Desired status of the IPv6 network protocol. Possible values: • open (1) - Administrative status changes to open. • close (2) - Administrative status changes to closed.
jnxPppIpv6ConfigInitiateIpv6	jnxPppIpv6ConfigEntry 2	Initiation of negotiation of the IPv6CP. Possible values: Enabled, Disabled.
jnxPppIpv6ConfigMaxIpv6cpRenegotiation	jnxPppIpv6ConfigEntry 3	Maximum number of permitted IPv6CP renegotiation attempts from peer.

Related Documentation

- PPP MIB Overview on page 589
- PPP MIB Textual Conventions on page 590
- Managed Objects for PPP Link Control Protocol Layer on page 591
- Managed Objects for PPP IP Network Control Protocol Layer on page 596
- Managed Objects for PPP OSI Network Control Protocol Layer on page 599
- Managed Objects for PPP Session Attributes on page 602
- Managed Objects for Multilink PPP Layer on page 604
- Managed Objects for PPP Interface Summary Counts on page 609

Managed Objects for PPP Global Attributes

Table 272 on page 617 contains information about the non-interface based object used in the enterprise-specific PPP MIB.

Table 272: jnxPppGlobal

Object	Object ID	Description
jnxPppPeerIpAddressOptional	jnxPppGlobalConfig 1	Enables you to ignore conflicts between the IP address requested by the PPP client and the address returned by the RADIUS server during IPNCP negotiation.

Related Documentation

- PPP MIB Overview on page 589
- PPP MIB Textual Conventions on page 590
- Managed Objects for PPP Link Control Protocol Layer on page 591
- Managed Objects for PPP IP Network Control Protocol Layer on page 596
- Managed Objects for PPP OSI Network Control Protocol Layer on page 599

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- Managed Objects for PPP Session Attributes on page 602
- Managed Objects for Multilink PPP Layer on page 604
- Managed Objects for PPP Interface Summary Counts on page 609
- Managed Objects for PPP IPv6 Network Control Protocol Layer on page 615

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CHAPTER 76

Interpreting the Enterprise-Specific PPPoE MIB

- PPPoE MIB on page 619
- PPPoE MIB Textual Conventions on page 620
- Managed Objects for PPPoE Underlying Interfaces Layer Functions on page 620
- Managed Objects for PPPoE Interfaces Layer Functions on page 625
- Managed Objects for PPPoE Summary Counts on page 627
- Managed Objects for PPPoE ServiceName Tables on page 628

PPPoE MIB

The Juniper Networks enterprise-specific PPPoE MIB, whose object ID is **{jnxPppoeMibRoot 1}**, extends SNMP support for Point-to-Point Protocol over Ethernet (PPPoE) modules in Junos OS.

The PPPoE MIB stores PPPoE-related information, for both PPPoE interfaces and PPPoE underlying interfaces, such as authentication type, status, interface characteristics, and statistics. You can view this information by using SNMP **get** and **get-next** requests.

For a downloadable version of this MIB, see http://www.juniper.net/techpubs/en_US/junos11.4/topics/reference/ mibs/mib-jnx-pppoe.txt .

For information about the enterprise-specific PPPoE MIB objects, see the following topics:

- PPPoE MIB Textual Conventions on page 620
- Managed Objects for PPPoE Underlying Interfaces Layer Functions on page 620
- Managed Objects for PPPoE Interfaces Layer Functions on page 625
- Managed Objects for PPPoE Summary Counts on page 627
- Managed Objects for PPPoE ServiceName Tables on page 628

Related Documentation

- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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PPPoE MIB Textual Conventions

Table 273 on page 620 contains information about the textual conventions used in the enterprise-specific PPPoE MIB.

Table 273: PPPoE MIB Textual Conventions

Object	Description	Syntax
jnxPPPoEServiceNameAction	Set of Service-Name action types.	This object uses the following integer values: • **0**–Drop (no PADO packet sent) • **1**–Terminate (PADO packet sent)

Related Documentation

- PPPoE MIB Overview on page 619
- Managed Objects for PPPoE Underlying Interfaces Layer Functions on page 620
- Managed Objects for PPPoE Interfaces Layer Functions on page 625
- Managed Objects for PPPoE Summary Counts on page 627
- Managed Objects for PPPoE ServiceName Tables on page 628

Managed Objects for PPPoE Underlying Interfaces Layer Functions

You can use the managed objects listed in Table 274 on page 620 along with the objects listed in **jnxPPPoEIfTable**, **jnxPPPoEIfStatsTable**, and **jnxPPPoEIfLockoutTable** to retrieve information about the PPPoE underlying interfaces.

The **jnxPPPoEIfLayer** object, whose object identifier is **{jnxPPPoEObjects 1}**, contains the object listed in Table 274 on page 620. Successive **Get** requests usually return different values to avoid collisions among clients seeking to create table entries simultaneously.

Table 274: jnxPPPoENextIfIndex

Object	Object ID	Description
jnxPPPoENextIfIndex	**jnxPPPoEIfLayer 1**	Coordinates allocation of ifIndex value for entries in **jnxPPPoEIfTable**.

For information about **jnxPPPoEIfTable**, **jnxPPPoEIfStatsTable**, and **jnxPPPoEIfLockoutTable**, see the following sections:

- jnxPPPoEIfTable on page 621
- jnxPPPoEIfStatsTable on page 622
- jnxPPPoEIfLockoutTable on page 624

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jnxPPPoEIfTable

jnxPPPoEIfTable, whose object identifier is **{jnxPPPoEIfLayer 2}**, contains **jnxPPPoEIfEntry**, which maps to the parameters for a specific PPPoE underlying interface.

Each **jnxPPPoEIfEntry**, whose object identifier is **{jnxPPPoEIfTable 1}**, contains the objects listed in Table 275 on page 621.

Table 275: jnxPPPoEIfTable

Object	Object ID	Description
jnxPPPoEIfIfIndex	jnxPPPoEIfEntry 1	IfIndex of the PPPoE underlying interface.
jnxPPPoEIfMaxNumSessions	jnxPPPoEIfEntry 2	Maximum number of sessions allowed on the PPPoE underlying interface. Zero indicates that the maximum number of sessions allowed is unlimited.
jnxPPPoEIfRowStatus	jnxPPPoEIfEntry 3	Row status for this entry. Possible values: • createAndGo • destroy
jnxPPPoEIfLowerIfIndex	jnxPPPoEIfEntry 4	IfIndex of an interface over which this PPPoE underlying interface is to be layered. Zero indicates that there is no layering.
jnxPPPoEIfAcName	jnxPPPoEIfEntry 5	Name of the AC-NAME tag sent in any PADO packet on this interface.
jnxPPPoEIfDupProtect	jnxPPPoEIfEntry 6	State of PPPoE duplicate protection. Possible values: • On - Allows duplicate MAC addresses. • Off - Does not allow duplicate MAC addresses.
jnxPPPoEIfPADIFlag	jnxPPPoEIfEntry 7	Action taken when the interface receives a PADI packet. The response can be a PADO packet.
jnxPPPoEIfAutoconfig	jnxPPPoEIfEntry 8	Whether the PPPoE interface is created dynamically or statically. Possible values: • enable (1) - dynamic • disable (2) - static
jnxPPPoEIfServiceNameTable	jnxPPPoEIfEntry 9	Name of the ServiceNameTable associated with this interface for PADI processing.

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Table 275: jnxPPPoEIfTable *(continued)*

Object	Object ID	Description
jnxPPPoEIfPadrRemoteCircuitIdcapture	**jnxPPPoEIfEntry 10**	Whether the remote-circuit-id string is captured and subsequently used as the NAS-Port-Id RADIUS attribute when it arrives as a tag in the PADR packet.
jnxPPPoEIfMtu	**jnxPPPoEIfEntry 11**	Initial maximum transmit unit (MTU) that the PPPoE underlying interface entity advertises to the remote entity. Possible values: • 1 - Local PPPoE entity uses the MTU value determined by its underlying media interface. • 2 - Local PPPoE entity uses a value determined by the PPPoE Max-Mtu-Tag transmitted from the client in the PADR packet. If no Max-Mtu-tag is received, the value defaults to a maximum of 1494.
jnxPPPoEIfLockoutMin	**jnxPPPoEIfEntry 12**	Minimum number of seconds used to specify the duration of the lockout of the client from recognition for the specified interface.
jnxPPPoEIfLockoutMax	**jnxPPPoeIfEntry 13**	Maximum number of seconds used to specify the duration of the lockout of the client from recognition for the specified interface.
jnxPPPoEIfDynamicProfile	**jnxPPPoEIfEntry 14**	Dynamic profile attachment to this interface.

jnxPPPoEIfStatsTable

jnxPPPoEIfStatsTable, whose object identifier is **{jnxPPPoEIfLayer 3}**, contains **jnxPPPoEIfStatsEntry**, which displays the statistics for the PPPoE underlying interface.

Each **jnxPPPoEIfStatsEntry**, whose object identifier is **{jnxPPPoEIfStatsTable 1}**, contains the objects listed in Table 276 on page 622.

Table 276: jnxPPPoEIfStatsTable

Object	Object ID	Description
jnxPPPoEIfStatsRxPADI	**jnxPPPoEIfStatsEntry 1**	Number of PADI packets received.
jnxPPPoEIfStatsTxPADO	**jnxPPPoEIfStatsEntry 2**	Number of PADO packets transmitted.
jnxPPPoEIfStatsRxPADR	**jnxPPPoEIfStatsEntry 3**	Number of PADR packets received.

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Table 276: jnxPPPoEIfStatsTable *(continued)*

Object	Object ID	Description
jnxPPPoEIfStatsTxPADS	jnxPPPoEIfStatsEntry 4	Number of PADS packets transmitted.
jnxPPPoEIfStatsRxPADT	jnxPPPoEIfStatsEntry 5	Number of PADT packets received.
jnxPPPoEIfStatsTxPADT	jnxPPPoEIfStatsEntry 6	Number of PADT packets transmitted.
jnxPPPoEIfStatsRxInvVersion	jnxPPPoEIfStatsEntry 7	Number of packets received with invalid version number.
jnxPPPoEIfStatsRxInvCode	jnxPPPoEIfStatsEntry 8	Number of packets received with invalid code.
jnxPPPoEIfStatsRxInvTags	jnxPPPoEIfStatsEntry 9	Number of packets received with invalid tags.
jnxPPPoEIfStatsRxInvSession	jnxPPPoEIfStatsEntry 10	Number of packets received with invalid session identifiers. NOTE: This object is now obsolete; PADI and PADR packets have separate counters.
jnxPPPoEIfStatsRxInvTypes	jnxPPPoEIfStatsEntry 11	Number of packets received with invalid types.
jnxPPPoEIfStatsRxInvPackets	jnxPPPoEIfStatsEntry 12	Number of invalid packets received.
jnxPPPoEIfStatsRxInsufficientResources	jnxPPPoeIfStatsEntry 13	Number of session requests that were not honored because of insufficient resources.
jnxPPPoEIfStatsTxPADM	jnxPPPoEIfStatsEntry 14	Number of PADM packets transmitted.
jnxPPPoEIfStatsTxPADN	jnxPPPoEIfStatsEntry 15	Number of PADN packets transmitted.
jnxPPPoEIfStatsRxInvTagLength	jnxPPPoEIfStatsEntry 16	Number of packets received with invalid tag lengths.
jnxPPPoEIfStatsRxInvLength	jnxPPPoEIfStatsEntry 17	Number of packets received with invalid length.
jnxPPPoEIfStatsRxInvPadISession	jnxPPPoEIfStatsEntry 18	Number of PADI packets received with invalid session identifiers.
jnxPPPoEIfStatsRxInvPadRSession	jnxPPPoEIfStatsEntry 19	Number of PADR packets received with invalid session identifiers.

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jnxPPPoEIfLockoutTable

jnxPPPoEIfLockoutTable, whose object identifier is **{jnxPPPoEIfLayer 4}**, enables you to lock out the client in the event of an error during the creation of the PPPoE underlying interface. To disable the ability to lock out the client from recognition, the value of **jnxPPPoEIfLockoutMax** and **jnxPPPoEIfLockoutMin** objects must be set to 0.

NOTE: None of the objects listed in Table 277 on page 624 are supported. These attributes return the default value.

Each **jnxPPPoEIfLockoutEntry**, whose object identifier is **{jnxPPPoEIfLockoutTable 1}**, contains the objects listed in Table 277 on page 624.

Table 277: jnxPPPoEIfLockoutTable

Object	Object ID	Description
jnxPPPoEIfLockoutClientAddress	jnxPPPoEIfLockoutEntry 1	Source MAC address of the client.
jnxPPPoEIfLockoutTime	jnxPPPoEIfLockoutEntry 2	Duration, in seconds, used to lock out the specified encapsulation type from recognition for the specified interface. Zero indicates that no lockout occurred for the encapsulation type for the specified interface.
jnxPPPoEIfLockoutElapsedTime	jnxPPPoEIfLockoutEntry 3	Duration, in seconds, of the elapsed time since the specified encapsulation type was locked out from recognition for the specified interface. Zero indicates that no lockout occurred for the encapsulation type for the specified interface.
jnxPPPoEIfLockoutNextTime	jnxPPPoEIfLockoutEntry 4	Duration, in seconds, to lock out the specified encapsulation type from recognition for the specified interface for the next event that results in a lockout condition. When **jnxPPPoEIfEnable** is enabled, a value of zero indicates that lockout is prevented from occurring for the encapsulation type for the specified interface.

Related Documentation

- PPPoE MIB Overview on page 619
- PPPoE MIB Textual Conventions on page 620
- Managed Objects for PPPoE Interfaces Layer Functions on page 625
- Managed Objects for PPPoE Summary Counts on page 627
- Managed Objects for PPPoE ServiceName Tables on page 628

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Managed Objects for PPPoE Interfaces Layer Functions

The **jnxPPPoESubIfLayer** object, whose object identifier is **{jnxPPPoEObjects 2}**, contains the object listed in Table 278 on page 625, along with the objects listed in **jnxPPPoESubIfTable** and **jnxPPPoESubIfQueueStatsTable**. You can use these managed objects to retrieve information about the PPPoE Interfaces layer.

For information about **jnxPPPoESubIfNextIndex**,**jnxPPPoESubIfTable**, and **jnxPPPoESubIfQueueStatsTable**, see the following sections:

- jnxPPPoESubIfNextIndex on page 625
- jnxPPPoESubIfTable on page 625
- jnxPPPoESubIfQueueStatsTable on page 626

jnxPPPoESubIfNextIndex

The **jnxPPPoESubIfNextIndex** object, whose object identifier is **{jnxPPPoESubIfLayer 1}** contains the objects listed in Table 278 on page 625. Successive **Get** requests usually return different values to avoid collisions among clients seeking to create table entries simultaneously.

Table 278: jnxPPPoESubIfNextIfIndex

Object	Object ID	Description
jnxPPPoESubIfNextIfIndex	**jnxPPPoESubIfLayer 1**	Coordinates allocation of ifIndex value for entries in **jnxPPPoESubIfTable**.

jnxPPPoESubIfTable

jnxPPPoESubIfTable, whose object identifier is **{jnxPPPoESubIfLayer 2}**, contains **jnxPPPoESubIfEntry**, which maps to the parameters of the specific PPPoE interface.

Each **jnxPPPoESubIfEntry**, whose object identifier is **{jnxPPPoESubIfTable 1}**, contains the objects listed in Table 279 on page 625.

Table 279: jnxPPPoESubIfTable

Object	Object ID	Description
jnxPPPoESubIfIndex	**jnxPPPoESubIfEntry 1**	IfIndex of the PPPoE interface.
jnxPPPoESubIfRowStatus	**jnxPPPoESubIfEntry 2**	Row status for this entry. Possible values: • createAndGo • destroy
jnxPPPoESubIfLowerIfIndex	**jnxPPPoESubIfEntry 3**	IfIndex of a PPPoE underlying interface over which this PPPoE interface is to be layered. Zero indicates that there is no layering.

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Table 279: jnxPPPoESubIfTable *(continued)*

Object	Object ID	Description
jnxPPPoESubIfId	jnxPPPoESubIfEntry 4	Identifier for the PPPoE interface.
jnxPPPoESubIfSessionId	jnxPPPoESubIfEntry 5	Session identifier of the PPPoE interface.
jnxPPPoESubIfMotm	jnxPPPoESubIfEntry 6	Message to send via a PADM packet on the interface when the interface transitions to the IfOperStatusUp state.
jnxPPPoESubIfUrl	jnxPPPoESubIfEntry 7	URL to be sent via a PADM packet on the interface when the interface transitions to the IfOperStatusUp state.

jnxPPPoESubIfQueueStatsTable

jnxPPPoESubIfQueueStatsTable, whose object identifier is **{jnxPPPoESubIfLayer 3}**, contains **jnxPPPoESubIfPerQueueStatsEntry** that displays the statistics for the PPPoE interface.

Each **jnxPPPoESubIfPerQueueStatsEntry**, whose object identifier is **{jnxPPPoESubIfQueueStatsTable 1}**, contains the objects listed in Table 280 on page 626.

Table 280: jnxPPPoESubIfQueueStatsTable

Object	Object ID	Description
jnxPPPoESubIfQueueIndex	jnxPPPoESubIfPerQueueuStatsEntry 1	Queue index of the queue configured on the PPPoE interface. Range of values is 0 through 7.
jnxPPPoESubIfQueueStatsPacketSent	jnxPPPoESubIfPerQueueStatsEntry 2	Number of packets sent per PPPoE session and per queue.
jnxPPPoESubIfQueueStatsBytesSent	jnxPPPoESubIfPerQueueStatsEntry 3	Number of bytes sent per PPPoE session and per queue.
jnxPPPoESubIfQueueStatsPacketDropped	jnxPPPoESubIfPerQueueStatsEntry 4	Number of packets dropped per PPPoE session and per queue.
jnxPPPoESubIfQueueStatsBytesDropped	jnxPPPoESubIfPerQueueStatsEntry 5	Number of bytes dropped per PPPoE session and per queue.
jnxPPPoESubIfQueueStatsActualBitRate	jnxPPPoESubIfPerQueueStatsEntry 6	Actual bit rate for per PPPoE session and per queue.
jnxPPPoESubIfQueueStatsActualDroppedBitRate	jnxPPPoESubIfPerQueueStatsEntry 7	Actual dropped bit rate per PPPoE session and per queue.

Related Documentation

- PPPoE MIB Overview on page 619

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- PPPoE MIB Textual Conventions on page 620
- Managed Objects for PPPoE Underlying Interfaces Layer Functions on page 620
- Managed Objects for PPPoE Summary Counts on page 627
- Managed Objects for PPPoE ServiceName Tables on page 628

Managed Objects for PPPoE Summary Counts

The managed objects listed in Table 281 on page 627 provide information about the number of PPPoE interfaces, PPPoE underlying interfaces configured or created, as well as the status of the interfaces, and underlying interfaces.

Each **jnxPPPoESummary**, whose object identifier is **{jnxPPPoEObjects 3}**, contains the objects listed in Table 281 on page 627.

Table 281: jnxPPPoESummaryCounts

Object	Object ID	Description
jnxPPPoEMajorInterfaceCount	jnxPPPoESummary 1	Total number of PPPoE underlying interfaces configured and created in the system.
jnxPPPoESummaryMajorIfAdminUp	jnxPPPoESummary 2	Total number of PPPoE underlying interfaces in the system with administrative status configured as up.
jnxPPPoESummaryMajorIfAdminDown	jnxPPPoESummary 3	Total number of PPPoE underlying interfaces in the system with administrative status configured as down.
jnxPPPoESummaryMajorIfOperUp	jnxPPPoESummary 4	Total number of PPPoE underlying interfaces in the system whose operating status is up.
jnxPPPoESummaryMajorIfOperDown	jnxPPPoESummary 5	Total number of PPPoE underlying interfaces in the system whose operational status is down.
jnxPPPoESummaryMajorIfLowerLayerDown	jnxPPPoESummary 6	Total number of PPPoE underlying interfaces in the system whose operational status is lowerLayerDown.
jnxPPPoESummaryMajorIfNotPresent	jnxPPPoESummary 7	Total number of PPPoE underlying interfaces whose operational state is notPresent.
jnxPPPoESummarySubInterfaceCount	jnxPPPoESummary 8	Total number of PPPoE interfaces configured in the system.
jnxPPPoESummarySubIfAdminUp	jnxPPPoESummary 9	Total number of PPPoE interfaces in the system whose administrative status is up.
jnxPPPoESummarySubIfAdminDown	jnxPPPoESummary 10	Total number of PPPoE interfaces in the system whose administrative status is down.
jnxPPPoESummarySubIfOperUp	jnxPPPoESummary 11	Total number of PPPoE interfaces in the system with an operational status of up.

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Table 281: jnxPPPoESummaryCounts *(continued)*

Object	Object ID	Description
jnxPPPoESummarySubIfOperDown	jnxPPPoESummary 12	Total number of PPPoE interfaces in the system whose operational status is down.
jnxPPPoESummarySubIfLowerLayerDown	jnxPPPoESummary 13	Total number of PPPoE interfaces in the system whose operational status is lowerLayerDown.
jnxPPPoESummarySubIfNotPresent	jnxPPPoESummary 14	Total number of PPPoE interfaces in the system whose operational status is notPresent.

Related Documentation

- PPPoE MIB Overview on page 619
- PPPoE MIB Textual Conventions on page 620
- Managed Objects for PPPoE Underlying Interfaces Layer Functions on page 620
- Managed Objects for PPPoE Interfaces Layer Functions on page 625
- Managed Objects for PPPoE ServiceName Tables on page 628

Managed Objects for PPPoE ServiceName Tables

You can use the managed objects listed in **jnxPPPoEServiceNameTableTable**, **jnxPPPoEServiceNameTable**, and **jnxPPPoEServiceNameAciAriTable** to retrieve information about the PPPoE service name tables.

For information about **jnxPPPoEServiceNameTableTable**, **jnxPPPoEServiceNameTable**, and **jnxPPPoEServiceNameAciAriTable**, see the following sections:

- jnxPPPoEServiceNameTableTable on page 628
- jnxPPPoEServiceNameTable on page 629
- jnxPPPoEServiceNameAciAriTable on page 629

jnxPPPoEServiceNameTableTable

jnxPPPoEServiceNameTableTable, whose object identifier is **{jnxPPPoEServices 1}**, contains **jnxPPPoEServiceNameTableEntry**, which maps to the objects in the service name table. The Empty service and Any service are automatically configured for each service name table created.

Each **jnxPPPoEServiceNameTableEntry**, whose object identifier is **{jnxPPPoEServiceNameTableTable 1}**, contains the objects listed in Table 282 on page 628.

Table 282: jnxPPPoEServiceNameTableTable

Object	Object ID	Description
jnxPPPoEServiceNameTableName	jnxPPPoEServiceNameTableEntry 1	Name of the service name table.

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Table 282: jnxPPPoEServiceNameTableTable *(continued)*

Object	Object ID	Description
jnxPPPoEServiceNameTableRowStatus	jnxPPPoEServiceNameTableEntry 2	Row status for this entry. Possible values: • createAndGo • destroy

jnxPPPoEServiceNameTable

jnxPPPoEServiceNameTable, whose object identifier is **{jnxPPPoEServices 2}**, contains **jnxPPPoEServiceNameEntry**, which maps to the parameters of the service name table.

Each **jnxPPPoEServiceNameEntry**, whose object identifier is **{jnxPPPoEServiceNameTable 1}**, contains the objects listed in Table 283 on page 629.

Table 283: jnxPPPoEServiceNameTable

Object	Object ID	Description
jnxPPPoEServiceName	jnxPPPoEServiceNameEntry 1	Service name tag value.
jnxPPPoEServiceNameAction	jnxPPPoEServiceNameEntry 2	Identifies the behavior when the service name tag is received in a PADI frame.
jnxPPPoEServiceNameDynamicProfile	jnxPPPoEServiceNameEntry 3	Name of the dynamic profile associated with the service name.
jnxPPPoEServiceNameRoutingInstance	jnxPPPoEServiceNameEntry 4	Routing instance associated with the service name.
jnxPPPoEServiceNameMaxSessions	jnxPPPoEServiceNameEntry 5	Maximum number of active PPPoE sessions that can be established with the specified service entry.
jnxPPPoEServiceNameRowStatus	jnxPPPoEServiceNameEntry 6	Row status for this entry. Possible values: • createAndGo • destroy

jnxPPPoEServiceNameAciAriTable

jnxPPPoEServiceNameAciAriTable, whose object identifier is **{jnxPPPoEServices 3}**, contains **jnxPPPoEServiceNameAciAriEntry**, which maps to the parameters in the PPPoE Service Name Agent Circuit Identifier (ACI)/ Agent Remote Identifier (ARI) tables.

Each **jnxPPPoEServiceNameAciAriEntry**, whose object identifier is **{jnxPPPoEServiceNameAriAriTable 1}**, contains the objects listed in Table 284 on page 630.

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Table 284: jnxPPPoEServiceNameAciAriTable

Object	Object ID	Description
jnxPPPoEServiceNameAgentCircuitId	jnxPPPoEServiceNameAciAriEntry 1	Agent Circuit Identifier tag values that the PPPoE client sends in the PADI or PADR control packet.
jnxPPPoEServiceNameAgentRemoteId	jnxPPPoEServiceNameAciAriEntry 2	Agent Circuit Identifier tag values that the PPPoE client sends in the PADI or PADR control packet.
jnxPPPoEServiceNameAciAriAction	jnxPPPoEServiceNameAciAriEntry 3	Identification of the behavior when the service name with ACI/ARI pair is received in a PADI frame.
jnxPPPoEServiceNameAciAriDynamicProfile	jnxPPPoEServiceNameAciAriEntry 4	Dynamic profile associated with a service name and ACI/ARI pair.
jnxPPPoEServiceNameAciAriRoutingInstance	jnxPPPoEServiceNameAciAriEntry 5	Routing instance associated with a service name and ACI/ARI pair.
jnxPPPoEServiceNameAciAriStaticInterface	jnxPPPoEServiceNameAciAriEntry 6	Static interface associated with matching ACI/ARI pair. A static interface can be configured only for an ACI/ARI pair.
jnxPPPoEServiceNameAciAriRowStatus	jnxPPPoEServiceNameAciAriEntry 7	Row status for this entry. Possible values: • createAndGo • destroy

Related Documentation

- PPPoE MIB Overview on page 619
- PPPoE MIB Textual Conventions on page 620
- Managed Objects for PPPoE Underlying Interfaces Layer Functions on page 620
- Managed Objects for PPPoE Interfaces Layer Functions on page 625
- Managed Objects for PPPoE Summary Counts on page 627

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PART 3

SNMP Notifications

- Juniper Networks Enterprise-Specific SNMP Traps on page 633
- Standard SNMP Traps on page 653

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CHAPTER 77

Juniper Networks Enterprise-Specific SNMP Traps

This chapter contains the following topics:

- Juniper Networks Enterprise-Specific SNMP Traps on page 633
- Juniper Networks Enterprise-Specific SNMP Version 1 Traps on page 634
- Juniper Networks Enterprise-Specific SNMP Version 2 Traps on page 641
- Juniper Networks Enterprise-Specific BGP Traps on page 648
- Juniper Networks Enterprise-Specific LDP Traps on page 649
- Juniper Networks Enterprise-Specific License MIB Traps on page 649
- Juniper Networks Enterprise-Specific MIMSTP Traps on page 649
- Juniper Networks Enterprise-Specific MPLS Traps on page 649
- Juniper Networks Enterprise-Specific Network Address Translation Resources-Monitoring Traps on page 650
- Juniper Networks Enterprise-Specific Traps on EX Series Ethernet Switches on page 650
- Juniper Networks Enterprise-Specific Traps on MX Series 3D Universal Edge Routers on page 650
- Raising Traps for Events Based on System Log Messages on page 651
- Spoofing Enterprise-Specific SNMP Traps on page 651

Juniper Networks Enterprise-Specific SNMP Traps

This topic provides pointers to the enterprise-specific SNMP traps supported by the Junos OS.

NOTE: **All enterprise-specific SNMP traps supported by the Junos OS can be sent in version 1, 2, and 3 formats.**

- Juniper Networks Enterprise-Specific SNMP Version 1 Traps on page 634
- Juniper Networks Enterprise-Specific SNMP Version 2 Traps on page 641
- Juniper Networks Enterprise-Specific BGP Traps on page 648

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- Juniper Networks Enterprise-Specific LDP Traps on page 649
- Juniper Networks Enterprise-Specific License MIB Notifications on page 569
- Juniper Networks Enterprise-Specific MIMSTP Traps on page 431
- Juniper Networks Enterprise-Specific MPLS Traps on page 409



NOTE: For scalability reasons, the MPLS traps are generated by the ingress router only. For information about disabling the generation of MPLS traps, see the *Junos OS MPLS Applications Configuration Guide*.

- Juniper Networks Enterprise-Specific Traps on EX Series Ethernet Switches on page 650
- Juniper Networks Enterprise-Specific Traps on MX Series 3D Universal Edge Routers on page 650

Related Documentation

- Standard SNMP Traps Supported on Devices Running Junos OS on page 653
- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7
- Configuring SNMP Trap Options and Groups on a Device Running Junos OS
- Managing Traps and Informs

Juniper Networks Enterprise-Specific SNMP Version 1 Traps

The Junos OS supports enterprise-specific SNMP version 1 traps shown in Table 285 on page 634. The traps are organized first by trap category and then by trap name. The system logging severity levels are listed for those traps that have them. Traps that do not have corresponding system logging severity levels are marked with an en dash (–).

For more information about system log messages, see the *Junos OS System Log Messages Reference*. For more information about configuring system logging, see the *Junos OS System Basics Configuration Guide*. To view the Juniper Networks enterprise-specific SNMP version 2 traps, see "Juniper Networks Enterprise-Specific SNMP Version 2 Traps" on page 641. For more information about chassis traps, see "Chassis Traps" on page 158.

Table 285: Juniper Networks Enterprise-Specific Supported SNMP Version 1 Traps

Defined in	Trap Name	Enterprise ID	Generic Trap Number	Specific Trap Number	System Logging Severity Level	System Log Tag	Supported On
Chassis Notifications (Alarm Conditions)							
Chassis MIB (**jnx-chassis.mib**)	**jnxPowerSupplyFailure**	1.3.6.1.4.1.2636.4.1	6	1	Warning	CHASSISD_SNMP_TRAP	All devices running Junos OS.

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Table 285: Juniper Networks Enterprise-Specific Supported SNMP Version 1 Traps *(continued)*

Defined in	Trap Name	Enterprise ID	Generic Trap Number	Specific Trap Number	System Logging Severity Level	System Log Tag	Supported On
	jnxFanFailure	1.3.6.1.4.1.2636.4.1	6	2	Critical	CHASSISD_ SNMP_ TRAP	All devices running Junos OS.
	jnxOverTemperature	1.3.6.1.4.1.2636.4.1	6	3	Alert	CHASSISD_ SNMP_ TRAP	All devices running Junos OS.
	jnxRedundancySwitchOver	1.3.6.1.4.1.2636.4.1	6	4	Critical	CHASSISD_ SNMP_ TRAP	All devices running Junos OS.
	jnxFruRemoval	1.3.6.1.4.1.2636.4.1	6	5	Notice	CHASSISD_ SNMP_ TRAP	All devices running Junos OS.
	jnxFruInsertion	1.3.6.1.4.1.2636.4.1	6	6	Notice	CHASSISD_ SNMP_ TRAP	All devices running Junos OS.
	jnxFruPowerOff	1.3.6.1.4.1.2636.4.1	6	7	Notice	CHASSISD_ SNMP_ TRAP	All devices running Junos OS.
	jnxFruPowerOn	1.3.6.1.4.1.2636.4.1	6	8	Notice	CHASSISD_ SNMP_ TRAP	All devices running Junos OS.
	jnxFruFailed	1.3.6.1.4.1.2636.4.1	6	9	Warning	CHASSISD_ SNMP_ TRAP	All devices running Junos OS.
	jnxFruOffline	1.3.6.1.4.1.2636.4.1	6	10	Notice	CHASSISD_ SNMP_ TRAP	All devices running Junos OS.
	jnxFruOnline	1.3.6.1.4.1.2636.4.1	6	11	Notice	CHASSISD_ SNMP_ TRAP	All devices running Junos OS.
	jnxFruCheck	1.3.6.1.4.1.2636.4.1	6	12	Warning	CHASSISD_ SNMP_ TRAP	All devices running Junos OS.
	jnxFEBSwitchover	1.3.6.1.4.1.2636.4.1	6	13	Warning	CHASSISD_ SNMP_ TRAP	All devices running Junos OS.

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Table 285: Juniper Networks Enterprise-Specific Supported SNMP Version 1 Traps *(continued)*

Defined in	Trap Name	Enterprise ID	Generic Trap Number	Specific Trap Number	System Logging Severity Level	System Log Tag	Supported On
	jnxHardDiskFailed	1.3.6.1.4.1.2636.4.1	6	14	Warning	CHASSISD_SNMP_TRAP	All devices running Junos OS.
	jnxHardDiskMissing	1.3.6.1.4.1.2636.4.1	6	15	Warning	CHASSISD_SNMP_TRAP	All devices running Junos OS.
	jnxPowerSupplyOk	1.3.6.1.4.1.2636.4.2	6	1	Critical	CHASSISD_SNMP_TRAP	All devices running Junos OS.
	jnxFanOK	1.3.6.1.4.1.2636.4.2	6	2	Critical	CHASSISD_SNMP_TRAP	All devices running Junos OS.
	jnxTemperatureOK	1.3.6.1.4.1.2636.4.2	6	3	Alert	CHASSISD_SNMP_TRAP	All devices running Junos OS.
Configuration Notifications							
Configuration Management MIB (**jnx-configmgmt.mib**)	**jnxCmCfgChange**	1.3.6.1.4.1.2636.4.5	6	1	–	–	All devices running Junos OS.
	jnxCmRescueChange	1.3.6.1.4.1.2636.4.5	6	2	–	–	All devices running Junos OS.

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Table 285: Juniper Networks Enterprise-Specific Supported SNMP Version 1 Traps *(continued)*

Defined in	Trap Name	Enterprise ID	Generic Trap Number	Specific Trap Number	System Logging Severity Level	System Log Tag	Supported On
Link Notifications							
Flow Collection Services MIB (**jnx-coll.mib**)	**jnxCollUnavailableDest**	1.3.6.1.4.1.2636.4.8	6	1	–	–	Devices that run Junos OS and have collector PICs installed.
	jnxCollUnavailableDestCleared	1.3.6.1.4.1.2636.4.8	6	2	–	–	Devices that run Junos OS and have collector PICs installed.
	jnxCollUnsuccessfulTransfer	1.3.6.1.4.1.2636.4.8	6	3	–	–	Devices that run Junos OS and have collector PICs installed.
	jnxCollFlowOverload	1.3.6.1.4.1.2636.4.8	6	4	–	–	Devices that run Junos OS and have collector PICs installed.
	jnxCollFlowOverloadCleared	1.3.6.1.4.1.2636.4.8	6	5	–	–	Devices that run Junos OS and have collector PICs installed.
	jnxCollMemoryUnavailable	1.3.6.1.4.1.2636.4.8	6	6	–	–	Devices that run Junos OS and have collector PICs installed.
	jnxCollMemoryAvailable	1.3.6.1.4.1.2636.4.8	6	7	–	–	Devices that run Junos OS and have collector PICs installed.
	jnxCollFtpSwitchover	1.3.6.1.4.1.2636.4.8	6	8	–	–	Devices that run Junos OS and have collector PICs installed.

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Table 285: Juniper Networks Enterprise-Specific Supported SNMP Version 1 Traps *(continued)*

Defined in	Trap Name	Enterprise ID	Generic Trap Number	Specific Trap Number	System Logging Severity Level	System Log Tag	Supported On
Passive Monitoring MIB (jnx-pmon.mib)	**jnxPMonOverloadSet**	1.3.6.1.4.1.2636.4.7.0.1	6	1	–	–	Devices that run Junos OS and have PICs that support passive monitoring installed.
	jnxPMonOverloadCleared	1.3.6.1.4.1.2636.4.7.0.2	6	2	–	–	Devices that run Junos OS and have PICs that support passive monitoring installed.
SONET APS MIB (**jnx-sonetaps.mib**)	**apsEventChannelMismatch**	1.3.6.1.4.1.2636.3.24.2	6	3	–	–	Devices that run Junos OS and have SONET PICs installed.
	apsEventPSBF	1.3.6.1.4.1.2636.3.24.2	6	4	–	–	Devices that run Junos OS and have SONET PICs installed.
	apsEventFEPLF	1.3.6.1.4.1.2636.3.24.2	6	5	–	–	Devices that run Junos OS and have SONET PICs installed.
Remote Operations							
PING MIB (**jnx-ping.mib**)	**jnxPingRttThresholdExceeded**	1.3.6.1.4.1.2636.4.9	6	1	–	–	All devices running Junos OS.
	jnxPingRttStdDevThreshold Exceeded	1.3.6.1.4.1.2636.4.9	6	2	–	–	All devices running Junos OS.
	jnxPingRttJitterThreshold Exceeded	1.3.6.1.4.1.2636.4.9	6	3	–	–	All devices running Junos OS.
	jnxPingEgressThreshold Exceeded	1.3.6.1.4.1.2636.4.9	6	4	–	–	All devices running Junos OS.

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Table 285: Juniper Networks Enterprise-Specific Supported SNMP Version 1 Traps *(continued)*

Defined in	Trap Name	Enterprise ID	Generic Trap Number	Specific Trap Number	System Logging Severity Level	System Log Tag	Supported On
	jnxPingEgressStdDev ThresholdExceeded	1.3.6.1.4.1.2636.4.9	6	5	–	–	All devices running Junos OS.
	jnxPingEgressJitter ThresholdExceeded	1.3.6.1.4.1.2636.4.9	6	6	–	–	All devices running Junos OS.
	jnxPingIngressThreshold Exceeded	1.3.6.1.4.1.2636.4.9	6	7	–	–	All devices running Junos OS.
	jnxPingIngressStddevThreshold Exceeded	1.3.6.1.4.1.2636.4.9	6	8	–	–	All devices running Junos OS.
	jnxPingIngressJitterThreshold Exceeded	1.3.6.1.4.1.2636.4.9	6	9	–	–	All devices running Junos OS.
Routing Notifications							
BFD Experimental MIB (**jnx-bfd-exp.mib**)	**bfdSessUp**	1.3.6.1.4.1. 2636.5.3.1	6	1	–	–	All devices running Junos OS.
	bfdSessDown	1.3.6.1.4.1. 2636.5.3.1	6	2	–	–	All devices running Junos OS.
LDP MIB (**jnx-ldp.mib**)	**jnxLdpLspUp**	1.3.6.1.4.1.2636.4.4	6	1	–	–	M, T, and MX Series routers.
	jnxLdpLspDown	1.3.6.1.4.1.2636.4.4	6	2	–	–	M, T, and MX Series routers.
	jnxLdpSesUp	1.3.6.1.4.1.2636.4.4	6	3	–	–	M, T, and MX Series routers.
	jnxLdpSesDown	1.3.6.1.4.1.2636.4.4	6	4	–	–	M, T, and MX Series routers.

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Table 285: Juniper Networks Enterprise-Specific Supported SNMP Version 1 Traps *(continued)*

Defined in	Trap Name	Enterprise ID	Generic Trap Number	Specific Trap Number	System Logging Severity Level	System Log Tag	Supported On
MPLS MIB (**jnx-mpls.mib**)	**mplsLspUp** (Deprecated)	1.3.6.1.4.1.2636.3.2.4	6	1	–	–	
	mplsLspDown (Deprecated)	1.3.6.1.4.1.2636.3.2.4	6	2	–	–	
	mplsLspChange (Deprecated)	1.3.6.1.4.1.2636.3.2.4	6	3	–	–	
	mplsLspPathDown (Deprecated)	1.3.6.1.4.1.2636.3.2.4	6	4	–	–	
VPN MIB (**jnx-vpn.mib**)	**jnxVpnIfUp**	1.3.6.1.4.1.2636.3.26	6	1	–	–	M, T, and MX Series routers.
	jnxVpnIfDown	1.3.6.1.4.1.2636.3.26	6	2	–	–	M, T, and MX Series routers.
	jnxVpnPwUp	1.3.6.1.4.1.2636.3.26	6	3	–	–	M, T, and MX Series routers.
	jnxVpnPwDown	1.3.6.1.4.1.2636.3.26	6	4	–	–	M, T, and MX Series routers.
RMON Alarms							
RMON MIB (**jnx-rmon.mib**)	**jnxRmonAlarmGetFailure**	1.3.6.1.4.1.2636.4.3	6	1	–	–	All devices running Junos OS.
	jnxRmonGetOk	1.3.6.1.4.1.2636.4.3	6	2	–	–	All devices running Junos OS.
SONET Alarms							
SONET MIB (**jnx-sonet.mib**)	**jnxSonetAlarmSet**	1.3.6.1.4.1.2636.4.6	6	1	–	–	Devices that run Junos OS and have SONET PICs installed.
	jnxSonetAlarmCleared	1.3.6.1.4.1.2636.4.6	6	2	–	–	Devices that run Junos OS and have SONET PICs installed.

Related Documentation

- Juniper Networks Enterprise-Specific SNMP Traps on page 633
- Standard SNMP Traps Supported on Devices Running Junos OS on page 653
- Juniper Networks Enterprise-Specific MIBs on page 30

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- Standard SNMP MIBs Supported by Junos OS on page 7
- Configuring SNMP Trap Options and Groups on a Device Running Junos OS
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Juniper Networks Enterprise-Specific SNMP Version 2 Traps

The Junos OS supports the enterprise-specific SNMP version 2 traps shown in Table 286 on page 641. The traps are organized first by trap category and then by trap name. The system logging severity levels are listed for those traps that have them. Traps that do not have corresponding system logging severity levels are marked with an en dash (–).

For more information about system messages, see the *Junos OS System Log Messages Reference*. For more information about configuring system logging, see the *Junos OS System Basics Configuration Guide*. To view the Juniper Networks enterprise-specific SNMP version 1 traps, see "Juniper Networks Enterprise-Specific SNMP Version 1 Traps" on page 634. For more information about chassis traps, see "Chassis Traps" on page 158.

Table 286: Juniper Networks Enterprise-Specific Supported SNMP Version 2 Traps

Source MIB	Trap Name	snmpTrapOID	System Logging Severity Level	System Log Tag	Supported On
Chassis (Alarm Conditions) Notifications					
Chassis MIB (**jnx-chassis.mib**)	**jnxPowerSupplyFailure**	1.3.6.1.4.1.2636.4.1.1	Alert	CHASSISD_SNMP_TRAP	All devices running Junos OS.
	jnxFanFailure	1.3.6.1.4.1.2636.4.1.2	Critical	CHASSISD_SNMP_TRAP	All devices running Junos OS.
	jnxOverTemperature	1.3.6.1.4.1.2636.4.1.3	Critical	CHASSISD_SNMP_TRAP	All devices running Junos OS.
	jnxRedundancySwitchOver	1.3.6.1.4.1.2636.4.1.4	Critical	CHASSISD_SNMP_TRAP	All devices running Junos OS.
	jnxFruRemoval	1.3.6.1.4.1.2636.4.1.5	Notice	CHASSISD_SNMP_TRAP	All devices running Junos OS.
	jnxFruInsertion	1.3.6.1.4.1.2636.4.1.6	Notice	CHASSISD_SNMP_TRAP	All devices running Junos OS.
	jnxFruPowerOff	1.3.6.1.4.1.2636.4.1.7	Notice	CHASSISD_SNMP_TRAP	All devices running Junos OS.
	jnxFruPowerOn	1.3.6.1.4.1.2636.4.1.8	Notice	CHASSISD_SNMP_TRAP	All devices running Junos OS.

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Table 286: Juniper Networks Enterprise-Specific Supported SNMP Version 2 Traps *(continued)*

Source MIB	Trap Name	snmpTrapOID	System Logging Severity Level	System Log Tag	Supported On
	jnxFruFailed	1.3.6.1.4.1.2636.4.1.9	Warning	CHASSISD_SNMP_TRAP	All devices running Junos OS.
	jnxFruOffline	1.3.6.1.4.1.2636.4.1.10	Notice	CHASSISD_SNMP_TRAP	All devices running Junos OS.
	jnxFruOnline	1.3.6.1.4.1.2636.4.1.11	Notice	CHASSISD_SNMP_TRAP	All devices running Junos OS.
	jnxFruCheck	1.3.6.1.4.1.2636.4.1.12	Notice	CHASSISD_SNMP_TRAP	All devices running Junos OS.
	jnxFEBSwitchover	1.3.6.1.4.1.2636.4.1.13	Notice	CHASSISD_SNMP_TRAP	All devices running Junos OS.
	jnxHardDiskFailed	1.3.6.1.4.1.2636.4.1.14	Notice	CHASSISD_SNMP_TRAP	All devices running Junos OS.
	jnxHardDiskMissing	1.3.6.1.4.1.2636.4.1.15	Notice	CHASSISD_SNMP_TRAP	All devices running Junos OS.
	jnxPowerSupplyOK	1.3.6.1.4.1.2636.4.2.1	Critical	CHASSISD_SNMP_TRAP	All devices running Junos OS.
	jnxFanOK	1.3.6.1.4.1.2636.4.2.2	Critical	CHASSISD_SNMP_TRAP	All devices running Junos OS.
	jnxTemperatureOK	1.3.6.1.4.1.2636.4.2.3	Alert	CHASSISD_SNMP_TRAP	All devices running Junos OS.
Configuration Notifications					
Configuration Management MIB (**jnx-cfgmgmt.mib**)	**jnxCmCfgChange**	1.3.6.1.4.1.2636.4.5.0.1	–	–	All devices running Junos OS.
	jnxCmRescueChange	1.3.6.1.4.1.2636.4.5.0.2	–	–	All devices running Junos OS.
Link Notifications					

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Table 286: Juniper Networks Enterprise-Specific Supported SNMP Version 2 Traps *(continued)*

Source MIB	Trap Name	snmpTrapOID	System Logging Severity Level	System Log Tag	Supported On
Flow Collection Services MIB (**jnx-coll.mib**)	**jnxCollUnavailableDest**	1.3.6.1.4.1.2636.4.8.0.1	–	–	Devices that run Junos OS and have collector PICs installed.
	jnxCollUnavailableDestCleared	1.3.6.1.4.1.2636.4.8.0.2	–	–	Devices that run Junos OS and have collector PICs installed.
	jnxCollUnsuccessfulTransfer	1.3.6.1.4.1.2636.4.8.0.3	–	–	Devices that run Junos OS and have collector PICs installed.
	jnxCollFlowOverload	1.3.6.1.4.1.2636.4.8.0.4	–	–	Devices that run Junos OS and have collector PICs installed.
	jnxCollFlowOverloadCleared	1.3.6.1.4.1.2636.4.8.0.5	–	–	Devices that run Junos OS and have collector PICs installed.
	jnxCollMemoryUnavailable	1.3.6.1.4.1.2636.4.8.0.6	–	–	Devices that run Junos OS and have collector PICs installed.
	jnxCollMemoryAvailable	1.3.6.1.4.1.2636.4.8.0.7	–	–	Devices that run Junos OS and have collector PICs installed.
	jnxCollFtpSwitchover	1.3.6.1.4.1.2636.4.8.0.8	–	–	Devices that run Junos OS and have collector PICs installed.
PMON MIB (**jnx-pmon.mib**)	**jnxPMonOverloadSet**	1.3.6.1.4.1.2636.4.7.0.1	–	–	Devices that run Junos OS and have PICs that support passive monitoring installed.
	jnxPMonOverloadCleared	1.3.6.1.4.1.2636.4.7.0.2	–	–	Devices that run Junos OS and have PICs that support passive monitoring installed.

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Table 286: Juniper Networks Enterprise-Specific Supported SNMP Version 2 Traps *(continued)*

Source MIB	Trap Name	snmpTrapOID	System Logging Severity Level	System Log Tag	Supported On
SONET APS MIB (**jnx-sonetaps.mib**)	**apsEventChannelMismatch**	1.3.6.1.4.1.2636.3.24.2.0.3	–	–	Devices that run Junos OS and have SONET PICs installed.
	apsEventPSBF	1.3.6.1.4.1.2636.3.24.2.0.4	–	–	Devices that run Junos OS and have SONET PICs installed.
	apsEventFEPLF	1.3.6.1.4.1.2636.3.24.2.0.5	–	–	Devices that run Junos OS and have SONET PICs installed.
Remote Operations Notifications					
PING MIB (**jnx-ping.mib**)	**jnxPingRttThreshold Exceeded**	1.3.6.1.4.1.2636.4.9.0.1	–	–	All devices running Junos OS.
	jnxPingRttStdDevThreshold Exceeded	1.3.6.1.4.1.2636.4.9.0.2	–	–	All devices running Junos OS.
	jnxPingRttJitterThreshold Exceeded	1.3.6.1.4.1.2636.4.9.0.3	–	–	All devices running Junos OS.
	jnxPingEgressThreshold Exceeded	1.3.6.1.4.1.2636.4.9.0.4	–	–	All devices running Junos OS.
	jnxPingEgressStdDevThreshold Exceeded	1.3.6.1.4.1.2636.4.9.0.5	–	–	All devices running Junos OS.
	jnxPingEgressJitterThreshold Exceeded	1.3.6.1.4.1.2636.4.9.0.6	–	–	All devices running Junos OS.
	jnxPingIngressThreshold Exceeded	1.3.6.1.4.1.2636.4.9.0.7	–	–	All devices running Junos OS.
	jnxPingIngressStddevThreshold Exceeded	1.3.6.1.4.1.2636.4.9.0.8	–	–	All devices running Junos OS.
	jnxPingIngressJitterThreshold Exceeded	1.3.6.1.4.1.2636.4.9.0.9	–	–	All devices running Junos OS.
Routing Notifications					
BFD Experimental MIB (**jnx-bfd-exp.mib**)	**bfdSessUp**	1.3.6.1.4.1.2636.5.3.1.0.1	–	–	All devices running Junos OS.
	bfdSessDown	1.3.6.1.4.1.2636.5.3.1.0.2	–	–	All devices running Junos OS.

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Table 286: Juniper Networks Enterprise-Specific Supported SNMP Version 2 Traps *(continued)*

Source MIB	Trap Name	snmpTrapOID	System Logging Severity Level	System Log Tag	Supported On
BGP4 V2 MIB (**jnx-bgpmib2.mib**)	**jnxBgpM2Established**	1.3.6.1.4.1.2636.5.1.1.1.0.1	–	–	All devices running Junos OS.
	jnxBgpM2BackwardTransition	1.3.6.1.4.1.2636.5.1.1.1.0.2	–	–	All devices running Junos OS.
DHCP MIB (**jnx-dhcp.mib**)	**jnxJdhcpLocalServer DuplicateClient**	1.3.6.1.4.1.2636.3.61.61.1.3.1	–	–	All devices running Junos OS.
	jnxJdhcpLocalServer InterfaceLimitExceeded	1.3.6.1.4.1.2636.3.61.61.1.3.2	–	–	All devices running Junos OS.
	jnxJdhcpLocalServer InterfaceLimitAbated	1.3.6.1.4.1.2636.3.61.61.1.3.3	–	–	All devices running Junos OS.
	jnxJdhcpLocalServer Health	1.3.6.1.4.1.2636.3.61.61.1.3.4	–	–	All devices running Junos OS.
	jnxJdhcpRelayInterface LimitExceeded	1.3.6.1.4.1.2636.3.61.61.2.3.1	–	–	All devices running Junos OS.
	jnxJdhcpRelayInterface LimitAbated	1.3.6.1.4.1.2636.3.61.61.2.3.2	–	–	All devices running Junos OS.
DHCPv6MIB (**jnx-dhcpv6.mib**)	**jnxJdhcpv6LocalServer InterfaceLimitExceeded**	1.3.6.1.4.1.2636.3.62.62.2.3.1	–	–	All devices running Junos OS.
	jnxJdhcpv6LocalServer InterfaceLimitAbated	1.3.6.1.4.1.2636.3.62.62.2.3.2	–	–	All devices running Junos OS.
	jnxJdhcpv6LocalServer Health	1.3.6.1.4.1.2636.3.62.62.2.3.3	–	–	All devices running Junos OS.
LDP MIB (**jnx-ldp.mib**)	**jnxLdpLspUp**	1.3.6.1.4.1.2626.4.4.0.1	–	–	M, T, and MX Series routers.
	jnxLdpLspDown	1.3.6.1.4.1.2626.4.4.0.2	–	–	M, T, and MX Series routers.
	jnxLdpSesUp	1.3.6.1.4.1.2626.4.4.0.3	–	–	M, T, and MX Series routers.
	jnxLdpSesDown	1.3.6.1.4.1.2626.4.4.0.4	–	–	M, T, and MX Series routers.

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Table 286: Juniper Networks Enterprise-Specific Supported SNMP Version 2 Traps *(continued)*

Source MIB	Trap Name	snmpTrapOID	System Logging Severity Level	System Log Tag	Supported On
MPLS MIB (**jnx-mpls.mib**)	**mplsLspUp** (Deprecated)	1.3.6.1.4.1.2636.3.2.4.1	–	–	
	mplsLspInfoUp	1.3.6.1.4.1.2636.3.2.0.1	–	–	M, T, and MX Series routers.
	mplsLspDown (Deprecated)	1.3.6.1.4.1.2636.3.2.4.2	–	–	
	mplsLspInfoDown	1.3.6.1.4.1.2636.3.2.0.2	–	–	M, T, and MX Series routers.
	mplsLspChange (Deprecated)	1.3.6.1.4.1.2636.3.2.4.3	–	–	
	mplsLspInfoChange	1.3.6.1.4.1.2636.3.2.0.3	–	–	M, T, and MX Series routers.
	mplsLspPathDown (Deprecated)	1.3.6.1.4.1.2636.3.2.4.4	–	–	
	mplsLspInfoPathDown	1.3.6.1.4.1.2636.3.2.0.4	–	–	M, T, and MX Series routers.
	mplsLspInfoPathUp	1.3.6.1.4.1.2636.3.2.0.5	–	–	M, T, and MX Series routers.
VPN MIB (**jnx-vpn.mib**)	**jnxVpnIfUp**	1.3.6.1.4.1.2636.3.26.0.1	–	–	M, T, and MX Series routers.
	jnxVpnIfDown	1.3.6.1.4.1.2636.3.26.0.2	–	–	M, T, and MX Series routers.
	jnxVpnPwUp	1.3.6.1.4.1.2636.3.26.0.3	–	–	M, T, and MX Series routers.
	jnxVpnPwDown	1.3.6.1.4.1.2636.3.26.0.4	–	–	M, T, and MX Series routers.

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Table 286: Juniper Networks Enterprise-Specific Supported SNMP Version 2 Traps *(continued)*

Source MIB	Trap Name	snmpTrapOID	System Logging Severity Level	System Log Tag	Supported On
AAA MIB (**jnx-user-aaa.mib**)	**jnxAccessAuthAddress PoolHighThreshold**	1.3.6.1.4.1.2636.3.51.1.0.5	–	–	J Series and SRX Series devices.
	jnxAccessAuthAddress PoolAbateThreshold	1.3.6.1.4.1.2636.3.51.1.0.6	–	–	J Series and SRX Series devices.
	jnxAccessAuthAddress PoolOutOfAddresses	1.3.6.1.4.1.2636.3.51.1.0.7	–	–	J Series and SRX Series devices.
	jnxAccessAuthAddress PoolOutOfMemory	1.3.6.1.4.1.2636.3.51.1.0.8	–	–	J Series and SRX Series devices.
	jnxAccessAuthService Up	1.3.6.1.4.1.2636.3.51. 1.0.1	–	–	J Series and SRX Series devices.
	jnxAccessAuthService Down	1.3.6.1.4.1.2636.3.51. 1.0.2	–	–	J Series and SRX Series devices.
	jnxAccessAuthServer Disabled	1.3.6.1.4.1.2636.3.51. 1.0.3	–	–	J Series and SRX Series devices.
	jnxAccessAuthServer Enabled	1.3.6.1.4.1.2636.3.51. 1.0.4	–	–	J Series and SRX Series devices.
	jnxJsFwAuthFailure	1.3.6.1.4.1.2636.3.39.1.2. 1.0.1	–	–	J Series and SRX Series devices.
Access Authentication Methods MIB (**jnx-js-auth.mib**)	**jnxJsFwAuthServiceUp**	1.3.6.1.4.1.2636.3.39.1.2. 1.0.2	–	–	J Series and SRX Series devices.
	jnxJsFwAuthServiceDown	1.3.6.1.4.1.2636.3.39.1.2. 1.0.3	–	–	J Series and SRX Series devices.
	jnxJsFwAuthCapacityExceeded	1.3.6.1.4.1.2636.3.39.1.2. 1.0.4	–	–	J Series and SRX Series devices.
	jnxJsNatAddrPool ThresholdStatus	1.3.6.1.4.1.2636.3.39.1.7. 1.0.1	–	–	J Series and SRX Series devices.
Network Address Translation Resources-Monitoring MIB (**jnxNatMIB**)	**jnxNatAddrPoolUtil**	1.3.6.1.4.1.2636.3.59.1.2.1	–	–	M Series and MX Series routers
	jnxNatTrapSrcPoolName	1.3.6.1.4.1.2636.3.59.1.2.2	–	–	M Series and MX Series routers
	jnxNatAddrPoolThresholdStatus	1.3.6.1.4.1.2636.3.59.1.0.1	–	–	M Series and MX Series routers

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Table 286: Juniper Networks Enterprise-Specific Supported SNMP Version 2 Traps *(continued)*

Source MIB	Trap Name	snmpTrapOID	System Logging Severity Level	System Log Tag	Supported On
Network Address Translation MIB (**jnx-js-nat.mib**)	**jnxJsScreen Attack**	1.3.6.1.4.1.2636.3.39.1.8.1.0.1	Warning	RT_SCREEN_ICMP, RT_SCREEN_IP, RT_SCREEN_SESSION_LIMIT, RT_SCREEN_TCP, RT_SCREEN_UDP	J Series and SRX Series devices.
Security Screening Objects MIB (**jnx-js-screening.mib**)	**jnxJsScreenCfg Change**	1.3.6.1.4.1.2636.3.39.1.8.1.0.2	–	–	J Series and SRX Series devices.
RMON Alarms					
RMON MIB (**jnx-rmon.mib**)	**jnxRmonGetOk**	1.3.6.1.4.1.2636.4.3.0.2	–	–	All devices running Junos OS.
SONET Alarms					
SONET MIB (**jnx-sonet.mib**)	**jnxSonetAlarm Cleared**	1.3.6.1.4.1.2636.4.6.0.2	–	–	Devices that run Junos OS and have SONET PICs installed.

Related Documentation

- Juniper Networks Enterprise-Specific SNMP Traps on page 633
- Standard SNMP Traps Supported on Devices Running Junos OS on page 653
- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7
- Configuring SNMP Trap Options and Groups on a Device Running Junos OS
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Juniper Networks Enterprise-Specific BGP Traps

Junos OS Release 10.0 introduced two enterprise-specific traps, **jnxBgpM2Established** and **jnxBgpM2BackwardTransition**, to support BGP clients that follow IPv6 addressing. The previous versions of the Junos OS supported only the standard BGP traps, **BgpM2Established** and **BgpM2BackwardTransition**. The standard BGP traps support only IPv4 addresses, and return 0.0.0.0 as the IP address of the BGP remote peer if the remote peer is configured with an IPv6 address. The newly introduced enterprise-specific traps support IPv6 addressing and contain the proper IPv6 address of the remote peer if the remote peer is configured with an IPv6 address. However, the Junos OS continues to

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support the standard traps and generates them when the BGP clients are configured with IPv4 addresses.

Related Documentation

- Juniper Networks Enterprise-Specific SNMP Traps on page 633
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Juniper Networks Enterprise-Specific LDP Traps

For information about the enterprise-specific LDP traps, see "LDP MIB" on page 517.

Disabling LDP Traps

You can disable the LDP LSP notifications by including the **trap disable** statement at the **[show protocols ldp log-updown]** hierarchy level.

Related Documentation

- Juniper Networks Enterprise-Specific SNMP Traps on page 633
- Standard SNMP Traps Supported on Devices Running Junos OS on page 653
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Juniper Networks Enterprise-Specific License MIB Traps

For information about Juniper Networks enterprise-specific License MIB traps, see "Juniper Networks Enterprise-Specific License MIB Notifications" on page 569.

Juniper Networks Enterprise-Specific MIMSTP Traps

For information about Juniper Networks enterprise-specific MIMSTP traps that are supported on MX Series routers, see "Juniper Networks Enterprise-Specific MIMSTP Traps" on page 431.

Juniper Networks Enterprise-Specific MPLS Traps

For information about Juniper Networks enterprise-specific MPLS traps, see "Juniper Networks Enterprise-Specific MPLS Traps" on page 409.

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Juniper Networks Enterprise-Specific Network Address Translation Resources-Monitoring Traps

For information about Juniper Networks enterprise-specific Network Address Translation Resources-Monitoring traps, see ."Juniper Networks Enterprise-Specific NAT Traps" on page 486.

Juniper Networks Enterprise-Specific Traps on EX Series Ethernet Switches

EX Series Ethernet Switches support the following enterprise-specific traps:

- **jnxVccpPortUp**
- **jnxVccpPortDown**

Related Documentation

- Juniper Networks Enterprise-Specific SNMP Traps on page 633
- Standard SNMP Traps Supported on Devices Running Junos OS on page 653
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Juniper Networks Enterprise-Specific Traps on MX Series 3D Universal Edge Routers

On MX Series routers, SNMP traps are generated when the MAC address table on a logical interface or on a bridging-domain reaches its maximum number of entries. You can enable or disable the MAC address learning feature and also configure the maximum number of MAC entries that a logical interface or bridging-domain can store in the MAC address table.

The following traps, defined in the L2ALD MIB, **jnxl2ald.mib**, are generated when the respective MAC limit is reached:

- **jnxl2aldRoutingInstMacLimit**—Generated when the number of MAC addresses for the given routing instance, **jnxl2aldRoutingInst**, exceeds the set limit.
- **jnxl2aldInterfaceMacLimit**—Generated when the number of MAC addresses for the given physical interface exceeds the configured limit.
- **jnxl2aldGlobalMacLimit**—Generated when the number of MAC addresses for the entire system exceeds the configured limit.

For information about Juniper Networks enterprise-specific MIMSTP traps that are supported on MX Series routers, see "Juniper Networks Enterprise-Specific MIMSTP Traps" on page 431.

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Related Documentation

- Juniper Networks Enterprise-Specific SNMP Traps on page 633
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Raising Traps for Events Based on System Log Messages

Event policies can include an action that raises traps for events based on system log messages. This feature enables notification of an SNMP trap-based application when an important system log message occurs. You can convert any system log message (for which there are no corresponding traps) into a trap. This feature is valuable for customers who use network management system traps rather than system log messages to monitor their networks.

For information about converting system log messages into traps, see the *Junos OS Configuration and Operations Automation Guide*. For information about the System Log MIB that provides support for this feature, see "System Log MIB" on page 309.

Related Documentation

- Juniper Networks Enterprise-Specific SNMP Traps on page 633
- Standard SNMP Traps Supported on Devices Running Junos OS on page 653
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Spoofing Enterprise-Specific SNMP Traps

You can use the **request snmp spoof-trap** operational mode command to mimic SNMP trap behavior. The contents of the traps (the values and instances of the objects carried in the trap) can be specified on the command line or they can be spoofed automatically. This feature is useful if you want to trigger SNMP traps from routers and ensure they are processed correctly within your existing network management infrastructure, but find it difficult to simulate the error conditions that trigger many of the traps on the router. For more information, see the *Junos OS System Basics and Services Command Reference*.

Related Documentation

- Juniper Networks Enterprise-Specific SNMP Traps on page 633
- Standard SNMP Traps Supported on Devices Running Junos OS on page 653
- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

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CHAPTER 78

Standard SNMP Traps

- Standard SNMP Traps Supported on Devices Running Junos OS on page 653
- Standard SNMP Version 1 Traps on page 654
- Standard SNMP Version 2 Traps on page 657
- Standard SNMP Traps on EX Series Ethernet Switches on page 661
- Unsupported Standard SNMP Traps on page 662
- Spoofing Standard SNMP Traps on page 665

Standard SNMP Traps Supported on Devices Running Junos OS

This topic provides pointers to the standard SNMP traps supported by the Junos OS.



NOTE: For scalability reasons, the MPLS traps are generated by the ingress router only. For information about disabling the generation of MPLS traps, see the *Junos OS MPLS Applications Configuration Guide*.

- Standard SNMP Version 1 Traps on page 654
- Standard SNMP Version 2 Traps on page 657
- Standard SNMP Traps on EX Series Ethernet Switches on page 661
- Unsupported Standard SNMP Traps on page 662

Related Documentation

- Juniper Networks Enterprise-Specific SNMP Traps on page 633
- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7
- Configuring SNMP Trap Options and Groups on a Device Running Junos OS
- Managing Traps and Informs

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Standard SNMP Version 1 Traps

Table 287 on page 654 provides an overview of the standard traps for SNMPv1. The traps are organized first by trap category and then by trap name, and include their enterprise ID, generic trap number, and specific trap number. The system logging severity levels are listed for those traps that have them with their corresponding system log tag. Traps that do not have corresponding system logging severity levels are marked with an en dash (–) in the table.

For more information about system log messages, see the *Junos OS System Log Messages Reference*. For more information about configuring system logging, see the *Junos OS System Basics Configuration Guide*.

Table 287: Standard Supported SNMP Version 1 Traps

Defined in	Trap Name	Enterprise ID	Generic Trap Number	Specific Trap Number	System Logging Severity Level	Syslog Tag	Supported On
Startup Notifications							
RFC 1215, *Conventions for Defining Traps for Use with the SNMP*	**authenticationFailure**	1.3.6.1.4.1.2636	4	0	Notice	SNMPD_ TRAP_ GEN_FAILURE	All devices running Junos OS.
	coldStart	1.3.6.1.4.1.2636	0	0	Critical	SNMPD_TRAP_ COLD_START	All devices running Junos OS.
	warmStart	1.3.6.1.4.1.2636	1	0	Error	SNMPD_TRAP_ WARM_START	All devices running Junos OS.
Link Notifications							
RFC 1215, *Conventions for Defining Traps for Use with the SNMP*	**linkDown**	1.3.6.1.4.1.2636	2	0	Warning	SNMP_ TRAP_ LINK_DOWN	All devices running Junos OS.
	linkUp	1.3.6.1.4.1.2636	3	0	Info	SNMP_TRAP_ LINK_UP	All devices running Junos OS.
Remote Operations Notifications							
RFC 2925, *Definitions of Managed Objects for Remote Ping, Traceroute, and Lookup Operations*	**pingProbeFailed**	1.3.6.1.2.1.80.0	6	1	Info	SNMP_TRAP _PING_ PROBE_ FAILED	All devices running Junos OS.
	pingTestFailed	1.3.6.1.2.1.80.0	6	2	Info	SNMP_TRAP_ PING_TEST _FAILED	All devices running Junos OS.
	pingTestCompleted	1.3.6.1.2.1.80.0	6	3	Info	SNMP_TRAP_ PING_TEST_ COMPLETED	All devices running Junos OS.
	traceRoutePathChange	1.3.6.1.2.1.81.0	6	1	Info	SNMP_TRAP_ TRACE_ROUTE_ PATH_CHANGE	All devices running Junos OS.

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Table 287: Standard Supported SNMP Version 1 Traps *(continued)*

Defined in	Trap Name	Enterprise ID	Generic Trap Number	Specific Trap Number	System Logging Severity Level	Syslog Tag	Supported On
	traceRouteTestFailed	1.3.6.1.2.1.81.0	6	2	Info	SNMP_TRAP_ TRACE_ROUTE_ TEST_FAILED	All devices running Junos OS.
	traceRouteTestCompleted	1.3.6.1.2.1.81.0	6	3	Info	SNMP_TRAP_ TRACE_ROUTE_ TEST_COMPLETED	All devices running Junos OS.
RMON Alarms							
RFC 2819a, *RMON MIB*	**fallingAlarm**	1.3.6.1.2.1.16	6	2	–	–	All devices running Junos OS.
	risingAlarm	1.3.6.1.2.1.16	6	1	–	–	All devices running Junos OS.
Routing Notifications							
BGP 4 MIB	**bgpEstablished**	1.3.6.1.2.1.15.7	6	1	–	–	M, T, MX, J, EX, and SRX for branch devices.
	bgpBackwardTransition	1.3.6.1.2.1.15.7	6	2	–	–	M, T, MX, J, EX, and SRX for branch devices.

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Table 287: Standard Supported SNMP Version 1 Traps *(continued)*

Defined in	Trap Name	Enterprise ID	Generic Trap Number	Specific Trap Number	System Logging Severity Level	Syslog Tag	Supported On
OSPF TRAP MIB	**ospfVirtIfStateChange**	1.3.6.1.2.1.14.16.2	6	1	–	–	M, T, MX, J, EX, and SRX for branch devices.
	ospfNbrStateChange	1.3.6.1.2.1.14.16.2	6	2	–	–	M, T, MX, J, EX, and SRX for branch devices.
	ospfVirtNbrStateChange	1.3.6.1.2.1.14.16.2	6	3	–	–	M, T, MX, J, EX, and SRX for branch devices.
	ospfIfConfigError	1.3.6.1.2.1.14.16.2	6	4	–	–	M, T, MX, J, EX, and SRX for branch devices.
	ospfVirtIfConfigError	1.3.6.1.2.1.14.16.2	6	5	–	–	M, T, MX, J, EX, and SRX for branch devices.
	ospfIfAuthFailure	1.3.6.1.2.1.14.16.2	6	6	–	–	M, T, MX, J, EX, and SRX for branch devices.
	ospfVirtIfAuthFailure	1.3.6.1.2.1.14.16.2	6	7	–	–	M, T, MX, J, EX, and SRX for branch devices.
	ospfIfRxBadPacket	1.3.6.1.2.1.14.16.2	6	8	–	–	M, T, MX, J, EX, and SRX for branch devices.
	ospfVirtIfRxBadPacket	1.3.6.1.2.1.14.16.2	6	9	–	–	M, T, MX, J, EX, and SRX for branch devices.
	ospfTxRetransmit	1.3.6.1.2.1.14.16.2	6	10	–	–	M, T, MX, J, EX, and SRX for branch devices.
	ospfVirtIfTxRetransmit	1.3.6.1.2.1.14.16.2	6	11	–	–	M, T, MX, J, EX, and SRX for branch devices.
	ospfMaxAgeLsa	1.3.6.1.2.1.14.16.2	6	13	–	–	M, T, MX, J, EX, and SRX for branch devices.

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Table 287: Standard Supported SNMP Version 1 Traps *(continued)*

Defined in	Trap Name	Enterprise ID	Generic Trap Number	Specific Trap Number	System Logging Severity Level	Syslog Tag	Supported On
	ospfIfStateChange	1.3.6.1.2.1.14.16.2	6	16	–	–	M, T, MX, J, EX, and SRX for branch devices.
VRRP Notifications							
RFC 2787, *Definitions of Managed Objects for the Virtual Router Redundancy Protocol*	**vrrpTrapNewMaster**	1.3.6.1.2.1.68	6	1	Warning	VRRPD_NEW MASTER_TRAP	All devices running Junos OS.
	vrrpTrapAuthFailure	1.3.6.1.2.1.68	6	2	Warning	VRRPD_AUTH_ FAILURE_TRAP	All devices running Junos OS.

Related Documentation

- Juniper Networks Enterprise-Specific SNMP Traps on page 633
- Standard SNMP Traps Supported on Devices Running Junos OS on page 653
- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7
- Configuring SNMP Trap Options and Groups on a Device Running Junos OS
- Managing Traps and Informs

Standard SNMP Version 2 Traps

Table 288 on page 658 provides an overview of the standard SNMPv2 traps supported by the Junos OS. The traps are organized first by trap category and then by trap name and include their **snmpTrapOID**. The system logging severity levels are listed for those traps that have them with their corresponding system log tag. Traps that do not have corresponding system logging severity levels are marked with an en dash (–) in the table.

For more information about system log messages, see the *Junos OS System Log Messages Reference*. For more information about configuring system logging, see the *Junos OS System Basics Configuration Guide*.

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Table 288: Standard Supported SNMP Version 2 Traps

Defined in	Trap Name	snmpTrapOID	System Logging Severity Level	Syslog Tag	Supported On
Startup Notifications					
RFC 1907, *Management Information Base for Version 2 of the Simple Network Management Protocol (SNMPv2)*	**coldStart**	1.3.6.1.6.3.1.1.5.1	Critical	SNMPD_TRAP_COLD_START	All devices running Junos OS.
	warmStart	1.3.6.1.6.3.1.1.5.2	Error	SNMPD_TRAP_WARM_START	All devices running Junos OS.
	authenticationFailure	1.3.6.1.6.3.1.1.5.5	Notice	SNMPD_TRAP_GEN_FAILURE	All devices running Junos OS.
Link Notifications					
RFC 2863, *The Interfaces Group MIB*	**linkDown**	1.3.6.1.6.3.1.1.5.3	Warning	SNMP_TRAP_LINK_DOWN	All devices running Junos OS.
	linkUp	1.3.6.1.6.3.1.1.5.4	Info	SNMP_TRAP_LINK_UP	All devices running Junos OS.
Remote Operations Notifications					
RFC 2925, *Definitions of Managed Objects for Remote Ping, Traceroute, and Lookup Operations*	**pingProbeFailed**	1.3.6.1.2.1.80.0.1	Info	SNMP_TRAP_PING_PROBE_FAILED	All devices running Junos OS.
	pingTestFailed	1.3.6.1.2.1.80.0.2	Info	SNMP_TRAP_PING_TEST_FAILED	All devices running Junos OS.
	pingTestCompleted	1.3.6.1.2.1.80.0.3	Info	SNMP_TRAP_PING_TEST_COMPLETED	All devices running Junos OS.
	traceRoutePathChange	1.3.6.1.2.1.81.0.1	Info	SNMP_TRAP_TRACE_ROUTE_PATH_CHANGE	All devices running Junos OS.
	traceRouteTestFailed	1.3.6.1.2.1.81.0.2	Info	SNMP_TRAP_TRACE_ROUTE_TEST_FAILED	All devices running Junos OS.
	traceRouteTestCompleted	1.3.6.1.2.1.81.0.3	Info	SNMP_TRAP_TRACE_ROUTE_TEST_COMPLETED	All devices running Junos OS.
RMON Alarms					
RFC 2819a, *RMON MIB*	**fallingAlarm**	1.3.6.1.2.1.16.0.1	–	–	All devices running Junos OS.
	risingAlarm	1.3.6.1.2.1.16.0.2	–	–	All devices running Junos OS.

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Table 288: Standard Supported SNMP Version 2 Traps *(continued)*

Defined in	Trap Name	snmpTrapOID	System Logging Severity Level	Syslog Tag	Supported On
Routing Notifications					
BGP 4 MIB	**bgpEstablished**	1.3.6.1.2.1.15.7.1	–	–	M, MX, T, EX, J, and SRX for branch devices.
	bgpBackwardTransition	1.3.6.1.2.1.15.7.2	–	–	M, MX, T, EX, J, and SRX for branch devices.
OSPF Trap MIB	**ospfVirtIfStateChange**	1.3.6.1.2.1.14.16.2.1	–	–	M, MX, T, EX, J, and SRX for branch devices.
	ospfNbrStateChange	1.3.6.1.2.1.14.16.2.2	–	–	M, MX, T, EX, J, and SRX for branch devices.
	ospfVirtNbrStateChange	1.3.6.1.2.1.14.16.2.3	–	–	M, MX, T, EX, J, and SRX for branch devices.
	ospfIfConfigError	1.3.6.1.2.1.14.16.2.4	–	–	M, MX, T, EX, J, and SRX for branch devices.
	ospfVirtIfConfigError	1.3.6.1.2.1.14.16.2.5	–	–	M, MX, T, EX, J, and SRX for branch devices.
	ospfIfAuthFailure	1.3.6.1.2.1.14.16.2.6	–	–	M, MX, T, EX, J, and SRX for branch devices.
	ospfVirtIfAuthFailure	1.3.6.1.2.1.14.16.2.7	–	–	M, MX, T, EX, J, and SRX for branch devices.
	ospfIfRxBadPacket	1.3.6.1.2.1.14.16.2.8	–	–	M, MX, T, EX, J, and SRX for branch devices.
	ospfVirtIfRxBadPacket	1.3.6.1.2.1.14.16.2.9	–	–	M, MX, T, EX, J, and SRX for branch devices.
	ospfTxRetransmit	1.3.6.1.2.1.14.16.2.10	–	–	M, MX, T, EX, J, and SRX for branch devices.

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Table 288: Standard Supported SNMP Version 2 Traps *(continued)*

Defined in	Trap Name	snmpTrapOID	System Logging Severity Level	Syslog Tag	Supported On
	ospfVirtIfTxRetransmit	1.3.6.1.2.1.14.16.2.11	–	–	M, MX, T, EX, J, and SRX for branch devices.
	ospfMaxAgeLsa	1.3.6.1.2.1.14.16.2.13	–	–	M, MX, T, EX, J, and SRX for branch devices.
	ospfIfStateChange	1.3.6.1.2.1.14.16.2.16	–	–	M, MX, T, EX, J, and SRX for branch devices.
RFC 2787, *Definitions of Managed Objects for the Virtual Router Redundancy Protocol*	**vrrpTrapNewMaster**	1.3.6.1.2.1.68.0.1	Warning	VRRPD_ NEWMASTER_TRAP	All devices running Junos OS.
	vrrpTrapAuthFailure	1.3.6.1.2.1.68.0.2	Warning	VRRPD_AUTH_ FAILURE_ TRAP	All devices running Junos OS.

The Junos OS also supports the following standard SNMP version 2 traps:

- SNMP Version 2 MPLS Traps on page 660

SNMP Version 2 MPLS Traps

The Junos OS supports the MPLS SNMP version 2 traps defined in RFC 3812, *Multiprotocol Label Switching (MPLS) Traffic Engineering (TE) Management Information Base*.

You can disable the MPLS traps by including the **no-trap** option at the **[edit protocol mpls log-updown]** hierarchy level. For information about disabling the generation of MPLS traps, see the *Junos OS MPLS Applications Configuration Guide*.

The Junos OS supports the following MPLS traps:

- **mplsTunnelUp**—Generated when an **mplsTunnelOperStatus** object for one of the configured tunnels leaves the **down** state and transitions into another state, other than the **notPresent** state.
- **mplsTunnelDown**—Generated when an **mplsTunnelOperStatus** object for one of the configured tunnels enters the **down** state from a state other than the **notPresent** state.



NOTE: When an LSP flaps, only the ingress and egress routers of that LSP generate the **mplsTunnelUp** and **mplsTunnelDown** traps. Previously, all the routers associated with an LSP—that is, the ingress, egress, and transit routers—used to generate the traps when the LSP flaps.

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- **mplsTunnelRerouted**—Generated when a tunnel is rerouted.
- **mplsTunnelReoptimized**—Generated when a tunnel is reoptimized.



NOTE: In Junos OS Release 8.3 and earlier, **mplsTunnelReoptimized** was generated every time the optimization timer expired; that is, when the optimization timer exceeded the value set for the **optimize-timer** statement at the **[edit protocols mpls label-switched-path path-name]** hierarchy level. However, in Release 8.4 and later, this trap is generated only when the path is reoptimized, and not when the optimization timer expires.

Related Documentation

- Juniper Networks Enterprise-Specific SNMP Traps on page 633
- Standard SNMP Traps Supported on Devices Running Junos OS on page 653
- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7
- Configuring SNMP Trap Options and Groups on a Device Running Junos OS
- Managing Traps and Informs

Standard SNMP Traps on EX Series Ethernet Switches

Apart from the standard SNMP traps listed in the preceding sections, the Junos OS also supports the following standard traps on the EX Series Ethernet Switches:

- **ptopoConfigChange**—Generated when the value of **ptopoLastChangeTime** changes. Enables a network management system to trigger physical topology table maintenance polls (RFC 2622, *PTOPO MIB*).
- **pethPsePortOnOffNotification**—Generated when the power supply status of a PSE port changes. Indicates whether the PSE port is supplying power to the PD port or not (RFC 3621, *Power Ethernet MIB*).
- **pethMainPowerUsageOnNotification**—Generated when the PSE threshold indicator is on (RFC 3621, *Power Ethernet MIB*).
- **pethMainPowerUsageOffNotification**—Generated when the PSE threshold indicator is off (RFC 3621, *Power Ethernet MIB*).

Related Documentation

- Juniper Networks Enterprise-Specific SNMP Traps on page 633
- Standard SNMP Traps Supported on Devices Running Junos OS on page 653
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- Standard SNMP MIBs Supported by Junos OS on page 7
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Unsupported Standard SNMP Traps

Standard SNMP traps that are defined in MIBs supported by the Junos OS but are not generated by the Junos OS are shown in Table 289 on page 663.

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Table 289: Unsupported Standard SNMP Traps

MIB	Trap Name	Description
isismib.mib	isisDatabaseOverload	Generated when the system enters or leaves the overload state.
	isisManualAddressDrops	Generated when one of the manual **areaAddresses** assigned to the system is ignored when computing routes.
	isisCorruptedLSPDetected	Generated when an LSP stored in memory becomes corrupted.
	isisAttemptToExceedMaxSequence	Generated when the sequence number on a generated LSP wraps the 32-bit sequence counter and the number is purged.
	isisIDLenMismatch	Generated when a protocol data unit (PDU) is received with a different value for the system ID length. This trap includes an index to identify the circuit where the PDU was received and the PDU header.
	isisMaxAreaAddressesMismatch	Generated when a PDU with a different value for the maximum area addresses is received.
	isisOwnLSPPurge	Generated when a PDU is received with a system ID and zero age. This notification includes the circuit index if available.
	isisSequenceNumberSkip	Generated when an LSP is received with a system ID and different contents, indicating the LSP might require a higher sequence number.
	isisAuthenticationTypeFailure	Generated when a PDU with the wrong authentication type field is received.
	isisAuthenticationFailure	Generated when a PDU with an incorrect authentication information field is received.
	isisVersionSkew	Generated when a hello PDU from an IS running a different version of the protocol is received.
	isisAreaMismatch	Generated when a hello PDU from an IS which does not share any area address is received.
	isisRejectedAdjacency	Generated when a hello PDU from an IS is received, but no adjacency is established because of a lack of resources.
	isisLSPTooLargeToPropagate	Generated when a link-state PDU that is larger than the **dataLinkBlockSize** for a circuit is attempted, but not propagated.
	isisOriginatingLSPBufferSizeMismatch	

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Table 289: Unsupported Standard SNMP Traps *(continued)*

MIB	Trap Name	Description
		Generated when a Level 1 link-state PDU or Level 2 link-state PDU is received that is larger than the local value for originating **L1LSPBufferSize** or originating **L2LSPBufferSize**, respectively, or when a Level 1 link-state PDU or Level 2 link-state PDU is received containing the originating **LSPBufferSize** option and the value in the PDU option field does not match the local value for originating **L1LSPBufferSize** or originating **L2LSPBufferSize**, respectively.
	isisProtocolsSupportedMismatch	Generated when a nonpseudonode, segment 0 link-state PDU is received that has no matching protocols.
l3vpnmib.mib	**mplsVrfIfUp**	Generated when the **ifOperStatus** of an interface associated with a VRF table changes to the **up(1)** state, or when an interface with **ifOperStatus = up(1)** is associated with a VRF table.
	mplsVrfIfDown	Generated when the **ifOperStatus** of an interface associated with a VRF table changes to the **down(1)** state, or when an interface with **ifOperStatus = up(1)** state is disassociated from a VRF table.
	mplsNumVrfRouteMidThreshExceeded	Generated when the number of routes contained by the specified VRF table exceeds the value indicated by **mplsVrfMidRouteThreshold**.
	mplsNumVrfRouteMaxThreshExceeded	Generated when the number of routes contained by the specified VRF table reaches or attempts to exceed the maximum allowed value as indicated by **mplsVrfMaxRouteThreshold**.
	mplsNumVrfSecIllglLblThrshExcd	Generated when the number of illegal label violations on a VRF table as indicated by **mplsVpnVrfSecIllegalLblVltns** has exceeded **mplsVpnVrfSecIllegalLblRcvThrsh**.
msdpmib.mib	**msdpEstablished**	Generated when the Multicast Source Discovery Protocol (MSDP) finite state machine (FSM) enters the **Established** state.
	msdpBackwardTransition	Generated when the MSDP FSM moves from a higher numbered state to a lower numbered state.
ospf2trap.mib	**ospfOriginateLsa**	Generated when a new LSA is originated by the router because of a topology change.
	ospfLsdbOverflow	Generated when the number of LSAs in the router's link-state database exceeds the value of **ospfExtLsdbLimit**.
	ospfLsdbApproachingOverflow	Generated when the number of LSAs in the router's link-state database exceeds 90% of the value of **ospfExtLsdbLimit**.

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Table 289: Unsupported Standard SNMP Traps *(continued)*

MIB	Trap Name	Description
rfc1747.mib	sdlcPortStatusChange	Generated when the state of an SDLC port transitions to active or inactive.
	sdlcLSStatusChange	Generated when the state of an SDLC link station transitions to contacted or disconnected.
rfc2115a.mib	frDLCIStatusChange	Generated when a virtual circuit changes state (has been created or invalidated, or has toggled between the active and inactive states).
rfc2662.mib	adslAtucRateChangeTrap	Generated when the ATUCs transmit rate has changed (RADSL mode only).
	adslAtucPerfLofsThreshTrap	Generated when the loss of framing 15-minute interval threshold is reached.
	adslAtucInitFailureTrap	Generated when ATUC initialization fails.
	adslAturPerfLprsThreshTrap	Generated when the loss of power 15-minute interval threshold is reached.
	adslAturRateChangeTrap	Generated when the ATURs transmit rate changes (RADSL mode only).
rfc3020.mib	mfrMibTrapBundleLinkMismatch	Generated when a bundle link mismatch is detected.
rfc3813.mib	mplsXCUp	Generated when **mplsXCOperStatus** for one or more contiguous entries in **mplsXCTable** enters the **up(1)** state from some other state.
	mplsXCDown	Generated when **mplsXCOperStatus** for one or more contiguous entries in **mplsXCTable** enters the **down(2)** state from some other state.

Related Documentation

- Juniper Networks Enterprise-Specific SNMP Traps on page 633
- Standard SNMP Traps Supported on Devices Running Junos OS on page 653
- Juniper Networks Enterprise-Specific MIBs on page 30
- Standard SNMP MIBs Supported by Junos OS on page 7

Spoofing Standard SNMP Traps

You can use the **request snmp spoof-trap** operational mode command to mimic SNMP trap behavior. The contents of the traps (the values and instances of the objects carried in the trap) can be specified on the command line or they can be spoofed automatically. This feature is useful if you want to trigger SNMP traps from routers and ensure they are processed correctly within your existing network management infrastructure, but find it

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difficult to simulate the error conditions that trigger many of the traps on the router. For more information, see the *Junos OS System Basics and Services Command Reference*.

Related Documentation

- Juniper Networks Enterprise-Specific SNMP Traps on page 633
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PART 4

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